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One Shot Handwritten Notes

CLASS 12 MATHS

KHATAM IN 199 PAGES

Session 2026-27



FREE e-book

Formulas | Most Asked Qs | Memes | Mock Papers

By Shivam Bhaiya

One Shot handwritten notes | All 13 chapters | Formulas + Tricks | Most Important Questions

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Theory 80 marks
+ 20 internal assessment

KHATAM IN 199 PAGES

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5	Vectors, 3D Geometry, LPP
6	Probability + Formula Sheet + Most Asked Bank
7	Mock Papers 1 + 2 (3 hours each) + Last-night revision



Shivam Bhaiya says

Class 12 Maths mein **Calculus = 35 marks**. Pehle Calculus pakka karo, phir Vectors + 3D, phir baaki. Har chapter: notes, Formula Box, solved examples khud karo, phir Most Important Questions bina dekhe.

Board paper pattern (80 marks, 3 hrs)

Section	Type	Marks
A	18 MCQ + 2 A-R	20 x 1
B	5 very short	5 x 2
C	6 short	6 x 3
D	4 long	4 x 5
E	3 case based	3 x 4

Unit-wise weightage

Unit	Marks
Relations & Functions	8
Algebra	10
Calculus	35
Vectors & 3D	14
Linear Programming	5
Probability	8

Is book ko kaise padhein

- ✓ Purple **Formula Box** = chapter ke saare formulas.
- ✓ Blue box = solved example, steps ke saath.
- ✓ Orange "Yaad karo" = trick; red dashed = common mistakes.
- ✓ Answers green mein: pehle khud solve karo, phir milao.

Is book mein kya kya hai

13 chapters, CBSE 2026-27	Notes + Formula Box
740+ Most Important Qs	solutions green mein
540+ solved examples	step by step
75+ Yaad karo tricks	+ 30 memes
200+ Most Asked Qs	board pattern bank
2 full mock papers	80 marks, with answers

Most asked: show R is an equivalence relation, prove f is one-one and onto

RELATIONS & FUNCTIONS

1. Relation: the Basics

For sets A and B, the **Cartesian product** $A \times B = \{(a, b) : a \in A,\ b \in B\}$. If $n(A) = m$ and $n(B) = n$, then $n(A \times B) = mn$.

A **relation** R from A to B is any subset of $A \times B$. A relation **on** A (or "in A") is a subset of $A \times A$. If $(a, b) \in R$ we write $a\,R\,b$: "a is related to b".

- → **Domain** = set of first entries, **Range** = set of second entries, **Codomain** = B.
- → Number of relations from A to B = 2^{mn}; on A with $n(A) = n$: 2^{n^2}.

Key terms: three special relations

- → **Empty relation** $R = \phi$: no element is related to any element (e.g. $R = \{(a, b) : a - b = 10\}$ on $\{1, 2, 3\}$).
- → **Universal relation** $R = A \times A$: every element is related to every element (e.g. $R = \{(a, b) : |a - b| \geq 0\}$).
- → **Identity relation** $I_A = \{(a, a) : a \in A\}$: each element related only to itself.

Empty and universal relations are called **trivial relations**.

2. Reflexive, Symmetric, Transitive

Let R be a relation on a set A. Then R is:

Type	Condition (for all elements of A)
Reflexive	$(a, a) \in R$ for **every** $a \in A$
Symmetric	$(a, b) \in R \Rightarrow (b, a) \in R$
Transitive	$(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$

Definition: Equivalence relation

R is an **equivalence relation** if it is **reflexive, symmetric and transitive** (all three).

Yaad karo: "RST = Ra Sa Ta"

Equivalence ke liye teeno chahiye: **R**eflexive (khud se), **S**ymmetric (ulta bhi), **T**ransitive (chain poori). Ek bhi fail, toh equivalence **nahi**.

How to check on a finite set (listed pairs)

- → **Reflexive:** all diagonal pairs $(1, 1), (2, 2), \ldots$ present? One missing = not reflexive.
- → **Symmetric:** for each (a, b) with $a \neq b$, is (b, a) there?
- → **Transitive:** for every chain $(a, b), (b, c)$, is (a, c) there? No chain at all = transitive (vacuously).

Solved Example 1

$A = \{1, 2, 3\}$, $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3)\}$. Check reflexive, symmetric, transitive.

1. $(1, 1), (2, 2), (3, 3) \in R$, so **reflexive**.
2. $(1, 2) \in R$ but $(2, 1)$ is not in R, so **not symmetric**.
3. $(1, 2), (2, 3) \in R$ but $(1, 3)$ is not in R, so **not transitive**.

✔ Reflexive only (not an equivalence relation)

Solved Example 2

Check $R = \{(a, b) : a \leq b^2\}$ on $\mathbb{R}$ for reflexivity, symmetry and transitivity.

1. Take $a = \frac{1}{2}$: $\frac{1}{2} \leq \frac{1}{4}$ is false, so $(\frac{1}{2}, \frac{1}{2})$ is not in R: **not reflexive**.
2. $(1, 2) \in R$ as $1 \leq 4$, but $(2, 1)$ is not in R as $2 \leq 1$ is false: **not symmetric**.
3. $(5, -3) \in R$ ($5 \leq 9$), $(-3, 1) \in R$ ($-3 \leq 1$), but $(5, 1)$ is not in R ($5 \leq 1$ false): **not transitive**.

✔ Neither reflexive, nor symmetric, nor transitive

Solved Example 3

Show that $R = \{(a, b) : 1 + ab > 0\}$ on $\mathbb{R}$ is reflexive and symmetric but not transitive.

1. Reflexive: $1 + a \times a = 1 + a^2 > 0$ for every real a.
2. Symmetric: $1 + ab > 0 \Rightarrow 1 + ba > 0$.
3. Take $a = 2$, $b = -\frac{1}{4}$, $c = -2$: $1 + ab = \frac{1}{2} > 0$ and $1 + bc = \frac{3}{2} > 0$, but $1 + ac = -3 < 0$.

✔ Reflexive, symmetric, not transitive

Solved Example 4

$A = \{1, 2, 3, 4, 5, 6\}$, $R = \{(a, b) : b = a + 1\}$. Write R in roster form and check its properties.

Quick fact On a set with n elements there are n pairs on the diagonal.

1. $R = \{(1,2),(2,3),(3,4),(4,5),(5,6)\}$.
2. $(1,1)$ is not in R: not reflexive. $(1,2) \in R$, $(2,1)$ is not in R: not symmetric.
3. $(1,2),(2,3) \in R$ but $(1,3)$ is not in R: not transitive.

✔ None of the three properties

Solved Example 5

$R = \{(a,b) : a + 2b = 10\}$ is a relation on $\mathbb{N}$. Write R, its domain and range, and check its properties.

1. $b = 1,2,3,4$ give $a = 8,6,4,2$: $R = \{(8,1),(6,2),(4,3),(2,4)\}$.
2. Domain $= \{2,4,6,8\}$, range $= \{1,2,3,4\}$.
3. $(1,1)$ missing: not reflexive. $(8,1) \in R$, $(1,8)$ not $(1 + 16 \neq 10)$: not symmetric.
4. $(6,2),(2,4) \in R$ but $(6,4)$ not $(6 + 8 \neq 10)$: not transitive.

✔ None of reflexive, symmetric, transitive

5-mark answer format (equivalence)

1. **Reflexive:** "Let $a \in A$. Since ..., $(a,a) \in R$."
2. **Symmetric:** "Let $(a,b) \in R \Rightarrow \ldots \Rightarrow (b,a) \in R$."
3. **Transitive:** "Let $(a,b),(b,c) \in R \Rightarrow$ ... add/multiply ... $\Rightarrow (a,c) \in R$."
4. Conclude: "Hence R is an equivalence relation." Then the class.

Solved Example 6

$A = \{1,2,3,4\}$, $R = \{(1,1),(2,2),(3,3),(4,4),(1,2),(1,3),(3,2)\}$. Check reflexive, symmetric and transitive.

1. All four (a,a) present: reflexive.
2. $(1,2) \in R$, $(2,1)$ is not in R: not symmetric.
3. Chains: $(1,3),(3,2) \Rightarrow (1,2) \in R$ ✓; chains through (a,a) give back the same pair. Every chain closes.

✔ Reflexive and transitive, not symmetric



Shivam Bhaiya says
Counter-example dene mein **specific numbers** likho, jaise $(5,-3)$ aur $(-3,1)$. "Generally true nahi hai" likhoge toh marks katenge. Ek solid counter-example = poore marks.

3. Equivalence Class

If R is an equivalence relation on A, the **equivalence class** of a is $[a] = \{x \in A : (x,a) \in R\}$

= all elements related to a.

→ Two classes are either identical or disjoint.

→ The union of all classes is A: the classes form a **partition** of A.

→ For "$a - b$ divisible by n" on $\mathbb{Z}$: exactly n classes $[0],[1],\ldots,[n-1]$.

Solved Example 7

Show that $R = \{(a,b) : a - b \text{ is divisible by } 5\}$ on $\mathbb{Z}$ is an equivalence relation. Write its classes.

1. Reflexive: $a - a = 0 = 5 \times 0$, so $(a,a) \in R$.
2. Symmetric: $a - b = 5k \Rightarrow b - a = 5(-k)$, so $(b,a) \in R$.
3. Transitive: $a - b = 5k, b - c = 5m \Rightarrow a - c = 5(k + m)$, so $(a,c) \in R$.
4. Classes by remainder on division by 5: $[0],[1],[2],[3],[4]$, e.g. $[2] = \{\ldots,-8,-3,2,7,12,\ldots\}$.

✔ Equivalence relation with 5 classes

Solved Example 8

$A = \{0,1,2,\ldots,10\}$, $R = \{(a,b) : |a - b| \text{ is a multiple of } 3\}$. Show R is an equivalence relation and find $[2]$ and the set of elements related to 1.

1. $|a - a| = 0$ is a multiple of 3: reflexive.
2. $|a - b| = |b - a|$: symmetric.
3. $a - b = 3k, b - c = 3m \Rightarrow a - c = 3(k + m)$: transitive.
4. $[2]$ = numbers in A leaving remainder 2 on division by 3: $\{2,5,8\}$; related to 1: $\{1,4,7,10\}$.

✔ $[2] = \{2,5,8\}$, related to 1: $\{1,4,7,10\}$



Meme samjho
Transitive check mein koi chain $(a,b),(b,c)$ hi nahi mili? Toh fail kaise karega? **Smart!**
Asli baat: A relation with no such chain is transitive (vacuously), e.g. $R = \{(1,2)\}$ on $\{1,2,3\}$ is transitive.

Solved Example 9

Let $A = \{1,2,\ldots,9\}$ and R on $A \times A$: $(a,b)\,R\,(c,d)$ iff $a + d = b + c$. Show R is an equivalence relation and find $[(3,5)]$.

Quick fact Identity relation is reflexive, symmetric and transitive.

1. Reflexive: $a+b=b+a$, so $(a,b)R(a,b)$.
2. Symmetric: $a+d=b+c \Rightarrow c+b=d+a$, so $(c,d)R(a,b)$.
3. Transitive: $a+d=b+c$ and $c+f=d+e$. Add: $a+d+c+f=b+c+d+e \Rightarrow a+f=b+e$.
4. $(x,y)R(3,5) \iff x+5=y+3 \iff y=x+2$.

✔ $[(3,5)] = \{(1,3),(2,4),(3,5),(4,6),(5,7),(6,8),(7,9)\}$

Solved Example 10

Find the smallest equivalence relation on $A=\{1,2,3\}$ containing $(1,2)$. How many equivalence relations on A contain $(1,2)$?

1. Must contain $(1,1),(2,2),(3,3)$ (reflexive) and $(2,1)$ (symmetric).
2. $R_1=\{(1,1),(2,2),(3,3),(1,2),(2,1)\}$ is already transitive.
3. Adding $(2,3)$ forces $(3,2),(1,3),(3,1)$, giving $A\times A$. No other option.

✔ Smallest $=R_1$; number of such relations $=2$

Solved Example 11

L = set of all lines in a plane, $R=\{(L_1,L_2): L_1 \parallel L_2\}$. Show R is an equivalence relation and find all lines related to $y=2x+4$.

1. Every line is parallel to itself (convention): reflexive.
2. $L_1 \parallel L_2 \Rightarrow L_2 \parallel L_1$: symmetric.
3. $L_1 \parallel L_2$ and $L_2 \parallel L_3 \Rightarrow L_1 \parallel L_3$: transitive.
4. Lines parallel to $y=2x+4$ have slope 2.

✔ Equivalence; related lines: $y=2x+c, c\in\mathbb{R}$

Solved Example 12

T = set of triangles, T_1RT_2 iff T_1 is similar to T_2. Which of $T_1(3,4,5)$, $T_2(5,12,13)$, $T_3(6,8,10)$ are related?

1. Similar triangles have proportional sides.
2. $\frac{6}{3}=\frac{8}{4}=\frac{10}{5}=2$: $T_1 \sim T_3$.
3. $\frac{5}{3}\neq\frac{12}{4}$: T_2 is not similar to T_1 (or T_3).

✔ Only T_1 and T_3 are related

Quick fact Empty relation on a non-empty set: symmetric, transitive, not reflexive.

Solved Example 13

For $A=\{a,b,c\}$ find the number of (i) relations (ii) reflexive relations (iii) symmetric relations (iv) relations that are reflexive and symmetric.

1. (i) $2^9=512$
2. (ii) $2^{9-3}=2^6=64$
3. (iii) $2^{\frac{3\times4}{2}}=2^6=64$
4. (iv) $2^{\frac{3\times2}{2}}=2^3=8$

✔ 512, 64, 64, 8

4. Counting Relations

Formula Box 1: Relations

- → Relations from A to B: 2^{mn}
- → Relations on A (n elements): 2^{n^2}
- → Reflexive relations: $2^{n^2-n}=2^{n(n-1)}$
- → Symmetric relations: $2^{\frac{n(n+1)}{2}}$
- → Reflexive and symmetric: $2^{\frac{n(n-1)}{2}}$
- → Equivalence relations on 3 elements: 5 (on 2 elements: 2)
- → Equivalence classes form a partition of A

Yaad karo: "Diagonal fix, baaki free"

Reflexive mein n diagonal pairs **fix** (hone hi chahiye), baaki n^2-n pairs **free** (ho ya na ho): 2^{n^2-n}.
Symmetric mein (a,b), (b,a) ka **joda** ek saath chalta hai: diagonal n + jode $\frac{n(n-1)}{2}$ = $\frac{n(n+1)}{2}$ choices.

Ready table: standard relations

Relation	Refl.	Symm.	Trans.
$a=b$, "parallel to", "similar to"	Yes	Yes	Yes
$a\le b$ on $\mathbb{R}$, "divides" on $\mathbb{N}$	Yes	No	Yes
$a<b$ on $\mathbb{R}$	No	No	Yes
$L_1\perp L_2$	No	Yes	No
$\lvert a-b\rvert\le 1, 1+ab>0$	Yes	Yes	No
$b=a+1$	No	No	No
Empty ϕ (on $A\neq\phi$)	No	Yes	Yes
Universal $A\times A$	Yes	Yes	Yes

↳ "has the same birthday as" (on people): equivalence; har birthday wala group ek class hai.

5. Functions

A relation f from A to B is a **function** if every element of A has exactly one image in B. Write $f: A\to B$.

→ A = domain, B = codomain, $f(A) = \{f(x) : x \in A\}$ = range ($\subseteq B$).

→ Two elements may share an image; one element can never have two images.

Is it a function? (quick checks)

→ **Ordered pairs:** no first entry repeats with different second entries, and every element of A appears.

→ **Graph:** every vertical line cuts it at most once (vertical line test).

→ Circle $x^2 + y^2 = 4$ is **not** a function of x ($x = 0$ gives $y = \pm 2$).

Solved Example 14

$A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$. Which are functions from A to B? $f_1 = \{(1, 4), (2, 5), (3, 6)\}$, $f_2 = \{(1, 4), (1, 5), (2, 6), (3, 6)\}$, $f_3 = \{(1, 5), (2, 5)\}$.

1. f_1: each element has exactly one image: function (bijective).
2. f_2: 1 has two images (4 and 5): **not** a function.
3. f_3: 3 has no image: **not** a function.

✔ Only f_1

Yaad karo: "Vertical se function, Horizontal se one-one"

Vertical line test = Valid function hai ya nahi.
Horizontal line test = one-one (ek baar kaate).
Horizontal line agar kahin graph ko **miss** kare (codomain mein), toh onto nahi.

Definition: Types of functions

→ **One-one (injective):** different inputs give different outputs: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$.

→ **Many-one:** not one-one (two inputs share an image).

→ **Onto (surjective):** range = codomain: every $y \in B$ has some $x \in A$ with $f(x) = y$.

→ **Into:** not onto (some element of B is left out).

→ **Bijective:** one-one and onto.

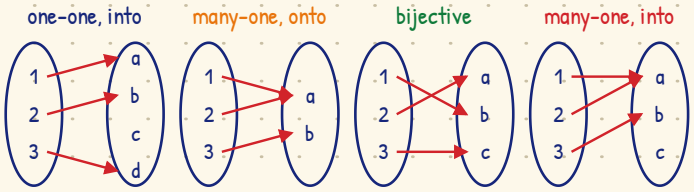


Fig: arrow diagrams of the four types

Quick fact Universal relation $A \times A$ is always an equivalence relation.

Quick fact Constant function on a set with 2 or more elements is never one-one.

Solved Example 15

$A = \{1, 2, 3\}$, $B = \{a, b, c\}$. Classify $f = \{(1, a), (2, c), (3, b)\}$ and $g = \{(1, a), (2, a), (3, c)\}$.

1. f: all images different (one-one) and all of a, b, c used (onto).
2. g: $g(1) = g(2) = a$, not one-one; b has no pre-image, not onto.

✔ f bijective; g neither

6. How to Prove One-one and Onto

One-one: assume f(a) = f(b), simplify to a = b

↓

Onto: take any y in the codomain, solve y = f(x) for x

↓

check that this x lies in the domain and f(x) = y

↳ Not one-one dikhana ho: do alag inputs dikhao jinka output same ho, jaise $f(-2) = f(2)$.

↳ Not onto dikhana ho: codomain ka ek element dikhao jiska koi pre-image nahi.

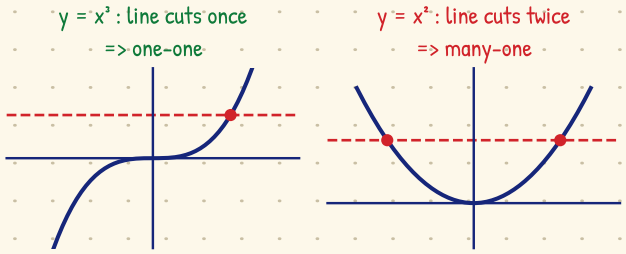


Fig: horizontal line test: cuts the graph at most once means one-one

Solved Example 16

Show that $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 4x - 7$ is bijective.

1. One-one: $4x_1 - 7 = 4x_2 - 7 \Rightarrow x_1 = x_2$.
2. Onto: let $y \in \mathbb{R}$, $y = 4x - 7 \Rightarrow x = \frac{y+7}{4} \in \mathbb{R}$.
3. $f\left(\frac{y+7}{4}\right) = y + 7 - 7 = y$, so every y has a pre-image.

✔ f is one-one and onto, hence bijective

Solved Example 17

Show that $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2 + 1$ is neither one-one nor onto.

1. $f(-3) = 10 = f(3)$ but $-3 \neq 3$: **not one-one.**
2. $x^2 + 1 \geq 1$, so $y = 0$ (or any $y < 1$) has no pre-image: **not onto.**

✔ Neither one-one nor onto (range $= [1, \infty)$)

Solved Example 18

Show that $f:\mathbb{R}-\{2\}\to\mathbb{R}-\{1\}$, $f(x)=\dfrac{x-3}{x-2}$ is one-one and onto.

1. One-one: $\frac{x_1-3}{x_1-2}=\frac{x_2-3}{x_2-2}\Rightarrow x_1x_2-2x_1-3x_2+6=x_1x_2-3x_1-2x_2+6\Rightarrow x_1=x_2$.
2. Onto: $y=\frac{x-3}{x-2}\Rightarrow xy-2y=x-3\Rightarrow x=\dfrac{2y-3}{y-1}$, defined as $y\neq 1$.
3. Also $x\neq 2$ (else $2y-3=2y-2$, impossible) and $f(x)=y$ on substituting.

✔ f is bijective

Solved Example 19

Show that $f:\mathbb{R}\to\mathbb{R}$, $f(x)=x|x|$ is bijective.

1. $f(x)=x^2$ for $x\geq 0$ and $-x^2$ for $x<0$; $f(x)$ has the same sign as x.
2. One-one: if $f(x_1)=f(x_2)$, both x_1,x_2 have the same sign, then $x_1^2=x_2^2\Rightarrow x_1=x_2$.
3. Onto: $y\geq 0$: take $x=\sqrt{y}$; $y<0$: take $x=-\sqrt{-y}$. Then $f(x)=y$.

✔ f is bijective

Solved Example 20

$f:\mathbb{N}\to\mathbb{N}$, $f(n)=\frac{n+1}{2}$ if n is odd, $f(n)=\frac{n}{2}$ if n is even. Is f one-one? onto?

1. $f(1)=1$ and $f(2)=1$: **not one-one.**
2. For any $k\in\mathbb{N}$: $f(2k)=k$, so every k has a pre-image: **onto.**

✔ Many-one and onto (not bijective)

Solved Example 21

Show that $f:\mathbb{N}\to\mathbb{Z}$, $f(n)=\frac{n-1}{2}$ for odd n, $f(n)=-\frac{n}{2}$ for even n, is bijective.

1. Odd n: $1,3,5,\cdots\mapsto 0,1,2,\ldots$ (non-negative); even n: $2,4,6,\cdots\mapsto -1,-2,-3,\ldots$ (negative).
2. One-one: same sign forces the same branch, and each branch is linear, so $f(n_1)=f(n_2)\Rightarrow n_1=n_2$.
3. Onto: $k\geq 0$ comes from $n=2k+1$; $k<0$ comes from $n=-2k$.

✔ f is bijective

Solved Example 22

Check $f:\mathbb{R}\to[-1,1]$, $f(x)=\sin x$ for one-one and onto.

1. $\sin 0=\sin\pi=0$: not one-one.
2. Range of $\sin x$ is $[-1,1]$ = codomain: onto.
3. (On $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]\to[-1,1]$ it becomes bijective: this is why $\sin^{-1}$ uses that branch, see Ch 2.)

✔ Onto but not one-one

Finding the range (needed for onto)

→ Put $y=f(x)$ and solve for x; the values of y for which a valid x exists form the range.

→ Quadratic in x? Use discriminant ≥ 0.

→ Squares/modulus: $x^2\geq 0$, $|x|\geq 0$; $\frac{x^2}{1+x^2}\in[0,1)$.

Solved Example 23

Find the range of $f(x)=\dfrac{x^2}{1+x^2}$, $x\in\mathbb{R}$. Is $f:\mathbb{R}\to\mathbb{R}$ onto?

1. $y=\frac{x^2}{1+x^2}\Rightarrow x^2=\dfrac{y}{1-y}$ (need $y\neq 1$).
2. $x^2\geq 0\Rightarrow\dfrac{y}{1-y}\geq 0\Rightarrow 0\leq y<1$.
3. Range $[0,1)\neq\mathbb{R}$.

✔ Range $=[0,1)$; not onto (also not one-one: $f(1)=f(-1)$)

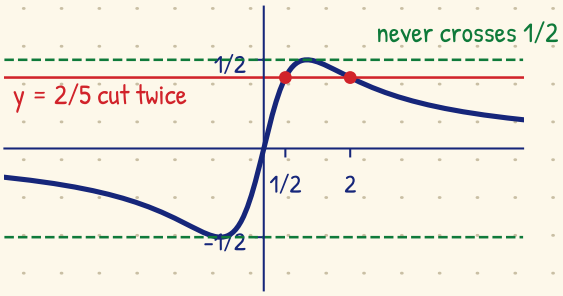


Fig: $y=\frac{x}{x^2+1}$: the line $y=\frac{2}{5}$ cuts it twice (not one-one) and no point goes above $\frac{1}{2}$ (not onto $\mathbb{R}$)

7. Standard Functions and Counting

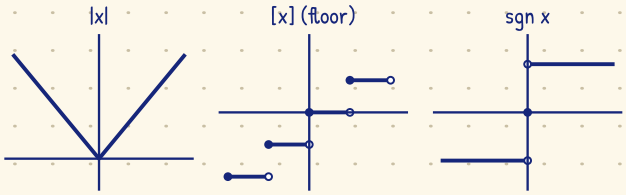


Fig: modulus, greatest integer (floor) and signum functions

$f:\mathbb{R}\to\mathbb{R}$	One-one?	Onto?
$ax+b\,(a\neq 0)$, x^3	Yes	Yes
x^2, $\lvert x\rvert$	No	No
$[x]$ (floor), $\operatorname{sgn}x$	No	No

$f : \mathbb{R} \to \mathbb{R}$	One-one?	Onto?
e^x	Yes	No (range $(0, \infty)$)

↳Domain/codomain badlo toh answer badal jaata hai: x^2 on $\mathbb{N} \to \mathbb{N}$ one-one hai (not onto); $[0, \infty) \to [0, \infty)$ par bijective.

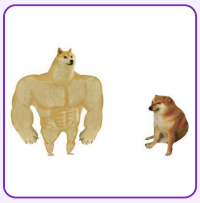
Solved Example 24

$f : \mathbb{N} \to \mathbb{N}$ is given by $f(1) = f(2) = 1$ and $f(n) = n - 1$ for $n > 2$. Show f is onto but not one-one.

1. $f(1) = f(2) = 1$ with $1 \neq 2$: not one-one.
2. For any $k \in \mathbb{N}$: $f(k+1) = k$ (as $k + 1 \geq 2$; for $k = 1$, $f(2) = 1$).
3. So every natural number has a pre-image.

✔ Onto, not one-one

Meme samjho



Bijective function: har input ka apna alag output, codomain mein koi akela nahi. Many-one into: **dono kaam adhoore.**

Asli baat: Bijective = one-one + onto. For finite A, B with $n(A) = n(B)$, one-one $\iff$ onto.

Formula Box 2: Functions

→ Functions from A (m el.) to B (n el.): n^m

→ One-one functions ($m \leq n$): ${}^nP_m = \dfrac{n!}{(n-m)!}$, else 0

→ Onto functions to a 2-element set: $2^m - 2$

→ Onto functions ($m \geq n$): $\sum_{r=0}^{n} (-1)^r \, {}^nC_r \, (n-r)^m$

→ Bijections ($m = n$): $n!$

→ Finite A, $f : A \to A$: one-one $\iff$ onto

→ One-one test: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

→ Onto test: range = codomain

Solved Example 25

Show that $f : \mathbb{R} - \{0\} \to \mathbb{R} - \{0\}$, $f(x) = \dfrac{1}{x}$ is bijective. Is it still onto if the domain is changed to $\mathbb{N}$?

1. One-one: $\frac{1}{x_1} = \frac{1}{x_2} \Rightarrow x_1 = x_2$.
2. Onto: for $y \neq 0$, $x = \dfrac{1}{y}$ is non-zero and $f(x) = y$.
3. Domain $\mathbb{N}$: still one-one, but $y = 2$ needs $x = 0.5$, which is not a natural number: not onto.

✔ Bijective; on $\mathbb{N}$: one-one, not onto

Solved Example 26

$A = \{1, 2, 3\}$, $B = \{p, q\}$. Find the number of (i) functions $A \to B$ (ii) one-one functions $A \to B$ (iii) onto functions $A \to B$ (iv) functions $B \to A$ that are one-one.

1. (i) $2^3 = 8$
2. (ii) 3 elements cannot go to 2 distinct images: 0
3. (iii) all except the 2 constant ones: $8 - 2 = 6$
4. (iv) ${}^3P_2 = 6$

✔ 8, 0, 6, 6

Solved Example 27

Find the number of (i) one-one functions from $\{a, b, c\}$ to $\{1, 2, 3, 4\}$ (ii) onto functions from $\{1, 2, 3, 4\}$ to $\{a, b, c\}$.

1. (i) ${}^4P_3 = 4 \times 3 \times 2 = 24$.
2. (ii) All functions $3^4 = 81$; remove those missing at least one image: ${}^3C_1 \, 2^4 - {}^3C_2 \, 1^4 = 48 - 3 = 45$.
3. Onto $= 81 - 45 = 36$.

✔ 24 and 36

Yaad karo: "One-one = Ek ka ek, Onto = Koi na chhoote"

One-one: ek input, ek apna output (koi share nahi).
Onto: codomain ka koi bhi member **chhoot** na jaaye.
Dono = **Bijective** (dono kaam).

Yaad karo: "Gin ke pakdo"

$n(A) > n(B)$: one-one **impossible**. $n(A) < n(B)$: onto **impossible**. Bijection sirf tab jab $n(A) = n(B)$.

8. Mixed Board-level Practice

Solved Example 28

$A = \{1, 2, 3, 4, 5, 6\}$, $R = \{(a, b) : b \text{ is divisible by } a\}$. Check reflexive, symmetric, transitive.

1. a divides a: reflexive.
2. $(2, 4) \in R$ but 2 is not divisible by 4: not symmetric.
3. $a \mid b$, $b \mid c \Rightarrow b = ak$, $c = bm = a(km)$, so $a \mid c$: transitive.

✔ Reflexive and transitive, not symmetric

Solved Example 29

$A = \{1,2,3,4,5\}$, $R = \{(a,b) : |a^2 - b^2| < 8\}$. Is R an equivalence relation?

1. $|a^2 - a^2| = 0 < 8$: reflexive. $|a^2 - b^2| = |b^2 - a^2|$: symmetric.
2. $(1,2)$: $|1-4| = 3 < 8$; $(2,3)$: $|4-9| = 5 < 8$.
3. But $(1,3)$: $|1-9| = 8$, not < 8: not transitive.

✔ Not an equivalence relation (fails transitivity)

Solved Example 30

$f : \{-1,0,1,2\} \to \{0,1,4\}$, $f(x) = x^2$. Is f one-one? onto?

1. $f(-1) = f(1) = 1$: not one-one.
2. Range $= \{1,0,1,4\} = \{0,1,4\}$ = codomain: onto.

✔ Onto but not one-one

Solved Example 31

Show that $R = \{(a,b) : a - b \text{ is rational}\}$ on $\mathbb{R}$ is an equivalence relation. Is $\sqrt{2}$ related to $1+\sqrt{2}$? to $\sqrt{3}$?

1. $a - a = 0$ rational: reflexive. $a - b = q \Rightarrow b - a = -q$ rational: symmetric.
2. $a - b = q_1, b - c = q_2 \Rightarrow a - c = q_1 + q_2$ rational: transitive.
3. $(1+\sqrt{2}) - \sqrt{2} = 1$ rational: related. $\sqrt{3} - \sqrt{2}$ is irrational: not related.

✔ Equivalence; $[\sqrt{2}] = \{\sqrt{2} + q : q \in \mathbb{Q}\}$

Chapter 1 in 8 lines (quick revision)

- Relation = subset of $A \times B$; on A: 2^{n^2} relations.
- Reflexive: all (a,a); Symmetric: pairs both ways; Transitive: every chain closes.
- Equivalence = R + S + T; classes are disjoint and cover A.
- Function: every input exactly one output.
- One-one: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$; Onto: range = codomain.
- Bijective = one-one + onto; finite equal-size sets: one-one $\iff$ onto.
- Counts: n^m functions, ${}^nP_m$ one-one, $n!$ bijections, $2^m - 2$ onto (to 2 elements).
- Composition, inverse, binary operations: NOT in syllabus.

Common doubts, clear answers

- **Har function relation hai?** Yes; but a relation is a function only if each input has exactly one image.
- **Empty relation transitive hai?** Yes (no chain to break) and symmetric too; not reflexive on a non-empty set.
- **Reflexive check mein sirf kuch (a,a) kaafi?** No, every element of the set needs its own (a,a).
- **Onto prove karte waqt x domain mein hai ye likhna zaroori?** Yes, that line carries a mark.

Common mistakes

- Calling a relation reflexive when just one pair like $(3,3)$ is missing: every a needs (a,a).
- Proving onto without checking that the found x lies in the domain (e.g. $x = \frac{y}{2}$ fails for $\mathbb{N} \to \mathbb{N}$).
- Writing "$f(x) = x^2$ is not one-one" for domain $\mathbb{N}$: on $\mathbb{N}$ it IS one-one.
- Confusing range with codomain: onto means range = codomain.
- Spending time on composition, inverse functions or binary operations: deleted from the 2026-27 syllabus.

Most Asked in Boards

- Show that the given relation is an equivalence relation, find an equivalence class (5M)
- Check a relation for reflexive, symmetric, transitive with counter-examples (2-3M)
- Prove f is one-one and onto (bijective), often with $\mathbb{R} - \{a\}$ domains (3M)
- Show a function is neither one-one nor onto (2M)
- MCQ: number of relations/functions/bijections, smallest equivalence relation
- Assertion-Reason on one-one/onto of x^2, $2x$, $|x|$
- Case study: students/seats, relations in a family or on roll numbers (4M)



Shivam Bhaiya says

Ye chapter **scoring** hai: 5 marker equivalence relation ka format ratt lo: Reflexive, Symmetric, Transitive, phir class. Har step mein "$\Rightarrow (b,a) \in R$" wali line zaroor likho.

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. The number of relations on a set with 3 elements is 1M Most asked

(a) 9 (b) 64 (c) 512 (d) 8

Ans. $2^{3^2} = 2^9 = 512$. (c)

Q2. $R = \{(1,1),(2,2),(3,3),(1,3)\}$ on $\{1,2,3\}$ is 1M
(a) equivalence (b) reflexive, transitive, not symmetric (c) symmetric only (d) reflexive only

> **Ans.** All diagonal pairs present: reflexive. $(1,3)$ but no $(3,1)$: not symmetric. Only chains $(1,1),(1,3)$ etc. give $(1,3)$: transitive. **(b)**

Q3. The number of equivalence relations on $\{1,2,3\}$ is 1M Board fav
(a) 1 (b) 2 (c) 3 (d) 5

> **Ans.** Partitions of a 3-element set: $\{123\}$, $\{1\}\{23\}$, $\{2\}\{13\}$, $\{3\}\{12\}$, $\{1\}\{2\}\{3\}$ = 5. **(d)**

Q4. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3$ is 1M
(a) one-one, not onto (b) onto, not one-one (c) bijective (d) neither

> **Ans.** $x_1^3 = x_2^3 \Rightarrow x_1 = x_2$ and every real y has cube root $y^{1/3}$. **(c)**

Q5. If $n(A) = 4$, the number of bijections from A to A is 1M
(a) 16 (b) 24 (c) 256 (d) 4

> **Ans.** $4! = 24$. **(b)**

Q6. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = |x|$ is 1M
(a) bijective (b) one-one only (c) onto only (d) neither one-one nor onto

> **Ans.** $f(-1) = f(1)$: not one-one; -1 has no pre-image: not onto. **(d)**

Q7. **A:** $f : \mathbb{N} \to \mathbb{N}$, $f(x) = 2x$ is one-one but not onto. **R:** The range of f is the set of even natural numbers, which is not $\mathbb{N}$. 1M Must do
(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

> **Ans.** $2x_1 = 2x_2 \Rightarrow x_1 = x_2$ (one-one); 1, 3, 5 have no pre-image, exactly as R says. **(a)**

Q8. **A:** $R = \{(1,2),(2,1)\}$ on $\{1,2\}$ is symmetric and transitive. **R:** R is transitive if $(a,b),(b,c) \in R \Rightarrow (a,c) \in R$. 1M
(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

> **Ans.** $(1,2),(2,1) \in R$ needs $(1,1) \in R$, which is missing: not transitive, so A is false; R is the correct definition. **(d)**

Q9. The number of onto functions from $\{1,2,3,4\}$ to $\{a,b\}$ is 1M Board fav
(a) 16 (b) 14 (c) 8 (d) 12

> **Ans.** $2^4 - 2 = 14$. **(b)**

Q10. The number of symmetric relations on a set with 3 elements is 1M
(a) 64 (b) 512 (c) 8 (d) 27

> **Ans.** $2^{\frac{3\times4}{2}} = 2^6 = 64$. **(a)**

Quick fact Identity relation $\{(a,a) : a \in A\}$ is reflexive, symmetric and transitive: an equivalence relation.

Q11. **A:** $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ is not one-one. **R:** $f(-x) = f(x)$ for all $x \in \mathbb{R}$. 1M
(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

> **Ans.** $f(-2) = f(2) = 4$ with $-2 \neq 2$: A true, and R is exactly the reason. **(a)**

Q12. $R = \{(1,2),(2,3)\}$ on $\{1,2,3\}$. The minimum number of ordered pairs to be added to make R an equivalence relation is 1M HOTS
(a) 3 (b) 5 (c) 7 (d) 9

> **Ans.** Need $(1,1),(2,2),(3,3),(2,1),(3,2)$, then transitivity forces $(1,3),(3,1)$: 7 pairs. **(c)**

Q13. $R = \{(1,1),(2,2),(3,3),(1,2),(2,1),(2,3),(3,2)\}$ on $\{1,2,3\}$ is 1M
(a) an equivalence relation (b) reflexive, symmetric, not transitive (c) transitive only (d) symmetric only

> **Ans.** Reflexive and symmetric; $(1,2),(2,3) \in R$ but $(1,3)$ is not in R, so not transitive. **(b)**

Q14. The number of functions from $\{1,2\}$ to $\{a,b,c\}$ is 1M
(a) 6 (b) 8 (c) 5 (d) 9

> **Ans.** $3^2 = 9$. **(d)**

Q15. **A:** The identity relation on a non-empty set A is an equivalence relation. **R:** The identity relation is reflexive, symmetric and transitive. 1M
(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

> **Ans.** I_A contains every (a,a), each pair reverses to itself and chains stay (a,a). R explains A. **(a)**

Q16. If $f : \mathbb{R} \to \mathbb{R}$ is $f(x) = 3x - 5$, then the pre-image of 7 is 1M
(a) 16 (b) 4 (c) 2 (d) 12

> **Ans.** $3x - 5 = 7 \Rightarrow x = 4$. **(b)**

Very short (2 marks)

Q17. Show that $f : \mathbb{N} \to \mathbb{N}$, $f(x) = 3x + 2$ is one-one but not onto. 2M

> **Ans.** $3x_1 + 2 = 3x_2 + 2 \Rightarrow x_1 = x_2$: one-one. $f(x) \geq 5$, so $1 \in \mathbb{N}$ has no pre-image ($x = -\frac{1}{3}$ is not in $\mathbb{N}$): not onto.

Q18. Give a relation on $A = \{1,2,3\}$ which is symmetric and transitive but not reflexive. 2M HOTS

> **Ans.** $R = \{(1,1),(1,2),(2,1),(2,2)\}$: symmetric (pairs come both ways), transitive (all chains stay inside), not reflexive as $(3,3)$ is not in R.

Q19. Show that $R = \{(a,b) : a \leq b\}$ on $\mathbb{R}$ is reflexive and transitive but not symmetric. 2M

> **Ans.** $a \leq a$: reflexive. $a \leq b,\ b \leq c \Rightarrow a \leq c$: transitive. $(2,5) \in R$ but $5 \leq 2$ is false: not symmetric.

Q20. Show that $f:\mathbb{R}\to\mathbb{R}$, $f(x)=[x]$ (greatest integer) is neither one-one nor onto. 2M

> **Ans.** $[1.2]=[1.7]=1$: not one-one. Range $=\mathbb{Z}$, so 0.5 has no pre-image: not onto.

Q22. If $R=\{(x,y):x+2y=8\}$ is a relation on $\mathbb{N}$, write its range and check whether R is symmetric. 2M Most asked

> **Ans.** $y=1,2,3$ give $x=6,4,2$: $R=\{(6,1),(4,2),(2,3)\}$, range $=\{1,2,3\}$. $(1,6)$ is not in R $(1+12\neq 8)$: not symmetric.

Q21. Show that $f:\mathbb{R}\to\mathbb{R}$, $f(x)=\dfrac{2x-1}{3}$ is bijective. 2M

> **Ans.** $\frac{2x_1-1}{3}=\frac{2x_2-1}{3}\Rightarrow x_1=x_2$. For $y\in\mathbb{R}$, $x=\frac{3y+1}{2}\in\mathbb{R}$ and $f(x)=\frac{3y+1-1}{3}=y$. Bijective.

Q23. Show that $f:\mathbb{Z}\to\mathbb{Z}$, $f(x)=x^2$ is neither one-one nor onto. 2M

> **Ans.** $f(-1)=f(1)=1$: not one-one. $f(x)\geq 0$, so -1 (also 2, a non-square) has no pre-image: not onto.

Q24. Let $A=\{1,2,3\}$. Write the smallest relation on A that is reflexive and symmetric and contains $(1,3)$. Is it transitive? 2M

> **Ans.** $R=\{(1,1),(2,2),(3,3),(1,3),(3,1)\}$. Chains: $(1,3),(3,1)\Rightarrow(1,1)$ ✓; $(3,1),(1,3)\Rightarrow(3,3)$ ✓. So it is transitive too (an equivalence relation).

Q25. If $A=\{1,2,3,4\}$ and $R=\{(1,2),(2,2),(1,1),(4,4),(1,3),(3,3),(3,2)\}$, show that R is reflexive and transitive but not symmetric. 2M

> **Ans.** All (a,a) present: reflexive. $(1,2)\in R$ but $(2,1)$ is not: not symmetric. Only non-trivial chain $(1,3),(3,2)$ gives $(1,2)\in R$: transitive.

Q26. If $f:\mathbb{R}\to\mathbb{R}$, $f(x)=x^2$, write the pre-images of 9 and of -4. 2M

> **Ans.** $x^2=9\Rightarrow x=\pm 3$, pre-images $\{-3,3\}$. $x^2=-4$ has no real solution: no pre-image (empty set). This also shows f is neither one-one nor onto.

Q27. How many equivalence relations on $\{a,b\}$ are there? List them. 2M

> **Ans.** Two: $\{(a,a),(b,b)\}$ (identity) and $\{(a,a),(b,b),(a,b),(b,a)\}$ (universal).

Short answer (3 marks)

Q28. Show that $f:\mathbb{R}-\{-\frac{1}{2}\}\to\mathbb{R}-\{\frac{3}{2}\}$, $f(x)=\dfrac{3x}{2x+1}$ is one-one and onto. 3M Board fav

> **Ans.** One-one: $\frac{3x_1}{2x_1+1}=\frac{3x_2}{2x_2+1}\Rightarrow 6x_1x_2+3x_1=6x_1x_2+3x_2\Rightarrow x_1=x_2$. Onto: $y=\frac{3x}{2x+1}\Rightarrow x=\dfrac{y}{3-2y}$, defined for $y\neq\frac{3}{2}$; $x=-\frac{1}{2}$ would give $-3+2y=2y$, impossible. And $f(x)=y$. Hence bijective.

Q30. Show that $R=\{(x,y):x-y\in\mathbb{Z}\}$ on $\mathbb{R}$ is an equivalence relation. Find $\left[\frac{1}{2}\right]$. 3M

> **Ans.** $x-x=0\in\mathbb{Z}$; $x-y=k\Rightarrow y-x=-k\in\mathbb{Z}$; $x-y=k$, $y-z=m\Rightarrow x-z=k+m\in\mathbb{Z}$. So equivalence. $\left[\frac{1}{2}\right]=\{n+\frac{1}{2}:n\in\mathbb{Z}\}=\{\ldots,-\frac{1}{2},\frac{1}{2},\frac{3}{2},\ldots\}$.

Q29. Show that $f:\mathbb{R}\to\mathbb{R}$, $f(x)=\dfrac{x}{x^2+1}$ is neither one-one nor onto. 3M HOTS

> **Ans.** $f(2)=\frac{2}{5}=f\left(\frac{1}{2}\right)$: not one-one. $y=\frac{x}{x^2+1}\Rightarrow yx^2-x+y=0$, real x needs $1-4y^2\geq 0$, i.e. $|y|\leq\frac{1}{2}$. So $y=1$ has no pre-image: not onto.

Q31. Show that $f:\mathbb{N}\to\mathbb{N}$, $f(n)=n+1$ if n is odd and $f(n)=n-1$ if n is even, is bijective. 3M Most asked

> **Ans.** Odd $n\mapsto$ even, even $n\mapsto$ odd. If $f(n_1)=f(n_2)$, both have the same parity, so $n_1\pm 1=n_2\pm 1\Rightarrow n_1=n_2$: one-one. Onto: odd m comes from $m+1$ (even), even m comes from $m-1$ (odd). Bijective.

Q32. B = set of books in a library, $R=\{(x,y):$ x and y have the same number of pages$\}$. Show R is an equivalence relation. 3M

> **Ans.** A book has as many pages as itself (reflexive); same pages for x,y means same for y,x (symmetric); x,y same and y,z same $\Rightarrow x,z$ same (transitive). Hence equivalence.

Q33. Show that $R=\{(a,b):|a-b|\leq 1\}$ on $\mathbb{R}$ is reflexive and symmetric but not transitive. 3M

> **Ans.** $|a-a|=0\leq 1$: reflexive. $|a-b|=|b-a|$: symmetric. $(1,2),(2,2.8)\in R$ but $|1-2.8|=1.8>1$: not transitive.

Q34. Let $Y=\{5x+2:x\in\mathbb{N}\}=\{7,12,17,\ldots\}$. Show that $f:\mathbb{N}\to Y$, $f(x)=5x+2$ is bijective. 3M

> **Ans.** One-one: $5x_1+2=5x_2+2\Rightarrow x_1=x_2$. Onto: any $y\in Y$ is $5x+2$ for some $x\in\mathbb{N}$ by the definition of Y; explicitly $x=\frac{y-2}{5}\in\mathbb{N}$. Bijective.

Q35. Show that $f:\mathbb{R}\to\mathbb{R}$, $f(x)=\begin{cases}x+2, & x\geq 0\\ x-2, & x<0\end{cases}$ is one-one but not onto. 3M HOTS

> **Ans.** $x\geq 0$ gives $f(x)\geq 2$; $x<0$ gives $f(x)<-2$. Equal images force the same branch, then $x_1=x_2$: one-one. Values in $[-2,2)$, e.g. 0, have no pre-image: not onto.

Q36. Show that $R=\{(a,b):a-b$ is divisible by $3\}$ on $A=\{1,2,\ldots,10\}$ is an equivalence relation and write all its equivalence classes. 3M Most asked

> **Ans.** $a-a=0$; $3\mid(a-b)\Rightarrow 3\mid(b-a)$; $3\mid(a-b)$, $3\mid(b-c)\Rightarrow 3\mid(a-c)$. Classes: $\{1,4,7,10\}$, $\{2,5,8\}$, $\{3,6,9\}$.

Q37. Find the number of all one-one functions from $A=\{1,2,3\}$ to itself, and show that each of them is also onto. 3M

(contd.) One-one functions $= 3! = 6$. Each sends 3 elements to 3 different images in a 3-element set, so all of $\{1,2,3\}$ is covered: onto. (Finite equal-size sets: one-one $\iff$ onto.)

Q38. Show that $f:\mathbb{N}\to\mathbb{N}$, $f(x)=x^2+x+1$ is one-one but not onto. 3M Board fav

Ans. $x_1^2+x_1+1=x_2^2+x_2+1\Rightarrow(x_1-x_2)(x_1+x_2+1)=0$; $x_1+x_2+1>0$, so $x_1=x_2$: one-one. $f(x)\geq f(1)=3$, so $1,2$ have no pre-image: not onto.

Q39. Check whether $R=\{(a,b):a\leq b^3\}$ on $\mathbb{R}$ is reflexive, symmetric or transitive. 3M Most asked

Ans. $(\frac{1}{2},\frac{1}{2})$: $\frac{1}{2}\leq\frac{1}{8}$ false, not reflexive. $(1,2)$: $1\leq 8$ true, but $(2,1)$: $2\leq 1$ false, not symmetric. $(3,\frac{3}{2})$: $3\leq\frac{27}{8}$ true; $(\frac{3}{2},\frac{6}{5})$: $\frac{3}{2}\leq\frac{216}{125}$ true; but $(3,\frac{6}{5})$: $3\leq\frac{216}{125}$ false, not transitive.

Long answer (5 marks)

Q40. Let $A=\{0,1,2,\ldots,12\}$ and $R=\{(a,b):|a-b|$ is a multiple of $4\}$. Show R is an equivalence relation. Find all elements related to 1 and the class $[2]$. 5M Most asked

Ans. Reflexive: $|a-a|=0=4\times 0$. Symmetric: $|a-b|=|b-a|$. Transitive: $a-b=4k$, $b-c=4m\Rightarrow a-c=4(k+m)$. Related to 1: $\{1,5,9\}$. $[2]=\{2,6,10\}$. (Other classes: $\{0,4,8,12\}$, $\{3,7,11\}$.)

Q41. Let $A=\mathbb{N}\times\mathbb{N}$ and $(a,b)\,R\,(c,d)$ iff $ad=bc$. Show that R is an equivalence relation. Write three elements of $[(2,3)]$. 5M

Ans. Reflexive: $ab=ba$. Symmetric: $ad=bc\Rightarrow cb=da$. Transitive: $ad=bc$ and $cf=de$; multiply: $adcf=bcde\Rightarrow af=be$ (cancel $cd\neq 0$). Hence equivalence. $(x,y)R(2,3)\iff 3x=2y$: $(2,3),(4,6),(6,9)$.

Q42. Show that $f:[0,\infty)\to[2,\infty)$, $f(x)=4x^2+4x+2$ is bijective. 5M Board fav

Ans. $f(x)=(2x+1)^2+1$. One-one: $(2x_1+1)^2=(2x_2+1)^2$, both bases positive $\Rightarrow x_1=x_2$. Onto: $y=(2x+1)^2+1\Rightarrow x=\dfrac{\sqrt{y-1}-1}{2}$; for $y\geq 2$, $\sqrt{y-1}\geq 1$ so $x\geq 0$, and $f(x)=y$. Bijective.

Q43. Show that $R=\{(a,b):a-b$ is even$\}$ on $\mathbb{Z}$ is an equivalence relation. Show that all elements of $\{1,3,5,7\}$ are related to each other, no element of $\{1,3,5,7\}$ is related to any element of $\{2,4,6\}$, and find the classes. 5M Most asked

Ans. $a-a=0$ even; $a-b$ even $\Rightarrow b-a$ even; $a-b$, $b-c$ even $\Rightarrow a-c=(a-b)+(b-c)$ even. Odd minus odd is even: $\{1,3,5,7\}$ all related. Odd minus even is odd: none related. Classes: $[0]$ = even integers, $[1]$ = odd integers.

Q44. A = set of points in a plane, $R=\{(P,Q):OP=OQ\}$ where O is the origin. Show R is an equivalence relation and describe the class of $P(3,4)$. 5M Board fav

Ans. (contd.) $OP=OP$: reflexive. $OP=OQ\Rightarrow OQ=OP$: symmetric. $OP=OQ$, $OQ=OS\Rightarrow OP=OS$: transitive. $OP=\sqrt{9+16}=5$, so $[P]$ = all points at distance 5 from O: the circle $x^2+y^2=25$.

Q45. Let $f:\mathbb{R}-\{3\}\to\mathbb{R}-\{2\}$, $f(x)=\dfrac{2x+1}{x-3}$. (i) Show that f is one-one. (ii) Show that f is onto. (iii) Find x such that $f(x)=5$. 5M Must do

Ans. (i) $\frac{2x_1+1}{x_1-3}=\frac{2x_2+1}{x_2-3}\Rightarrow 2x_1x_2-6x_1+x_2-3=2x_1x_2-6x_2+x_1-3\Rightarrow 7x_2=7x_1\Rightarrow x_1=x_2$. (ii) $y=\frac{2x+1}{x-3}\Rightarrow x=\dfrac{3y+1}{y-2}$, defined for $y\neq 2$; $x=3$ would need $3y+1=3y-6$, impossible; and $f(x)=y$. (iii) $x=\frac{15+1}{5-2}=\frac{16}{3}$.

Case based (4 marks)

Q46. Roll numbers. A teacher puts students with roll numbers $A=\{1,2,\ldots,8\}$ in groups by the relation $R=\{(a,b)\}$ where a and b leave the same remainder on division by 3. (i) Is R reflexive? (1) (ii) Find $[2]$. (1) (iii) Show R is an equivalence relation and write all the groups. (2) 4M Must do

Ans. (i) Yes, a and a leave the same remainder. (ii) $[2]=\{2,5,8\}$. (iii) Same remainder works both ways (symmetric) and through a chain (transitive), with reflexive: equivalence. Groups: $\{1,4,7\}$, $\{2,5,8\}$, $\{3,6\}$.

Q47. Seats in a hall. Four students $A=\{s_1,s_2,s_3,s_4\}$ are to be seated on chairs $B=\{c_1,\ldots,c_5\}$, one student per chair. (i) How many one-one functions $A\to B$? (1) (ii) Is an onto function $A\to B$ possible? (1) (iii) If one chair is removed, find the number of bijections $A\to B$ and the number of functions $A\to B$ that are not one-one. (2) 4M

Ans. (i) ${}^5P_4=120$. (ii) No, since $n(A)=4<5=n(B)$. (iii) Bijections $=4!=24$; functions $=4^4=256$, not one-one $=256-24=232$.

Trap $R=\{(1,2)\}$ on $\{1,2\}$ is transitive (no pair (b,c) to break it) but not reflexive, not symmetric.

Quick check $f(x)=x^2$ on $\mathbb{R}\to\mathbb{R}$: not one-one ($f(-1)=f(1)$), not onto (-1 has no pre-image).

Yaad rakho $f(x)=ax+b$, $a\neq 0$, on $\mathbb{R}$ is always a bijection.

Apne notes / galtiyan yahan likho

Most asked: principal value questions, domain and range of ITF

Inverse Trigonometric Functions

1. Why Inverse? Principal Branch

A function has an inverse only if it is **one-one and onto** (bijective). $\sin x$ on $\mathbb{R}$ is many-one ($\sin 0 = \sin \pi$), so we **restrict its domain** to an interval where it is bijective.

- $\sin : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \to [-1, 1]$ is bijective, so $\sin^{-1} : [-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ exists.
- That chosen interval is the **principal value branch**; the output is the **principal value**.
- $y = \sin^{-1} x \iff \sin y = x$ and $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Definition

$\sin^{-1} x$ is the **angle in the principal range** whose sine is x. It is also written $\arcsin x$.

↳ $\sin^{-1} x \neq \frac{1}{\sin x}$. Power -1 yahan inverse function hai, reciprocal nahi: $(\sin x)^{-1} = \operatorname{cosec} x$.

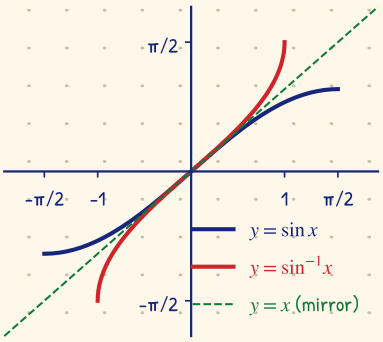


Fig: graph of $\sin^{-1} x$ is the mirror image of the restricted $\sin x$ in the line $y = x$

2. Domain and Range Table (most important)

Function	Domain	Range (principal branch)
$\sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$
$\sec^{-1} x$	$\mathbb{R} - (-1, 1)$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$
$\operatorname{cosec}^{-1} x$	$\mathbb{R} - (-1, 1)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

↳ $\mathbb{R} - (-1, 1)$ means $x \leq -1$ or $x \geq 1$, i.e. $|x| \geq 1$.

Quick fact $\sin^{-1} 1 = \frac{\pi}{2}$, $\cos^{-1} 1 = 0$, $\cos^{-1}(-1) = \pi$.

Quick fact $\sin^{-1} 0 = 0$ but $\cos^{-1} 0 = \frac{\pi}{2}$.

Yaad karo: "STC right side, CCS upar"

Sin, Tan, Cosec (STC): range $-\frac{\pi}{2}$ se $\frac{\pi}{2}$ (IV + I quadrant, "right side"). Cos, Cot, Sec (CCS): range 0 se π (I + II quadrant, "upar wala half"). Brackets: jo value function le hi nahi sakta wahan open/hata do ($\tan$ at $\pm\frac{\pi}{2}$, $\sec$ at $\frac{\pi}{2}$, cosec at 0).

3. Graphs of the Six ITFs

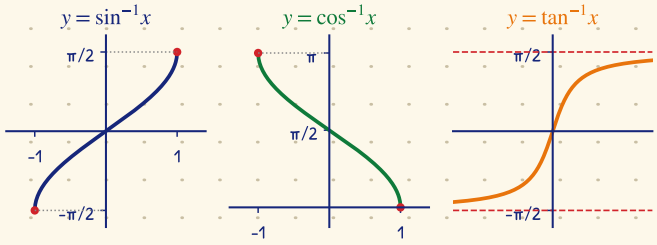


Fig: $\sin^{-1} x$ and $\tan^{-1} x$ increase; $\cos^{-1} x$ decreases

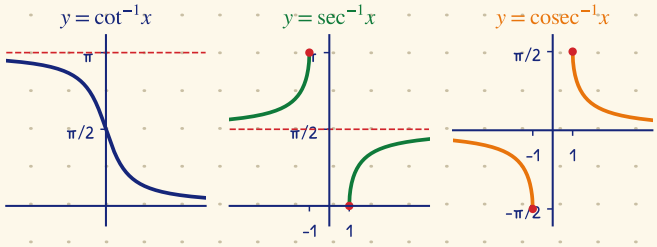


Fig: $\cot^{-1} x$, $\sec^{-1} x$, $\operatorname{cosec}^{-1} x$ (dashed lines = asymptotes)

- $\sin^{-1} x$, $\tan^{-1} x$: **increasing**; $\cos^{-1} x$, $\cot^{-1} x$: **decreasing**.
- $\tan^{-1} x \to \pm\frac{\pi}{2}$ as $x \to \pm\infty$ but never reaches them.
- $\sec^{-1} x$ has a gap at $\frac{\pi}{2}$ (asymptote), $\operatorname{cosec}^{-1} x$ at 0.



Shivam Bhaiya says

Range wali table **ek baar likh ke dekho, roz.** Is chapter ke 80 percent questions sirf isi table se solve hote hain. Principal value galat range mein aayi = zero marks.

How to draw an ITF graph in the exam

1. Mark the domain end points on the x-axis (± 1 for $\sin^{-1}$, $\cos^{-1}$).
2. Mark the range end values on the y-axis ($\pm\frac{\pi}{2}$ or $0, \pi$); draw dotted asymptotes for $\tan^{-1}$, $\cot^{-1}$.
3. Plot 3 points (ends and middle), join smoothly; label the curve.

Solved Example 1

From the graphs: for which x is (i) $\sin^{-1} x = \cos^{-1} x$

(ii) $\tan^{-1} x = 0$ (iii) $\cot^{-1} x = \frac{\pi}{2}$?

1. (i) Both equal $\frac{\pi}{4}$ at $x = \frac{1}{\sqrt{2}}$ (rising curve meets falling curve once).
2. (ii) $\tan^{-1} x = 0 \iff x = \tan 0 = 0$.
3. (iii) $\cot^{-1} x = \frac{\pi}{2} \iff x = \cot \frac{\pi}{2} = 0$.

✔ (i) $x = \frac{1}{\sqrt{2}}$ (ii) $x = 0$ (iii) $x = 0$

Solved Example 2

Evaluate $\operatorname{cosec}^{-1}(-\sqrt{2}) + \cot^{-1}(1) - \sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$.

1. $\operatorname{cosec}^{-1}(-\sqrt{2}) = -\frac{\pi}{4}$, $\cot^{-1} 1 = \frac{\pi}{4}$, $\sec^{-1} \frac{2}{\sqrt{3}} = \frac{\pi}{6}$.
2. $-\frac{\pi}{4} + \frac{\pi}{4} - \frac{\pi}{6}$

✔ $-\frac{\pi}{6}$

4. Finding Principal Values

Step 1: ignore the sign, find the basic angle from the standard table

↓

Step 2: place it in the **principal range** of that function

↓

Step 3: check: the trig value of your answer gives back the given number

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	n.d.

Ready reckoner 1: $\sin^{-1}$ and $\cos^{-1}$

x	$\sin^{-1} x$	$\cos^{-1} x$
-1	$-\frac{\pi}{2}$	π
$-\frac{\sqrt{3}}{2}$	$-\frac{\pi}{3}$	$\frac{5\pi}{6}$
$-\frac{1}{\sqrt{2}}$	$-\frac{\pi}{4}$	$\frac{3\pi}{4}$
$-\frac{1}{2}$	$-\frac{\pi}{6}$	$\frac{2\pi}{3}$
0	0	$\frac{\pi}{2}$
$\frac{1}{2}$	$\frac{\pi}{6}$	$\frac{\pi}{3}$
$\frac{1}{\sqrt{2}}$	$\frac{\pi}{4}$	$\frac{\pi}{4}$
$\frac{\sqrt{3}}{2}$	$\frac{\pi}{3}$	$\frac{\pi}{6}$
1	$\frac{\pi}{2}$	0

Ready reckoner 2: $\tan^{-1}$ and $\cot^{-1}$

x	$\tan^{-1} x$	$\cot^{-1} x$
$-\sqrt{3}$	$-\frac{\pi}{3}$	$\frac{5\pi}{6}$
-1	$-\frac{\pi}{4}$	$\frac{3\pi}{4}$
$-\frac{1}{\sqrt{3}}$	$-\frac{\pi}{6}$	$\frac{2\pi}{3}$
0	0	$\frac{\pi}{2}$
$\frac{1}{\sqrt{3}}$	$\frac{\pi}{6}$	$\frac{\pi}{3}$
1	$\frac{\pi}{4}$	$\frac{\pi}{4}$
$\sqrt{3}$	$\frac{\pi}{3}$	$\frac{\pi}{6}$

↳Pattern dekho: $\cos^{-1}$ column = $\frac{\pi}{2}$ minus $\sin^{-1}$ column (same for cot and tan). Ye sirf values yaad rakhne ka shortcut hai.

Yaad karo: "Negative ka jugaad"

Negative value aaye toh range se socho: **sin, tan, cosec** mein angle bhi negative: $\sin^{-1}(-x) = -\sin^{-1} x$. **cos, cot, sec** mein angle π se minus: $\cos^{-1}(-x) = \pi - \cos^{-1} x$. Jaise $\cos^{-1}\left(-\frac{1}{2}\right) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$.

Solved Example 3

Find the principal value of $\sin^{-1}\left(-\frac{1}{2}\right)$.

1. Let $\sin^{-1}\left(-\frac{1}{2}\right) = y$, so $\sin y = -\frac{1}{2}$ with $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.
2. $\sin\left(-\frac{\pi}{6}\right) = -\frac{1}{2}$ and $-\frac{\pi}{6}$ lies in the range.

✔ $-\frac{\pi}{6}$

Solved Example 4

Find the principal value of $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$.

1. Need $y \in [0, \pi]$ with $\cos y = -\frac{\sqrt{3}}{2}$.
2. Basic angle $\frac{\pi}{6}$; cosine negative in the II quadrant: $y = \pi - \frac{\pi}{6}$.

✔ $\frac{5\pi}{6}$

Solved Example 5

Find the principal values of (i) $\tan^{-1}(-1)$ (ii) $\cot^{-1}(-\sqrt{3})$ (iii) $\operatorname{cosec}^{-1}(-2)$ (iv) $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$.

1. (i) range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$: $\tan\left(-\frac{\pi}{4}\right) = -1$, answer $-\frac{\pi}{4}$.
2. (ii) range $(0, \pi)$: $\cot\frac{5\pi}{6} = -\sqrt{3}$, answer $\frac{5\pi}{6}$.
3. (iii) $\operatorname{cosec}\left(-\frac{\pi}{6}\right) = -2$, answer $-\frac{\pi}{6}$.
4. (iv) $\sec\frac{\pi}{6} = \frac{2}{\sqrt{3}}$, answer $\frac{\pi}{6}$.

✔ $-\frac{\pi}{4}$, $\frac{5\pi}{6}$, $-\frac{\pi}{6}$, $\frac{\pi}{6}$

Quick fact $\frac{\pi}{6} = 30°$, $\frac{\pi}{4} = 45°$, $\frac{\pi}{3} = 60°$.

Solved Example 6

Evaluate $\tan^{-1}\sqrt{3} - \sec^{-1}(-2)$.

1. $\tan^{-1}\sqrt{3} = \frac{\pi}{3}$.
2. $\sec^{-1}(-2)$: need $y \in [0,\pi]$, $\cos y = -\frac{1}{2}$, so $y = \frac{2\pi}{3}$.
3. $\frac{\pi}{3} - \frac{2\pi}{3}$

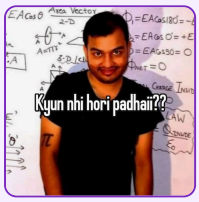
✔ $-\frac{\pi}{3}$

Solved Example 7

Evaluate $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) + 2\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$.

1. $\cos^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{6}$, $\sin^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{3}$.
2. $\frac{\pi}{6} + \frac{2\pi}{3} = \frac{\pi+4\pi}{6}$

✔ $\frac{5\pi}{6}$



Meme samjho

$\cos^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{3}$ likh diya aur pooch rahe ho marks **kyun nahi aaye??**

Asli baat: $-\frac{\pi}{3}$ is not in $[0,\pi]$. Correct answer: $\pi - \frac{\pi}{3} = \frac{2\pi}{3}$.

Solved Example 8

Evaluate $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$.

1. $\frac{\pi}{4} + \frac{2\pi}{3} - \frac{\pi}{6}$
2. LCM 12: $\frac{3\pi+8\pi-2\pi}{12} = \frac{9\pi}{12}$

✔ $\frac{3\pi}{4}$

Solved Example 9

Find the value of $\sin\left(\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right)$.

1. $\sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$.
2. $\sin\left(\frac{\pi}{3} + \frac{\pi}{6}\right) = \sin\frac{\pi}{2}$

✔ 1

Principal value drill (cover the answers and try)

Find	Ans	Find	Ans
$\sin^{-1}\frac{1}{\sqrt{2}}$	$\frac{\pi}{4}$	$\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$	$\frac{3\pi}{4}$
$\sin^{-1}(-1)$	$-\frac{\pi}{2}$	$\cos^{-1}(-1)$	π
$\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$	$-\frac{\pi}{6}$	$\cot^{-1}(-1)$	$\frac{3\pi}{4}$
$\sec^{-1}(-\sqrt{2})$	$\frac{3\pi}{4}$	$\operatorname{cosec}^{-1}\sqrt{2}$	$\frac{\pi}{4}$
$\sec^{-1}\left(-\frac{2}{\sqrt{3}}\right)$	$\frac{5\pi}{6}$	$\operatorname{cosec}^{-1}\left(-\frac{2}{\sqrt{3}}\right)$	$-\frac{\pi}{3}$
$\cot^{-1}\sqrt{3}$	$\frac{\pi}{6}$	$\tan^{-1}(-\sqrt{3})$	$-\frac{\pi}{3}$

Solved Example 10

Find the value of $\tan^{-1}\left[2\sin\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)\right]$.

1. $\cos^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{6}$, so $2\cos^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{3}$.
2. $2\sin\frac{\pi}{3} = 2\times\frac{\sqrt{3}}{2} = \sqrt{3}$.
3. $\tan^{-1}\sqrt{3} = \frac{\pi}{3}$.

✔ $\frac{\pi}{3}$

Solved Example 11

Evaluate (i) $\sin^{-1}\left(\cos\frac{3\pi}{4}\right)$ (ii) $\cos^{-1}\left(\sin\frac{4\pi}{3}\right)$ (iii) $\tan^{-1}\left(\tan\frac{7\pi}{6}\right)$.

1. (i) $\cos\frac{3\pi}{4} = -\frac{1}{\sqrt{2}}$, so $\sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) = -\frac{\pi}{4}$.
2. (ii) $\sin\frac{4\pi}{3} = -\frac{\sqrt{3}}{2}$, so $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \frac{5\pi}{6}$.
3. (iii) $\tan\frac{7\pi}{6} = \tan\left(\pi+\frac{\pi}{6}\right) = \tan\frac{\pi}{6}$, and $\frac{\pi}{6}$ is in range: $\frac{\pi}{6}$.

✔ $-\frac{\pi}{4}$, $\frac{5\pi}{6}$, $\frac{\pi}{6}$

5. ITF of a Trig Value (two cases)

→ $\sin(\sin^{-1}x) = x$ for **every** $x \in [-1,1]$ (similarly $\cos(\cos^{-1}x) = x$, $\tan(\tan^{-1}x) = x$ for all real x).

→ $\sin^{-1}(\sin x) = x$ only if $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$;
$\cos^{-1}(\cos x) = x$ only if $x \in [0,\pi]$;
$\tan^{-1}(\tan x) = x$ only if $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

→ Outside the range: first find the trig value, then take its principal value.

Solved Example 12

Evaluate (i) $\sin^{-1}\left(\sin\frac{3\pi}{5}\right)$ (ii) $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$ (iii) $\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$.

1. (i) $\frac{3\pi}{5} > \frac{\pi}{2}$; $\sin\frac{3\pi}{5} = \sin\left(\pi - \frac{3\pi}{5}\right) = \sin\frac{2\pi}{5}$ and $\frac{2\pi}{5}$ is in range: $\frac{2\pi}{5}$.
2. (ii) $\frac{7\pi}{6} > \pi$; $\cos\frac{7\pi}{6} = \cos\left(2\pi - \frac{7\pi}{6}\right) = \cos\frac{5\pi}{6}$: $\frac{5\pi}{6}$.
3. (iii) $\tan\frac{3\pi}{4} = -1$, so the value is $\tan^{-1}(-1) = -\frac{\pi}{4}$.

✔ $\frac{2\pi}{5}$, $\frac{5\pi}{6}$, $-\frac{\pi}{4}$

Solved Example 13

If $\sin^{-1}x = y$, find the range of y and hence the values of x for which $\sin^{-1}x > \frac{\pi}{6}$.

1. $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.
2. $\sin^{-1}$ is increasing: $\sin^{-1} x > \frac{\pi}{6} = \sin^{-1}\frac{1}{2} \iff x > \frac{1}{2}$.
3. With $x \le 1$ (domain).

✔ $x \in \left(\frac{1}{2}, 1\right]$

Solved Example 14

Evaluate $\cos^{-1}\left(-\frac{1}{2}\right) - 2\sin^{-1}\left(\frac{1}{2}\right) + 3\tan^{-1}(-1)$.

1. $\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$, $2\sin^{-1}\frac{1}{2} = \frac{\pi}{3}$, $3\tan^{-1}(-1) = -\frac{3\pi}{4}$.
2. $\frac{2\pi}{3} - \frac{\pi}{3} - \frac{3\pi}{4} = \frac{\pi}{3} - \frac{3\pi}{4} = \frac{4\pi - 9\pi}{12}$

✔ $-\frac{5\pi}{12}$

Common mistakes

→ Writing $\sin^{-1}\left(\sin\frac{2\pi}{3}\right) = \frac{2\pi}{3}$: it is $\frac{\pi}{3}$ (answer must lie in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$).

→ Giving $\cos^{-1}$ or $\cot^{-1}$ of a negative number as a negative angle.

→ Treating $\sin^{-1} x$ as $\frac{1}{\sin x}$.

→ Forgetting that $\sin^{-1} 2$, $\cos^{-1}(-1.5)$ do **not exist** (outside $[-1, 1]$).

→ Writing $\tan^{-1} x$ range as closed $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$; it is open.

Trig vs ITF: inputs and outputs swap

	$\sin x$ (restricted)	$\sin^{-1} x$
Input	angle in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	number in $[-1, 1]$
Output	number in $[-1, 1]$	angle in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
Graph	original	mirror in $y = x$

Solved Example 15

Find the principal value of $\cos^{-1}\left(\cos\frac{2\pi}{3}\right) + \sin^{-1}\left(\sin\frac{2\pi}{3}\right)$.

1. $\frac{2\pi}{3} \in [0, \pi]$, so $\cos^{-1}\left(\cos\frac{2\pi}{3}\right) = \frac{2\pi}{3}$.
2. $\frac{2\pi}{3}$ is not in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$: $\sin\frac{2\pi}{3} = \sin\frac{\pi}{3}$, so it gives $\frac{\pi}{3}$.
3. Sum $= \frac{2\pi}{3} + \frac{\pi}{3}$

✔ π

6. Domain of Composite ITF Expressions

Whatever sits inside $\sin^{-1}$ or $\cos^{-1}$ must lie in $[-1, 1]$; inside $\sec^{-1}$ or $\operatorname{cosec}^{-1}$ the inside u must satisfy $|u| \ge 1$; inside $\tan^{-1}$, $\cot^{-1}$ anything real.

Solved Example 16

Find the domain of (i) $\sin^{-1}(2x - 1)$ (ii) $\cos^{-1}(x^2 - 4)$.

1. (i) $-1 \le 2x - 1 \le 1 \Rightarrow 0 \le 2x \le 2 \Rightarrow 0 \le x \le 1$.
2. (ii) $-1 \le x^2 - 4 \le 1 \Rightarrow 3 \le x^2 \le 5$.
3. So $\sqrt{3} \le |x| \le \sqrt{5}$.

✔ (i) $[0, 1]$ (ii) $[-\sqrt{5}, -\sqrt{3}] \cup [\sqrt{3}, \sqrt{5}]$

Solved Example 17

Find the domain of $f(x) = \sec^{-1}(2x + 1)$.

1. Need $|2x + 1| \ge 1$.
2. $2x + 1 \ge 1 \Rightarrow x \ge 0$, or $2x + 1 \le -1 \Rightarrow x \le -1$.

✔ $(-\infty, -1] \cup [0, \infty)$

Yaad karo: "Andar wale ko sandwich karo"

$\sin^{-1}(u)$, $\cos^{-1}(u)$: u ko -1 aur 1 ke beech **sandwich** karo: $-1 \le u \le 1$, phir x nikaalo. $\sec^{-1}$, $\operatorname{cosec}^{-1}$ mein ulta: $u \le -1$ ya $u \ge 1$.

Solved Example 18

Find the domain of (i) $\sin^{-1}\sqrt{x}$ (ii) $\sqrt{\sin^{-1} x}$.

1. (i) Need $x \ge 0$ (for the root) and $\sqrt{x} \le 1$, i.e. $0 \le x \le 1$.
2. (ii) Need $x \in [-1, 1]$ and $\sin^{-1} x \ge 0$; $\sin^{-1} x \ge 0$ exactly when $x \ge 0$.

✔ (i) $[0, 1]$ (ii) $[0, 1]$

Solved Example 19

Find the range of $\cos^{-1} x$ when $x \in \left[-\frac{1}{2}, 1\right]$.

1. $\cos^{-1}$ is decreasing, so the largest value is at $x = -\frac{1}{2}$ and the smallest at $x = 1$.
2. $\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$, $\cos^{-1} 1 = 0$.

✔ $\left[0, \frac{2\pi}{3}\right]$

7. Range Problems and the Triangle Method

→ If $y = a\sin^{-1} x + b$ $(a > 0)$: $y \in \left[-\frac{a\pi}{2} + b, \frac{a\pi}{2} + b\right]$.

→ $\sin^{-1} x$ and $\tan^{-1} x$ are increasing, so on $[-1, 1]$ the sum $\sin^{-1} x + \tan^{-1} x$ is least at $x = -1$ and greatest at $x = 1$.

→ **Triangle method:** to find $\tan(\cos^{-1}\frac{3}{5})$, let $\theta = \cos^{-1}\frac{3}{5}, \theta \in [0, \frac{\pi}{2}]$ (as $\frac{3}{5} > 0$), draw a right triangle with base 3, hypotenuse 5, perpendicular 4.

Solved Example 20

Find (i) the range of $y = 3\cos^{-1}x - \pi$ (ii) the greatest value of $\sin^{-1}x + \tan^{-1}x$.

1. (i) $0 \le \cos^{-1}x \le \pi \Rightarrow 0 \le 3\cos^{-1}x \le 3\pi \Rightarrow -\pi \le y \le 2\pi$.
2. (ii) both increase, greatest at $x = 1$: $\frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}$.

✔ (i) $[-\pi, 2\pi]$ (ii) $\frac{3\pi}{4}$

Solved Example 21

Find $\tan\left(\cos^{-1}\frac{3}{5}\right)$ and $\sin\left(\cos^{-1}\frac{3}{5}\right)$.

1. Let $\theta = \cos^{-1}\frac{3}{5}$, $\cos\theta = \frac{3}{5}$, θ in I quadrant.
2. Perpendicular $= \sqrt{5^2 - 3^2} = 4$.
3. $\tan\theta = \frac{4}{3}$, $\sin\theta = \frac{4}{5}$.

✔ $\frac{4}{3}$ and $\frac{4}{5}$

Solved Example 22

Find (i) $\sin\left(\tan^{-1}\frac{3}{4}\right)$ (ii) $\cos\left(\sin^{-1}\left(-\frac{5}{13}\right)\right)$ (iii) $\cot\left(\cos^{-1}\left(-\frac{4}{5}\right)\right)$.

1. (i) θ in I quadrant, perpendicular 3, base 4, hypotenuse 5: $\sin\theta = \frac{3}{5}$.
2. (ii) $\theta \in [-\frac{\pi}{2}, 0)$, so $\cos\theta > 0$: $\cos\theta = \sqrt{1 - \frac{25}{169}} = \frac{12}{13}$.
3. (iii) $\theta \in (\frac{\pi}{2}, \pi)$: $\cos\theta = -\frac{4}{5}$, $\sin\theta = \frac{3}{5}$, $\cot\theta = -\frac{4}{3}$.

✔ $\frac{3}{5}$, $\frac{12}{13}$, $-\frac{4}{3}$

Solved Example 23

HOTS: Find $\sin^{-1}(\sin 10)$ (10 in radians).

1. $\pi \approx 3.14$, so $3\pi \approx 9.42 < 10 < \frac{7\pi}{2} \approx 11.0$.
2. $\sin 10 = \sin(3\pi + (10 - 3\pi)) = -\sin(10 - 3\pi) = \sin(3\pi - 10)$.
3. $3\pi - 10 \approx -0.58$ lies in $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

✔ $3\pi - 10$

Graph facts in one table

ITF	Nature	Key points
$\sin^{-1}x$	increasing	$(-1, -\frac{\pi}{2}), (0,0), (1, \frac{\pi}{2})$
$\cos^{-1}x$	decreasing	$(-1, \pi), (0, \frac{\pi}{2}), (1, 0)$
$\tan^{-1}x$	increasing	$(0,0)$; asymptotes $y = \pm\frac{\pi}{2}$
$\cot^{-1}x$	decreasing	$(0, \frac{\pi}{2})$; asymptotes $y = 0, \pi$

Yaad karo: "Circle pe range dekho"

Unit circle banao. **Right half** (IV + I quadrant, $-\frac{\pi}{2}$ to $\frac{\pi}{2}$) = sin, tan, cosec ki range. **Upper half** (I + II quadrant, 0 to π) = cos, cot, sec ki range. Answer hamesha usi half mein girna chahiye.

Formula Box: Inverse Trig

- $y = \sin^{-1}x \iff x = \sin y, y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$
- $y = \cos^{-1}x \iff x = \cos y, y \in [0, \pi]$
- $y = \tan^{-1}x \iff x = \tan y, y \in (-\frac{\pi}{2}, \frac{\pi}{2})$
- $\sin^{-1}(-x) = -\sin^{-1}x$, $\tan^{-1}(-x) = -\tan^{-1}x$, $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}x$
- $\cos^{-1}(-x) = \pi - \cos^{-1}x$, $\cot^{-1}(-x) = \pi - \cot^{-1}x$, $\sec^{-1}(-x) = \pi - \sec^{-1}x$
- $\sin(\sin^{-1}x) = x, x \in [-1, 1]$; $\sin^{-1}(\sin x) = x, x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$
- $\cos^{-1}(\cos x) = x$ only for $x \in [0, \pi]$
- $\sec^{-1}x = \cos^{-1}\frac{1}{x}$, $\operatorname{cosec}^{-1}x = \sin^{-1}\frac{1}{x}$ for $|x| \ge 1$

Yaad karo: "Check karo, phir likho"

Answer likhne se pehle **do check**: (1) kya angle principal range mein hai? (2) uska sin/cos/tan wapas diya hua number deta hai? Dono haan = pakka sahi.

8. ITF in Real Life: Angles

Whenever an angle is found from two lengths, the answer is an ITF: angle of elevation $\theta = \tan^{-1}\frac{\text{height}}{\text{distance}}$; angle of a ladder with the ground $\theta = \cos^{-1}\frac{\text{foot distance}}{\text{length}}$. Such angles lie in $(0, \frac{\pi}{2})$, which is inside every principal branch, so no sign trouble.

Solved Example 24

A pole 20 m tall casts a shadow $20\sqrt{3}$ m long. Find the angle of elevation of the sun using an ITF.

1. $\tan\theta = \frac{20}{20\sqrt{3}} = \frac{1}{\sqrt{3}}$.
2. $\theta = \tan^{-1}\frac{1}{\sqrt{3}} = \frac{\pi}{6}$.

✔ $\frac{\pi}{6}$ $(30°)$

Solved Example 25

A kite string 100 m long makes an angle θ with the ground, and the kite is $50\sqrt{2}$ m high. Find θ. If the kite rises to $50\sqrt{3}$

m (string still 100 m), find the increase in θ.

1. $\sin\theta = \frac{50\sqrt{2}}{100} = \frac{1}{\sqrt{2}} \Rightarrow \theta = \sin^{-1}\frac{1}{\sqrt{2}} = \frac{\pi}{4}$.
2. New: $\sin^{-1}\frac{50\sqrt{3}}{100} = \sin^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{3}$.
3. Increase $= \frac{\pi}{3} - \frac{\pi}{4}$.

✔ $\theta = \frac{\pi}{4}$; increase $= \frac{\pi}{12}\ (15^\circ)$

Common doubts, clear answers

→ **Answer degree mein likh sakte hain?** Radians ($\frac{\pi}{6}$) preferred; degrees only if the question uses them.

→ $\sin^{-1}x$ **aur** $\sin x^{-1}$ **same?** No. $\sin x^{-1} = \sin\frac{1}{x}$; $\sin^{-1}x$ is the inverse function.

→ $\cos^{-1}(-0.5)$ **ka answer** $\frac{4\pi}{3}$ **bhi toh sahi hai?** $\cos\frac{4\pi}{3} = -0.5$ true, but $\frac{4\pi}{3}$ is not in $[0,\pi]$; principal value is $\frac{2\pi}{3}$.

→ **Kya** $\tan^{-1}x$ **kabhi** $\frac{\pi}{2}$ **hoga?** Never; it only approaches $\frac{\pi}{2}$ as $x \to \infty$.

Most Asked in Boards

→ Principal value of $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$, $\sec^{-1}$ of a standard negative value (1M MCQ)

→ Evaluate a sum like $\tan^{-1}1 + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$ (2M)

→ $\sin^{-1}\left(\sin\frac{3\pi}{5}\right)$ type: answer outside the principal range (1M)

→ Domain of $\sin^{-1}(2x-1)$, $\cos^{-1}(x^2-4)$ type (2M)

→ Range/principal branch of a given ITF, graph reading (MCQ, AR)

→ Case study on angles of elevation, ladders, ramps expressed with ITF (4M)



Shivam Bhaiya says

Unit I (Ch 1 + Ch 2) ke total **8 marks** hain aur ye chapter chhota hai: 1-2 MCQ + ek 2 marker pakka. Table + negative wala jugaad + domain sandwich, bas itna clear rakho.



Meme samjho

Main jab $\tan^{-1}$ ki range $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ **open bracket** mein likhta hoon: haan, ye attention to detail hai.

Asli baat: $\tan\left(\pm\frac{\pi}{2}\right)$ is not defined, so $\pm\frac{\pi}{2}$ are never attained: the range is open.

Quick fact $\tan^{-1}1 = \frac{\pi}{4}$, $\tan^{-1}0 = 0$, $\cot^{-1}0 = \frac{\pi}{2}$.

Trap $\cos^{-1}\left(\cos\frac{7\pi}{6}\right) = \frac{5\pi}{6}$, not $\frac{7\pi}{6}$ (range $[0,\pi]$).

Most Important Questions

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. The principal value of $\cos^{-1}\left(-\frac{1}{2}\right)$ is 1M Most asked

(a) $-\frac{\pi}{3}$ (b) $\frac{\pi}{3}$ (c) $\frac{2\pi}{3}$ (d) $\frac{4\pi}{3}$

Ans. Need $y \in [0,\pi]$ with $\cos y = -\frac{1}{2}$: $y = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$. **(c)**

Q2. The range of $\sec^{-1}x$ is 1M

(a) $[0,\pi]$ (b) $[0,\pi] - \left\{\frac{\pi}{2}\right\}$ (c) $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ (d) $(0,\pi)$

Ans. Principal branch of $\sec^{-1}$: $[0,\pi] - \left\{\frac{\pi}{2}\right\}$. **(b)**

Q3. The domain of $\sin^{-1}(3x)$ is 1M Board fav

(a) $\left[-\frac{1}{3},\frac{1}{3}\right]$ (b) $[-3,3]$ (c) $[-1,1]$ (d) $\left[0,\frac{1}{3}\right]$

Ans. $-1 \le 3x \le 1 \Rightarrow -\frac{1}{3} \le x \le \frac{1}{3}$. **(a)**

Q4. $\sin^{-1}\left(\sin\frac{4\pi}{5}\right)$ equals 1M

(a) $\frac{4\pi}{5}$ (b) $\frac{\pi}{5}$ (c) $-\frac{\pi}{5}$ (d) $\frac{3\pi}{5}$

Ans. $\frac{4\pi}{5}$ is not in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$; $\sin\frac{4\pi}{5} = \sin\frac{\pi}{5}$, so the value is $\frac{\pi}{5}$. **(b)**

Q5. $\tan^{-1}\sqrt{3} - \cot^{-1}(-\sqrt{3})$ equals 1M Most asked

(a) π (b) 0 (c) $\frac{\pi}{2}$ (d) $-\frac{\pi}{2}$

Ans. $\frac{\pi}{3} - \frac{5\pi}{6} = -\frac{\pi}{2}$. **(d)**

Q6. $\cos^{-1}\left(\cos\frac{13\pi}{6}\right)$ equals 1M

(a) $\frac{\pi}{6}$ (b) $\frac{13\pi}{6}$ (c) $\frac{7\pi}{6}$ (d) $\frac{5\pi}{6}$

Ans. $\cos\frac{13\pi}{6} = \cos\left(2\pi + \frac{\pi}{6}\right) = \cos\frac{\pi}{6}$, and $\frac{\pi}{6} \in [0,\pi]$. **(a)**

Q7. **A:** The principal value of $\sin^{-1}\left(\sin\frac{2\pi}{3}\right)$ is $\frac{2\pi}{3}$. **R:** $\sin^{-1}(\sin x) = x$ for $x \in \left[-\frac{\pi}{2},\frac{\pi}{2}\right]$. 1M Must do

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. $\frac{2\pi}{3}$ is outside the range; the value is $\frac{\pi}{3}$. A is false, R is true. **(d)**

Q8. **A:** $\cos^{-1}(-x) = \pi - \cos^{-1}x$ for $x \in [-1,1]$. **R:** The principal range of $\cos^{-1}x$ is $[0,\pi]$, symmetric about $\frac{\pi}{2}$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. If $\cos^{-1}x = \theta \in [0,\pi]$, then $\cos(\pi-\theta) = -x$ with $\pi - \theta \in [0,\pi]$, which is why the result holds. **(a)**

Q9. The principal value of $\sin^{-1}\left(\cos\frac{2\pi}{3}\right)$ is 1M

(a) $-\frac{\pi}{6}$ (b) $\frac{5\pi}{6}$ (c) $\frac{2\pi}{3}$ (d) $\frac{\pi}{6}$

Ans. $\cos\frac{2\pi}{3} = -\frac{1}{2}$, $\sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$. **(a)**

Q10. Which of the following is NOT defined? 1M

(a) $\tan^{-1}100$ (b) $\sec^{-1}(-3)$ (c) $\sin^{-1}1.5$ (d) $\cot^{-1}0$

Ans. $\sin^{-1}$ needs input in $[-1,1]$; 1.5 is outside. **(c)**

Q11. The domain of $\cos^{-1}(2x-3)$ is 1M Board fav

(a) $[-1,1]$ (b) $[1,2]$ (c) $[2,3]$ (d) $[0,2]$

Ans. $-1 \le 2x-3 \le 1 \Rightarrow 2 \le 2x \le 4 \Rightarrow 1 \le x \le 2$. **(b)**

Q12. If $\sin^{-1} x = \dfrac{\pi}{5}$ for some $x \in (-1,1)$, then the value of $\cos^{-1} x$ is 1M HOTS

(a) $\frac{\pi}{5}$ (b) $\frac{7\pi}{10}$ (c) $\frac{\pi}{10}$ (d) $\frac{3\pi}{10}$

Ans. $x = \sin\frac{\pi}{5} = \cos\left(\frac{\pi}{2} - \frac{\pi}{5}\right) = \cos\frac{3\pi}{10}$, and $\frac{3\pi}{10} \in [0,\pi]$. So $\cos^{-1} x = \frac{3\pi}{10}$. **(d)**

Q13. If $\cos^{-1} x = \dfrac{2\pi}{3}$, then x equals 1M

(a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) $\frac{\sqrt{3}}{2}$ (d) $-\frac{\sqrt{3}}{2}$

Ans. $x = \cos\frac{2\pi}{3} = -\frac{1}{2}$. **(b)**

Q14. The greatest value of $\tan^{-1} x$ on the interval $[-1,\sqrt{3}]$ is 1M

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $-\frac{\pi}{4}$

Ans. $\tan^{-1}$ is increasing: greatest at $x = \sqrt{3}$, value $\frac{\pi}{3}$. **(c)**

Q15. A: $\sin^{-1}\left(\frac{3}{2}\right)$ is not defined. R: The domain of $\sin^{-1} x$ is $[-1,1]$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. $\frac{3}{2} > 1$ is outside the domain, exactly as R says. **(a)**

Q16. The value of $\sin\left(\cos^{-1}\dfrac{4}{5}\right)$ is 1M

(a) $\frac{3}{5}$ (b) $\frac{4}{5}$ (c) $\frac{3}{4}$ (d) $\frac{5}{3}$

Ans. θ in I quadrant, $\cos\theta = \frac{4}{5} \Rightarrow \sin\theta = \frac{3}{5}$. **(a)**

Q17. $\sin^{-1} x = \cos^{-1} x$ holds for $x =$ 1M

(a) 0 (b) $\frac{1}{\sqrt{2}}$ (c) 1 (d) $\frac{1}{2}$

Ans. Both are $\frac{\pi}{4}$ at $x = \frac{1}{\sqrt{2}}$. **(b)**

Q18. The principal value of $\tan^{-1}\left(\tan\dfrac{9\pi}{8}\right)$ is 1M

(a) $\frac{9\pi}{8}$ (b) $-\frac{\pi}{8}$ (c) $\frac{\pi}{8}$ (d) $\frac{7\pi}{8}$

Ans. $\tan\frac{9\pi}{8} = \tan\left(\pi + \frac{\pi}{8}\right) = \tan\frac{\pi}{8}$ and $\frac{\pi}{8} \in \left(-\frac{\pi}{2},\frac{\pi}{2}\right)$. **(c)**

Very short (2 marks)

Q19. Find the value of $\cot^{-1}\left(-\dfrac{1}{\sqrt{3}}\right) + \operatorname{cosec}^{-1}(-2) + \sec^{-1}(2)$. 2M Board fav

Ans. $\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$, $\operatorname{cosec}^{-1}(-2) = -\frac{\pi}{6}$, $\sec^{-1} 2 = \frac{\pi}{3}$. Sum $= \frac{4\pi-\pi+2\pi}{6} = \frac{5\pi}{6}$.

Q20. Find the domain of $f(x) = \sin^{-1}(x^2-3)$. 2M

Ans. $-1 \le x^2-3 \le 1 \Rightarrow 2 \le x^2 \le 4 \Rightarrow \sqrt{2} \le |x| \le 2$. Domain $= [-2,-\sqrt{2}] \cup [\sqrt{2},2]$.

Q21. Evaluate $\cos^{-1}\left(\cos\dfrac{5\pi}{4}\right)$. 2M

Ans. $\frac{5\pi}{4} > \pi$. $\cos\frac{5\pi}{4} = \cos\left(2\pi - \frac{5\pi}{4}\right) = \cos\frac{3\pi}{4}$ and $\frac{3\pi}{4} \in [0,\pi]$. Value $= \frac{3\pi}{4}$.

Q22. Find the value of $\sin\left(2\sin^{-1}\dfrac{3}{5}\right)$. 2M HOTS

Ans. Let $\theta = \sin^{-1}\frac{3}{5}$, $\sin\theta = \frac{3}{5}$, $\cos\theta = \frac{4}{5}$ (θ in I quadrant). $\sin 2\theta = 2 \times \frac{3}{5} \times \frac{4}{5} = \frac{24}{25}$.

Q23. Find the value of $\tan^{-1}\left[2\cos\left(2\sin^{-1}\dfrac{1}{2}\right)\right]$. 2M Most asked

Ans. $\sin^{-1}\frac{1}{2} = \frac{\pi}{6}$, $2 \times \frac{\pi}{6} = \frac{\pi}{3}$, $2\cos\frac{\pi}{3} = 1$, $\tan^{-1} 1 = \frac{\pi}{4}$.

Q24. Find the principal value of $\cos^{-1}\left(\cos\left(-\dfrac{\pi}{3}\right)\right)$ and of $\sin^{-1}\left(\sin\left(-\dfrac{\pi}{3}\right)\right)$. 2M

Ans. $\cos\left(-\frac{\pi}{3}\right) = \frac{1}{2} \Rightarrow \cos^{-1}\frac{1}{2} = \frac{\pi}{3}$ (not $-\frac{\pi}{3}$). $-\frac{\pi}{3} \in \left[-\frac{\pi}{2},\frac{\pi}{2}\right]$, so $\sin^{-1}\left(\sin\left(-\frac{\pi}{3}\right)\right) = -\frac{\pi}{3}$.

Q25. Find the range of $f(x) = 2\tan^{-1} x + \dfrac{\pi}{2}$. 2M

Ans. $-\frac{\pi}{2} < \tan^{-1} x < \frac{\pi}{2} \Rightarrow -\pi < 2\tan^{-1} x < \pi \Rightarrow -\frac{\pi}{2} < f(x) < \frac{3\pi}{2}$. Range $\left(-\frac{\pi}{2},\frac{3\pi}{2}\right)$.

Q26. Write the principal value of $\sec^{-1}(-\sqrt{2}) + \operatorname{cosec}^{-1}(-\sqrt{2})$. 2M

Ans. $\sec^{-1}(-\sqrt{2}) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$, $\operatorname{cosec}^{-1}(-\sqrt{2}) = -\frac{\pi}{4}$. Sum $= \frac{\pi}{2}$.

Q27. If $\sin^{-1}(x^2-7x+12) = 0$, find x. 2M

Ans. $\sin^{-1} u = 0 \iff u = 0$: $x^2-7x+12 = 0 \Rightarrow (x-3)(x-4) = 0 \Rightarrow x = 3$ or 4.

Q28. Write the interval in which $\cos^{-1} x$ takes values when $x \in \left[0,\dfrac{\sqrt{3}}{2}\right]$. 2M

Ans. $\cos^{-1}$ is decreasing: $\cos^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{6}$, $\cos^{-1} 0 = \frac{\pi}{2}$. Values: $\left[\frac{\pi}{6},\frac{\pi}{2}\right]$.

Q29. Find the domain of $f(x) = \cos^{-1}(|x|-2)$. 2M

Ans. $-1 \le |x|-2 \le 1 \Rightarrow 1 \le |x| \le 3$. Domain $= [-3,-1] \cup [1,3]$.

Q30. Evaluate $\tan\left(\sin^{-1}\dfrac{8}{17}\right)$. 2M

Ans. $\sin\theta = \frac{8}{17}$, θ in I quadrant, base $= \sqrt{289-64} = 15$. $\tan\theta = \frac{8}{15}$.

Q31. Find the value of $\cos\left(\sec^{-1} 2 + \operatorname{cosec}^{-1}(-2)\right)$. 2M

Ans. $\sec^{-1} 2 = \frac{\pi}{3}$, $\operatorname{cosec}^{-1}(-2) = -\frac{\pi}{6}$. Sum $= \frac{\pi}{6}$, and $\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$.

Short answer (3 marks)

Q32. Find the domain of $f(x) = \cos^{-1}\left(\dfrac{x-1}{2}\right) + \sin^{-1}(x-2)$. 3M

Ans. $-1 \le \frac{x-1}{2} \le 1 \Rightarrow -1 \le x \le 3$. $-1 \le x-2 \le 1 \Rightarrow 1 \le x \le 3$. Common part: $[1,3]$.

Q33. Find the domain of $f(x) = \sin^{-1}(1-2x) + \sec^{-1}(3x)$. 3M Board fav

(contd.) $-1 \le 1-2x \le 1 \Rightarrow 0 \le x \le 1$. $|3x| \ge 1 \Rightarrow x \le -\frac{1}{3}$ or $x \ge \frac{1}{3}$. Common: $\left[\frac{1}{3}, 1\right]$.

Q34. Evaluate $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) + \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) - \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$. 3M Most asked

Ans. $-\frac{\pi}{3} + \left(\pi - \frac{\pi}{4}\right) - \left(-\frac{\pi}{6}\right) = -\frac{\pi}{3} + \frac{3\pi}{4} + \frac{\pi}{6}$. With LCM 12 the sum is $(-4+9+2)\frac{\pi}{12} = \frac{7\pi}{12}$.

Q35. A tower is 30 m high. Find, using ITF, the angles of elevation of its top from points 30 m and $10\sqrt{3}$ m from its foot, and the difference of the two angles. 3M Must do

Ans. $\tan^{-1}\frac{30}{30} = \tan^{-1}1 = \frac{\pi}{4}$; $\tan^{-1}\frac{30}{10\sqrt{3}} = \tan^{-1}\sqrt{3} = \frac{\pi}{3}$. Difference $= \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}$.

Q36. Evaluate $\sin^{-1}\left(\sin\frac{5\pi}{6}\right) + \cos^{-1}\left(\cos\frac{5\pi}{3}\right) + \tan^{-1}\left(\tan\frac{2\pi}{3}\right)$. 3M Most asked

Ans. $\sin\frac{5\pi}{6} = \sin\frac{\pi}{6} \Rightarrow \frac{\pi}{6}$. $\cos\frac{5\pi}{3} = \cos\frac{\pi}{3} \Rightarrow \frac{\pi}{3}$. $\tan\frac{2\pi}{3} = -\sqrt{3} \Rightarrow -\frac{\pi}{3}$. Sum $= \frac{\pi}{6}$.

Q37. Find the values of x for which $\cos^{-1}x > \frac{\pi}{3}$ and those for which $\tan^{-1}x < -\frac{\pi}{4}$. 3M HOTS

Ans. $\cos^{-1}$ is decreasing: $\cos^{-1}x > \frac{\pi}{3} = \cos^{-1}\frac{1}{2} \iff x < \frac{1}{2}$; with domain: $x \in \left[-1, \frac{1}{2}\right)$. $\tan^{-1}$ is increasing: $\tan^{-1}x < -\frac{\pi}{4} = \tan^{-1}(-1) \iff x < -1$, i.e. $x \in (-\infty, -1)$.

Q38. Find the value of $\cos\left(\tan^{-1}\frac{5}{12}\right) + \sin\left(\cos^{-1}\left(-\frac{3}{5}\right)\right)$. 3M HOTS

Ans. $\tan\alpha = \frac{5}{12}$, α in I quadrant, hypotenuse 13: $\cos\alpha = \frac{12}{13}$. $\cos\beta = -\frac{3}{5}$, $\beta \in \left(\frac{\pi}{2}, \pi\right)$, $\sin\beta = \frac{4}{5}$. Sum $= \frac{12}{13} + \frac{4}{5} = \frac{60+52}{65} = \frac{112}{65}$.

Long answer (5 marks)

Q39. (i) Draw the graph of $y = \cos^{-1}x$ and write its domain and range. (ii) Find the principal values of $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ and $\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$ and mark them on the graph. (iii) Is $\cos^{-1}$ increasing or decreasing? Hence compare $\cos^{-1}(0.2)$ and $\cos^{-1}(0.7)$. 5M

Ans. (i) Graph from $(-1, \pi)$ through $\left(0, \frac{\pi}{2}\right)$ to $(1, 0)$; domain $[-1, 1]$, range $[0, \pi]$. (ii) $\frac{5\pi}{6}$ at $x = -\frac{\sqrt{3}}{2}$, $\frac{\pi}{4}$ at $x = \frac{1}{\sqrt{2}}$. (iii) Decreasing, so $0.2 < 0.7 \Rightarrow \cos^{-1}(0.2) > \cos^{-1}(0.7)$.

Q40. (i) Find the domain of $f(x) = \sin^{-1}(2x-1) + \cos^{-1}(1-x)$. (ii) Evaluate $f(1)$ and $f\left(\frac{1}{2}\right)$. (iii) Find the principal value of $\tan^{-1}\left(\tan\frac{5\pi}{6}\right) + \cos^{-1}\left(\cos\frac{13\pi}{6}\right)$. 5M Board fav

Ans. (i) $-1 \le 2x-1 \le 1 \Rightarrow [0, 1]$; $-1 \le 1-x \le 1 \Rightarrow [0, 2]$; domain $= [0, 1]$. (ii) $f(1) = \sin^{-1}1 + \cos^{-1}0 = \frac{\pi}{2} + \frac{\pi}{2} = \pi$; $f\left(\frac{1}{2}\right) = \sin^{-1}0 + \cos^{-1}\frac{1}{2} = \frac{\pi}{3}$

(contd.). (iii) $\tan\frac{5\pi}{6} = -\frac{1}{\sqrt{3}} \Rightarrow -\frac{\pi}{6}$; $\cos\frac{13\pi}{6} = \cos\frac{\pi}{6} \Rightarrow \frac{\pi}{6}$. Sum $= 0$.

Case based (4 marks)

Q41. Ladder on a wall. A 10 m ladder rests against a vertical wall with its foot 5 m from the wall. Let θ be the angle it makes with the ground. (i) Write θ using an inverse trigonometric function. (1) (ii) If the foot is moved to $5\sqrt{2}$ m from the wall, find the new angle. (1) (iii) Write the domain and range of the function used, and find the angle when the top of the ladder is $5\sqrt{3}$ m high, using $\sin^{-1}$. (2) 4M Must do

Ans. (i) $\cos\theta = \frac{5}{10}$, so $\theta = \cos^{-1}\frac{1}{2} = \frac{\pi}{3}$. (ii) $\theta = \cos^{-1}\frac{5\sqrt{2}}{10} = \cos^{-1}\frac{1}{\sqrt{2}} = \frac{\pi}{4}$. (iii) $\cos^{-1}$: domain $[-1, 1]$, range $[0, \pi]$. $\sin\theta = \frac{5\sqrt{3}}{10} = \frac{\sqrt{3}}{2}$, $\theta = \sin^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{3}$.

Q42. Graph in class. A teacher draws the graph of $y = \sin^{-1}x$ on the board. (i) Find the domain of $y = \sin^{-1}(2x)$. (1) (ii) Find the principal value of $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$. (1) (iii) Find the range of $y = 2\sin^{-1}x + \frac{\pi}{2}$ and the value of x for which $y = \frac{\pi}{2}$. (2) 4M

Ans. (i) $-1 \le 2x \le 1 \Rightarrow \left[-\frac{1}{2}, \frac{1}{2}\right]$. (ii) $-\frac{\pi}{3}$. (iii) $-\pi \le 2\sin^{-1}x \le \pi$, so $y \in \left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$; $y = \frac{\pi}{2} \Rightarrow \sin^{-1}x = 0 \Rightarrow x = 0$.

Quick fact $\sec^{-1}1 = 0$, $\sec^{-1}(-1) = \pi$, $\operatorname{cosec}^{-1}1 = \frac{\pi}{2}$.

Quick fact $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}$ is a result you may use, but proofs of such identities are out of the 2026-27 syllabus.

Quick check $\sin^{-1}2$ is not defined: domain of $\sin^{-1}$ is $[-1, 1]$.

Yaad rakho Principal value answer hamesha range ke andar: pehle range likho, phir angle.

Apne notes / galtiyan yahan likho

MATRICES

Most asked: symmetric + skew sum, $A^2 - kA + I = O$ type

Chapter at a glance (board weight: Unit Algebra = 10 marks)

- → **Basics:** order, a_{ij}, types, equality
- → **Operations:** $A \pm B$, kA, AB (row into column)
- → **Transpose:** $(AB)' = B'A'$, symmetric and skew-symmetric
- → **Inverse:** definition, uniqueness, $(AB)^{-1} = B^{-1}A^{-1}$

↳ Out of syllabus 2026-27: elementary row/column operations and inverse by elementary operations.

1. What is a Matrix?

A **matrix** is an ordered rectangular array of numbers (or functions) arranged in rows and columns. A matrix with m rows and n columns has **order** $m \times n$ and has mn elements.

$$A = [a_{ij}]_{m\times n} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

- → a_{ij} = element in the i-th row and j-th column (row first, column later).
- → Rows are **horizontal**, columns are **vertical**.
- → If a matrix has N elements, its possible orders are all pairs $m \times n$ with $mn = N$.
- → Number of $m \times n$ matrices with each entry chosen from k values $= k^{mn}$.

Yaad karo: "RC Cola" = Row phir Column

Order hamesha Rows × Columns. a_{23} matlab **2nd row, 3rd column**. Pehle khade log (row) gino, phir khambe (column).

Solved Example 1

(i) Find all possible orders of a matrix having 18 elements. (ii) How many 3×3 matrices are possible if every entry is 0 or 1?

1. (i) $18 = 1\times 18 = 2\times 9 = 3\times 6 = 6\times 3 = 9\times 2 = 18\times 1$
2. (ii) 9 entries, each has 2 choices: 2^9

✔ **(i) 6 orders:** $1\times 18, 18\times 1, 2\times 9, 9\times 2, 3\times 6, 6\times 3$ **(ii) 512**

Solved Example 2

Construct a 2×3 matrix $A = [a_{ij}]$ whose elements are $a_{ij} = \dfrac{(i-2j)^2}{2}$.

1. $a_{11} = \frac{(1-2)^2}{2} = \frac{1}{2}$, $a_{12} = \frac{(1-4)^2}{2} = \frac{9}{2}$, $a_{13} = \frac{(1-6)^2}{2} = \frac{25}{2}$
2. $a_{21} = \frac{(2-2)^2}{2} = 0$, $a_{22} = \frac{(2-4)^2}{2} = 2$, $a_{23} = \frac{(2-6)^2}{2} = 8$

✔

$$A = \begin{bmatrix} \frac{1}{2} & \frac{9}{2} & \frac{25}{2} \\ 0 & 2 & 8 \end{bmatrix}$$

Solved Example 3

Construct a 3×2 matrix with $a_{ij} = \dfrac{1}{2}|i-3j|$.

1. $a_{11} = \frac{1}{2}|1-3| = 1$, $a_{12} = \frac{1}{2}|1-6| = \frac{5}{2}$
2. $a_{21} = \frac{1}{2}|2-3| = \frac{1}{2}$, $a_{22} = \frac{1}{2}|2-6| = 2$
3. $a_{31} = \frac{1}{2}|3-3| = 0$, $a_{32} = \frac{1}{2}|3-6| = \frac{3}{2}$

✔

$$A = \begin{bmatrix} 1 & \frac{5}{2} \\ \frac{1}{2} & 2 \\ 0 & \frac{3}{2} \end{bmatrix}$$

2. Types of Matrices

Type	Condition	Example
Row	only 1 row ($1 \times n$)	$\begin{bmatrix} 2 & -1 & 5 \end{bmatrix}$
Column	only 1 column ($m \times 1$)	$\begin{bmatrix} 3 \\ 7 \end{bmatrix}$
Square	$m = n$	$\begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}$
Diagonal	square, $a_{ij} = 0$ for $i \neq j$	$\begin{bmatrix} 3 & 0 \\ 0 & -5 \end{bmatrix}$
Scalar	diagonal with all diagonal entries equal (k)	$\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$
Identity I	scalar with $k = 1$	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
Zero O	every element 0	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

↳ Diagonal elements of a square matrix: $a_{11}, a_{22}, a_{33}, \dots$ (where $i = j$). Every identity matrix is scalar, every scalar matrix is diagonal, but not conversely.

Equality of Matrices

$A = [a_{ij}]$ and $B = [b_{ij}]$ are **equal** if (i) they have the same order and (ii) $a_{ij} = b_{ij}$ for every i, j (corresponding elements equal).

Solved Example 4

Find x, y, z if $\begin{bmatrix} x+y & 2 \\ 5+z & xy \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 8 \end{bmatrix}$.

1. Equate corresponding elements: $x + y = 6$, $xy = 8$, $5 + z = 5$
2. $z = 0$. x, y are roots of $t^2 - 6t + 8 = 0 \Rightarrow t = 2, 4$

✔ $x = 2, y = 4, z = 0$ **or** $x = 4, y = 2, z = 0$

Solved Example 5

Find a, b, c, d if $\begin{bmatrix} a-b & 2a+c \\ 2a-b & 3c+d \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 0 & 13 \end{bmatrix}$.

1. $a - b = -1$ and $2a - b = 0$: subtract $\Rightarrow a = 1$, so $b = 2$
2. $2a + c = 5 \Rightarrow c = 3$
3. $3c + d = 13 \Rightarrow d = 4$

✔ $a = 1, b = 2, c = 3, d = 4$

Exam tip

Equality questions: pehle **order same** hai ye likho, phir har corresponding element ki equation banao. Do unknown ke liye do equations ko add/subtract karo, guess mat karo.

3. Addition and Scalar Multiplication

Addition is defined only for matrices of the same order: $A + B = [a_{ij} + b_{ij}]$. **Scalar multiple**: $kA = [k\, a_{ij}]$ (multiply EVERY element by k). $-A = (-1)A$ and $A - B = A + (-1)B$.

Properties (same order matrices)

- → Commutative: $A + B = B + A$
- → Associative: $(A + B) + C = A + (B + C)$
- → Additive identity: $A + O = A$; additive inverse: $A + (-A) = O$
- → $k(A + B) = kA + kB$, $(k + l)A = kA + lA$

Quick fact A 1×1 matrix $[a]$ behaves just like the number a.

Yaad rakho Diagonal of a skew-symmetric matrix is always 0.

Solved Example 6

If $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 1 \\ 0 & 3 \end{bmatrix}$, find X such that $2A + 3X = 5B$.

1. $3X = 5B - 2A = \begin{bmatrix} 20 & 5 \\ 0 & 15 \end{bmatrix} - \begin{bmatrix} 2 & -4 \\ 6 & 0 \end{bmatrix}$
2. $3X = \begin{bmatrix} 18 & 9 \\ -6 & 15 \end{bmatrix}$

✔ $X = \begin{bmatrix} 6 & 3 \\ -2 & 5 \end{bmatrix}$

Solved Example 7

Find X and Y if $X + Y = \begin{bmatrix} 5 & 2 \\ 0 & 9 \end{bmatrix}$ and $X - Y = \begin{bmatrix} 3 & 6 \\ 0 & -1 \end{bmatrix}$.

1. Add: $2X = \begin{bmatrix} 8 & 8 \\ 0 & 8 \end{bmatrix} \Rightarrow X = \begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix}$
2. Subtract: $2Y = \begin{bmatrix} 2 & -4 \\ 0 & 10 \end{bmatrix} \Rightarrow Y = \begin{bmatrix} 1 & -2 \\ 0 & 5 \end{bmatrix}$

✔ $X = \begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix}$, $Y = \begin{bmatrix} 1 & -2 \\ 0 & 5 \end{bmatrix}$

Solved Example 8

Find X if $Y = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$ and $2X + Y = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$.

1. $2X = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} -2 & -2 \\ -4 & -2 \end{bmatrix}$

✔ $X = \begin{bmatrix} -1 & -1 \\ -2 & -1 \end{bmatrix}$

4. Multiplication of Matrices

AB is defined only when number of columns of A = number of rows of B. If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$ and

$$c_{ik} = \sum_{j=1}^{n} a_{ij}\, b_{jk} = a_{i1}b_{1k} + a_{i2}b_{2k} + \cdots + a_{in}b_{nk}$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ 8 & 9 \end{bmatrix}$$

Row 1 of A Col 1 of B entry c11

c11 = (1)(2) + (2)(0) + (-1)(1) = 1

A is 2x3, B is 3x2, so AB is 2x2

Fig: row of A into column of B (multiply pairwise, then add)

Yaad karo: "Andar wale same, bahar wale answer"

$(m \times \underline{n})(\underline{n} \times p)$: beech ke (andar ke) dono numbers **same** hone chahiye, aur bahar wale do numbers $m \times p$ answer ka order hain. Row ko column mein "daalo": **Row into Column.**

Solved Example 9

If $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 3 \end{bmatrix}$, find AB and BA. Is $AB = BA$?

1. A is 2×3, B is 3×2: AB is 2×2, BA is 3×3
2. $AB = \begin{bmatrix} 2+0-1 & 1-2-3 \\ 6+0+2 & 3+0+6 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ 8 & 9 \end{bmatrix}$
3. $BA = \begin{bmatrix} 2+3 & 4+0 & -2+2 \\ 0-3 & 0+0 & 0-2 \\ 1+9 & 2+0 & -1+6 \end{bmatrix}$

✔ $AB = \begin{bmatrix} 1 & -4 \\ 8 & 9 \end{bmatrix}$, $BA = \begin{bmatrix} 5 & 4 & 0 \\ -3 & 0 & -2 \\ 10 & 2 & 5 \end{bmatrix}$; orders differ, so $AB \neq BA$

Solved Example 10

If $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$, find AB and BA.

1. Row 1 of A into columns of B: $(1+4+0,\ 0+2+0,\ 1+0+0) = (5, 2, 1)$
2. Row 2: $(0+2+0,\ 0+1+9,\ 0+0+3) = (2, 10, 3)$
3. Row 3: $(2+0+0,\ 0+0+3,\ 2+0+1) = (2, 3, 3)$
4. Similarly $BA = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 5 & 3 \\ 2 & 3 & 10 \end{bmatrix}$

✔ $AB = \begin{bmatrix} 5 & 2 & 1 \\ 2 & 10 & 3 \\ 2 & 3 & 3 \end{bmatrix}$, so $AB \neq BA$ even for square matrices

Example of zero product: $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, but neither matrix is zero.

Recall $A'A$ and AA' are always symmetric.

Board tip For $f(A)$, replace the constant c by cI.

Properties of Multiplication

- → **NOT commutative**: in general $AB \neq BA$ (even when both exist)
- → Associative: $(AB)C = A(BC)$
- → Distributive: $A(B+C) = AB + AC$, $(A+B)C = AC + BC$
- → Identity: $AI = IA = A$ (for square A)
- → $AB = O$ does **not** imply $A = O$ or $B = O$
- → $AB = AC$ does **not** imply $B = C$ (no cancellation in general)

Meme samjho



Do non-zero matrices ka product zero matrix aa gaya? **Hain? Ye kaise ho gaya!**

Asli baat: In matrices $AB = O$ is possible with $A \neq O$ and $B \neq O$. So never "cancel" a matrix like a number.

Solved Example 11

If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, show that $(A+B)^2 \neq A^2 + 2AB + B^2$.

1. $A + B = \begin{bmatrix} 1 & 3 \\ 4 & 4 \end{bmatrix} \Rightarrow (A+B)^2 = \begin{bmatrix} 13 & 15 \\ 20 & 28 \end{bmatrix}$
2. $A^2 = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$, $AB = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$, $B^2 = I$
3. $A^2 + 2AB + B^2 = \begin{bmatrix} 7+4+1 & 10+2 \\ 15+8 & 22+6+1 \end{bmatrix} = \begin{bmatrix} 12 & 12 \\ 23 & 29 \end{bmatrix}$

✔ Not equal, because $(A+B)^2 = A^2 + AB + BA + B^2$ and $AB \neq BA$ here ($BA = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$)

↳ So for matrices: $(A+B)^2 = A^2 + AB + BA + B^2$ and $(A+B)(A-B) = A^2 - AB + BA - B^2$. The usual algebra identities hold only when $AB = BA$.

Numbers	Matrices
$ab = ba$ always	$AB \neq BA$ in general
$ab = 0 \Rightarrow a = 0$ or $b = 0$	$AB = O$ possible with $A, B \neq O$
$ab = ac, a \neq 0 \Rightarrow b = c$	no cancellation in general
$(a+b)^2 = a^2 + 2ab + b^2$	$(A+B)^2 = A^2 + AB + BA + B^2$

Numbers	Matrices
every $a \neq 0$ has $\frac{1}{a}$	only some square matrices have A^{-1}

Powers: for a square matrix, $A^2 = AA$, $A^3 = A^2A$. If $f(x) = x^2 - 5x + 7$, then $f(A) = A^2 - 5A + 7I$ (the constant becomes a multiple of I).

Solved Example 12

If $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$, show that $A^2 - 6A + 5I = O$. Hence find A^{-1}.

1. $A^2 = \begin{bmatrix} 4+3 & 6+12 \\ 2+4 & 3+16 \end{bmatrix} = \begin{bmatrix} 7 & 18 \\ 6 & 19 \end{bmatrix}$
2. $A^2 - 6A + 5I = \begin{bmatrix} 7-12+5 & 18-18+0 \\ 6-6+0 & 19-24+5 \end{bmatrix} = O$
3. Multiply by A^{-1}: $A - 6I + 5A^{-1} = O \Rightarrow A^{-1} = \frac{1}{5}(6I - A)$

✔ $A^{-1} = \frac{1}{5}\begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$

Solved Example 13

If $A = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}$, find k so that $A^2 = kA - 2I$.

1. $A^2 = \begin{bmatrix} 16-6 & -12+3 \\ 8-2 & -6+1 \end{bmatrix} = \begin{bmatrix} 10 & -9 \\ 6 & -5 \end{bmatrix}$
2. $kA - 2I = \begin{bmatrix} 4k-2 & -3k \\ 2k & -k-2 \end{bmatrix}$
3. Compare: $4k - 2 = 10$, $-3k = -9$, $2k = 6 \Rightarrow k = 3$ (all agree)

✔ $k = 3$

Solved Example 14

If $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, find A^2, A^3 and guess A^n.

1. $A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$
2. $A^3 = A^2A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$
3. Pattern: the top-right entry increases by 1 each time (provable by induction)

✔ $A^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$

Do you know? If A is $m \times n$, then AA' is $m \times m$ and $A'A$ is $n \times n$.

Check trick Every identity matrix is scalar, every scalar matrix is diagonal.

Solved Example 15

A trust invests Rs 30,000 in two bonds paying 5% and 7% interest per year. Using matrix multiplication, find how much is put in each bond to get Rs 1,800 interest a year.

1. Let Rs x go in the 5% bond, Rs $(30000 - x)$ in the 7% bond
2. Interest $= \begin{bmatrix} x & 30000 - x \end{bmatrix}\begin{bmatrix} 0.05 \\ 0.07 \end{bmatrix} = 0.05x + 2100 - 0.07x$
3. $2100 - 0.02x = 1800 \Rightarrow x = 15000$

✔ Rs 15,000 in each bond

Solved Example 16

Show that $A = \begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix}$ is idempotent ($A^2 = A$) and find $(I + A)^3 - 7A$.

1. $A^2 = \begin{bmatrix} 4-2 & -4+2 \\ 2-1 & -2+1 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix} = A$
2. $(I + A)^3 = I + 3A + 3A^2 + A^3$ (I and A commute)
3. With $A^2 = A^3 = A$: $(I + A)^3 = I + 7A$

✔ $(I + A)^3 - 7A = I$

Yaad karo: 2x2 ka shortcut "T-D rule"

Any 2×2 matrix satisfies $A^2 - (\text{trace})A + (\det A)I = O$, where trace = sum of diagonal. E.g. $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$: trace 6, $\det = 8 - 3 = 5$, so $A^2 - 6A + 5I = O$. Answer check karne ke liye best!

Solved Example 17

A stationery seller has two shops. Shop I sells 10 pens, 20 notebooks and 5 bags in a day, Shop II sells 15 pens, 10 notebooks and 8 bags. Prices are Rs 10, Rs 40 and Rs 300 each. Find the day's sales of each shop using matrices.

1. Quantity matrix $Q = \begin{bmatrix} 10 & 20 & 5 \\ 15 & 10 & 8 \end{bmatrix}$, price matrix $P = \begin{bmatrix} 10 \\ 40 \\ 300 \end{bmatrix}$
2. $QP = \begin{bmatrix} 100 + 800 + 1500 \\ 150 + 400 + 2400 \end{bmatrix}$

✔ Shop I: Rs 2400, Shop II: Rs 2950



Shivam Bhaiya says
"Matrix multiplication mein sabse zyada marks **order check** na karne se katte hain. Pehle order likho, phir multiply karo. Aur $A^2 - kA + I$ wale sawal 2-3 marks ke pakke aate hain, T-D rule se answer cross check kar lo."

5. Transpose of a Matrix

The **transpose** A^T (or A') is obtained by **interchanging rows and columns**. If $A = [a_{ij}]_{m\times n}$ then $A' = [a_{ji}]_{n\times m}$.

$$A = \begin{bmatrix} 2 & -1 & 5 \\ 0 & 3 & 4 \end{bmatrix} \Rightarrow A' = \begin{bmatrix} 2 & 0 \\ -1 & 3 \\ 5 & 4 \end{bmatrix}$$

Properties of Transpose

- $(A')' = A$
- $(kA)' = kA'$
- $(A+B)' = A' + B'$
- $(AB)' = B'A'$ **(order reverses)**
- $(ABC)' = C'B'A'$

Meme samjho



$(AB)'$ ka answer $A'B'$ nahi, $B'A'$ hota hai? **Achha?!**
Asli baat: Transpose of a product reverses the order: $(AB)' = B'A'$. Same reversal for inverse: $(AB)^{-1} = B^{-1}A^{-1}$.

Solved Example 18

If $A = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 & 1 \end{bmatrix}$, verify that $(AB)' = B'A'$.

1. $AB = \begin{bmatrix} -1 & 2 & 1 \\ 2 & -4 & -2 \\ -3 & 6 & 3 \end{bmatrix} \Rightarrow (AB)' = \begin{bmatrix} -1 & 2 & -3 \\ 2 & -4 & 6 \\ 1 & -2 & 3 \end{bmatrix}$

2. $B'A' = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 3 \end{bmatrix} = \begin{bmatrix} -1 & 2 & -3 \\ 2 & -4 & 6 \\ 1 & -2 & 3 \end{bmatrix}$

✔ $(AB)' = B'A'$, **verified**

6. Symmetric and Skew-Symmetric Matrices

Definitions (square matrix A)

- **Symmetric**: $A' = A$, i.e. $a_{ij} = a_{ji}$ (mirror image about the diagonal)
- **Skew-symmetric**: $A' = -A$, i.e. $a_{ij} = -a_{ji}$
- In a skew-symmetric matrix every **diagonal element is 0** (since $a_{ii} = -a_{ii}$)

$$\begin{bmatrix} 2 & 5 & -1 \\ 5 & 0 & 4 \\ -1 & 4 & 7 \end{bmatrix} \qquad \begin{bmatrix} 0 & 3 & -2 \\ -3 & 0 & 6 \\ 2 & -6 & 0 \end{bmatrix}$$

Symmetric Skew

Fig: mirror about the leading diagonal (same colour = mirror pair)

Solved Example 19

If $A = \begin{bmatrix} 0 & 2x-1 & 4 \\ 3 & 0 & y+2 \\ -4 & 5 & 0 \end{bmatrix}$ is skew-symmetric, find x and y.

1. Skew: $a_{12} = -a_{21} \Rightarrow 2x - 1 = -3 \Rightarrow x = -1$
2. $a_{23} = -a_{32} \Rightarrow y + 2 = -5 \Rightarrow y = -7$
3. Check: $a_{13} = 4 = -a_{31}$, diagonal all 0

✔ $x = -1, y = -7$

Theorems

1. For any square A: $A + A'$ is **symmetric** and $A - A'$ is **skew-symmetric**.
   Proof: $(A + A')' = A' + A = A + A'$; $(A - A')' = A' - A = -(A - A')$.
2. Every square matrix is **uniquely** the sum of a symmetric and a skew-symmetric matrix:

$$A = \underbrace{\tfrac{1}{2}(A + A')}_{P\ (\text{symmetric})} + \underbrace{\tfrac{1}{2}(A - A')}_{Q\ (\text{skew})}$$

Solved Example 20

Find x, y, z if $A = \begin{bmatrix} 1 & x & 4 \\ 3 & 2 & z \\ y & -5 & 0 \end{bmatrix}$ is symmetric.

1. Symmetric $\Rightarrow a_{ij} = a_{ji}$
2. $a_{12} = a_{21} \Rightarrow x = 3$; $a_{13} = a_{31} \Rightarrow y = 4$; $a_{23} = a_{32} \Rightarrow z = -5$

✔ $x = 3, y = 4, z = -5$

→ If A, B symmetric (same order): AB is symmetric $\iff AB = BA$.

→ $A'A$ and AA' are always symmetric. A matrix that is both symmetric and skew-symmetric is O.

Solved Example 21

For $A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$, verify that $A + A'$ is symmetric and $A - A'$ is skew-symmetric.

1. $A' = \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix}$, $A + A' = \begin{bmatrix} 4 & 7 \\ 7 & 10 \end{bmatrix}$: mirror entries equal, symmetric
2. $A - A' = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$: its transpose is $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = -(A - A')$, skew

✔ Both verified

Solved Example 22

Express $A = \begin{bmatrix} 3 & 5 & -1 \\ 1 & -2 & 4 \\ 7 & 0 & 6 \end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrix.

1. $A' = \begin{bmatrix} 3 & 1 & 7 \\ 5 & -2 & 0 \\ -1 & 4 & 6 \end{bmatrix}$
2. $P = \frac{1}{2}(A + A') = \frac{1}{2}\begin{bmatrix} 6 & 6 & 6 \\ 6 & -4 & 4 \\ 6 & 4 & 12 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 3 \\ 3 & -2 & 2 \\ 3 & 2 & 6 \end{bmatrix}$
3. $Q = \frac{1}{2}(A - A') = \frac{1}{2}\begin{bmatrix} 0 & 4 & -8 \\ -4 & 0 & 4 \\ 8 & -4 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 & -4 \\ -2 & 0 & 2 \\ 4 & -2 & 0 \end{bmatrix}$
4. $P' = P$ (symmetric), $Q' = -Q$ (skew), and $P + Q = A$

✔ $A = P + Q$ with P, Q as above

Yaad karo: "Sym = Same, Skew = Sign badlo, diagonal Zero"

Symmetric: mirror wale elements **same**. **Skew**: mirror wale elements ka **sign ulta**, aur diagonal pe sirf **0**. P mein **plus** $(A + A')$, Q mein **minus** $(A - A')$, dono ko $\frac{1}{2}$.

Solved Example 23

If A is a skew-symmetric matrix and B is any square matrix of the same order, show that $B'AB$ is skew-symmetric.

1. $(B'AB)' = B'A'(B')'$ (reversal rule for three matrices)
2. $= B'(-A)B = -(B'AB)$ since $A' = -A$ and $(B')' = B$

✔ $B'AB$ is skew-symmetric

7. Invertible Matrices

Definition

A square matrix A of order n is **invertible** if there is a square matrix B of the same order with $AB = BA = I_n$. Then B is the **inverse** of A, written A^{-1}. A non-square matrix has no inverse.

Theorem 1: Inverse is unique

Let B and C both be inverses of A. Then $AB = BA = I$ and $AC = CA = I$.

$B = BI = B(AC) = (BA)C = IC = C$. Hence $B = C$: the inverse, if it exists, is unique.

Theorem 2: $(AB)^{-1} = B^{-1}A^{-1}$

$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$.

Similarly $(B^{-1}A^{-1})(AB) = I$. By uniqueness, $(AB)^{-1} = B^{-1}A^{-1}$.

Also: $(A^{-1})^{-1} = A$, $(A')^{-1} = (A^{-1})'$, $(kA)^{-1} = \frac{1}{k}A^{-1}$ $(k \neq 0)$. To check that $B = A^{-1}$, just show $AB = I$ (and $BA = I$).

↳ Quick 2×2 inverse (full method with adjoint in Determinants chapter): $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, if $ad - bc \neq 0$. Diagonal swap, off-diagonal sign change!

Check your inverse in 10 seconds

Found A^{-1}? Multiply $A \times A^{-1}$ in your head for just the (1,1) and (1,2) entries. Agar 1 aur 0 aaye, answer almost pakka sahi hai. Board mein ye check rough mein karo.

Solved Example 24

If $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ satisfies $A^2 = 2A + 3I$, find A^{-1} using this equation.

1. Multiply both sides by A^{-1}: $A = 2I + 3A^{-1}$
2. $3A^{-1} = A - 2I = \begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$

✔ $A^{-1} = \frac{1}{3}\begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$

Solved Example 25

If $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$, show that $A^2 - A - 2I = O$ and hence find A^{-1}.

1. $A^2 = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} = A + 2I$
2. So $A^2 - A - 2I = O$. Multiply by A^{-1}: $A - I - 2A^{-1} = O$

✔ $A^{-1} = \frac{1}{2}(A - I) = \frac{1}{2}\begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$

Solved Example 26

Show that $B = \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix}$ is the inverse of $A = \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}$.

1. $AB = \begin{bmatrix} 15-14 & -6+6 \\ 35-35 & -14+15 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
2. $BA = \begin{bmatrix} 15-14 & 10-10 \\ -21+21 & -14+15 \end{bmatrix} = I$

✔ $AB = BA = I$, so $B = A^{-1}$

Solved Example 27

If $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, find $(AB)^{-1}$.

1. $A^{-1} = \frac{1}{1}\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$, $B^{-1} = \frac{1}{1}\begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$
2. $(AB)^{-1} = B^{-1}A^{-1} = \begin{bmatrix} 1+2 & -1-4 \\ 0-1 & 0+2 \end{bmatrix}$

✔ $(AB)^{-1} = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}$

↳ Yaad rahe: A^{-1} exists only for a **square** matrix with $|A| \neq 0$ (non-singular). Full test in the Determinants chapter.

Formula Box 1: Basics

- Order $m \times n$: m rows, n columns, mn elements
- $A = B \iff$ same order and $a_{ij} = b_{ij}$
- $A \pm B = [a_{ij} \pm b_{ij}]$ (same order only), $kA = [ka_{ij}]$
- $(m \times n)(n \times p) = (m \times p)$, $c_{ik} = \sum_j a_{ij}b_{jk}$
- $AB \neq BA$ in general; $AI = IA = A$
- $A^2 - (\mathrm{tr}A)A + (\det A)I = O$ for 2×2

Formula Box 2: Transpose and Inverse

- $(A')' = A$, $(kA)' = kA'$, $(A+B)' = A' + B'$, $(AB)' = B'A'$
- Symmetric: $A' = A$; skew: $A' = -A$, $a_{ii} = 0$
- $A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$
- $AA^{-1} = A^{-1}A = I$; inverse is unique
- $(AB)^{-1} = B^{-1}A^{-1}$, $(A^{-1})^{-1} = A$, $(A^{-1})' = (A')^{-1}$
- 2×2: $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Solved Example 28

Verify $(AB)^{-1} = B^{-1}A^{-1}$ for $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$.

1. $A^{-1} = \frac{1}{5-6}\begin{bmatrix} 5 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$
2. $AB = \begin{bmatrix} 4 & 3 \\ 11 & 8 \end{bmatrix} \Rightarrow (AB)^{-1} = \frac{1}{32-33}\begin{bmatrix} 8 & -3 \\ -11 & 4 \end{bmatrix} = \begin{bmatrix} -8 & 3 \\ 11 & -4 \end{bmatrix}$
3. $B^{-1}A^{-1} = \begin{bmatrix} -5-3 & 2+1 \\ 5+6 & -2-2 \end{bmatrix} = \begin{bmatrix} -8 & 3 \\ 11 & -4 \end{bmatrix}$

✔ Both equal, verified

Results you can quote in proofs

- $(A^{-1})' = (A')^{-1}$: so inverse of a symmetric matrix is symmetric.
- $(A^n)' = (A')^n$ and $(A^n)^{-1} = (A^{-1})^n$.
- $I^{-1} = I$ and $I' = I$.

Board answer-writing: Symmetric + Skew (full marks format)

1. Write A' clearly.
2. $P = \frac{1}{2}(A + A')$: show the sum matrix, then halve.
3. Prove $P' = P$ (one line).
4. $Q = \frac{1}{2}(A - A')$: show the difference, then halve.
5. Prove $Q' = -Q$ (one line).
6. Conclude: $A = P + Q$ = symmetric + skew-symmetric.

Common mistakes

→ Multiplying without checking order: $(2 \times 3)(2 \times 3)$ is NOT defined.
→ Writing $(AB)' = A'B'$ or $(AB)^{-1} = A^{-1}B^{-1}$: order must reverse.
→ Writing $A^2 - 5A + 7$ instead of $A^2 - 5A + 7I$ (a number can't be added to a matrix).
→ Cancelling: $AB = AC \Rightarrow B = C$ is wrong in general.
→ Forgetting the $\frac{1}{2}$ in P and Q, or putting non-zero diagonal in a skew matrix.

Ek minute test (answers at the end)

1. Order of AB if A is 4×2, B is 2×5?
2. Diagonal elements of a skew-symmetric matrix?
3. $(ABC)' = ?$
4. Can a 2×3 matrix have an inverse?
5. If $A^2 = I$, then $A^{-1} = ?$
6. Is every diagonal matrix symmetric?

Answers: 1) 4×5 2) all 0 3) $C'B'A'$ 4) No, only square matrices 5) A 6) Yes, $A' = A$

Most Asked in Boards

→ Express a 3×3 matrix as sum of symmetric + skew-symmetric (3-5M)
→ Show $A^2 - kA + lI = O$ / find k, hence A^{-1} (2-3M)
→ Construct $A = [a_{ij}]$ from a formula (1-2M)
→ Find unknowns from matrix equality / matrix equation (2M)
→ Order of AB / when a product is defined (MCQ)
→ Skew-symmetric conditions, diagonal = 0 (MCQ, AR)
→ Verify $(AB)' = B'A'$, $(AB)^{-1} = B^{-1}A^{-1}$ (2-3M)
→ Real-life case: sales/cost matrix product (4M case based)

Yaad karo: Inverse ka "Uni-Rev" rule

Inverse **Uni**que hota hai (ek hi), aur product ka inverse ya transpose **Rev**erse order mein: $(AB)^{-1} = B^{-1}A^{-1}$, $(AB)' = B'A'$. Jaise moze pehle, joote baad mein pehno, toh utaarte waqt joote pehle!



Shivam Bhaiya says
"Symmetric + skew wala 3x3 question board ka favourite hai. Last step mein $P' = P$, $Q' = -Q$ aur $P + Q = A$ zaroor likhna, isi line ke marks hote hain."

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. If A is a 3×4 matrix and B is a matrix such that both $A'B$ and BA' are defined, then the order of B is 1M Board fav
(a) 3×3 (b) 4×4 (c) 3×4 (d) 4×3

Ans. (c) A' is 4×3. $A'B$ defined $\Rightarrow$ B has 3 rows; BA' defined $\Rightarrow$ B has 4 columns. So B is 3×4.

Q2. The number of all possible matrices of order 2×3 with each entry 0 or 1 is 1M
(a) 6 (b) 12 (c) 32 (d) 64

Ans. (d) 6 entries, 2 choices each: $2^6 = 64$.

Q3. If $A = \begin{bmatrix} 0 & a & -3 \\ 2 & 0 & -1 \\ b & 1 & 0 \end{bmatrix}$ is skew-symmetric, then 1M Most asked
(a) $a = 2, b = -3$ (b) $a = -2, b = 3$ (c) $a = -2, b = -3$ (d) $a = 2, b = 3$

Ans. (b) $a_{12} = -a_{21} \Rightarrow a = -2; a_{31} = -a_{13} \Rightarrow b = 3$.

Q4. If A is a square matrix with $A^2 = A$, then $(I + A)^3 - 7A$ equals 1M HOTS
(a) A (b) $I - A$ (c) I (d) $3A$

Ans. (c) $(I + A)^3 = I + 3A + 3A^2 + A^3 = I + 3A + 3A + A = I + 7A$, so the answer is I.

Q5. If $A = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$ is such that $A^2 = I$, then 1M
(a) $1 + \alpha^2 + \beta\gamma = 0$ (b) $1 - \alpha^2 + \beta\gamma = 0$ (c) $1 - \alpha^2 - \beta\gamma = 0$ (d) $1 + \alpha^2 - \beta\gamma = 0$

Ans. (c) $A^2 = (\alpha^2 + \beta\gamma)I = I \Rightarrow 1 - \alpha^2 - \beta\gamma = 0$.

Q6. If $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ and $A + A' = I$, then θ ($0 < \theta < \frac{\pi}{2}$) is 1M Board fav
(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$

(contd.) (b) $A + A' = \begin{bmatrix} 2\cos\theta & 0 \\ 0 & 2\cos\theta \end{bmatrix} = I \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$.

Q7. **A:** For any square matrix A, $A + A'$ is symmetric. **R:** $(A + A')' = A' + (A')' = A' + A$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) Both true and R gives $(A + A')' = A + A'$, which is the definition of symmetric.

Q8. **A:** If A and B are 2×2 matrices with $AB = O$, then $A = O$ or $B = O$. **R:** Matrix multiplication is not commutative. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (d) A is false (e.g. $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = O$); R is true.

Q9. A square matrix $A = [a_{ij}]_{3\times3}$ with $a_{ij} = i^2 - j^2$ is 1M Most asked

(a) symmetric (b) skew-symmetric (c) identity (d) neither

Ans. (b) $a_{ji} = j^2 - i^2 = -a_{ij}$ and $a_{ii} = 0$, so A is skew-symmetric.

Q10. If $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, then A^5 is 1M

(a) $10I$ (b) $5A$ (c) 2^5A (d) $32I$

Ans. (d) $A = 2I$, so $A^5 = 2^5 I^5 = 32I$.

Q11. If A and B are invertible matrices of the same order, then $(AB)^{-1}$ equals 1M Must do

(a) $A^{-1}B^{-1}$ (b) AB^{-1} (c) $B^{-1}A^{-1}$ (d) BA

Ans. (c) Reversal law: $(AB)^{-1} = B^{-1}A^{-1}$.

Q12. If A is 2×3 and B is 3×4, the number of elements in AB is 1M

(a) 6 (b) 8 (c) 12 (d) 24

Ans. (b) AB is 2×4: 8 elements.

Q13. **A:** A matrix which is both symmetric and skew-symmetric is a zero matrix. **R:** If $A' = A$ and $A' = -A$, then $A = -A$, i.e. $2A = O$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) Both true and R explains A.

Q14. If $A = [a_{ij}]_{2\times2}$ with $a_{ij} = i + j$, then A is 1M

(a) symmetric (b) skew-symmetric (c) diagonal (d) identity

Ans. (a) $A = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$ and $a_{ij} = a_{ji}$, so $A' = A$: symmetric.

Q15. If $\begin{bmatrix} x & 2 \end{bmatrix}\begin{bmatrix} 3 \\ 4 \end{bmatrix} = [17]$, then x is 1M

(a) 1 (b) 2 (c) 3 (d) 4

Ans. (c) $3x + 8 = 17 \Rightarrow x = 3$.

Q16. If A is a symmetric matrix, then A^3 is 1M

(a) symmetric (b) skew-symmetric (c) zero matrix (d) identity

(contd.) (a) $(A^3)' = (A')^3 = A^3$, so symmetric.

Q17. The number of matrices of order 3×2 whose each entry is 0, 1 or 2 is 1M

(a) 18 (b) 64 (c) 216 (d) 729

Ans. (d) 6 entries, 3 choices each: $3^6 = 729$.

Q18. If $A = \begin{bmatrix} 0 & 2 \\ 3 & -4 \end{bmatrix}$ and $kA = \begin{bmatrix} 0 & 3a \\ 2b & 24 \end{bmatrix}$, then (k, a, b) is 1M

(a) $(-6, 4, 9)$ (b) $(6, -4, -9)$ (c) $(-6, -4, -9)$ (d) $(-6, 12, 18)$

Ans. (c) $-4k = 24 \Rightarrow k = -6$; $2k = 3a \Rightarrow a = -4$; $3k = 2b \Rightarrow b = -9$.

Very short (2 marks)

Q19. Find x and y if $2\begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$. 2M Most asked

Ans. Equate: $2x + 3 = 7 \Rightarrow x = 2$; $2(y-3) + 2 = 14 \Rightarrow 2y = 18 \Rightarrow y = 9$. (Check: $10 - 4 = 6$, $14 + 1 = 15$.) $x = 2, y = 9$.

Q20. For $A = \begin{bmatrix} 2 & 4 \\ 5 & 6 \end{bmatrix}$, show that $A - A'$ is skew-symmetric. 2M

Ans. $A - A' = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Its transpose is $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = -(A - A')$. Hence skew-symmetric.

Q21. Find the matrix X if $X + \begin{bmatrix} 4 & 6 \\ -3 & 7 \end{bmatrix} = \begin{bmatrix} 3 & -6 \\ 5 & -8 \end{bmatrix}$. 2M

Ans. $X = \begin{bmatrix} 3-4 & -6-6 \\ 5+3 & -8-7 \end{bmatrix} = \begin{bmatrix} -1 & -12 \\ 8 & -15 \end{bmatrix}$.

Q22. If A and B are symmetric matrices of the same order, show that AB is symmetric if and only if $AB = BA$. 2M Board fav

Ans. $(AB)' = B'A' = BA$. So AB symmetric $\iff (AB)' = AB \iff BA = AB$.

Q23. If $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, show that $A^2 = 2A + 3I$. 2M

Ans. $A^2 = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$; $2A + 3I = \begin{bmatrix} 2+3 & 4 \\ 4 & 2+3 \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} = A^2$.

Q24. If $A' = \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$, find $(A + 2B)'$. 2M

Ans. $A = \begin{bmatrix} 3 & -1 & 0 \\ 4 & 2 & 1 \end{bmatrix}$, $A + 2B = \begin{bmatrix} 1 & 3 & 2 \\ 6 & 6 & 7 \end{bmatrix}$, so

(contd.) $(A+2B)' = \begin{bmatrix}1 & 6\\ 3 & 6\\ 2 & 7\end{bmatrix}$.

Q25. If $A = \begin{bmatrix}1 & -1\\ 3 & -1\end{bmatrix}$, $B = \begin{bmatrix}a & 1\\ b & -1\end{bmatrix}$ and $(A+B)^2 = A^2 + B^2$, find a and b. 2M HOTS

Ans. Need $AB + BA = O$. $AB = \begin{bmatrix}a-b & 2\\ 3a-b & 4\end{bmatrix}$, $BA = \begin{bmatrix}a+3 & -a-1\\ 3b-3 & 1-b\end{bmatrix}$. Sum $= \begin{bmatrix}2a-b+3 & 1-a\\ 3a+2b-3 & 5-b\end{bmatrix} = O \Rightarrow a=1, b=5$.

Q26. If $A = \begin{bmatrix}5 & 3\\ -1 & -2\end{bmatrix}$, show that $A^2 - 3A - 7I = O$. 2M Must do

Ans. $A^2 = \begin{bmatrix}22 & 9\\ -3 & 1\end{bmatrix}$, $3A + 7I = \begin{bmatrix}15+7 & 9\\ -3 & -6+7\end{bmatrix} = \begin{bmatrix}22 & 9\\ -3 & 1\end{bmatrix} = A^2$. Hence $A^2 - 3A - 7I = O$.

Q27. Find the values of x and y if $\begin{bmatrix}x+2y & -y\\ 3x & 4\end{bmatrix} = \begin{bmatrix}-4 & 3\\ 6 & 4\end{bmatrix}$. 2M

Ans. $-y = 3 \Rightarrow y = -3$; $3x = 6 \Rightarrow x = 2$; check $x + 2y = 2 - 6 = -4$. So $x = 2, y = -3$.

Q28. Show that $A = \begin{bmatrix}0 & 1 & -2\\ -1 & 0 & 3\\ 2 & -3 & 0\end{bmatrix}$ is skew-symmetric and write A^2's (1,1) entry. 2M

Ans. $A' = \begin{bmatrix}0 & -1 & 2\\ 1 & 0 & -3\\ -2 & 3 & 0\end{bmatrix} = -A$, so skew-symmetric.
$(A^2)_{11} = 0(0) + 1(-1) + (-2)(2) = -5$.

Q29. Find A if $2A - 3B + 5C = O$, where $B = \begin{bmatrix}1 & 2\\ 0 & -1\end{bmatrix}$ and $C = \begin{bmatrix}1 & 0\\ 3 & 1\end{bmatrix}$. 2M

Ans. $2A = 3B - 5C = \begin{bmatrix}3-5 & 6-0\\ 0-15 & -3-5\end{bmatrix} = \begin{bmatrix}-2 & 6\\ -15 & -8\end{bmatrix}$, so $A = \begin{bmatrix}-1 & 3\\ -\frac{15}{2} & -4\end{bmatrix}$.

Q30. If $A = \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$, find $A^2 - 5A$. 2M

Ans. $A^2 = \begin{bmatrix}7 & 10\\ 15 & 22\end{bmatrix}$, $5A = \begin{bmatrix}5 & 10\\ 15 & 20\end{bmatrix}$, so $A^2 - 5A = \begin{bmatrix}2 & 0\\ 0 & 2\end{bmatrix} = 2I$. (T-D check: $A^2 - 5A - 2I = O$, trace 5, $\det = -2$.)

Q31. If $A = \begin{bmatrix}4 & x+2\\ 2x-3 & x+1\end{bmatrix}$ is symmetric, find x. 2M Must do

Ans. Symmetric $\Rightarrow a_{12} = a_{21}$: $x + 2 = 2x - 3 \Rightarrow x = 5$.

Q32. Give an example of a non-zero 2×2 matrix A with $A^2 = O$. 2M HOTS

Ans. Yes. $A = \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}$: $A^2 = \begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix} = O$ though $A \neq O$.

Short answer (3 marks)

Q33. Find x if $\begin{bmatrix}1 & x & 1\end{bmatrix}\begin{bmatrix}1 & 2 & 0\\ 2 & 0 & 1\\ 1 & 0 & 2\end{bmatrix}\begin{bmatrix}1\\ 1\\ x\end{bmatrix} = O$. 3M Must do

Ans. First product: $\begin{bmatrix}1+2x+1 & 2+0+0 & 0+x+2\end{bmatrix} = \begin{bmatrix}2x+2 & 2 & x+2\end{bmatrix}$. Then $(2x+2)(1) + 2(1) + (x+2)x = x^2 + 4x + 4 = (x+2)^2 = 0 \Rightarrow x = -2$.

Q34. If $A = \begin{bmatrix}1 & -1\\ 2 & 3\end{bmatrix}$ and $B = \begin{bmatrix}2 & 0\\ 1 & 4\end{bmatrix}$, verify that $(AB)' = B'A'$. 3M

Ans. $AB = \begin{bmatrix}2-1 & 0-4\\ 4+3 & 0+12\end{bmatrix} = \begin{bmatrix}1 & -4\\ 7 & 12\end{bmatrix}$, so $(AB)' = \begin{bmatrix}1 & 7\\ -4 & 12\end{bmatrix}$. $B'A' = \begin{bmatrix}2 & 1\\ 0 & 4\end{bmatrix}\begin{bmatrix}1 & 2\\ -1 & 3\end{bmatrix} = \begin{bmatrix}1 & 7\\ -4 & 12\end{bmatrix}$. Verified.

Q35. If $A = \begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$, find k such that $A^2 - 4A + kI = O$. 3M Most asked

Ans. $A^2 = \begin{bmatrix}5 & -4\\ -4 & 5\end{bmatrix}$, $4A = \begin{bmatrix}8 & -4\\ -4 & 8\end{bmatrix}$. $A^2 - 4A = \begin{bmatrix}-3 & 0\\ 0 & -3\end{bmatrix} = -3I$. So $-3I + kI = O \Rightarrow k = 3$.

Q36. Express $A = \begin{bmatrix}4 & 7\\ 1 & -2\end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrix. 3M

Ans. $A' = \begin{bmatrix}4 & 1\\ 7 & -2\end{bmatrix}$. $P = \frac{1}{2}(A + A') = \begin{bmatrix}4 & 4\\ 4 & -2\end{bmatrix}$ (symmetric), $Q = \frac{1}{2}(A - A') = \begin{bmatrix}0 & 3\\ -3 & 0\end{bmatrix}$ (skew). $P + Q = A$.

Q37. If $A = \begin{bmatrix}1 & 0 & 2\\ 2 & 1 & 0\\ 0 & 1 & 1\end{bmatrix}$, find $A^2 - 3A + 2I$. 3M

Ans. $A^2 = \begin{bmatrix}1 & 2 & 4\\ 4 & 1 & 4\\ 2 & 2 & 1\end{bmatrix}$. $A^2 - 3A + 2I = \begin{bmatrix}1-3+2 & 2-0 & 4-6\\ 4-6 & 1-3+2 & 4-0\\ 2-0 & 2-3 & 1-3+2\end{bmatrix} = \begin{bmatrix}0 & 2 & -2\\ -2 & 0 & 4\\ 2 & -1 & 0\end{bmatrix}$.

Q38. If $A = \begin{bmatrix}2 & 3\\ -1 & 4\end{bmatrix}$, find $A^2 - 6A + 11I$ and hence find A^{-1}. 3M Most asked

(contd.) $A^2 = \begin{bmatrix} 1 & 18 \\ -6 & 13 \end{bmatrix}$; $A^2 - 6A + 11I = \begin{bmatrix} 1-12+11 & 18-18 \\ -6+6 & 13-24+11 \end{bmatrix} = O$. So $A - 6I + 11A^{-1} = O \Rightarrow A^{-1} = \frac{1}{11}(6I - A) = \frac{1}{11}\begin{bmatrix} 4 & -3 \\ 1 & 2 \end{bmatrix}$.

Q39. For any square matrix A, show that AA' and $A'A$ are symmetric. Verify AA' for $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$. 3M

Ans. $(AA')' = (A')'A' = AA'$ and $(A'A)' = A'(A')' = A'A$, so both symmetric. Here $AA' = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 6 & 9 \end{bmatrix}$, which is symmetric.

Q40. If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, find $A^2 - 5A + 7I$. Hence find A^{-1} in terms of A and I. 3M Board fav

Ans. $A^2 = \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}$; $A^2 - 5A + 7I = \begin{bmatrix} 8-15+7 & 5-5 \\ -5+5 & 3-10+7 \end{bmatrix} = O$. Then $A - 5I + 7A^{-1} = O \Rightarrow A^{-1} = \frac{1}{7}(5I - A) = \frac{1}{7}\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$.

Q41. If $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$, find AB, BA and check whether $(AB)' = B'A'$. 3M

Ans. $AB = [-1+0+6] = [5]$ (1×1). $BA = \begin{bmatrix} -1 & -2 & -3 \\ 0 & 0 & 0 \\ 2 & 4 & 6 \end{bmatrix}$ (3×3). $(AB)' = [5]$ and $B'A' = \begin{bmatrix} -1 & 0 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = [5]$. Verified.

Q42. If $A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$, show that $A^2 - 2A + I = O$ and use it to find A^{-1} and A^3. 3M HOTS

Ans. $A^2 = \begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix}$, $A^2 - 2A + I = \begin{bmatrix} 3-4+1 & -2+2 \\ 2-2 & -1+1 \end{bmatrix} = O$. So $A^{-1} = 2I - A = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$. Also $A^2 = 2A - I \Rightarrow A^3 = 2A^2 - A = 2(2A - I) - A = 3A - 2I = \begin{bmatrix} 4 & -3 \\ 3 & -2 \end{bmatrix}$.

Long answer (5 marks)

Q43. Express $A = \begin{bmatrix} 2 & 4 & -6 \\ 7 & 3 & 5 \\ 1 & -2 & 4 \end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrix. 5M Board fav

(contd.) $A' = \begin{bmatrix} 2 & 7 & 1 \\ 4 & 3 & -2 \\ -6 & 5 & 4 \end{bmatrix}$.

$P = \frac{1}{2}(A + A') = \frac{1}{2}\begin{bmatrix} 4 & 11 & -5 \\ 11 & 6 & 3 \\ -5 & 3 & 8 \end{bmatrix} = \begin{bmatrix} 2 & \frac{11}{2} & \frac{-5}{2} \\ \frac{11}{2} & 3 & \frac{3}{2} \\ \frac{-5}{2} & \frac{3}{2} & 4 \end{bmatrix}$, $P' = P$.

$Q = \frac{1}{2}(A - A') = \frac{1}{2}\begin{bmatrix} 0 & -3 & -7 \\ 3 & 0 & 7 \\ 7 & -7 & 0 \end{bmatrix}$, $Q' = -Q$.

Adding, $P + Q = A$. Hence A = symmetric P + skew-symmetric Q.

Q44. If $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{bmatrix}$, show that $A^3 - 6A^2 + 10A - 4I = O$. Hence find A^{-1}. 5M HOTS

Ans. $A^2 = \begin{bmatrix} 2 & 0 & 4 \\ 0 & 4 & 0 \\ 4 & 0 & 10 \end{bmatrix}$, $A^3 = A^2A = \begin{bmatrix} 6 & 0 & 14 \\ 0 & 8 & 0 \\ 14 & 0 & 34 \end{bmatrix}$.

$A^3 - 6A^2 + 10A - 4I$: entry (1,1) $= 6 - 12 + 10 - 4 = 0$, (1,3) $= 14 - 24 + 10 - 0 = 0$, (2,2) $= 8 - 24 + 20 - 4 = 0$, (3,3) $= 34 - 60 + 30 - 4 = 0$, rest 0. So it is O.

Multiply by A^{-1}: $A^2 - 6A + 10I - 4A^{-1} = O \Rightarrow A^{-1} = \frac{1}{4}(A^2 - 6A + 10I) = \frac{1}{4}\begin{bmatrix} 6 & 0 & -2 \\ 0 & 2 & 0 \\ -2 & 0 & 2 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} & 0 & -\frac{1}{2} \\ 0 & \frac{1}{2} & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$.

Q45. If $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$, show that $F(x)F(y) = F(x+y)$. Hence show that $[F(x)]^{-1} = F(-x)$. 5M Board fav

Ans. Write $c_1 = \cos x, s_1 = \sin x, c_2 = \cos y, s_2 = \sin y$.

$$F(x)F(y) = \begin{bmatrix} c_1c_2 - s_1s_2 & -(s_1c_2 + c_1s_2) & 0 \\ s_1c_2 + c_1s_2 & c_1c_2 - s_1s_2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now $c_1c_2 - s_1s_2 = \cos(x+y)$ and $s_1c_2 + c_1s_2 = \sin(x+y)$, so $F(x)F(y) = F(x+y)$.

Put $y = -x$: $F(x)F(-x) = F(0) = I$; similarly $F(-x)F(x) = I$. So $[F(x)]^{-1} = F(-x)$.

Q46. A farmer sells wheat, rice and maize in three markets. Quantities sold (quintals) are: Market I: 10, 5, 8; Market II: 6, 12, 4; Market III: 8, 6, 10. Selling price per quintal is Rs 2500, Rs 3000, Rs 2000 and cost price is Rs 2000, Rs 2400, Rs 1500 respectively. Using matrices, find the total revenue and total profit of each market. 5M Must do

Ans. $Q = \begin{bmatrix} 10 & 5 & 8 \\ 6 & 12 & 4 \\ 8 & 6 & 10 \end{bmatrix}$, $S = \begin{bmatrix} 2500 \\ 3000 \\ 2000 \end{bmatrix}$,

(contd.) $S-C=\begin{bmatrix}500\\600\\500\end{bmatrix}$.

Revenue $QS=\begin{bmatrix}25000+15000+16000\\15000+36000+8000\\20000+18000+20000\end{bmatrix}=\begin{bmatrix}56000\\59000\\58000\end{bmatrix}$.

Profit $Q(S-C)=\begin{bmatrix}5000+3000+4000\\3000+7200+2000\\4000+3600+5000\end{bmatrix}=\begin{bmatrix}12000\\12200\\12600\end{bmatrix}$.

Revenue: Rs 56,000, Rs 59,000, Rs 58,000; Profit: Rs 12,000, Rs 12,200, Rs 12,600.

Q47. If $A=\begin{bmatrix}2&0&1\\1&1&0\\0&2&1\end{bmatrix}$, find $A^2-4A+3I$. Hence find the matrix X such that $A^2-4A+3I+X=O$. 5M Most asked

Ans. $A^2=\begin{bmatrix}4+0+0&0+0+2&2+0+1\\2+1+0&0+1+0&1+0+0\\0+2+0&0+2+2&0+0+1\end{bmatrix}=\begin{bmatrix}4&2&3\\3&1&1\\2&4&1\end{bmatrix}$.

$4A=\begin{bmatrix}8&0&4\\4&4&0\\0&8&4\end{bmatrix}$, $3I=\begin{bmatrix}3&0&0\\0&3&0\\0&0&3\end{bmatrix}$.

$A^2-4A+3I=\begin{bmatrix}-1&2&-1\\-1&0&1\\2&-4&0\end{bmatrix}$.

So $X=-(A^2-4A+3I)=\begin{bmatrix}1&-2&1\\1&0&-1\\-2&4&0\end{bmatrix}$.

Q48. Prove that every square matrix A can be expressed **uniquely** as the sum of a symmetric and a skew-symmetric matrix. 5M Board fav

Ans. Existence: $P=\frac{1}{2}(A+A')$ gives $P'=\frac{1}{2}(A'+A)=P$; $Q=\frac{1}{2}(A-A')$ gives $Q'=\frac{1}{2}(A'-A)=-Q$; and $P+Q=A$.

Uniqueness: let $A=R+S$ with $R'=R, S'=-S$. Then $A'=R'+S'=R-S$. Adding and subtracting: $R=\frac{1}{2}(A+A')=P$ and $S=\frac{1}{2}(A-A')=Q$. So the expression is unique.

Case based (4 marks)

Q49. Case: Rotating a game sprite. A game designer rotates points using $R(\theta)=\begin{bmatrix}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{bmatrix}$. (i) Write $R\left(\frac{\pi}{2}\right)$. (ii) Show that $R\left(\frac{\pi}{2}\right)$ is skew-symmetric. (iii) Show that $R(\theta)R(\theta)'=I$ and hence write $R(\theta)^{-1}$. 4M HOTS

(contd.) (i) $R(\frac{\pi}{2})=\begin{bmatrix}0&-1\\1&0\end{bmatrix}$. (ii) Its transpose is $\begin{bmatrix}0&1\\-1&0\end{bmatrix}=-R(\frac{\pi}{2})$, so skew-symmetric. (iii) With $c=\cos\theta, s=\sin\theta$: $RR'=\begin{bmatrix}c^2+s^2&cs-sc\\sc-cs&s^2+c^2\end{bmatrix}=I$. So $R(\theta)^{-1}=R(\theta)'$.

Q50. Case: Two branches of a stationery shop. In a day, Branch 1 sells 30 notebooks, 50 pens and 20 files; Branch 2 sells 25 notebooks, 40 pens and 30 files. Selling prices are Rs 40, Rs 10, Rs 25 and cost prices are Rs 30, Rs 6, Rs 18 respectively. Let $S=\begin{bmatrix}30&50&20\\25&40&30\end{bmatrix}$. (i) If P is the 3×1 selling-price matrix, what is the order of SP? (ii) Find the revenue of Branch 1. (iii) Find the profit of each branch using matrices. 4M Must do

Ans. (i) $(2\times3)(3\times1)$: order 2×1. (ii) $30(40)+50(10)+20(25)=1200+500+500=$ Rs 2200. (iii) Profit per item $P-C=\begin{bmatrix}10\\4\\7\end{bmatrix}$; $S(P-C)=\begin{bmatrix}300+200+140\\250+160+210\end{bmatrix}=\begin{bmatrix}640\\620\end{bmatrix}$: Branch 1 Rs 640, Branch 2 Rs 620.

Q51. Case: Marks of three friends. Aman, Riya and Kabir scored marks in Maths, Physics and Chemistry given by $M=\begin{bmatrix}80&70&90\\75&85&80\\90&60&70\end{bmatrix}$ (rows: Aman, Riya, Kabir; columns: M, P, C). The school calculates a weighted score using weights $W=\begin{bmatrix}0.5\\0.3\\0.2\end{bmatrix}$. (i) What is the order of MW? (ii) Find the weighted score of Riya. (iii) Who has the highest weighted score? Show working. 4M Board fav

Ans. (i) $(3\times3)(3\times1)$: 3×1. (ii) Riya: $75(0.5)+85(0.3)+80(0.2)=37.5+25.5+16=79$. (iii) Aman: $40+21+18=79$; Kabir: $45+18+14=77$. $MW=\begin{bmatrix}79\\79\\77\end{bmatrix}$: Aman and Riya tie at 79 (highest).

Apne notes / galtiyan yahan likho

Most asked: inverse by adjoint, solve system by $X = A^{-1}B$

DETERMINANTS

Chapter at a glance (Unit Algebra = 10 marks)

- → **Value** of a 2×2 and 3×3 determinant
- → **Area** of a triangle, collinearity, line through 2 points
- → **Minors, cofactors, adjoint, inverse**
- → **System of equations:** consistency and $X = A^{-1}B$

↳ Out of syllabus 2026-27: properties of determinants (as a topic) and Cramer's rule.

1. Determinant of a Square Matrix

Every square matrix A has a number attached to it called its **determinant**, written $|A|$ or $\det A$. Only square matrices have determinants.

Matrix A	Determinant $\lvert A\rvert$
an arrangement (no value)	a single number
any order $m \times n$	only square
written in [ ] or ( )	written in | |
kA: k multiplies every element	$\lvert kA\rvert = k^n \lvert A\rvert$

→ 1×1: $|[a]| = a$

→ 2×2: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$ (down-diagonal minus up-diagonal)

3×3 (expand along R_1): for $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$,

$$\Delta = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

+ − +	sign of position (i, j)
− + −	= (-1) to the power i+j
+ − +	corners and centre: +

Fig: sign chart for expanding a 3x3 determinant

You may expand along **any row or column**; the value is the same. Choose the row/column with the most zeros to save time.

Quick fact $|I| = 1$ and $|O| = 0$ for any order.

Yaad rakho Determinant of a diagonal matrix = product of its diagonal entries.

Yaad karo: Sign chart "Plus se shuru, chess board"

3×3 ke signs chessboard jaise: Row 1: $+ \ - \ +$, Row 2: $- \ + \ -$, Row 3: $+ \ - \ +$. Top-left hamesha $+$, phir har kadam pe sign badlo.

Solved Example 1

Evaluate (i) $\begin{vmatrix} 3 & -2 \\ 5 & 4 \end{vmatrix}$ (ii) $\begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix}$.

1. (i) $3(4) - (-2)(5) = 12 + 10$
2. (ii) $\cos^2\theta + \sin^2\theta$

✔ (i) 22 (ii) 1

Solved Example 2

Evaluate $\Delta = \begin{vmatrix} 2 & -1 & 3 \\ 1 & 0 & 4 \\ -2 & 5 & 1 \end{vmatrix}$ by expanding along R_2.

1. Signs of R_2: $- \ + \ -$. The middle element is 0, so that term vanishes
2. $\Delta = -1 \begin{vmatrix} -1 & 3 \\ 5 & 1 \end{vmatrix} + 0 - 4 \begin{vmatrix} 2 & -1 \\ -2 & 5 \end{vmatrix}$
3. $= -1(-1 - 15) - 4(10 - 2) = 16 - 32$

✔ $\Delta = -16$

Solved Example 3

Find x if $\begin{vmatrix} 2x & 5 \\ 8 & x \end{vmatrix} = \begin{vmatrix} 6 & 5 \\ 8 & 3 \end{vmatrix}$.

1. LHS $= 2x^2 - 40$, RHS $= 18 - 40 = -22$
2. $2x^2 - 40 = -22 \Rightarrow x^2 = 9$

✔ $x = \pm 3$

Solved Example 4

Evaluate $\begin{vmatrix} 1 & 2 & 4 \\ -1 & 3 & 0 \\ 4 & 1 & 0 \end{vmatrix}$ smartly.

1. C_3 has two zeros: expand along C_3 (signs $+, -, +$)
2. $= 4 \begin{vmatrix} -1 & 3 \\ 4 & 1 \end{vmatrix} - 0 + 0 = 4(-1 - 12)$

✔ $= -52$

Recall $|A^{-1}|\,|A| = 1$ for every invertible A.

Do you know? A skew-symmetric matrix of odd order always has determinant 0.

Solved Example 5

Find x if $\begin{vmatrix} 1 & 2 & x \\ 1 & 1 & 1 \\ 2 & 1 & -1 \end{vmatrix} = 0$.

1. Expand along R_1: $1(-1-1) - 2(-1-2) + x(1-2)$
2. $= -2 + 6 - x = 4 - x = 0$

✔ $x = 4$

Singular and non-singular

A square matrix A is **singular** if $|A| = 0$ and **non-singular** if $|A| \neq 0$. Only a non-singular matrix has an inverse.

Formula Box 1: Results on determinant values

- → $|A'| = |A|$
- → $|kA| = k^n|A|$ for A of order n (NOT $k|A|$)
- → $|AB| = |A||B|$
- → $|A^{-1}| = \dfrac{1}{|A|}$
- → $|\text{adj } A| = |A|^{n-1}$
- → $|A^m| = |A|^m$

Solved Example 6

A is a square matrix of order 3 with $|A| = 4$. Find (i) $|3A|$ (ii) $|\text{adj } A|$ (iii) $|A^{-1}|$ (iv) $|AA'|$.

1. (i) $|3A| = 3^3|A| = 27 \times 4$
2. (ii) $|\text{adj } A| = |A|^{3-1} = 4^2$
3. (iii) $|A^{-1}| = \frac{1}{4}$
4. (iv) $|AA'| = |A||A'| = 4 \times 4$

✔ (i) 108 (ii) 16 (iii) $\frac{1}{4}$ (iv) 16



Meme samjho

$|2A|$ for a 3×3 matrix is $8|A|$, not $2|A|$? ACHHAA!!

Asli baat: k multiplies every one of the n rows, so $|kA| = k^n|A|$. For order 3, $|2A| = 2^3|A| = 8|A|$.

Check trick After A^{-1}, multiply row 1 of A with column 1 of A^{-1}: it must give 1.

Board tip Always write "$|A| \neq 0$, so A^{-1} exists" before finding the inverse.

Yaad rakho $\text{adj}(\text{adj } A) = |A|^{n-2}A$ for $|A| \neq 0$.

Solved Example 7

Verify $|AB| = |A||B|$ for $A = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$.

1. $|A| = 8 - 3 = 5$, $|B| = 3 + 2 = 5$
2. $AB = \begin{bmatrix} 4 & 1 \\ 11 & 9 \end{bmatrix}$, $|AB| = 36 - 11 = 25$

✔ $|AB| = 25 = 5 \times 5 = |A||B|$

2. Area of a Triangle

Area of the triangle with vertices (x_1, y_1), (x_2, y_2), (x_3, y_3):

$$\Delta = \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|$$

- → Area is always positive: take the **absolute value**.
- → If area is given, use **both signs**: $\det = \pm 2\Delta$ (two answers usually).
- → Three points are collinear $\iff$ the determinant $= 0$.
- → Line through (x_1, y_1), (x_2, y_2): $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$.

Solved Example 8

Find the area of the triangle with vertices $A(2, 1)$, $B(7, 1)$ and $C(5, 5)$.

1. $\Delta = \frac{1}{2}\left|\begin{vmatrix} 2 & 1 & 1 \\ 7 & 1 & 1 \\ 5 & 5 & 1 \end{vmatrix}\right|$
2. $= \frac{1}{2}|2(1-5) - 1(7-5) + 1(35-5)| = \frac{1}{2}|-8 - 2 + 30|$
3. Check: base $AB = 5$, height $= 4$, area $= \frac{1}{2}(5)(4) = 10$ ✓

✔ Area = 10 sq units

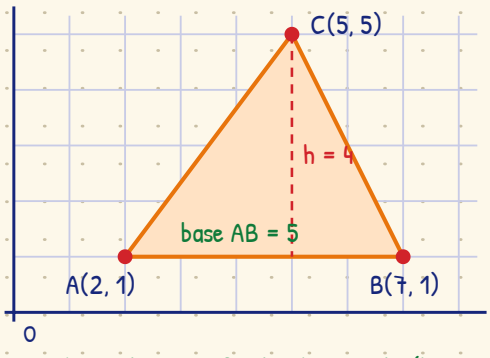


Fig: triangle ABC of Solved Example (base 5, height 4)

Recall $\text{adj } I = I$ and $\text{adj } O = O$.

Solved Example 9

Find the area of the triangle with vertices $(1,4)$, $(5,3)$, $(2,1)$.

1. $\begin{vmatrix}1&4&1\\5&3&1\\2&1&1\end{vmatrix} = 1(3-1)-4(5-2)+1(5-6) = 2-12-1 = -11$
2. Area is never negative: take the absolute value

✔ Area $= \frac{1}{2}|-11| = \frac{11}{2}$ sq units

Solved Example 10

Find k if the area of the triangle with vertices $(k,0)$, $(6,0)$, $(0,3)$ is 9 sq units.

1. $\frac{1}{2}\left|\begin{vmatrix}k&0&1\\6&0&1\\0&3&1\end{vmatrix}\right| = 9 \Rightarrow |k(0-3)-0+1(18-0)| = 18$
2. $|18-3k| = 18 \Rightarrow 18-3k = 18$ or $18-3k = -18$

✔ $k = 0$ or $k = 12$

Solved Example 11

Find k if the area of the triangle with vertices $(1,-2)$, $(4,3)$, $(k,3)$ is 10 sq units.

1. $\frac{1}{2}|1(3-3)+2(4-k)+1(12-3k)| = 10$
2. $|8-2k+12-3k| = |20-5k| = 20$
3. $20-5k = 20$ or $20-5k = -20$

✔ $k = 0$ or $k = 8$

Solved Example 12

Show that the points $(1,-1)$, $(2,1)$ and $(4,5)$ are collinear.

1. $\begin{vmatrix}1&-1&1\\2&1&1\\4&5&1\end{vmatrix} = 1(1-5)+1(2-4)+1(10-4)$
2. $= -4-2+6 = 0$

✔ Determinant is 0, so the points are collinear



Shivam Bhaiya says

"Area wale sawal mein k ki **do values** aati hain ($\pm$ wala case). Ek value likh ke chhod di toh 1 mark gaya. Aur collinear points ka matlab area zero, bas!"

Board tip Show the cofactor list clearly: 1 mark is often for cofactors alone.

Solved Example 13

Using determinants, find the equation of the line joining $(2,3)$ and $(4,7)$.

1. Let (x,y) be any point on it: $\begin{vmatrix}x&y&1\\2&3&1\\4&7&1\end{vmatrix} = 0$
2. $x(3-7)-y(2-4)+1(14-12) = 0 \Rightarrow -4x+2y+2 = 0$

✔ $y = 2x-1$

3. Minors and Cofactors

Definitions

→ **Minor** M_{ij} of a_{ij}: the determinant left after deleting row i and column j.

→ **Cofactor** $A_{ij} = (-1)^{i+j}M_{ij}$ (minor with the sign from the chart).

→ $|A|$ = sum of (elements of any row or column) $\times$ (their own cofactors): $|A| = a_{11}A_{11}+a_{12}A_{12}+a_{13}A_{13}$.

→ Elements of one row $\times$ cofactors of a **different** row, added, give **0**.

Solved Example 14

For $A = \begin{bmatrix}1&3&-2\\4&-1&0\\2&5&3\end{bmatrix}$, find the cofactors of the elements of R_1 and hence $|A|$. Also verify $a_{11}A_{21}+a_{12}A_{22}+a_{13}A_{23} = 0$.

1. $A_{11} = +\begin{vmatrix}-1&0\\5&3\end{vmatrix} = -3$, $A_{12} = -\begin{vmatrix}4&0\\2&3\end{vmatrix} = -12$, $A_{13} = +\begin{vmatrix}4&-1\\2&5\end{vmatrix} = 22$
2. $|A| = 1(-3)+3(-12)+(-2)(22) = -3-36-44 = -83$
3. $A_{21} = -\begin{vmatrix}3&-2\\5&3\end{vmatrix} = -19$, $A_{22} = +\begin{vmatrix}1&-2\\2&3\end{vmatrix} = 7$, $A_{23} = -\begin{vmatrix}1&3\\2&5\end{vmatrix} = 1$
4. $1(-19)+3(7)+(-2)(1) = -19+21-2 = 0$ ✓

✔ $|A| = -83$, and the mixed sum is 0

Check trick $(\text{adj}\ A)A$ must also equal $|A|I$; use it to catch sign slips.

Do you know? Collinear points give area 0, so they cannot form a triangle.

Solved Example 15

Write the minors and cofactors of the elements of $\begin{vmatrix} 5 & -2 \\ 3 & 7 \end{vmatrix}$ and verify $\Delta = a_{11}A_{11} + a_{12}A_{12}$.

1. $M_{11} = 7, M_{12} = 3, M_{21} = -2, M_{22} = 5$
2. $A_{11} = 7, A_{12} = -3, A_{21} = 2, A_{22} = 5$
3. $a_{11}A_{11} + a_{12}A_{12} = 5(7) + (-2)(-3) = 41$, and $\Delta = 35 + 6 = 41$

✔ Verified: $\Delta = 41$

Solved Example 16

Find the cofactors of the elements of C_2 of $\Delta = \begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$ and evaluate Δ along C_2.

1. $A_{12} = -\begin{vmatrix} 6 & 4 \\ 1 & -7 \end{vmatrix} = -(-42 - 4) = 46$
2. $A_{22} = +\begin{vmatrix} 2 & 5 \\ 1 & -7 \end{vmatrix} = -14 - 5 = -19$
3. $A_{32} = -\begin{vmatrix} 2 & 5 \\ 6 & 4 \end{vmatrix} = -(8 - 30) = 22$
4. $\Delta = (-3)(46) + 0(-19) + 5(22) = -138 + 110$

✔ $\Delta = -28$

Yaad karo: "Minor = Mitao, Cofactor = Chart ka sign"

Minor: row aur column **mitao** (delete), bacha hua determinant. **Cofactor**: minor ke saath **chart ka sign** lagao: $(-1)^{i+j}$. $i + j$ even toh same, odd toh ulta.

4. Adjoint of a Matrix

Definition

adj A = transpose of the cofactor matrix: if $C = [A_{ij}]$ then adj $A = C'$.

2×2 **shortcut**: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ (swap the diagonal, change sign of the other two).

Key Theorem

$$A(\text{adj } A) = (\text{adj } A)A = |A|\, I$$

Also $|\text{adj } A| = |A|^{n-1}$, $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$, $\text{adj}(A') = (\text{adj } A)'$.

Quick fact If two rows of A are equal, $|A| = 0$.

Yaad rakho Area answer: always write "sq units".

Solved Example 17

Find adj A for $A = \begin{bmatrix} 3 & -4 \\ 2 & 5 \end{bmatrix}$ and verify $A(\text{adj } A) = |A|I$.

1. adj $A = \begin{bmatrix} 5 & 4 \\ -2 & 3 \end{bmatrix}$, $|A| = 15 + 8 = 23$
2. $A(\text{adj } A) = \begin{bmatrix} 15+8 & 12-12 \\ 10-10 & 8+15 \end{bmatrix} = \begin{bmatrix} 23 & 0 \\ 0 & 23 \end{bmatrix}$

✔ $= 23I = |A|I$, verified

Solved Example 18

For $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$, find adj A and verify $A(\text{adj } A) = |A|I$.

1. $|A| = 2(6-1) - 1(2-0) + 0 = 8$
2. Cofactors: $A_{11} = 5, A_{12} = -2, A_{13} = 1, A_{21} = -2, A_{22} = 4, A_{23} = -2, A_{31} = 1, A_{32} = -2, A_{33} = 5$
3. adj $A = \begin{bmatrix} 5 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 5 \end{bmatrix}$ (here C is symmetric, so $C' = C$)
4. $A(\text{adj } A) = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix}$

✔ $= 8I = |A|I$, verified

↳ Notice: when A is symmetric, its cofactor matrix is symmetric too, so adj A is also symmetric. Good self-check in exams!

5. Inverse of a Matrix

Theorem

A square matrix A is invertible if and only if A is non-singular ($|A| \neq 0$), and then

$$A^{-1} = \frac{1}{|A|}\,\text{adj } A$$

↳ Steps: (1) find $|A|$, stop if 0; (2) all 9 cofactors; (3) transpose to get adj A; (4) divide by $|A|$; (5) check one entry of $AA^{-1} = I$.

Recall $X = A^{-1}B$: A^{-1} on the LEFT of B.

Board tip For a 2x2 system you may use the quick inverse; still write $|A| \neq 0$.

2x2 inverse in 5 seconds

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$: swap a and d, change signs of b and c, divide by $ad - bc$. E.g. $\begin{bmatrix} 4 & 7 \\ 1 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} 2 & -7 \\ -1 & 4 \end{bmatrix}$ since $ad - bc = 1$.

Solved Example 19

Find A^{-1} for $A = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 1 & -2 \\ 3 & 0 & 1 \end{bmatrix}$.

1. $|A| = 2(1-0) + 1(1+6) + 1(0-3) = 2 + 7 - 3 = 6 \neq 0$, so A^{-1} exists
2. $A_{11} = 1, A_{12} = -7, A_{13} = -3$; $A_{21} = 1, A_{22} = -1, A_{23} = -3$; $A_{31} = 1, A_{32} = 5, A_{33} = 3$
3. $\text{adj } A = \begin{bmatrix} 1 & 1 & 1 \\ -7 & -1 & 5 \\ -3 & -3 & 3 \end{bmatrix}$

✔ $A^{-1} = \frac{1}{6}\begin{bmatrix} 1 & 1 & 1 \\ -7 & -1 & 5 \\ -3 & -3 & 3 \end{bmatrix}$

Solved Example 20

For $A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$, show $A^2 - 4A + I = O$ and find A^{-1}. Check with the adjoint method.

1. $A^2 = \begin{bmatrix} 11 & 8 \\ 4 & 3 \end{bmatrix}$, $A^2 - 4A + I = \begin{bmatrix} 11 - 12 + 1 & 8 - 8 \\ 4 - 4 & 3 - 4 + 1 \end{bmatrix} = O$
2. So $A^{-1} = 4I - A = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$
3. Adjoint: $|A| = 1$, $\text{adj } A = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$ ✓

✔ $A^{-1} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$

Yaad karo: "Inverse = Adj upon Det" (A-u-D)

$A^{-1} = \dfrac{\textbf{Adj}}{\textbf{Det}}$. Pehle **Det** nikalo: zero aaya toh inverse hai hi nahi, aage time waste mat karo. Cofactors likhte waqt **transpose** bhoolna sabse common galti hai.

6. Solving a System of Linear Equations

The system $a_1x + b_1y + c_1z = d_1$, $a_2x + b_2y + c_2z = d_2$, $a_3x + b_3y + c_3z = d_3$ is written as $AX = B$:

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

Case	Result
$\lvert A\rvert \neq 0$	**consistent**, unique solution $X = A^{-1}B$
$\lvert A\rvert = 0$, $(\text{adj } A)B \neq O$	**inconsistent** (no solution)
$\lvert A\rvert = 0$, $(\text{adj } A)B = O$	may be consistent (infinitely many) or inconsistent: check the equations

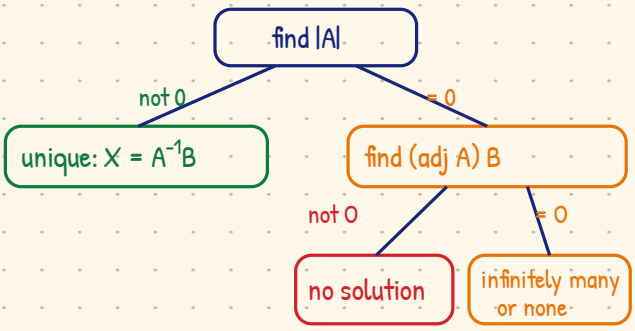


Fig: decision chart for the system AX = B

Consistent vs inconsistent

A system is **consistent** if it has at least one solution, and **inconsistent** if it has none. Order matters: $X = A^{-1}B$, never BA^{-1} (B is a column).

Solved Example 21

Solve by the matrix method: $2x + 3y = 8$, $x - 2y = -3$.

1. $A = \begin{bmatrix} 2 & 3 \\ 1 & -2 \end{bmatrix}$, $B = \begin{bmatrix} 8 \\ -3 \end{bmatrix}$, $|A| = -4 - 3 = -7 \neq 0$
2. $A^{-1} = \frac{1}{-7}\begin{bmatrix} -2 & -3 \\ -1 & 2 \end{bmatrix}$
3. $X = A^{-1}B = \frac{1}{-7}\begin{bmatrix} -16 + 9 \\ -8 - 6 \end{bmatrix} = \frac{1}{-7}\begin{bmatrix} -7 \\ -14 \end{bmatrix}$

✔ $x = 1, y = 2$

Solved Example 22

The cost of 4 kg onions and 3 kg wheat is Rs 220; the cost of 2 kg onions and 5 kg wheat is Rs 250. Find the cost per kg of each using matrices.

1. $4x + 3y = 220$, $2x + 5y = 250$; $|A| = 20 - 6 = 14$
2. $A^{-1} = \frac{1}{14}\begin{bmatrix} 5 & -3 \\ -2 & 4 \end{bmatrix}$, $X = \frac{1}{14}\begin{bmatrix} 1100 - 750 \\ -440 + 1000 \end{bmatrix} = \frac{1}{14}\begin{bmatrix} 350 \\ 560 \end{bmatrix}$

✔ Onions Rs 25/kg, wheat Rs 40/kg

Do you know? $|kI_n| = k^n$.

Solved Example 23

Solve: $x+y+z=6, 2x-y+z=3, x+2y-z=2$.

1. $A=\begin{bmatrix}1&1&1\\2&-1&1\\1&2&-1\end{bmatrix}$, $|A|=1(1-2)-1(-2-1)+1(4+1)=-1+3+5=7$
2. $\text{adj } A=\begin{bmatrix}-1&3&2\\3&-2&1\\5&-1&-3\end{bmatrix}$
3. $X=\frac{1}{7}\begin{bmatrix}-1&3&2\\3&-2&1\\5&-1&-3\end{bmatrix}\begin{bmatrix}6\\3\\2\end{bmatrix}=\frac{1}{7}\begin{bmatrix}-6+9+4\\18-6+2\\30-3-6\end{bmatrix}=\frac{1}{7}\begin{bmatrix}7\\14\\21\end{bmatrix}$

✔ $x=1, y=2, z=3$

Solved Example 24

Examine the consistency of (i) $x+2y=3, 2x+4y=7$ (ii) $x+2y=3, 2x+4y=6$.

1. Both: $A=\begin{bmatrix}1&2\\2&4\end{bmatrix}$, $|A|=4-4=0$, $\text{adj } A=\begin{bmatrix}4&-2\\-2&1\end{bmatrix}$
2. (i) $(\text{adj } A)B=\begin{bmatrix}12-14\\-6+7\end{bmatrix}=\begin{bmatrix}-2\\1\end{bmatrix}\neq O$
3. (ii) $(\text{adj } A)B=\begin{bmatrix}12-12\\-6+6\end{bmatrix}=O$; second equation is just 2 times the first

✔ (i) inconsistent (ii) consistent, infinitely many solutions

Solved Example 25

Show that the system $x+y+z=3, 2x+2y+2z=5, x-y+z=1$ is inconsistent.

1. $A=\begin{bmatrix}1&1&1\\2&2&2\\1&-1&1\end{bmatrix}$: $R_2=2R_1$, so $|A|=0$
2. $\text{adj } A=\begin{bmatrix}4&-2&0\\0&0&0\\-4&2&0\end{bmatrix}$, $(\text{adj } A)B=\begin{bmatrix}4&-2&0\\0&0&0\\-4&2&0\end{bmatrix}\begin{bmatrix}3\\5\\1\end{bmatrix}=\begin{bmatrix}2\\0\\-2\end{bmatrix}\neq O$

✔ Inconsistent: no solution (also clear since $2(x+y+z)$ cannot be both 6 and 5)

Solved Example 26

If $A=\begin{bmatrix}1&1&1\\1&-1&1\\2&1&-1\end{bmatrix}$ and $B=\begin{bmatrix}0&2&2\\3&-3&0\\3&1&-2\end{bmatrix}$, find AB. Hence solve $x+y+z=6, x-y+z=2, 2x+y-z=1$.

1. $AB=\begin{bmatrix}6&0&0\\0&6&0\\0&0&6\end{bmatrix}=6I\Rightarrow A\left(\frac{1}{6}B\right)=I\Rightarrow A^{-1}=\frac{1}{6}B$
2. The system is $AX=C$ with $C=\begin{bmatrix}6\\2\\1\end{bmatrix}$, so $X=\frac{1}{6}BC$
3. $X=\frac{1}{6}\begin{bmatrix}0+4+2\\18-6+0\\18+2-2\end{bmatrix}=\frac{1}{6}\begin{bmatrix}6\\12\\18\end{bmatrix}$

✔ $x=1, y=2, z=3$

Yaad karo: "Product = kI wala sawal"

Agar $AB=kI$ diya ho, toh $A^{-1}=\frac{1}{k}B$. Equations ka coefficient matrix dekho: agar wo A hai toh $X=\frac{1}{k}BC$; agar wo B hai (ya A') toh dhyaan se sahi matrix choose karo.

Solved Example 27

The sum of three numbers is 10. The first number plus twice the third is 12, and the sum of the second and third is 8. Find the numbers using matrices.

1. $x+y+z=10, x+2z=12, y+z=8$; $A=\begin{bmatrix}1&1&1\\1&0&2\\0&1&1\end{bmatrix}$, $|A|=1(0-2)-1(1-0)+1(1-0)=-2$
2. $\text{adj } A=\begin{bmatrix}-2&0&2\\-1&1&-1\\1&-1&-1\end{bmatrix}$
3. $X=\frac{1}{-2}\begin{bmatrix}-20+0+16\\-10+12-8\\10-12-8\end{bmatrix}=\frac{1}{-2}\begin{bmatrix}-4\\-6\\-10\end{bmatrix}$

✔ The numbers are 2, 3, 5



Shivam Bhaiya says

"$X=A^{-1}B$ wala 5 marks ka sawal har saal aata hai. Steps likho: $|A|$, cofactors, adj, A^{-1}, phir X. Answer mil jaye toh ek equation mein daal ke check karo, 10 second lagta hai."

Solved Example 28

A school gives cash prizes for Discipline (Rs x), Sports (Rs y) and Academics (Rs z) per student. Batch I: one of each prize, total Rs 1200. Batch II: 2 Discipline, 1 Sports, 3 Academics, total Rs 2500. Batch III: 3 Discipline, 2 Sports, 1 Academics, total Rs 2200. Find each prize using matrices.

1. $x+y+z=1200$, $2x+y+3z=2500$, $3x+2y+z=2200$; $A=\begin{bmatrix}1&1&1\\2&1&3\\3&2&1\end{bmatrix}$
2. $|A|=1(1-6)-1(2-9)+1(4-3)=-5+7+1=3$, $\text{adj } A=\begin{bmatrix}-5&1&2\\7&-2&-1\\1&1&-1\end{bmatrix}$
3. $X=\frac{1}{3}\begin{bmatrix}-5&1&2\\7&-2&-1\\1&1&-1\end{bmatrix}\begin{bmatrix}1200\\2500\\2200\end{bmatrix}=\frac{1}{3}\begin{bmatrix}900\\1200\\1500\end{bmatrix}$

✔ Discipline Rs 300, Sports Rs 400, Academics Rs 500

Formula Box 2: Whole chapter

- $\begin{vmatrix}a&b\\c&d\end{vmatrix}=ad-bc$
- M_{ij} = minor, $A_{ij}=(-1)^{i+j}M_{ij}$
- $|A|=\sum$ (element $\times$ own cofactor) along any row/column
- Area $=\frac{1}{2}\,|\det[x_i\ \ y_i\ \ 1]|$; collinear $\iff \det=0$
- $\text{adj } A=[A_{ij}]'$; $A(\text{adj } A)=|A|I$
- $A^{-1}=\frac{1}{|A|}\text{adj } A$ $(|A|\neq 0)$
- $AX=B$, $|A|\neq 0\Rightarrow X=A^{-1}B$

Board answer-writing: $X=A^{-1}B$ (full marks format)

1. Write $AX=B$ with A, X, B shown.
2. $|A|$ = value $\neq 0$, so A^{-1} exists (unique solution).
3. List all 9 cofactors.
4. adj A (transpose!), then $A^{-1}=\frac{1}{|A|}\text{adj } A$.
5. $X=A^{-1}B$: show the multiplication.
6. Write x, y, z clearly in a box.

Quick fact $|A^2|=|A|^2$, so $|A^2|$ is never negative for real A.

Common mistakes

- $|kA|=k|A|$ (wrong): correct is $k^n|A|$.
- Forgetting the sign $(-1)^{i+j}$ while writing cofactors.
- Writing the cofactor matrix as adj A without **transposing**.
- Writing $X=BA^{-1}$ instead of $X=A^{-1}B$.
- In area questions, taking only one value of k (forgetting $\pm$).
- Finding A^{-1} without first checking $|A|\neq 0$.

Ek minute test (answers at the end)

1. $|A|=3$, order 2. $|2A|=?$
2. Can a 2×3 matrix have a determinant?
3. Cofactor sign at position (2, 3)?
4. $|A|=0$: does A^{-1} exist?
5. Area of triangle with collinear vertices?
6. A order 3, $|A|=2$: $|\text{adj } A|=?$

Answers: 1) 12 2) No, only square 3) minus ($2+3$ odd) 4) No, singular 5) 0 6) $2^2=4$

Most Asked in Boards

- Solve a system of 3 equations by matrix method $X=A^{-1}B$ (5M)
- Given $AB=kI$, find A^{-1} and solve a system (5M)
- Find A^{-1} by adjoint for a 3×3 matrix (3-5M)
- $|kA|$, $|\text{adj } A|$, $|A^{-1}|$ value based MCQs (1M)
- Area of triangle / find k / collinearity (2-3M)
- Equation of a line using determinants (2M)
- Consistency of a system (2-3M, AR)
- Word problem (prizes, prices) converted to $AX=B$ (5M or case based)

Yaad karo: Consistency ka "Zero-Zero" rule

$|A|\neq 0$: **pakka ek** solution. $|A|=0$ aur $(\text{adj } A)B\neq O$: **koi solution nahi**. $|A|=0$ aur $(\text{adj } A)B=O$: "zero-zero" case, equations dekh ke decide karo (usually infinitely many).

Meme samjho



Examiner: "Inverse kaise nikala?"
Student: "Pehle $|A|$ check kiya, zero nahi tha, phir adj upon det."
Asli baat: A^{-1} exists only when $|A|\neq 0$, and then $A^{-1}=\frac{1}{|A|}\text{adj } A$.

Most Important Questions

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. If $\begin{vmatrix} x & 2 \\ 3 & x \end{vmatrix} = \begin{vmatrix} 4 & 2 \\ 3 & 4 \end{vmatrix}$, then x is 1M Board fav

(a) 4 (b) -4 (c) ± 4 (d) ± 2

Ans. (c) $x^2 - 6 = 16 - 6 = 10 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4$.

Q2. If A is a square matrix of order 3 with $|A| = 5$, then $|\text{adj } A|$ is 1M Most asked

(a) 5 (b) 15 (c) 25 (d) 125

Ans. (c) $|A|^{n-1} = 5^2 = 25$.

Q3. If A is a 2×2 matrix with $|A| = -3$, then $|2A|$ is 1M Must do

(a) -6 (b) 6 (c) 12 (d) -12

Ans. (d) $2^2(-3) = -12$.

Q4. If A is invertible of order 3 and $|A| = 8$, then $|A^{-1}|$ is 1M

(a) 8 (b) $\frac{1}{8}$ (c) $\frac{1}{64}$ (d) 64

Ans. (b) $\frac{1}{|A|} = \frac{1}{8}$.

Q5. The value of $\begin{vmatrix} \cos 15^\circ & \sin 15^\circ \\ \sin 75^\circ & \cos 75^\circ \end{vmatrix}$ is 1M

(a) 0 (b) 1 (c) $\frac{1}{2}$ (d) $\frac{\sqrt{3}}{2}$

Ans. (a) $\cos 15^\circ \cos 75^\circ - \sin 15^\circ \sin 75^\circ = \cos 90^\circ = 0$.

Q6. The area of the triangle with vertices $(0,0)$, $(4,0)$, $(0,6)$ is 1M

(a) 24 (b) 12 (c) 10 (d) 6

Ans. (b) $\frac{1}{2}|0 + 0 + 1(24 - 0)| = 12$.

Q7. If the points $(2,-3)$, $(k,-1)$, $(0,4)$ are collinear, then k is 1M

(a) $\frac{10}{7}$ (b) $\frac{7}{10}$ (c) $-\frac{10}{7}$ (d) 0

Ans. (a) $2(-1-4) + 3(k-0) + 1(4k-0) = 7k - 10 = 0 \Rightarrow k = \frac{10}{7}$.

Q8. For $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, the cofactor A_{21} is 1M

(a) 2 (b) -2 (c) 3 (d) -3

Ans. (b) $A_{21} = (-1)^{2+1} M_{21} = -(2) = -2$.

Q9. For a system $AX = B$, if $|A| = 0$ and $(\text{adj } A)B \neq O$, the system is 1M Board fav

(a) consistent, unique (b) consistent, infinitely many (c) inconsistent (d) cannot say

Ans. (c) inconsistent: no solution.

Q10. If $A(\text{adj } A) = 10I$ for a 3×3 matrix A, then $|\text{adj } A|$ is 1M HOTS

(a) 10 (b) 30 (c) 1000 (d) 100

Ans. (d) $|A| = 10$, so $|\text{adj } A| = 10^2 = 100$.

Q11. The matrix $\begin{bmatrix} k & 4 \\ 1 & k \end{bmatrix}$ is singular when k is 1M

(a) 4 (b) 2 only (c) ± 2 (d) 0

Ans. (c) $k^2 - 4 = 0 \Rightarrow k = \pm 2$.

Q12. If A is a skew-symmetric matrix of order 3, then $|A|$ is 1M HOTS

(a) 0 (b) 1 (c) -1 (d) $|A|^2$

Ans. (a) $|A| = |A'| = |-A| = (-1)^3|A| = -|A| \Rightarrow |A| = 0$.

Q13. **A:** If $|A| = 0$, then A^{-1} does not exist. **R:** $A^{-1} = \frac{1}{|A|}\text{adj } A$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) Both true, and R explains A (division by 0 is not possible).

Q14. **A:** The area of the triangle with vertices $(1,1)$, $(2,2)$, $(3,3)$ is 0. **R:** The determinant formed by three collinear points is 0. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) The points lie on $y = x$, so they are collinear and the determinant is 0. R explains A.

Q15. **A:** For a 3×3 matrix A, $|3A| = 3|A|$. **R:** For a square matrix of order n, $|kA| = k^n|A|$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (d) A is false ($|3A| = 27|A|$), R is true.

Q16. The roots of $\begin{vmatrix} x & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{vmatrix} = 0$ are 1M HOTS

(a) $1, 2$ (b) $1, -2$ (c) $-1, 2$ (d) $0, 1$

Ans. (b) Expanding: $x^3 - 3x + 2 = (x-1)^2(x+2) = 0$, so $x = 1, 1, -2$.

Q17. If $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$, then $|A|$ is 1M

(a) 9 (b) 0 (c) 24 (d) 12

Ans. (c) Diagonal matrix: product of diagonal entries $= 2 \times 3 \times 4 = 24$.

Q18. If $|A| = 2$ and $|B| = -3$ for 3×3 matrices, then $|2AB|$ is 1M Most asked

(a) -48 (b) -12 (c) 48 (d) -6

Ans. (a) $2^3|A||B| = 8(2)(-3) = -48$.

Q19. If A is non-singular of order 3, then $|A^{-1}\text{ adj } A|$ equals 1M HOTS

(a) 1 (b) $|A|^2$ (c) $\frac{1}{|A|}$ (d) $|A|$

Ans. (d) $|A^{-1}||\text{adj } A| = \frac{1}{|A|}|A|^2 = |A|$.

Yaad rakho Cofactor sign at (i,j): plus when $i + j$ is even.

Q20. If $A=\begin{bmatrix}1&2\\4&2\end{bmatrix}$, then $|2A|$ is 1M

(a) -12 (b) 24 (c) -24 (d) -6

Ans. (c) $|A|=2-8=-6$, $|2A|=2^2(-6)=-24$.

Q21. If $\Delta=\begin{vmatrix}5&3&8\\2&0&1\\1&2&3\end{vmatrix}$, the cofactor of a_{23} is 1M

(a) 7 (b) -7 (c) 13 (d) -13

Ans. (b) $A_{23}=-\begin{vmatrix}5&3\\1&2\end{vmatrix}=-(10-3)=-7$.

Q22. The equation of the line through $(0,0)$ and $(3,5)$ using determinants is 1M

(a) $5x-3y=0$ (b) $3x-5y=0$ (c) $5x+3y=0$ (d) $x=y$

Ans. (a) $\begin{vmatrix}x&y&1\\0&0&1\\3&5&1\end{vmatrix}=x(0-5)-y(0-3)+0=0\Rightarrow 5x=3y$.

Q23. If $\begin{vmatrix}1&2&3\\2&x&3\\3&4&6\end{vmatrix}=0$, then x is 1M

(a) 1 (b) 3 (c) 2 (d) 4

Ans. (c) $1(6x-12)-2(12-9)+3(8-3x)=6x-12-6+24-9x=6-3x=0\Rightarrow x=2$.

Very short (2 marks)

Q24. Find x if $\begin{vmatrix}3&x\\x&1\end{vmatrix}=\begin{vmatrix}3&2\\4&1\end{vmatrix}$. 2M

Ans. $3-x^2=3-8=-5\Rightarrow x^2=8\Rightarrow x=\pm 2\sqrt{2}$.

Q25. Find the area of the triangle with vertices $(-2,-3)$, $(3,2)$ and $(-1,-8)$. 2M Most asked

Ans. $\frac{1}{2}\,|-2(2+8)+3(3+1)+1(-24+2)|=\frac{1}{2}|-20+12-22|=\frac{1}{2}(30)=15$ sq units.

Q26. Using determinants, find the equation of the line joining $(1,2)$ and $(3,6)$. 2M

Ans. $\begin{vmatrix}x&y&1\\1&2&1\\3&6&1\end{vmatrix}=0\Rightarrow x(2-6)-y(1-3)+(6-6)=0\Rightarrow -4x+2y=0$, i.e. $y=2x$.

Q27. If A is a 3×3 matrix with $|A|=-2$, find $|\text{adj}(\text{adj}\,A)|$. 2M HOTS

Ans. $|\text{adj}\,A|=|A|^2=4$. Then $|\text{adj}(\text{adj}\,A)|=|\text{adj}\,A|^2=16$.

Q28. Find k if the area of the triangle with vertices $(1,k)$, $(-2,1)$, $(0,3)$ is 6 sq units. 2M

Ans. $\frac{1}{2}|1(1-3)-k(-2-0)+1(-6-0)|=6\Rightarrow |2k-8|=12\Rightarrow k=10$ or $k=-2$.

Q29. Show that $A=\begin{bmatrix}2&3\\4&6\end{bmatrix}$ is singular and that the system $2x+3y=5$, $4x+6y=7$ is inconsistent. 2M

(contd.) $|A|=12-12=0$: singular. $(\text{adj}\,A)B=\begin{bmatrix}6&-3\\-4&2\end{bmatrix}\begin{bmatrix}5\\7\end{bmatrix}=\begin{bmatrix}9\\-6\end{bmatrix}\neq O$: inconsistent.

Q30. Write the adjoint of $A=\begin{bmatrix}2&-4\\3&5\end{bmatrix}$ and find A^{-1}. 2M

Ans. $\text{adj}\,A=\begin{bmatrix}5&4\\-3&2\end{bmatrix}$, $|A|=10+12=22$, $A^{-1}=\frac{1}{22}\begin{bmatrix}5&4\\-3&2\end{bmatrix}$.

Q31. Find the minor and cofactor of each element of $\begin{bmatrix}2&-3\\4&5\end{bmatrix}$. 2M

Ans. $M_{11}=5, M_{12}=4, M_{21}=-3, M_{22}=2$.
Cofactors: $A_{11}=5, A_{12}=-4, A_{21}=3, A_{22}=2$.

Q32. If $\begin{vmatrix}2x&4\\x&3\end{vmatrix}=10$, find x. 2M

Ans. $6x-4x=2x=10\Rightarrow x=5$.

Q33. If A is a square matrix of order 3 with $|A|=3$, find $|A\,\text{adj}\,A|$. 2M Board fav

Ans. $A\,\text{adj}\,A=|A|I=3I$, so $|3I|=3^3=27$.

Q34. Show that $\begin{vmatrix}a&b\\c&d\end{vmatrix}$ changes sign when the two rows are interchanged (check with $a=1,b=2,c=3,d=4$). 2M

Ans. $ad-bc$ becomes $cb-da=-(ad-bc)$. Check: $\begin{vmatrix}1&2\\3&4\end{vmatrix}=-2$, $\begin{vmatrix}3&4\\1&2\end{vmatrix}=2$.

Q35. If $A=\begin{bmatrix}3&1\\-1&2\end{bmatrix}$, find $|A|$, adj A and A^{-1}. 2M Must do

Ans. $|A|=6+1=7$, $\text{adj}\,A=\begin{bmatrix}2&-1\\1&3\end{bmatrix}$, $A^{-1}=\frac{1}{7}\begin{bmatrix}2&-1\\1&3\end{bmatrix}$.

Q36. Find the value of $\begin{vmatrix}102&18&36\\1&3&4\\17&3&6\end{vmatrix}$. 2M HOTS

Ans. Expand along R_1: $102(18-12)-18(6-68)+36(3-51)=612+1116-1728=0$. (Note $R_1=6R_3$, so 0 is expected.)

Q37. If $A=\begin{bmatrix}2&3\\5&-2\end{bmatrix}$, show that $A^{-1}=\frac{1}{19}A$. 2M Board fav

Ans. $|A|=-4-15=-19$, $\text{adj}\,A=\begin{bmatrix}-2&-3\\-5&2\end{bmatrix}=-A$. So $A^{-1}=\frac{-A}{-19}=\frac{1}{19}A$.

Q38. If A is a 3×3 matrix with $|A|=4$, find $|2\,\text{adj}\,A|$ and $|(\text{adj}\,A)^{-1}|$. 2M Most asked

Ans. $|\text{adj}\,A|=4^2=16$. $|2\,\text{adj}\,A|=2^3(16)=128$.
$|(\text{adj}\,A)^{-1}|=\frac{1}{16}$.

Short answer (3 marks)

Q39. Solve by matrix method: $3x+2y=7, 5x-y=3$. 3M Must do

Ans. $|A| = -3-10 = -13$. $A^{-1} = \frac{1}{-13}\begin{bmatrix}-1 & -2\\ -5 & 3\end{bmatrix}$.
$X = \frac{1}{-13}\begin{bmatrix}-7-6\\ -35+9\end{bmatrix} = \frac{1}{-13}\begin{bmatrix}-13\\ -26\end{bmatrix}: x=1, y=2.$

Q40. If $A=\begin{bmatrix}3 & 2\\ 1 & 1\end{bmatrix}$ and $B=\begin{bmatrix}2 & 1\\ 1 & 1\end{bmatrix}$, verify $(AB)^{-1}=B^{-1}A^{-1}$. 3M

Ans. $AB=\begin{bmatrix}8 & 5\\ 3 & 2\end{bmatrix}, |AB|=1, (AB)^{-1}=\begin{bmatrix}2 & -5\\ -3 & 8\end{bmatrix}$.
$A^{-1}=\begin{bmatrix}1 & -2\\ -1 & 3\end{bmatrix}, B^{-1}=\begin{bmatrix}1 & -1\\ -1 & 2\end{bmatrix}, B^{-1}A^{-1} = \begin{bmatrix}1+1 & -2-3\\ -1-2 & 2+6\end{bmatrix}=\begin{bmatrix}2 & -5\\ -3 & 8\end{bmatrix}$. Verified.

Q41. Examine the consistency of $x+3y=5, 2x+6y=8$. 3M

Ans. $|A|=\begin{vmatrix}1 & 3\\ 2 & 6\end{vmatrix}=0$. $(\text{adj } A)B=\begin{bmatrix}6 & -3\\ -2 & 1\end{bmatrix}\begin{bmatrix}5\\ 8\end{bmatrix} = \begin{bmatrix}6\\ -2\end{bmatrix} \neq O$. Hence inconsistent (lines are parallel).

Q42. Find adj A for $A=\begin{bmatrix}1 & 0 & 2\\ 2 & 1 & 0\\ 1 & -1 & 1\end{bmatrix}$ and verify that $A(\text{adj } A)=|A|I$. 3M Board fav

Ans. $|A| = 1(1-0)-0+2(-2-1) = -5$. Cofactors: $A_{11}=1, A_{12}=-2, A_{13}=-3, A_{21}=-2, A_{22}=-1, A_{23}=1, A_{31}=-2, A_{32}=4, A_{33}=1$. adj $A=\begin{bmatrix}1 & -2 & -2\\ -2 & -1 & 4\\ -3 & 1 & 1\end{bmatrix}$. $A(\text{adj } A)=\begin{bmatrix}-5 & 0 & 0\\ 0 & -5 & 0\\ 0 & 0 & -5\end{bmatrix} = -5I$. Verified.

Q43. Find the values of k for which the points $(k,2), (3,4), (5,6)$ are collinear, and find the line on which they lie. 3M

Ans. $\begin{vmatrix}k & 2 & 1\\ 3 & 4 & 1\\ 5 & 6 & 1\end{vmatrix} = k(4-6)-2(3-5)+1(18-20) = -2k+4-2=0 \Rightarrow k=1$. Line through $(3,4), (5,6)$: $\begin{vmatrix}x & y & 1\\ 3 & 4 & 1\\ 5 & 6 & 1\end{vmatrix}=0 \Rightarrow -2x+2y-2=0$, i.e. $y=x+1$.

Q44. Find x if $\begin{vmatrix}1 & 2 & x\\ 1 & 1 & 1\\ 2 & 1 & -1\end{vmatrix}=0$ and then evaluate the determinant with $x=0$. 3M

Ans. $4-x=0 \Rightarrow x=4$ (Solved Example). With $x=0$: $1(-1-1)-2(-1-2)+0 = -2+6=4$.

Q45. Find the area of the triangle with vertices $(3,8), (-4,2), (5,1)$ and check whether the points are collinear. 3M Most asked

(contd.) $\frac{1}{2}|3(2-1)-8(-4-5)+1(-4-10)| = \frac{1}{2}|3+72-14| = \frac{61}{2}$ sq units. Area $\neq 0$, so not collinear.

Q46. If $A=\begin{bmatrix}1 & \tan x\\ -\tan x & 1\end{bmatrix}$, show that $A'A^{-1}=\begin{bmatrix}\cos 2x & -\sin 2x\\ \sin 2x & \cos 2x\end{bmatrix}$. 3M HOTS

Ans. $|A| = 1+\tan^2 x = \sec^2 x$. $A^{-1} = \cos^2 x\begin{bmatrix}1 & -\tan x\\ \tan x & 1\end{bmatrix}$. $A'A^{-1} = \cos^2 x\begin{bmatrix}1 & -\tan x\\ \tan x & 1\end{bmatrix}\begin{bmatrix}1 & -\tan x\\ \tan x & 1\end{bmatrix} = \cos^2 x\begin{bmatrix}1-\tan^2 x & -2\tan x\\ 2\tan x & 1-\tan^2 x\end{bmatrix}$. Since $\cos^2 x(1-\tan^2 x) = \cos 2x$ and $2\tan x\cos^2 x = \sin 2x$, done.

Q47. If $A=\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$, verify $A(\text{adj } A) = (\text{adj } A)A = |A|I$ and find A^{-1}. 3M

Ans. $|A|=3$, adj $A=\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$. $A(\text{adj } A) = \begin{bmatrix}4-1 & 2-2\\ -2+2 & -1+4\end{bmatrix} = 3I$; similarly $(\text{adj } A)A = 3I$.
$A^{-1} = \frac{1}{3}\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$.

Q48. For $A=\begin{bmatrix}2 & 3\\ 4 & 5\end{bmatrix}$, verify (i) $|A'|=|A|$ (ii) $|A^{-1}| = \frac{1}{|A|}$ (iii) $|\text{adj } A| = |A|$. 3M

Ans. $|A| = 10-12=-2$. (i) $A'=\begin{bmatrix}2 & 4\\ 3 & 5\end{bmatrix}, |A'|=-2$.
(ii) $A^{-1} = -\frac{1}{2}\begin{bmatrix}5 & -3\\ -4 & 2\end{bmatrix}, |A^{-1}| = \frac{1}{4}(10-12) = -\frac{1}{2}$. (iii) $|\text{adj } A| = 10-12=-2=|A|^{2-1}$.

Q49. Show that the system $x+y+z=6, x+2y+3z=14, 2x+3y+4z=20$ has infinitely many solutions and write them. 3M HOTS

Ans. $|A| = \begin{vmatrix}1 & 1 & 1\\ 1 & 2 & 3\\ 2 & 3 & 4\end{vmatrix} = 1(8-9)-1(4-6)+1(3-4)=0$. adj $A=\begin{bmatrix}-1 & -1 & 1\\ 2 & 2 & -2\\ -1 & -1 & 1\end{bmatrix}$, $(\text{adj } A)B = \begin{bmatrix}-6-14+20\\ 12+28-40\\ -6-14+20\end{bmatrix} = O$. Eq. 3 = Eq. 1 + Eq. 2, so put $z=k$: $y=8-2k, x=k-2$ (any real k).

Q50. If $A=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$ and $B=\begin{bmatrix}2 & 0\\ 1 & 3\end{bmatrix}$, verify $|AB| = |A||B|$. 3M

Ans. $|A| = 4-6=-2, |B|=6$. $AB=\begin{bmatrix}4 & 6\\ 10 & 12\end{bmatrix}$, $|AB| = 48-60=-12=(-2)(6)$. Verified.

Long answer (5 marks)

Q51. Solve by matrix method: $2x - y + z = 3$, $x + y - 2z = -3$, $3x + z = 6$. 5M Most asked

Ans. $A = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 1 & -2 \\ 3 & 0 & 1 \end{bmatrix}$, $|A| = 6 \neq 0$ (Solved Example above). $\text{adj } A = \begin{bmatrix} 1 & 1 & 1 \\ -7 & -1 & 5 \\ -3 & -3 & 3 \end{bmatrix}$.

$X = A^{-1}B = \frac{1}{6}\begin{bmatrix} 1 & 1 & 1 \\ -7 & -1 & 5 \\ -3 & -3 & 3 \end{bmatrix}\begin{bmatrix} 3 \\ -3 \\ 6 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 3-3+6 \\ -21+3+30 \\ -9+9+18 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 6 \\ 12 \\ 18 \end{bmatrix}$.

$x = 1, y = 2, z = 3$.

Q52. If $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 2 \\ 3 & 1 & 1 \end{bmatrix}$, find A^{-1}. Hence solve $x + 2y - z = -1$, $y + 2z = 1$, $3x + y + z = 6$. 5M Board fav

Ans. $|A| = 1(1-2) - 2(0-6) - 1(0-3) = -1 + 12 + 3 = 14$. $\text{adj } A = \begin{bmatrix} -1 & -3 & 5 \\ 6 & 4 & -2 \\ -3 & 5 & 1 \end{bmatrix}$, so $A^{-1} = \frac{1}{14}\begin{bmatrix} -1 & -3 & 5 \\ 6 & 4 & -2 \\ -3 & 5 & 1 \end{bmatrix}$.

$X = \frac{1}{14}\begin{bmatrix} -1 & -3 & 5 \\ 6 & 4 & -2 \\ -3 & 5 & 1 \end{bmatrix}\begin{bmatrix} -1 \\ 1 \\ 6 \end{bmatrix} = \frac{1}{14}\begin{bmatrix} 1-3+30 \\ -6+4-12 \\ 3+5+6 \end{bmatrix} = \frac{1}{14}\begin{bmatrix} 28 \\ -14 \\ 14 \end{bmatrix}$.

$x = 2, y = -1, z = 1$.

Q53. Find A^{-1} for $A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$ and verify that $AA^{-1} = I$. 5M

Ans. $|A| = 3(2-1) - 1(1-2) + 2(1-4) = 3 + 1 - 6 = -2$. Cofactors: $A_{11} = 1, A_{12} = 1, A_{13} = -3, A_{21} = 1, A_{22} = -1, A_{23} = -1, A_{31} = -3, A_{32} = -1, A_{33} = 5$.

$\text{adj } A = \begin{bmatrix} 1 & 1 & -3 \\ 1 & -1 & -1 \\ -3 & -1 & 5 \end{bmatrix}$, $A^{-1} = -\frac{1}{2}\begin{bmatrix} 1 & 1 & -3 \\ 1 & -1 & -1 \\ -3 & -1 & 5 \end{bmatrix}$.

Check row 1 of A with column 1 of adj: $3(1) + 1(1) + 2(-3) = -2$; so $A(\text{adj } A) = -2I$ and $AA^{-1} = I$.

Q54. Using matrices, solve: $x - y + 2z = 7$, $3x + 4y - 5z = -5$, $2x - y + 3z = 12$. 5M Board fav

(contd.) $A = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 4 & -5 \\ 2 & -1 & 3 \end{bmatrix}$, $|A| = 1(12-5) + 1(9+10) + 2(-3-8) = 7 + 19 - 22 = 4 \neq 0$.

Cofactors: $A_{11} = 7, A_{12} = -19, A_{13} = -11; A_{21} = 1, A_{22} = -1, A_{23} = -1; A_{31} = -3, A_{32} = 11, A_{33} = 7$.

$\text{adj } A = \begin{bmatrix} 7 & 1 & -3 \\ -19 & -1 & 11 \\ -11 & -1 & 7 \end{bmatrix}$.

$X = \frac{1}{4}\begin{bmatrix} 49-5-36 \\ -133+5+132 \\ -77+5+84 \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 8 \\ 4 \\ 12 \end{bmatrix}$: $x = 2, y = 1, z = 3$.

Q55. The cost of 2 kg apples, 6 dozen bananas and 1 kg oranges is Rs 560; of 1 kg apples, 12 dozen bananas and 2 kg oranges is Rs 760; of 3 kg apples and 2 kg oranges is Rs 520. Find the price of each using matrices. 5M Must do

Ans. $2a + 6b + o = 560$, $a + 12b + 2o = 760$, $3a + 2o = 520$. $A = \begin{bmatrix} 2 & 6 & 1 \\ 1 & 12 & 2 \\ 3 & 0 & 2 \end{bmatrix}$, $|A| = 2(24) - 6(2-6) + 1(0-36) = 48 + 24 - 36 = 36$.

$\text{adj } A = \begin{bmatrix} 24 & -12 & 0 \\ 4 & 1 & -3 \\ -36 & 18 & 18 \end{bmatrix}$.

$X = \frac{1}{36}\begin{bmatrix} 13440 - 9120 + 0 \\ 2240 + 760 - 1560 \\ -20160 + 13680 + 9360 \end{bmatrix} = \frac{1}{36}\begin{bmatrix} 4320 \\ 1440 \\ 2880 \end{bmatrix}$.

Apples Rs 120/kg, bananas Rs 40/dozen, oranges Rs 80/kg.

Q56. If $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, find A^{-1}. Using $(A')^{-1} = (A^{-1})'$, solve $2x + y = 4$, $x + z = 4$, $x + y + z = 6$. 5M HOTS

Ans. $|A| = 2(0-1) - 1(1-0) + 1(1-0) = -2$.

$\text{adj } A = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 2 & -1 \\ 1 & -2 & -1 \end{bmatrix}$, $A^{-1} = -\frac{1}{2}\begin{bmatrix} -1 & 0 & 1 \\ -1 & 2 & -1 \\ 1 & -2 & -1 \end{bmatrix}$.

The coefficient matrix is $A' = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, so $X = (A^{-1})'B = -\frac{1}{2}\begin{bmatrix} -1 & -1 & 1 \\ 0 & 2 & -2 \\ 1 & -1 & -1 \end{bmatrix}\begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}$

$= -\frac{1}{2}\begin{bmatrix} -4-4+6 \\ 0+8-12 \\ 4-4-6 \end{bmatrix} = -\frac{1}{2}\begin{bmatrix} -2 \\ -4 \\ -6 \end{bmatrix}$: $x = 1, y = 2, z = 3$.

Q57. Solve using matrices: $x + 2y - z = 1$, $2x - y + z = 3$, $x + y + 2z = 6$. 5M Must do

(contd.) $A=\begin{bmatrix}1&2&-1\\2&-1&1\\1&1&2\end{bmatrix}$, $|A|=1(-2-1)-2(4-1)-1(2+1)=-3-6-3=-12$.
Cofactors: $A_{11}=-3, A_{12}=-3, A_{13}=3; A_{21}=-5, A_{22}=3, A_{23}=1; A_{31}=1, A_{32}=-3, A_{33}=-5$.
$\text{adj } A=\begin{bmatrix}-3&-5&1\\-3&3&-3\\3&1&-5\end{bmatrix}$.
$X=\frac{1}{-12}\begin{bmatrix}-3-15+6\\-3+9-18\\3+3-30\end{bmatrix}=\frac{1}{-12}\begin{bmatrix}-12\\-12\\-24\end{bmatrix}: x=1, y=1, z=2.$

Q58. For $A=\begin{bmatrix}2&1&1\\1&2&1\\1&1&2\end{bmatrix}$, show that $A^3-6A^2+9A-4I=O$. Hence find A^{-1} and check it by the adjoint method. 5M HOTS

Ans. $A^2=\begin{bmatrix}6&5&5\\5&6&5\\5&5&6\end{bmatrix}$, $A^3=A^2A=\begin{bmatrix}22&21&21\\21&22&21\\21&21&22\end{bmatrix}$.
Entry (1,1): $22-36+18-4=0$; entry (1,2): $21-30+9-0=0$; by symmetry every entry is 0, so $A^3-6A^2+9A-4I=O$.
Multiply by A^{-1}: $A^2-6A+9I=4A^{-1}\Rightarrow A^{-1}=\frac{1}{4}\begin{bmatrix}3&-1&-1\\-1&3&-1\\-1&-1&3\end{bmatrix}$.
Adjoint check: $|A|=4$, $\text{adj } A=\begin{bmatrix}3&-1&-1\\-1&3&-1\\-1&-1&3\end{bmatrix}$ ✓

Q59. Find A^{-1} for $A=\begin{bmatrix}1&1&1\\1&2&-3\\2&-1&3\end{bmatrix}$ and hence solve $x+y+z=6, x+2y-3z=-4, 2x-y+3z=9$. 5M Board fav

Ans. $|A|=1(6-3)-1(3+6)+1(-1-4)=3-9-5=-11$.
Cofactors: $A_{11}=3, A_{12}=-9, A_{13}=-5; A_{21}=-4, A_{22}=1, A_{23}=3; A_{31}=-5, A_{32}=4, A_{33}=1$.
$\text{adj } A=\begin{bmatrix}3&-4&-5\\-9&1&4\\-5&3&1\end{bmatrix}$, $A^{-1}=-\frac{1}{11}\begin{bmatrix}3&-4&-5\\-9&1&4\\-5&3&1\end{bmatrix}$.
$X=-\frac{1}{11}\begin{bmatrix}18+16-45\\-54-4+36\\-30-12+9\end{bmatrix}=-\frac{1}{11}\begin{bmatrix}-11\\-22\\-33\end{bmatrix}: x=1, y=2, z=3.$

Recall Line through 2 points: put (x, y) in the first row of the determinant.

Case based (4 marks)

Q60. Case: A triangular park. A society plans a triangular park with corners $A(1,2)$, $B(5,2)$ and $C(3,6)$ (units in 10 m). (i) Find the area of the park using determinants. (ii) A lamp post is at $P(3,2)$. Show that A, P, B are collinear. (iii) If the corner C is moved to $(k,6)$, show that the area does not change for any k. 4M Board fav

Ans. (i) $\frac{1}{2}|1(2-6)-2(5-3)+1(30-6)|=\frac{1}{2}|-4-4+24|=8$ sq units. (ii) $\begin{vmatrix}1&2&1\\3&2&1\\5&2&1\end{vmatrix}=1(0)-2(3-5)+1(6-10)=0$: collinear. (iii) $\frac{1}{2}|1(2-6)-2(5-k)+1(30-2k)|=\frac{1}{2}|-4-10+2k+30-2k|=\frac{1}{2}(16)=8$, free of k (base AB fixed, height always 4).

Q61. Case: Three friends shopping. Riya buys 1 pen, 1 notebook, 1 file for Rs 6 (in tens of rupees); Aman buys 2 pens, 1 notebook, 3 files for Rs 13 (tens); Kabir buys 3 pens, 2 notebooks, 1 file for Rs 10 (tens). Let the prices be x, y, z (tens of Rs). (i) Write the system as $AX=B$. (ii) Find $|A|$. (iii) Find the prices using $X=A^{-1}B$. 4M Must do

Ans. (i) $A=\begin{bmatrix}1&1&1\\2&1&3\\3&2&1\end{bmatrix}$, $B=\begin{bmatrix}6\\13\\10\end{bmatrix}$. (ii) $|A|=3$. (iii) $X=\frac{1}{3}\begin{bmatrix}-5&1&2\\7&-2&-1\\1&1&-1\end{bmatrix}\begin{bmatrix}6\\13\\10\end{bmatrix}=\frac{1}{3}\begin{bmatrix}-30+13+20\\42-26-10\\6+13-10\end{bmatrix}=\frac{1}{3}\begin{bmatrix}3\\6\\9\end{bmatrix}$: pen Rs 10, notebook Rs 20, file Rs 30.

Q62. Case: Secret messages. A club encodes a pair of letters (A = 1, B = 2, ..., Z = 26) as a column X and sends $C=AX$, where $A=\begin{bmatrix}2&1\\1&1\end{bmatrix}$. (i) Find $|A|$. (ii) Find A^{-1}. (iii) A received code is $C=\begin{bmatrix}13\\8\end{bmatrix}$. Decode the two letters. 4M HOTS

Ans. (i) $|A|=2-1=1$. (ii) $A^{-1}=\begin{bmatrix}1&-1\\-1&2\end{bmatrix}$. (iii) $X=A^{-1}C=\begin{bmatrix}13-8\\-13+16\end{bmatrix}=\begin{bmatrix}5\\3\end{bmatrix}$: letters E and C.

Check trick Got x, y, z? Put them back in ALL three equations.

Board tip Write "consistent with unique solution" when $|A|\neq 0$.

Quick fact $|A|$ of a triangular matrix = product of diagonal entries.

CONTINUITY & DIFFERENTIABILITY

Most asked: find k for continuity, log differentiation, $\frac{d^2y}{dx^2}$ proofs

1. Continuity at a Point

Roughly: f is continuous at $x = a$ if its graph can be drawn without lifting the pen near $x = a$. Board answers need the limit test below.

Definition: Continuity at $x = a$

A real function f is **continuous at** $x = a$ (a point of its domain) if

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = f(a)$$

i.e. **LHL = RHL = value**. If any one of these fails (or a limit does not exist), f is **discontinuous** at a.

LHL $= \lim_{h \to 0} f(a-h)$, **RHL** $= \lim_{h \to 0} f(a+h)$, $h > 0$.

A function is **continuous** if it is continuous at every point of its domain.

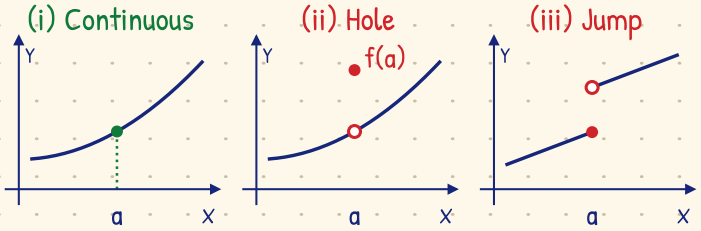


Fig: (i) no break (ii) hole: limit exists but differs from $f(a)$ (iii) jump: LHL and RHL differ

Yaad karo: "L = R = V"

Left limit = Right limit = Value. Teeno barabar, tabhi continuous. Piecewise function mein teeno alag-alag likho, phir equate karo.

Solved Example 1

Check the continuity of $f(x) = \begin{cases} 2x+3, & x \le 2 \\ 3x+1, & x > 2 \end{cases}$ at $x = 2$.

1. LHL $= \lim_{x \to 2^-} (2x+3) = 7$
2. RHL $= \lim_{x \to 2^+} (3x+1) = 7$
3. $f(2) = 2(2)+3 = 7$
4. LHL = RHL = $f(2) = 7$

✔ f is continuous at $x = 2$

Quick fact $\frac{d}{dx}|x| = \frac{|x|}{x}$, $x \neq 0$

Quick fact Polynomials, $\sin x$, $\cos x$, e^x are differentiable everywhere.

Solved Example 2

Is $f(x) = \begin{cases} \frac{x^2-9}{x-3}, & x \neq 3 \\ 5, & x = 3 \end{cases}$ continuous at $x = 3$?

1. For $x \neq 3$: $\frac{(x-3)(x+3)}{x-3} = x+3$
2. $\lim_{x \to 3} f(x) = 6$
3. But $f(3) = 5 \neq 6$

✔ Discontinuous at $x = 3$ (removable: redefine $f(3) = 6$)

2. Find k / a, b for Continuity

Most common board question: a piecewise function with unknown constants is **given continuous**. Write LHL, RHL, $f(a)$ at each joining point and equate. Two joining points give two equations for a, b.

Solved Example 3

Find k so that $f(x) = \begin{cases} kx+1, & x \le 5 \\ 3x-5, & x > 5 \end{cases}$ is continuous at $x = 5$.

1. LHL $= f(5) = 5k+1$
2. RHL $= \lim_{x \to 5^+} (3x-5) = 10$
3. Continuity: $5k+1 = 10$

✔ $k = \frac{9}{5}$

Solved Example 4

Find k if $f(x) = \begin{cases} \frac{1-\cos 4x}{x^2}, & x \neq 0 \\ k, & x = 0 \end{cases}$ is continuous at $x = 0$.

1. $1 - \cos 4x = 2\sin^2 2x$
2. $\lim_{x \to 0} \frac{2\sin^2 2x}{x^2} = 2 \times 4 \lim_{x \to 0} \left(\frac{\sin 2x}{2x}\right)^2 = 8$
3. Continuity: $k = f(0) = 8$

✔ $k = 8$

Quick fact $\frac{d}{dx} \log|\sec x| = \tan x$

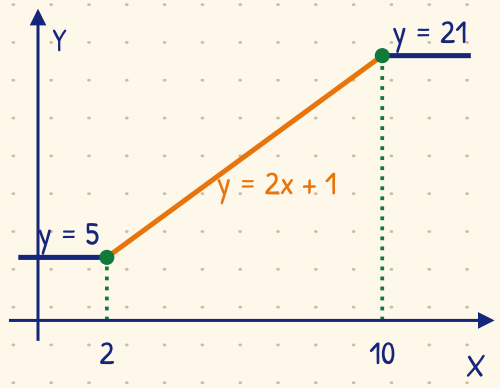


Fig: with $a = 2, b = 1$ the three pieces join without a break

Solved Example 5

Find a and b if $f(x) = \begin{cases} 5, & x \le 2 \\ ax + b, & 2 < x < 10 \\ 21, & x \ge 10 \end{cases}$ is continuous everywhere.

1. At $x = 2$: LHL $= 5$, RHL $= 2a + b$, so $2a + b = 5$
2. At $x = 10$: LHL $= 10a + b$, RHL $= f(10) = 21$, so $10a + b = 21$
3. Subtract: $8a = 16 \Rightarrow a = 2$, then $b = 1$

✔ $a = 2,\ b = 1$

Formula Box 1: Limits you need

- $\lim_{x\to 0} \dfrac{\sin x}{x} = 1,\ \lim_{x\to 0} \dfrac{\tan x}{x} = 1$
- $\lim_{x\to 0} \dfrac{\sin ax}{bx} = \dfrac{a}{b}$
- $\lim_{x\to 0} \dfrac{1 - \cos ax}{x^2} = \dfrac{a^2}{2}$
- $\lim_{x\to 0} \dfrac{e^x - 1}{x} = 1$
- $\lim_{x\to 0} \dfrac{\log(1+x)}{x} = 1$
- $\lim_{x\to a} \dfrac{x^n - a^n}{x - a} = na^{n-1}$
- $1 - \cos 2\theta = 2\sin^2\theta$
- $1 + \cos 2\theta = 2\cos^2\theta$

Quick fact $\dfrac{d}{dx}(e^{ax}) = ae^{ax}$

Quick fact Differentiable $\Rightarrow$ continuous; never the other way round.

Quick fact $\dfrac{d}{dx}\log|\sin x| = \cot x,\ \dfrac{d}{dx}\log|\cos x| = -\tan x$

Solved Example 6

Find k if $f(x) = \begin{cases} \dfrac{\sin 5x}{3x} + \cos x, & x \ne 0 \\ k, & x = 0 \end{cases}$ is continuous at $x = 0$.

1. $\lim_{x\to 0} \dfrac{\sin 5x}{3x} = \dfrac{5}{3},\ \lim_{x\to 0}\cos x = 1$
2. Limit $= \dfrac{5}{3} + 1 = \dfrac{8}{3}$
3. Set $k = \dfrac{8}{3}$

✔ $k = \dfrac{8}{3}$

Solved Example 7

Find k if $f(x) = \begin{cases} \dfrac{\sqrt{1+kx} - \sqrt{1-kx}}{x}, & -1 \le x < 0 \\ \dfrac{2x+1}{x-1}, & 0 \le x \le 1 \end{cases}$ is continuous at $x = 0$.

1. RHL $= f(0) = \dfrac{0+1}{0-1} = -1$
2. LHL: rationalise, $\dfrac{(1+kx)-(1-kx)}{x\left(\sqrt{1+kx}+\sqrt{1-kx}\right)} = \dfrac{2k}{\sqrt{1+kx}+\sqrt{1-kx}} \to \dfrac{2k}{2} = k$
3. $k = -1$

✔ $k = -1$

Solved Example 8

Find λ if $f(x) = \begin{cases} \lambda(x^2 - 2x), & x \le 0 \\ 4x + 1, & x > 0 \end{cases}$ is continuous at $x = 0$. What happens at $x = 0$?

1. LHL $= f(0) = \lambda(0) = 0$
2. RHL $= \lim_{x\to 0^+}(4x+1) = 1$
3. $0 \ne 1$ for every λ

✔ No value of λ works: f is discontinuous at 0 for all λ

Answer writing format (3 marks)

1. Write LHL $= \lim_{h\to 0} f(a-h)$ and simplify.
2. Write RHL $= \lim_{h\to 0} f(a+h)$ and simplify.
3. Write $f(a)$ from the "$=$" piece.
4. Conclude: "Since LHL = RHL = $f(a)$, f is continuous at $x = a$."

Quick fact $\dfrac{dy}{dx} \times \dfrac{dx}{dy} = 1$

3. Algebra of Continuous Functions

Theorem (no proof needed)

If f and g are continuous at a, then $f+g$, $f-g$, fg and $\frac{f}{g}$ (where $g(a) \neq 0$) are continuous at a. If g is continuous at a and f at $g(a)$, then $f \circ g$ is continuous at a.

Function	Continuous on
Polynomial, $\lvert x\rvert$, $\sin x$, $\cos x$, e^x	all of $\mathbb{R}$
Rational $\frac{p(x)}{q(x)}$	wherever $q(x) \neq 0$
$\tan x$, $\sec x$	$x \neq (2n+1)\frac{\pi}{2}$
$\cot x$, $\csc x$	$x \neq n\pi$
$\log x$	$x > 0$
$[x]$ (greatest integer)	discontinuous at **every integer**

So $\sin\lvert x\rvert$, $\lvert\cos x\rvert$, e^{x^2}, $\cos(x^2)$ are continuous everywhere (composition of continuous functions).

Solved Example 9

Find all points of discontinuity of $f(x) = \begin{cases} x+1, & x \geq 1 \\ x^2+1, & x < 1 \end{cases}$

1. For $x > 1$ and $x < 1$, f is a polynomial: continuous
2. At $x = 1$: LHL $= 1+1 = 2$, RHL $= 1+1 = 2$, $f(1) = 2$
3. So continuous at $x = 1$ also

✔ No point of discontinuity

Solved Example 10

Show that $f(x) = \lvert x-3\rvert$ is continuous at $x = 3$.

1. $f(x) = 3-x$ for $x < 3$ and $x-3$ for $x \geq 3$
2. LHL $= \lim_{x\to 3^-}(3-x) = 0$, RHL $= \lim_{x\to 3^+}(x-3) = 0$
3. $f(3) = 0$

✔ LHL = RHL = $f(3)$, so continuous at $x = 3$

Quick fact $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$, so their derivatives add to 0.

Quick fact $\frac{d}{dx}\sin^2 x = \sin 2x$, $\frac{d}{dx}\cos^2 x = -\sin 2x$

Solved Example 11

Discuss the continuity of $f(x) = \begin{cases} \frac{\lvert x\rvert}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ at $x = 0$.

1. For $x > 0$: $f(x) = 1$, so RHL $= 1$
2. For $x < 0$: $f(x) = -1$, so LHL $= -1$
3. LHL $\neq$ RHL

✔ Discontinuous at $x = 0$ (a jump of 2)

Solved Example 12

Show that $f(x) = \begin{cases} x\sin\frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ is continuous at $x = 0$.

1. $\left\lvert\sin\frac{1}{x}\right\rvert \leq 1$, so $\left\lvert x\sin\frac{1}{x}\right\rvert \leq \lvert x\rvert$
2. As $x \to 0$, $\lvert x\rvert \to 0$, so $\lim_{x\to 0} x\sin\frac{1}{x} = 0$ (sandwich)
3. $f(0) = 0$

✔ Limit $= f(0) = 0$: continuous at 0

Solved Example 13

Is $f(x) = [x]$ continuous at $x = 2$? At $x = 2.5$?

1. At 2: LHL $= \lim_{x\to 2^-}[x] = 1$, RHL $= 2$
2. LHL $\neq$ RHL, so discontinuous at 2
3. At 2.5: $[x] = 2$ near 2.5 on both sides, $f(2.5) = 2$

✔ Discontinuous at 2; continuous at 2.5

Yaad karo: "Integer pe toot, beech mein chhoot"

$[x]$ har **integer** pe toot-ta hai (LHL $= n-1$, RHL $= n$), aur do integers ke **beech** constant hai, isliye wahan continuous.

4. Differentiability

Definition: Derivative at $x = a$

$$f'(a) = \lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$$

f is **differentiable at a** if LHD = RHD (both finite):

$$\text{LHD} = \lim_{h\to 0}\frac{f(a-h)-f(a)}{-h}$$

$$\text{RHD} = \lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$$

Quick fact $\frac{d}{dx}\tan^{-1}(e^x) = \frac{e^x}{1+e^{2x}}$

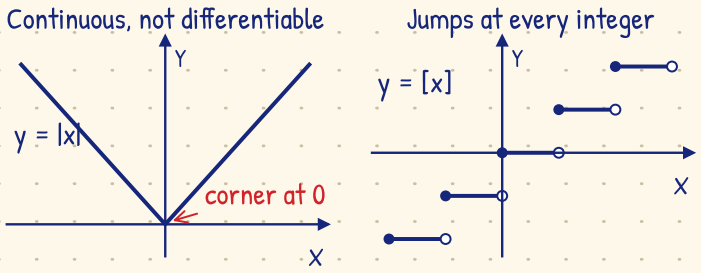


Fig: $|x|$ has a corner at 0; $[x]$ jumps at each integer

Theorem

If f is differentiable at a, then f is continuous at a.

The converse is **NOT** true: $|x|$ is continuous at 0 but not differentiable there.

Contrapositive: discontinuous at $a \Rightarrow$ not differentiable at a.

Solved Example 14

Show that $f(x) = |x-2|$ is continuous but not differentiable at $x = 2$.

1. Continuity: LHL $=$ RHL $= f(2) = 0$ (as in Example 8)
2. RHD $= \lim_{h\to 0} \dfrac{|h|-0}{h} = \lim_{h\to 0} \dfrac{h}{h} = 1$
3. LHD $= \lim_{h\to 0} \dfrac{|-h|-0}{-h} = \lim_{h\to 0} \dfrac{h}{-h} = -1$
4. LHD $\neq$ RHD

✔ Continuous, but not differentiable at $x = 2$

Solved Example 15

Is $f(x) = x|x|$ differentiable at $x = 0$?

1. RHD $= \lim_{h\to 0} \dfrac{h|h|-0}{h} = \lim_{h\to 0} |h| = 0$
2. LHD $= \lim_{h\to 0} \dfrac{(-h)|-h|-0}{-h} = \lim_{h\to 0} |h| = 0$
3. LHD = RHD = 0

✔ Yes, differentiable at 0 and $f'(0) = 0$

Solved Example 16

Find a, b if $f(x) = \begin{cases} x^2 + ax + b, & x < 1 \\ 3x + 2, & x \geq 1 \end{cases}$ is differentiable at $x = 1$.

1. Differentiable $\Rightarrow$ continuous: $1 + a + b = 5$, so $a + b = 4$
2. LHD $= \dfrac{d}{dx}(x^2 + ax + b)$ at $1 = 2 + a$; RHD $= 3$
3. $2 + a = 3 \Rightarrow a = 1$, then $b = 3$

✔ $a = 1,\ b = 3$

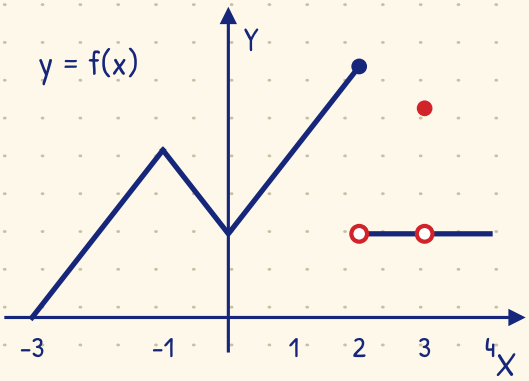


Fig: for the graph-reading example below



Shivam Bhaiya says

Differentiability wale sawal mein pehle **continuity check** karo. Agar function continuous hi nahi, to seedha likh do "not differentiable", LHD/RHD nikalne ki zarurat hi nahi. Board mein yahi 1 line pura mark dilati hai.

Solved Example 17 (graph reading)

The graph of f on $[-3, 4]$ is shown. Find the points where f is (i) discontinuous, (ii) continuous but not differentiable.

1. $x = 2$: jump (LHL $= 3$, RHL $= 1$), discontinuous
2. $x = 3$: hole, limit $= 1$ but $f(3) = 2.5$, discontinuous
3. $x = -1$ and $x = 0$: unbroken but sharp corners (slopes $1, -1$ and $-1, 1$)

✔ (i) $x = 2, 3$ (ii) $x = -1, 0$

Solved Example 18

Find the points of discontinuity of $f(x) = \dfrac{x+2}{x^2 - 4x + 3}$.

1. Rational function: continuous wherever the denominator $\neq 0$
2. $x^2 - 4x + 3 = (x-1)(x-3) = 0$ at $x = 1, 3$

✔ Discontinuous only at $x = 1$ and $x = 3$

Solved Example 19

Is $f(x) = |x| + |x-1|$ continuous at $x = 1$? Differentiable at $x = 1$?

1. Near $x = 1$: $f(x) = 1$ for $0 \leq x \leq 1$ and $f(x) = 2x - 1$ for $x > 1$
2. LHL $=$ RHL $= f(1) = 1$: continuous
3. LHD $= 0$, RHD $= 2$: not equal

✔ Continuous but not differentiable at $x = 1$

5. Standard Derivatives

Formula Box 2: Basic derivatives

- $\frac{d}{dx}(x^n) = nx^{n-1}$, $\frac{d}{dx}(c) = 0$
- $\frac{d}{dx}(\sin x) = \cos x$
- $\frac{d}{dx}(\cos x) = -\sin x$
- $\frac{d}{dx}(\tan x) = \sec^2 x$
- $\frac{d}{dx}(\cot x) = -\csc^2 x$
- $\frac{d}{dx}(\sec x) = \sec x \tan x$
- $\frac{d}{dx}(\csc x) = -\csc x \cot x$
- $\frac{d}{dx}(e^x) = e^x$, $\frac{d}{dx}(a^x) = a^x \log a$
- $\frac{d}{dx}(\log x) = \frac{1}{x}$, $\frac{d}{dx}(\log_a x) = \frac{1}{x \log a}$
- $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$, $\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$

Formula Box 3: Inverse trig derivatives

- $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$
- $\frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$
- $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}$
- $\frac{d}{dx}(\csc^{-1} x) = \frac{-1}{|x|\sqrt{x^2-1}}$
- "co" wale = minus of the partner

Formula Box 4: Rules

- Product: $(uv)' = u'v + uv'$
- Quotient: $\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$
- Chain: $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$
- Parametric: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$, $\frac{dx}{dt} \neq 0$
- Log diff: $y = u^v \Rightarrow \frac{1}{y}\frac{dy}{dx} = \frac{d}{dx}(v \log u)$
- Second order: $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$

Yaad karo: "Co = Cut (minus)"

Jis function ke naam mein **co** hai ($\cos$, $\cot$, $\csc$, $\cos^{-1}$, $\cot^{-1}$, $\csc^{-1}$), uske derivative mein **minus** aata hai. Baaki sab positive.

6. Chain Rule

For a **function of a function**, differentiate from **outside to inside** and multiply: $\frac{d}{dx} f(g(x)) = f'(g(x)) \times g'(x)$.

Yaad karo: "Pyaaz ki tarah chheelo"

Onion jaisa: bahar ki layer ka derivative, phir andar wali ka, phir aur andar ki, sab **multiply**. $\sin(x^2)$: $\cos(x^2) \times 2x$. Andar wala (x^2) bahar wale mein waise hi rehta hai.

Solved Example 20

Find $\frac{dy}{dx}$ if $y = \sin\left(\cos(x^2)\right)$.

1. Outer: $\cos(\cos x^2)$
2. Middle: $\frac{d}{dx}\cos(x^2) = -\sin(x^2) \times 2x$
3. Multiply the layers

✔ $\frac{dy}{dx} = -2x \sin(x^2) \cos(\cos x^2)$

Solved Example 21

Differentiate $y = \sqrt{\tan\sqrt{x}}$ w.r.t. x.

1. $\frac{dy}{dx} = \frac{1}{2\sqrt{\tan\sqrt{x}}} \times \frac{d}{dx}(\tan\sqrt{x})$
2. $= \frac{1}{2\sqrt{\tan\sqrt{x}}} \times \sec^2\sqrt{x} \times \frac{1}{2\sqrt{x}}$

✔ $\frac{dy}{dx} = \frac{\sec^2\sqrt{x}}{4\sqrt{x}\sqrt{\tan\sqrt{x}}}$

Solved Example 22

Differentiate $y = \cos^3(5x+2)$.

1. $y = [\cos(5x+2)]^3$
2. $\frac{dy}{dx} = 3\cos^2(5x+2) \times (-\sin(5x+2)) \times 5$

✔ $\frac{dy}{dx} = -15\cos^2(5x+2)\sin(5x+2)$

Quick fact $\frac{d}{dx}(x^x) = x^x(1+\log x)$: yaad kar lo, MCQ mein aata hai.

Quick fact $[x]$ is continuous at every non-integer point.

Solved Example 23

Differentiate $y = \sec(\tan\sqrt{x})$.

1. $y_1 = \sec(\tan\sqrt{x})\tan(\tan\sqrt{x}) \times \sec^2\sqrt{x} \times \frac{1}{2\sqrt{x}}$

✔ $\frac{dy}{dx} = \frac{\sec(\tan\sqrt{x})\tan(\tan\sqrt{x})\sec^2\sqrt{x}}{2\sqrt{x}}$

Solved Example 24

Differentiate $y = \frac{\sin(ax+b)}{\cos(cx+d)}$.

1. Write $y = \sin(ax+b)\sec(cx+d)$ (product is easier than quotient)
2. $y_1 = a\cos(ax+b)\sec(cx+d) + \sin(ax+b) \times c\sec(cx+d)\tan(cx+d)$

✔ $\frac{dy}{dx} = \sec(cx+d)\left[a\cos(ax+b) + c\sin(ax+b)\tan(cx+d)\right]$

Solved Example 25

If $f(x) = \sqrt{x^2+9}$, find $f'(4)$.

1. $f'(x) = \frac{2x}{2\sqrt{x^2+9}} = \frac{x}{\sqrt{x^2+9}}$
2. $f'(4) = \frac{4}{\sqrt{25}}$

✔ $f'(4) = \frac{4}{5}$

7. Implicit Differentiation

When y is not separated (e.g. $x^2+xy+y^2=7$), differentiate **both sides** w.r.t. x, treating y as a function of x: $\frac{d}{dx}(y^2) = 2y\frac{dy}{dx}$, $\frac{d}{dx}(xy) = y + x\frac{dy}{dx}$. Then collect $\frac{dy}{dx}$ terms.

Solved Example 26

Find $\frac{dy}{dx}$ if $x^2+xy+y^2=7$.

1. $2x + \left(y + x\frac{dy}{dx}\right) + 2y\frac{dy}{dx} = 0$
2. $\frac{dy}{dx}(x+2y) = -(2x+y)$

✔ $\frac{dy}{dx} = -\frac{2x+y}{x+2y}$

Quick fact $\frac{d}{dx}e^{\sqrt{x}} = \frac{e^{\sqrt{x}}}{2\sqrt{x}}$

Quick fact $\frac{d}{dx}\log(x^2+1) = \frac{2x}{x^2+1}$

Solved Example 27

Find $\frac{dy}{dx}$ if $\sin^2 x + \cos^2 y = 1$.

1. $2\sin x\cos x - 2\cos y\sin y\frac{dy}{dx} = 0$
2. $\sin 2x = \sin 2y\,\frac{dy}{dx}$

✔ $\frac{dy}{dx} = \frac{\sin 2x}{\sin 2y}$

Solved Example 28

If $y = \sqrt{x+\sqrt{x+\sqrt{x+\ldots\infty}}}$, show that $\frac{dy}{dx} = \frac{1}{2y-1}$.

1. The infinite part is y itself: $y = \sqrt{x+y}$
2. Square: $y^2 = x+y$
3. Differentiate: $2y\,y' = 1+y' \Rightarrow y'(2y-1) = 1$

✔ $\frac{dy}{dx} = \frac{1}{2y-1}$ (Proved)

Solved Example 29

Find $\frac{dy}{dx}$ if $\sin(xy) + \frac{x}{y} = x^2 - y$.

1. $\cos(xy)(y+xy_1) + \frac{y-xy_1}{y^2} = 2x - y_1$
2. Collect y_1: $y_1\left[x\cos(xy) - \frac{x}{y^2} + 1\right] = 2x - y\cos(xy) - \frac{1}{y}$

✔ $\frac{dy}{dx} = \frac{2xy^2 - y^3\cos(xy) - y}{xy^2\cos(xy) - x + y^2}$

Solved Example 30

If $y = x\sin y$, prove that $\frac{dy}{dx} = \frac{\sin y}{1 - x\cos y}$.

1. $y_1 = \sin y + x\cos y\,y_1$
2. $y_1(1 - x\cos y) = \sin y$

✔ Proved

Yaad karo: "y dikhe to y' chipkao"

Implicit mein jahan bhi y ka koi function differentiate karo, peeche $\frac{dy}{dx}$ **chipkana** mat bhoolo: $\frac{d}{dx}\sin y = \cos y\,\frac{dy}{dx}$, $\frac{d}{dx}e^y = e^y\frac{dy}{dx}$.

8. Derivatives of Inverse Trig (Substitution)

Directly use Formula Box 3 with chain rule, OR first

simplify by a trig substitution, which turns a scary expression into a multiple of θ.

Expression has	Put	Becomes
$\frac{2x}{1+x^2}$	$x=\tan\theta$	$\sin 2\theta$
$\frac{1-x^2}{1+x^2}$	$x=\tan\theta$	$\cos 2\theta$
$\frac{2x}{1-x^2}$	$x=\tan\theta$	$\tan 2\theta$
$3x-4x^3$	$x=\sin\theta$	$\sin 3\theta$
$4x^3-3x$	$x=\cos\theta$	$\cos 3\theta$
$\frac{3x-x^3}{1-3x^2}$	$x=\tan\theta$	$\tan 3\theta$
$\sqrt{1-x^2}$	$x=\sin\theta$ or $\cos\theta$	$\cos\theta$ or $\sin\theta$
$\sqrt{\frac{1-\cos x}{1+\cos x}}$	half angle	$\tan\frac{x}{2}$

↳Note: substitution answers hold on the stated interval (e.g. $-1<x<1$ for $x=\tan\theta$ with $\sin 2\theta$). Board pe interval diya ho to wahi maan ke chalo.

Solved Example 31

Find $\frac{dy}{dx}$ if $y=\sin^{-1}\left(\frac{2x}{1+x^2}\right)$, $-1<x<1$.

1. Put $x=\tan\theta$: $\frac{2\tan\theta}{1+\tan^2\theta}=\sin 2\theta$
2. $y=\sin^{-1}(\sin 2\theta)=2\theta$ (as $-\frac{\pi}{2}<2\theta<\frac{\pi}{2}$)
3. $y=2\tan^{-1}x$

✔ $\frac{dy}{dx}=\frac{2}{1+x^2}$

Solved Example 32

Differentiate $y=\cos^{-1}(4x^3-3x)$, $\frac{1}{2}<x<1$.

1. Put $x=\cos\theta$, $0<\theta<\frac{\pi}{3}$: $4\cos^3\theta-3\cos\theta=\cos 3\theta$
2. $y=\cos^{-1}(\cos 3\theta)=3\theta=3\cos^{-1}x$ (since $0<3\theta<\pi$)

✔ $\frac{dy}{dx}=\frac{-3}{\sqrt{1-x^2}}$

Solved Example 33

Find $\frac{dy}{dx}$ if $y=\tan^{-1}\left(\sqrt{\frac{1-\cos x}{1+\cos x}}\right)$, $0<x<\pi$.

1. $\frac{1-\cos x}{1+\cos x}=\frac{2\sin^2\frac{x}{2}}{2\cos^2\frac{x}{2}}=\tan^2\frac{x}{2}$
2. $y=\tan^{-1}\left(\tan\frac{x}{2}\right)=\frac{x}{2}$

✔ $\frac{dy}{dx}=\frac{1}{2}$

Solved Example 34

Differentiate $y=\tan^{-1}\left(\frac{\cos x-\sin x}{\cos x+\sin x}\right)$, $-\frac{\pi}{4}<x<\frac{\pi}{4}$.

1. Divide by $\cos x$: $\frac{1-\tan x}{1+\tan x}=\tan\left(\frac{\pi}{4}-x\right)$
2. $y=\frac{\pi}{4}-x$

✔ $\frac{dy}{dx}=-1$

Solved Example 35

Differentiate $y=\sin^{-1}\left(2x\sqrt{1-x^2}\right)$, $-\frac{1}{\sqrt{2}}<x<\frac{1}{\sqrt{2}}$.

1. Put $x=\sin\theta$, $-\frac{\pi}{4}<\theta<\frac{\pi}{4}$: $2\sin\theta\cos\theta=\sin 2\theta$
2. $y=\sin^{-1}(\sin 2\theta)=2\theta=2\sin^{-1}x$

✔ $\frac{dy}{dx}=\frac{2}{\sqrt{1-x^2}}$

Solved Example 36

Find $\frac{dy}{dx}$ if $y=\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$, $-\frac{1}{\sqrt{3}}<x<\frac{1}{\sqrt{3}}$.

1. Put $x=\tan\theta$, $-\frac{\pi}{6}<\theta<\frac{\pi}{6}$: expression $=\tan 3\theta$
2. $y=3\theta=3\tan^{-1}x$

✔ $\frac{dy}{dx}=\frac{3}{1+x^2}$

Solved Example 37

Differentiate $\sec^{-1}\left(\frac{1}{2x^2-1}\right)$, $0<x<\frac{1}{\sqrt{2}}$.

1. $\sec^{-1}\frac{1}{u}=\cos^{-1}u$, so $y=\cos^{-1}(2x^2-1)$
2. Put $x=\cos\theta$, $\frac{\pi}{4}<\theta<\frac{\pi}{2}$: $2\cos^2\theta-1=\cos 2\theta$, $y=2\theta=2\cos^{-1}x$

✔ $\frac{dy}{dx}=\frac{-2}{\sqrt{1-x^2}}$

Yaad karo: "Tan wala table, teen line"

$\frac{2x}{1+x^2}\to\sin 2\theta$, $\frac{1-x^2}{1+x^2}\to\cos 2\theta$, $\frac{2x}{1-x^2}\to\tan 2\theta$: teeno mein $x=\tan\theta$. Neeche $1+x^2$ dikhe to seedha $\tan\theta$ socho.

Solved Example 38

Differentiate $\sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)$, $0<x<1$.

1. Put $x=\tan\theta$, $0<\theta<\frac{\pi}{4}$: $\frac{1-x^2}{1+x^2}=\cos 2\theta=\sin\left(\frac{\pi}{2}-2\theta\right)$
2. $\frac{\pi}{2}-2\theta\in\left(0,\frac{\pi}{2}\right)$, so $y=\frac{\pi}{2}-2\tan^{-1}x$

✔ $\frac{dy}{dx}=\frac{-2}{1+x^2}$

Meme samjho



$\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ dekh ke darr gaye? $x=\tan\theta$ bolo: **jalwa hai hamara!**

Asli baat: With $x=\tan\theta$, $\frac{2x}{1-x^2}=\tan 2\theta$, so for $-1<x<1$ the function is just $2\tan^{-1}x$ and its derivative is $\frac{2}{1+x^2}$.

9. Exponential and Log Functions

→ $\frac{d}{dx}e^{f(x)}=e^{f(x)}f'(x)$, e.g. $\frac{d}{dx}e^{\sin x}=e^{\sin x}\cos x$

→ $\frac{d}{dx}\log f(x)=\frac{f'(x)}{f(x)}$, e.g. $\frac{d}{dx}\log(\sec x+\tan x)=\sec x$

→ $\log(mn)=\log m+\log n$, $\log\frac{m}{n}=\log m-\log n$, $\log m^n=n\log m$

→ $e^{\log x}=x$, $\log e=1$, $\log 1=0$ ($\log$ means base e)

Solved Example 39

Differentiate $y=e^x\log(\sin 2x)$.

1. Product rule: $y'=e^x\log(\sin 2x)+e^x\times\frac{1}{\sin 2x}\times 2\cos 2x$

✔ $\frac{dy}{dx}=e^x\left[\log(\sin 2x)+2\cot 2x\right]$

Solved Example 40

Find $\frac{dy}{dx}$ if $y=\log(\log x)$, $x>1$.

1. $y'=\frac{1}{\log x}\times\frac{1}{x}$

✔ $\frac{dy}{dx}=\frac{1}{x\log x}$

Quick fact A rational function is continuous at every point of its domain.

Solved Example 41

Find $\frac{dy}{dx}$ if $y=e^{x^2}\cos(\log x)$, $x>0$.

1. $y_1=2xe^{x^2}\cos(\log x)+e^{x^2}(-\sin(\log x))\frac{1}{x}$

✔ $\frac{dy}{dx}=e^{x^2}\left[2x\cos(\log x)-\frac{\sin(\log x)}{x}\right]$

Solved Example 42

Differentiate $y=\log_7(\log x)$, $x>1$.

1. $y=\frac{\log(\log x)}{\log 7}$ (change of base)
2. $y_1=\frac{1}{\log 7}\times\frac{1}{\log x}\times\frac{1}{x}$

✔ $\frac{dy}{dx}=\frac{1}{x\log x\log 7}$

10. Logarithmic Differentiation

Use when (i) **variable power** $[f(x)]^{g(x)}$, like x^x, $(\sin x)^{\cos x}$; or (ii) a big **product/quotient** of many factors. Take log both sides, then differentiate.

Method for $y=u^v$ (u, v functions of x)

1. $\log y=v\log u$
2. $\frac{1}{y}\frac{dy}{dx}=v\times\frac{u'}{u}+v'\log u$
3. $\frac{dy}{dx}=u^v\left[\frac{v\,u'}{u}+v'\log u\right]$

Solved Example 43

Differentiate $y=x^{\sin x}$, $x>0$.

1. $\log y=\sin x\log x$
2. $\frac{1}{y}\frac{dy}{dx}=\cos x\log x+\frac{\sin x}{x}$

✔ $\frac{dy}{dx}=x^{\sin x}\left[\cos x\log x+\frac{\sin x}{x}\right]$

Solved Example 44

Find $\frac{dy}{dx}$ if $y=(\sin x)^x+x^{\cos x}$.

1. Let $u=(\sin x)^x$, $v=x^{\cos x}$, so $y'=u'+v'$ (NEVER take log of a sum)
2. $\log u=x\log\sin x\Rightarrow u'=(\sin x)^x\left[\log\sin x+x\cot x\right]$
3. $\log v=\cos x\log x\Rightarrow v'=x^{\cos x}\left[\frac{\cos x}{x}-\sin x\log x\right]$

✔ $y'=(\sin x)^x\left[\log\sin x+x\cot x\right]+x^{\cos x}\left[\frac{\cos x}{x}-\sin x\log x\right]$

Solved Example 45

If $x^y = e^{x-y}$, prove that $\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$.

1. Take log: $y \log x = x - y \Rightarrow y(1+\log x) = x$
2. $y = \frac{x}{1+\log x}$
3. Quotient: $y' = \frac{(1+\log x)\times 1 - x \times \frac{1}{x}}{(1+\log x)^2} = \frac{\log x}{(1+\log x)^2}$

✔ Proved

Solved Example 46

Differentiate $y = \frac{(x-1)(x-2)}{(x-3)(x-4)}$ by taking logs ($x > 4$).

1. $\log y = \log(x-1) + \log(x-2) - \log(x-3) - \log(x-4)$
2. $\frac{1}{y}\frac{dy}{dx} = \frac{1}{x-1} + \frac{1}{x-2} - \frac{1}{x-3} - \frac{1}{x-4}$

✔ $\frac{dy}{dx} = y\left[\frac{1}{x-1} + \frac{1}{x-2} - \frac{1}{x-3} - \frac{1}{x-4}\right]$

Yaad karo: "Power mein x? Log lagao, jhatka hatao"

Jab **base aur power dono** mein x ho, seedha formula nahi chalta. log lo, power neeche aa jayegi. Lekin **sum** ho ($u+v$) to pehle tod do, phir har part ka alag log.

Solved Example 47

Find $\frac{dy}{dx}$ if $y = x^x + 2^x$, $x > 0$.

1. $\frac{d}{dx}x^x = x^x(1+\log x)$ (log differentiation)
2. $\frac{d}{dx}2^x = 2^x \log 2$ (formula, no log needed)

✔ $\frac{dy}{dx} = x^x(1+\log x) + 2^x \log 2$

Solved Example 48

If $x^y = y^x$, find $\frac{dy}{dx}$.

1. $y \log x = x \log y$
2. $y_1 \log x + \frac{y}{x} = \log y + \frac{x}{y} y_1$
3. $y_1\left(\log x - \frac{x}{y}\right) = \log y - \frac{y}{x}$

✔ $\frac{dy}{dx} = \frac{y(x\log y - y)}{x(y\log x - x)}$

Solved Example 49

Differentiate $y = \sqrt{\frac{(x-1)(x-2)}{(x-3)(x-4)(x-5)}}$, $x > 5$.

1. $\log y = \frac{1}{2}\left[\log(x-1) + \log(x-2) - \log(x-3) - \log(x-4) - \log(x-5)\right]$
2. Differentiate both sides

✔ $\frac{dy}{dx} = \frac{y}{2}\left[\frac{1}{x-1} + \frac{1}{x-2} - \frac{1}{x-3} - \frac{1}{x-4} - \frac{1}{x-5}\right]$

11. Parametric Differentiation

If $x = f(t)$, $y = g(t)$ (parameter t), then $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$.

For the second derivative: $\frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right) \times \frac{dt}{dx}$.

Trap: $\frac{d^2y}{dx^2} \neq \frac{d^2y/dt^2}{d^2x/dt^2}$

Second derivative mein $\frac{dt}{dx}$ **se multiply** karna zaroori hai. Seedha $y''(t) \div x''(t)$ likha to answer galat.

Solved Example 50

Find $\frac{dy}{dx}$ if $x = a(\theta + \sin\theta)$, $y = a(1-\cos\theta)$.

1. $\frac{dx}{d\theta} = a(1+\cos\theta)$, $\frac{dy}{d\theta} = a\sin\theta$
2. $\frac{dy}{dx} = \frac{\sin\theta}{1+\cos\theta} = \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}}$

✔ $\frac{dy}{dx} = \tan\frac{\theta}{2}$

Solved Example 51

If $x = 3t^2$, $y = 2t^3$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

1. $\frac{dx}{dt} = 6t$, $\frac{dy}{dt} = 6t^2$, so $\frac{dy}{dx} = t$
2. $\frac{d^2y}{dx^2} = \frac{d}{dt}(t) \times \frac{dt}{dx} = 1 \times \frac{1}{6t}$

✔ $\frac{dy}{dx} = t$, $\frac{d^2y}{dx^2} = \frac{1}{6t}$

Quick fact $\frac{d}{dx}\sqrt{\sin x} = \frac{\cos x}{2\sqrt{\sin x}}$

Quick fact Sum, difference, product of continuous functions is continuous.

Solved Example 52

If $x = a\cos^3\theta$, $y = a\sin^3\theta$, find $\frac{dy}{dx}$ at $\theta = \frac{\pi}{4}$.

1. $\frac{dx}{d\theta} = -3a\cos^2\theta\sin\theta$, $\frac{dy}{d\theta} = 3a\sin^2\theta\cos\theta$
2. $\frac{dy}{dx} = -\frac{\sin\theta}{\cos\theta} = -\tan\theta$
3. At $\theta = \frac{\pi}{4}$: -1

✔ $\frac{dy}{dx} = -\tan\theta$; value $= -1$

Solved Example 53

Differentiate $\sin^2 x$ w.r.t. $e^{\cos x}$.

1. Let $u = \sin^2 x$, $v = e^{\cos x}$; find $\frac{du}{dv} = \frac{du/dx}{dv/dx}$
2. $\frac{du}{dx} = 2\sin x\cos x$, $\frac{dv}{dx} = -\sin x\, e^{\cos x}$

✔ $\frac{du}{dv} = \frac{-2\cos x}{e^{\cos x}}$

Solved Example 54

If $x = a\sec\theta$, $y = b\tan\theta$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

1. $\frac{dx}{d\theta} = a\sec\theta\tan\theta$, $\frac{dy}{d\theta} = b\sec^2\theta$
2. $\frac{dy}{dx} = \frac{b\sec\theta}{a\tan\theta} = \frac{b}{a}\csc\theta$
3. $y_2 = -\frac{b}{a}\csc\theta\cot\theta \times \frac{1}{a\sec\theta\tan\theta} = -\frac{b}{a^2}\cot^3\theta$

✔ $\frac{dy}{dx} = \frac{b}{a}\csc\theta$, $\frac{d^2y}{dx^2} = -\frac{b\cot^3\theta}{a^2}$

Solved Example 55

If $x = \sin t$, $y = \cos 2t$, find $\frac{dy}{dx}$.

1. $\frac{dx}{dt} = \cos t$, $\frac{dy}{dt} = -2\sin 2t = -4\sin t\cos t$
2. $\frac{dy}{dx} = \frac{-4\sin t\cos t}{\cos t}$

✔ $\frac{dy}{dx} = -4\sin t = -4x$

Solved Example 56

If $x = \sqrt{a^{\sin^{-1}t}}$, $y = \sqrt{a^{\cos^{-1}t}}$, show that $\frac{dy}{dx} = -\frac{y}{x}$.

1. $xy = \sqrt{a^{\sin^{-1}t + \cos^{-1}t}} = \sqrt{a^{\pi/2}}$, a constant
2. Differentiate $xy = c$: $y + x\,y_1 = 0$

✔ $\frac{dy}{dx} = -\frac{y}{x}$ (Proved)

Yaad karo: "Upar t, neeche t, beech mein x"

Parametric: $\frac{dy}{dx} = \frac{\text{upar ka } t\text{-derivative}}{\text{neeche ka } t\text{-derivative}}$. Doosre derivative ke liye phir se t se differentiate karo aur $\frac{dx}{dt}$ se divide.

12. Second Order Derivatives

$y_2 = \frac{d^2y}{dx^2} = \frac{d}{dx}(y_1)$. Board favourite: "If $y = \ldots$, prove $(\ldots)y_2 + (\ldots)y_1 + (\ldots)y = 0$". Method: find y_1, clear fractions/roots first, then differentiate again and substitute back.

Solved Example 57

If $y = 3\cos(\log x) + 4\sin(\log x)$, show that $x^2y_2 + xy_1 + y = 0$.

1. $y_1 = \frac{-3\sin(\log x) + 4\cos(\log x)}{x}$
2. So $xy_1 = -3\sin(\log x) + 4\cos(\log x)$
3. Differentiate: $xy_2 + y_1 = \frac{-3\cos(\log x) - 4\sin(\log x)}{x} = -\frac{y}{x}$
4. Multiply by x: $x^2y_2 + xy_1 = -y$

✔ $x^2y_2 + xy_1 + y = 0$ (Proved)

Solved Example 58

If $y = e^{m\sin^{-1}x}$, show that $(1-x^2)y_2 - xy_1 - m^2y = 0$.

1. $y_1 = e^{m\sin^{-1}x} \times \frac{m}{\sqrt{1-x^2}} = \frac{my}{\sqrt{1-x^2}}$
2. $\sqrt{1-x^2}\,y_1 = my$; square: $(1-x^2)y_1^2 = m^2y^2$
3. Differentiate: $(1-x^2)2y_1y_2 - 2xy_1^2 = 2m^2yy_1$
4. Divide by $2y_1$

✔ $(1-x^2)y_2 - xy_1 - m^2y = 0$ (Proved)

Solved Example 59

If $y = A\sin 3x + B\cos 3x$, show that $\frac{d^2y}{dx^2} + 9y = 0$.

1. $y_1 = 3A\cos 3x - 3B\sin 3x$
2. $y_2 = -9A\sin 3x - 9B\cos 3x = -9y$

✔ $y_2 + 9y = 0$ (Proved)

Quick fact $\frac{d}{dx}(\sin x\cos x) = \cos 2x$

Solved Example 60

If $y = (\tan^{-1} x)^2$, show that $(1+x^2)^2 y_2 + 2x(1+x^2)y_1 = 2$.

1. $y_1 = \dfrac{2\tan^{-1} x}{1+x^2} \Rightarrow (1+x^2)y_1 = 2\tan^{-1} x$
2. Differentiate: $(1+x^2)y_2 + 2xy_1 = \dfrac{2}{1+x^2}$
3. Multiply by $(1+x^2)$

✔ $(1+x^2)^2 y_2 + 2x(1+x^2)y_1 = 2$ **(Proved)**

Solved Example 61

If $y = e^x \sin x$, show that $y_2 - 2y_1 + 2y = 0$.

1. $y_1 = e^x \sin x + e^x \cos x$
2. $y_2 = e^x \sin x + e^x \cos x + e^x \cos x - e^x \sin x = 2e^x \cos x$
3. $y_2 - 2y_1 + 2y = 2e^x \cos x - 2e^x \sin x - 2e^x \cos x + 2e^x \sin x$

✔ $= 0$ **(Proved)**

Solved Example 62

If $y = \sin^{-1} x$, show that $(1-x^2)y_2 = xy_1$.

1. $y_1 = \dfrac{1}{\sqrt{1-x^2}} \Rightarrow \sqrt{1-x^2}\, y_1 = 1$
2. Differentiate: $\sqrt{1-x^2}\, y_2 - \dfrac{x}{\sqrt{1-x^2}} y_1 = 0$
3. Multiply by $\sqrt{1-x^2}$

✔ $(1-x^2)y_2 = xy_1$ **(Proved)**

Solved Example 63

Find $\dfrac{d^2y}{dx^2}$ if $y = \log(\log x)$.

1. $y_1 = \dfrac{1}{x \log x} = (x \log x)^{-1}$
2. $y_2 = -(x \log x)^{-2}(\log x + 1)$

✔ $\dfrac{d^2y}{dx^2} = -\dfrac{1+\log x}{(x \log x)^2}$



Shivam Bhaiya says

Second derivative proof mein **square karke root hatao** ya cross multiply karo, **phir** dobara differentiate karo. Seedha y_1 ko differentiate karoge to quotient rule ka jungle ban jayega. Ye 3 marks ka pakka sawal hai.

Yaad karo: "Saaf karo, phir maaro"

y_1 nikalte hi **saaf karo** (denominator ko left mein le jao, root ho to square karo), **phir** doosra derivative **maaro**. Last step: original y wapas substitute.

Common mistakes

- Checking only LHL = RHL and forgetting to compare with $f(a)$.
- Saying "continuous $\Rightarrow$ differentiable" ($|x|$ at 0 is the counter-example).
- Forgetting the inner derivative in chain rule: $\dfrac{d}{dx}\sin 3x = 3\cos 3x$, not $\cos 3x$.
- Taking log of a sum $(\sin x)^x + x^{\cos x}$ in one go.
- Parametric $\dfrac{d^2y}{dx^2}$: forgetting to multiply by $\dfrac{dt}{dx}$.
- Writing $\dfrac{d}{dx}(a^x) = xa^{x-1}$ (power rule needs a constant power).

Most Asked in Boards

- Find k (or a, b) so that a piecewise function is continuous (3M/5M)
- Check continuity/differentiability of $|x-a|$ type functions (2M/3M)
- Logarithmic differentiation: u^v, $u+v$ forms, $x^y = e^{x-y}$ types (3M/5M)
- Inverse trig derivative by substitution $x = \tan\theta$ etc. (2M/3M)
- Parametric $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$ (3M)
- Prove a second order relation like $(1-x^2)y_2 - xy_1 = \ldots$ (3M/5M)
- Implicit differentiation, including infinite root forms (2M/3M)



Meme samjho

Student: $\log(u+v) = \log u + \log v$ likh diya. Examiner: **kya tapleek hai apko?** **Asli baat:** $\log(mn) = \log m + \log n$, but $\log(u+v)$ does NOT split. For $y = u+v$, differentiate u and v separately, each by logs.

13. Extra Practice: Board Mix

Solved Example 64

Find k if $f(x) = \begin{cases} \dfrac{e^{3x}-1}{4x}, & x \neq 0 \\ k, & x = 0 \end{cases}$ is continuous at 0.

1. $\lim_{x\to 0} \dfrac{e^{3x}-1}{4x} = \dfrac{3}{4}\lim_{x\to 0}\dfrac{e^{3x}-1}{3x} = \dfrac{3}{4}$

✔ $k = \dfrac{3}{4}$

Solved Example 65

Differentiate $y = e^{\sin^{-1} x} + \log(\tan^{-1} x)$.

1. $\frac{d}{dx} e^{\sin^{-1} x} = \frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}}$
2. $\frac{d}{dx} \log(\tan^{-1} x) = \frac{1}{\tan^{-1} x} \times \frac{1}{1+x^2}$

✔ $\frac{dy}{dx} = \frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}} + \frac{1}{(1+x^2)\tan^{-1} x}$

Solved Example 66

If $y = \sqrt{\sin x + y}$, find $\frac{dy}{dx}$.

1. $y^2 = \sin x + y$
2. $2y\, y_1 = \cos x + y_1$

✔ $\frac{dy}{dx} = \frac{\cos x}{2y - 1}$

Solved Example 67

If $y = \tan^{-1} x$, find $\frac{d^2y}{dx^2}$ in terms of y alone.

1. $y_1 = \frac{1}{1+x^2} = \frac{1}{1+\tan^2 y} = \cos^2 y$
2. $y_2 = \frac{d}{dx} \cos^2 y = -2 \cos y \sin y\, y_1 = -\sin 2y \cos^2 y$

✔ $\frac{d^2y}{dx^2} = -\sin 2y \cos^2 y$

Solved Example 68

Differentiate $(\log x)^{\cos x}$ w.r.t. x, $x > 1$.

1. $\log y = \cos x \log(\log x)$
2. $\frac{y_1}{y} = -\sin x \log(\log x) + \frac{\cos x}{x \log x}$

✔ $\frac{dy}{dx} = (\log x)^{\cos x} \left[\frac{\cos x}{x \log x} - \sin x \log(\log x)\right]$

Solved Example 69

If $x = \cos t + \log \tan \frac{t}{2}$, $y = \sin t$, find $\frac{dy}{dx}$ at $t = \frac{\pi}{3}$.

1. $\frac{dx}{dt} = -\sin t + \frac{\sec^2 \frac{t}{2}}{2 \tan \frac{t}{2}} = -\sin t + \frac{1}{\sin t} = \frac{\cos^2 t}{\sin t}$
2. $\frac{dy}{dt} = \cos t$, so $\frac{dy}{dx} = \frac{\cos t \sin t}{\cos^2 t} = \tan t$
3. At $t = \frac{\pi}{3}$: $\sqrt{3}$

✔ $\frac{dy}{dx} = \tan t = \sqrt{3}$

Daily habit

Har din **5 derivatives** bina dekhe likho: $\sin^{-1}$, $\tan^{-1}$, $\sec x$, a^x, $\log_a x$. Formula pakka to chapter ka aadha kaam ho gaya. Exam mein galti formula se nahi, **chain rule bhoolne** se hoti hai.

Solved Example 70

If $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$, show that $(1-x^2)y_1 - xy = 1$.

1. $\sqrt{1-x^2}\, y = \sin^{-1} x$
2. Differentiate: $\sqrt{1-x^2}\, y_1 - \frac{x}{\sqrt{1-x^2}} y = \frac{1}{\sqrt{1-x^2}}$
3. Multiply by $\sqrt{1-x^2}$

✔ $(1-x^2)y_1 - xy = 1$ **(Proved)**

Solved Example 71

Find $\frac{dy}{dx}$ if $y = (x \cos x)^x + (x \sin x)^{1/x}$.

1. $u = (x \cos x)^x$: $\log u = x \log x + x \log \cos x$, so $u_1 = u\,[\log x + 1 + \log \cos x - x \tan x]$
2. $v = (x \sin x)^{1/x}$: $\log v = \frac{\log x + \log \sin x}{x}$, so $v_1 = v \times \frac{1 + x \cot x - \log(x \sin x)}{x^2}$

✔ $y_1 = u_1 + v_1$ **with** u_1, v_1 **above**

Solved Example 72

If $x = 2\cos\theta - \cos 2\theta$, $y = 2\sin\theta - \sin 2\theta$, find $\frac{dy}{dx}$ at $\theta = \frac{\pi}{2}$.

1. $\frac{dx}{d\theta} = -2\sin\theta + 2\sin 2\theta$, $\frac{dy}{d\theta} = 2\cos\theta - 2\cos 2\theta$
2. At $\frac{\pi}{2}$: $\frac{dx}{d\theta} = -2$, $\frac{dy}{d\theta} = 0 - 2(-1) = 2$

✔ $\frac{dy}{dx} = \frac{2}{-2} = -1$

Solved Example 73

Show that $f(x) = \begin{cases} 3x - 2, & 0 < x \le 1 \\ 2x^2 - x, & 1 < x \le 2 \\ 5x - 4, & x > 2 \end{cases}$ is continuous at $x = 1$ and $x = 2$.

1. At 1: LHL $= f(1) = 1$, RHL $= 2 - 1 = 1$
2. At 2: LHL $= f(2) = 8 - 2 = 6$, RHL $= 10 - 4 = 6$

✔ **Continuous at both** $x = 1$ **and** $x = 2$

Solved Example 74

Find $\frac{dy}{dx}$ if $y = \cos^{-1}\left(\frac{2x}{1+x^2}\right)$, $-1 < x < 1$.

1. Put $x = \tan\theta$: $\cos^{-1}(\sin 2\theta) = \frac{\pi}{2} - \sin^{-1}(\sin 2\theta)$
2. $= \frac{\pi}{2} - 2\theta = \frac{\pi}{2} - 2\tan^{-1} x$ (as $|2\theta| < \frac{\pi}{2}$)

✔ $\frac{dy}{dx} = \frac{-2}{1+x^2}$

Solved Example 75

Differentiate $y = \frac{1}{2}\log\left(\frac{1+\sin x}{1-\sin x}\right)$.

1. $y = \frac{1}{2}[\log(1+\sin x) - \log(1-\sin x)]$
2. $y_1 = \frac{1}{2}\left[\frac{\cos x}{1+\sin x} + \frac{\cos x}{1-\sin x}\right] = \frac{1}{2} \times \frac{2\cos x}{\cos^2 x}$

✔ $\frac{dy}{dx} = \sec x$

Solved Example 76

Find $\frac{d^2y}{dx^2}$ if $y = e^{-x}\cos x$, and show $y_2 + 2y_1 + 2y = 0$.

1. $y_1 = -e^{-x}\cos x - e^{-x}\sin x = -e^{-x}(\cos x + \sin x)$
2. $y_2 = e^{-x}(\cos x + \sin x) - e^{-x}(\cos x - \sin x) = 2e^{-x}\sin x$
3. $y_2 + 2y_1 + 2y = 2e^{-x}\sin x - 2e^{-x}\cos x - 2e^{-x}\sin x + 2e^{-x}\cos x = 0$

✔ $y_2 = 2e^{-x}\sin x$; **relation proved**

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. The greatest integer function $f(x) = [x]$ is discontinuous at 1M

(a) every integer (b) only $x = 0$ (c) no point (d) every non-integer

Ans. (a) It jumps at every integer.

Q2. If $f(x) = \begin{cases} kx^2, & x \le 2 \\ 3, & x > 2 \end{cases}$ is continuous at $x = 2$, then $k =$ 1M Board fav

(a) $\frac{3}{4}$ (b) $\frac{4}{3}$ (c) $\frac{3}{2}$ (d) 1

Ans. (a) $4k = 3 \Rightarrow k = \frac{3}{4}$.

Q3. $\frac{d}{dx}\left(e^{3\log x}\right) =$ 1M

(a) $3x^2$ (b) e^{3x} (c) $\frac{3}{x}$ (d) x^3

Ans. (a) $e^{3\log x} = x^3$, derivative $3x^2$.

Q4. If $y = \log(\cos e^x)$, then $\frac{dy}{dx} =$ 1M

(a) $-e^x\tan(e^x)$ (b) $e^x\tan(e^x)$ (c) $-\tan(e^x)$ (d) $e^x\cot(e^x)$

Ans. (a) $-\frac{\sin e^x}{\cos e^x} \times e^x$

Q5. The number of points where $f(x) = |x| + |x-1|$ is NOT differentiable is 1M Most asked

(a) 0 (b) 1 (c) 2 (d) 3

Ans. (c) 2, at the corners $x = 0$ and $x = 1$.

Q6. If $x = t^2$, $y = t^3$, then $\frac{d^2y}{dx^2} =$ 1M

(a) $\frac{3}{2}$ (b) $\frac{3}{4t}$ (c) $\frac{3}{2t}$ (d) $\frac{3t}{2}$

Ans. (b) $\frac{dy}{dx} = \frac{3t}{2}$, so $y_2 = \frac{3}{2} \times \frac{1}{2t} = \frac{3}{4t}$.

Q7. $\frac{d}{dx}(x^x) =$ 1M

(a) $x^x(1+\log x)$ (b) $x\,x^{x-1}$ (c) $x^x\log x$ (d) x^x

Ans. (a) $\log y = x\log x \Rightarrow y_1 = x^x(1+\log x)$.

Q8. A: $f(x) = |x|$ is continuous at $x = 0$. R: Every differentiable function is continuous. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (b) Both are true, but $|x|$ is not differentiable at 0, so R does not explain A.

Q9. A: If $y = \log x^2$ $(x > 0)$, then $\frac{dy}{dx} = \frac{2}{x}$. R: $\log m^n = n\log m$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) $y = 2\log x$, $y_1 = \frac{2}{x}$; R is exactly the reason.

Q10. If $y = \sin^{-1} x$, then $(1-x^2)y_2 - xy_1 =$ 1M

(a) 0 (b) 1 (c) -1 (d) y

Ans. (a) $\sqrt{1-x^2}\,y_1 = 1$; differentiate: $\sqrt{1-x^2}\,y_2 - \frac{x}{\sqrt{1-x^2}}y_1 = 0$, multiply by $\sqrt{1-x^2}$: 0.

Q11. If $f(x) = \begin{cases} \frac{\sin 3x}{x}, & x \ne 0 \\ k, & x = 0 \end{cases}$ is continuous at 0, then $k =$ 1M Board fav

(a) 0 (b) 1 (c) 3 (d) $\frac{1}{3}$

Ans. (c) $\lim_{x\to 0}\frac{\sin 3x}{x} = 3$.

Q12. $\frac{d}{dx}\left(\tan^{-1}\frac{2x}{1-x^2}\right)$ for $|x| < 1$ is 1M

(a) $\frac{1}{1+x^2}$ (b) $\frac{2}{1+x^2}$ (c) $\frac{-2}{1+x^2}$ (d) $\frac{2}{1-x^2}$

Ans. (b) It equals $2\tan^{-1} x$, derivative $\frac{2}{1+x^2}$.

Q13. The derivative of $\cos^{-1} x$ w.r.t. $\sqrt{1-x^2}$ $(0<x<1)$ is 1M HOTS

(a) $\frac{1}{x}$ (b) $-\frac{1}{x}$ (c) x (d) $\sqrt{1-x^2}$

Ans. (a) $\dfrac{-1/\sqrt{1-x^2}}{-x/\sqrt{1-x^2}} = \dfrac{1}{x}$.

Q14. A: $f(x)=[x]$ is not differentiable at $x=2$. R: $[x]$ is discontinuous at $x=2$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) Both true; discontinuity at 2 forces non-differentiability, so R explains A.

Q15. If $y=\log\tan\left(\dfrac{\pi}{4}+\dfrac{x}{2}\right)$, then $\dfrac{dy}{dx}=$ 1M HOTS

(a) $\csc x$ (b) $\sec x$ (c) $\tan x$ (d) $\cos x$

Ans. (b) $\dfrac{\frac{1}{2}\sec^2\left(\frac{\pi}{4}+\frac{x}{2}\right)}{\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)} = \dfrac{1}{\sin\left(\frac{\pi}{2}+x\right)} = \sec x$.

Q16. The function $f(x)=\dfrac{x^2-1}{x^2-5x+6}$ is discontinuous at 1M

(a) only 2 (b) only 3 (c) 1 and -1 (d) 2 and 3

Ans. (d) Denominator $(x-2)(x-3)=0$ at $x=2,3$.

Very short (2 marks)

Q17. Find $\dfrac{dy}{dx}$ if $y=\sin^{-1}(x\sqrt{x})$, $0\le x<1$. 2M

Ans. $y_1 = \dfrac{1}{\sqrt{1-x^3}}\times\dfrac{3}{2}\sqrt{x} = \dfrac{3\sqrt{x}}{2\sqrt{1-x^3}}$.

Q18. Show that $f(x)=|x-5|$ is not differentiable at $x=5$. 2M Board fav

Ans. RHD $=\lim_{h\to0}\dfrac{|h|}{h}=1$, LHD $=\lim_{h\to0}\dfrac{|-h|}{-h}=-1$. LHD $\neq$ RHD, so not differentiable at 5.

Q19. Find $\dfrac{dy}{dx}$ if $2x+3y=\sin y$. 2M

Ans. $2+3y_1=\cos y\, y_1 \Rightarrow y_1=\dfrac{2}{\cos y-3}$.

Q20. Find $\dfrac{d^2y}{dx^2}$ if $y=x^3\log x$. 2M

Ans. $y_1=3x^2\log x+x^2$, $y_2=6x\log x+3x+2x=6x\log x+5x$.

Q21. Differentiate $\cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right)$, $0<x<1$. 2M Most asked

Ans. Put $x=\tan\theta$: $\cos^{-1}(\cos 2\theta)=2\theta=2\tan^{-1}x$.

Derivative $=\dfrac{2}{1+x^2}$.

Q22. Find k if $f(x)=\begin{cases}\dfrac{k\cos x}{\pi-2x}, & x\neq\frac{\pi}{2}\\ 5, & x=\frac{\pi}{2}\end{cases}$ is continuous at $x=\dfrac{\pi}{2}$. 2M Board fav

Ans. Put $x=\frac{\pi}{2}-h$: $\dfrac{k\sin h}{2h}\to\dfrac{k}{2}$. So $\dfrac{k}{2}=5$, $k=10$.

Q23. If $y=\log\left(x+\sqrt{x^2+a^2}\right)$, show that $\dfrac{dy}{dx}=\dfrac{1}{\sqrt{x^2+a^2}}$. 2M

Ans. $y_1=\dfrac{1}{x+\sqrt{x^2+a^2}}\left(1+\dfrac{x}{\sqrt{x^2+a^2}}\right)=\dfrac{1}{x+\sqrt{x^2+a^2}}\times\dfrac{\sqrt{x^2+a^2}+x}{\sqrt{x^2+a^2}}=\dfrac{1}{\sqrt{x^2+a^2}}$.

Q24. Find $\dfrac{dy}{dx}$ if $y=x^2e^x\sin x$. 2M

Ans. Product of three: $y_1=e^x\left[2x\sin x+x^2\sin x+x^2\cos x\right]$.

Q25. If $x=at^2$, $y=2at$, find $\dfrac{d^2y}{dx^2}$. 2M Must do

Ans. $\dfrac{dy}{dx}=\dfrac{2a}{2at}=\dfrac{1}{t}$; $y_2=-\dfrac{1}{t^2}\times\dfrac{1}{2at}=-\dfrac{1}{2at^3}$.

Q26. Find $\dfrac{dy}{dx}$ if $y=\cos^{-1}(\sin x)$, $0<x<\pi/2$. 2M

Ans. $\sin x=\cos\left(\dfrac{\pi}{2}-x\right)$ and $\dfrac{\pi}{2}-x\in\left(0,\dfrac{\pi}{2}\right)$, so $y=\dfrac{\pi}{2}-x$, $y_1=-1$.

Q27. Find $\dfrac{dy}{dx}$ if $xy=e^{x-y}$. 2M Board fav

Ans. $\log x+\log y=x-y\Rightarrow\dfrac{1}{x}+\dfrac{y_1}{y}=1-y_1\Rightarrow y_1=\dfrac{y(x-1)}{x(y+1)}$.

Q28. If $y=5x^3-\dfrac{2}{x}$, find $\dfrac{d^2y}{dx^2}$. 2M

Ans. $y_1=15x^2+\dfrac{2}{x^2}$, $y_2=30x-\dfrac{4}{x^3}$.

Q29. Find $\dfrac{dy}{dx}$ if $y=\sin^{-1}x+\sin^{-1}\sqrt{1-x^2}$, $0<x<1$. 2M HOTS

Ans. Put $x=\sin\theta$: $\sqrt{1-x^2}=\cos\theta=\sin\left(\frac{\pi}{2}-\theta\right)$, so $y=\theta+\frac{\pi}{2}-\theta=\frac{\pi}{2}$, hence $y_1=0$.

Q30. Check the differentiability of $f(x)=\begin{cases}1+x, & x\le 2\\ 5-x, & x>2\end{cases}$ at $x=2$. 2M Most asked

Ans. Continuity: $3=3=f(2)$. LHD $=1$, RHD $=-1$. Not equal, so continuous but not differentiable at $x=2$.

Q31. If $y=x^3\log\left(\dfrac{1}{x}\right)$, find $\dfrac{d^2y}{dx^2}$ at $x=1$. 2M

Ans. $y=-x^3\log x$, $y_1=-3x^2\log x-x^2$, $y_2=-6x\log x-3x-2x=-6x\log x-5x$. At $x=1$: -5.

Short answer (3 marks)

Q32. Find $\dfrac{dy}{dx}$ if $y=(\log x)^x+x^{\log x}$, $x>1$. 3M Most asked

Ans. $u=(\log x)^x$: $\log u=x\log(\log x)\Rightarrow u_1=(\log x)^x\left[\log(\log x)+\dfrac{1}{\log x}\right]$. $v=x^{\log x}$:

(contd.) $\log v = (\log x)^2 \Rightarrow v_1 = x^{\log x} \times \frac{2\log x}{x}$. $y_1 = u_1 + v_1$.

Q33. If $x = a(\cos t + t\sin t)$, $y = a(\sin t - t\cos t)$, find $\frac{d^2y}{dx^2}$. 3M Board fav

Ans. $\frac{dx}{dt} = at\cos t$, $\frac{dy}{dt} = at\sin t$, so $\frac{dy}{dx} = \tan t$. $y_2 = \sec^2 t \times \frac{1}{at\cos t} = \frac{\sec^3 t}{at}$.

Q34. If $y = (\sin^{-1} x)^2$, show that $(1-x^2)y_2 - xy_1 = 2$. 3M Must do

Ans. $y_1 = \frac{2\sin^{-1} x}{\sqrt{1-x^2}} \Rightarrow (1-x^2)y_1^2 = 4y$. Differentiate: $(1-x^2)2y_1y_2 - 2xy_1^2 = 4y_1$. Divide by $2y_1$: $(1-x^2)y_2 - xy_1 = 2$.

Q35. Find a, b so that $f(x) = \begin{cases} x^2 + ax + b, & 0 \le x < 2 \\ 3x + 2, & 2 \le x \le 4 \\ 2ax + 5b, & 4 < x \le 8 \end{cases}$ is continuous on $[0, 8]$. 3M Board fav

Ans. At $x = 2$: $4 + 2a + b = 8 \Rightarrow 2a + b = 4$. At $x = 4$: $14 = 8a + 5b$. Then $14 = 8a + 5(4 - 2a) = 20 - 2a \Rightarrow a = 3, b = -2$.

Q36. Differentiate $\tan^{-1}\left(\frac{\sqrt{1+x^2} - 1}{x}\right)$, $x \neq 0$. 3M

Ans. Put $x = \tan\theta$: $\frac{\sec\theta - 1}{\tan\theta} = \frac{1 - \cos\theta}{\sin\theta} = \tan\frac{\theta}{2}$. So $y = \frac{\theta}{2} = \frac{1}{2}\tan^{-1} x$, $y_1 = \frac{1}{2(1+x^2)}$.

Q37. Find $\frac{dy}{dx}$ if $x^3 + y^3 = 3xy$. 3M

Ans. $3x^2 + 3y^2y_1 = 3(y + xy_1) \Rightarrow y_1(y^2 - x) = y - x^2 \Rightarrow y_1 = \frac{y - x^2}{y^2 - x}$.

Q38. If $e^y(x+1) = 1$, show that $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$. 3M

Ans. $e^y = \frac{1}{x+1} \Rightarrow y = -\log(x+1)$, $y_1 = -\frac{1}{x+1}$, $y_2 = \frac{1}{(x+1)^2} = (y_1)^2$.

Q39. If $y = 3e^{2x} + 2e^{3x}$, prove that $y_2 - 5y_1 + 6y = 0$. 3M

Ans. $y_1 = 6e^{2x} + 6e^{3x}$, $y_2 = 12e^{2x} + 18e^{3x}$. $y_2 - 5y_1 + 6y = (12 - 30 + 18)e^{2x} + (18 - 30 + 12)e^{3x} = 0$.

Q40. Show that $f(x) = \begin{cases} x^2, & x \le 1 \\ 2x - 1, & x > 1 \end{cases}$ is differentiable at $x = 1$. 3M

Ans. Continuity: $1 = 1 = f(1)$. LHD $= 2x|_1 = 2$, RHD $= 2$. Equal, so differentiable and $f'(1) = 2$.

Q41. If $y = \sin(\log x)$, prove that $x^2y_2 + xy_1 + y = 0$. 3M

Ans. $xy_1 = \cos(\log x)$. Differentiate: $xy_2 + y_1 = -\frac{\sin(\log x)}{x} = -\frac{y}{x}$. Multiply by x: $x^2y_2 + xy_1 + y = 0$.

Q42. Find the value of k for which $f(x) = \begin{cases} \frac{1 - \cos kx}{x\sin x}, & x \neq 0 \\ \frac{1}{2}, & x = 0 \end{cases}$ is continuous at $x = 0$. 3M Board fav

Ans. $\frac{2\sin^2\frac{kx}{2}}{x\sin x} = 2\left(\frac{\sin\frac{kx}{2}}{\frac{kx}{2}}\right)^2 \frac{k^2}{4} \times \frac{x}{\sin x} \to \frac{k^2}{2}$. So $\frac{k^2}{2} = \frac{1}{2}$, $k = \pm 1$.

Q43. If $x = a\left(\cos t + \log\tan\frac{t}{2}\right)$, $y = a\sin t$, find $\frac{d^2y}{dx^2}$. 3M Board fav

Ans. As in the solved example, $\frac{dx}{dt} = \frac{a\cos^2 t}{\sin t}$, $\frac{dy}{dt} = a\cos t$, so $\frac{dy}{dx} = \tan t$. $y_2 = \sec^2 t \times \frac{\sin t}{a\cos^2 t} = \frac{\sec^4 t\sin t}{a}$.

Q44. Find $\frac{dy}{dx}$ if $y^x = x^y$. 3M

Ans. $x\log y = y\log x \Rightarrow \log y + \frac{x}{y}y_1 = y_1\log x + \frac{y}{x} \Rightarrow y_1 = \frac{y(y - x\log y)}{x(x - y\log x)}$.

Q45. Show that $f(x) = e^x\sin x$ is differentiable everywhere and find the points in $[0, \pi]$ where $f'(x) = 0$. 3M

Ans. Product of two functions differentiable on $\mathbb{R}$. $f'(x) = e^x(\sin x + \cos x) = 0 \Rightarrow \tan x = -1 \Rightarrow x = \frac{3\pi}{4}$.

Long answer (5 marks)

Q46. Find $\frac{dy}{dx}$ if $(\cos x)^y = (\cos y)^x$. 5M Most asked

Ans. $y\log\cos x = x\log\cos y$. Differentiate: $y_1\log\cos x - y\tan x = \log\cos y - x\tan y\, y_1$. So $y_1 = \frac{\log\cos y + y\tan x}{\log\cos x + x\tan y}$.

Q47. Show that $f(x) = |x-1| + |x-2|$ is continuous everywhere but not differentiable at $x = 1$ and $x = 2$. 5M

Ans. $f(x) = 3 - 2x$ $(x < 1)$, 1 $(1 \le x \le 2)$, $2x - 3$ $(x > 2)$. Each piece is a polynomial; at 1: LHL $=$ RHL $= f(1) = 1$; at 2: all $= 1$, so continuous everywhere. At 1: LHD $= -2$, RHD $= 0$. At 2: LHD $= 0$, RHD $= 2$. So not differentiable at $x = 1, 2$.

Q48. If $y = \left(x + \sqrt{1+x^2}\right)^n$, show that $(1+x^2)y_2 + xy_1 = n^2y$. 5M HOTS

(contd.) $y_1 = n\left(x+\sqrt{1+x^2}\right)^{n-1}\left(1+\dfrac{x}{\sqrt{1+x^2}}\right) = \dfrac{ny}{\sqrt{1+x^2}}$. So $(1+x^2)y_1^2 = n^2y^2$. Differentiate: $2xy_1^2 + (1+x^2)2y_1y_2 = 2n^2yy_1$; divide by $2y_1$: $(1+x^2)y_2 + xy_1 = n^2y$.

Q49. Find $\dfrac{dy}{dx}$ if $y = x^{\sin x} + (\sin x)^{\cos x}$, $0 < x < \pi$. 5M Board fav

Ans. $u = x^{\sin x}$: $u_1 = x^{\sin x}\left[\cos x \log x + \dfrac{\sin x}{x}\right]$. $v = (\sin x)^{\cos x}$: $\log v = \cos x \log \sin x \Rightarrow v_1 = (\sin x)^{\cos x}\left[\dfrac{\cos^2 x}{\sin x} - \sin x \log \sin x\right]$. $y_1 = u_1 + v_1$.

Q50. If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, prove that $\dfrac{dy}{dx} = \sqrt{\dfrac{1-y^2}{1-x^2}}$. 5M HOTS

Ans. Put $x = \sin A$, $y = \sin B$: $\cos A + \cos B = a(\sin A - \sin B)$, i.e. $2\cos\frac{A+B}{2}\cos\frac{A-B}{2} = 2a\cos\frac{A+B}{2}\sin\frac{A-B}{2}$, so $\cot\frac{A-B}{2} = a$, $A - B = 2\cot^{-1}a$. Thus $\sin^{-1}x - \sin^{-1}y = $ const; differentiate: $\dfrac{1}{\sqrt{1-x^2}} - \dfrac{y_1}{\sqrt{1-y^2}} = 0$, giving the result.

Case based (4 marks)

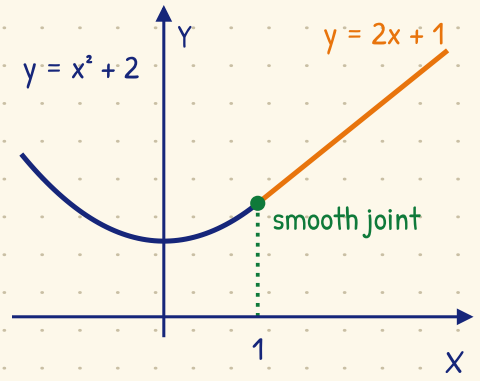


Fig: Q case 1, the road with $a = 2, b = 1$

Q51. Smooth road joint. An engineer joins two road pieces along $f(x) = \begin{cases} x^2+2, & x \le 1 \\ ax+b, & x > 1 \end{cases}$. (i) Condition for no break at $x = 1$. (1) (ii) For no sharp turn (differentiable at 1), find a. (1) (iii) Find b and $f'(x)$ for $x > 1$; is f' continuous at 1? (2) 4M

Ans. (i) $1 + 2 = a + b$, i.e. $a + b = 3$. (ii) LHD $= 2x|_{x=1} = 2$, RHD $= a$, so $a = 2$. (iii) $b = 1$; $f'(x) = 2$ for $x > 1$; $f'(1^-) = 2 = f'(1^+)$, so yes.

Q52. Drone path. A drone moves so that at time t its position is $x = t^2 + 1$, $y = 2t^3 - 1$. (i) Find $\dfrac{dx}{dt}$ at $t = 1$. (1) (ii) Find $\dfrac{dy}{dx}$ in terms of t. (1) (iii) Find $\dfrac{d^2y}{dx^2}$ at $t = 1$. (2) 4M

Ans. (i) $2t = 2$. (ii) $\dfrac{6t^2}{2t} = 3t$. (iii)

(contd.) $y_2 = \dfrac{d}{dt}(3t) \times \dfrac{1}{2t} = \dfrac{3}{2t}$; at $t = 1$: $\dfrac{3}{2}$.

Q53. Cooling tea. The temperature of tea (in °C) t minutes after pouring is $T = 25 + 70e^{-t/10}$. (i) Is T continuous for $t \ge 0$? Why? (1) (ii) Find $\dfrac{dT}{dt}$. (1) (iii) Find $\dfrac{d^2T}{dt^2}$ and the rate of cooling at $t = 10$ min. (2) 4M

Ans. (i) Yes: sum of a constant and an exponential, both continuous. (ii) $\dfrac{dT}{dt} = -7e^{-t/10}$. (iii) $\dfrac{d^2T}{dt^2} = \dfrac{7}{10}e^{-t/10}$; at $t = 10$: $\dfrac{dT}{dt} = -\dfrac{7}{e} \approx -2.58$, so cooling at about 2.58° C per min.

Ek nazar mein: Continuity & Differentiability

Type of sawal	Method	Marks
Find k, a, b	LHL = RHL = $f(a)$	2-5
Differentiable?	continuity, then LHD = RHD	2-3
Inverse trig	$x = \tan\theta$ etc.	2-3
u^v, $u+v$	log differentiation	3-5
Parametric	$\dfrac{dy/dt}{dx/dt}$	2-3
y_2 relation	clear, differentiate again	3-5

Last minute: 8 one-liners

1. Continuous at a: LHL = RHL = $f(a)$.
2. Differentiable $\Rightarrow$ continuous, not conversely.
3. $|x-a|$: continuous, not differentiable at a.
4. $[x]$: discontinuous at every integer.
5. Chain rule: outside to inside, multiply.
6. Variable power: take log first.
7. Parametric: $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$; for y_2 multiply by $\dfrac{dt}{dx}$.
8. y_2 proofs: clear roots and fractions, then differentiate.

Apne notes / galtiyan yahan likho

Most asked: max/min word problems (5M), intervals of increasing/decreasing

APPLICATION OF DERIVATIVES

Is chapter mein kya hai (2026-27)

→ **In syllabus:** rate of change, increasing/decreasing functions, maxima and minima (first and second derivative tests, absolute max/min on a closed interval, word problems).

→ **Deleted:** tangents and normals, approximations. Inko padhne ki zaroorat nahi.

1. Rate of Change of Quantities

If $y = f(x)$, then $\frac{dy}{dx}$ is the rate of change of y with respect to x. Its value at $x = x_0$ is $\left(\frac{dy}{dx}\right)_{x=x_0}$.

→ If both x and y change with time t, use the **chain rule**:

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$

→ Rate **positive** $\Rightarrow$ quantity is **increasing**; rate **negative** $\Rightarrow$ **decreasing**.

→ Units: (unit of y) per (unit of x), e.g. cm^2/s, cm^3/s, m/s.

→ **Marginal cost** $MC = \frac{dC}{dx}$ and **marginal revenue** $MR = \frac{dR}{dx}$, where x = number of items.

Formula Box 1: shapes for rate questions

→ Circle: $A = \pi r^2$, $C = 2\pi r$

→ Sphere: $V = \frac{4}{3}\pi r^3$, $S = 4\pi r^2$

→ Cube (edge x): $V = x^3$, $S = 6x^2$

→ Cylinder: $V = \pi r^2 h$, CSA $= 2\pi rh$, TSA $= 2\pi r(r + h)$

→ Cone: $V = \frac{1}{3}\pi r^2 h$, CSA $= \pi rl$, $l^2 = r^2 + h^2$

→ Rectangle: $A = xy$, $P = 2(x + y)$; square: $A = x^2$

→ Related rates: $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$

Yaad karo: "Pehle differentiate, phir value daalo"

Given value (like $r = 6$) ko **differentiate karne ke baad** hi put karo. Pehle daal diya toh constant ka derivative 0 aa jayega aur answer zero ho jayega.

Quick fact $f'(x) > 0$ on an interval means the graph rises from left to right there.

Solved Example 1

Find the rate of change of the area of a circle with respect to its radius r when $r = 6$ cm.

1. $A = \pi r^2 \Rightarrow \frac{dA}{dr} = 2\pi r$
2. At $r = 6$: $\frac{dA}{dr} = 2\pi(6) = 12\pi$

✔ 12π cm^2/cm

Solved Example 2

The radius of a spherical balloon increases at 0.4 cm/s. Find the rate at which its volume increases when $r = 10$ cm.

1. $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$
2. $= 4\pi(100)(0.4) = 160\pi$

✔ 160π cm^3/s

Solved Example 3

The volume of a cube increases at 9 cm^3/s. How fast is its surface area increasing when the edge is 10 cm?

1. $V = x^3 \Rightarrow \frac{dV}{dt} = 3x^2 \frac{dx}{dt} \Rightarrow 9 = 300 \frac{dx}{dt} \Rightarrow \frac{dx}{dt} = \frac{3}{100}$
2. $S = 6x^2 \Rightarrow \frac{dS}{dt} = 12x\frac{dx}{dt} = 120 \times \frac{3}{100} = 3.6$

✔ 3.6 cm^2/s

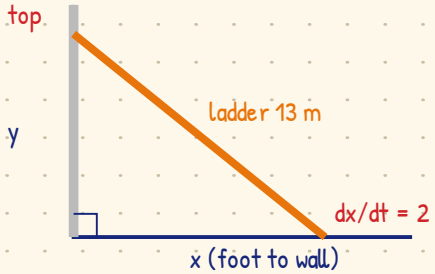


Fig: $x^2 + y^2 = 13^2$ at every instant

Solved Example 4

A 13 m ladder leans against a wall. Its foot is pulled away from the wall at 2 m/s. How fast is the top sliding down when the foot is 5 m from the wall?

Quick fact Sphere: $\frac{dV}{dr} = 4\pi r^2$ = its surface area.

1. $x^2 + y^2 = 169$; at $x = 5$, $y = 12$
2. Differentiate w.r.t. t: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$
3. $\frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt} = -\frac{5}{12} \times 2 = -\frac{5}{6}$
4. Negative sign: height is decreasing

✔ Top slides down at $\frac{5}{6}$ m/s

Solved Example 5

Sand pours at 18 cm^3/s and forms a cone whose height is always one-third of its base radius. How fast is the height increasing when $h = 3$ cm?

1. $h = \frac{r}{3} \Rightarrow r = 3h$, so $V = \frac{1}{3}\pi(3h)^2 h = 3\pi h^3$
2. $\frac{dV}{dt} = 9\pi h^2 \frac{dh}{dt} \Rightarrow 18 = 9\pi(9)\frac{dh}{dt}$
3. $\frac{dh}{dt} = \frac{18}{81\pi} = \frac{2}{9\pi}$

✔ $\frac{2}{9\pi}$ cm/s

Solved Example 6

(i) Total revenue $R(x) = 3x^2 + 36x + 5$. Find MR at $x = 5$. (ii) Total cost $C(x) = 0.01x^3 - 0.3x^2 + 15x + 400$. Find MC at $x = 10$.

1. (i) $MR = \frac{dR}{dx} = 6x + 36$; at $x = 5$: $30 + 36 = 66$
2. (ii) $MC = 0.03x^2 - 0.6x + 15$; at $x = 10$: $3 - 6 + 15 = 12$

✔ MR = Rs 66, MC = Rs 12

Solved Example 7

The length x of a rectangle decreases at 3 cm/min and its width y increases at 2 cm/min. Find the rates of change of its perimeter and area when $x = 10$ cm, $y = 6$ cm.

1. $P = 2(x + y) \Rightarrow \frac{dP}{dt} = 2\left(\frac{dx}{dt} + \frac{dy}{dt}\right) = 2(-3 + 2) = -2$
2. $A = xy \Rightarrow \frac{dA}{dt} = x\frac{dy}{dt} + y\frac{dx}{dt} = 10(2) + 6(-3) = 2$

✔ Perimeter decreases at 2 cm/min; area increases at 2 cm^2/min

Quick fact Circle: $\frac{dA}{dr} = 2\pi r$ = its circumference.

Quick fact Max of $a\sin x + b\cos x$ is $\sqrt{a^2 + b^2}$.

Solved Example 8

Water is poured at 8π cm^3/s into an inverted cone whose base radius always equals its depth ($r = h$). How fast is the water level rising when $h = 4$ cm?

1. $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi h^3$
2. $\frac{dV}{dt} = \pi h^2 \frac{dh}{dt} \Rightarrow 8\pi = 16\pi\frac{dh}{dt}$

✔ Level rises at $\frac{1}{2}$ cm/s

Solved Example 9

A particle moves along the curve $y = \frac{x^3}{3} + 2$. Find the points where its y-coordinate changes 4 times as fast as its x-coordinate.

1. $\frac{dy}{dt} = x^2\frac{dx}{dt}$; we need $\frac{dy}{dt} = 4\frac{dx}{dt}$, so $x^2 = 4 \Rightarrow x = \pm 2$
2. $x = 2$: $y = \frac{8}{3} + 2 = \frac{14}{3}$; $x = -2$: $y = -\frac{8}{3} + 2 = -\frac{2}{3}$

✔ Points $\left(2, \frac{14}{3}\right)$ and $\left(-2, -\frac{2}{3}\right)$

Related rates: the link equation

Question	Link equation	Differentiate
Balloon (sphere)	$V = \frac{4}{3}\pi r^3$	$\dot{V} = 4\pi r^2 \dot{r}$
Ripple (circle)	$A = \pi r^2$	$\dot{A} = 2\pi r\dot{r}$
Cube	$V = x^3$, $S = 6x^2$	$\dot{V} = 3x^2\dot{x}$
Ladder	$x^2 + y^2 = l^2$	$x\dot{x} + y\dot{y} = 0$
Shadow	similar triangles	$s = kx$, $\dot{s} = k\dot{x}$
Cone of sand	$V = \frac{1}{3}\pi r^2 h$ + given r-h link	one variable first

↳ Here $\dot{V}$ means $\frac{dV}{dt}$ (rate with time). Board mein $\frac{dV}{dt}$ hi likhna.

Solved Example 10

Each side of an equilateral triangle increases at 2 cm/s. Find the rate of increase of its area when a side is 10 cm.

1. $A = \frac{\sqrt{3}}{4}a^2 \Rightarrow \frac{dA}{dt} = \frac{\sqrt{3}}{2}a\frac{da}{dt}$
2. $= \frac{\sqrt{3}}{2}(10)(2) = 10\sqrt{3}$

✔ $10\sqrt{3}$ cm^2/s

Quick fact Fixed perimeter rectangle: the square has the largest area.

Solved Example 11

The radius of a circle increases at 0.7 cm/s. Find the rate of increase of its circumference.

1. $C = 2\pi r \Rightarrow \dfrac{dC}{dt} = 2\pi \dfrac{dr}{dt} = 2\pi(0.7)$

✔ 1.4π cm/s (same for every radius)

Solved Example 12

Water flows into a cylindrical tank of radius 2 m at 3π m^3/min. How fast is the water level rising?

1. $V = \pi r^2 h = 4\pi h \Rightarrow \dfrac{dV}{dt} = 4\pi \dfrac{dh}{dt}$
2. $3\pi = 4\pi \dfrac{dh}{dt} \Rightarrow \dfrac{dh}{dt} = \dfrac{3}{4}$

✔ $\dfrac{3}{4}$ m/min (constant, since r is fixed)

2. Increasing and Decreasing Functions

Definitions (on an interval I)

- → **Increasing:** $x_1 < x_2 \Rightarrow f(x_1) \le f(x_2)$
- → **Strictly increasing:** $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$
- → **Decreasing:** $x_1 < x_2 \Rightarrow f(x_1) \ge f(x_2)$
- → **Strictly decreasing:** $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$

Theorem (derivative test)

Let f be continuous on $[a, b]$ and differentiable on (a, b).

- → $f'(x) > 0$ for all $x \in (a, b) \Rightarrow f$ is strictly increasing on $[a, b]$
- → $f'(x) < 0$ for all $x \in (a, b) \Rightarrow f$ is strictly decreasing on $[a, b]$
- → $f'(x) = 0$ for all $x \in (a, b) \Rightarrow f$ is constant

↳ $f(x) = x^3$ is strictly increasing on R even though $f'(0) = 0$: if $f' \ge 0$ and $f' = 0$ only at isolated points, f is still strictly increasing.

Method: intervals of increasing/decreasing

Find $f'(x)$, factorise → $f'(x) = 0$: critical points → mark on number line → sign of f' in each part → + inc, − dec

Yaad karo: "Wavy curve: right se plus"

$f'(x)$ ko factors mein todo, roots number line pe lagao. **Sabse right** wale interval mein sign leading coefficient ka (usually +), phir har simple root pe sign **badlo**. Squared factor $(x-a)^2$ pe sign **nahi** badalta.

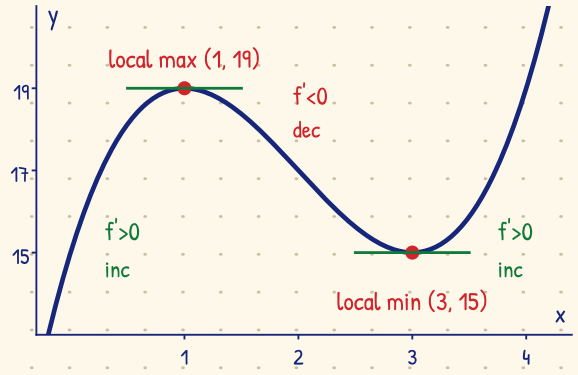


Fig: $f(x) = x^3 - 6x^2 + 9x + 15$: rises, falls, rises again

Solved Example 13

Find the intervals in which $f(x) = 2x^3 - 9x^2 + 12x + 5$ is strictly increasing or strictly decreasing.

1. $f'(x) = 6x^2 - 18x + 12 = 6(x-1)(x-2)$
2. $f'(x) = 0 \Rightarrow x = 1, 2$
3. $(-\infty, 1)$: $f' > 0$; $(1, 2)$: $f' < 0$; $(2, \infty)$: $f' > 0$

✔ Strictly increasing on $(-\infty, 1) \cup (2, \infty)$, strictly decreasing on $(1, 2)$

Solved Example 14

Find the intervals in which $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is strictly increasing or decreasing.

1. $f'(x) = 12x^3 - 12x^2 - 24x = 12x(x+1)(x-2)$
2. Critical points: $x = -1, 0, 2$
3. Signs: $(-\infty, -1)$: $-$; $(-1, 0)$: $+$; $(0, 2)$: $-$; $(2, \infty)$: $+$

✔ Inc on $(-1, 0) \cup (2, \infty)$; dec on $(-\infty, -1) \cup (0, 2)$

Solved Example 15

Find the intervals in $[0, 2\pi]$ where $f(x) = \sin x + \cos x$ is strictly increasing or decreasing.

1. $f'(x) = \cos x - \sin x = 0 \Rightarrow \tan x = 1 \Rightarrow x = \dfrac{\pi}{4}, \dfrac{5\pi}{4}$
2. At $x = 0$: $f' = 1 > 0$; at $x = \pi$: $f' = -1 < 0$; at $x = \dfrac{3\pi}{2}$: $f' = 1 > 0$

✔ Inc on $\left[0, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right]$; dec on $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$

Quick fact e^x and $\log x$ are strictly increasing wherever defined.

Quick fact $x + \frac{1}{x} \ge 2$ for $x > 0$ (min at $x = 1$).

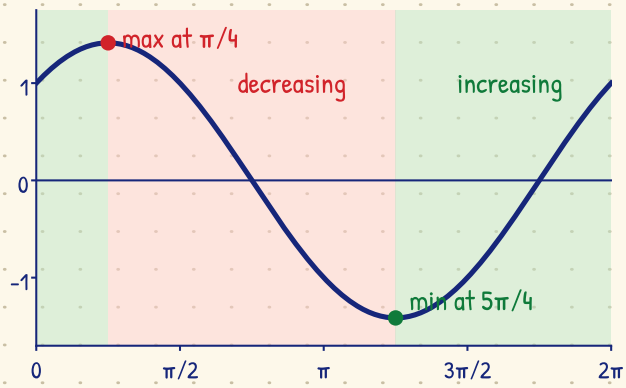


Fig: $\sin x + \cos x$ on $[0, 2\pi]$: shaded green = increasing

Solved Example 16

Show that $f(x) = x^3 - 6x^2 + 12x - 18$ is strictly increasing on R.

1. $f'(x) = 3x^2 - 12x + 12 = 3(x-2)^2 \geq 0$ for all x
2. $f'(x) = 0$ only at the single point $x = 2$

✔ So f is strictly increasing on R

Solved Example 17

Find the least value of a such that $f(x) = x^2 + ax + 5$ is strictly increasing on $(2, 3)$.

1. $f'(x) = 2x + a$; on $(2,3)$ its least value is just above $4 + a$
2. Need $2x + a > 0$ for all $x \in (2,3) \Rightarrow 4 + a \geq 0 \Rightarrow a \geq -4$

✔ Least value of a is -4

Standard functions: where they rise and fall

Function	Increasing	Decreasing
e^x, $\log x$	whole domain	never
e^{-x}, $\frac{1}{x}$	never	each interval of domain
$\sin x$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	$\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$
$\cos x$	$(\pi, 2\pi)$	$(0, \pi)$
$\tan x$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	never
x^2	$(0, \infty)$	$(-\infty, 0)$
x^3	R	never

Solved Example 19

Find the intervals in which $f(x) = x^3(x-2)^2$ is strictly increasing or decreasing.

1. $f'(x) = 3x^2(x-2)^2 + 2x^3(x-2) = x^2(x-2)(5x-6)$
2. Roots $0, \frac{6}{5}, 2$; x^2 is squared, so no sign change at 0
3. Signs: $(-\infty, 0)$: $+$, $\left(0, \frac{6}{5}\right)$: $+$, $\left(\frac{6}{5}, 2\right)$: $-$, $(2, \infty)$: $+$

✔ Increasing on $\left(-\infty, \frac{6}{5}\right) \cup (2, \infty)$; decreasing on $\left(\frac{6}{5}, 2\right)$

Quick fact A constant function is both increasing and decreasing (not strictly).

Solved Example 18

Find the intervals in which $f(x) = x^2 e^{-x}$ is strictly increasing or decreasing.

1. $f'(x) = 2xe^{-x} - x^2e^{-x} = e^{-x}\,x(2-x)$; $e^{-x} > 0$ always
2. $f' = 0 \Rightarrow x = 0, 2$; signs: $(-\infty, 0)$: $-$, $(0, 2)$: $+$, $(2, \infty)$: $-$

✔ Increasing on $(0, 2)$; decreasing on $(-\infty, 0) \cup (2, \infty)$

Solved Example 20

Show that $f(x) = \tan x - x$ is strictly increasing on $\left(0, \frac{\pi}{2}\right)$.

1. $f'(x) = \sec^2 x - 1 = \tan^2 x$
2. $\tan x > 0$ on $\left(0, \frac{\pi}{2}\right)$, so $f'(x) > 0$ there

✔ f is strictly increasing on $\left(0, \frac{\pi}{2}\right)$

Solved Example 21

Find the intervals in $\left[0, \frac{\pi}{2}\right]$ where $f(x) = \sin^4 x + \cos^4 x$ is increasing or decreasing.

1. $f'(x) = 4\sin^3 x \cos x - 4\cos^3 x \sin x = -4 \sin x \cos x \cos 2x = -\sin 4x$
2. On $\left(0, \frac{\pi}{4}\right)$: $4x \in (0, \pi)$, $\sin 4x > 0$, so $f' < 0$
3. On $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$: $4x \in (\pi, 2\pi)$, $\sin 4x < 0$, so $f' > 0$

✔ Decreasing on $\left[0, \frac{\pi}{4}\right]$, increasing on $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$

Yaad karo: "Plus = pahaad chadho"

$f' > 0$: graph **upar chadh raha** (increasing). $f' < 0$: graph **neeche utar raha** (decreasing). Jahan chadhai khatam ho ke utraai shuru ho, wahi **peak = max**.

3. Maxima and Minima

Theorem

If f has a local max or min at c and $f'(c)$ exists, then $f'(c) = 0$. Converse is false: $f'(c) = 0$ does not guarantee a max or min.

Key terms

- **Maximum value** of f on I: $f(c)$ with $f(c) \geq f(x)$ for all $x \in I$ (c = point of maximum). Minimum similarly.
- **Local maximum** at c: $f(c) \geq f(x)$ for all x in some interval $(c-h, c+h)$. Local minimum: $f(c) \leq f(x)$ there.
- **Critical point:** c where $f'(c) = 0$ or f is not differentiable.
- **Point of inflection:** $f'(c) = 0$ but no max/min there (e.g. x^3 at 0).

First derivative test

- f' changes from + to − at $c \Rightarrow$ local maximum
- f' changes from − to + $\Rightarrow$ local minimum
- No change of sign $\Rightarrow$ neither (point of inflection)



Fig: sign of f' just left and just right of c

Second derivative test

Let $f'(c) = 0$.

- $f''(c) < 0 \Rightarrow$ local maximum at c
- $f''(c) > 0 \Rightarrow$ local minimum at c
- $f''(c) = 0 \Rightarrow$ test fails; use the first derivative test (e.g. $f(x) = x^4$ has a minimum at 0 though $f''(0) = 0$)

Yaad karo: "MAX ko MINUS, MIN ko PLUS"

Second derivative test: **MAX**imum ke liye $f''(c)$ **MINUS**, **MIN**imum ke liye $f''(c)$ **PLUS**. Shape se yaad karo: $f'' < 0$ = udaas chehra $\frown$ = top = max; $f'' > 0$ = smile $\smile$ = bottom = min.

Absolute max/min on a closed interval $[a, b]$

A continuous function on $[a, b]$ always has an absolute max and an absolute min.

Find critical points in (a, b) → compute f there and at a, b → largest = abs max, smallest = abs min

Solved Example 22

Find the local maxima and minima of $f(x) = x^3 - 6x^2 + 9x + 15$ (graph in Section 2).

1. $f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$, so $x = 1, 3$
2. $f''(x) = 6x - 12$: $f''(1) = -6 < 0$, $f''(3) = 6 > 0$
3. $f(1) = 1 - 6 + 9 + 15 = 19$, $f(3) = 27 - 54 + 27 + 15 = 15$

✔ Local max value 19 at $x = 1$; local min value 15 at $x = 3$

Formula Box 2: increasing, max and min

- $f'(x) > 0$ on $(a, b) \Rightarrow$ strictly increasing; $f'(x) < 0 \Rightarrow$ strictly decreasing
- Critical point: $f'(c) = 0$ or $f'(c)$ does not exist
- 1st test: $+ \to -$ max, $- \to +$ min, no change: neither
- 2nd test: $f'(c) = 0$, $f''(c) < 0$ max, $f''(c) > 0$ min, $f''(c) = 0$ fails
- On $[a, b]$: compare $f(a)$, $f(b)$, $f(\text{critical pts})$
- $a \sin x + b \cos x$ has max $\sqrt{a^2 + b^2}$, min $-\sqrt{a^2 + b^2}$
- $ax^2 + bx + c$ $(a > 0)$ has min at $x = -\dfrac{b}{2a}$

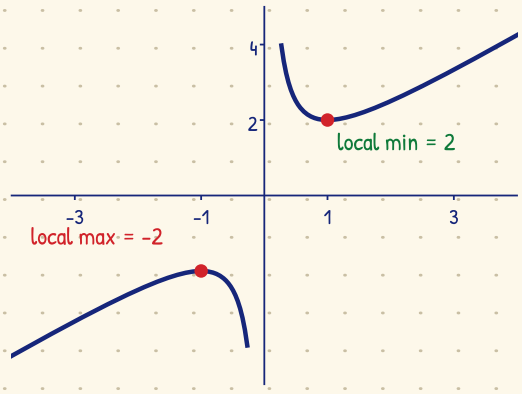


Fig: $y = x + \frac{1}{x}$: local max value -2 is LESS than local min value 2

Solved Example 23

Find the local maximum and minimum values of $f(x) = x + \dfrac{1}{x}$, $x \neq 0$.

1. $f'(x) = 1 - \dfrac{1}{x^2} = 0 \Rightarrow x = \pm 1$
2. $f''(x) = \dfrac{2}{x^3}$: $f''(1) = 2 > 0$ (min), $f''(-1) = -2 < 0$ (max)
3. $f(1) = 2$, $f(-1) = -2$

✔ Local max $= -2$ at $x = -1$, local min $= 2$ at $x = 1$ (local max can be less than local min!)

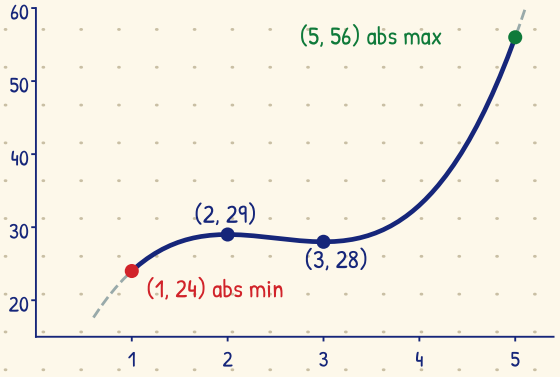


Fig: on $[1,5]$ the end points decide the answer

Solved Example 24

Find the absolute maximum and minimum values of $f(x) = 2x^3 - 15x^2 + 36x + 1$ on $[1,5]$.

1. $f'(x) = 6x^2 - 30x + 36 = 6(x-2)(x-3) \Rightarrow x = 2, 3$ (both in $[1,5]$)
2. $f(1) = 24,\ f(2) = 29,\ f(3) = 28,\ f(5) = 56$

✔ Abs max $= 56$ at $x = 5$; abs min $= 24$ at $x = 1$

Solved Example 25

Find the absolute max and min of $f(x) = \sin x + \cos x$ on $[0, \pi]$.

1. $f'(x) = \cos x - \sin x = 0 \Rightarrow x = \dfrac{\pi}{4}$
2. $f(0) = 1,\ f\left(\frac{\pi}{4}\right) = \sqrt{2},\ f(\pi) = -1$

✔ Abs max $\sqrt{2}$ at $x = \frac{\pi}{4}$; abs min -1 at $x = \pi$

Solved Example 26

Use the first derivative test to find the local max and min of $f(x) = (x-1)(x+2)^2$.

1. $f'(x) = (x+2)^2 + 2(x-1)(x+2) = (x+2)(3x) = 3x(x+2)$
2. Critical points $x = -2, 0$. Signs: $(-\infty, -2)$: $+$, $(-2, 0)$: $-$, $(0, \infty)$: $+$
3. At -2: $+ \to -$ (max), $f(-2) = 0$. At 0: $- \to +$ (min), $f(0) = -4$

✔ Local max value 0 at $x = -2$; local min value -4 at $x = 0$

Solved Example 27

Find the local maximum value of $f(x) = xe^{-x}$.

1. $f'(x) = e^{-x} - xe^{-x} = e^{-x}(1-x) = 0 \Rightarrow x = 1$
2. $f''(x) = e^{-x}(x-2)$; $f''(1) = -\dfrac{1}{e} < 0$: local max
3. $f(1) = \dfrac{1}{e}$

✔ Local max value $\dfrac{1}{e}$ at $x = 1$

Solved Example 28

Find the absolute maximum and minimum of $f(x) = 6x - x^2$ on $[0,5]$.

1. $f'(x) = 6 - 2x = 0 \Rightarrow x = 3$
2. $f(0) = 0,\ f(3) = 9,\ f(5) = 5$

✔ Abs max 9 at $x = 3$; abs min 0 at $x = 0$

Solved Example 29

Find the absolute maximum and minimum values of $f(x) = x^3 - 3x$ on $[-2, 3]$.

1. $f'(x) = 3x^2 - 3 = 0 \Rightarrow x = \pm 1$
2. $f(-2) = -2,\ f(-1) = 2,\ f(1) = -2,\ f(3) = 18$

✔ Abs max 18 at $x = 3$; abs min -2 at $x = -2$ and $x = 1$

Which test when?

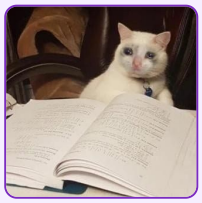
Situation	Use
f'' easy to find, $f''(c) \neq 0$	2nd derivative test
$f''(c) = 0$ or f'' messy	1st derivative test
f not differentiable at c (like $\lvert x \rvert$)	1st derivative test / graph
Closed interval $[a,b]$, "absolute"	table of values (CE)



Shivam Bhaiya says

Absolute max/min mein **end points ki value** nikalna kabhi mat bhoolna. Upar wale example mein answer $x = 5$ pe aaya, jo critical point hai hi nahi! Board mein ye 3 marks ka pakka sawal hai.

Meme samjho



Jab $[1,5]$ pe absolute max nikala, end points check hi nahi kiye, aur answer galat aa gaya

Asli baat: On a closed interval, compare f at the critical points **and** at both end points; the largest value is the absolute max.

Quick fact If f is strictly increasing, $-f$ is strictly decreasing.

Quick fact MR $= \frac{dR}{dx}$, MC $= \frac{dC}{dx}$: just derivatives.

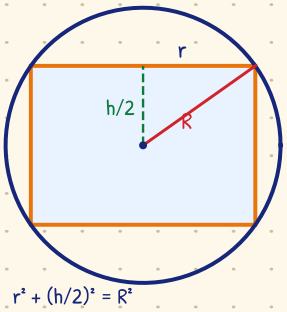
4. Word Problems on Maxima and Minima (5 marks)

Steps for every max/min word problem

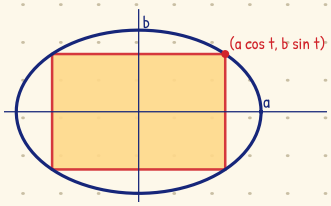
1. Draw a figure, name the variables (x, r, h ...).
2. Write the quantity to optimise (area, volume, cost) as Q.
3. Use the given condition to write Q in **one** variable.
4. $\frac{dQ}{dx} = 0$ gives the critical value.
5. Show $\frac{d^2Q}{dx^2} < 0$ (max) or > 0 (min).
6. Final answer with units.

Formula Box 3: standard results (prove before using)

- → Fixed sum $x + y = k$: product xy is max when $x = y = \frac{k}{2}$
- → Rectangle with fixed perimeter: max area when it is a **square**
- → Rectangle inscribed in a circle: max area when it is a **square**
- → Closed cylinder, fixed volume, least surface: $h = 2r$
- → Open cylinder (no lid), fixed volume, least surface: $h = r$
- → Closed cylinder, fixed surface, max volume: $h = 2r$
- → Cylinder in sphere of radius R, max volume: $h = \frac{2R}{\sqrt{3}}$
- → Cone in sphere of radius R, max volume: $h = \frac{4R}{3}$, $V = \frac{8}{27} \times$ sphere
- → Cone with given slant height l, max volume: $h = \frac{l}{\sqrt{3}}$, $\tan\alpha = \sqrt{2}$



Cylinder in sphere



Rectangle in ellipse

Yaad karo: "Height = Diameter"

Closed tin (cylinder) mein kam se kam metal lagana ho to **height = diameter** ($h = 2r$). Dhakkan hi nahi (open top) to **height = radius** ($h = r$). Isliye cold drink ki can lagbhag square dikhti hai side se.

Solved Example 30

Find two positive numbers whose sum is 18 such that the product of one and the square of the other is maximum.

1. Let the numbers be x and $18 - x$; $P = x^2(18 - x) = 18x^2 - x^3$
2. $\frac{dP}{dx} = 36x - 3x^2 = 3x(12 - x) = 0 \Rightarrow x = 12$
3. $\frac{d^2P}{dx^2} = 36 - 6x = -36 < 0$ at $x = 12$: maximum
4. $P = 144 \times 6 = 864$

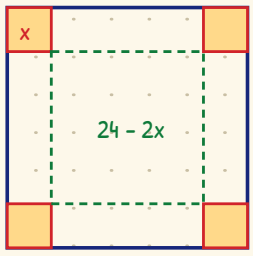
✔ The numbers are 12 and 6 (max product $12^2 \times 6 = 864$)

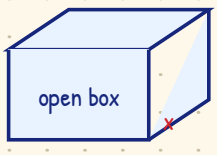
Solved Example 31

An open box is made from a square sheet of side 24 cm by cutting equal squares from the corners and folding up. Find the side of the square cut off for maximum volume.

1. Side of square cut $= x$; base $(24 - 2x)$, height x, $0 < x < 12$
2. $V = x(24 - 2x)^2 = 4x^3 - 96x^2 + 576x$
3. $V' = 12x^2 - 192x + 576 = 12(x - 4)(x - 12) = 0 \Rightarrow x = 4$ ($x = 12$ gives no box)
4. $V'' = 24x - 192 = -96 < 0$ at $x = 4$: max; $V = 4 \times 16^2 = 1024$

✔ Cut squares of side 4 cm; max volume 1024 cm^3





base (24-2x)

Fig: cut squares of side x, fold the flaps up

↳Domain check: box from a square of side a needs $0 < x < \frac{a}{2}$. A root outside this (like $x = 12$ above) is rejected.

Solved Example 32

Show that a closed right circular cylinder of given volume V has the least total surface area when its height equals its diameter.

Quick fact Every local max/min of a differentiable f is a critical point.

1. $V = \pi r^2 h \Rightarrow h = \dfrac{V}{\pi r^2}$
2. $S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + \dfrac{2V}{r}$
3. $\dfrac{dS}{dr} = 4\pi r - \dfrac{2V}{r^2} = 0 \Rightarrow V = 2\pi r^3$
4. $\dfrac{d^2S}{dr^2} = 4\pi + \dfrac{4V}{r^3} > 0$: minimum
5. Then $h = \dfrac{2\pi r^3}{\pi r^2} = 2r$

✔ Least surface when $h = 2r$, i.e. height = diameter. Hence proved.

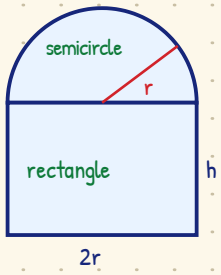


Fig: Norman window, perimeter $2r + 2h + \pi r$

Solved Example 33

A window is a rectangle surmounted by a semicircle. Its perimeter is 20 m. Find the dimensions that let in the most light (maximum area).

1. Width $2r$, height h: $2r + 2h + \pi r = 20 \Rightarrow h = \dfrac{20 - (2+\pi)r}{2}$
2. $A = 2rh + \frac{1}{2}\pi r^2 = 20r - 2r^2 - \frac{\pi}{2}r^2$
3. $\dfrac{dA}{dr} = 20 - (4+\pi)r = 0 \Rightarrow r = \dfrac{20}{4+\pi}$
4. $\dfrac{d^2A}{dr^2} = -(4+\pi) < 0$: maximum; then $h = \dfrac{20}{4+\pi}$

✔ Width $= \dfrac{40}{4+\pi}$ m, height of rectangle $= \dfrac{20}{4+\pi}$ m

Solved Example 34

Find the point on the curve $y^2 = 2x$ nearest to the point $(1, 4)$.

1. Point $\left(\frac{y^2}{2}, y\right)$; $D = d^2 = \left(\frac{y^2}{2} - 1\right)^2 + (y-4)^2$
2. $\dfrac{dD}{dy} = y\left(y^2 - 2\right) + 2(y-4) = y^3 - 8 = 0 \Rightarrow y = 2$
3. $\dfrac{d^2D}{dy^2} = 3y^2 = 12 > 0$: minimum; $x = \dfrac{4}{2} = 2$

✔ Nearest point $(2, 2)$, least distance $\sqrt{5}$

Solved Example 35

A box without a lid is made from a 16 cm $\times$ 10 cm sheet by cutting equal squares from the corners. Find the maximum volume.

1. $V = x(16-2x)(10-2x) = 4x^3 - 52x^2 + 160x$, $0 < x < 5$
2. $V' = 12x^2 - 104x + 160 = 4(3x-20)(x-2) = 0 \Rightarrow x = 2$ ($\frac{20}{3} > 5$ rejected)
3. $V'' = 24x - 104 = -56 < 0$ at $x = 2$: max; $V = 2 \times 12 \times 6$

✔ Cut squares of side 2 cm; max volume 144 cm^3

Solved Example 36

A wire 20 m long is cut into two pieces: one is bent into a square, the other into a circle. Where should it be cut so that the total area is least?

1. Square side x, circle radius r: $4x + 2\pi r = 20 \Rightarrow r = \dfrac{10 - 2x}{\pi}$
2. $A = x^2 + \pi r^2 = x^2 + \dfrac{(10-2x)^2}{\pi}$
3. $A' = 2x - \dfrac{4(10-2x)}{\pi} = 0 \Rightarrow (\pi+4)x = 20 \Rightarrow x = \dfrac{20}{\pi+4}$
4. $A'' = 2 + \dfrac{8}{\pi} > 0$: minimum

✔ Square piece $= 4x = \dfrac{80}{\pi+4}$ m, circle piece $= \dfrac{20\pi}{\pi+4}$ m

Solved Example 37

Show that the semi-vertical angle of a cone of given slant height l and maximum volume is $\tan^{-1}\sqrt{2}$.

1. $r^2 = l^2 - h^2$; $V = \frac{\pi}{3}(l^2 - h^2)h = \frac{\pi}{3}(l^2 h - h^3)$
2. $V' = \frac{\pi}{3}(l^2 - 3h^2) = 0 \Rightarrow h = \dfrac{l}{\sqrt{3}}$; $V'' = -2\pi h < 0$: max
3. $r = \sqrt{l^2 - \frac{l^2}{3}} = \dfrac{\sqrt{2}\,l}{\sqrt{3}}$, so $\tan\alpha = \dfrac{r}{h} = \sqrt{2}$

✔ Semi-vertical angle $\alpha = \tan^{-1}\sqrt{2}$. Hence proved.

Solved Example 38

Find the area of the largest rectangle that can be inscribed in the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ (sides parallel to the axes).

1. Corner $(a\cos\theta, b\sin\theta)$, $0<\theta<\frac{\pi}{2}$: $A = 4ab\sin\theta\cos\theta = 2ab\sin 2\theta$
2. $\frac{dA}{d\theta} = 4ab\cos 2\theta = 0 \Rightarrow \theta = \frac{\pi}{4}$
3. $\frac{d^2A}{d\theta^2} = -8ab\sin 2\theta = -8ab < 0$: max

✔ Largest area $= 2ab$ (sides $\sqrt{2}a$ and $\sqrt{2}b$)

Solved Example 39

For a shop, revenue $R(x) = 120x - x^2$ and cost $C(x) = 20x + 600$ (in Rs) for x items. Find the number of items for maximum profit and the profit.

1. $P = R - C = -x^2 + 100x - 600$
2. $P' = -2x + 100 = 0 \Rightarrow x = 50$; $P'' = -2 < 0$: max
3. $P(50) = -2500 + 5000 - 600 = 1900$

✔ 50 items, maximum profit Rs 1900



Shivam Bhaiya says

5 marks wale word problem mein last line zaroor likho: "$\frac{d^2V}{dx^2} < 0$, hence volume is **maximum**". Figure, variable, ek variable mein function, $f' = 0$, phir f'' test: yahi 5 steps, yahi 5 marks.



Meme samjho

Jab question bole "show that the volume is maximum" aur tumne pehle hi $V'' < 0$ dikha diya

Asli baat: The second derivative test line is the **proof** of maximum/minimum in word problems; without it the answer is incomplete.

5. More Board-Level Solved Problems

Solved Example 40

Find the absolute max and min of $f(x) = 2\sin x + \cos 2x$ on $\left[0, \frac{\pi}{2}\right]$.

1. $f'(x) = 2\cos x - 2\sin 2x = 2\cos x(1 - 2\sin x)$
2. $f' = 0 \Rightarrow x = \frac{\pi}{6}$ or $x = \frac{\pi}{2}$
3. $f(0) = 1$, $f\left(\frac{\pi}{6}\right) = 1 + \frac{1}{2} = \frac{3}{2}$, $f\left(\frac{\pi}{2}\right) = 2 - 1 = 1$

✔ Abs max $\frac{3}{2}$ at $x = \frac{\pi}{6}$; abs min 1 at $x = 0$ and $x = \frac{\pi}{2}$

Quick fact Speed $= \frac{ds}{dt}$; at rest when it is 0.

Solved Example 41

Show that of all right triangles with a given hypotenuse h, the isosceles one has the maximum area.

1. Legs x and $\sqrt{h^2 - x^2}$; maximise $Z = A^2 = \frac{1}{4}x^2(h^2 - x^2) = \frac{1}{4}(h^2x^2 - x^4)$
2. $Z' = \frac{1}{4}(2h^2x - 4x^3) = 0 \Rightarrow x^2 = \frac{h^2}{2}$
3. $Z'' = \frac{1}{4}(2h^2 - 12x^2) = -h^2 < 0$: max; other leg $= \sqrt{h^2 - \frac{h^2}{2}} = \frac{h}{\sqrt{2}}$

✔ Both legs $= \frac{h}{\sqrt{2}}$: isosceles. Hence proved.

Yaad karo: "Root hatao, square karo"

Area/distance mein square root aaye (jaise $\sqrt{h^2 - x^2}$), to A ki jagah $Z = A^2$ ko maximise karo. $A > 0$ hai, isliye jahan A^2 max wahin A max. Differentiation bahut aasaan ho jata hai.

Solved Example 42

A rectangle is inscribed in a semicircle of radius r with one side on the diameter. Find its maximum area.

1. Corner $(r\cos\theta, r\sin\theta)$: width $2r\cos\theta$, height $r\sin\theta$
2. $A = 2r^2\sin\theta\cos\theta = r^2\sin 2\theta$
3. $A' = 2r^2\cos 2\theta = 0 \Rightarrow \theta = \frac{\pi}{4}$; $A'' = -4r^2 < 0$: max

✔ Max area r^2 (width $\sqrt{2}r$, height $\frac{r}{\sqrt{2}}$)

Solved Example 43

Show that a closed cylinder of given total surface area S has maximum volume when its height equals its diameter.

1. $S = 2\pi r^2 + 2\pi rh \Rightarrow h = \frac{S - 2\pi r^2}{2\pi r}$
2. $V = \pi r^2 h = \frac{r(S - 2\pi r^2)}{2} = \frac{Sr}{2} - \pi r^3$
3. $V' = \frac{S}{2} - 3\pi r^2 = 0 \Rightarrow S = 6\pi r^2$; $V'' = -6\pi r < 0$: max
4. $h = \frac{6\pi r^2 - 2\pi r^2}{2\pi r} = 2r$

✔ Max volume when $h = 2r$. Hence proved.

Solved Example 45

Find the local maximum and minimum values of $f(x) = x^4 - 4x^3 + 4x^2 + 5$.

1. $f'(x) = 4x^3 - 12x^2 + 8x = 4x(x-1)(x-2)$, so $x = 0, 1, 2$
2. $f''(x) = 12x^2 - 24x + 8$: $f''(0) = 8 > 0$, $f''(1) = -4 < 0$, $f''(2) = 8 > 0$
3. $f(0) = 5$, $f(1) = 6$, $f(2) = 16 - 32 + 16 + 5 = 5$

✔ Local min 5 at $x = 0$ and $x = 2$; local max 6 at $x = 1$

Solved Example 44

A particle moves so that its distance from a fixed point is $s = t^3 - 6t^2 + 9t$ m at time t s. Find when it is at rest and when its velocity is decreasing.

1. Velocity $v = \frac{ds}{dt} = 3t^2 - 12t + 9 = 3(t-1)(t-3)$; at rest when $v = 0$: $t = 1, 3$
2. $\frac{dv}{dt} = 6t - 12 < 0$ for $t < 2$

✔ At rest at $t = 1$ s and $t = 3$ s; velocity decreases for $0 \le t < 2$

Common mistakes

1. Value (like $r = 6$) put **before** differentiating: rate becomes 0. Differentiate first.
2. Absolute max/min on $[a, b]$ without checking the **end points**.
3. Taking $f'(c) = 0$ as proof of max/min (x^3 at 0 has neither).
4. Word problem ends at $f' = 0$; the f'' test (max/min proof) is missing.
5. Rate came negative but answer written as "increasing"; also missing units like cm^3/s.

Yaad karo: "CE: Critical + End points"

Closed interval $[a, b]$ dikhe to **CE** yaad karo: **C**ritical points aur **E**nd points, sab ki value table mein likho. Sabse badi = abs max, sabse chhoti = abs min. Koi derivative test nahi chahiye.

Most Asked in Boards

1. Rate of change: balloon, cube, cone of sand, ladder, shadow (2-3M)
2. Find intervals where f is strictly increasing/decreasing (3M)
3. Show f is increasing/decreasing on a given interval (2M)
4. Local max/min by first or second derivative test (3M)
5. Absolute max/min on a closed interval (2-3M)
6. Open box, tank cost, window, cylinder/cone word problems (5M)
7. Prove a standard result: rectangle in circle, cone/cylinder in sphere (5M)
8. Case-based: fencing, profit, box (4M)

Khud karo: 8 quick practice questions

1. Rate of change of volume of a cube w.r.t. edge x at $x = 3$.
2. Rate of change of area of a circle w.r.t. diameter D.
3. Interval where $f(x) = x^2 - 2x$ is strictly increasing.
4. Is $f(x) = \cos x$ increasing or decreasing on $(0, \pi)$?
5. Critical points of $f(x) = x^3 - 12x$.
6. Local min value of $f(x) = x^2 - 8x + 20$.
7. Max value of $5\sin x - 12\cos x$.
8. Abs max of $f(x) = x^2$ on $[-3, 2]$.

Answers: 1) 27 2) $\frac{\pi D}{2}$ 3) $(1, \infty)$ 4) decreasing 5) $x = \pm 2$ 6) 4 7) 13 8) 9 (at $x = -3$)

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. The rate of change of the area of a circle with respect to its radius at $r = 5$ cm is 1M

(a) 10π (b) 25π (c) 5π (d) 100π

Ans. (a) $\frac{dA}{dr} = 2\pi r = 10\pi$ cm^2/cm.

Q2. The edge of a cube increases at 2 cm/s. When the edge is 4 cm, its volume increases at 1M Board fav

(a) 48 (b) 96 (c) 32 (d) 192

Ans. (b) $\frac{dV}{dt} = 3x^2\frac{dx}{dt} = 3(16)(2) = 96$ cm^3/s.

Q3. $f(x) = x^3 + 3x$ is 1M

(a) strictly increasing on R (b) strictly decreasing on R (c) neither (d) constant

Ans. (a) $f'(x) = 3x^2 + 3 > 0$ for all x.

Q4. $f(x) = x^2 - 4x + 7$ is strictly decreasing in 1M

(a) $(2, \infty)$ (b) $(-\infty, 2)$ (c) $(-\infty, 4)$ (d) R

Ans. (b) $f'(x) = 2x - 4 < 0$ for $x < 2$.

Q5. The maximum value of $\sin x \cos x$ is 1M Most asked

(a) 1 (b) $\frac{1}{2}$ (c) $\sqrt{2}$ (d) 2

Ans. (b) $\sin x \cos x = \frac{1}{2}\sin 2x \le \frac{1}{2}$, attained at $x = \frac{\pi}{4}$.

Q6. The minimum value of $x^2 + 6x + 13$ is 1M
(a) 13 (b) -3 (c) 4 (d) 0
Ans. (c) $(x+3)^2 + 4 \geq 4$, at $x = -3$.

Q7. The point of local maximum of $f(x) = 3x^2 - 2x^3$ is 1M
(a) $x = 0$ (b) $x = 1$ (c) $x = -1$ (d) $x = \frac{1}{2}$
Ans. (b) $f' = 6x(1-x) = 0 \Rightarrow x = 0, 1$; $f'' = 6 - 12x$, $f''(1) = -6 < 0$.

Q8. $f(x) = 2x^3 - 15x^2 + 36x + 4$ is strictly decreasing in 1M
(a) $(-\infty, 2)$ (b) $(2, 3)$ (c) $(3, \infty)$ (d) R
Ans. (b) $f' = 6(x-2)(x-3) < 0$ only between 2 and 3.

Q9. The maximum value of $3\sin x + 4\cos x$ is 1M Most asked
(a) 7 (b) 1 (c) 5 (d) 4
Ans. (c) $\sqrt{3^2 + 4^2} = 5$.

Q10. $f(x) = ax + b$ is strictly decreasing on R if 1M
(a) $a = 0$ (b) $a > 0$ (c) $a < 0$ (d) $b < 0$
Ans. (c) $f'(x) = a < 0$.

Q11. The least value of $e^x + e^{-x}$ is 1M
(a) 0 (b) 2 (c) 1 (d) e
Ans. (b) $f' = e^x - e^{-x} = 0 \Rightarrow x = 0$, $f''(0) = 2 > 0$; $f(0) = 2$.

Q12. If $y = x^3$ and x increases at 3 units/s, then y increases at $x = 2$ at the rate 1M
(a) 12 (b) 18 (c) 27 (d) 36
Ans. (d) $\frac{dy}{dt} = 3x^2 \times 3 = 36$.

Q13. The number of critical points of $f(x) = x^3 - 3x^2 + 3x$ is 1M HOTS
(a) 0 (b) 1 (c) 2 (d) 3
Ans. (b) $f' = 3(x-1)^2 = 0$ only at $x = 1$ (a point of inflection).

Q14. A: $f(x) = e^x$ is strictly increasing on R. R: If $f'(x) > 0$ for all $x \in$ R, then f is strictly increasing on R. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true
Ans. (a) $f'(x) = e^x > 0$; R is the test that proves A.

Q15. A: $f(x) = x^3$ has a local extremum at $x = 0$. R: For $f(x) = x^3$, $f'(0) = 0$. 1M HOTS
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true
Ans. (d) A is false ($f' = 3x^2$ does not change sign at 0), R is true.

Very short (2 marks)

Q16. A stone dropped in a pond makes circular ripples. The radius grows at 5 cm/s. How fast is the enclosed area growing when $r = 8$ cm? 2M
(contd.) $\frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2\pi(8)(5) = 80\pi$ cm^2/s.

Q17. Show that $f(x) = \log(\cos x)$ is strictly decreasing on $(0, \frac{\pi}{2})$. 2M Must do
Ans. $f'(x) = \frac{-\sin x}{\cos x} = -\tan x < 0$ on $(0, \frac{\pi}{2})$, so f is strictly decreasing.

Q18. Find the intervals in which $f(x) = x^2 - 6x + 5$ is strictly increasing or decreasing. 2M
Ans. $f'(x) = 2x - 6$. Decreasing on $(-\infty, 3)$, increasing on $(3, \infty)$.

Q19. Show that $f(x) = 4x - \sin 2x$ is strictly increasing on R. 2M
Ans. $f'(x) = 4 - 2\cos 2x \geq 4 - 2 = 2 > 0$ for all x. So strictly increasing on R.

Q20. Find the absolute maximum and minimum values of $f(x) = 2x^3 - 6x + 11$ on $[0, 2]$. 2M Board fav
Ans. $f' = 6x^2 - 6 = 0 \Rightarrow x = 1$ (in $[0, 2]$). $f(0) = 11$, $f(1) = 7$, $f(2) = 15$. Abs max 15 at $x = 2$, abs min 7 at $x = 1$.

Q21. Find the rate of change of the volume of a sphere with respect to its surface area when $r = 2$ cm. 2M HOTS
Ans. $\frac{dV}{dS} = \frac{dV/dr}{dS/dr} = \frac{4\pi r^2}{8\pi r} = \frac{r}{2} = 1$ cm^3/cm^2.

Q22. Show that $f(x) = e^{2x}$ is strictly increasing on R. 2M
Ans. $f'(x) = 2e^{2x} > 0$ for every real x. Hence strictly increasing on R.

Q23. Find the maximum profit if $P(x) = 41 + 24x - 18x^2$. 2M
Ans. $P' = 24 - 36x = 0 \Rightarrow x = \frac{2}{3}$; $P'' = -36 < 0$. $P_{max} = 41 + 16 - 8 = 49$.

Q24. Find the intervals in which $f(x) = 5 + 36x - 3x^2$ is increasing or decreasing. 2M
Ans. $f'(x) = 36 - 6x$: increasing on $(-\infty, 6)$, decreasing on $(6, \infty)$.

Q25. Show that $f(x) = (x-1)^3$ is strictly increasing on R. 2M
Ans. $f'(x) = 3(x-1)^2 \geq 0$, and $f' = 0$ only at the single point $x = 1$. Hence strictly increasing on R.

Q26. Find the intervals in which $f(x) = \log(1 + x^2)$ is increasing or decreasing. 2M
Ans. $f'(x) = \frac{2x}{1+x^2}$: negative for $x < 0$, positive for $x > 0$. Decreasing on $(-\infty, 0)$, increasing on $(0, \infty)$.

Q27. The radius of a cylinder of fixed height 10 cm increases at 2 cm/s. Find the rate of change of its volume when $r = 5$ cm. 2M

Quick fact Cube: $\frac{dV}{dx} = 3x^2 = \frac{1}{2} \times$ surface area.

(contd.) $V = 10\pi r^2 \Rightarrow \frac{dV}{dt} = 20\pi r\frac{dr}{dt} = 20\pi(5)(2) = 200\pi$ cm^3/s.

Short answer (3 marks)

Q28. Find the intervals in which $f(x) = 2x^3 - 3x^2 - 12x + 7$ is strictly increasing or strictly decreasing. 3M Most asked

Ans. $f'(x) = 6x^2 - 6x - 12 = 6(x-2)(x+1)$, zero at $x = -1, 2$. Signs: $+, -, +$. Increasing on $(-\infty, -1) \cup (2, \infty)$, decreasing on $(-1, 2)$.

Q29. Find the local maximum and minimum values of $f(x) = x^3 - 3x^2 - 9x + 4$. 3M

Ans. $f' = 3(x-3)(x+1)$, so $x = -1, 3$. $f'' = 6x - 6$: $f''(-1) = -12 < 0$ max, $f''(3) = 12 > 0$ min. Local max $f(-1) = 9$; local min $f(3) = -23$.

Q30. Find the points of local max/min of $f(x) = \sin x - \cos x$, $0 < x < 2\pi$, and the values. 3M

Ans. $f' = \cos x + \sin x = 0 \Rightarrow x = \frac{3\pi}{4}, \frac{7\pi}{4}$. $f'' = \cos x - \sin x$: $f''\left(\frac{3\pi}{4}\right) = -\sqrt{2} < 0$, $f''\left(\frac{7\pi}{4}\right) = \sqrt{2} > 0$. Local max $\sqrt{2}$ at $\frac{3\pi}{4}$, local min $-\sqrt{2}$ at $\frac{7\pi}{4}$.

Q31. Find two positive numbers x and y with $x + y = 40$ such that xy^3 is maximum. 3M

Ans. $P = (40-y)y^3$, $P' = 120y^2 - 4y^3 = 4y^2(30 - y) = 0 \Rightarrow y = 30$. $P'' = 240y - 12y^2 = -3600 < 0$ at $y = 30$: max. So $x = 10, y = 30$.

Q32. A man 1.5 m tall walks away from a 4.5 m high lamp post at 4 km/h. How fast does his shadow lengthen? 3M Board fav

Ans. Let distance from post $= x$, shadow $= s$. Similar triangles: $\frac{s}{1.5} = \frac{x+s}{4.5} \Rightarrow 3s = 1.5x \Rightarrow s = \frac{x}{2}$. So $\frac{ds}{dt} = \frac{1}{2} \times 4 = 2$ km/h.

Q33. Find two positive numbers whose product is 64 and whose sum is minimum. 3M

Ans. $S = x + \frac{64}{x}$, $S' = 1 - \frac{64}{x^2} = 0 \Rightarrow x = 8$; $S'' = \frac{128}{x^3} > 0$: min. Numbers 8 and 8, least sum 16.

Q34. Show that $f(x) = x^3 - 3x^2 + 6x - 100$ is strictly increasing on R. 3M Must do

Ans. $f'(x) = 3x^2 - 6x + 6 = 3\left[(x-1)^2 + 1\right] \geq 3 > 0$ for all x. Hence strictly increasing on R.

Q35. Find the local minimum value of $f(x) = x\log x$, $x > 0$. 3M Must do

Ans. $f'(x) = \log x + 1 = 0 \Rightarrow x = \frac{1}{e}$. $f''(x) = \frac{1}{x} = e > 0$: min. Local min value $f\left(\frac{1}{e}\right) = -\frac{1}{e}$.

Q36. Find the intervals in which $f(x) = \frac{x}{x^2+1}$ is strictly increasing or decreasing. 3M

Ans. $f'(x) = \frac{(x^2+1) - 2x^2}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}$

(contd.) . Increasing on $(-1, 1)$; decreasing on $(-\infty, -1) \cup (1, \infty)$.

Q37. Find the absolute max and min of $f(x) = \cos^2 x + \sin x$ on $[0, \pi]$. 3M Board fav

Ans. $f' = \cos x(1 - 2\sin x) = 0 \Rightarrow x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$. $f(0) = 1, f\left(\frac{\pi}{6}\right) = \frac{5}{4}, f\left(\frac{\pi}{2}\right) = 1, f\left(\frac{5\pi}{6}\right) = \frac{5}{4}$, $f(\pi) = 1$. Abs max $\frac{5}{4}$ (at $\frac{\pi}{6}, \frac{5\pi}{6}$), abs min 1 (at $0, \frac{\pi}{2}, \pi$).

Q38. Show that $f(x) = \log(1+x) - \frac{x}{1+x}$ is increasing for $x > 0$ and decreasing for $-1 < x < 0$. 3M

Ans. $f'(x) = \frac{1}{1+x} - \frac{1}{(1+x)^2} = \frac{x}{(1+x)^2}$. Denominator > 0, so $f' > 0$ for $x > 0$ and $f' < 0$ for $-1 < x < 0$.

Q39. Find the local max and min of $f(x) = x\sqrt{4-x}$, $x < 4$. 3M

Ans. $f'(x) = \sqrt{4-x} - \frac{x}{2\sqrt{4-x}} = \frac{8-3x}{2\sqrt{4-x}} = 0 \Rightarrow x = \frac{8}{3}$. f' changes $+ \to -$: local max $f\left(\frac{8}{3}\right) = \frac{8}{3}\sqrt{\frac{4}{3}} = \frac{16}{3\sqrt{3}}$. No local min.

Q40. The volume of a spherical balloon decreases at 50 cm^3/min as air leaks. How fast is its radius decreasing when $r = 5$ cm? 3M

Ans. $\frac{dV}{dt} = 4\pi r^2\frac{dr}{dt} \Rightarrow -50 = 4\pi(25)\frac{dr}{dt} \Rightarrow \frac{dr}{dt} = -\frac{1}{2\pi}$. Radius decreases at $\frac{1}{2\pi}$ cm/min.

Long answer (5 marks)

Q41. Show that of all rectangles inscribed in a given circle of radius a, the square has the maximum area. 5M Most asked

Ans. Sides $2a\cos\theta, 2a\sin\theta$: $A = 4a^2\sin\theta\cos\theta = 2a^2\sin 2\theta$. $\frac{dA}{d\theta} = 4a^2\cos 2\theta = 0 \Rightarrow \theta = \frac{\pi}{4}$; $\frac{d^2A}{d\theta^2} = -8a^2\sin 2\theta = -8a^2 < 0$: max. At $\theta = \frac{\pi}{4}$ both sides $= \sqrt{2}a$: a square.

Q42. Show that the height of the cylinder of maximum volume inscribed in a sphere of radius R is $\frac{2R}{\sqrt{3}}$. Also find the maximum volume. 5M Board fav

Ans. Height h, radius r: $r^2 = R^2 - \frac{h^2}{4}$. $V = \pi\left(R^2h - \frac{h^3}{4}\right)$, $V' = \pi\left(R^2 - \frac{3h^2}{4}\right) = 0 \Rightarrow h = \frac{2R}{\sqrt{3}}$; $V'' = -\frac{3\pi h}{2} < 0$: max. $V_{max} = \frac{4\pi R^3}{3\sqrt{3}}$.

Q43. Show that the volume of the largest cone that can be inscribed in a sphere of radius R is $\frac{8}{27}$ of the volume of the sphere. 5M HOTS

Ans. Cone height h, base radius r: $r^2 = h(2R - h)$. $V = \frac{\pi}{3}(2Rh^2 - h^3)$, $V' = \frac{\pi}{3}(4Rh - 3h^2) = 0 \Rightarrow h = \frac{4R}{3}$; $V'' = \frac{\pi}{3}(4R - 6h) = -\frac{4\pi R}{3} < 0$: max.

(contd.) $V = \frac{\pi}{3} \times \frac{8R^2}{9} \times \frac{4R}{3} = \frac{32\pi R^3}{81} = \frac{8}{27} \times \frac{4}{3}\pi R^3$.

Q44. An open tank with a rectangular base is 3 m deep and holds 75 m^3. The base costs Rs 100 per m^2 and the sides Rs 50 per m^2. Find the cost of the cheapest tank. 5M Must do

Ans. Base $l \times b$, $3lb = 75 \Rightarrow lb = 25$. Cost $C = 100(25) + 50 \times 2 \times 3(l + b) = 2500 + 300\left(l + \frac{25}{l}\right)$. $C' = 300\left(1 - \frac{25}{l^2}\right) = 0 \Rightarrow l = 5$, $C'' = \frac{15000}{l^3} > 0$: min. $l = b = 5$, cost $= 2500 + 3000 =$ Rs 5500.

Q45. A box without a lid is made from a 30 cm $\times$ 14 cm sheet by cutting equal squares from the corners. Find the side of the square for maximum volume and the volume. 5M Most asked

Ans. $V = x(30 - 2x)(14 - 2x) = 4x^3 - 88x^2 + 420x$. $V' = 12x^2 - 176x + 420 = 4(3x - 35)(x - 3) = 0 \Rightarrow x = 3$ ($\frac{35}{3} > 7$ rejected). $V'' = 24x - 176 = -104 < 0$: max. $V = 3 \times 24 \times 8 = 576$ cm^3.

Q46. The sum of the perimeters of a circle and a square is k. Prove that the sum of their areas is least when the side of the square is twice the radius of the circle. 5M Board fav

Ans. $2\pi r + 4x = k \Rightarrow x = \frac{k - 2\pi r}{4}$. $A = \pi r^2 + \frac{(k - 2\pi r)^2}{16}$. $A' = 2\pi r - \frac{\pi(k - 2\pi r)}{4} = 0 \Rightarrow k - 2\pi r = 8r$, i.e. $4x = 8r \Rightarrow x = 2r$. $A'' = 2\pi + \frac{\pi^2}{2} > 0$: least.

Q47. A window has the shape of a rectangle with an equilateral triangle on top. The perimeter is 12 m. Find the width for maximum area. 5M HOTS

Ans. Width x, rectangle height y: $3x + 2y = 12 \Rightarrow y = \frac{12 - 3x}{2}$. $A = xy + \frac{\sqrt{3}}{4}x^2 = 6x - \frac{3}{2}x^2 + \frac{\sqrt{3}}{4}x^2$. $A' = 6 - 3x + \frac{\sqrt{3}}{2}x = 0 \Rightarrow x = \frac{12}{6 - \sqrt{3}}$ m; $A'' = -3 + \frac{\sqrt{3}}{2} < 0$: max.

Q48. Find the point on the parabola $x^2 = 4y$ nearest to the point $(0, 5)$ and the least distance. 5M Board fav

Ans. Point $(x, \frac{x^2}{4})$: $D = x^2 + \left(\frac{x^2}{4} - 5\right)^2$. With $y = \frac{x^2}{4}$: $D = 4y + (y - 5)^2$, $\frac{dD}{dy} = 4 + 2(y - 5) = 0 \Rightarrow y = 3$, $\frac{d^2D}{dy^2} = 2 > 0$: min. $x^2 = 12 \Rightarrow x = \pm 2\sqrt{3}$. Points $(\pm 2\sqrt{3}, 3)$, least distance $\sqrt{12 + 4} = 4$.

Q49. Show that a right circular cone of given volume and least curved surface area has altitude equal to $\sqrt{2}$ times the radius of its base. 5M Board fav

Ans. $V = \frac{1}{3}\pi r^2 h \Rightarrow h = \frac{3V}{\pi r^2}$. Minimise $Z = S^2 = \pi^2 r^2(r^2 + h^2) = \pi^2 r^4 + \frac{9V^2}{r^2}$. $Z' = 4\pi^2 r^3 - \frac{18V^2}{r^3} = 0 \Rightarrow 9V^2 = 2\pi^2 r^6$, i.e. $\pi^2 r^4 h^2 = 2\pi^2 r^6 \Rightarrow h = \sqrt{2}r$. $Z'' = 12\pi^2 r^2 + \frac{54V^2}{r^4} > 0$: least.

Q50. Find the maximum area of an isosceles triangle inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with its vertex at $(a, 0)$. 5M HOTS

Ans. Base ends $(a\cos\theta, \pm b\sin\theta)$: $A = \frac{1}{2}(2b\sin\theta)(a - a\cos\theta) = ab\sin\theta(1 - \cos\theta)$. $A' = ab(\cos\theta - \cos 2\theta) = 0 \Rightarrow \theta = \frac{2\pi}{3}$; $A'' = ab(-\sin\theta + 2\sin 2\theta) = -\frac{3\sqrt{3}}{2}ab < 0$: max. $A_{max} = ab \times \frac{\sqrt{3}}{2} \times \frac{3}{2} = \frac{3\sqrt{3}}{4}ab$.

Case based (4 marks)

Q51. Fencing by the river. A farmer has 600 m of fencing to enclose a rectangular field on three sides; the fourth side is a straight river. Let x m be each side perpendicular to the river. (i) Write the area $A(x)$. (1) (ii) Find the critical point of A. (1) (iii) Find the dimensions and the maximum area, showing it is a maximum. (2) 4M Board fav

Ans. (i) $A = x(600 - 2x) = 600x - 2x^2$. (ii) $A' = 600 - 4x = 0 \Rightarrow x = 150$. (iii) $A'' = -4 < 0$: max. Field 150 m $\times$ 300 m, max area 45000 m^2.

Q52. Shop profit. A shop's monthly profit (in Rs hundred) when it sells x hundred items is $P(x) = -5x^2 + 120x + 2500$. (i) Find the marginal profit $P'(x)$ at $x = 10$. (1) (ii) For what x is the profit maximum? (1) (iii) Find the maximum profit and justify it is maximum. (2) 4M

Ans. (i) $P'(x) = -10x + 120$; $P'(10) = 20$. (ii) $P' = 0 \Rightarrow x = 12$ (1200 items). (iii) $P'' = -10 < 0$: max. $P(12) = -720 + 1440 + 2500 = 3220$, i.e. Rs 3,22,000.

Q53. Tin can. A company makes closed cylindrical cans of volume 128π cm^3 and wants to use the least metal. (i) Express the total surface area S in terms of r. (1) (ii) Find r for which $\frac{dS}{dr} = 0$. (1) (iii) Find the height and the least surface area, and show it is least. (2) 4M Must do

Ans. (i) $h = \frac{128}{r^2}$, $S = 2\pi r^2 + \frac{256\pi}{r}$. (ii) $S' = 4\pi r - \frac{256\pi}{r^2} = 0 \Rightarrow r^3 = 64 \Rightarrow r = 4$ cm. (iii) $S'' = 4\pi + \frac{512\pi}{r^3} > 0$: least. $h = 8$ cm $= 2r$, $S = 32\pi + 64\pi = 96\pi$ cm^2.

Q54. Balloon festival. A spherical balloon is being inflated so that its volume increases at 100 cm^3/s. (i) Write $\frac{dV}{dt}$ in terms of r and $\frac{dr}{dt}$. (1) (ii) Find $\frac{dr}{dt}$ when $r = 5$ cm. (1) (iii) Find the rate of increase of its surface area at that instant. (2) 4M Board fav

Ans. (i) $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. (ii) $100 = 4\pi(25)\frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{1}{\pi}$ cm/s. (iii) $\frac{dS}{dt} = 8\pi r \frac{dr}{dt} = 40$ cm^2/s.

Q55. Gift box. A gift box without a lid is made from an 18 cm $\times$ 18 cm card by cutting equal squares of side x cm from the corners. (i) Write the volume $V(x)$. (1) (ii) Find x for which $V'(x) = 0$ in the valid range. (1) (iii) Find the maximum volume and justify. (2) 4M Most asked

Ans. (i) $V = x(18-2x)^2 = 4x^3 - 72x^2 + 324x$, $0 < x < 9$. (ii) $V' = 12x^2 - 144x + 324 = 12(x-3)(x-9) = 0 \Rightarrow x = 3$. (iii) $V'' = 24x - 144 = -72 < 0$ at $x = 3$: max. $V = 3 \times 12^2 = 432$ cm^3.

Ek nazar mein: Application of Derivatives

Type of sawal	Method	Marks
Rate of change	chain rule, differentiate then put value	2-3
Intervals inc/dec	f' factors, sign chart	3
Local max/min	$f' = 0$, then f'' sign	3
Abs max/min on $[a, b]$	critical + end points	2-3
Word problem	one variable, $f' = 0$, f'' test	5

Last minute: 8 one-liners

1. $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$; negative rate = decreasing.
2. $f' > 0$ strictly increasing, $f' < 0$ strictly decreasing.
3. Critical point: $f' = 0$ or f' does not exist.
4. 1st test: $+ \to -$ max, $- \to +$ min.
5. 2nd test: $f'' < 0$ max, $f'' > 0$ min, $f'' = 0$ fails.
6. Closed interval: check end points too.
7. Closed cylinder least surface: $h = 2r$.
8. Local max value can be smaller than a local min value.

5-mark answer ka format (copy this layout)

1. **Let** ... (define variables with units, draw the figure).
2. **Given** condition as an equation; express one variable in terms of the other.
3. **Function** to optimise in one variable, with its domain.
4. **Differentiate:** $\frac{dQ}{dx} = 0$ gives $x = \ldots$ (reject values outside the domain).
5. **Test:** $\frac{d^2Q}{dx^2}$ at that x is < 0 (max) or > 0 (min).
6. **Conclusion:** "Hence Q is maximum when ...", value with units.

Quick fact $|x|$ has a local min at 0, though $f'(0)$ does not exist.

Khud karo 2: board-level practice (answers given)

1. The side of a square increases at 3 cm/s. Rate of increase of its area when side $= 10$ cm. [60 cm^2/s]
2. Intervals where $f(x) = x^3 - 12x + 5$ is decreasing. [$(-2, 2)$]
3. Local max and min values of $f(x) = x^3 - 12x + 5$. [max 21 at $x = -2$, min -11 at $x = 2$]
4. Abs max and min of $f(x) = x^2 - 4x + 3$ on $[0, 3]$. [max 3 at $x = 0$, min -1 at $x = 2$]
5. Two positive numbers with sum 20 and maximum product. [10, 10; product 100]
6. Show $f(x) = e^x + x$ is strictly increasing on R. [$f' = e^x + 1 > 0$]
7. Rectangle of perimeter 36 cm with maximum area. [square of side 9 cm, area 81 cm^2]
8. Max value of $f(x) = \sin x + \sqrt{3}\cos x$. [2]

Yaad karo: "RIM: Rate, Interval, Max-min"

Is chapter ke teen hi kaam hain: **R**ate ($\frac{dy}{dt}$ by chain rule), **I**nterval (sign of f'), **M**ax-min ($f' = 0$, then f'' or end points). Har sawal padh ke pehle decide karo: R, I ya M?

Khud karo 3: one-minute drills

1. Rate of change of area of circle w.r.t. r at $r = 4$ [8π]
2. Interval where $f(x) = 6 - x^2$ is decreasing [$(0, \infty)$]
3. Local min value of $f(x) = x^2 - 6x + 10$ [1 at $x = 3$]
4. Abs max of $f(x) = x^3$ on $[-2, 1]$ [1 at $x = 1$]
5. Two numbers with sum 12 and maximum product [$6, 6$ (product 36)]

Quick fact Open box from square of side a: cut $x = \frac{a}{6}$ for max volume.

Quick fact $f(x) = x^3$ is strictly increasing on $\mathbb{R}$ even though $f'(0) = 0$.

Trap Critical point does not mean max/min: check sign change of f' or f''.

Quick check $f(x) = e^x$ has no maximum or minimum on $\mathbb{R}$.

Most asked: partial fractions, by parts, definite integral properties (5M)

INTEGRALS

1. Integration: Reverse of Differentiation

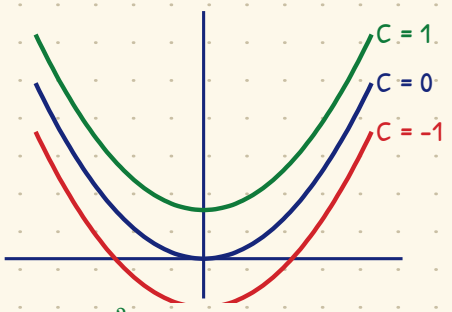
If $\frac{d}{dx}F(x) = f(x)$, then we write

$$\int f(x)\,dx = F(x) + C$$

Integration is the inverse process of differentiation. $F(x)$ is called an antiderivative (primitive) of $f(x)$.

Key terms

- **Integrand:** the function $f(x)$ being integrated.
- **Variable of integration:** x in dx.
- **Constant of integration C:** any real number. $\frac{d}{dx}(x^2+5) = \frac{d}{dx}(x^2-7) = 2x$, so $\int 2x\,dx = x^2 + C$.
- **Indefinite integral:** the whole family $F(x)+C$. Geometrically, $y = F(x)+C$ is a family of parallel curves (each one shifted up/down).



$\int 2x\,dx = x^2 + C$: a family of parallel curves

Properties of indefinite integrals

1. $\frac{d}{dx}\int f(x)\,dx = f(x)$ and $\int f'(x)\,dx = f(x) + C$
2. $\int [f(x) \pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$
3. $\int k\,f(x)\,dx = k\int f(x)\,dx$ (k constant)
4. Two antiderivatives of the same function differ only by a constant.

↳ Product ka integral = integrals ka product? **Bilkul nahi!** $\int fg\,dx \neq \int f\,dx \times \int g\,dx$

Try it (i) $\int x\sqrt{x+1}\,dx$ (ii) $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}dx$

Ans: (i) $\frac{2}{5}(x+1)^{5/2} - \frac{2}{3}(x+1)^{3/2}$ (put $x+1=t$) (ii) $2e^{\sqrt{x}}$, $+C$ for indefinite

Quick fact $\int \log x\,dx = x\log x - x + C$ (take 1 as second function).

Formula Box 1: Basic integrals

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C,\ n \neq -1$
- $\int 1\,dx = x + C$
- $\int \frac{1}{x}dx = \log|x| + C$
- $\int e^x dx = e^x + C$
- $\int a^x dx = \frac{a^x}{\log a} + C$
- $\int \frac{1}{\sqrt{x}}dx = 2\sqrt{x} + C$

Formula Box 2: Trigonometric

- $\int \sin x\,dx = -\cos x + C$
- $\int \cos x\,dx = \sin x + C$
- $\int \sec^2 x\,dx = \tan x + C$
- $\int \operatorname{cosec}^2 x\,dx = -\cot x + C$
- $\int \sec x\tan x\,dx = \sec x + C$
- $\int \operatorname{cosec} x\cot x\,dx = -\operatorname{cosec} x + C$

Formula Box 3: Inverse trig + linear rule

- $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1}x + C$
- $\int \frac{dx}{1+x^2} = \tan^{-1}x + C$
- $\int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1}x + C$
- Linear rule: if $\int f(x)dx = F(x)$, then $\int f(ax+b)\,dx = \frac{F(ax+b)}{a} + C$

Yaad karo: Linear andar? a se divide!

Andar $ax+b$ ho to normal formula lagao, phir **coefficient of x se divide** karo: $\int e^{3x}dx = \frac{e^{3x}}{3}$, $\int \cos(5x+2)dx = \frac{\sin(5x+2)}{5}$, $\int \frac{dx}{2x+1} = \frac{1}{2}\log|2x+1|$.

Try it (i) $\int \sin x \sin 2x\,dx$ (ii) $\int \frac{dx}{1+\cos x}$

Ans: (i) $\frac{1}{2}\sin x - \frac{1}{6}\sin 3x$ (ii) $\tan\frac{x}{2}$, $+C$ for indefinite

Quick fact $\int_a^a f(x)\,dx = 0$ and $\int_a^b f = -\int_b^a f$.

Solved Example 1

Evaluate $\int \frac{x^4+3x^2-5}{x^2}\,dx$.

1. Split: $\frac{x^4+3x^2-5}{x^2} = x^2+3-5x^{-2}$
2. Integrate term by term: $\frac{x^3}{3}+3x-5\times\frac{x^{-1}}{-1}$

✔ $\frac{x^3}{3}+3x+\frac{5}{x}+C$

Solved Example 2

Evaluate $\int \frac{3-2\sin x}{\cos^2 x}\,dx$.

1. $= \int 3\sec^2 x\,dx - 2\int \frac{\sin x}{\cos x}\times\frac{1}{\cos x}\,dx$
2. $= 3\int \sec^2 x\,dx - 2\int \sec x\tan x\,dx$

✔ $3\tan x - 2\sec x + C$

Solved Example 3

Evaluate $\int \frac{dx}{\sin^2 x\cos^2 x}$.

1. Write $1=\sin^2 x+\cos^2 x$: $\frac{\sin^2 x+\cos^2 x}{\sin^2 x\cos^2 x} = \sec^2 x+\operatorname{cosec}^2 x$
2. Integrate: $\tan x+(-\cot x)$

✔ $\tan x-\cot x+C$

Solved Example 4

Find $f(x)$ if $f'(x)=4x^3-\frac{3}{x^4}$ and $f(2)=0$.

1. $f(x)=\int(4x^3-3x^{-4})dx = x^4+x^{-3}+C$
2. $f(2)=16+\frac{1}{8}+C=0 \Rightarrow C=-\frac{129}{8}$

✔ $f(x)=x^4+\frac{1}{x^3}-\frac{129}{8}$

2. Integration by Substitution

If the integrand contains a function $g(x)$ **and its derivative** $g'(x)$, put

$$g(x)=t \;\Rightarrow\; g'(x)\,dx=dt$$

The integral becomes simpler in t. Integrate, then put back $t=g(x)$.

Spot $g(x)$ whose derivative is present → put $g(x)=t$, find dx → integrate in t → back to x

Try it (i) $\int \frac{dx}{x^2+x+1}$ (ii) $\int \frac{\log x}{x}dx$

Ans: (i) $\frac{2}{\sqrt{3}}\tan^{-1}\frac{2x+1}{\sqrt{3}}$ (ii) $\frac{(\log x)^2}{2}$, $+C$ for indefinite

Three super results (from substitution)

→ $\int \frac{f'(x)}{f(x)}\,dx = \log|f(x)|+C$

→ $\int [f(x)]^n f'(x)\,dx = \frac{[f(x)]^{n+1}}{n+1}+C, n\neq -1$

→ $\int \frac{f'(x)}{\sqrt{f(x)}}\,dx = 2\sqrt{f(x)}+C$

Formula Box 4: tan, cot, sec, cosec

→ $\int \tan x\,dx = \log|\sec x|+C = -\log|\cos x|+C$

→ $\int \cot x\,dx = \log|\sin x|+C$

→ $\int \sec x\,dx = \log|\sec x+\tan x|+C$

→ $\int \operatorname{cosec} x\,dx = \log|\operatorname{cosec} x-\cot x|+C$

↳ Proof of $\int\tan x\,dx$: $\int\frac{\sin x}{\cos x}dx$, put $\cos x=t$, $-\sin x\,dx=dt \Rightarrow -\int\frac{dt}{t}=-\log|\cos x|$.

Yaad karo: Upar derivative, neeche function = log

Numerator agar denominator ka **derivative** hai (ya uska multiple), to seedha log |Denominator|. Jaise $\int\frac{2x+3}{x^2+3x+7}dx=\log|x^2+3x+7|+C$.

Solved Example 5

Evaluate: (i) $\int x\,e^{x^2}dx$ (ii) $\int\frac{(\log x)^2}{x}dx$ (iii) $\int\frac{e^{\tan^{-1}x}}{1+x^2}dx$

1. (i) $x^2=t$, $2x\,dx=dt$: $\frac{1}{2}\int e^t dt=\frac{1}{2}e^{x^2}+C$
2. (ii) $\log x=t$, $\frac{dx}{x}=dt$: $\int t^2dt=\frac{t^3}{3}=\frac{(\log x)^3}{3}+C$
3. (iii) $\tan^{-1}x=t$, $\frac{dx}{1+x^2}=dt$: $\int e^t dt = e^{\tan^{-1}x}+C$

✔ (i) $\frac{1}{2}e^{x^2}$ (ii) $\frac{(\log x)^3}{3}$ (iii) $e^{\tan^{-1}x}$, each $+C$

Solved Example 6

Evaluate $\int\frac{dx}{x+x\log x}$.

1. $= \int\frac{dx}{x(1+\log x)}$. Put $1+\log x=t$, $\frac{dx}{x}=dt$
2. $= \int\frac{dt}{t}=\log|t|$

✔ $\log|1+\log x|+C$

Quick fact $\int e^{ax}dx=\frac{e^{ax}}{a}+C$, $\int\sin ax\,dx=-\frac{\cos ax}{a}+C$.

Solved Example 7

Evaluate $\int \frac{\sin x}{\sin(x-a)}\,dx$. (Board fav)

1. Put $x-a=t \Rightarrow x=t+a,\ dx=dt$
2. $\int \frac{\sin(t+a)}{\sin t}dt = \int \frac{\sin t\cos a+\cos t\sin a}{\sin t}dt$
3. $= \int (\cos a+\sin a\cot t)\,dt = t\cos a + \sin a\log|\sin t|$
4. $t=x-a$; the constant $-a\cos a$ merges into C

✔ $x\cos a+\sin a\log|\sin(x-a)|+C$

Solved Example 8

Evaluate $\int \frac{e^{2x}-1}{e^{2x}+1}\,dx$.

1. Divide num and den by e^x: $\int \frac{e^x-e^{-x}}{e^x+e^{-x}}dx$
2. Numerator = derivative of denominator, so use $\int \frac{f'}{f} = \log|f|$

✔ $\log(e^x+e^{-x})+C$

Solved Example 9

Evaluate $\int \frac{\sqrt{\tan x}}{\sin x\cos x}\,dx$.

1. Divide num and den by $\cos^2 x$: $\int \frac{\sqrt{\tan x}\,\sec^2 x}{\tan x}dx = \int \frac{\sec^2 x}{\sqrt{\tan x}}dx$
2. Put $\tan x=t$, $\sec^2 x\,dx=dt$: $\int t^{-1/2}dt = 2\sqrt{t}$

✔ $2\sqrt{\tan x}+C$

Solved Example 10

Evaluate $\int \frac{dx}{\sqrt{x}+x}$.

1. $= \int \frac{dx}{\sqrt{x}(1+\sqrt{x})}$. Put $1+\sqrt{x}=t$, $\frac{dx}{2\sqrt{x}}=dt$
2. $= \int \frac{2\,dt}{t} = 2\log|t|$

✔ $2\log(1+\sqrt{x})+C$

Try it (i) $\int \frac{x\,dx}{(x+1)(x+2)}$ (ii) $\int xe^{-x}dx$

Ans: (i) $2\log|x+2|-\log|x+1|$ (ii) $-e^{-x}(x+1)$, $+C$ for indefinite

Quick fact Definite integral of an odd function over $[-a,a]$ is always 0.

Standard trig substitutions (for roots)

Expression	Put	Root becomes
$\sqrt{a^2-x^2}$	$x=a\sin\theta$	$a\cos\theta$
$\sqrt{a^2+x^2}$	$x=a\tan\theta$	$a\sec\theta$
$\sqrt{x^2-a^2}$	$x=a\sec\theta$	$a\tan\theta$

Example: $\int \frac{dx}{(x^2+a^2)^{3/2}}$, put $x=a\tan\theta$:

$\int \frac{a\sec^2\theta\,d\theta}{a^3\sec^3\theta} = \frac{\sin\theta}{a^2} = \frac{x}{a^2\sqrt{x^2+a^2}}+C$

3. Using Trigonometric Identities

Powers and products of sin, cos cannot be integrated directly. First **convert them into sums** using identities, then integrate each term.

Formula Box 5: Identities you need

- $\sin^2 x = \frac{1-\cos 2x}{2}$
- $\cos^2 x = \frac{1+\cos 2x}{2}$
- $\sin^3 x = \frac{3\sin x-\sin 3x}{4}$
- $\cos^3 x = \frac{3\cos x+\cos 3x}{4}$
- $2\sin A\cos B = \sin(A+B)+\sin(A-B)$
- $2\cos A\cos B = \cos(A+B)+\cos(A-B)$
- $2\sin A\sin B = \cos(A-B)-\cos(A+B)$
- $\tan^2 x = \sec^2 x-1$, $\cot^2 x = \operatorname{cosec}^2 x-1$
- $1+\cos 2x = 2\cos^2 x$, $1-\cos 2x = 2\sin^2 x$

Solved Example 11

Evaluate $\int \sin 3x\cos 2x\,dx$.

1. $\sin 3x\cos 2x = \frac{1}{2}[\sin 5x+\sin x]$
2. $\frac{1}{2}\left[-\frac{\cos 5x}{5}-\cos x\right]$

✔ $-\frac{\cos 5x}{10}-\frac{\cos x}{2}+C$

Solved Example 12

Evaluate $\int \cos^3 x\,dx$.

1. $\cos^3 x = (1-\sin^2 x)\cos x$. Put $\sin x=t$, $\cos x\,dx=dt$
2. $\int (1-t^2)dt = t-\frac{t^3}{3}$

✔ $\sin x-\frac{\sin^3 x}{3}+C$

Solved Example 13

Evaluate $\int \frac{\sin^2 x}{1+\cos x}\,dx$.

1. $\sin^2 x = 1-\cos^2 x = (1-\cos x)(1+\cos x)$
2. So integrand $= 1-\cos x$

✔ $x - \sin x + C$

Solved Example 14

Evaluate $\int \frac{dx}{1+\sin x}$.

1. Multiply by $(1-\sin x)$: $\frac{1-\sin x}{1-\sin^2 x} = \frac{1-\sin x}{\cos^2 x}$
2. $= \sec^2 x - \sec x \tan x$

✔ $\tan x - \sec x + C$

Solved Example 15

Evaluate $\int \sin^4 x\,dx$. (HOTS)

1. $\sin^4 x = \left(\frac{1-\cos 2x}{2}\right)^2 = \frac{1}{4}(1 - 2\cos 2x + \cos^2 2x)$
2. $\cos^2 2x = \frac{1+\cos 4x}{2}$, so $\sin^4 x = \frac{3}{8} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x$
3. Integrate each term

✔ $\frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$

Yaad karo: Odd power = ek tod do

$\sin^3 x$, $\cos^5 x$ jaisi **odd power**: ek factor alag karo, baaki ko $1-\cos^2$ ya $1-\sin^2$ mein badlo, phir substitution. **Even power**: $\cos 2x$ wali identity se power girao.

4. Special Integrals (6 Standard Forms)

These six results are proved by substitution / partial fractions. Learn them by heart: half of the board's 2 and 3 mark integrals use them.

Formula Box 6: Without root

$$\to \int \frac{dx}{x^2-a^2} = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right| + C$$

$$\to \int \frac{dx}{a^2-x^2} = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + C$$

$$\to \int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$$

Try it (i) $\int \cot^2 x\,dx$ (ii) $\int \frac{x^2}{1+x^2}dx$

Ans: (i) $-\cot x - x$ (ii) $x - \tan^{-1} x$, $+C$ for indefinite

Formula Box 7: With root

$$\to \int \frac{dx}{\sqrt{x^2-a^2}} = \log\left|x+\sqrt{x^2-a^2}\right| + C$$

$$\to \int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + C$$

$$\to \int \frac{dx}{\sqrt{x^2+a^2}} = \log\left|x+\sqrt{x^2+a^2}\right| + C$$

Yaad karo: Jo aage, wo upar aage

Log wale do formulas mein confusion? Denominator mein jo **pehle** likha hai, wahi numerator ke log mein **pehle**: $x^2-a^2 \to \frac{x-a}{x+a}$, $a^2-x^2 \to \frac{a+x}{a-x}$. Aur root wale: **sirf** a^2-x^2 **pe** $\sin^{-1}$, baaki dono pe $\log|x+\sqrt{\ }|$.

Solved Example 16

Evaluate: (i) $\int \frac{dx}{9x^2-4}$ (ii) $\int \frac{dx}{\sqrt{4-9x^2}}$ (iii) $\int \frac{x\,dx}{x^4+1}$

1. (i) $9x^2-4 = 9\left[x^2 - \left(\frac{2}{3}\right)^2\right]$, so $I = \frac{1}{9}\times \frac{1}{2\times\frac{2}{3}}\log\left|\frac{x-\frac{2}{3}}{x+\frac{2}{3}}\right|$
2. (ii) $\frac{1}{3}\int \frac{dx}{\sqrt{(2/3)^2-x^2}} = \frac{1}{3}\sin^{-1}\frac{3x}{2}$
3. (iii) $x^2 = t$, $x\,dx = \frac{dt}{2}$: $\frac{1}{2}\int \frac{dt}{t^2+1} = \frac{1}{2}\tan^{-1} t$

✔ (i) $\frac{1}{12}\log\left|\frac{3x-2}{3x+2}\right|$ (ii) $\frac{1}{3}\sin^{-1}\frac{3x}{2}$ (iii) $\frac{1}{2}\tan^{-1} x^2$, $+C$

Type 1: $\int \frac{dx}{ax^2+bx+c}$ and $\int \frac{dx}{\sqrt{ax^2+bx+c}}$

Method: take a out, then complete the square:

$$x^2 + \frac{b}{a}x + \frac{c}{a} = \left(x+\frac{b}{2a}\right)^2 \pm k^2$$

Now it is one of the six standard forms with $x+\frac{b}{2a}$ in place of x.

Solved Example 17

Evaluate $\int \frac{dx}{x^2+6x+13}$.

1. $x^2+6x+13 = (x+3)^2+4 = (x+3)^2+2^2$
2. Use $\int \frac{dx}{X^2+a^2}$ with $X = x+3$, $a=2$

✔ $\frac{1}{2}\tan^{-1}\frac{x+3}{2} + C$

Quick fact $\int \frac{dx}{x\log x} = \log|\log x| + C$.

Solved Example 18

Evaluate $\int \frac{dx}{\sqrt{8+2x-x^2}}$.

1. $8+2x-x^2 = 8-(x^2-2x) = 8-[(x-1)^2-1] = 9-(x-1)^2$
2. $\int \frac{dx}{\sqrt{3^2-(x-1)^2}} = \sin^{-1}\frac{x-1}{3}$

✔ $\sin^{-1}\frac{x-1}{3}+C$

Solved Example 19

Evaluate $\int \frac{dx}{2x^2-3x+1}$.

1. $= \frac{1}{2}\int \frac{dx}{x^2-\frac{3}{2}x+\frac{1}{2}} = \frac{1}{2}\int \frac{dx}{\left(x-\frac{3}{4}\right)^2-\left(\frac{1}{4}\right)^2}$
2. $= \frac{1}{2}\times\frac{1}{2\times\frac{1}{4}}\log\left|\frac{x-\frac{3}{4}-\frac{1}{4}}{x-\frac{3}{4}+\frac{1}{4}}\right| = \log\left|\frac{x-1}{x-\frac{1}{2}}\right|$
3. $= \log\left|\frac{2(x-1)}{2x-1}\right|$; the $\log 2$ goes into C

✔ $\log\left|\frac{x-1}{2x-1}\right|+C$

Solved Example 20

Evaluate $\int \frac{dx}{\sqrt{x^2+4x+1}}$.

1. $x^2+4x+1 = (x+2)^2-3 = (x+2)^2-(\sqrt{3})^2$
2. Use $\int \frac{dX}{\sqrt{X^2-a^2}} = \log|X+\sqrt{X^2-a^2}|$

✔ $\log|x+2+\sqrt{x^2+4x+1}|+C$

Type 2: $\int \frac{px+q}{ax^2+bx+c}dx$ **and** $\int \frac{px+q}{\sqrt{ax^2+bx+c}}dx$

Method: write numerator = $A\times$(derivative of the quadratic) $+B$:

$$px+q = A(2ax+b)+B$$

Find A, B by comparing coefficients. The A part gives log (or $2\sqrt{\ }$), the B part is Type 1.

Try it (i) $\int_0^2 \sqrt{4-x^2}\,dx$ (ii) $\int_1^2 \log x\,dx$

Ans: (i) π (quarter circle of radius 2) (ii) $2\log 2-1$, $+C$ for indefinite

Quick fact $\frac{d}{dx}\int_0^x f(t)\,dt = f(x)$ (first FTC).

Solved Example 21

Evaluate $\int \frac{x+2}{x^2+2x+5}\,dx$.

1. $x+2 = A(2x+2)+B$: $2A=1$, $2A+B=2$ $\Rightarrow A=\frac{1}{2}$, $B=1$
2. $I = \frac{1}{2}\int \frac{2x+2}{x^2+2x+5}dx + \int \frac{dx}{(x+1)^2+2^2}$
3. $= \frac{1}{2}\log(x^2+2x+5)+\frac{1}{2}\tan^{-1}\frac{x+1}{2}$

✔ $\frac{1}{2}\log(x^2+2x+5)+\frac{1}{2}\tan^{-1}\frac{x+1}{2}+C$

Solved Example 22

Evaluate $\int \frac{3x+1}{\sqrt{x^2+2x+5}}\,dx$. (Board fav)

1. $3x+1 = A(2x+2)+B$: $A=\frac{3}{2}$, $B=1-3=-2$
2. $I = \frac{3}{2}\int \frac{2x+2}{\sqrt{x^2+2x+5}}dx - 2\int \frac{dx}{\sqrt{(x+1)^2+2^2}}$
3. First: $\int \frac{f'}{\sqrt{f}} = 2\sqrt{f}$. Second: $\log|X+\sqrt{X^2+a^2}|$

✔ $3\sqrt{x^2+2x+5} - 2\log|x+1+\sqrt{x^2+2x+5}|+C$

Solved Example 23

Evaluate $\int \frac{\cos x}{\sin^2 x+4\sin x+5}\,dx$.

1. Put $\sin x = t$, $\cos x\,dx = dt$: $\int \frac{dt}{t^2+4t+5} = \int \frac{dt}{(t+2)^2+1}$
2. $= \tan^{-1}(t+2)$

✔ $\tan^{-1}(\sin x+2)+C$

Check karo

Special integrals mein sabse zyada marks **completing the square** mein katte hain. Har baar check karo: $(x+3)^2+4$ ko expand karke wapas $x^2+6x+13$ aata hai na? 10 second ka check, 3 marks safe.

5. Integration by Partial Fractions

Used for a **rational function** $\frac{P(x)}{Q(x)}$. If it is **improper** (degree of $P \geq$ degree of Q), first **divide** to get polynomial + proper fraction. Then factorise $Q(x)$ and split as in the table.

Quick fact $\int \sqrt{x}\,dx = \frac{2}{3}x^{3/2}+C$.

↳Last row: x^2+bx+c cannot be factorised further (discriminant negative). Uske upar hamesha $Bx+C$.

Form of the fraction	Partial fractions
$\frac{px+q}{(x-a)(x-b)}$	$\frac{A}{x-a}+\frac{B}{x-b}$
$\frac{px+q}{(x-a)^2}$	$\frac{A}{x-a}+\frac{B}{(x-a)^2}$
$\frac{px^2+qx+r}{(x-a)(x-b)(x-c)}$	$\frac{A}{x-a}+\frac{B}{x-b}+\frac{C}{x-c}$
$\frac{px^2+qx+r}{(x-a)^2(x-b)}$	$\frac{A}{x-a}+\frac{B}{(x-a)^2}+\frac{C}{x-b}$
$\frac{px^2+qx+r}{(x-a)(x^2+bx+c)}$	$\frac{A}{x-a}+\frac{Bx+C}{x^2+bx+c}$

Yaad karo: Haath rakho, x daalo (cover-up)

Distinct linear factors ho to A nikalne ke liye: denominator mein $(x-a)$ pe **haath rakho**, baaki expression mein $x=a$ **daalo**. $\frac{3x-1}{(x-1)(x+2)}$: $A=\frac{3-1}{1+2}=\frac{2}{3}$, $B=\frac{-6-1}{-2-1}=\frac{7}{3}$. Repeated factor ke liye sirf highest power wala aise milta hai.

Solved Example 24

Evaluate $\int \frac{3x-1}{(x-1)(x+2)}\,dx$.

1. $\frac{3x-1}{(x-1)(x+2)}=\frac{A}{x-1}+\frac{B}{x+2}$, so $3x-1=A(x+2)+B(x-1)$
2. $x=1$: $2=3A\Rightarrow A=\frac{2}{3}$. $x=-2$: $-7=-3B\Rightarrow B=\frac{7}{3}$

✔ $\frac{2}{3}\log|x-1|+\frac{7}{3}\log|x+2|+C$

Solved Example 25

Evaluate $\int \frac{x}{(x-1)^2(x+2)}\,dx$.

1. $x=A(x-1)(x+2)+B(x+2)+C(x-1)^2$
2. $x=1$: $1=3B\Rightarrow B=\frac{1}{3}$. $x=-2$: $-2=9C\Rightarrow C=-\frac{2}{9}$
3. Compare x^2: $0=A+C\Rightarrow A=\frac{2}{9}$
4. $\int \frac{dx}{(x-1)^2}=-\frac{1}{x-1}$

✔ $\frac{2}{9}\log\left|\frac{x-1}{x+2}\right|-\frac{1}{3(x-1)}+C$

Try it (i) $\int_0^1 \frac{x}{(1+x)^2}dx$ (ii) $\int e^{2x+3}dx$

Ans: (i) $\log 2-\frac{1}{2}$ (ii) $\frac{1}{2}e^{2x+3}$, $+C$ for indefinite

Solved Example 26

Evaluate $\int \frac{dx}{(x+1)(x^2+1)}$. (Board fav)

1. $1=A(x^2+1)+(Bx+C)(x+1)$. $x=-1$: $1=2A\Rightarrow A=\frac{1}{2}$
2. x^2: $0=A+B\Rightarrow B=-\frac{1}{2}$. Constant: $1=A+C\Rightarrow C=\frac{1}{2}$
3. $I=\frac{1}{2}\int\frac{dx}{x+1}-\frac{1}{4}\int\frac{2x\,dx}{x^2+1}+\frac{1}{2}\int\frac{dx}{x^2+1}$

✔ $\frac{1}{2}\log|x+1|-\frac{1}{4}\log(x^2+1)+\frac{1}{2}\tan^{-1}x+C$

Solved Example 27

Evaluate $\int \frac{x^2}{(x^2+1)(x^2+4)}\,dx$.

1. Only x^2 appears, so for splitting put $x^2=y$: $\frac{y}{(y+1)(y+4)}=\frac{-1/3}{y+1}+\frac{4/3}{y+4}$
2. $I=-\frac{1}{3}\int\frac{dx}{x^2+1}+\frac{4}{3}\int\frac{dx}{x^2+2^2}$

✔ $-\frac{1}{3}\tan^{-1}x+\frac{2}{3}\tan^{-1}\frac{x}{2}+C$

Solved Example 28

Evaluate $\int \frac{x^2+1}{x^2-5x+6}\,dx$. (improper)

1. Divide: $\frac{x^2+1}{x^2-5x+6}=1+\frac{5x-5}{(x-2)(x-3)}$
2. $5x-5=A(x-3)+B(x-2)$: $x=2\Rightarrow A=-5$, $x=3\Rightarrow B=10$

✔ $x-5\log|x-2|+10\log|x-3|+C$

Solved Example 29

Evaluate $\int \frac{\sec^2 x}{(\tan x+1)(\tan x+2)}\,dx$.

1. Put $\tan x=t$, $\sec^2 x\,dx=dt$: $\int\frac{dt}{(t+1)(t+2)}=\int\left(\frac{1}{t+1}-\frac{1}{t+2}\right)dt$
2. $=\log|t+1|-\log|t+2|$

✔ $\log\left|\frac{\tan x+1}{\tan x+2}\right|+C$

Try it (i) $\int \frac{dx}{\sqrt{x^2+6x+10}}$ (ii) $\int_0^{\pi}\sin^{-1}(\cos x)\,dx$

Ans: (i) $\log|x+3+\sqrt{x^2+6x+10}|$ (ii) 0 (it equals $\frac{\pi}{2}-x$ there), $+C$ for indefinite

Quick fact $\int_0^{\pi}\sin x\,dx=2$, $\int_0^{\pi/2}\sin x\,dx=1$.

Meme samjho



Ek mushkil fraction ko **chutki** mein 3 simple fractions mein tod diya. Perfectly balanced, jaisa hona chahiye.

Asli baat: Partial fractions ek bada rational function chhote parts mein todta hai, jinka integral seedha $\log$ ya $\tan^{-1}$ hota hai. Pehle check karo fraction proper hai.

6. Integration by Parts

For a **product of two functions** (where substitution does not work):

$$\int u\,v\,dx = u\int v\,dx - \int\left[\frac{du}{dx}\int v\,dx\right]dx$$

u = **first function** (gets differentiated), v = **second function** (gets integrated). Choose u by ILATE: the one coming **first** in the list is u.

Inverse trig → Log → Algebraic → Trig → Exponential

Formula Box 8: By parts results

- $\int u\,v\,dx = u\int v\,dx - \int(u'\int v\,dx)\,dx$
- $\int e^x[f(x)+f'(x)]\,dx = e^x f(x)+C$
- $\int\sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right|+C$
- $\int\sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\log\left|x+\sqrt{x^2+a^2}\right|+C$
- $\int\sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}+C$

Yaad karo: ILATE + "1" wala jugaad

Akela $\log x$, $\tan^{-1}x$, $\sin^{-1}x$ ho to usko 1 se multiply karo: $u=\log x, v=1$. Aur $x^n e^x$ jaise mein baar baar parts lagana hai: signs $+,-,+,-$ alternate karte hain ($\int x^2e^x = e^x(x^2-2x+2)$).

Solved Example 31

Evaluate $\int x^2e^x\,dx$.

1. $u=x^2$: $I = x^2e^x - \int 2x\,e^x dx$
2. Again: $\int 2xe^x dx = 2xe^x - 2e^x$
3. $I = x^2e^x - 2xe^x + 2e^x$

✔ $e^x(x^2-2x+2)+C$

Solved Example 30

Evaluate $\int x\cos 3x\,dx$.

1. ILATE: $u=x$ (A), $v=\cos 3x$ (T). $\int\cos 3x\,dx = \frac{\sin 3x}{3}$
2. $I = x\frac{\sin 3x}{3} - \int 1\times\frac{\sin 3x}{3}dx = \frac{x\sin 3x}{3} + \frac{\cos 3x}{9}$

✔ $\frac{x\sin 3x}{3}+\frac{\cos 3x}{9}+C$

Solved Example 32

Evaluate: (i) $\int\log x\,dx$ (ii) $\int\tan^{-1}x\,dx$ (iii) $\int\sin^{-1}x\,dx$

1. (i) $u=\log x, v=1$: $x\log x - \int x\times\frac{1}{x}dx = x\log x - x$
2. (ii) $x\tan^{-1}x - \int\frac{x}{1+x^2}dx = x\tan^{-1}x - \frac{1}{2}\log(1+x^2)$
3. (iii) $x\sin^{-1}x - \int\frac{x}{\sqrt{1-x^2}}dx = x\sin^{-1}x + \sqrt{1-x^2}$

✔ (i) $x\log x - x$ (ii) $x\tan^{-1}x - \frac{1}{2}\log(1+x^2)$ (iii) $x\sin^{-1}x+\sqrt{1-x^2}, +C$

Solved Example 33

Evaluate $\int x^2\log x\,dx$.

1. ILATE: Log comes before Algebraic, so $u=\log x$, $v=x^2$
2. $I = \log x\times\frac{x^3}{3} - \int\frac{1}{x}\times\frac{x^3}{3}dx = \frac{x^3}{3}\log x - \frac{x^3}{9}$

✔ $\frac{x^3}{3}\log x - \frac{x^3}{9}+C$

The $e^x[f+f']$ result

$$\int e^x\,[f(x)+f'(x)]\,dx = e^x f(x)+C$$

Proof: $\frac{d}{dx}[e^x f(x)] = e^x f(x)+e^x f'(x)$. Spot karo: bracket mein ek function aur **uska derivative**.

Quick fact $\int\frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a}+C$: the only root form with $\sin^{-1}$.

Quick fact $\int\operatorname{cosec}x\,dx = \log\left|\tan\frac{x}{2}\right|+C$ (same as $\log|\operatorname{cosec}x-\cot x|$).

Solved Example 34

Evaluate $\int e^x \frac{1+x}{(2+x)^2}\,dx$. (Most asked)

1. Make the pair: $\frac{1+x}{(2+x)^2} = \frac{(2+x)-1}{(2+x)^2} = \frac{1}{2+x} - \frac{1}{(2+x)^2}$
2. $f(x) = \frac{1}{x+2}$, $f'(x) = -\frac{1}{(x+2)^2}$: exactly $f + f'$

✔ $\frac{e^x}{x+2} + C$

Solved Example 35

Evaluate: (i) $\int e^x(\sin x + \cos x)\,dx$ (ii) $\int e^x\left(\frac{1}{x} - \frac{1}{x^2}\right)dx$ (iii) $\int e^x(\tan x + \sec^2 x)\,dx$

1. (i) $f = \sin x$, $f' = \cos x$ (ii) $f = \frac{1}{x}$, $f' = -\frac{1}{x^2}$ (iii) $f = \tan x$, $f' = \sec^2 x$

✔ (i) $e^x \sin x$ (ii) $\frac{e^x}{x}$ (iii) $e^x \tan x$, each $+C$

Solved Example 36

Evaluate $\int e^x \sin x\,dx$.

1. $I = e^x \sin x - \int e^x \cos x\,dx$ (taking $u = \sin x$)
2. $\int e^x \cos x\,dx = e^x \cos x + \int e^x \sin x\,dx = e^x \cos x + I$
3. $I = e^x \sin x - e^x \cos x - I \Rightarrow 2I = e^x(\sin x - \cos x)$

✔ $\frac{e^x}{2}(\sin x - \cos x) + C$

Solved Example 37

Evaluate $\int \sqrt{x^2+4x+8}\,dx$.

1. $x^2+4x+8 = (x+2)^2 + 2^2$. Use $\int \sqrt{X^2+a^2}\,dX$, $X = x+2$, $a = 2$
2. $= \frac{x+2}{2}\sqrt{x^2+4x+8} + \frac{4}{2}\log\left|x+2+\sqrt{x^2+4x+8}\right|$

✔ $\frac{x+2}{2}\sqrt{x^2+4x+8} + 2\log\left|x+2+\sqrt{x^2+4x+8}\right| + C$



Shivam Bhaiya says

By parts mein pehla step hi sab kuch hai: **ILATE likh ke u choose karo.** Galat u liya to integral aur mushkil ho jayega. Aur e^x dikhte hi pehle check karo: kya bracket $f + f'$ hai? Wo 4 line ka sawal 1 line mein khatam.

Solved Example 38

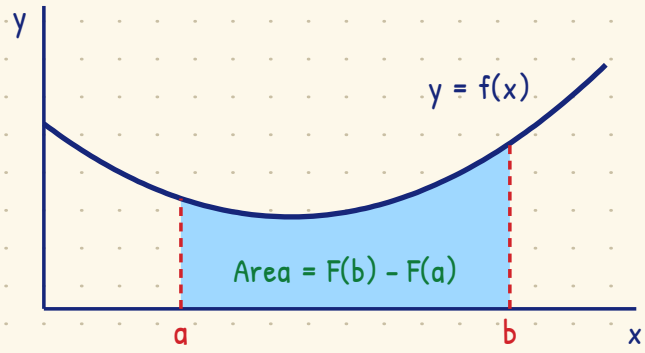
Evaluate $\int \sqrt{1-4x^2}\,dx$.

1. $= 2\int \sqrt{\left(\frac{1}{2}\right)^2 - x^2}\,dx =$ $2\left[\frac{x}{2}\sqrt{\frac{1}{4}-x^2} + \frac{1}{8}\sin^{-1} 2x\right]$
2. $x\sqrt{\frac{1}{4}-x^2} = \frac{x}{2}\sqrt{1-4x^2}$

✔ $\frac{x}{2}\sqrt{1-4x^2} + \frac{1}{4}\sin^{-1} 2x + C$

7. Definite Integrals and the FTC

$\int_a^b f(x)\,dx$ is a **definite integral**: a number, not a family of functions. a = lower limit, b = upper limit.



For $f(x) \geq 0$, $\int_a^b f(x)\,dx$ = area under the curve from $x = a$ to $x = b$

Fundamental Theorem of Calculus (statements, no proof)

→ **First FTC:** if $A(x) = \int_a^x f(t)\,dt$ (area function), then $A'(x) = f(x)$ for all x in $[a,b]$.

→ **Second FTC:** if F is any antiderivative of a continuous f on $[a,b]$, then

$$\int_a^b f(x)\,dx = \Big[F(x)\Big]_a^b = F(b) - F(a)$$

Steps to evaluate $\int_a^b f(x)\,dx$

1. Find the indefinite integral $F(x)$ (no need of C).
2. Put upper limit minus lower limit: $F(b) - F(a)$.
3. With **substitution**, change the limits also (then no need to go back to x).



Meme samjho

Definite integral mein C se kehna padta hai: "samay aa gaya hai... **vida** lene ka." **Asli baat:** $[F(x)+C]_a^b = F(b) + C - F(a) - C = F(b) - F(a)$. Constant cancel ho jata hai, isliye definite integral mein C nahi likhte.

Solved Example 39

Evaluate $\int_0^{\pi/2} \cos^2 x \, dx$.

1. $\cos^2 x = \dfrac{1+\cos 2x}{2}$
2. $\left[\dfrac{x}{2} + \dfrac{\sin 2x}{4}\right]_0^{\pi/2} = \left(\dfrac{\pi}{4} + 0\right) - 0$

✔ $\dfrac{\pi}{4}$

Solved Example 40

Evaluate $\int_0^1 \dfrac{x}{x^2+1} \, dx$.

1. Put $x^2+1 = t$, $2x\,dx = dt$. Limits: $x=0 \Rightarrow t=1$; $x=1 \Rightarrow t=2$
2. $\dfrac{1}{2}\int_1^2 \dfrac{dt}{t} = \dfrac{1}{2}\Big[\log t\Big]_1^2 = \dfrac{1}{2}(\log 2 - 0)$

✔ $\dfrac{1}{2}\log 2$

Solved Example 41

Evaluate $\int_0^1 x\,e^x \, dx$.

1. By parts: $\int xe^x dx = xe^x - e^x$
2. $\Big[e^x(x-1)\Big]_0^1 = e^1(0) - e^0(-1) = 0 + 1$

✔ 1

Solved Example 42

Evaluate $\int_0^{\pi/2} \dfrac{\sin x}{1+\cos^2 x} \, dx$.

1. Put $\cos x = t$, $-\sin x\,dx = dt$. $x=0 \Rightarrow t=1$, $x=\frac{\pi}{2} \Rightarrow t=0$
2. $-\int_1^0 \dfrac{dt}{1+t^2} = \int_0^1 \dfrac{dt}{1+t^2} = \Big[\tan^{-1} t\Big]_0^1$

✔ $\dfrac{\pi}{4}$

Solved Example 43

Evaluate $\int_1^2 \dfrac{dx}{x(x+1)}$.

1. $\dfrac{1}{x(x+1)} = \dfrac{1}{x} - \dfrac{1}{x+1}$, so $F(x) = \log\left|\dfrac{x}{x+1}\right|$
2. $F(2) - F(1) = \log\dfrac{2}{3} - \log\dfrac{1}{2} = \log\dfrac{4}{3}$

✔ $\log\dfrac{4}{3}$

Yaad karo: Limits badlo, x bhool jao

Definite integral mein substitution kiya? To **limits bhi naye variable mein badlo** ($x=0 \to t=1$ etc.). Phir wapas x mein jaane ki zaroorat nahi. Purane limits t mein daalna = galat answer.

8. Properties of Definite Integrals

These properties turn "impossible looking" definite integrals into 3-line answers. Board ka 5 marker yahin se aata hai.

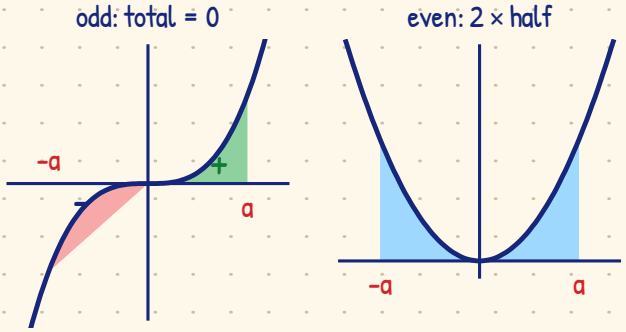
Formula Box 9: Properties P0 to P4

- P_0: $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$ (dummy variable)
- P_1: $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$; so $\int_a^a f(x)\,dx = 0$
- P_2: $\int_a^b f\,dx = \int_a^c f\,dx + \int_c^b f\,dx$ (split at c)
- P_3: $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$
- P_4: $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$ (King property)

Formula Box 10: Properties P5 to P7

- P_5: $\int_0^{2a} f(x)\,dx = \int_0^a f(x)\,dx + \int_0^a f(2a-x)\,dx$
- P_6: $\int_0^{2a} f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx, & f(2a-x) = f(x) \\ 0, & f(2a-x) = -f(x) \end{cases}$
- P_7: $\int_{-a}^a f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx, & f \text{ even} \\ 0, & f \text{ odd} \end{cases}$
- Even: $f(-x) = f(x)$ (x^2, $\cos x$, $|x|$). Odd: $f(-x) = -f(x)$ (x^3, $\sin x$, $\tan x$)

↳ Numbering alag books mein thoda alag ho sakta hai. Answer mein property ka **formula likho**, number nahi: examiner formula dekhta hai.



Odd function: areas cancel. Even function: double the half.

Type A: King property (P_3 / P_4), "add karo, 2I banao"

When $f(a+b-x)$ swaps $\sin \leftrightarrow \cos$ (or $x \leftrightarrow a+b-x$), write I twice and **add**: the denominators match and $2I$

becomes easy.

Solved Example 44

Evaluate $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx$. (Most asked)

1. $I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}dx$ ...(1)
2. By P_4 $(x \to \frac{\pi}{2}-x)$: $I = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}dx$ ...(2)
3. (1)+(2): $2I = \int_0^{\pi/2} 1\,dx = \frac{\pi}{2}$

✔ $I = \frac{\pi}{4}$

Solved Example 45

Evaluate $\int_2^8 \frac{\sqrt{x}}{\sqrt{x}+\sqrt{10-x}}\,dx$.

1. $a+b=10$. By P_3: $I = \int_2^8 \frac{\sqrt{10-x}}{\sqrt{10-x}+\sqrt{x}}dx$
2. Add: $2I = \int_2^8 1\,dx = 8-2 = 6$

✔ $I = 3$

Solved Example 46

Evaluate $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\cot x}}$.

1. $\frac{1}{1+\sqrt{\cot x}} = \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}$
2. $a+b=\frac{\pi}{2}$, so P_3 swaps sin and cos: $I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}dx$
3. Add: $2I = \frac{\pi}{3}-\frac{\pi}{6} = \frac{\pi}{6}$

✔ $I = \frac{\pi}{12}$

Yaad karo: a+b ka aadha

$\int_a^b \frac{f(x)}{f(x)+f(a+b-x)}dx = \frac{b-a}{2}$. MCQ mein seedha lagao: Ex 45 $= \frac{8-2}{2} = 3$, Ex 46 $= \frac{1}{2}\left(\frac{\pi}{3}-\frac{\pi}{6}\right) = \frac{\pi}{12}$. Board ke 3/5 marker mein steps likhna zaroori.

↳ Check before using: denominator $f(x)+f(a+b-x)$ jaisa hona chahiye, tabhi add karne pe $2I = \int_a^b 1\,dx$ banta hai.

Quick fact $\int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} \cos^n x\,dx$ (King property).

Type B: Odd/even (P_7) and modulus (P_2)

Solved Example 47

Evaluate $\int_{-\pi/4}^{\pi/4} (x^3 + x\cos x + \tan^5 x + 1)\,dx$.

1. x^3, $x\cos x$, $\tan^5 x$ are all **odd**, so their integrals over $[-\frac{\pi}{4}, \frac{\pi}{4}]$ are 0 (P_7)
2. Left: $\int_{-\pi/4}^{\pi/4} 1\,dx = \frac{\pi}{4}+\frac{\pi}{4}$

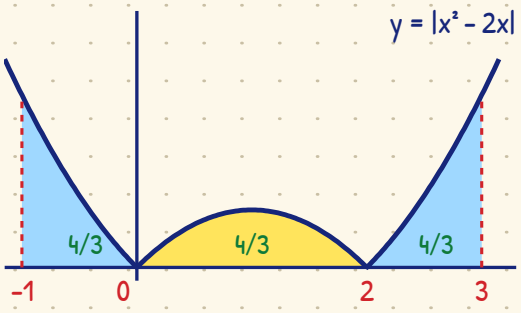
✔ $\frac{\pi}{2}$

Solved Example 48

Evaluate $\int_{-1}^{3} |x^2-2x|\,dx$. (Board fav)

1. $x^2-2x = x(x-2)$: negative on $(0,2)$, positive outside. Split at 0 and 2 (P_2)
2. $\int_{-1}^{0}(x^2-2x)dx = \left[\frac{x^3}{3}-x^2\right]_{-1}^{0} = \frac{4}{3}$
3. $\int_0^2 (2x-x^2)dx = 4-\frac{8}{3} = \frac{4}{3}$, $\int_2^3 (x^2-2x)dx = 0-\left(-\frac{4}{3}\right) = \frac{4}{3}$

✔ $\frac{4}{3}+\frac{4}{3}+\frac{4}{3} = 4$



Ex 48: the part below the axis flips up, so split at 0 and 2

Type C: x in numerator (P_4 removes it)

Solved Example 49

Evaluate $\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}\,dx$. (5M)

1. P_4: $I = \int_0^{\pi} \frac{(\pi-x)\sin x}{1+\cos^2 x}dx$ (as $\sin(\pi-x) = \sin x$, $\cos^2(\pi-x) = \cos^2 x$)
2. Add: $2I = \pi\int_0^{\pi} \frac{\sin x}{1+\cos^2 x}dx$. Put $\cos x = t$: $= \pi\int_{-1}^{1} \frac{dt}{1+t^2}$
3. $= \pi\left[\tan^{-1} t\right]_{-1}^{1} = \pi\left(\frac{\pi}{4}+\frac{\pi}{4}\right) = \frac{\pi^2}{2}$

✔ $I = \frac{\pi^2}{4}$

Solved Example 50

Evaluate $\int_0^{\pi} \frac{x\,dx}{4\cos^2 x + 9\sin^2 x}$. (HOTS)

1. P_4 and add: $2I = \pi \int_0^{\pi} \frac{dx}{4\cos^2 x + 9\sin^2 x}$; by P_6: $I = \pi \int_0^{\pi/2} \frac{dx}{4\cos^2 x + 9\sin^2 x}$
2. Divide by $\cos^2 x$, put $\tan x = t$: $I = \pi \int_0^{\infty} \frac{dt}{4 + 9t^2} = \frac{\pi}{9} \int_0^{\infty} \frac{dt}{t^2 + (2/3)^2}$
3. $= \frac{\pi}{9} \times \frac{3}{2}\left[\tan^{-1} \frac{3t}{2}\right]_0^{\infty} = \frac{\pi}{6} \times \frac{\pi}{2}$

✔ $I = \frac{\pi^2}{12}$

Type D: the famous log integral

Solved Example 51

Evaluate $\int_0^{\pi/2} \log \sin x\,dx$. (5M, Most asked)

1. $I = \int_0^{\pi/2} \log \sin x\,dx = \int_0^{\pi/2} \log \cos x\,dx$ (by P_4)
2. Add: $2I = \int_0^{\pi/2} \log(\sin x \cos x)dx = \int_0^{\pi/2} \log \sin 2x\,dx - \frac{\pi}{2}\log 2$
3. In $\int_0^{\pi/2} \log \sin 2x\,dx$ put $2x = t$: $= \frac{1}{2}\int_0^{\pi} \log \sin t\,dt = \frac{1}{2} \times 2\int_0^{\pi/2} \log \sin t\,dt = I$ (by P_6)
4. So $2I = I - \frac{\pi}{2}\log 2$

✔ $I = -\frac{\pi}{2}\log 2$ (same for $\int_0^{\pi/2} \log \cos x\,dx$)



Shivam Bhaiya says

Properties wale sawal mein pehli line hamesha likho: "Using $\int_0^a f(x)dx = \int_0^a f(a-x)dx$". Ye ek line hi step marks dila deti hai, chahe aage calculation mein galti ho jaye.

9. More Board Types (with tricks)

Type E: $\int \frac{dx}{a\sin^2 x + b\cos^2 x}$

Divide num and den by $\cos^2 x$, then put $\tan x = t$.

Quick fact $\int \frac{dx}{a^2 + b^2x^2} = \frac{1}{ab}\tan^{-1}\frac{bx}{a} + C$.

Quick fact $\int \frac{dx}{\sqrt{x}} = 2\sqrt{x} + C$.

Quick fact $\int \tan^2 x\,dx = \tan x - x + C$.

Solved Example 52

Evaluate $\int \frac{dx}{\sin^2 x + 4\cos^2 x}$.

1. $= \int \frac{\sec^2 x\,dx}{\tan^2 x + 4}$. Put $\tan x = t$, $\sec^2 x\,dx = dt$
2. $\int \frac{dt}{t^2 + 2^2} = \frac{1}{2}\tan^{-1}\frac{t}{2}$

✔ $\frac{1}{2}\tan^{-1}\left(\frac{\tan x}{2}\right) + C$

Type F: $\int \frac{\sin x}{\sin x + \cos x}dx$: Num $= A$ Den $+ B$ Den′

Solved Example 53

Evaluate $\int \frac{\sin x}{\sin x + \cos x}\,dx$.

1. Write $\sin x = A(\sin x + \cos x) + B(\cos x - \sin x)$
2. Compare: $\sin x$: $A - B = 1$; $\cos x$: $A + B = 0$ $\Rightarrow A = \frac{1}{2}, B = -\frac{1}{2}$
3. $I = \frac{1}{2}\int dx - \frac{1}{2}\int \frac{\cos x - \sin x}{\sin x + \cos x}dx$

✔ $\frac{x}{2} - \frac{1}{2}\log|\sin x + \cos x| + C$

Type G: $\int \frac{x^2 + 1}{x^4 + 1}dx$ (divide by x^2)

Solved Example 54

Evaluate $\int \frac{x^2 + 1}{x^4 + 1}\,dx$. (HOTS)

1. Divide by x^2: $\int \frac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2}}dx = \int \frac{1 + \frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + 2}dx$
2. Put $x - \frac{1}{x} = t$, $\left(1 + \frac{1}{x^2}\right)dx = dt$: $\int \frac{dt}{t^2 + (\sqrt{2})^2} = \frac{1}{\sqrt{2}}\tan^{-1}\frac{t}{\sqrt{2}}$

✔ $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^2 - 1}{\sqrt{2}\,x}\right) + C$

↳ For $\frac{x^2-1}{x^4+1}$: same trick but put $x + \frac{1}{x} = t$, answer has log.

Quick fact $\int \sin^2 x\,dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$.

Quick fact $\int \cos^2 x\,dx = \frac{x}{2} + \frac{\sin 2x}{4} + C$.

Quick fact $\int_0^{\pi/2} \log \sin x\,dx = \int_0^{\pi/2} \log \cos x\,dx = -\frac{\pi}{2}\log 2$.

Solved Example 55

Evaluate $\int \frac{dx}{x(x^4+1)}$.

1. Multiply num and den by x^3: $\int \frac{x^3\,dx}{x^4(x^4+1)}$. Put $x^4 = t$, $x^3dx = \frac{dt}{4}$
2. $\frac{1}{4}\int \frac{dt}{t(t+1)} = \frac{1}{4}\int \left(\frac{1}{t} - \frac{1}{t+1}\right) dt$

✔ $\frac{1}{4}\log\left(\frac{x^4}{x^4+1}\right) + C$

Type H: piecewise function and P_4 on polynomials

Solved Example 56

If $f(x) = 2x+1$ for $x \le 1$ and $f(x) = 3x^2$ for $x > 1$, find $\int_0^2 f(x)\,dx$.

1. Split at $x = 1$ (P_2): $\int_0^1 (2x+1)dx + \int_1^2 3x^2 dx$
2. $= \left[x^2 + x\right]_0^1 + \left[x^3\right]_1^2 = 2 + 7$

✔ 9

Solved Example 57

Evaluate $\int_0^1 x(1-x)^5\,dx$.

1. By P_4 ($x \to 1-x$): $I = \int_0^1 (1-x)x^5 dx = \int_0^1 (x^5 - x^6)dx$
2. $= \frac{1}{6} - \frac{1}{7}$

✔ $\frac{1}{42}$

Solved Example 58

Evaluate $\int_0^{\pi} |\cos x|\,dx$.

1. $\cos x \ge 0$ on $[0, \frac{\pi}{2}]$, ≤ 0 on $[\frac{\pi}{2}, \pi]$
2. $= \int_0^{\pi/2} \cos x\,dx - \int_{\pi/2}^{\pi} \cos x\,dx = 1 - (0 - 1)$

✔ 2

Quick fact $\int_0^a \frac{f(x)}{f(x)+f(a-x)}dx = \frac{a}{2}$.

Quick fact $|x|$ is even, so $\int_{-a}^{a} |x|dx = a^2$.

10. Proofs Board Can Ask

Proof 1: $\int \frac{dx}{x^2-a^2}$

$\frac{1}{x^2-a^2} = \frac{1}{(x-a)(x+a)} = \frac{1}{2a}\left[\frac{1}{x-a} - \frac{1}{x+a}\right]$

$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\left[\log|x-a| - \log|x+a|\right] = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right| + C$

Proof 2: $\int \frac{dx}{x^2+a^2}$

Put $x = a\tan\theta$, $dx = a\sec^2\theta\,d\theta$, $x^2 + a^2 = a^2\sec^2\theta$.

$\int \frac{a\sec^2\theta\,d\theta}{a^2\sec^2\theta} = \frac{1}{a}\theta = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$

Proof 3: $\int \frac{dx}{\sqrt{x^2+a^2}}$

Put $x = a\tan\theta$: $\int \frac{a\sec^2\theta\,d\theta}{a\sec\theta} = \int \sec\theta\,d\theta = \log|\sec\theta + \tan\theta|$

$= \log\left|\frac{\sqrt{x^2+a^2}}{a} + \frac{x}{a}\right| = \log\left|x + \sqrt{x^2+a^2}\right| - \log a$; the $-\log a$ goes into C.

Proof 4: $\int \sqrt{a^2-x^2}\,dx$ (by parts)

$I = x\sqrt{a^2-x^2} - \int x \times \frac{-x}{\sqrt{a^2-x^2}}dx = x\sqrt{a^2-x^2} - \int \frac{(a^2-x^2)-a^2}{\sqrt{a^2-x^2}}dx$

$I = x\sqrt{a^2-x^2} - I + a^2\sin^{-1}\frac{x}{a} \Rightarrow I = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$

11. Extra Practice: Board + Competency

Solved Example 59

Evaluate $\int \frac{(x+1)(x+\log x)^2}{x}\,dx$.

1. $= \int \left(1 + \frac{1}{x}\right)(x+\log x)^2 dx$. Put $x + \log x = t$, $\left(1+\frac{1}{x}\right)dx = dt$
2. $\int t^2 dt = \frac{t^3}{3}$

✔ $\frac{(x+\log x)^3}{3} + C$

Quick fact $\int \frac{dx}{1+\sin x} = \tan x - \sec x + C$.

Solved Example 60

Evaluate $\int \frac{dx}{1+\tan x}$.

1. $= \int \frac{\cos x}{\cos x+\sin x}dx$. Write $\cos x = A(\cos x+\sin x)+B(\cos x-\sin x)$
2. $A+B=1, A-B=0 \Rightarrow A=B=\frac{1}{2}$
3. $I = \frac{1}{2}\int dx + \frac{1}{2}\int \frac{\cos x-\sin x}{\cos x+\sin x}dx$

✔ $\frac{x}{2}+\frac{1}{2}\log|\cos x+\sin x|+C$

Solved Example 61

Evaluate $\int \frac{\sin 2x}{\sin^4 x+\cos^4 x}\,dx$.

1. Divide by $\cos^4 x$: $\int \frac{2\tan x\sec^2 x}{\tan^4 x+1}dx$
2. Put $\tan^2 x = t$, $2\tan x\sec^2 x\,dx = dt$: $\int \frac{dt}{t^2+1} = \tan^{-1} t$

✔ $\tan^{-1}(\tan^2 x)+C$

Solved Example 62

Evaluate $\int \sqrt{\frac{1-x}{1+x}}\,dx$.

1. Multiply inside by $\frac{1-x}{1-x}$: $\int \frac{1-x}{\sqrt{1-x^2}}dx = \int \frac{dx}{\sqrt{1-x^2}} - \int \frac{x\,dx}{\sqrt{1-x^2}}$
2. $= \sin^{-1} x + \sqrt{1-x^2}$

✔ $\sin^{-1} x+\sqrt{1-x^2}+C$

Solved Example 63

Evaluate $\int e^x \frac{x^2+1}{(x+1)^2}\,dx$. (Most asked)

1. $\frac{x^2+1}{(x+1)^2} = \frac{(x^2-1)+2}{(x+1)^2} = \frac{x-1}{x+1} + \frac{2}{(x+1)^2}$
2. $f = \frac{x-1}{x+1}$, $f' = \frac{2}{(x+1)^2}$: form $e^x[f+f']$

✔ $e^x\frac{x-1}{x+1}+C$

Quick fact $\int \frac{dx}{(x+a)(x+b)} = \frac{1}{b-a}\log\left|\frac{x+a}{x+b}\right|+C$, $a \neq b$.

Quick fact $\int e^x(f+f')dx = e^x f + C$: look for the pair!

Solved Example 64

Evaluate $\int \frac{\log x}{(1+\log x)^2}\,dx$.

1. Put $\log x = t$, $x = e^t$, $dx = e^t dt$: $\int e^t\frac{t}{(1+t)^2}dt = \int e^t\left[\frac{1}{1+t}-\frac{1}{(1+t)^2}\right]dt$
2. $= \frac{e^t}{1+t}$

✔ $\frac{x}{1+\log x}+C$

Solved Example 65

Evaluate $\int \cos 2x\cos 4x\cos 6x\,dx$.

1. $\cos 2x\cos 4x = \frac{1}{2}(\cos 6x+\cos 2x)$
2. Product $= \frac{1}{2}\cos^2 6x + \frac{1}{2}\cos 2x\cos 6x = \frac{1}{4}(1+\cos 12x+\cos 8x+\cos 4x)$

✔ $\frac{1}{4}\left[x+\frac{\sin 4x}{4}+\frac{\sin 8x}{8}+\frac{\sin 12x}{12}\right]+C$

Solved Example 66

Evaluate $\int \frac{x+4}{\sqrt{5-4x-x^2}}\,dx$.

1. $x+4 = -\frac{1}{2}(-4-2x)+2$; and $5-4x-x^2 = 9-(x+2)^2$
2. $I = -\frac{1}{2}\int \frac{-4-2x}{\sqrt{5-4x-x^2}}dx + 2\int \frac{dx}{\sqrt{3^2-(x+2)^2}}$

✔ $-\sqrt{5-4x-x^2}+2\sin^{-1}\frac{x+2}{3}+C$

Solved Example 67

Evaluate $\int \frac{dx}{x^2(x^2+1)}$.

1. $\frac{1}{x^2(x^2+1)} = \frac{1}{x^2}-\frac{1}{x^2+1}$

✔ $-\frac{1}{x}-\tan^{-1}x+C$

Solved Example 68

Evaluate $\int \frac{x^3}{x^4+1}\,dx$ and $\int e^{3\log x}\,dx$.

1. First: numerator $= \frac{1}{4}$ derivative of x^4+1: $\frac{1}{4}\log(x^4+1)$
2. Second: $e^{3\log x} = e^{\log x^3} = x^3$, so $\frac{x^4}{4}$

✔ $\frac{1}{4}\log(x^4+1)+C$ and $\frac{x^4}{4}+C$

Quick fact $\int \sec x\tan x\,dx = \sec x + C$.

Solved Example 69

Evaluate $\int_1^2 \frac{x+3}{x(x+2)}\,dx$.

1. $\frac{x+3}{x(x+2)} = \frac{3/2}{x} - \frac{1/2}{x+2}$
2. $\left[\frac{3}{2}\log x - \frac{1}{2}\log(x+2)\right]_1^2 = \frac{3}{2}\log 2 - \frac{1}{2}\log 4 + \frac{1}{2}\log 3$

✔ $\frac{1}{2}\log 2 + \frac{1}{2}\log 3 = \frac{1}{2}\log 6$

Solved Example 70

Evaluate $\int_0^2 x\sqrt{2-x}\,dx$.

1. P_4: $I = \int_0^2 (2-x)\sqrt{x}\,dx = \int_0^2 (2x^{1/2} - x^{3/2})dx$
2. $= \left[\frac{4}{3}x^{3/2} - \frac{2}{5}x^{5/2}\right]_0^2 = \frac{8\sqrt{2}}{3} - \frac{8\sqrt{2}}{5}$

✔ $\frac{16\sqrt{2}}{15}$

Solved Example 71

Evaluate $\int_0^{\pi/2} \sin 2x\, \log(\tan x)\,dx$.

1. P_4: $\sin 2(\frac{\pi}{2} - x) = \sin 2x$, $\log\tan(\frac{\pi}{2} - x) = \log\cot x = -\log\tan x$
2. So $I = -I \Rightarrow 2I = 0$

✔ 0

Solved Example 72

Evaluate $\int_{-\pi}^{\pi} (1-x^2)\sin x\cos^2 x\,dx$.

1. $g(x) = (1-x^2)\sin x\cos^2 x$: $g(-x) = (1-x^2)(-\sin x)\cos^2 x = -g(x)$, odd
2. By P_7, the integral is 0

✔ 0

Solved Example 73

Evaluate $\int_0^{\pi/2} \frac{\sin x\cos x}{\sin^4 x + \cos^4 x}\,dx$.

1. From Ex 61 (half of it): $\int \frac{\sin x\cos x}{\sin^4 x+\cos^4 x}dx = \frac{1}{2}\tan^{-1}(\tan^2 x)$
2. As $x \to \frac{\pi}{2}$, $\tan^2 x \to \infty$: $\frac{1}{2}\left(\frac{\pi}{2} - 0\right)$

✔ $\frac{\pi}{4}$

Quick fact $\int_0^{\pi} \cos x\,dx = 0$ but $\int_0^{\pi} |\cos x|dx = 2$.



Shivam Bhaiya says

Integrals ka sabse bada secret: **practice**. Har din 10 sawal, alag alag type ke. Jab sawal dekhte hi method dimaag mein aa jaye (substitution? parts? partial?), samjho chapter khatam. Ye 22 page ka chapter boards mein lagbhag 10 marks laata hai.

12. Definite Integrals: More Practice

Solved Example 74

Evaluate $\int_0^{\pi/2} \sin^3 x\,dx$.

1. $\sin^3 x = \frac{3\sin x - \sin 3x}{4}$
2. $\left[-\frac{3\cos x}{4} + \frac{\cos 3x}{12}\right]_0^{\pi/2} = 0 - \left(-\frac{3}{4} + \frac{1}{12}\right)$

✔ $\frac{2}{3}$

Solved Example 75

Evaluate $\int_0^1 \sin^{-1} x\,dx$.

1. From Ex 32 (iii): $\int \sin^{-1} x\,dx = x\sin^{-1} x + \sqrt{1-x^2}$
2. $\left[x\sin^{-1} x + \sqrt{1-x^2}\right]_0^1 = \left(\frac{\pi}{2} + 0\right) - (0+1)$

✔ $\frac{\pi}{2} - 1$

Solved Example 76

Evaluate $\int_0^a \frac{dx}{x + \sqrt{a^2-x^2}}$. (HOTS)

1. Put $x = a\sin\theta$, $dx = a\cos\theta\,d\theta$; limits $0 \to \frac{\pi}{2}$
2. $I = \int_0^{\pi/2} \frac{\cos\theta}{\sin\theta + \cos\theta}d\theta$
3. King property: same as $\int_0^{\pi/2} \frac{\sin\theta}{\sin\theta+\cos\theta}d\theta$; add: $2I = \frac{\pi}{2}$

✔ $\frac{\pi}{4}$

Solved Example 77

Evaluate $\int_{-2}^{2} |x+1|\,dx$.

1. $x+1 < 0$ for $x < -1$. Split at -1
2. $\int_{-2}^{-1} -(x+1)dx + \int_{-1}^{2} (x+1)dx = \frac{1}{2} + \frac{9}{2}$

✔ 5

Quick fact $\int \frac{dx}{x^2} = -\frac{1}{x} + C$.

Solved Example 78

Evaluate $\int_{-1}^{1} e^{|x|}\,dx$.

1. $e^{|x|}$ is even: $= 2\int_0^1 e^x dx$
2. $= 2\left[e^x\right]_0^1 = 2(e-1)$

✔ $2(e-1)$

Solved Example 79

Evaluate $\int_1^e \frac{(\log x)^2}{x}\,dx$ and $\int_0^{2\pi} |\sin x|\,dx$.

1. First: $\log x = t$, limits $0 \to 1$: $\int_0^1 t^2 dt = \frac{1}{3}$
2. Second: $|\sin x|$ repeats every π: $2\int_0^{\pi} \sin x\,dx = 2 \times 2 = 4$

✔ $\frac{1}{3}$ and 4

Solved Example 80

Evaluate $\int_0^{\pi/2} \sqrt{1-\sin 2x}\,dx$.

1. $1 - \sin 2x = (\cos x - \sin x)^2$, so $\sqrt{1-\sin 2x} = |\cos x - \sin x|$ (careful: modulus!)
2. $\cos x \geq \sin x$ on $[0, \frac{\pi}{4}]$: $\int_0^{\pi/4} (\cos x - \sin x)dx = \left[\sin x + \cos x\right]_0^{\pi/4} = \sqrt{2} - 1$
3. $\int_{\pi/4}^{\pi/2} (\sin x - \cos x)dx = \left[-\cos x - \sin x\right]_{\pi/4}^{\pi/2} = -1 + \sqrt{2}$

✔ $2\sqrt{2} - 2$

Solved Example 81

Evaluate $\int_0^1 x\tan^{-1} x\,dx$ and $\int_{-1}^{2} x|x|\,dx$.

1. First (from Q30): $\left[\frac{x^2+1}{2}\tan^{-1} x - \frac{x}{2}\right]_0^1 = \frac{\pi}{4} - \frac{1}{2}$
2. Second: $\int_{-1}^{0} (-x^2)dx + \int_0^2 x^2 dx = -\frac{1}{3} + \frac{8}{3}$

✔ $\frac{\pi}{4} - \frac{1}{2}$ and $\frac{7}{3}$

Try it (i) $\int \frac{\sec^2 x}{3+\tan x}dx$ (ii) $\int \cos^3 x \sin x\,dx$

Ans: (i) $\log|3+\tan x|$ (ii) $-\frac{\cos^4 x}{4}$, $+C$ for indefinite

Quick fact $\int \frac{dx}{x^3} = -\frac{1}{2x^2} + C$.

Solved Example 82

Evaluate $\int_0^{\pi/4} \frac{\sin 2x}{\sin^4 x + \cos^4 x}\,dx$.

1. From Ex 61: antiderivative $= \tan^{-1}(\tan^2 x)$
2. $\tan^{-1}(1) - \tan^{-1}(0)$

✔ $\frac{\pi}{4}$

13. Topper Style: Full 5-Mark Answers

Board mein **step marks** milte hain. Neeche dekho ek 5 marker kaise likhna hai: har important line pe mark ka hint diya hai.

Solved Example 83 (5 marks)

Find $\int \frac{x^3}{(x-1)(x^2+1)}\,dx$.

1. Improper (degree 3 = 3), so divide: $\frac{x^3}{(x-1)(x^2+1)} = 1 + \frac{x^2-x+1}{(x-1)(x^2+1)}$ (1)
2. $\frac{x^2-x+1}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$; $x=1$: $1 = 2A \Rightarrow A = \frac{1}{2}$
3. x^2: $1 = A + B \Rightarrow B = \frac{1}{2}$; constant: $1 = A - C \Rightarrow C = -\frac{1}{2}$ (2)
4. $I = \int dx + \frac{1}{2}\int \frac{dx}{x-1} + \frac{1}{4}\int \frac{2x\,dx}{x^2+1} - \frac{1}{2}\int \frac{dx}{x^2+1}$ (1)
5. $= x + \frac{1}{2}\log|x-1| + \frac{1}{4}\log(x^2+1) - \frac{1}{2}\tan^{-1} x + C$ (1)

✔ $x + \frac{1}{2}\log|x-1| + \frac{1}{4}\log(x^2+1) - \frac{1}{2}\tan^{-1} x + C$

Try it (i) $\int \frac{dx}{\sqrt{4-x^2}}$ (ii) $\int \frac{dx}{4-x^2}$

Ans: (i) $\sin^{-1}\frac{x}{2}$ (ii) $\frac{1}{4}\log\left|\frac{2+x}{2-x}\right|$, $+C$ for indefinite

Try it (i) $\int_0^1 \frac{dx}{1+x^2}$ (ii) $\int_0^{\pi/4} \tan x\,dx$

Ans: (i) $\frac{\pi}{4}$ (ii) $\frac{1}{2}\log 2$, $+C$ for indefinite

Try it (i) $\int x\log x\,dx$ (ii) $\int \frac{dx}{x^2-x}$

Ans: (i) $\frac{x^2}{2}\log x - \frac{x^2}{4}$ (ii) $\log\left|\frac{x-1}{x}\right|$, $+C$ for indefinite

Solved Example 84 (5 marks)

Evaluate $\int_0^{\pi/2} \frac{x}{\sin x + \cos x}\,dx$.

1. Using $\int_0^a f(x)dx = \int_0^a f(a-x)dx$: $I = \int_0^{\pi/2} \frac{\frac{\pi}{2} - x}{\cos x + \sin x}dx$ (1)
2. Add: $2I = \frac{\pi}{2}\int_0^{\pi/2} \frac{dx}{\sin x + \cos x}$ (1)
3. $\sin x + \cos x = \sqrt{2}\sin\left(x + \frac{\pi}{4}\right)$, so $2I = \frac{\pi}{2\sqrt{2}}\int_0^{\pi/2} \operatorname{cosec}\left(x + \frac{\pi}{4}\right)dx$ (1)
4. $= \frac{\pi}{2\sqrt{2}}\left[\log\left|\operatorname{cosec}\left(x+\frac{\pi}{4}\right) - \cot\left(x+\frac{\pi}{4}\right)\right|\right]_0^{\pi/2} = \frac{\pi}{2\sqrt{2}}\log\frac{\sqrt{2}+1}{\sqrt{2}-1}$ (1)
5. $\frac{\sqrt{2}+1}{\sqrt{2}-1} = (\sqrt{2}+1)^2$, so $2I = \frac{\pi}{\sqrt{2}}\log(\sqrt{2}+1)$ (1)

✔ $I = \frac{\pi}{2\sqrt{2}}\log(\sqrt{2}+1)$

14. Two More Types Boards Love

Solved Example 85

Evaluate $\int (x+1)\sqrt{x^2+4x+5}\,dx$.

1. $x + 1 = \frac{1}{2}(2x+4) - 1$
2. $\frac{1}{2}\int (2x+4)\sqrt{x^2+4x+5}\,dx = \frac{1}{2}\times\frac{2}{3}(x^2+4x+5)^{3/2}$ (put $x^2+4x+5 = t$)
3. $\int \sqrt{(x+2)^2+1}\,dx = \frac{x+2}{2}\sqrt{x^2+4x+5} + \frac{1}{2}\log\left|x+2+\sqrt{x^2+4x+5}\right|$

✔ $\frac{1}{3}(x^2+4x+5)^{3/2} - \frac{x+2}{2}\sqrt{x^2+4x+5} - \frac{1}{2}\log\left|x+2+\sqrt{x^2+4x+5}\right| + C$

Try it (i) $\int_0^{\pi} \sin^2\frac{x}{2}\,dx$ (ii) $\int \frac{2^x}{1}dx$

Ans: (i) $\frac{\pi}{2}$ (ii) $\frac{2^x}{\log 2}$, $+C$ for indefinite

Try it (i) $\int \frac{e^x}{1+e^{2x}}dx$ (ii) $\int_{-2}^{2} x^3\cos x\,dx$

Ans: (i) $\tan^{-1}e^x$ (ii) 0 (odd), $+C$ for indefinite

Quick fact $\int \cos 2x\,dx = \frac{\sin 2x}{2} + C$.

Solved Example 86

Evaluate $\int \frac{x^2}{x^4+x^2-2}\,dx$.

1. $x^4+x^2-2 = (x^2+2)(x^2-1)$. With $x^2 = y$: $\frac{y}{(y+2)(y-1)} = \frac{2/3}{y+2} + \frac{1/3}{y-1}$
2. $I = \frac{2}{3}\int \frac{dx}{x^2+(\sqrt{2})^2} + \frac{1}{3}\int \frac{dx}{x^2-1}$

✔ $\frac{\sqrt{2}}{3}\tan^{-1}\frac{x}{\sqrt{2}} + \frac{1}{6}\log\left|\frac{x-1}{x+1}\right| + C$

Solved Example 87

Evaluate $\int_0^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x}\,dx$.

1. King property: $I = \int_0^{\pi/2} \frac{\cos^2 x}{\cos x + \sin x}dx$. Add: $2I = \int_0^{\pi/2} \frac{dx}{\sin x + \cos x}$
2. From Ex 84: $\int_0^{\pi/2} \frac{dx}{\sin x + \cos x} = \sqrt{2}\log(\sqrt{2}+1)$

✔ $\frac{1}{\sqrt{2}}\log(\sqrt{2}+1)$

15. Spot the Method: 10 Second Drill

Board mein pehle 10 second mein method pehchaano. Ye table cover karke khud try karo, phir check karo.

Integral	Method	Answer ($+C$)
$\int \sin^5 x\cos x\,dx$	$\sin x = t$	$\frac{\sin^6 x}{6}$
$\int \frac{dx}{x^2+4x+8}$	complete square	$\frac{1}{2}\tan^{-1}\frac{x+2}{2}$
$\int xe^{2x}dx$	by parts	$\frac{e^{2x}}{4}(2x-1)$
$\int \frac{dx}{(x+1)(x+3)}$	partial fractions	$\frac{1}{2}\log\left\|\frac{x+1}{x+3}\right\|$
$\int \sec^4 x\,dx$	$\sec^2 = 1+\tan^2$, $\tan x = t$	$\tan x + \frac{\tan^3 x}{3}$
$\int e^x(\cot x - \operatorname{cosec}^2 x)dx$	$e^x[f+f']$	$e^x\cot x$
$\int \frac{dx}{\sqrt{9-4x^2}}$	special form	$\frac{1}{2}\sin^{-1}\frac{2x}{3}$
$\int \cos^2 3x\,dx$	identity	$\frac{x}{2} + \frac{\sin 6x}{12}$
$\int \frac{\log x}{x^2}dx$	by parts	$-\frac{\log x}{x} - \frac{1}{x}$
$\int \frac{2x+1}{x^2+x+5}dx$	$\frac{f'}{f}$	$\log(x^2+x+5)$

Rapid fire: bolo, likho mat (answers neeche)

1. $\int \frac{dx}{x}$?
2. $\int \sec^2 x\,dx$?

3. $\int_{-3}^{3} x^5\,dx$?
4. $\int \frac{f'(x)}{f(x)}\,dx$?
5. $\int e^x(\sin x + \cos x)\,dx$?
6. $\int_0^{\pi/2} \frac{\cos x}{\sin x+\cos x}\,dx$?
7. $\int a^x\,dx$?
8. $\int \frac{dx}{\sqrt{1-x^2}}$?

Answers: (1) $\log|x|$ (2) $\tan x$ (3) 0 (4) $\log|f(x)|$ (5) $e^x \sin x$ (6) $\frac{\pi}{4}$ (7) $\frac{a^x}{\log a}$ (8) $\sin^{-1} x$ (each indefinite one $+C$)

16. Summary: Which Method When?

Step 1: Standard form? Use Formula Box 1 to 3 directly

↓

Step 2: Function + its derivative present? **Substitution**

↓

Step 3: Powers/products of sin, cos? **Identities**

↓

Step 4: Quadratic in denominator / under root? **Special integrals** (complete the square)

↓

Step 5: Rational function, factorable denominator? **Partial fractions**

↓

Step 6: Product of two different types? **By parts** (ILATE)

Most Asked in Boards

- Integrate by **partial fractions**, incl. repeated or quadratic factor (3M/5M)
- $\int \frac{px+q}{\sqrt{ax^2+bx+c}}\,dx$ or $\int \frac{px+q}{ax^2+bx+c}\,dx$ type (3M)
- **By parts**, especially $\int e^x[f+f']\,dx$ and $\int e^x \sin x\,dx$ (2M/3M)
- King property: $\int_0^{\pi/2} \frac{\sin^n x}{\sin^n x+\cos^n x}\,dx$ type (3M)
- $\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}\,dx$ or $\int_0^{\pi/2} \log\sin x\,dx$ (5M)
- Definite integral with modulus $|\ |$, split into parts (3M)
- Odd/even function MCQ, $\int_{-a}^{a}$ (1M)
- Substitution with change of limits (2M)

Common mistakes

- Forgetting $+C$ in indefinite integrals (and writing $+C$ in definite ones).
- $\int \frac{dx}{x^2-a^2}$ vs $\int \frac{dx}{a^2-x^2}$: swapping $\frac{x-a}{x+a}$ and $\frac{a+x}{a-x}$.
- Substitution in definite integral without changing the limits.
- Partial fractions on an **improper** fraction without dividing first.
- Wrong u in by parts (ignoring ILATE), or losing the minus sign in $-\int(u'\int v)$.

→ Writing $\int \frac{1}{x}dx = \log x$ instead of $\log|x|$.

Ek nazar mein: Integrals

Sawal mein dikhe	Method	Marks
$f(x)$ and $f'(x)$ both	substitution	2-3
$\sin^2, \cos^3, \sin A\cos B$	identities	2
ax^2+bx+c in den	complete the square	3
$\frac{P(x)}{Q(x)}$, Q factorises	partial fractions	3-5
$x\sin x, x^2e^x, \log x$	by parts (ILATE)	3
$\int_0^a$ with $f(a-x)$ nice	King property	3-5
$\int_{-a}^{a}$	odd/even check	1-3

Last minute: 10 one-liners

1. $\int x^n\,dx = \frac{x^{n+1}}{n+1}$ for $n \neq -1$; for $n=-1$ it is $\log|x|$.
2. $\int \frac{f'}{f} = \log|f|$, $\int \frac{f'}{\sqrt{f}} = 2\sqrt{f}$.
3. $\int \tan x = \log|\sec x|$, $\int \sec x = \log|\sec x+\tan x|$.
4. Linear inside: divide by the coefficient of x.
5. Complete the square before using special forms.
6. Improper fraction: divide first, then partial fractions.
7. ILATE decides the first function.
8. $\int e^x[f+f']dx = e^x f$.
9. $\int_a^b \frac{f(x)}{f(x)+f(a+b-x)}dx = \frac{b-a}{2}$.
10. $\int_{-a}^{a}(\text{odd}) = 0$, $\int_{-a}^{a}(\text{even}) = 2\int_0^a$.

Most Important Questions

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. $\int e^x(\cos x - \sin x)\,dx$ is equal to 1M Board fav

(a) $e^x\cos x + C$ (b) $e^x\sin x + C$ (c) $-e^x\cos x + C$ (d) $-e^x\sin x + C$

Ans. (a) Here $f=\cos x$, $f'=-\sin x$, so $e^x\cos x + C$.

Q2. $\int \frac{dx}{x^2+2x+2}$ equals 1M

(a) $\tan^{-1}(x+1)+C$ (b) $2\tan^{-1}(x+1)+C$ (c) $\log|x^2+2x+2|+C$ (d) $\tan^{-1}x+C$

Ans. (a) $= \int \frac{dx}{(x+1)^2+1} = \tan^{-1}(x+1)+C$.

Q3. $\int_{-1}^{1} x^3|x|\,dx$ is 1M

(a) 0 (b) 1 (c) 2 (d) $\frac{1}{2}$

Ans. (a) $x^3|x|$ is odd, so the integral is 0.

Q4. $\int \dfrac{dx}{x\log x}\ (x>1)$ equals 1M

(a) $\log|\log x|+C$ (b) $\log x+C$ (c) $\frac{(\log x)^2}{2}+C$ (d) $\frac{1}{\log x}+C$

Ans. (a) Put $\log x=t$: $\int \frac{dt}{t}=\log|\log x|+C$.

Q5. $\int_0^{\pi/2}\dfrac{\sin^7 x}{\sin^7 x+\cos^7 x}\,dx$ is 1M Most asked

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) 0 (d) $\frac{\pi}{8}$

Ans. (b) King property: $2I=\frac{\pi}{2}$, so $I=\frac{\pi}{4}$.

Q6. $\int x^2e^{x^3}\,dx$ equals 1M

(a) $\frac{1}{3}e^{x^3}+C$ (b) $\frac{x^3}{3}e^{x^3}+C$ (c) $e^{x^3}+C$ (d) $\frac{1}{2}e^{x^2}+C$

Ans. (a) Put $x^3=t$, $x^2dx=\frac{dt}{3}$: $\frac{1}{3}e^{x^3}+C$.

Q7. $\int \dfrac{dx}{e^x+e^{-x}}$ equals 1M

(a) $\tan^{-1}(e^x)+C$ (b) $\tan^{-1}(e^{-x})+C$ (c) $\log(e^x+e^{-x})+C$ (d) $\log(e^x-e^{-x})+C$

Ans. (a) $=\int\frac{e^x\,dx}{e^{2x}+1}$, put $e^x=t$: $\tan^{-1}(e^x)+C$.

Q8. If $\int_0^a 3x^2\,dx=8$, then a is 1M

(a) 2 (b) 3 (c) 4 (d) 8

Ans. (a) $a^3=8\Rightarrow a=2$.

Q9. $\int\dfrac{\cos 2x}{(\cos x+\sin x)^2}\,dx$ equals 1M

(a) $\frac{-1}{\cos x+\sin x}+C$ (b) $\log|\cos x-\sin x|+C$ (c) $\log|\cos x+\sin x|+C$ (d) $\frac{1}{(\cos x+\sin x)^2}+C$

Ans. (c) $\cos 2x=(\cos x-\sin x)(\cos x+\sin x)$, integrand $=\frac{\cos x-\sin x}{\cos x+\sin x}=\frac{f'}{f}$: $\log|\cos x+\sin x|+C$.

Q10. For $0<x<\frac{\pi}{2}$, $\int\sqrt{1+\sin 2x}\,dx$ equals 1M

(a) $\sin x+\cos x+C$ (b) $\sin x-\cos x+C$ (c) $\cos x-\sin x+C$ (d) $\frac{(1+\sin 2x)^{3/2}}{3}+C$

Ans. (b) $1+\sin 2x=(\sin x+\cos x)^2$, so integrand $=\sin x+\cos x$; integral $=\sin x-\cos x+C$.

Q11. $\int_0^2|x-1|\,dx$ is 1M

(a) 0 (b) 1 (c) 2 (d) $\frac{1}{2}$

Ans. (b) $\int_0^1(1-x)dx+\int_1^2(x-1)dx=\frac{1}{2}+\frac{1}{2}=1$.

Q12. An antiderivative of $\sqrt{x}+\dfrac{1}{\sqrt{x}}$ is 1M

(a) $\frac{1}{3}x^{1/3}+2x^{1/2}$ (b) $\frac{2}{3}x^{2/3}+\frac{1}{2}x^2$ (c) $\frac{2}{3}x^{3/2}+2x^{1/2}$ (d) $\frac{3}{2}x^{3/2}+\frac{1}{2}x^{1/2}$

Ans. (c) $\frac{2}{3}x^{3/2}+2x^{1/2}+C$.

Q13. $\int\dfrac{dx}{x^2-9}$ equals 1M

(a) $\frac{1}{3}\tan^{-1}\frac{x}{3}+C$ (b) $\frac{1}{6}\log\left|\frac{x+3}{x-3}\right|+C$ (c) $\frac{1}{3}\log\left|\frac{x-3}{x+3}\right|+C$ (d) $\frac{1}{6}\log\left|\frac{x-3}{x+3}\right|+C$

Ans. (d) $\frac{1}{2\times 3}\log\left|\frac{x-3}{x+3}\right|+C$.

Q15. A: $\int\sec x\,dx=\log|\sec x+\tan x|+C$. R: $\int\tan x\,dx=\log|\sec x|+C$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

(contd.) (b) Both are true results, but R is not the reason for A.

Q14. A: $\int_{-\pi/2}^{\pi/2}\sin^3 x\,dx=0$. R: $\sin^3 x$ is an odd function. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) Both true; odd function over $[-a,a]$ gives 0, so R explains A.

Q16. A: $\int_{-1}^{1}|x|\,dx=0$. R: $|x|$ is an even function. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (d) A is false: $\int_{-1}^{1}|x|dx=2\int_0^1 x\,dx=1$. R is true.

Very short (2 marks)

Q17. Find $\int\dfrac{\sin x-\cos x}{\sqrt{1+\sin 2x}}\,dx,\ 0<x<\frac{\pi}{2}$. 2M Board fav

Ans. $\sqrt{1+\sin 2x}=\sin x+\cos x$. Put $\sin x+\cos x=t$, $(\cos x-\sin x)dx=dt$: $-\int\frac{dt}{t}=-\log|\sin x+\cos x|+C$.

Q18. Find $\int\dfrac{dx}{x^2-6x+13}$. 2M

Ans. $x^2-6x+13=(x-3)^2+2^2$, so $\frac{1}{2}\tan^{-1}\frac{x-3}{2}+C$.

Q19. Find $\int x\sec^2 x\,dx$. 2M

Ans. $u=x, v=\sec^2 x$: $x\tan x-\int\tan x\,dx=x\tan x+\log|\cos x|+C$.

Q20. Evaluate $\int_0^{\pi/4}\tan^2 x\,dx$. 2M

Ans. $\int_0^{\pi/4}(\sec^2 x-1)dx=\left[\tan x-x\right]_0^{\pi/4}=1-\frac{\pi}{4}$.

Q21. Find $\int\dfrac{3x^2}{x^6+1}\,dx$. 2M

Ans. Put $x^3=t, 3x^2dx=dt$: $\int\frac{dt}{t^2+1}=\tan^{-1}(x^3)+C$.

Q22. Find $\int e^x\dfrac{1+x\log x}{x}\,dx$. 2M Most asked

Ans. $=\int e^x\left(\log x+\frac{1}{x}\right)dx; f=\log x, f'=\frac{1}{x}$: $e^x\log x+C$.

Q23. Evaluate $\int_0^{\pi/2}\dfrac{\cos x}{(1+\sin x)(2+\sin x)}\,dx$. 2M

Ans. $\sin x=t$, limits $0\to 1$: $\int_0^1\left(\frac{1}{1+t}-\frac{1}{2+t}\right)dt=\left[\log\frac{1+t}{2+t}\right]_0^1=\log\frac{2}{3}-\log\frac{1}{2}=\log\frac{4}{3}$.

Q24. Find $\int\dfrac{2x+1}{(x+1)(x-2)}\,dx$. 2M

Ans. Cover-up: $A=\frac{-2+1}{-1-2}=\frac{1}{3}, B=\frac{4+1}{2+1}=\frac{5}{3}$. Answer $\frac{1}{3}\log|x+1|+\frac{5}{3}\log|x-2|+C$.

Q25. Evaluate $\int_1^e\log x\,dx$. 2M

(contd.) $\left[x\log x - x\right]_1^e = (e-e)-(0-1) = 1.$

Q26. Find $\displaystyle\int \frac{dx}{\sqrt{x+2}-\sqrt{x+1}}$. 2M

Ans. Rationalise: denominator becomes $(x+2)-(x+1)=1$. $\int(\sqrt{x+2}+\sqrt{x+1})dx = \frac{2}{3}(x+2)^{3/2} + \frac{2}{3}(x+1)^{3/2} + C$.

Q27. Evaluate $\displaystyle\int_{-1}^{1} \log\left(\frac{2-x}{2+x}\right) dx$. 2M

Ans. $f(-x) = \log\frac{2+x}{2-x} = -f(x)$, so f is odd and the integral is 0.

Short answer (3 marks)

Q28. Find $\displaystyle\int \frac{x^2+2x+3}{(x+1)(x^2+1)}\,dx$. 3M Must do

Ans. $\frac{A}{x+1} + \frac{Bx+C}{x^2+1}$: $x=-1$: $2=2A \Rightarrow A=1$; x^2: $1 = A+B \Rightarrow B=0$; constant: $3 = A+C \Rightarrow C=2$. Answer $\log|x+1| + 2\tan^{-1}x + C$.

Q29. Find $\displaystyle\int \frac{2x+5}{\sqrt{x^2+4x+3}}\,dx$. 3M

Ans. $2x+5 = (2x+4)+1$. $\int\frac{2x+4}{\sqrt{x^2+4x+3}}dx = 2\sqrt{x^2+4x+3}$; $\int\frac{dx}{\sqrt{(x+2)^2-1}} = \log|x+2+\sqrt{x^2+4x+3}|$. Answer $2\sqrt{x^2+4x+3} + \log\left|x+2+\sqrt{x^2+4x+3}\right| + C$.

Q30. Find $\int x\tan^{-1}x\,dx$. 3M Board fav

Ans. $u=\tan^{-1}x, v=x$: $\frac{x^2}{2}\tan^{-1}x - \frac{1}{2}\int\frac{x^2}{1+x^2}dx = \frac{x^2}{2}\tan^{-1}x - \frac{1}{2}(x-\tan^{-1}x)$. Answer $\frac{x^2+1}{2}\tan^{-1}x - \frac{x}{2} + C$.

Q31. Find $\int e^{2x}\sin x\,dx$. 3M

Ans. Parts twice (take $\sin x$ as u): $I = \frac{e^{2x}}{2}\sin x - \frac{1}{2}\left[\frac{e^{2x}}{2}\cos x + \frac{1}{2}I\right]$, so $\frac{5}{4}I = \frac{e^{2x}}{4}(2\sin x - \cos x)$. Answer $\frac{e^{2x}}{5}(2\sin x - \cos x) + C$.

Q32. Evaluate $\displaystyle\int_0^{\pi/2} x^2\sin x\,dx$. 3M

Ans. $\int x^2\sin x\,dx = -x^2\cos x + 2x\sin x + 2\cos x$. At $\frac{\pi}{2}$: $0+\pi+0$; at 0: 2. Answer $\pi-2$.

Q33. Find $\int\sqrt{x^2-8x+7}\,dx$. 3M

Ans. $x^2-8x+7 = (x-4)^2-3^2$. Answer $\frac{x-4}{2}\sqrt{x^2-8x+7} - \frac{9}{2}\log\left|x-4+\sqrt{x^2-8x+7}\right| + C$.

Q34. Evaluate $\displaystyle\int_0^4 (|x-1|+|x-3|)\,dx$. 3M Most asked

Ans. $\int_0^4|x-1|dx = \frac{1}{2}+\frac{9}{2} = 5$; $\int_0^4|x-3|dx = \frac{9}{2}+\frac{1}{2} = 5$. Answer 10.

Q35. Evaluate $\displaystyle\int_0^{\pi/2}\frac{\sin x-\cos x}{1+\sin x\cos x}\,dx$. 3M

Ans. By P_4, $I = \int_0^{\pi/2}\frac{\cos x-\sin x}{1+\cos x\sin x}dx = -I$. So $2I=0$, $I=0$.

Q36. Find $\displaystyle\int\frac{e^x}{(1+e^x)(2+e^x)}\,dx$. 3M

Ans. $e^x = t$: $\int\frac{dt}{(1+t)(2+t)} = \log\left|\frac{1+t}{2+t}\right|$. Answer $\log\left(\frac{1+e^x}{2+e^x}\right) + C$.

Q37. Evaluate $\displaystyle\int_0^{\pi/4}\log(1+\tan x)\,dx$. 3M HOTS

Ans. P_4: $1+\tan\left(\frac{\pi}{4}-x\right) = 1+\frac{1-\tan x}{1+\tan x} = \frac{2}{1+\tan x}$. So $I = \int_0^{\pi/4}[\log 2 - \log(1+\tan x)]dx = \frac{\pi}{4}\log 2 - I$. Answer $I = \frac{\pi}{8}\log 2$.

Q38. Find $\displaystyle\int\frac{x^3}{(x-1)(x-2)}\,dx$. 3M

Ans. Divide: $\frac{x^3}{x^2-3x+2} = x+3+\frac{7x-6}{(x-1)(x-2)}$; $A = \frac{7-6}{1-2} = -1$, $B = \frac{14-6}{2-1} = 8$. Answer $\frac{x^2}{2}+3x-\log|x-1| + 8\log|x-2| + C$.

Long answer (5 marks)

Q39. Evaluate $\displaystyle\int_0^{\pi}\frac{x}{1+\sin x}\,dx$. 5M Must do

Ans. P_4: $I = \int_0^{\pi}\frac{\pi-x}{1+\sin x}dx$. Add: $2I = \pi\int_0^{\pi}\frac{dx}{1+\sin x} = \pi\int_0^{\pi}\frac{1-\sin x}{\cos^2 x}dx = \pi\left[\tan x - \sec x\right]_0^{\pi} = \pi[(0+1)-(0-1)] = 2\pi$. Answer $I=\pi$.

Q40. Evaluate $\displaystyle\int_0^{\pi}\log(1+\cos x)\,dx$. 5M HOTS

Ans. $1+\cos x = 2\cos^2\frac{x}{2}$: $I = \pi\log 2 + 2\int_0^{\pi}\log\cos\frac{x}{2}dx$. Put $\frac{x}{2}=t$: $= \pi\log 2 + 4\int_0^{\pi/2}\log\cos t\,dt = \pi\log 2 + 4\left(-\frac{\pi}{2}\log 2\right)$. Answer $-\pi\log 2$.

Q41. Find $\displaystyle\int\left(\sqrt{\tan x}+\sqrt{\cot x}\right)dx$. 5M

Ans. $= \int\frac{\sin x+\cos x}{\sqrt{\sin x\cos x}}dx$. Put $\sin x-\cos x = t$: $(\cos x+\sin x)dx = dt$, $t^2 = 1-2\sin x\cos x \Rightarrow \sin x\cos x = \frac{1-t^2}{2}$. $I = \int\frac{\sqrt{2}\,dt}{\sqrt{1-t^2}} = \sqrt{2}\sin^{-1}t$. Answer $\sqrt{2}\sin^{-1}(\sin x-\cos x) + C$.

Q42. Find $\displaystyle\int\frac{dx}{x^4-1}$. 5M

Ans. $\frac{1}{(x^2-1)(x^2+1)} = \frac{1}{2}\left[\frac{1}{x^2-1}-\frac{1}{x^2+1}\right]$. Answer $\frac{1}{4}\log\left|\frac{x-1}{x+1}\right| - \frac{1}{2}\tan^{-1}x + C$.

Q43. Evaluate $\displaystyle\int_0^{\pi/4}\frac{\sin x+\cos x}{16+9\sin 2x}\,dx$. 5M

Ans. Put $\sin x-\cos x = t$; $\sin 2x = 1-t^2$; limits $-1\to 0$. $I = \int_{-1}^0\frac{dt}{25-9t^2} = \frac{1}{9}\int_{-1}^0\frac{dt}{(5/3)^2-t^2} = \frac{1}{30}\left[\log\frac{5+3t}{5-3t}\right]_{-1}^0 = \frac{1}{30}\left(0-\log\frac{2}{8}\right)$. Answer $\frac{1}{30}\log 4$.

Q44. Find $\displaystyle\int\frac{x^2+1}{(x-1)^2(x+3)}\,dx$. 5M Board fav

Ans. $\frac{A}{x-1}+\frac{B}{(x-1)^2}+\frac{C}{x+3}$: $x=1$: $2=4B\Rightarrow B=\frac{1}{2}$; $x=-3$: $10=16C\Rightarrow C=\frac{5}{8}$; x^2: $1=A+C\Rightarrow A=\frac{3}{8}$. Answer $\frac{3}{8}\log|x-1| - \frac{1}{2(x-1)} + \frac{5}{8}\log|x+3| + C$.

Q45. Evaluate $\int_0^{\pi/2} 2\sin x\cos x\tan^{-1}(\sin x)\,dx$. 5M

Ans. Put $\sin x = t$, limits $0\to 1$: $\int_0^1 2t\tan^{-1}t\,dt$. By parts: $= \left[t^2\tan^{-1}t\right]_0^1 - \int_0^1 \frac{t^2}{1+t^2}dt = \frac{\pi}{4} - \left(1-\frac{\pi}{4}\right)$. Answer $\frac{\pi}{2}-1$.

Case based (4 marks)

Q46. Splitting a fraction. In class, the teacher asks to integrate $f(x) = \dfrac{2x}{(x^2+1)(x^2+3)}$. (i) Write $f(x) = A\,\frac{2x}{x^2+1} + B\,\frac{2x}{x^2+3}$; find A, B. (1) (ii) Find $\int f(x)\,dx$. (1) (iii) Evaluate $\int_0^1 f(x)\,dx$. (2) 4M

Ans. (i) $\frac{1}{(t+1)(t+3)} = \frac{1}{2}\left[\frac{1}{t+1} - \frac{1}{t+3}\right]$ with $t = x^2$, so $A = \frac{1}{2}, B = -\frac{1}{2}$. (ii) $\frac{1}{2}\log\frac{x^2+1}{x^2+3} + C$. (iii) $\frac{1}{2}\log\frac{2}{4} - \frac{1}{2}\log\frac{1}{3} = \frac{1}{2}\log\frac{3}{2}$.

Q47. The King property. Ananya learns $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$. (i) Find $\int_0^{\pi/2} \dfrac{\cos^5 x}{\sin^5 x + \cos^5 x}\,dx$. (1) (ii) Find $\int_0^1 x(1-x)^9\,dx$. (1) (iii) Using the property, evaluate $\int_0^{\pi} x\sin x\,dx$. (2) 4M Must do

Ans. (i) $2I = \frac{\pi}{2}$, $I = \frac{\pi}{4}$. (ii) $\int_0^1 (1-x)x^9\,dx = \frac{1}{10} - \frac{1}{11} = \frac{1}{110}$. (iii) $I = \int_0^{\pi}(\pi - x)\sin x\,dx \Rightarrow 2I = \pi\int_0^{\pi}\sin x\,dx = 2\pi$, so $I = \pi$.

Q48. Area function. A graphing app shows the area function $A(x) = \int_1^x \left(t^2 + \dfrac{1}{t}\right)dt$ for $x \ge 1$. (i) Find $A'(x)$. (1) (ii) Find $A(1)$. (1) (iii) Find $A(2)$. (2) 4M

Ans. (i) By the first FTC, $A'(x) = x^2 + \frac{1}{x}$. (ii) $A(1) = \int_1^1(\dots) = 0$. (iii) $\left[\frac{t^3}{3} + \log t\right]_1^2 = \frac{8}{3} + \log 2 - \frac{1}{3} = \frac{7}{3} + \log 2$.

More MCQ + Assertion-Reason (1 mark)

Q49. $\int \dfrac{dx}{1+e^{-x}}$ equals 1M

(a) $\log(1+e^{-x}) + C$ (b) $x + e^x + C$ (c) $\log(1+e^x) + C$ (d) $\tan^{-1}e^x + C$

Ans. (c) $= \int \frac{e^x}{e^x+1}dx = \log(1+e^x) + C$.

Q50. $\int_0^{\pi/2} \cos x\, e^{\sin x}\,dx$ is 1M

(a) e (b) $e-1$ (c) $e+1$ (d) 1

Ans. (b) $\sin x = t$: $\int_0^1 e^t dt = e - 1$.

Q51. $\int \dfrac{dx}{\sin x\cos x}$ equals 1M

(a) $\log|\tan x| + C$ (b) $\log|\sin x\cos x| + C$ (c) $\log|\sin 2x| + C$ (d) $\tan x + C$

Ans. (a) $= \int \frac{\sec^2 x}{\tan x}dx = \log|\tan x| + C$.

Q53. $\int_2^3 \dfrac{dx}{x^2-1}$ is 1M

(a) $\frac{1}{2}\log\frac{3}{2}$ (b) $\log\frac{3}{2}$ (c) $\frac{1}{2}\log 6$ (d) $\log 2$

(contd.) (a) $\frac{1}{2}\left[\log\frac{x-1}{x+1}\right]_2^3 = \frac{1}{2}\left(\log\frac{1}{2} - \log\frac{1}{3}\right) = \frac{1}{2}\log\frac{3}{2}$.

Q52. $\int \dfrac{x\,dx}{\sqrt{1-x^2}}$ equals 1M

(a) $\sin^{-1}x + C$ (b) $\sqrt{1-x^2} + C$ (c) $\frac{1}{2}\sqrt{1-x^2} + C$ (d) $-\sqrt{1-x^2} + C$

Ans. (d) $1 - x^2 = t$: $-\frac{1}{2}\int t^{-1/2}dt = -\sqrt{1-x^2} + C$.

Q54. $\int e^x \sec x(1+\tan x)\,dx$ equals 1M

(a) $e^x\cos x + C$ (b) $e^x\sec x + C$ (c) $e^x\sin x + C$ (d) $e^x\tan x + C$

Ans. (b) $f = \sec x$, $f' = \sec x\tan x$: $e^x\sec x + C$.

Q55. A: $\int \dfrac{dx}{x^2+4} = \dfrac{1}{2}\tan^{-1}\dfrac{x}{2} + C$. R: $\int \dfrac{dx}{x^2+a^2} = \dfrac{1}{a}\tan^{-1}\dfrac{x}{a} + C$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) A follows from R with $a = 2$.

Q56. A: $\int_0^{\pi}\sin^2 x\,dx = 2\int_0^{\pi/2}\sin^2 x\,dx$. R: if $f(2a-x) = f(x)$ then $\int_0^{2a} f(x)dx = 2\int_0^a f(x)dx$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) $\sin^2(\pi - x) = \sin^2 x$, so R (with $a = \frac{\pi}{2}$) gives A.

Q57. A: $\int xe^x\,dx = xe^x + C$. R: $\int e^x[f(x) + f'(x)]dx = e^x f(x) + C$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (d) A is false: $\int xe^x dx = e^x(x-1) + C$. R is true.

More 2 and 3 markers

Q58. Find $\int \dfrac{\sin^{-1}x}{\sqrt{1-x^2}}\,dx$. 2M

Ans. $\sin^{-1}x = t$, $\frac{dx}{\sqrt{1-x^2}} = dt$: $\int t\,dt = \frac{(\sin^{-1}x)^2}{2} + C$.

Q59. Find $\int \dfrac{\cos x - \cos 2x}{1-\cos x}\,dx$. 2M

Ans. $\cos x - \cos 2x = 1 + \cos x - 2\cos^2 x = (1-\cos x)(1+2\cos x)$. Integrand $= 1 + 2\cos x$. Answer $x + 2\sin x + C$.

Q60. Evaluate $\int_0^1 xe^{x^2}\,dx$. 2M

Ans. $x^2 = t$, limits $0\to 1$: $\frac{1}{2}\int_0^1 e^t dt = \frac{e-1}{2}$.

Q61. Find $\int \dfrac{dx}{x(x^2+1)}$. 2M

Ans. $\frac{1}{x(x^2+1)} = \frac{1}{x} - \frac{x}{x^2+1}$. Answer $\log|x| - \frac{1}{2}\log(x^2+1) + C$.

Q62. Find $\int \dfrac{x^2}{(x-1)(x-2)(x-3)}\,dx$. 3M Board fav

Ans. Cover-up: $A = \frac{1}{(-1)(-2)} = \frac{1}{2}$, $B = \frac{4}{(1)(-1)} = -4$, $C = \frac{9}{(2)(1)} = \frac{9}{2}$. Answer

(contd.) $\frac{1}{2}\log|x-1| - 4\log|x-2| + \frac{9}{2}\log|x-3| + C$.

Q63. Find $\int x \sin x \cos x\, dx$. 3M

Ans. $= \frac{1}{2}\int x \sin 2x\, dx = \frac{1}{2}\left[-\frac{x\cos 2x}{2} + \frac{\sin 2x}{4}\right]$. Answer $-\frac{x\cos 2x}{4} + \frac{\sin 2x}{8} + C$.

Q64. Evaluate $\int_0^{\pi/2} \frac{dx}{1+\tan x}$. 3M

Ans. $= \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x}dx$. P_4 gives $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x}dx$. Add: $2I = \frac{\pi}{2}$. Answer $\frac{\pi}{4}$.

Q65. Find $\int e^x \frac{2+\sin 2x}{1+\cos 2x}\, dx$. 3M Most asked

Ans. $\frac{2+2\sin x\cos x}{2\cos^2 x} = \sec^2 x + \tan x$; $f = \tan x$, $f' = \sec^2 x$. Answer $e^x \tan x + C$.

Q66. Find $\int \frac{dx}{3+2\sin^2 x}$. 3M

Ans. Divide by $\cos^2 x$: $\int \frac{\sec^2 x\, dx}{3\sec^2 x + 2\tan^2 x} = \int \frac{\sec^2 x\, dx}{3+5\tan^2 x}$. $\tan x = t$: $\frac{1}{5}\int \frac{dt}{t^2+\frac{3}{5}} = \frac{1}{5}\sqrt{\frac{5}{3}}\tan^{-1}\left(\sqrt{\frac{5}{3}}\,t\right)$. Answer $\frac{1}{\sqrt{15}}\tan^{-1}\left(\sqrt{\frac{5}{3}}\tan x\right) + C$.

More 5 markers + case based

Q67. Evaluate $\int_0^{\pi} x \sin x \cos^2 x\, dx$. 5M

Ans. P_4: $I = \int_0^{\pi}(\pi - x)\sin x\cos^2 x\, dx$. Add: $2I = \pi\int_0^{\pi}\sin x\cos^2 x\, dx$; $\cos x = t$: $= \pi\int_{-1}^{1} t^2 dt = \frac{2\pi}{3}$. Answer $I = \frac{\pi}{3}$.

Q68. Find $\int \frac{2x+3}{x^3 - x^2 - x + 1}\, dx$. 5M Must do

Ans. $x^3 - x^2 - x + 1 = (x-1)^2(x+1)$. $2x+3 = A(x-1)(x+1) + B(x+1) + C(x-1)^2$: $x=1$: $B = \frac{5}{2}$; $x=-1$: $1 = 4C$, $C = \frac{1}{4}$; x^2: $A = -\frac{1}{4}$. Answer $-\frac{1}{4}\log|x-1| - \frac{5}{2(x-1)} + \frac{1}{4}\log|x+1| + C$.

Q69. Filling a tank. Water flows into a tank at the rate $r(t) = 3t^2 + 2t$ litres per minute. (i) Find $\int r(t)\, dt$. (1) (ii) How much water enters in the first 2 minutes? (1) (iii) If the tank had 5 litres at $t = 0$, how much does it have at $t = 3$? (2) 4M

Ans. (i) $t^3 + t^2 + C$. (ii) $\int_0^2 (3t^2+2t)dt = 8 + 4 = 12$ litres. (iii) $5 + \int_0^3 (3t^2+2t)dt = 5 + 27 + 9 = 41$ litres.

Practice set: try first, then see the answer

Q70. $\frac{d}{dx}\int_0^x e^{t^2}\, dt$ equals 1M

(a) e^{t^2} (b) $2xe^{x^2}$ (c) e^{x^2} (d) $e^{x^2} - 1$

Ans. (c) By the first FTC, e^{x^2}.

Q71. $\int_0^{\pi/2} \log(\tan x)\, dx$ is 1M

(a) 0 (b) $\frac{\pi}{2}\log 2$ (c) $-\frac{\pi}{2}\log 2$ (d) 1

Ans. (a) P_4 gives $\log\cot x = -\log\tan x$, so $I = -I$, $I = 0$.

Q72. Find $\int \frac{dx}{x^2 - 4x + 3}$. 2M

Ans. $(x-2)^2 - 1^2$: $\frac{1}{2}\log\left|\frac{x-2-1}{x-2+1}\right| = \frac{1}{2}\log\left|\frac{x-3}{x-1}\right| + C$.

Q73. Evaluate $\int_0^1 \log\left(\frac{1}{x} - 1\right)dx$. 2M

Ans. $f(x) = \log\frac{1-x}{x}$, $f(1-x) = \log\frac{x}{1-x} = -f(x)$. By P_4, $I = -I$, so $I = 0$.

Q74. Find $\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x\cos^2 x}\, dx$. 3M

Ans. $= \int\left(\frac{\sin x}{\cos^2 x} + \frac{\cos x}{\sin^2 x}\right)dx = \int(\sec x\tan x + \operatorname{cosec} x\cot x)dx = \sec x - \operatorname{cosec} x + C$.

Q75. Find $\int x^3 e^{x^2}\, dx$. 3M

Ans. $x^2 = t$, $x\, dx = \frac{dt}{2}$: $\frac{1}{2}\int te^t dt = \frac{1}{2}e^t(t-1)$. Answer $\frac{1}{2}e^{x^2}(x^2-1) + C$.

Q76. Evaluate $\int_0^{\pi/4} \frac{\sin x + \cos x}{3 + \sin 2x}\, dx$. 3M

Ans. $\sin x - \cos x = t$, $\sin 2x = 1 - t^2$, limits $-1 \to 0$: $\int_{-1}^{0} \frac{dt}{4-t^2} = \frac{1}{4}\left[\log\frac{2+t}{2-t}\right]_{-1}^{0} = \frac{1}{4}\left(0 - \log\frac{1}{3}\right) = \frac{1}{4}\log 3$.

Q77. Find $\int \sec^3 x\, dx$. 3M HOTS

Ans. $u = \sec x$, $v = \sec^2 x$: $I = \sec x\tan x - \int \sec x\tan^2 x\, dx = \sec x\tan x - \int \sec x(\sec^2 x - 1)dx = \sec x\tan x - I + \log|\sec x + \tan x|$. Answer $\frac{1}{2}\sec x\tan x + \frac{1}{2}\log|\sec x + \tan x| + C$.

Q78. Find $\int e^x \frac{1-\sin x}{1-\cos x}\, dx$. 3M

Ans. $\frac{1 - 2\sin\frac{x}{2}\cos\frac{x}{2}}{2\sin^2\frac{x}{2}} = \frac{1}{2}\operatorname{cosec}^2\frac{x}{2} - \cot\frac{x}{2}$. With $f = -\cot\frac{x}{2}$, $f' = \frac{1}{2}\operatorname{cosec}^2\frac{x}{2}$. Answer $-e^x\cot\frac{x}{2} + C$.

Q79. Evaluate $\int_1^3 |x^2 - 4|\, dx$. 3M

Ans. Split at 2: $\int_1^2 (4 - x^2)dx + \int_2^3 (x^2 - 4)dx = \left(4 - \frac{7}{3}\right) + \left(\frac{19}{3} - 4\right) = \frac{5}{3} + \frac{7}{3} = 4$.

One more case study (4 marks)

Q80. Cost of production. A small factory finds that the marginal cost of making x notebooks is $C'(x) = 2x + \frac{1}{x+1}$ (in Rs hundred). (i) Find the cost function $C(x)$ if $C(0) = 5$. (2) (ii) Find the extra cost of raising production from $x = 1$ to $x = 3$. (2) 4M Must do

Ans. (i) $C(x) = x^2 + \log(x+1) + K$; $C(0) = 0 + 0 + K = 5$, so $C(x) = x^2 + \log(x+1) + 5$. (ii) $\int_1^3 C'(x)dx = \left[x^2 + \log(x+1)\right]_1^3 = (9 + \log 4) - (1 + \log 2) = 8 + \log 2$ (Rs hundred).

Last practice round

Q81. $\int \frac{\sin^2 x - \cos^2 x}{\sin^2 x\cos^2 x}\, dx$ equals 1M

(a) $\tan x - \cot x + C$ (b) $\tan x + \cot x + C$ (c) $-\tan x + \cot x + C$ (d) $\sin x + \cos x + C$

(contd.) (b) $= \int(\sec^2 x - \operatorname{cosec}^2 x)dx = \tan x + \cot x + C$.

Q82. $\int_0^1 |2x-1|\,dx$ is **1M**

(a) 0 (b) 1 (c) $\frac{1}{2}$ (d) 2

Ans. (c) Split at $\frac{1}{2}$: $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.

Q83. Find $\int \frac{1-\cos 2x}{1+\cos 2x}\,dx$. **2M**

Ans. $= \int \frac{2\sin^2 x}{2\cos^2 x}dx = \int(\sec^2 x - 1)dx = \tan x - x + C$.

Q84. Evaluate $\int_0^{\pi/2} e^x(\sin x + \cos x)\,dx$. **2M**

Ans. $\left[e^x \sin x\right]_0^{\pi/2} = e^{\pi/2}$.

Q85. Find $\int x^2 \tan^{-1} x\,dx$. **3M**

Ans. $u = \tan^{-1} x$: $\frac{x^3}{3}\tan^{-1} x - \frac{1}{3}\int \frac{x^3}{1+x^2}dx$; $\frac{x^3}{1+x^2} = x - \frac{x}{1+x^2}$. Answer $\frac{x^3}{3}\tan^{-1} x - \frac{x^2}{6} + \frac{1}{6}\log(1+x^2) + C$.

Q86. Find $\int \frac{dx}{e^x - 1}$. **3M**

Ans. $= \int \frac{e^{-x}dx}{1-e^{-x}}$; put $1 - e^{-x} = t, e^{-x}dx = dt$: $\log|1 - e^{-x}| + C$, i.e. $\log|e^x - 1| - x + C$.

Q87. Evaluate $\int_0^{\pi/2} x \sin x \cos x\,dx$. **3M**

Ans. $= \frac{1}{2}\int_0^{\pi/2} x \sin 2x\,dx = \frac{1}{2}\left[-\frac{x\cos 2x}{2} + \frac{\sin 2x}{4}\right]_0^{\pi/2} = \frac{1}{2} \times \frac{\pi}{4} = \frac{\pi}{8}$.

Q88. Evaluate $\int_0^{\pi/2} \log(\sin 2x)\,dx$. **5M** HOTS

Ans. $\log \sin 2x = \log 2 + \log \sin x + \log \cos x$. So $I = \frac{\pi}{2}\log 2 + \left(-\frac{\pi}{2}\log 2\right) + \left(-\frac{\pi}{2}\log 2\right) = -\frac{\pi}{2}\log 2$ (using Ex 51).

Q89. Moving ball. A ball rolls with velocity $v(t) = 6t - t^2$ m/s for $0 \le t \le 6$. (i) Distance covered in the first 3 s. (1) (ii) Distance in the full 6 s. (1) (iii) If its position $s(0) = 2$ m, find $s(t)$ and $s(3)$. (2) **4M**

Ans. (i) $\int_0^3 (6t - t^2)dt = 27 - 9 = 18$ m. (ii) $\int_0^6 (6t - t^2)dt = 108 - 72 = 36$ m ($v \ge 0$ throughout). (iii) $s(t) = 3t^2 - \frac{t^3}{3} + 2$; $s(3) = 27 - 9 + 2 = 20$ m.

Q90. If $\int f(x)\,dx = g(x) + C$, then $\int f(2x+3)\,dx$ equals **1M**

(a) $g(2x+3) + C$ (b) $\frac{1}{2}g(2x+3) + C$ (c) $2g(2x+3) + C$ (d) $\frac{1}{3}g(2x+3) + C$

Ans. (b) Linear rule: divide by the coefficient of x, so $\frac{1}{2}g(2x+3) + C$.

Q91. Find $\int \frac{x^3 - x^2 + x - 1}{x - 1}\,dx$. **2M**

Ans. $x^3 - x^2 + x - 1 = (x-1)(x^2+1)$, so integrand $= x^2 + 1$. Answer $\frac{x^3}{3} + x + C$.

Q92. Find $\int \frac{dx}{x\log x\,(2 + \log x)}$. **3M**

Ans. $\log x = t$, $\frac{dx}{x} = dt$: $\int \frac{dt}{t(t+2)} = \frac{1}{2}\int\left(\frac{1}{t} - \frac{1}{t+2}\right)dt$. Answer $\frac{1}{2}\log\left|\frac{\log x}{2+\log x}\right| + C$.

Q93. Evaluate $\int_0^{\pi/2} \frac{\sin x \cos x}{\cos^2 x + 3\cos x + 2}\,dx$. **5M** Must do

Ans. $\cos x = t$, $-\sin x\,dx = dt$, limits $1 \to 0$: $I = \int_0^1 \frac{t\,dt}{(t+1)(t+2)} = \int_0^1 \left(\frac{-1}{t+1} + \frac{2}{t+2}\right)dt = \left[2\log(t+2) - \log(t+1)\right]_0^1 = 2\log 3 - \log 2 - 2\log 2$. Answer $\log\frac{9}{8}$.

Q94. Find $\int \frac{2x}{(1+x^2)^2}\,dx$. **2M**

Ans. $1 + x^2 = t$: $\int t^{-2}dt = -\frac{1}{t} = -\frac{1}{1+x^2} + C$.

Q95. Evaluate $\int_0^1 x^2 e^x\,dx$. **3M**

Ans. From Ex 31: $\left[e^x(x^2 - 2x + 2)\right]_0^1 = e(1) - 1(2) = e - 2$.

Q96. Find $\int \frac{x + \sin x}{1 + \cos x}\,dx$. **3M** HOTS

Ans. $\frac{x}{2\cos^2\frac{x}{2}} + \frac{2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}} = \frac{x}{2}\sec^2\frac{x}{2} + \tan\frac{x}{2}$. By parts on the first: $\int \frac{x}{2}\sec^2\frac{x}{2}dx = x\tan\frac{x}{2} - \int\tan\frac{x}{2}dx$, which cancels the second. Answer $x\tan\frac{x}{2} + C$.

Q97. Evaluate $\int_0^{\pi/4} \sin^2 x \cos^2 x\,dx$. **3M**

Ans. $\sin^2 x\cos^2 x = \frac{\sin^2 2x}{4} = \frac{1-\cos 4x}{8}$. $\left[\frac{x}{8} - \frac{\sin 4x}{32}\right]_0^{\pi/4} = \frac{\pi}{32}$.

Q98. Find $\int \frac{dx}{\sin^4 x + \cos^4 x}$. **5M** HOTS

Ans. Divide by $\cos^4 x$: $\int \frac{\sec^4 x\,dx}{\tan^4 x + 1} = \int \frac{(1+\tan^2 x)\sec^2 x\,dx}{1+\tan^4 x}$. Put $\tan x = t$: $\int \frac{1+t^2}{1+t^4}dt$, which is Ex 54. Answer $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\tan^2 x - 1}{\sqrt{2}\tan x}\right) + C$.

Chapter checklist: tick karo

- → ☐ 30 standard integrals yaad (Formula Box 1 to 4)
- → ☐ Substitution: f and f' pehchaan leta hoon
- → ☐ 6 special integrals + complete the square
- → ☐ $(px+q)$ type: A(derivative) $+ B$
- → ☐ Partial fractions: all 5 forms, improper pehle divide
- → ☐ By parts: ILATE, $e^x[f + f']$, the 3 root formulas
- → ☐ Definite: change of limits, modulus split
- → ☐ Properties: King, odd/even, $\log\sin x$ wala

Quick fact $\int_0^1 x^n dx = \frac{1}{n+1}$.

Most asked: area of circle/ellipse, parabola cut by a line

APPLICATION OF INTEGRALS

1. Area Under a Simple Curve

A definite integral adds up thin strips. The area bounded by a curve, an axis and two lines is found by choosing the right strip and the right limits. (Syllabus 2026-27: area under simple curves only; area between two curves is deleted.)

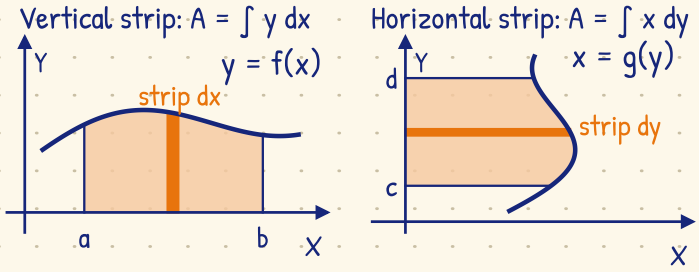


Fig: vertical strip (width dx) and horizontal strip (width dy)

Key results

→ Area bounded by $y = f(x)$, the **x-axis** and $x = a$, $x = b$ ($f \geq 0$):

$$A = \int_a^b y\,dx = \int_a^b f(x)\,dx$$

→ Area bounded by $x = g(y)$, the **y-axis** and $y = c$, $y = d$:

$$A = \int_c^d x\,dy = \int_c^d g(y)\,dy$$

→ If the curve is **below** the x-axis ($f \leq 0$):

$$A = \left|\int_a^b f(x)\,dx\right|$$

→ Part above, part below: split at the zero and add the absolute values.

Formula Box 1: Integrals you need

→ $\int x^n\,dx = \dfrac{x^{n+1}}{n+1} + C$

→ $\int \sqrt{x}\,dx = \dfrac{2}{3}x^{3/2} + C$

→ $\int \sqrt{a^2 - x^2}\,dx = \dfrac{x}{2}\sqrt{a^2 - x^2} + \dfrac{a^2}{2}\sin^{-1}\dfrac{x}{a} + C$

→ $\int_0^a \sqrt{a^2 - x^2}\,dx = \dfrac{\pi a^2}{4}$ (quarter circle)

→ $\int_a^b f(x)\,dx = F(b) - F(a)$

Quick fact Area is always positive: if $\int$ comes negative, take its modulus.

Formula Box 2: Standard areas (verify in 1 line)

→ Circle $x^2 + y^2 = a^2$: $A = \pi a^2$

→ Ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$: $A = \pi ab$

→ Parabola $y^2 = 4ax$ and its latus rectum $x = a$: $A = \dfrac{8a^2}{3}$

→ Parabola $y^2 = 4ax$ and line $x = k$: $A = \dfrac{8}{3}\sqrt{a}\,k^{3/2}$

→ Symmetry: find one part, multiply by 2 or 4

Yaad karo: "Pehle Figure, Phir Formula"

Figure banao, Strip lagao (vertical ya horizontal), Limits padho figure se, phir integrate. "FSL": Figure, Strip, Limits. Bina figure ke area ka sawal mat chhedo.

2. Parabola Problems

Solved Example 1

Find the area of the region bounded by $y^2 = 4x$, the lines $x = 1$, $x = 4$ and the x-axis in the first quadrant.

1. First quadrant: $y = 2\sqrt{x}$
2. $A = \int_1^4 2\sqrt{x}\,dx = 2 \times \dfrac{2}{3}\left[x^{3/2}\right]_1^4$
3. $= \dfrac{4}{3}(8 - 1) = \dfrac{28}{3}$

✔ Area $= \dfrac{28}{3}$ sq units

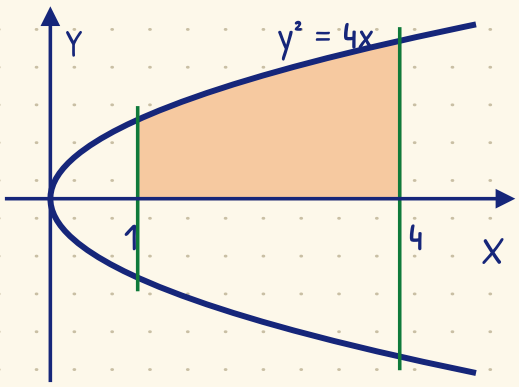


Fig: Example 1, shaded region between $x = 1$ and $x = 4$

Quick fact $\int_0^a \sqrt{a^2 - x^2}\,dx = \dfrac{\pi a^2}{4}$

Quick fact Ellipse area πab; circle is the case $a = b$.

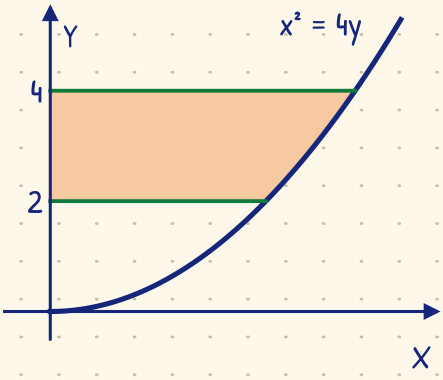


Fig: Example 2, horizontal strips from $y = 2$ to $y = 4$

Solved Example 2

Find the area bounded by $x^2 = 4y$, the lines $y = 2$, $y = 4$ and the y-axis in the first quadrant.

1. Horizontal strip: $x = 2\sqrt{y}$
2. $A = \int_2^4 2\sqrt{y}\,dy = \frac{4}{3}\left[y^{3/2}\right]_2^4$
3. $= \frac{4}{3}\left(8 - 2\sqrt{2}\right)$

✔ Area $= \frac{32 - 8\sqrt{2}}{3}$ sq units

Solved Example 3

Find the area of the region bounded by the parabola $y^2 = 12x$ and its latus rectum.

1. $4a = 12 \Rightarrow a = 3$; latus rectum is $x = 3$, ends $(3, \pm 6)$
2. By symmetry about the x-axis: $A = 2\int_0^3 \sqrt{12x}\,dx = 2 \times 2\sqrt{3} \times \frac{2}{3}\left[x^{3/2}\right]_0^3$
3. $= \frac{8\sqrt{3}}{3} \times 3\sqrt{3} = 24$

✔ Area $= 24$ sq units (check: $\frac{8a^2}{3} = \frac{8 \times 9}{3} = 24$)

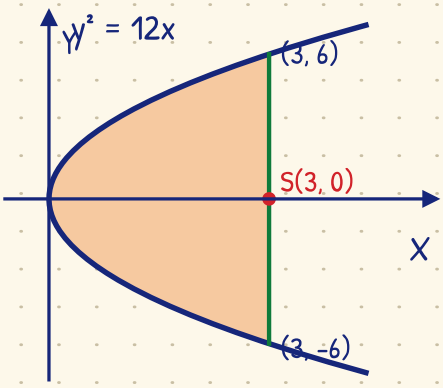


Fig: Example 3, region cut off by the latus rectum $x = 3$



Shivam Bhaiya says

Area ke sawal mein **figure + shading** ke alag se marks hote hain. Ek saaf figure, limits likhi hui, aur last mein **"sq units"** zaroor likhna. Ye 3-5 marks ka scoring sawal hai, chhodna mat.

Solved Example 4

Find the area bounded by the curve $x^2 = y$, the lines $y = 1$, $y = 4$ and the y-axis (first quadrant).

1. $x = \sqrt{y}$ (horizontal strip)
2. $A = \int_1^4 \sqrt{y}\,dy = \frac{2}{3}\left[y^{3/2}\right]_1^4 = \frac{2}{3}(8 - 1)$

✔ Area $= \frac{14}{3}$ sq units

3. Circle and Ellipse

Both are symmetric about both axes, so find the **first-quadrant** area A_1 and multiply by 4. Use $\int \sqrt{a^2 - x^2}\,dx$ from Formula Box 1.

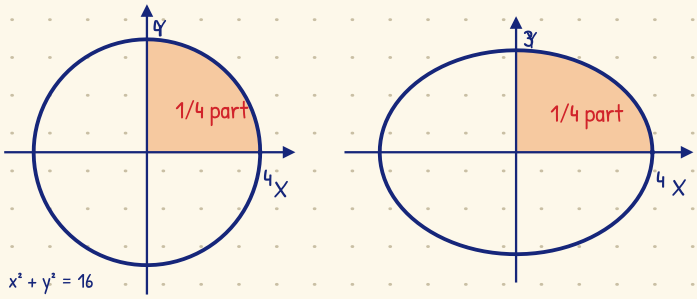


Fig: circle of radius 4 and ellipse with $a = 4, b = 3$; shaded = first quadrant part

Solved Example 5

Find the area enclosed by the circle $x^2 + y^2 = 16$.

1. $y = \sqrt{16 - x^2}$ in the first quadrant
2. $A_1 = \int_0^4 \sqrt{16 - x^2}\,dx = \left[\frac{x}{2}\sqrt{16 - x^2} + 8\sin^{-1}\frac{x}{4}\right]_0^4$
3. $= 0 + 8 \times \frac{\pi}{2} - 0 = 4\pi$
4. Total $= 4A_1$

✔ Area $= 16\pi$ sq units

Solved Example 6

Find the area of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ by integration.

1. $y = \frac{3}{4}\sqrt{16 - x^2}$ (first quadrant)
2. $A_1 = \frac{3}{4}\int_0^4 \sqrt{16 - x^2}\,dx = \frac{3}{4} \times 4\pi = 3\pi$
3. Total $= 4 \times 3\pi$

✔ Area $= 12\pi$ sq units $(= \pi ab$ ✓$)$

Quick fact $\int_0^k \sqrt{x}\,dx = \frac{2}{3}k^{3/2}$

Quick fact $y^2 = 4ax$: focus $(a, 0)$, latus rectum $x = a$, length $4a$.

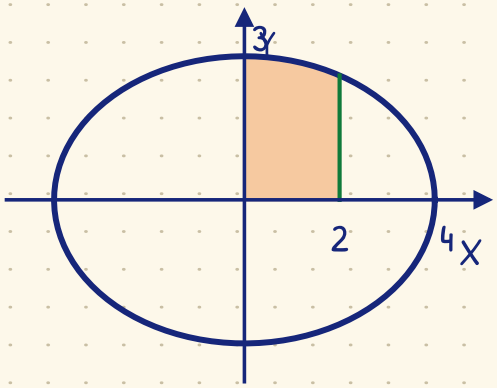


Fig: Example 7, part of the ellipse between $x = 0$ and $x = 2$

Solved Example 7

Find the area of the region bounded by the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and the ordinates $x = 0$ and $x = 2$ in the first quadrant.

1. $A = \frac{3}{4}\int_0^2 \sqrt{16 - x^2}\,dx =$ $\frac{3}{4}\left[\frac{x}{2}\sqrt{16 - x^2} + 8\sin^{-1}\frac{x}{4}\right]_0^2$
2. $= \frac{3}{4}\left[\sqrt{12} + 8 \times \frac{\pi}{6}\right] = \frac{3}{4}\left[2\sqrt{3} + \frac{4\pi}{3}\right]$

✔ Area $= \frac{3\sqrt{3}}{2} + \pi$ sq units

Solved Example 8

Find the area of the smaller part of the circle $x^2 + y^2 = 16$ cut off by the line $x = 2\sqrt{2}$.

1. Symmetric about the x-axis: $A = 2\int_{2\sqrt{2}}^4 \sqrt{16 - x^2}\,dx$
2. $= 2\left[\frac{x}{2}\sqrt{16 - x^2} + 8\sin^{-1}\frac{x}{4}\right]_{2\sqrt{2}}^4$
3. $= 2\left[4\pi - \left(\sqrt{2} \times 2\sqrt{2} + 8 \times \frac{\pi}{4}\right)\right] =$ $2\left[4\pi - 4 - 2\pi\right]$

✔ Area $= 4\pi - 8$ sq units $= 8\left(\frac{\pi}{2} - 1\right)$

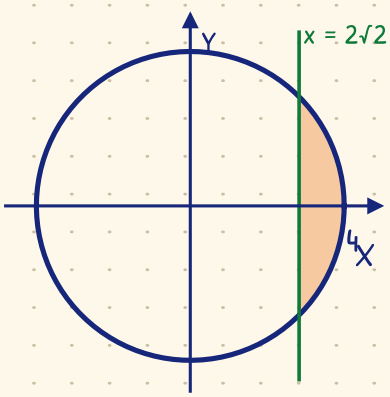


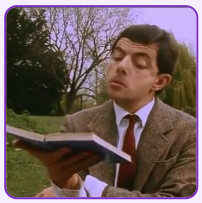
Fig: Example 8, the smaller segment (shaded) cut off by the vertical line

Quick fact 1 sq unit = area of a 1×1 square on the graph.

Yaad karo: "Chaar guna chamatkar"

Circle/ellipse poore ka area: sirf **pehle quadrant** ka nikalo aur $\times 4$. $\int_0^a \sqrt{a^2 - x^2}\,dx = \frac{\pi a^2}{4}$ yaad ho to 2 line mein answer: circle πa^2, ellipse πab.

Meme samjho



Question padha, figure nahi banaya, seedha integrate... **ab limits kahan se laaun?**

Asli baat: Limits figure se aati hain: the curve meets the line or axis at those points. For the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ the x-limits are 0 to 4.

4. Lines and Sign Care

A straight line is the easiest "curve". The only trap: where the line goes **below the x-axis** the integral is negative, so split at the point where $y = 0$ and take the absolute value of each part.

Solved Example 9

Find the area of the region bounded by the line $y = 3x + 2$, the x-axis and the ordinates $x = -1$ and $x = 1$.

1. $y = 0$ at $x = -\frac{2}{3}$; below the axis on $\left[-1, -\frac{2}{3}\right]$
2. $A_1 = \left|\int_{-1}^{-2/3}(3x + 2)\,dx\right| =$ $\left|\left[\frac{3x^2}{2} + 2x\right]_{-1}^{-2/3}\right| = \left|-\frac{1}{6}\right| = \frac{1}{6}$
3. $A_2 = \int_{-2/3}^{1}(3x + 2)\,dx = \frac{7}{2} + \frac{2}{3} = \frac{25}{6}$
4. $A = A_1 + A_2 = \frac{26}{6}$

✔ Area $= \frac{13}{3}$ sq units

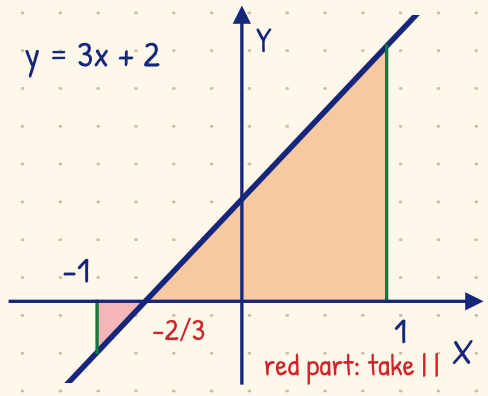


Fig: Example 9, red part below the axis, orange part above

Quick fact $x^2 = 4ay$ opens upward; use horizontal strips with the y-axis.

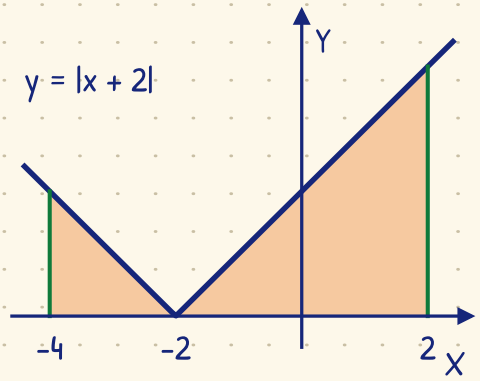


Fig: two triangles under $y = |x + 2|$

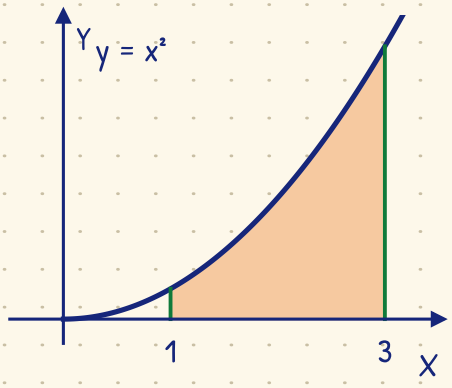


Fig: area under $y = x^2$ from $x = 1$ to $x = 3$

Solved Example 10

Find the area bounded by $y = |x + 2|$, the x-axis and the lines $x = -4$ and $x = 2$.

1. $y = -(x + 2)$ for $x < -2$ and $y = x + 2$ for $x \geq -2$ (both above the axis)
2. $A = \int_{-4}^{-2} -(x + 2)\,dx + \int_{-2}^{2} (x + 2)\,dx$
3. $= 2 + 8$ (two triangles: $\frac{1}{2} \times 2 \times 2$ and $\frac{1}{2} \times 4 \times 4$)

✔ Area $= 10$ sq units

Solved Example 11

Find the area bounded by $y = x$, the x-axis and the lines $x = -2$, $x = 2$. Is it equal to $\int_{-2}^{2} x\,dx$?

1. $\int_{-2}^{2} x\,dx = 0$, but area is never zero here
2. Area $= \left|\int_{-2}^{0} x\,dx\right| + \int_{0}^{2} x\,dx = 2 + 2$

✔ Area $= 4$ sq units (integral $= 0$, area $= 4$)

Yaad karo: "Neeche gaya to Mod Lagaya"

Curve x-axis ke **neeche** gaya, integral negative aayega. Wahan $|\ |$ **(mod)** lagao. Upar-neeche dono ho to zero point pe tod ke alag-alag jodo.

5. Extra Practice

Solved Example 12

Find the area of the region bounded by the parabola $y^2 = 8x$ and the line $x = 2$.

1. Symmetric: $A = 2\int_0^2 \sqrt{8x}\,dx = 2 \times 2\sqrt{2} \times \frac{2}{3}\left[x^{3/2}\right]_0^2$
2. $= \frac{8\sqrt{2}}{3} \times 2\sqrt{2} = \frac{32}{3}$

✔ Area $= \frac{32}{3}$ sq units

Quick fact Triangle check: area under a line = $\frac{1}{2} \times$ base $\times$ height.

Solved Example 13

Find the area of the region bounded by $y = x^2$, the x-axis and the lines $x = 1$, $x = 3$.

1. $A = \int_1^3 x^2\,dx = \left[\frac{x^3}{3}\right]_1^3 = 9 - \frac{1}{3}$

✔ Area $= \frac{26}{3}$ sq units

Solved Example 14

Using integration, find the area of the circle $x^2 + y^2 = a^2$ lying in the first quadrant.

1. $A = \int_0^a \sqrt{a^2 - x^2}\,dx = \left[\frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}\right]_0^a$
2. $= \frac{a^2}{2} \times \frac{\pi}{2}$

✔ Area $= \frac{\pi a^2}{4}$ sq units

Solved Example 15

The area bounded by $y^2 = 4x$ and the line $x = k$ is $\frac{64}{3}$. Find k.

1. Area $= 2\int_0^k 2\sqrt{x}\,dx = \frac{8}{3}k^{3/2}$
2. $\frac{8}{3}k^{3/2} = \frac{64}{3} \Rightarrow k^{3/2} = 8$

✔ $k = 4$

Quick fact $\sin^{-1} 1 = \frac{\pi}{2}$, $\sin^{-1}\frac{1}{2} = \frac{\pi}{6}$, $\sin^{-1}\frac{1}{\sqrt{2}} = \frac{\pi}{4}$

Quick fact Semicircle $y = \sqrt{a^2 - x^2}$: area $\frac{\pi a^2}{2}$

Quick fact $\int_0^a \sqrt{ax}\,dx = \frac{2}{3}a^2$

6. More Solved Problems

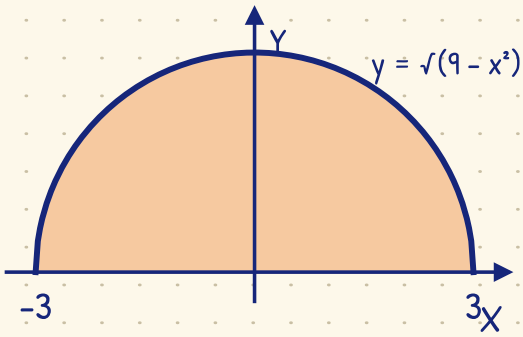


Fig: upper half of the circle of radius 3

Solved Example 16

Find the area of the region bounded by the curve $y = \sqrt{9 - x^2}$ and the x-axis.

1. $y = \sqrt{9 - x^2} \Rightarrow x^2 + y^2 = 9, y \geq 0$: upper semicircle
2. $A = \int_{-3}^{3} \sqrt{9 - x^2}\, dx = 2\int_{0}^{3} \sqrt{9 - x^2}\, dx = 2 \times \frac{9\pi}{4}$

✔ Area $= \frac{9\pi}{2}$ sq units

Solved Example 17

Find the area bounded by $x^2 = 16y$, the lines $y = 1$, $y = 4$ and the y-axis in the first quadrant.

1. $x = 4\sqrt{y}$
2. $A = \int_{1}^{4} 4\sqrt{y}\, dy = \frac{8}{3}\left[y^{3/2}\right]_{1}^{4} = \frac{8}{3}(8 - 1)$

✔ Area $= \frac{56}{3}$ sq units

Solved Example 18

Find the area of the region in the first quadrant bounded by the circle $x^2 + y^2 = 8$ and the lines $x = 0$, $x = 2$.

1. $A = \int_{0}^{2} \sqrt{8 - x^2}\, dx = \left[\frac{x}{2}\sqrt{8 - x^2} + 4\sin^{-1}\frac{x}{2\sqrt{2}}\right]_{0}^{2}$
2. $= \frac{2}{2} \times 2 + 4\sin^{-1}\frac{1}{\sqrt{2}} = 2 + 4 \times \frac{\pi}{4}$

✔ Area $= 2 + \pi$ sq units

Quick fact Area of a semi-ellipse $= \frac{\pi ab}{2}$

Quick fact $x^2 = 4ay$ and $y = a$: area $= \frac{8a^2}{3}$ too (same shape, turned).

Quick fact $\int_{a}^{b} c\, dx = c(b - a)$: area of a rectangle.

Solved Example 19

Find the area of the upper half of the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$.

1. $y = \frac{2}{3}\sqrt{9 - x^2}, -3 \leq x \leq 3$
2. $A = 2 \times \frac{2}{3}\int_{0}^{3} \sqrt{9 - x^2}\, dx = \frac{4}{3} \times \frac{9\pi}{4}$

✔ Area $= 3\pi$ sq units $(= \frac{1}{2}\pi ab$ ✓$)$

Solved Example 20

Find the area bounded by $y = 2x + 1$, the x-axis and the lines $x = 0$, $x = 2$. Check by geometry.

1. $A = \int_{0}^{2}(2x + 1)\, dx = \left[x^2 + x\right]_{0}^{2} = 6$
2. Trapezium: parallel sides 1 and 5, width 2: $\frac{1}{2}(1 + 5) \times 2 = 6$ ✓

✔ Area $= 6$ sq units

Solved Example 21

Find the area bounded by $x^2 = 4y$, the y-axis and the line $y = 1$, in the first quadrant, using (a) horizontal strips (b) vertical strips.

1. (a) $\int_{0}^{1} 2\sqrt{y}\, dy = \frac{4}{3}$
2. (b) Region: $0 \leq x \leq 2$, between the curve $y = \frac{x^2}{4}$ and $y = 1$: rectangle 2×1 minus $\int_{0}^{2} \frac{x^2}{4}\, dx = \frac{2}{3}$, i.e. $2 - \frac{2}{3} = \frac{4}{3}$ ✓

✔ Area $= \frac{4}{3}$ sq units

Answer writing format (5 marks)

1. Draw the figure, shade the region, mark intersection points.
2. Write the strip: "$A = \int_{a}^{b} y\, dx$" with limits.
3. Put y from the curve, integrate with the formula.
4. Use symmetry if any ($\times 2$, $\times 4$).
5. Final line: "Required area $= \ldots$ sq units".



Meme samjho

Ellipse ka area πa^2 likh diya? **Look toh dekho apna!**

Asli baat: Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ has area πab, not πa^2. Only a circle ($a = b$) gives πa^2.

Solved Example 22

Find the area bounded by the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (general proof of πab).

1. $y=\frac{b}{a}\sqrt{a^2-x^2}$ in the first quadrant
2. $A=4\times\frac{b}{a}\int_0^a\sqrt{a^2-x^2}\,dx=4\times\frac{b}{a}\times\frac{\pi a^2}{4}$

✔ Area $=\pi ab$ sq units

Solved Example 23

Find the area of the region bounded by $y^2=4ax$ and its latus rectum (general result).

1. Latus rectum $x=a$; symmetric about the x-axis
2. $A=2\int_0^a 2\sqrt{a}\sqrt{x}\,dx=4\sqrt{a}\times\frac{2}{3}a^{3/2}$

✔ Area $=\frac{8a^2}{3}$ sq units

Solved Example 24

Find the area bounded by the line $x+2y=8$, the x-axis and the lines $x=2$, $x=4$.

1. $y=\frac{8-x}{2}$, positive on $[2,4]$
2. $A=\int_2^4\frac{8-x}{2}\,dx=\frac{1}{2}\left[8x-\frac{x^2}{2}\right]_2^4=\frac{1}{2}(24-14)$

✔ Area $=5$ sq units

Solved Example 25

Find the area of the region bounded by the circle $x^2+y^2=4$ and the lines $x=1$, $x=2$ in the first quadrant.

1. $A=\int_1^2\sqrt{4-x^2}\,dx=\left[\frac{x}{2}\sqrt{4-x^2}+2\sin^{-1}\frac{x}{2}\right]_1^2$
2. $=\left(0+2\times\frac{\pi}{2}\right)-\left(\frac{\sqrt{3}}{2}+2\times\frac{\pi}{6}\right)=\pi-\frac{\sqrt{3}}{2}-\frac{\pi}{3}$

✔ Area $=\frac{2\pi}{3}-\frac{\sqrt{3}}{2}$ sq units

Yaad karo: "sin inverse ki teen value"

Circle wale sawal mein limits pe yahi aate hain:
$\sin^{-1}1=\frac{\pi}{2}$, $\sin^{-1}\frac{1}{\sqrt{2}}=\frac{\pi}{4}$, $\sin^{-1}\frac{1}{2}=\frac{\pi}{6}$. "2, 4, 6": 1 pe $\frac{\pi}{2}$, $\frac{1}{\sqrt{2}}$ pe $\frac{\pi}{4}$, $\frac{1}{2}$ pe $\frac{\pi}{6}$.

Common mistakes

- → No figure: wrong limits and wrong strip (marks for figure lost too).
- → Forgetting $\times 2$ (symmetry about x-axis) for $y^2=4ax$ type regions.
- → Writing a negative area: take $|\ |$ for parts below the x-axis.
- → Using $y\,dx$ with y-limits (or $x\,dy$ with x-limits).
- → Missing "sq units", or using area between two curves (deleted) methods.

Most Asked in Boards

- → Area of circle or ellipse by integration (3M/5M)
- → Region bounded by $y^2=4ax$ (or $x^2=4ay$) and a line $x=k$ / $y=k$ (3M)
- → Smaller part of a circle cut off by a line $x=k$ (5M)
- → Area under a line between two ordinates, with sign care (2M/3M)
- → Part of an ellipse between ordinates $x=0$ and $x=k$ (3M)
- → Case study: arch, window or garden shaped as parabola/ellipse (4M)

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. The area enclosed by the circle $x^2+y^2=25$ is 1M
(a) 5π (b) 10π (c) 25π (d) $\frac{25\pi}{4}$
Ans. (c) $\pi a^2=25\pi$.

Q2. The area of the ellipse $\frac{x^2}{25}+\frac{y^2}{4}=1$ is 1M Board fav
(a) 20π (b) 10π (c) 7π (d) 100π
Ans. (b) $\pi ab=\pi\times5\times2=10\pi$.

Q3. The area bounded by $y=2x$, the x-axis and $x=3$ is 1M
(a) 6 (b) 9 (c) 18 (d) 3
Ans. (b) $\int_0^3 2x\,dx=9$ (a triangle $\frac{1}{2}\times3\times6$).

Q4. The area bounded by $y^2=4x$ and its latus rectum is 1M
(a) $\frac{8}{3}$ (b) $\frac{4}{3}$ (c) $\frac{16}{3}$ (d) $\frac{2}{3}$
Ans. (a) $a=1$: $\frac{8a^2}{3}=\frac{8}{3}$.

Q5. $\int_0^2\sqrt{4-x^2}\,dx$ equals 1M Most asked
(a) 4π (b) 2π (c) π (d) $\frac{\pi}{2}$
Ans. (c) Quarter of a circle of radius 2: $\frac{\pi\times4}{4}=\pi$.

Q6. **A:** The area bounded by $y = x$, the x-axis and $x = -1$, $x = 1$ is 1 sq unit. **R:** For a part below the x-axis, area is the absolute value of the integral. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) $\left|\int_{-1}^{0} x\,dx\right| + \int_{0}^{1} x\,dx = \frac{1}{2} + \frac{1}{2} = 1$; R is the reason.

Q7. **A:** The area of the ellipse $4x^2 + 9y^2 = 36$ is 6π. **R:** The area of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) $\frac{x^2}{9} + \frac{y^2}{4} = 1$, $\pi \times 3 \times 2 = 6\pi$; R explains A.

Q8. The area bounded by $y = \sqrt{16 - x^2}$ and the x-axis is 1M

(a) 16π (b) 8π (c) 4π (d) 2π

Ans. (b) Upper semicircle of radius 4: $\frac{16\pi}{2} = 8\pi$.

Q9. The area of the region bounded by $y = |x|$, the x-axis and $x = -2$, $x = 2$ is 1M Board fav

(a) 0 (b) 2 (c) 4 (d) 8

Ans. (c) Two triangles, each $\frac{1}{2} \times 2 \times 2 = 2$: total 4.

Q10. The area bounded by $y^2 = 4x$ and the line $x = 1$ (both sides of the x-axis) is 1M

(a) $\frac{8}{3}$ (b) $\frac{4}{3}$ (c) $\frac{2}{3}$ (d) 4

Ans. (a) $2\int_{0}^{1} 2\sqrt{x}\,dx = \frac{8}{3}$.

Q11. If the area of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ is 8π, then $b =$ 1M HOTS

(a) 1 (b) 2 (c) 4 (d) 8

Ans. (b) $\pi \times 4 \times b = 8\pi \Rightarrow b = 2$.

Very short (2 marks)

Q12. Find the area bounded by $y^2 = x$, the lines $x = 1$, $x = 4$ and the x-axis in the first quadrant. 2M

Ans. $A = \int_{1}^{4} \sqrt{x}\,dx = \frac{2}{3}\left[x^{3/2}\right]_{1}^{4} = \frac{2}{3}(8 - 1) = \frac{14}{3}$ sq units.

Q13. Find the area bounded by $y = \sqrt{x}$, the x-axis and the lines $x = 0$, $x = 9$. 2M

Ans. $A = \int_{0}^{9} \sqrt{x}\,dx = \frac{2}{3} \times 27 = 18$ sq units.

Q14. Find the area bounded by the line $y = x + 1$, the x-axis and the lines $x = 1$, $x = 3$. 2M Board fav

Ans. $A = \int_{1}^{3} (x + 1)\,dx = \left[\frac{x^2}{2} + x\right]_{1}^{3} = \frac{15}{2} - \frac{3}{2} = 6$ sq units.

Q15. Find the area of the ellipse $\frac{x^2}{4} + \frac{y^2}{1} = 1$ in the first quadrant. 2M

(contd.) $A_1 = \frac{1}{2}\int_{0}^{2} \sqrt{4 - x^2}\,dx = \frac{1}{2} \times \pi = \frac{\pi}{2}$ sq units.

Q16. Find the area bounded by $y = x^2$, the x-axis and $x = 0$, $x = 2$. 2M

Ans. $\int_{0}^{2} x^2\,dx = \frac{8}{3}$ sq units.

Q17. Find the area bounded by $y = 3x$, the x-axis and the lines $x = 1$, $x = 2$. 2M

Ans. $\int_{1}^{2} 3x\,dx = \frac{3}{2}(4 - 1) = \frac{9}{2}$ sq units.

Q18. Find the area of the region bounded by $x^2 = y$, the y-axis and the line $y = 9$ in the first quadrant. 2M

Ans. $x = \sqrt{y}$: $A = \int_{0}^{9} \sqrt{y}\,dy = \frac{2}{3}\left[y^{3/2}\right]_{0}^{9} = \frac{2}{3} \times 27 = 18$ sq units.

Short answer (3 marks)

Q19. Using integration, find the area of the ellipse $4x^2 + 9y^2 = 36$. 3M Most asked

Ans. $\frac{x^2}{9} + \frac{y^2}{4} = 1$, $y = \frac{2}{3}\sqrt{9 - x^2}$. $A = 4 \times \frac{2}{3}\int_{0}^{3} \sqrt{9 - x^2}\,dx = \frac{8}{3} \times \frac{9\pi}{4} = 6\pi$ sq units.

Q20. Find the area of the region bounded by $y^2 = 9x$, $x = 2$, $x = 4$ and the x-axis in the first quadrant. 3M Board fav

Ans. $A = \int_{2}^{4} 3\sqrt{x}\,dx = 2\left[x^{3/2}\right]_{2}^{4} = 2\left(8 - 2\sqrt{2}\right) = 16 - 4\sqrt{2}$ sq units.

Q21. Find the area bounded by $x^2 = 4y$, the line $y = 4$ and the y-axis in the first quadrant. 3M

Ans. $x = 2\sqrt{y}$: $A = \int_{0}^{4} 2\sqrt{y}\,dy = \frac{4}{3}\left[y^{3/2}\right]_{0}^{4} = \frac{32}{3}$ sq units.

Q22. Find the area of the region bounded by the line $y = 2x - 4$, the x-axis and the lines $x = 0$, $x = 3$. 3M Must do

Ans. $y = 0$ at $x = 2$. $A = \left|\int_{0}^{2} (2x - 4)\,dx\right| + \int_{2}^{3} (2x - 4)\,dx = |-4| + 1 = 5$ sq units.

Q23. Find the area of the region bounded by $x^2 = 9y$, the y-axis and the lines $y = 1$, $y = 4$ in the first quadrant. 3M

Ans. $x = 3\sqrt{y}$: $A = \int_{1}^{4} 3\sqrt{y}\,dy = 2\left[y^{3/2}\right]_{1}^{4} = 2(8 - 1) = 14$ sq units.

Q24. Find the area of the region bounded by the parabola $y^2 = 16x$ and the line $x = 4$. 3M Board fav

Ans. $A = 2\int_{0}^{4} 4\sqrt{x}\,dx = 8 \times \frac{2}{3} \times 8 = \frac{128}{3}$ sq units (check: latus rectum case $\frac{8a^2}{3}$, $a = 4$).

Q25. Find the area bounded by the line $y = 4 - x$, the x-axis and the lines $x = 0$, $x = 6$. 3M

(contd.) $y=0$ at $x=4$. $A=\int_0^4(4-x)\,dx+\left|\int_4^6(4-x)\,dx\right|=8+|-2|=10$ sq units.

Long answer (5 marks)

Q26. Find the area of the smaller part of the circle $x^2+y^2=a^2$ cut off by the line $x=\frac{a}{\sqrt{2}}$. 5M Board fav

Ans. $A=2\int_{a/\sqrt{2}}^{a}\sqrt{a^2-x^2}\,dx=$
$2\left[\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\frac{x}{a}\right]_{a/\sqrt{2}}^{a}=$
$2\left[\frac{\pi a^2}{4}-\frac{a^2}{4}-\frac{\pi a^2}{8}\right]=\frac{a^2}{2}\left(\frac{\pi}{2}-1\right)$ sq units.

Q27. Find the area of the region bounded by the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ and the ordinates $x=0$ and $x=3$ in the first quadrant. 5M

Ans. $y=\frac{4}{5}\sqrt{25-x^2}$. $A=$
$\frac{4}{5}\left[\frac{x}{2}\sqrt{25-x^2}+\frac{25}{2}\sin^{-1}\frac{x}{5}\right]_0^3=$
$\frac{4}{5}\left[6+\frac{25}{2}\sin^{-1}\frac{3}{5}\right]=\frac{24}{5}+10\sin^{-1}\frac{3}{5}$ sq units.

Q28. Using integration, find the area of the region bounded by the circle $x^2+y^2=9$ and the lines $x=0$, $x=\frac{3}{2}$ in the first quadrant. 5M Must do

Ans. $A=\int_0^{3/2}\sqrt{9-x^2}\,dx=$
$\left[\frac{x}{2}\sqrt{9-x^2}+\frac{9}{2}\sin^{-1}\frac{x}{3}\right]_0^{3/2}=\frac{3}{4}\times\frac{3\sqrt{3}}{2}+\frac{9}{2}\times\frac{\pi}{6}=\frac{9\sqrt{3}}{8}+\frac{3\pi}{4}$ sq units.

Case based (4 marks)

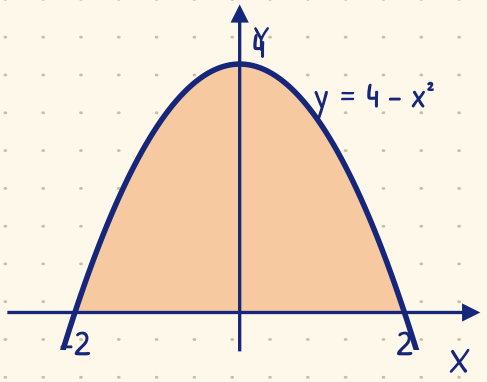


Fig: the arch $y=4-x^2$ (metres)

Q29. Garden arch. The opening of a garden gate is shaped like the parabola $y=4-x^2$ above the ground (x-axis), in metres. (i) Where does the arch meet the ground? (1) (ii) Find the height of the arch. (1) (iii) Find the area of the opening. (2) 4M

Ans. (i) $4-x^2=0\Rightarrow x=\pm2$, i.e. at $(-2,0)$ and $(2,0)$. (ii) At $x=0$: 4 m. (iii) $A=\int_{-2}^{2}(4-x^2)\,dx=$
$2\left[4x-\frac{x^3}{3}\right]_0^2=2\left(8-\frac{8}{3}\right)=\frac{32}{3}$ sq m.

Q30. Elliptical pond. A pond is shaped like $\frac{x^2}{36}+\frac{y^2}{16}=1$ (metres). (i) Write the lengths of its major and minor axes. (1) (ii) Write the integral for the area in the first quadrant. (1) (iii) Find the total area of the pond. (2) 4M

Ans. (i) $2a=12$ m, $2b=8$ m. (ii) $A_1=\frac{4}{6}\int_0^6\sqrt{36-x^2}\,dx$. (iii) $A_1=\frac{2}{3}\times9\pi=6\pi$, total $=24\pi$ sq m $(=\pi ab\checkmark)$.

Ek nazar mein: Area

Region	Set up	Answer
Circle $x^2+y^2=a^2$	$4\int_0^a\sqrt{a^2-x^2}\,dx$	πa^2
Ellipse	$4\frac{b}{a}\int_0^a\sqrt{a^2-x^2}\,dx$	πab
$y^2=4ax$, $x=a$	$2\int_0^a 2\sqrt{ax}\,dx$	$\frac{8a^2}{3}$
Line crossing axis	split, | |	sum of parts

Last minute: 6 one-liners

1. Figure, strip, limits, then integrate.
2. $y\,dx$ with x-limits; $x\,dy$ with y-limits.
3. Below the x-axis: take the absolute value.
4. Use symmetry: $\times2$ or $\times4$.
5. $\int_0^a\sqrt{a^2-x^2}\,dx=\frac{\pi a^2}{4}$.
6. Always end with "sq units".

Quick fact Area under $y=mx$ from 0 to $k=\frac{mk^2}{2}$.

Quick fact Area of circle $x^2+y^2=a^2$ in first quadrant $=\frac{\pi a^2}{4}$.

Apne notes / galtiyan yahan likho

Most asked: linear DE with IF, homogeneous DE by y = vx

DIFFERENTIAL EQUATIONS

1. What is a Differential Equation?

An equation involving the **independent variable** x, the **dependent variable** y and **derivatives of** y w.r.t. x is a **differential equation (DE)**. If there is only one independent variable it is an **ordinary** DE (the only kind in Class 12).

→ $\frac{dy}{dx} = \cos x$, $\frac{d^2y}{dx^2} + y = 0$, $x\frac{dy}{dx} + y = x^2$ are DEs.

→ Notation: $y' = \frac{dy}{dx}, y'' = \frac{d^2y}{dx^2}, y''' = \frac{d^3y}{dx^3}$.

Definition: Order and Degree

Order = order of the highest derivative present in the DE.

Degree = highest power of the highest-order derivative, only when the DE is a **polynomial in derivatives** ($y', y'', \ldots$). If a derivative sits inside $\sin, \cos, e^{(\,)}, \log$ etc., the degree is not defined.

↳ Order aur degree hamesha positive integers hote hain.

Order pehle dekho (sabse badi derivative), degree uski power.

Differential equation	Order	Degree
$\frac{dy}{dx} = e^x$	1	1
$\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^4 + y = 0$	2	3
$y''' + 2y'' + y' = 0$	3	1
$y' + \sin(y') = 0$	1	not defined
$(y'')^2 + \cos(y') = 0$	2	not defined

Yaad karo: "O = Oonchi derivative, D = Dum (power)"

Order = sabse Oonchi derivative kaunsi hai. Degree = us derivative ki power ka Dum. Pehle radicals/fractions hatao (polynomial banao), tabhi power padho. Derivative $\sin/e/\log$ ke andar? Degree **nahi**.

Solved Example 1

Find the order and degree (if defined) of

$$\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} = \frac{d^2y}{dx^2}.$$

1. Highest derivative is $\frac{d^2y}{dx^2}$, so **order = 2**.
2. Remove the fractional power: square both sides.
3. $\left[1 + (y')^2\right]^3 = (y'')^2$, a polynomial in y', y''.
4. Power of y'' is 2, so **degree = 2**.

✔ Order 2, degree 2

Solved Example 2

Find the order and degree of

$$\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^2 = x^2 \log\left(\frac{d^2y}{dx^2}\right)$$

1. Highest derivative: y'', so order = 2.
2. y'' appears inside $\log$, so the DE is not a polynomial in derivatives.

✔ Order 2, degree not defined

Solved Example 3

Find the order and degree of: (i) $\sqrt{1 + (y')^2} = 3y''$ (ii) $y = xy' + \frac{4}{y'}$ (iii) $(y''')^{2/3} = y'$

1. (i) Square: $1 + (y')^2 = 9(y'')^2$, so order 2, degree 2
2. (ii) Multiply by y': $yy' = x(y')^2 + 4$, so order 1, degree 2
3. (iii) Cube: $(y''')^2 = (y')^3$, so order 3, degree 2

✔ (i) 2, 2 (ii) 1, 2 (iii) 3, 2

Yaad karo: "Pehle saaf karo, phir power padho"

Fraction power ($\frac{1}{2}, \frac{2}{3}$) ya y' denominator mein ho to pehle square/cube/multiply karke **polynomial** banao. Degree hamesha **clean** form mein highest derivative ki power hai, kabhi fraction nahi.

2. Solutions: General and Particular

A function $y = \phi(x)$ that satisfies the DE (LHS = RHS after substituting) is a **solution**.

Key terms

→ **General solution:** contains arbitrary constants; number of constants = order of the DE. e.g. $y = a\cos x + b\sin x$ for $y'' + y = 0$.

→ **Particular solution:** obtained by giving particular values to the constants (using given conditions); it has **no** arbitrary constant. e.g. $y = 2\cos x$.

Quick fact $\frac{dy}{dx} = k$ (constant) gives $y = kx + C$, a family of parallel lines.

Yaad rakho $\int \frac{dx}{x} = \log|x| + C$; modulus mat bhoolo.

Recall $\int \tan x\,dx = \log|\sec x|$, $\int \cot x\,dx = \log|\sin x|$.

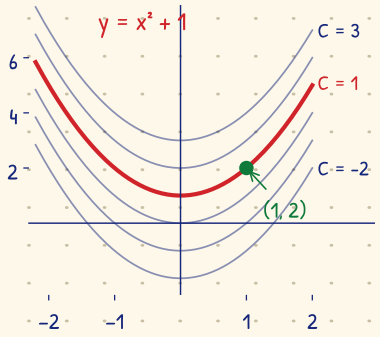


Fig: $\frac{dy}{dx} = 2x$ has general solution $y = x^2 + C$ (a family of curves). The condition $y(1) = 2$ picks one curve: $y = x^2 + 1$

Solved Example 4

Verify that $y = ae^{2x} + be^{-x}$ is a solution of $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 0$.

1. $y' = 2ae^{2x} - be^{-x}$, $y'' = 4ae^{2x} + be^{-x}$
2. LHS $= (4ae^{2x} + be^{-x}) - (2ae^{2x} - be^{-x}) - 2(ae^{2x} + be^{-x})$
3. $= (4 - 2 - 2)ae^{2x} + (1 + 1 - 2)be^{-x} = 0 =$ RHS ✓

✔ Verified; it is the general solution (2 constants, order 2)

Solved Example 5

Verify that $y = x \log|x| + Cx$ is a solution of $x\frac{dy}{dx} - y = x$ $(x \neq 0)$.

1. $\frac{dy}{dx} = \log|x| + 1 + C$
2. LHS $= x(\log|x| + 1 + C) - (x \log|x| + Cx) = x =$ RHS ✓

✔ Verified (1 constant, order 1, so general solution)

Solved Example 6

Verify that $xy = \log y + C$ is a solution of $\frac{dy}{dx} = \frac{y^2}{1 - xy}$ $(xy \neq 1)$.

1. Differentiate implicitly: $y + x\frac{dy}{dx} = \frac{1}{y}\frac{dy}{dx}$
2. $\frac{dy}{dx}\left(x - \frac{1}{y}\right) = -y \Rightarrow \frac{dy}{dx} \times \frac{xy - 1}{y} = -y$
3. $\frac{dy}{dx} = \frac{-y^2}{xy - 1} = \frac{y^2}{1 - xy}$ ✓

✔ Verified (implicit solution)

Do you know? $y = 0$ is always a solution of $\frac{dy}{dx} + Py = 0$ (trivial solution).

Quick fact Degree of a DE is never a fraction or negative.

Verify karne ka tareeka

1. Given y ko differentiate karo jitni order hai.
2. y, y', y'' DE ke LHS mein daalo.
3. Simplify karke RHS dikhao. Implicit solution ho to implicit differentiation karo.

Integration toolkit (needed in every DE)

- $\int x^n dx = \frac{x^{n+1}}{n+1}$ $(n \neq -1)$
- $\int \frac{dx}{x} = \log|x|$, $\int e^{ax} dx = \frac{e^{ax}}{a}$
- $\int a^x dx = \frac{a^x}{\log a}$, $\int \frac{dx}{1 + x^2} = \tan^{-1} x$
- $\int \frac{dx}{\sqrt{1 - x^2}} = \sin^{-1} x$
- $\int \frac{f'(x)}{f(x)} dx = \log|f(x)|$
- $\int \tan x \, dx = \log|\sec x|$
- $\int \cot x \, dx = \log|\sin x|$
- $\int \sec^2 x \, dx = \tan x$, $\int \operatorname{cosec}^2 x \, dx = -\cot x$
- By parts: $\int xe^x dx = (x - 1)e^x$
- $\int xe^{-x} dx = -(x + 1)e^{-x}$
- $\int e^x [f(x) + f'(x)] dx = e^x f(x)$
- $\int \frac{dx}{\sqrt{x^2 + a^2}} = \log\left|x + \sqrt{x^2 + a^2}\right|$

↳ Solution do tarah ka: **explicit** ($y = x^2 + C$, y akela) ya **implicit** ($x^2 - y^2 = Cx$). Dono valid hain, board mein implicit chhod sakte ho.

Solved Example 7

Solve $\frac{dy}{dx} = \sin^{-1} x$ (type $\frac{dy}{dx} = f(x)$: just integrate).

1. $y = \int \sin^{-1} x \, dx$; by parts with $\sin^{-1} x$ as first function
2. $= x \sin^{-1} x - \int \frac{x \, dx}{\sqrt{1 - x^2}} = x \sin^{-1} x + \sqrt{1 - x^2}$

✔ $y = x \sin^{-1} x + \sqrt{1 - x^2} + C$

3. Variables Separable Method

If $\frac{dy}{dx} = f(x)\, g(y)$, bring all y terms with dy and all x terms with dx, then integrate both sides.

$$\frac{dy}{g(y)} = f(x)\, dx \Rightarrow \int \frac{dy}{g(y)} = \int f(x)\, dx + C$$

↳ Constant sirf ek side lagao (RHS pe). Log wale answers

↳ mein C ko $\log|C|$ likhna allowed hai, answer neat ho jata hai.

Write as $f(x)\,g(y)$ → Separate y with dy → Integrate both sides → Add C once

Solved Example 8

Solve $\dfrac{dy}{dx} = \dfrac{1+y^2}{1+x^2}$.

1. Separate: $\dfrac{dy}{1+y^2} = \dfrac{dx}{1+x^2}$
2. Integrate: $\tan^{-1} y = \tan^{-1} x + C$

✔ $\tan^{-1} y = \tan^{-1} x + C$

Solved Example 9

Find the particular solution of $\dfrac{dy}{dx} = y\tan x$, given $y = 2$ when $x = 0$.

1. $\displaystyle\int \frac{dy}{y} = \int \tan x\,dx$
2. $\log|y| = \log|\sec x| + \log|C| \Rightarrow y = C\sec x$
3. $x = 0,\ y = 2$: $2 = C\sec 0 = C$

✔ $y = 2\sec x$

Solved Example 10

Solve $(1+e^{2x})\,dy + (1+y^2)\,e^x\,dx = 0$, given $y = \sqrt{3}$ when $x = 0$.

1. Separate: $\dfrac{dy}{1+y^2} = -\dfrac{e^x\,dx}{1+e^{2x}}$
2. RHS: put $e^x = t$, $e^x dx = dt$: $-\displaystyle\int \frac{dt}{1+t^2} = -\tan^{-1} t$
3. $\tan^{-1} y = -\tan^{-1}(e^x) + C$
4. $x = 0$: $\tan^{-1}\sqrt{3} = -\tan^{-1} 1 + C \Rightarrow \dfrac{\pi}{3} = -\dfrac{\pi}{4} + C \Rightarrow C = \dfrac{7\pi}{12}$

✔ $\tan^{-1} y + \tan^{-1}(e^x) = \dfrac{7\pi}{12}$

Solved Example 11

Solve $\dfrac{dy}{dx} = (1+x)(1+y^2)$.

1. $\dfrac{dy}{1+y^2} = (1+x)\,dx$
2. $\tan^{-1} y = x + \dfrac{x^2}{2} + C$

✔ $y = \tan\left(x + \dfrac{x^2}{2} + C\right)$

Check trick Answer ko differentiate karke DE mein daalo; LHS = RHS aaya to sahi.

Solved Example 12

Solve $x(1+y^2)\,dx - y(1+x^2)\,dy = 0$.

1. Separate and multiply by 2: $\displaystyle\int \frac{2x\,dx}{1+x^2} = \int \frac{2y\,dy}{1+y^2}$
2. $\log(1+x^2) = \log(1+y^2) + \log C$

✔ $1 + x^2 = C(1+y^2)$

Solved Example 13

Solve $\dfrac{dy}{dx} = e^{x+y} + x^2 e^y$.

1. Take e^y common: $\dfrac{dy}{dx} = e^y(e^x + x^2)$
2. $e^{-y}dy = (e^x + x^2)\,dx$
3. $-e^{-y} = e^x + \dfrac{x^3}{3} + C$

✔ $e^x + e^{-y} + \dfrac{x^3}{3} = C$

Solved Example 14

Find the particular solution of $(1+x^2)\dfrac{dy}{dx} = 1+y^2$, given $y = 1$ when $x = 0$.

1. $\displaystyle\int \frac{dy}{1+y^2} = \int \frac{dx}{1+x^2} \Rightarrow \tan^{-1} y = \tan^{-1} x + C$
2. $x = 0,\ y = 1$: $\dfrac{\pi}{4} = 0 + C$
3. $\tan^{-1} y - \tan^{-1} x = \dfrac{\pi}{4} \Rightarrow \dfrac{y-x}{1+xy} = 1$

✔ $y = \dfrac{1+x}{1-x}$

Solved Example 15

Solve $\dfrac{dy}{dx} = y\log y\ (y > 1)$.

1. $\displaystyle\int \frac{dy}{y\log y} = \int dx$; put $\log y = t$, $\dfrac{dy}{y} = dt$
2. $\log|\log y| = x + C_1 \Rightarrow \log y = Ce^x$

✔ $y = e^{Ce^x}$, i.e. $\log y = Ce^x$

Yaad rakho $e^{\log f(x)} = f(x)$, so I.F. often comes out as a simple function.

Quick fact Half-life problems: $N = N_0 e^{-kt}$, half-life $= \dfrac{\log 2}{k}$.

Board tip Write "which is a linear DE of the form $\frac{dy}{dx} + Py = Q$" before finding I.F.

Solved Example 16

Find the particular solution of $x(x-1)\dfrac{dy}{dx}=1$, given $y=0$ when $x=2$.

1. $dy=\dfrac{dx}{x(x-1)}=\left(\dfrac{1}{x-1}-\dfrac{1}{x}\right)dx$ (partial fractions)
2. $y=\log|x-1|-\log|x|+C=\log\left|\dfrac{x-1}{x}\right|+C$
3. $x=2,\ y=0: 0=\log\dfrac{1}{2}+C\Rightarrow C=\log 2$

✔ $y=\log\left|\dfrac{2(x-1)}{x}\right|$

Growth and decay problems

"Rate of change is proportional to the amount present" means $\dfrac{dN}{dt}=kN$ ($k>0$ growth, $k<0$ decay). Separating: $\dfrac{dN}{N}=k\,dt\Rightarrow N=N_0e^{kt}$.

Solved Example 17

A bacteria colony grows at a rate proportional to its size. 1000 bacteria become 2000 in 3 hours. How many will there be after 9 hours?

1. $\dfrac{dN}{dt}=kN\Rightarrow N=N_0e^{kt}$, with $N_0=1000$
2. $t=3: 2000=1000e^{3k}\Rightarrow e^{3k}=2$
3. $t=9: N=1000\,(e^{3k})^3=1000\times 2^3$

✔ 8000 bacteria

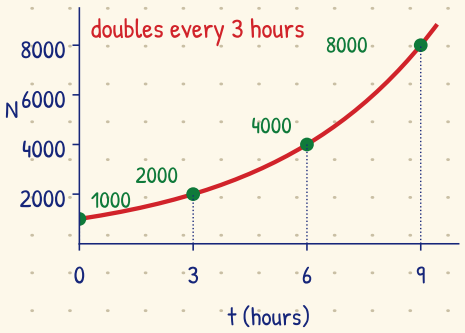


Fig: the colony doubles every 3 hours: 1000, 2000, 4000, 8000

Recall $\int e^x\left[f(x)+f'(x)\right]dx=e^xf(x)+C$ saves time in linear DEs.

Quick fact Order of $\dfrac{d^2y}{dx^2}=\left(\dfrac{dy}{dx}\right)^5$ is 2, degree 1.

Yaad rakho $\int e^x\sin x\,dx=\dfrac{e^x(\sin x-\cos x)}{2}$

Solved Example 18

Money in a bank grows continuously at $r\%$ per year. Find r if Rs 100 doubles in 8 years. (Use $\log_e 2=0.6931$.)

1. $\dfrac{dP}{dt}=\dfrac{r}{100}P\Rightarrow P=P_0e^{rt/100}$
2. Doubles in 8 years: $2=e^{8r/100}\Rightarrow\dfrac{8r}{100}=\log_e 2$
3. $r=\dfrac{100\times 0.6931}{8}=8.66$

✔ $r\approx 8.66\%$ per year

Solved Example 19

A town population grows continuously at 5% per year. In how many years will 20000 become 25000? (Use $\log_e 1.25=0.2231$.)

1. $\dfrac{dP}{dt}=0.05P\Rightarrow P=20000e^{0.05t}$
2. $25000=20000e^{0.05t}\Rightarrow e^{0.05t}=1.25$
3. $t=\dfrac{\log_e 1.25}{0.05}=20\times 0.2231=4.462$

✔ About 4.46 years

Solved Example 20

A radioactive sample decays at a rate proportional to the amount present. 100 g becomes 80 g in 2 years. How much is left after 6 years?

1. $N=N_0e^{-kt}; 80=100e^{-2k}\Rightarrow e^{-2k}=0.8$
2. $t=6: N=100\,(e^{-2k})^3=100\times 0.8^3=100\times 0.512$

✔ 51.2 g

Meme samjho



Exam mein variables separable DE aaya: x wale terms dx ke saath, y wale dy ke saath. "Humare liye kaam karoge?" Integration: haan bhai, ab toh karunga! **Asli baat:** Separate karne ke baad hi integrate karo. $\dfrac{dy}{dx}=f(x)g(y)$ ko $\dfrac{dy}{g(y)}=f(x)\,dx$ banaye bina integration possible nahi.

4. Homogeneous Differential Equations

Quick test: in $\dfrac{dy}{dx}=\dfrac{P(x,y)}{Q(x,y)}$, every term of P and Q has the **same total degree** (like x^2, xy, y^2). Also allowed: $\sin\dfrac{y}{x}$, $e^{y/x}$, $\log\dfrac{y}{x}$ (degree 0).

Recall $\int\sec x\,dx=\log|\sec x+\tan x|$

Definition: Homogeneous function

$F(x, y)$ is a **homogeneous function of degree** n if $F(\lambda x, \lambda y) = \lambda^n F(x, y)$ for every non-zero λ.

A DE $\frac{dy}{dx} = F(x, y)$ is **homogeneous** if $F(x, y)$ is homogeneous of degree zero, i.e. F can be written as $g\left(\frac{y}{x}\right)$.

Solved Example 21

Check whether $F(x, y) = x^2 + 3xy - y^2$ is homogeneous. Find its degree.

1. $F(\lambda x, \lambda y) = \lambda^2 x^2 + 3\lambda^2 xy - \lambda^2 y^2$
2. $= \lambda^2(x^2 + 3xy - y^2) = \lambda^2 F(x, y)$

✔ Homogeneous of degree 2

$\frac{dy}{dx} =$	Homogeneous?
$\frac{x^2 + y^2}{xy}$	Yes (deg 2 / deg 2)
$\frac{x + y}{x - y}$	Yes (deg 1 / deg 1)
$\frac{y}{x} + \sin\frac{y}{x}$	Yes (function of $\frac{y}{x}$)
$\frac{x^2 + y}{x}$	No (x^2 and y differ in degree)
$\frac{y + 1}{x}$	No (constant term 1)

Formula Box: Homogeneous DE

→ $\frac{dy}{dx} = g\left(\frac{y}{x}\right)$: put $y = vx$, $\frac{dy}{dx} = v + x\frac{dv}{dx}$

→ Then $x\frac{dv}{dx} = g(v) - v$, so separate:

$$\int \frac{dv}{g(v) - v} = \int \frac{dx}{x}$$

→ Finally replace $v = \frac{y}{x}$

→ $\frac{dx}{dy} = h\left(\frac{x}{y}\right)$: put $x = vy$, $\frac{dx}{dy} = v + y\frac{dv}{dy}$

Solved Example 23

Solve $\frac{dy}{dx} = \frac{y}{x} + \tan\frac{y}{x}$.

1. $y = vx$: $v + x\frac{dv}{dx} = v + \tan v \Rightarrow x\frac{dv}{dx} = \tan v$
2. $\int \cot v\, dv = \int \frac{dx}{x} \Rightarrow \log|\sin v| = \log|x| + \log|C|$
3. $\sin v = Cx$

✔ $\sin\frac{y}{x} = Cx$

Solved Example 22

Solve $(x^2 + y^2)\,dx = 2xy\,dy$.

1. $\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$ (degree 2 on top and bottom: homogeneous)
2. $y = vx$: $v + x\frac{dv}{dx} = \frac{1 + v^2}{2v} \Rightarrow x\frac{dv}{dx} = \frac{1 - v^2}{2v}$
3. $\int \frac{2v\,dv}{1 - v^2} = \int \frac{dx}{x} \Rightarrow -\log|1 - v^2| = \log|x| + \log|C_1|$
4. $x(1 - v^2) = C$; put $v = \frac{y}{x}$: $x\left(1 - \frac{y^2}{x^2}\right) = C$

✔ $x^2 - y^2 = Cx$

Solved Example 24

Find the particular solution of $x\frac{dy}{dx} = y + xe^{y/x}$, given $y = 0$ when $x = 1$.

1. $\frac{dy}{dx} = \frac{y}{x} + e^{y/x}$; put $y = vx$: $x\frac{dv}{dx} = e^v$
2. $\int e^{-v}dv = \int \frac{dx}{x} \Rightarrow -e^{-v} = \log|x| + C$
3. $x = 1,\ y = 0\ (v = 0)$: $-1 = 0 + C \Rightarrow C = -1$
4. $-e^{-y/x} = \log|x| - 1$

✔ $e^{-y/x} = 1 - \log|x|$

Solved Example 25

Solve $ye^{x/y}\,dx + \left(y - xe^{x/y}\right)dy = 0$, given $x = 0$ when $y = 1$.

1. $\frac{dx}{dy} = \frac{xe^{x/y} - y}{ye^{x/y}}$ (function of $\frac{x}{y}$), so put $x = vy$
2. $v + y\frac{dv}{dy} = \frac{ve^v - 1}{e^v} = v - e^{-v} \Rightarrow y\frac{dv}{dy} = -e^{-v}$
3. $\int e^v dv = -\int \frac{dy}{y} \Rightarrow e^v = -\log|y| + C$
4. $y = 1,\ x = 0$: $e^0 = 0 + C \Rightarrow C = 1$

✔ $e^{x/y} + \log|y| = 1$

Solved Example 27

Find the particular solution of $\frac{dy}{dx} = \frac{y}{x} + \sec\frac{y}{x}$, given $y = 0$ when $x = 1$.

1. $y = vx$: $x\frac{dv}{dx} = \sec v \Rightarrow \int \cos v\, dv = \int \frac{dx}{x}$
2. $\sin v = \log|x| + C$; $x = 1,\ v = 0$: $C = 0$

✔ $\sin\frac{y}{x} = \log|x|$

Solved Example 26

Solve $x\dfrac{dy}{dx} = y + \sqrt{x^2+y^2}$ $(x>0)$.

1. $\dfrac{dy}{dx} = \dfrac{y}{x} + \sqrt{1+\dfrac{y^2}{x^2}}$; put $y = vx$: $x\dfrac{dv}{dx} = \sqrt{1+v^2}$
2. $\displaystyle\int \frac{dv}{\sqrt{1+v^2}} = \int \frac{dx}{x} \Rightarrow \log\left|v+\sqrt{1+v^2}\right| = \log x + \log C$
3. $\dfrac{y}{x} + \dfrac{\sqrt{x^2+y^2}}{x} = Cx$

✔ $y + \sqrt{x^2+y^2} = Cx^2$

Solved Example 28

Solve $(x+2y)\,dx - x\,dy = 0$.

1. $\dfrac{dy}{dx} = \dfrac{x+2y}{x} = 1+2\dfrac{y}{x}$; $y = vx$: $x\dfrac{dv}{dx} = 1+v$
2. $\displaystyle\int \frac{dv}{1+v} = \int \frac{dx}{x} \Rightarrow \log|1+v| = \log|x| + \log|C|$
3. $1 + \dfrac{y}{x} = Cx$

✔ $x + y = Cx^2$

Solved Example 29

Solve $y^2\,dx + (x^2 - xy)\,dy = 0$.

1. $\dfrac{dy}{dx} = \dfrac{y^2}{xy-x^2} = \dfrac{v^2}{v-1}$ with $y = vx$
2. $x\dfrac{dv}{dx} = \dfrac{v^2}{v-1} - v = \dfrac{v}{v-1} \Rightarrow \displaystyle\int\left(1-\frac{1}{v}\right)dv = \int\frac{dx}{x}$
3. $v - \log|v| = \log|x| + C_1 \Rightarrow \dfrac{y}{x} = \log|y| + C_1$

✔ $y = Ce^{y/x}$

Solved Example 30

Find the particular solution of $2xy + y^2 - 2x^2\dfrac{dy}{dx} = 0$, given $y = 1$ when $x = 1$.

1. $\dfrac{dy}{dx} = \dfrac{2xy+y^2}{2x^2} = v + \dfrac{v^2}{2}$ with $y = vx$
2. $x\dfrac{dv}{dx} = \dfrac{v^2}{2} \Rightarrow \displaystyle\int\frac{2\,dv}{v^2} = \int\frac{dx}{x} \Rightarrow -\frac{2}{v} = \log|x| + C$
3. $-\dfrac{2x}{y} = \log|x| + C$; at $(1,1)$: $C = -2$

✔ $y = \dfrac{2x}{2-\log|x|}$

Solved Example 31

Solve $\left(1+e^{x/y}\right)dx + e^{x/y}\left(1-\dfrac{x}{y}\right)dy = 0$, given $x = 0$ when $y = 2$.

1. $\dfrac{dx}{dy} = -\dfrac{e^{x/y}(1-x/y)}{1+e^{x/y}}$, a function of $\dfrac{x}{y}$; put $x = vy$
2. $v + y\dfrac{dv}{dy} = -\dfrac{e^v(1-v)}{1+e^v} \Rightarrow y\dfrac{dv}{dy} = -\dfrac{v+e^v}{1+e^v}$
3. $\displaystyle\int\frac{(1+e^v)\,dv}{v+e^v} = -\int\frac{dy}{y} \Rightarrow \log|v+e^v| = -\log|y| + \log|C|$
4. $y\left(\dfrac{x}{y} + e^{x/y}\right) = C \Rightarrow x + ye^{x/y} = C$; at $(0,2)$: $C = 2$

✔ $x + ye^{x/y} = 2$

Yaad karo: "VX daalo, V kaato, separate karo"

Homogeneous ke 3 step: (1) $y = vx$ **daalo**, (2) dono side ka v **kaato** ($v + xv' = g(v)$ se $xv' = g(v) - v$), (3) **separate** karke integrate. Last mein $v = \dfrac{y}{x}$ wapas daalna mat bhoolna.



Meme samjho

$y = vx$ daala, phir bhi variables separate nahi ho rahe. "Kyu nahi ho rahi padhai?"

Asli baat: Sahi homogeneous DE mein v wala term hamesha cancel hota hai: $x\dfrac{dv}{dx} = g(v) - v$, jo separable hai. Agar nahi ho raha, to DE homogeneous tha hi nahi (degree check karo) ya $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$ galat likha.

Solved Example 32

Solve $\dfrac{dy}{dx} = \dfrac{y}{x} + \dfrac{y^2}{x^2}$.

1. $y = vx$: $v + x\dfrac{dv}{dx} = v + v^2 \Rightarrow x\dfrac{dv}{dx} = v^2$
2. $\displaystyle\int\frac{dv}{v^2} = \int\frac{dx}{x} \Rightarrow -\frac{1}{v} = \log|x| + C$
3. $-\dfrac{x}{y} = \log|x| + C$

✔ $y = -\dfrac{x}{\log|x| + C}$

Board tip Homogeneous answer mein v chhoda to answer incomplete; $v = \frac{y}{x}$ wapas daalo.

5. Linear Differential Equations

Definition: First order linear DE

$\frac{dy}{dx} + Py = Q$, where P, Q are **constants or functions of x only**. (y and $\frac{dy}{dx}$ appear only in first power, no product $y\frac{dy}{dx}$.)

Why does I.F. work?

With $\text{I.F.} = e^{\int P\,dx}$, we have $\frac{d}{dx}\left(e^{\int P\,dx}\right) = Pe^{\int P\,dx}$.

So

$$\frac{d}{dx}(y \times \text{I.F.}) = \text{I.F.}\left(\frac{dy}{dx} + Py\right) = Q \times \text{I.F.}$$

Integrating both sides gives the solution formula. The LHS becomes one perfect derivative.

Formula Box: Linear DE

- Integrating Factor: $\text{I.F.} = e^{\int P\,dx}$
- Solution: $y \times \text{I.F.} = \int (Q \times \text{I.F.})\,dx + C$
- Type 2 (P_1, Q_1 in y only): $\frac{dx}{dy} + P_1 x = Q_1$
- Then $\text{I.F.} = e^{\int P_1 dy}$ and
$$x \times \text{I.F.} = \int (Q_1 \times \text{I.F.})\,dy + C$$
- Useful: $e^{\log f(x)} = f(x), e^{-\log x} = \frac{1}{x}$, $e^{k\log x} = x^k$

P	$\int P\,dx$	I.F.
$\frac{1}{x}$	$\log x$	x
$-\frac{1}{x}$	$-\log x$	$\frac{1}{x}$
$\frac{2}{x}$	$2\log x$	x^2
$\tan x$	$\log \sec x$	$\sec x$
$\cot x$	$\log \sin x$	$\sin x$
$\frac{2x}{1+x^2}$	$\log(1+x^2)$	$1+x^2$
$\frac{1}{x\log x}$	$\log(\log x)$	$\log x$

DE	Linear?
$\frac{dy}{dx} + x^2 y = \sin x$	Yes, in y
$\frac{dy}{dx} + y^2 = x$	No (y^2)
$y\frac{dy}{dx} + x = 0$	No (product $y\,y'$); separable though
$\frac{dy}{dx} = \frac{1}{x+y^2}$	Yes in x: $\frac{dx}{dy} - x = y^2$
$\frac{dy}{dx} + y\sin y = x$	No ($\sin y$)

I.F. drill (1 mark each)

DE	I.F.
$\frac{dy}{dx} + \frac{3}{x}y = x$	x^3
$\frac{dy}{dx} - \frac{y}{x} = x^2$	$\frac{1}{x}$
$\frac{dy}{dx} + y\sec x = \tan x$	$\sec x + \tan x$
$\frac{dy}{dx} + 2xy = x$	e^{x^2}
$\frac{dx}{dy} + \frac{2}{y}x = y$	y^2
$\frac{dy}{dx} - y\cot x = \sin x$	$\text{cosec}\,x$

Yaad karo: "PQ ka IF, y-IF ka integral"

Pehle DE ko **standard form** $y' + Py = Q$ mein lao (y' ka coefficient 1 karo). Phir **IF** $= e^{\int P}$. Answer: y **into IF = integral of** (Q **into IF**) $+C$. y ke saath IF, Q ke saath bhi IF: dono jagah!

Solved Example 33

Solve $\frac{dy}{dx} + 2y = e^{-x}$.

1. $P = 2,\ Q = e^{-x}$; $\text{I.F.} = e^{\int 2\,dx} = e^{2x}$
2. $ye^{2x} = \int e^{-x}e^{2x}dx = \int e^x dx = e^x + C$

✔ $y = e^{-x} + Ce^{-2x}$

Solved Example 34

Solve $x\frac{dy}{dx} + 2y = x^2\ (x > 0)$.

1. Standard form (divide by x): $\frac{dy}{dx} + \frac{2}{x}y = x$
2. $\text{I.F.} = e^{\int \frac{2}{x}dx} = e^{2\log x} = x^2$
3. $yx^2 = \int x \times x^2\,dx = \frac{x^4}{4} + C$

✔ $y = \frac{x^2}{4} + \frac{C}{x^2}$

Do you know? Order 1 DE: general solution is a one-parameter family of curves.

Solved Example 35

Solve $\frac{dy}{dx} + 2y = 4x$.

1. I.F. $= e^{2x}$; $ye^{2x} = \int 4xe^{2x}dx$
2. By parts: $\int 4xe^{2x}dx = 2xe^{2x} - \int 2e^{2x}dx = (2x-1)e^{2x} + C$

✔ $y = 2x - 1 + Ce^{-2x}$

Solved Example 36

Solve $\frac{dy}{dx} - y = (1+x)e^x$.

1. I.F. $= e^{-x}$; $ye^{-x} = \int (1+x)\,dx = x + \frac{x^2}{2} + C$

✔ $y = e^x\left(x + \frac{x^2}{2} + C\right)$

Solved Example 37

Find the particular solution of $\frac{dy}{dx} + y\cot x = 2\cos x$, given $y = 0$ when $x = \frac{\pi}{2}$.

1. I.F. $= e^{\int \cot x\,dx} = e^{\log \sin x} = \sin x$
2. $y\sin x = \int 2\sin x\cos x\,dx = \sin^2 x + C$
3. $x = \frac{\pi}{2}$, $y = 0$: $0 = 1 + C \Rightarrow C = -1$
4. $y\sin x = \sin^2 x - 1 \Rightarrow y = \sin x - \operatorname{cosec} x$

✔ $y = \sin x - \operatorname{cosec} x$

Solved Example 38

Solve $y\,dx - (x + 3y^2)\,dy = 0$ $(y > 0)$.

1. In y the DE is not linear, so write $\frac{dx}{dy} = \frac{x + 3y^2}{y} \Rightarrow \frac{dx}{dy} - \frac{1}{y}x = 3y$
2. I.F. $= e^{-\int \frac{dy}{y}} = e^{-\log y} = \frac{1}{y}$
3. $x \times \frac{1}{y} = \int 3y \times \frac{1}{y}\,dy = 3y + C$

✔ $x = 3y^2 + Cy$

Check trick I.F. kabhi zero nahi hota; $e^{()}$ hamesha positive.

Quick fact $\frac{dy}{dx} = \frac{y}{x}$ gives $y = Cx$: straight lines through the origin.

Recall $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}$

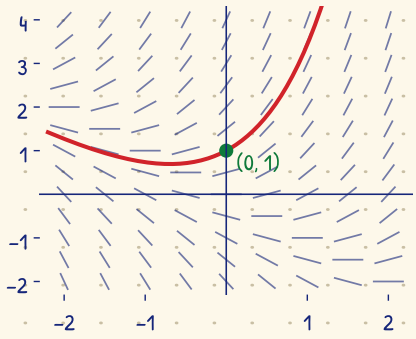


Fig: little dashes show the slope $x + y$ at each point; the red curve $y = 2e^x - x - 1$ follows them and passes through $(0, 1)$

Solved Example 39

Find the equation of the curve through $(0, 1)$ whose slope at any point (x, y) is $x + y$.

1. $\frac{dy}{dx} = x + y \Rightarrow \frac{dy}{dx} - y = x$; I.F. $= e^{-x}$
2. $ye^{-x} = \int xe^{-x}dx = -xe^{-x} - e^{-x} + C$ (by parts)
3. $y = -x - 1 + Ce^x$; at $(0, 1)$: $1 = -1 + C \Rightarrow C = 2$

✔ $y = 2e^x - x - 1$

Solved Example 40

Solve $(1 + x^2)\frac{dy}{dx} + 2xy = 3x^2$, given $y = 2$ when $x = 0$.

1. Standard form: $\frac{dy}{dx} + \frac{2x}{1+x^2}y = \frac{3x^2}{1+x^2}$
2. I.F. $= e^{\log(1+x^2)} = 1 + x^2$
3. $y(1 + x^2) = \int 3x^2dx = x^3 + C$; $x = 0$, $y = 2 \Rightarrow C = 2$

✔ $y = \frac{x^3 + 2}{1 + x^2}$

Solved Example 41

Solve $\cos^2 x\,\frac{dy}{dx} + y = 2\tan x\left(0 \le x < \frac{\pi}{2}\right)$.

1. Standard form: $\frac{dy}{dx} + \sec^2 x\,y = 2\tan x\sec^2 x$; I.F. $= e^{\tan x}$
2. $ye^{\tan x} = \int 2\tan x\sec^2 x\,e^{\tan x}dx$; put $\tan x = t$: $\int 2te^t dt = 2(t-1)e^t$
3. $ye^{\tan x} = 2(\tan x - 1)e^{\tan x} + C$

✔ $y = 2(\tan x - 1) + Ce^{-\tan x}$

Solved Example 42

Solve $(x+y)\dfrac{dy}{dx}=1$.

1. Flip: $\dfrac{dx}{dy}=x+y \Rightarrow \dfrac{dx}{dy}-x=y$ (linear in x); I.F. $=e^{-y}$
2. $xe^{-y}=\displaystyle\int ye^{-y}dy=-(y+1)e^{-y}+C$

✔ $x=Ce^{y}-y-1$

Solved Example 43

Find the particular solution of $\dfrac{dy}{dx}+2y\tan x=2\sin x$, given $y=0$ when $x=\dfrac{\pi}{3}$.

1. I.F. $=e^{2\log\sec x}=\sec^2 x$
2. $y\sec^2 x=\displaystyle\int 2\sin x\sec^2 x\,dx=$ $\displaystyle\int 2\sec x\tan x\,dx=2\sec x+C$
3. $x=\dfrac{\pi}{3}$: $0=2(2)+C\Rightarrow C=-4$
4. $y=2\cos x-4\cos^2 x$

✔ $y=2\cos x-4\cos^2 x$

Solved Example 44

Solve $y\,dx+(x-y^3)\,dy=0\ (y>0)$.

1. $\dfrac{dx}{dy}+\dfrac{1}{y}x=y^2$; I.F. $=e^{\log y}=y$
2. $xy=\displaystyle\int y^3dy=\frac{y^4}{4}+C$

✔ $xy=\dfrac{y^4}{4}+C$

Solved Example 45

Solve $\dfrac{dy}{dx}+y=\cos x$.

1. I.F. $=e^{x}$; $ye^{x}=\displaystyle\int e^{x}\cos x\,dx$
2. Standard result: $\displaystyle\int e^{x}\cos x\,dx=\frac{e^{x}(\sin x+\cos x)}{2}$

✔ $y=\dfrac{\sin x+\cos x}{2}+Ce^{-x}$

Yaad rakho Growth: $N=N_0e^{kt}$; doubling time $=\dfrac{\log 2}{k}$.

Board tip Particular solution = no C in the final answer.

Quick fact $\dfrac{dy}{dx}+Py=0$ is both linear and separable: $y=Ce^{-\int P\,dx}$.

Solved Example 46

Solve $\dfrac{dy}{dx}-y\tan x=-2\sin x$.

1. I.F. $=e^{-\int\tan x\,dx}=e^{-\log\sec x}=\cos x$
2. $y\cos x=\displaystyle\int -2\sin x\cos x\,dx=\cos^2 x+C$

✔ $y=\cos x+C\sec x$

Solved Example 47

Solve $(1-x^2)\dfrac{dy}{dx}-xy=1\ (-1<x<1)$. (HOTS)

1. $\dfrac{dy}{dx}-\dfrac{x}{1-x^2}y=\dfrac{1}{1-x^2}$; $\displaystyle\int -\frac{x}{1-x^2}dx=\frac{1}{2}\log(1-x^2)$
2. I.F. $=\sqrt{1-x^2}$
3. $y\sqrt{1-x^2}=\displaystyle\int\frac{dx}{\sqrt{1-x^2}}=\sin^{-1}x+C$

✔ $y=\dfrac{\sin^{-1}x+C}{\sqrt{1-x^2}}$

Solved Example 48

Solve $(1+y^2)\,dx=(2\tan^{-1}y-x)\,dy$.

1. $\dfrac{dx}{dy}+\dfrac{1}{1+y^2}x=\dfrac{2\tan^{-1}y}{1+y^2}$ (linear in x); I.F. $=e^{\tan^{-1}y}$
2. $xe^{\tan^{-1}y}=\displaystyle\int\frac{2\tan^{-1}y\,e^{\tan^{-1}y}}{1+y^2}dy$; put $\tan^{-1}y=t$: $\displaystyle\int 2te^{t}dt=2(t-1)e^{t}$

✔ $x=2(\tan^{-1}y-1)+Ce^{-\tan^{-1}y}$

Solved Example 49

Solve $e^{-y}\sec^2 y\,dy=dx+x\,dy$.

1. $\dfrac{dx}{dy}+x=e^{-y}\sec^2 y$ (linear in x); I.F. $=e^{\int 1\,dy}=e^{y}$
2. $xe^{y}=\displaystyle\int e^{-y}\sec^2 y\times e^{y}\,dy=\int\sec^2 y\,dy$

✔ $xe^{y}=\tan y+C$



Shivam Bhaiya says

Linear DE board mein **pakka 3 ya 5 marks** ka aata hai. IF nikalte hi ek line mein likho "I.F. $=e^{\int P\,dx}=\ldots$", aur answer ke saath $+C$. Sirf IF sahi likhne pe bhi step marks milte hain.

Recall $\displaystyle\int\operatorname{cosec}x\,dx=\log\left|\tan\frac{x}{2}\right|$

Recall $\displaystyle\int\frac{dx}{\sqrt{1+x^2}}=\log\left|x+\sqrt{1+x^2}\right|$

Solved Example 50

Find the particular solution of $x\dfrac{dy}{dx} + y = 2x\cos x + 2\sin x$, given $y = 1$ when $x = \dfrac{\pi}{2}$.

1. LHS $= \dfrac{d}{dx}(xy)$, so integrate directly (I.F. $= x$ gives the same)
2. $xy = \displaystyle\int (2x\cos x + 2\sin x)\,dx = 2x\sin x + C$ (since $\frac{d}{dx}(2x\sin x) = 2\sin x + 2x\cos x$)
3. At $x = \dfrac{\pi}{2}$: $\dfrac{\pi}{2} = \pi + C \Rightarrow C = -\dfrac{\pi}{2}$

✔ $y = 2\sin x - \dfrac{\pi}{2x}$

One DE, two methods

Some DEs fit two types. $\dfrac{dy}{dx} = \dfrac{x+y}{x}$ is homogeneous **and** linear; both give the same answer.

→ **Homogeneous:** $y = vx$ gives $x\dfrac{dv}{dx} = 1$, so $v = \log|x| + C$, i.e. $y = x\log|x| + Cx$.

→ **Linear:** $\dfrac{dy}{dx} - \dfrac{y}{x} = 1$, I.F. $= \dfrac{1}{x}$, so $\dfrac{y}{x} = \log|x| + C$. Same answer!

↳ Jo method jaldi lage, wahi lagao. Linear wala usually chhota hota hai.

6. Which Method? Identify First

Form of the DE	Method
$\dfrac{dy}{dx} = f(x)\,g(y)$	Variables separable
$\dfrac{dy}{dx} = g\left(\dfrac{y}{x}\right)$, same degree terms	Homogeneous: $y = vx$
$\dfrac{dy}{dx} + Py = Q$ (P, Q in x)	Linear: I.F. $= e^{\int P\,dx}$
$\dfrac{dx}{dy} + P_1x = Q_1$ (P_1, Q_1 in y)	Linear in x: I.F. $= e^{\int P_1 dy}$

Yaad karo: "S-H-L order mein try karo"

Pehle dekho Separate hota hai? Nahi to Homogeneous (sab terms ki same degree)? Nahi to Linear (y ki power 1)? Agar y^2 ya e^y hai to $\dfrac{dx}{dy}$ wala linear try karo.

Check trick Homogeneous test: put $x \to \lambda x, y \to \lambda y$; λ should cancel fully.

Yaad rakho $e^{\tan^{-1} y}$ type I.F. aaye to $t = \tan^{-1} y$ substitution lagao.

Quick fact Order of a DE = number of constants in its general solution.

Most Asked in Boards

→ Find order and degree (including "degree not defined") (1M)
→ Integrating factor of a given linear DE (1M)
→ Solve by variables separable, often with $\tan^{-1}$, e^x, $\log$ (2-3M)
→ Homogeneous DE: show it is homogeneous and solve (3-5M)
→ Linear DE $\dfrac{dy}{dx} + Py = Q$ with a particular condition (3-5M)
→ Linear in x: $\dfrac{dx}{dy} + P_1x = Q_1$ (3M)
→ Curve through a point with given slope (3M)
→ Case study on growth/decay or a moving body (4M)

Common doubts

→ **Can degree be 0?** No. A DE must contain a derivative, so degree ≥ 1 whenever defined.
→ **Why $+C$ only once?** C_1 on one side and C_2 on the other combine into one constant $C = C_2 - C_1$.
→ **Why $\log|C|$?** Any constant can be written as $\log|C|$; it lets us combine logs neatly.
→ **$\log|x|$ or $\log x$?** Use modulus in general; drop it when the question says $x > 0$.

Common mistakes

→ Taking the power of a lower derivative as the degree; or giving a degree when a derivative is inside $\sin$, e, $\log$.
→ Finding I.F. before making the coefficient of $\dfrac{dy}{dx}$ equal to 1.
→ Writing $y \times \text{I.F.} = \int Q\,dx$ (forgetting I.F. inside the integral) or forgetting $+C$.
→ Applying the condition before integrating, or not substituting $v = \dfrac{y}{x}$ back at the end.
→ $e^{-\log x} = -x$ (wrong!); correct: $e^{-\log x} = \dfrac{1}{x}$.



Shivam Bhaiya says

Particular solution wale sawal mein pehle **general solution** poora karo, phir condition daal ke C nikalo. Final answer ek box mein, bina C ke. Examiner sabse pehle final line hi dekhta hai.

Yaad karo: "e ki power mein log, dono gayab"

I.F. mein $e^{\log f} = f$: $e^{\log \sin x} = \sin x$, $e^{2\log x} = x^2$.

Minus aaye to ulta: $e^{-\log x} = \dfrac{1}{x}$, $e^{-\log \cos x} = \sec x$.

Power aage ho to andar le jao, phir e aur $\log$ cancel.

Ek nazar mein: Differential Equations

Type of sawal	Method	Marks
Order, degree	highest derivative, its power	1
Verify solution	differentiate, substitute	2
Separable	separate, integrate, $+C$	2-3
Homogeneous	$y = vx$, separate	3-5
Linear	I.F. $= e^{\int P\,dx}$	3-5

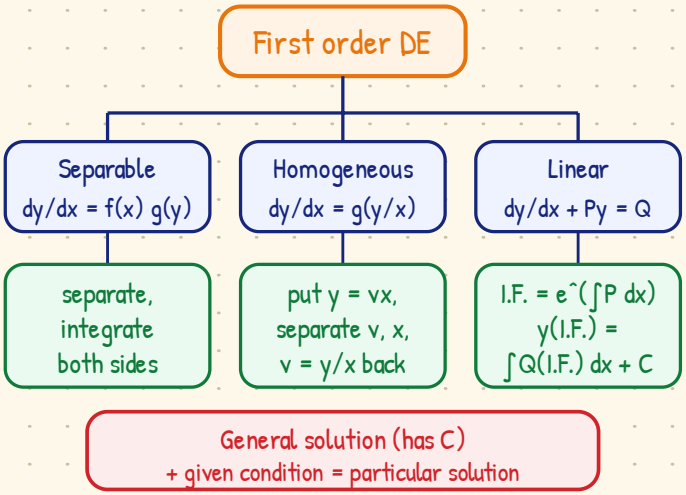


Fig: mind map of Chapter 9

5-mark answer kaise likhein

1. Write the DE in standard form and name its type ("homogeneous" / "linear").
2. Show the substitution or I.F. on its own line.
3. Integrate step by step; write $+C$ once.
4. Put the given condition, find C, and box the final particular solution.

Practice Set A: khud karo (answers green mein)

1. Order and degree of $y'' + (y')^2 + 2y = 0$
   Ans: 2, 1
2. Order and degree of $x(y'')^3 + (y')^4 = 0$
   Ans: 2, 3
3. $\dfrac{dy}{dx} = x^3 e^{-2y}$
   Ans: $2e^{2y} = x^4 + C$
4. $\dfrac{dy}{dx} = \dfrac{1-\cos x}{1+\cos x}$
   Ans: $y = 2\tan\dfrac{x}{2} - x + C$
5. $\dfrac{dy}{dx} = \sqrt{4-y^2}\ (-2 < y < 2)$
   Ans: $\sin^{-1}\dfrac{y}{2} = x + C$
6. $y\log y\,dx - x\,dy = 0$
   Ans: $\log y = Cx$

Practice Set B: homogeneous + linear

7. $(x+y)\,dy + (x-y)\,dx = 0$
   Ans: $\tan^{-1}\dfrac{y}{x} + \dfrac{1}{2}\log(x^2+y^2) = C$
8. $\dfrac{dy}{dx} - y = \cos x$
   Ans: $y = \dfrac{\sin x - \cos x}{2} + Ce^x$
9. $x\dfrac{dy}{dx} + y = x^3$
   Ans: $y = \dfrac{x^3}{4} + \dfrac{C}{x}$
10. $\dfrac{dy}{dx} + \dfrac{y}{x} = e^x$
    Ans: $xy = (x-1)e^x + C$
11. $\dfrac{dy}{dx} + 2y = \sin x$
    Ans: $y = \dfrac{2\sin x - \cos x}{5} + Ce^{-2x}$
12. $y^2\,dx + \left(x - \dfrac{1}{y}\right)dy = 0$
    Ans: $x = 1 + \dfrac{1}{y} + Ce^{1/y}$

Practice Set C: mixed revision

13. $\dfrac{dy}{dx} = \dfrac{2x+1}{2y}$
    Ans: $y^2 = x^2 + x + C$
14. $\dfrac{dy}{dx} + 4y = 8$
    Ans: $y = 2 + Ce^{-4x}$
15. $\dfrac{dy}{dx} = \dfrac{e^x}{1+e^x}$
    Ans: $y = \log(1+e^x) + C$
16. $x\dfrac{dy}{dx} = y + 3x$
    Ans: $y = 3x\log|x| + Cx$
17. $\dfrac{dy}{dx} - \dfrac{2y}{x} = x^2$
    Ans: $y = x^3 + Cx^2$
18. $\dfrac{dy}{dx} = \dfrac{y+1}{x+1}$
    Ans: $y + 1 = C(x+1)$

Quick fact $y = e^{mx}$ satisfies $y'' = m^2 y$.

Yaad rakho $y = A\cos x + B\sin x$ solves $y'' + y = 0$ (2 constants, order 2).

Board tip Write the type of DE in words before solving; it earns a step mark.

Last minute: 8 one-liners

1. Order = highest derivative; degree = its power (only if polynomial in derivatives).
2. General solution has as many constants as the order; particular has none.
3. Separable: $\int \frac{dy}{g(y)} = \int f(x)\,dx + C$.
4. Homogeneous: all terms same degree, put $y = vx$.
5. Linear: standard form first, I.F. $= e^{\int P\,dx}$.
6. $y \times$ I.F. $= \int Q \times$ I.F. $dx + C$.
7. y^2, e^y terms present? Try $\frac{dx}{dy} + P_1 x = Q_1$.
8. Put the condition only after integrating, to find C.

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. The order and degree of $\left(\frac{d^3y}{dx^3}\right)^2 + \left(\frac{d^2y}{dx^2}\right)^3 + y = 0$ are 1M Most asked

(a) 2, 3 (b) 3, 3 (c) 3, 2 (d) 2, 2

Ans. (c) Order 3 (highest derivative y'''), degree 2 (its power).

Q2. The degree of the DE $\frac{dy}{dx} + e^{dy/dx} = x$ is 1M Board fav

(a) 1 (b) 2 (c) 0 (d) not defined

Ans. (d) not defined, since $\frac{dy}{dx}$ appears in an exponent.

Q3. The number of arbitrary constants in the particular solution of a DE of order 3 is 1M

(a) 3 (b) 2 (c) 1 (d) 0

Ans. (d) 0. A particular solution has no arbitrary constant.

Q4. The integrating factor of $x\frac{dy}{dx} - y = x^3\ (x > 0)$ is 1M Board fav

(a) x (b) $\frac{1}{x}$ (c) $-x$ (d) $\log x$

Ans. (b) Standard form $\frac{dy}{dx} - \frac{1}{x}y = x^2$; I.F. $= e^{-\log x} = \frac{1}{x}$.

Q5. The integrating factor of $\frac{dy}{dx} + y\tan x = \sec x$ is 1M

(a) $\sec x$ (b) $\cos x$ (c) $\tan x$ (d) $e^{\tan x}$

Ans. (a) $e^{\int \tan x\,dx} = e^{\log \sec x} = \sec x$.

Q6. The general solution of $\frac{dy}{dx} = e^{x-y}$ is 1M

(a) $e^x + e^y = C$ (b) $e^y = e^x + C$ (c) $e^{-y} = e^x + C$ (d) $e^{x-y} = C$

Ans. (b) $e^y dy = e^x dx \Rightarrow e^y = e^x + C$.

Q7. Which of the following is a homogeneous DE? 1M

(a) $\frac{dy}{dx} = x + y^2$ (b) $\frac{dy}{dx} = \frac{x+1}{y}$ (c) $\frac{dy}{dx} = \frac{x^2+xy}{y^2}$ (d) $\frac{dy}{dx} = \frac{y}{x} + x$

Ans. (c) Every term (x^2, xy, y^2) has degree 2.

Q8. The general solution of $y\,dx - x\,dy = 0$ is 1M

(a) $y = Cx$ (b) $xy = C$ (c) $x + y = C$ (d) $y = Cx^2$

(contd.) (a) $\frac{dy}{y} = \frac{dx}{x} \Rightarrow \log|y| = \log|x| + \log|C| \Rightarrow y = Cx$.

Q9. A: The DE $\left(\frac{d^2y}{dx^2}\right)^2 + \sin\left(\frac{dy}{dx}\right) = 0$ has no defined degree. R: Degree is defined only when the DE is a polynomial in its derivatives. 1M

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. (a) Both true, and R explains A.

Q10. A: The integrating factor of $\frac{dy}{dx} + 3y = x$ is e^{3x}. R: For $\frac{dy}{dx} + Py = Q$, I.F. $= e^{\int P\,dx}$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. (a) Both true, and R explains A ($e^{\int 3\,dx} = e^{3x}$).

Q11. A: $y = Ce^{-x}$ is a solution of $\frac{dy}{dx} + y = 0$. R: The general solution of a DE of order n contains n arbitrary constants. 1M

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. (b) A true ($-Ce^{-x} + Ce^{-x} = 0$), R true, but R does not explain A.

Q12. The solution of $\frac{dy}{dx} + y = e^{-x}$ with $y(0) = 0$ is 1M

(a) $y = e^{-x}$ (b) $y = xe^{-x}$ (c) $y = xe^x$ (d) $y = (x+1)e^{-x}$

Ans. (b) I.F. $= e^x$: $ye^x = x + C$, $C = 0$, so $y = xe^{-x}$.

Q13. The integrating factor of $\frac{dy}{dx} - 3y = \sin 2x$ is 1M

(a) e^{3x} (b) $3x$ (c) e^{-3x} (d) e^{-3}

Ans. (c) $e^{\int -3\,dx} = e^{-3x}$.

Q14. The degree of $\left(\frac{d^2y}{dx^2}\right)^2 = \left(1 + \frac{dy}{dx}\right)^{1/2}$ is 1M HOTS

(a) 1 (b) 2 (c) 1/2 (d) 4

Ans. (d) Squaring: $(y'')^4 = 1 + y'$, so degree 4.

Q15. A: $\frac{dy}{dx} = \frac{x^2+y^2}{xy}$ is a homogeneous DE. R: $F(x,y) = \frac{x^2+y^2}{xy}$ satisfies $F(\lambda x, \lambda y) = \lambda^0 F(x,y)$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. (a) Both true, and R explains A.

Q16. A: The general solution of $\frac{dy}{dx} = e^{x+y}$ is $e^x + e^{-y} = C$. R: $e^{x+y} = e^x e^y$, so the variables can be separated. 1M

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. (a) Both true; $e^{-y}dy = e^x dx \Rightarrow -e^{-y} = e^x + C_1$, and R explains A.

Q17. $y = e^{2x}$ is a solution of 1M
(a) $y'' - 4y = 0$ (b) $y'' + 4y = 0$ (c) $y' + 2y = 0$ (d) $y'' - 2y = 0$

> **Ans.** (a) $y'' = 4e^{2x} = 4y$, so $y'' - 4y = 0$.

Q18. The general solution of $\frac{dy}{dx} + \frac{y}{x} = 0$ is 1M
(a) $y = Cx$ (b) $y = x + C$ (c) $xy = C$ (d) $y = Ce^x$

> **Ans.** (c) $\frac{dy}{y} = -\frac{dx}{x} \Rightarrow \log|y| = -\log|x| + \log|C| \Rightarrow xy = C$.

Very short (2 marks)

Q19. Solve $\frac{dy}{dx} = \frac{\cos^2 y}{\sin^2 x}$. 2M Board fav

> **Ans.** $\sec^2 y\,dy = \operatorname{cosec}^2 x\,dx \Rightarrow \tan y = -\cot x + C$, i.e. $\tan y + \cot x = C$.

Q20. Find the general solution of $\frac{dy}{dx} + y = 3$. 2M

> **Ans.** I.F. $= e^x$; $ye^x = \int 3e^x dx = 3e^x + C \Rightarrow y = 3 + Ce^{-x}$. (Also separable.)

Q21. Find the integrating factor of $(x \log x)\frac{dy}{dx} + y = 4\log x\ (x > 1)$ and solve. 2M Most asked

> **Ans.** Standard form: $\frac{dy}{dx} + \frac{1}{x\log x}y = \frac{4}{x}$. I.F. $= e^{\log(\log x)} = \log x$. $y\log x = \int \frac{4\log x}{x}dx = 2(\log x)^2 + C$.

Q22. Solve $\frac{dy}{dx} = 2^{x+y}$. 2M

> **Ans.** $2^{-y}dy = 2^x dx \Rightarrow \frac{-2^{-y}}{\log 2} = \frac{2^x}{\log 2} + C_1 \Rightarrow 2^x + 2^{-y} = C$.

Q23. Find the particular solution of $\frac{dy}{dx} = -4xy^2$, given $y = 1$ when $x = 0$. 2M

> **Ans.** $\int \frac{dy}{y^2} = -\int 4x\,dx \Rightarrow -\frac{1}{y} = -2x^2 + C; C = -1$. So $y = \frac{1}{1+2x^2}$.

Q24. Write the integrating factor of $(1 + x^2)\frac{dy}{dx} + y = \tan^{-1} x$. 2M Board fav

> **Ans.** $P = \frac{1}{1+x^2}$, I.F. $= e^{\tan^{-1} x}$.

Q25. Find the general solution of $\frac{dy}{dx} = e^{3x-2y} + x^2 e^{-2y}$. 2M Most asked

> **Ans.** $e^{2y}dy = (e^{3x} + x^2)\,dx \Rightarrow \frac{e^{2y}}{2} = \frac{e^{3x}}{3} + \frac{x^3}{3} + C$.

Q26. Solve $\log\left(\frac{dy}{dx}\right) = 2x + 3y$. 2M Board fav

> **Ans.** $\frac{dy}{dx} = e^{2x}e^{3y} \Rightarrow e^{-3y}dy = e^{2x}dx \Rightarrow -\frac{e^{-3y}}{3} = \frac{e^{2x}}{2} + C_1$, i.e. $3e^{2x} + 2e^{-3y} = C$.

Short answer (3 marks)

Q27. Solve $x\frac{dy}{dx} - y = x^2\cos x\ (x > 0)$. 3M Board fav

> **Ans.** $\frac{dy}{dx} - \frac{1}{x}y = x\cos x$; I.F. $= \frac{1}{x}$. $\frac{y}{x} = \int \cos x\,dx = \sin x + C \Rightarrow y = x\sin x + Cx$.

Q28. Show that $(x^2 - y^2)\,dx + 2xy\,dy = 0$ is homogeneous and solve it. 3M Most asked

> **Ans.** $\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$, degree 0, so homogeneous. $y = vx$: $x\frac{dv}{dx} = \frac{v^2-1}{2v} - v = -\frac{1+v^2}{2v}$. $\int \frac{2v\,dv}{1+v^2} = -\int \frac{dx}{x} \Rightarrow \log(1 + v^2) = -\log|x| + \log C \Rightarrow x(1 + v^2) = C$. So $x^2 + y^2 = Cx$.

Q29. Find the particular solution of $(2 - y)\frac{dy}{dx} = x + 1$, given $y = 1$ when $x = 0$. 3M

> **Ans.** $\int (2 - y)\,dy = \int (x + 1)\,dx \Rightarrow 2y - \frac{y^2}{2} = \frac{x^2}{2} + x + C$. At $(0, 1)$: $C = \frac{3}{2}$. So $4y - y^2 = x^2 + 2x + 3$, i.e. $(x + 1)^2 + (y - 2)^2 = 2$.

Q30. Solve $\frac{dy}{dx} + y\tan x = \cos^2 x$, given $y = 1$ when $x = 0$. 3M Must do

> **Ans.** I.F. $= \sec x$. $y\sec x = \int \cos^2 x \sec x\,dx = \int \cos x\,dx = \sin x + C$. $x = 0$: $1 = 0 + C$, $C = 1$. $y = \sin x\cos x + \cos x$.

Q31. Find the equation of the curve through the origin whose slope at any point (x, y) is $e^x - y$. 3M

> **Ans.** $\frac{dy}{dx} + y = e^x$; I.F. $= e^x$. $ye^x = \int e^{2x}dx = \frac{e^{2x}}{2} + C$. At $(0, 0)$: $C = -\frac{1}{2}$. $y = \frac{e^x - e^{-x}}{2}$.

Q32. Solve $x\frac{dy}{dx} + 2y = x^2\log x\ (x > 0)$. 3M

> **Ans.** I.F. $= x^2$. $yx^2 = \int x^3\log x\,dx = \frac{x^4}{4}\log x - \frac{x^4}{16} + C$. So $y = \frac{x^2}{16}(4\log x - 1) + \frac{C}{x^2}$.

Q33. Solve $x\frac{dy}{dx} = y(\log y - \log x + 2)\ (x, y > 0)$. 3M HOTS

> **Ans.** $y = vx$: $v + x\frac{dv}{dx} = v(\log v + 2) \Rightarrow x\frac{dv}{dx} = v(\log v + 1)$. $\int \frac{dv}{v(\log v+1)} = \int \frac{dx}{x} \Rightarrow \log|\log v + 1| = \log x + \log C$. So $\log\frac{y}{x} + 1 = Cx$.

Q34. Find the particular solution of $\frac{dy}{dx} + y\cot x = 6x\operatorname{cosec} x$, given $y = 0$ when $x = \frac{\pi}{2}$. 3M Board fav

> **Ans.** I.F. $= \sin x$. $y\sin x = \int 6x\,dx = 3x^2 + C$. At $x = \frac{\pi}{2}$: $0 = \frac{3\pi^2}{4} + C$. So $y\sin x = 3x^2 - \frac{3\pi^2}{4}$.

Q35. Solve $\frac{dy}{dx} = \frac{x(2\log x + 1)}{\sin y + y\cos y}\ (x > 0)$. 3M HOTS

> **Ans.** $\int (\sin y + y\cos y)\,dy = \int x(2\log x + 1)\,dx$. LHS $= y\sin y$ (since $\frac{d}{dy}(y\sin y) = \sin y + y\cos y$); RHS $= x^2\log x$ (since $\frac{d}{dx}(x^2\log x) = 2x\log x + x$). So $y\sin y = x^2\log x + C$.

Q36. Find the particular solution of $\frac{dy}{dx} = 1 + x + y + xy$, given $y = 0$ when $x = 0$. 3M Most asked

> **Ans.** $1 + x + y + xy = (1 + x)(1 + y)$. $\int \frac{dy}{1+y} = \int (1 + x)\,dx \Rightarrow \log|1 + y| = x + \frac{x^2}{2} + C; C = 0$. So $y = e^{x+\frac{x^2}{2}} - 1$.

Long answer (5 marks)

Q37. Solve $x\dfrac{dy}{dx} = y - x\tan\dfrac{y}{x}$, given $y = \dfrac{\pi}{2}$ when $x = 1$. 5M Board fav

Ans. Homogeneous: $\frac{dy}{dx} = \frac{y}{x} - \tan\frac{y}{x}$. $y = vx$: $x\frac{dv}{dx} = -\tan v$. $\int \cot v\,dv = -\int \frac{dx}{x} \Rightarrow \log|\sin v| = -\log|x| + \log|C| \Rightarrow x\sin\frac{y}{x} = C$. At $(1, \frac{\pi}{2})$: $C = 1$. Answer: $x\sin\frac{y}{x} = 1$.

Q38. Find the particular solution of $(1+x^2)\dfrac{dy}{dx} + 2xy = 6x^2$, given $y = 0$ when $x = 0$. 5M Most asked

Ans. Standard form: $\frac{dy}{dx} + \frac{2x}{1+x^2}y = \frac{6x^2}{1+x^2}$. I.F. $= 1 + x^2$. $y(1+x^2) = \int 6x^2dx = 2x^3 + C$. At $(0,0)$: $C = 0$. So $y = \frac{2x^3}{1+x^2}$.

Q39. Solve $x^2\,dy + (xy + y^2)\,dx = 0$, given $y = 1$ when $x = 1$. 5M HOTS

Ans. $\frac{dy}{dx} = -\frac{xy+y^2}{x^2}$ (homogeneous). $y = vx$: $x\frac{dv}{dx} = -(2v + v^2)$. $\int \frac{dv}{v(v+2)} = -\int \frac{dx}{x} \Rightarrow \frac{1}{2}\log\left|\frac{v}{v+2}\right| = -\log|x| + C_1 \Rightarrow \frac{y}{y+2x} = \frac{C}{x^2}$. At $(1,1)$: $C = \frac{1}{3}$. So $3x^2y = y + 2x$, i.e. $y = \frac{2x}{3x^2-1}$.

Q40. Solve $(x\,dy - y\,dx)\,y\sin\dfrac{y}{x} = (y\,dx + x\,dy)\,x\cos\dfrac{y}{x}$, given $y = \pi$ when $x = 1$. 5M Most asked

Ans. Divide by x^2 and put $y = vx$ ($dy = v\,dx + x\,dv$): it reduces to $\frac{v\sin v - \cos v}{v\cos v}\,dv = \frac{2\,dx}{x}$, i.e. $\left(\tan v - \frac{1}{v}\right)dv = \frac{2\,dx}{x}$. Integrate: $\log|\sec v| - \log|v| = 2\log|x| + \log|C_1| \Rightarrow \frac{\sec v}{v} = C_1x^2$, i.e. $xy\cos\frac{y}{x} = C$. At $(1,\pi)$: $C = \pi\cos\pi = -\pi$. Answer: $xy\cos\frac{y}{x} = -\pi$.

Q41. Find the particular solution of $\dfrac{dy}{dx} - 2y\cot x = \sin^2 x\cos x$, given $y = 2$ when $x = \dfrac{\pi}{2}$. 5M

Ans. I.F. $= e^{-2\log\sin x} = \frac{1}{\sin^2 x}$. $\frac{y}{\sin^2 x} = \int \cos x\,dx = \sin x + C$. At $x = \frac{\pi}{2}$: $2 = 1 + C$, $C = 1$. So $y = \sin^3 x + \sin^2 x$.

Q42. Solve $x\dfrac{dy}{dx} - y + x\sin\dfrac{y}{x} = 0$, given $y = \dfrac{\pi}{2}$ when $x = 1$. 5M

Ans. $y = vx$: $x\frac{dv}{dx} = -\sin v \Rightarrow \int \operatorname{cosec} v\,dv = -\int \frac{dx}{x}$. $\log\left|\tan\frac{v}{2}\right| = -\log|x| + \log|C| \Rightarrow x\tan\frac{y}{2x} = C$. At $(1, \frac{\pi}{2})$: $C = \tan\frac{\pi}{4} = 1$. Answer: $x\tan\frac{y}{2x} = 1$.

Q43. Find the particular solution of $\dfrac{dy}{dx} + y\cot x = 3x^2 + x^3\cot x$, given $y = 0$ when $x = \dfrac{\pi}{2}$. 5M Board fav

Ans. I.F. $= \sin x$. $y\sin x = \int (3x^2\sin x + x^3\cos x)\,dx = x^3\sin x + C$ (since $\frac{d}{dx}(x^3\sin x) = 3x^2\sin x + x^3\cos x$). At $x = \frac{\pi}{2}$: $0 = \frac{\pi^3}{8} + C$. So $y = x^3 - \frac{\pi^3}{8}\operatorname{cosec} x$.

Recall $\int \log x\,dx = x\log x - x + C$

Case based (4 marks)

Q44. Bank deposit. In a savings scheme, money grows continuously so that $\dfrac{dA}{dt} = \dfrac{5}{100}A$ (t in years). Rs 1000 is deposited at $t = 0$. (i) Write the order and degree of this DE. (1) (ii) Which method solves it? (1) (iii) Find $A(t)$ and the time in which the money doubles. Use $\log_e 2 = 0.6931$. (2) 4M

Ans. (i) Order 1, degree 1. (ii) Variables separable: $\frac{dA}{A} = \frac{dt}{20}$. (iii) $A = 1000e^{t/20}$. Doubling: $e^{t/20} = 2 \Rightarrow t = 20\log_e 2 = 13.86$ years (about 13.9 years).

Q45. Falling ball. The speed v (m/s) of a small ball falling through oil satisfies $\dfrac{dv}{dt} + v = 10$, with $v = 0$ at $t = 0$. (i) Name the type of this DE. (1) (ii) Find its integrating factor. (1) (iii) Find $v(t)$ and the speed at $t = \log_e 2$ s. (2) 4M

Ans. (i) Linear (also separable). (ii) I.F. $= e^t$. (iii) $ve^t = \int 10e^t dt = 10e^t + C$; $v(0) = 0 \Rightarrow C = -10$. $v = 10(1 - e^{-t})$; at $t = \log 2$: $v = 10\left(1 - \frac{1}{2}\right) = 5$ m/s. As t grows, $v \to 10$ m/s.

Q46. Radioactive decay. A radioactive substance decays at a rate proportional to the amount present: $\dfrac{dN}{dt} = -kN$. Its half-life is 5 years. (i) Write N in terms of N_0, k, t. (1) (ii) Find k. (1) (iii) What fraction of the substance is left after 15 years? (2) 4M Must do

Ans. (i) $N = N_0e^{-kt}$. (ii) $\frac{1}{2} = e^{-5k} \Rightarrow k = \frac{\log_e 2}{5}$. (iii) $\frac{N}{N_0} = e^{-15k} = \left(e^{-5k}\right)^3 = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$.

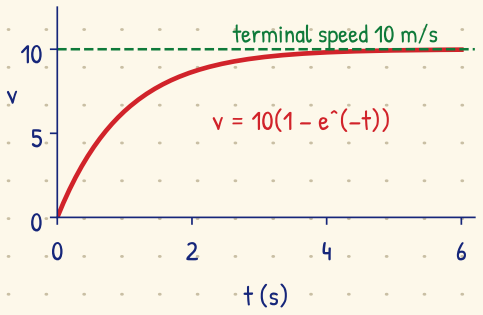


Fig (falling ball): speed rises fast and then levels off at 10 m/s

Check trick Separable answer check: differentiate implicitly, divide, compare.

Quick fact $\frac{d}{dx}(xy) = x\frac{dy}{dx} + y$: spot it, integrate directly.

Recall $\int e^{ax}\cos bx\,dx = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2}$

Yaad rakho $\frac{x\,dy - y\,dx}{x^2} = d\left(\frac{y}{x}\right)$

Most asked: projection, unit vector perpendicular to both, area of triangle by cross product

VECTOR ALGEBRA

1. Scalars, Vectors and Notation

A **scalar** has only magnitude (mass, time, speed, work). A **vector** has magnitude AND direction (displacement, velocity, force). It is drawn as a directed line segment $\overrightarrow{AB}$: A = initial point, B = terminal point.

→ Magnitude (length) of $\overrightarrow{AB}$ is written $|\overrightarrow{AB}|$ or $|\vec{a}|$; it is never negative.

→ **Position vector** of $P(x, y, z)$ w.r.t. origin O: $\overrightarrow{OP} = \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, and $|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$.

→ $\hat{i}, \hat{j}, \hat{k}$ are unit vectors along the X, Y, Z axes.

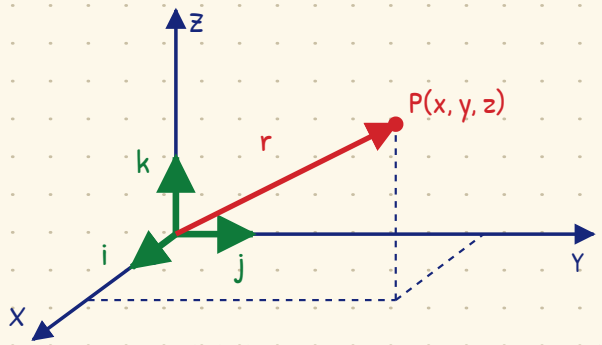


Fig: unit vectors $\hat{i}, \hat{j}, \hat{k}$ (green) along the axes; $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is the position vector of P

Types of vectors

Type	Meaning
Zero (null) $\vec{0}$	magnitude 0, direction not defined; initial = terminal point
Unit $\hat{a}$	magnitude 1; $\hat{a} = \dfrac{\vec{a}}{\lvert\vec{a}\rvert}$
Coinitial	same initial point
Collinear (parallel)	lie on parallel lines (or the same line), any magnitudes, same or opposite direction
Equal	same magnitude AND same direction (initial point may differ)
Negative of $\vec{a}$	same magnitude, opposite direction: $\overrightarrow{BA} = -\overrightarrow{AB}$

↳ Vectors in this chapter are **free vectors**: we can shift them parallel to themselves without changing them.

Quick fact $\hat{i} \cdot \hat{i} = 1$ but $\hat{i} \times \hat{i} = \vec{0}$.

Quick check Is $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{2}\hat{j} + \frac{1}{2}\hat{k}$ a unit vector? $\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1$, so yes.

Quick fact $|\hat{i} + \hat{j} + \hat{k}| = \sqrt{3}$, not 3.

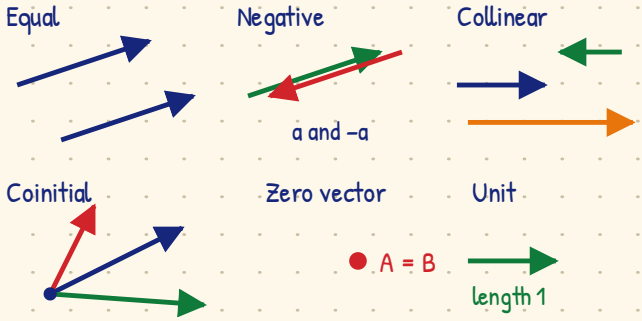


Fig: types of vectors at a glance

2. Addition and Scalar Multiple

Triangle law: place the tail of $\vec{b}$ at the head of $\vec{a}$; the side from the tail of $\vec{a}$ to the head of $\vec{b}$ is $\vec{a} + \vec{b}$. So $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$.

Parallelogram law: if $\vec{a}, \vec{b}$ are adjacent sides from one point, the diagonal from that point is $\vec{a} + \vec{b}$ (the other diagonal is $\vec{a} - \vec{b}$ or $\vec{b} - \vec{a}$).

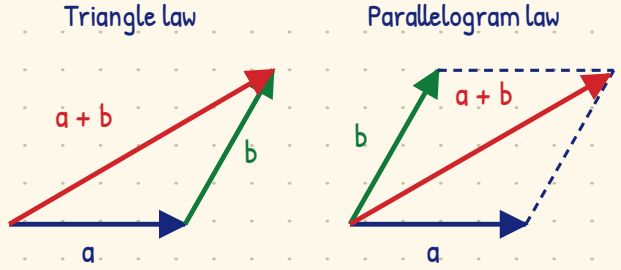


Fig: triangle law and parallelogram law give the same a + b

→ $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ (commutative); $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$ (associative)

→ $\vec{a} + \vec{0} = \vec{a}$; $\vec{a} + (-\vec{a}) = \vec{0}$

→ In a triangle ABC: $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \vec{0}$

→ $\lambda\vec{a}$ has magnitude $|\lambda||\vec{a}|$; same direction if $\lambda > 0$, opposite if $\lambda < 0$.

→ $\vec{a}$ and $\vec{b}$ ($\vec{b}$ non-zero) are collinear $\iff \vec{a} = \lambda\vec{b}$ for some scalar λ.

Scalar multiple: properties

→ $k(\vec{a} + \vec{b}) = k\vec{a} + k\vec{b}$; $(k + m)\vec{a} = k\vec{a} + m\vec{a}$; $k(m\vec{a}) = (km)\vec{a}$

→ $\vec{a} - \vec{b} = \vec{a} + (-\vec{b})$; $1\,\vec{a} = \vec{a}$, $0\,\vec{a} = \vec{0}$

3. Components, Direction Cosines and Ratios

Any vector in space: $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$. Here x, y, z are the **scalar components** and $x\hat{i}, y\hat{j}, z\hat{k}$ the **vector components**.

If $\vec{r}$ makes angles α, β, γ with the X, Y, Z axes, then $l = \cos\alpha, m = \cos\beta, n = \cos\gamma$ are its **direction cosines** (DCs) and

$$l = \frac{x}{|\vec{r}|},\ m = \frac{y}{|\vec{r}|},\ n = \frac{z}{|\vec{r}|},\quad l^2 + m^2 + n^2 = 1$$

Any numbers proportional to l, m, n (e.g. x, y, z themselves) are **direction ratios** (DRs).

Direction cosines l, m, n	Direction ratios a, b, c
unique (up to sign) for a line	infinitely many (any multiple)
$l^2 + m^2 + n^2 = 1$	$a^2 + b^2 + c^2$ need not be 1
$l = \pm\frac{a}{\sqrt{a^2+b^2+c^2}}$ etc.	$a = kl,\ b = km,\ c = kn$

Every vector can be written as $\vec{r} = |\vec{r}|\,(l\hat{i} + m\hat{j} + n\hat{k})$: magnitude × unit vector.

For $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$:

- $\vec{a} = \vec{b} \iff a_1 = b_1,\ a_2 = b_2,\ a_3 = b_3$
- $\vec{a} \pm \vec{b} = (a_1 \pm b_1)\hat{i} + (a_2 \pm b_2)\hat{j} + (a_3 \pm b_3)\hat{k}$
- Collinear $\iff \frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$
- Vector joining $P(x_1, y_1, z_1)$ to $Q(x_2, y_2, z_2)$:

$$\overrightarrow{PQ} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$$

("head minus tail")

Vectors in a plane (2D)

- $\vec{r} = x\hat{i} + y\hat{j}$, $|\vec{r}| = \sqrt{x^2 + y^2}$
- Angle with the +X axis: $\tan\theta = \frac{y}{x}$
- Unit vector making angle θ with +X axis: $\cos\theta\,\hat{i} + \sin\theta\,\hat{j}$

Solved Example 1

Find the unit vector making an angle of $30°$ with the positive X-axis, and the scalar components of the vector from $P(2, 1)$ to $Q(-5, 7)$.

1. Unit vector $= \cos 30°\hat{i} + \sin 30°\hat{j} = \frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}$
2. $\overrightarrow{PQ} = (-5 - 2)\hat{i} + (7 - 1)\hat{j} = -7\hat{i} + 6\hat{j}$

✔ $\frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}$; scalar components of $\overrightarrow{PQ}$: -7 and 6

Yaad karo: "Head minus Tail"

$\overrightarrow{AB}$ = (position vector of **B**) − (position vector of **A**).

Jahan arrow **pahunchta** hai, wahi pehle likho. $\overrightarrow{AB} = \vec{b} - \vec{a}$, kabhi $\vec{a} - \vec{b}$ nahi.

Solved Example 2

Find the magnitude of $\vec{a} = 2\hat{i} - 3\hat{j} + 6\hat{k}$ and a unit vector in its direction. Also write its direction cosines.

(contd.)

1. $|\vec{a}| = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$
2. $\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{2}{7}\hat{i} - \frac{3}{7}\hat{j} + \frac{6}{7}\hat{k}$
3. DCs are the components of $\hat{a}$. Check: $\frac{4+9+36}{49} = 1$ ✓

✔ $|\vec{a}| = 7$, $\hat{a} = \frac{1}{7}(2\hat{i} - 3\hat{j} + 6\hat{k})$, DCs $\frac{2}{7}, -\frac{3}{7}, \frac{6}{7}$

Solved Example 3

Find the direction cosines of the vector joining $A(1, 2, -3)$ to $B(-1, -2, 1)$, directed from A to B.

1. $\overrightarrow{AB} = (-1 - 1)\hat{i} + (-2 - 2)\hat{j} + (1 + 3)\hat{k} = -2\hat{i} - 4\hat{j} + 4\hat{k}$
2. $|\overrightarrow{AB}| = \sqrt{4 + 16 + 16} = 6$
3. DCs $= \left(\frac{-2}{6}, \frac{-4}{6}, \frac{4}{6}\right)$

✔ DCs: $-\frac{1}{3}, -\frac{2}{3}, \frac{2}{3}$

Solved Example 4

Find a vector of magnitude 10 units in the direction of $\vec{a} = 3\hat{i} - 4\hat{k}$.

1. $|\vec{a}| = \sqrt{9 + 0 + 16} = 5$, so $\hat{a} = \frac{1}{5}(3\hat{i} - 4\hat{k})$
2. Required vector $= 10\,\hat{a} = 2(3\hat{i} - 4\hat{k})$

✔ $6\hat{i} - 8\hat{k}$

Solved Example 5

Show that the points $A(2, 1, -1)$, $B(4, 4, 3)$ and $C(8, 10, 11)$ are collinear.

1. $\overrightarrow{AB} = 2\hat{i} + 3\hat{j} + 4\hat{k}$
2. $\overrightarrow{BC} = 4\hat{i} + 6\hat{j} + 8\hat{k} = 2\,\overrightarrow{AB}$
3. So $\overrightarrow{AB} \parallel \overrightarrow{BC}$ and B is common, hence A, B, C lie on one line.

✔ A, B, C are collinear

Solved Example 6

Find x, y, z if $(x + y)\hat{i} + (2x - z)\hat{j} + (y + z)\hat{k} = 5\hat{i} + 3\hat{j} + 4\hat{k}$. Also express $\vec{c} = 4\hat{i} + \hat{j}$ as a combination of $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = \hat{i} - \hat{j}$.

1. Equal vectors: $x + y = 5$, $2x - z = 3$, $y + z = 4$
2. Add 1st and 3rd: $x + 2y + z = 9$; with $z = 2x - 3$: $3x + 2y = 12$ and $x + y = 5 \Rightarrow x = 2, y = 3, z = 1$
3. $\vec{c} = p\vec{a} + q\vec{b}$: $p + q = 4$, $p - q = 1 \Rightarrow p = \frac{5}{2}$, $q = \frac{3}{2}$

✔ $x = 2, y = 3, z = 1$; $\vec{c} = \frac{5}{2}\vec{a} + \frac{3}{2}\vec{b}$

Quick fact $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \vec{0}$ in every triangle.

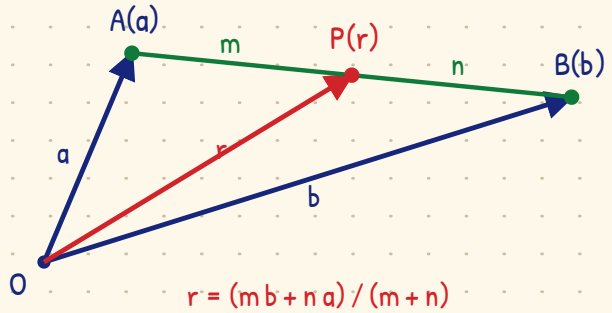


Meme samjho

Teacher: "$\overrightarrow{AB} \parallel \overrightarrow{BC}$, isliye A, B, C collinear." Student: **Guarantee deta hai??**
Asli baat: Yes, because the two parallel vectors share the point B, so they lie on ONE line. Parallel alone is not enough: always mention the common point.

4. Section Formula

If A, B have position vectors $\vec{a}, \vec{b}$ and P divides AB in the ratio $m : n$:



r = (m b + n a) / (m + n)

Fig: P divides AB internally in the ratio m : n

Formula Box: Section formula

- Internally: $\vec{r} = \dfrac{m\vec{b} + n\vec{a}}{m+n}$
- Externally: $\vec{r} = \dfrac{m\vec{b} - n\vec{a}}{m-n}$
- Mid-point: $\vec{r} = \dfrac{\vec{a}+\vec{b}}{2}$
- Centroid of triangle: $\dfrac{\vec{a}+\vec{b}+\vec{c}}{3}$

Yaad karo: "m goes with b" (cross multiply)

Ratio $m : n$ mein m ko B ke saath, n ko A ke saath. External mein sirf beech ka + sign $-$ ban jaata hai (numerator aur denominator dono mein).

Solved Example 7

Find the position vectors of the points which divide the join of $\vec{a} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{b} = -\hat{i} + \hat{j} + 3\hat{k}$ in the ratio $2 : 1$ (i) internally (ii) externally.

1. (i) $\vec{r} = \dfrac{2\vec{b} + \vec{a}}{3} = \dfrac{(-2\hat{i} + 2\hat{j} + 6\hat{k}) + (\hat{i} + 2\hat{j} - \hat{k})}{3}$
2. $= \dfrac{-\hat{i} + 4\hat{j} + 5\hat{k}}{3}$
3. (ii) $\vec{r} = \dfrac{2\vec{b} - \vec{a}}{2-1} = (-2\hat{i} + 2\hat{j} + 6\hat{k}) - (\hat{i} + 2\hat{j} - \hat{k})$

(contd.)
✔ (i) $-\frac{1}{3}\hat{i} + \frac{4}{3}\hat{j} + \frac{5}{3}\hat{k}$ (ii) $-3\hat{i} + 7\hat{k}$

Solved Example 8

Find the position vector of the centroid of the triangle with vertices $A(1,2,3)$, $B(-1,0,4)$, $C(3,1,-1)$. Also find the mid-point of BC.

1. Centroid $= \dfrac{\vec{a}+\vec{b}+\vec{c}}{3} = \dfrac{(1-1+3)\hat{i} + (2+0+1)\hat{j} + (3+4-1)\hat{k}}{3}$
2. $= \dfrac{3\hat{i} + 3\hat{j} + 6\hat{k}}{3} = \hat{i} + \hat{j} + 2\hat{k}$
3. Mid-point of BC $= \dfrac{\vec{b}+\vec{c}}{2} = \dfrac{2\hat{i} + \hat{j} + 3\hat{k}}{2}$

✔ Centroid $\hat{i} + \hat{j} + 2\hat{k}$, i.e. $G(1,1,2)$; mid-point of BC $(1, \frac{1}{2}, \frac{3}{2})$

Solved Example 9

Find the ratio in which $P(3,0,-1)$ divides the join of $A(1,-2,3)$ and $B(4,1,-3)$. Also find the fourth vertex D of the parallelogram ABCD if $C(2,5,1)$.

1. $\overrightarrow{AP} = 2\hat{i} + 2\hat{j} - 4\hat{k}$, $\overrightarrow{PB} = \hat{i} + \hat{j} - 2\hat{k}$
2. $\overrightarrow{AP} = 2\,\overrightarrow{PB}$ (same direction), so $AP : PB = 2 : 1$ internally
3. In a parallelogram the diagonals bisect each other: $\vec{a} + \vec{c} = \vec{b} + \vec{d}$
4. $\vec{d} = \vec{a} + \vec{c} - \vec{b} = (1+2-4)\hat{i} + (-2+5-1)\hat{j} + (3+1+3)\hat{k}$

✔ $2 : 1$ internally; $D(-1,2,7)$

5. Dot (Scalar) Product

Definition

$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$, where $0 \le \theta \le \pi$ is the angle between them. The answer is a scalar (number).

- $\cos\theta = \dfrac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$
- $\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$
- $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$, $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$
- $\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$ (for non-zero vectors)
- $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$; $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$; $\vec{a} \cdot \vec{a} = |\vec{a}|^2$
- $\theta = 0 \Rightarrow \vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|$; $\theta = \pi \Rightarrow \vec{a} \cdot \vec{b} = -|\vec{a}||\vec{b}|$
- θ acute $\iff \vec{a} \cdot \vec{b} > 0$; obtuse $\iff \vec{a} \cdot \vec{b} < 0$

Projection

Projection (scalar) of $\vec{a}$ on $\vec{b} = |\vec{a}| \cos\theta = \vec{a} \cdot \hat{b} = \dfrac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$

. Projection vector $= \left(\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\right)\vec{b}$.

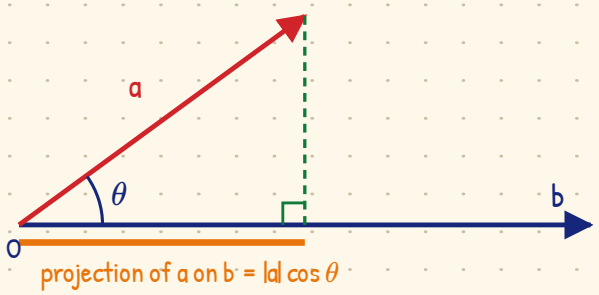


Fig: projection of a on b is the shadow of a on the line of b

Yaad karo: "Jis PAR projection, uska mod NEECHE"

Projection of $\vec{a}$ on $\vec{b} = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$. Line $\vec{b}$ par shadow gir rahi hai, toh $|\vec{b}|$ hi denominator mein. Ulta kiya toh answer galat.

Useful identities

→ $|\vec{a}+\vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2\,\vec{a}\cdot\vec{b}$
→ $|\vec{a}-\vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2\,\vec{a}\cdot\vec{b}$
→ $(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b}) = |\vec{a}|^2 - |\vec{b}|^2$
→ $|\vec{a}+\vec{b}| = |\vec{a}-\vec{b}| \iff \vec{a}\perp\vec{b}$
→ Triangle inequality: $|\vec{a}+\vec{b}| \le |\vec{a}| + |\vec{b}|$; Cauchy-Schwarz: $|\vec{a}\cdot\vec{b}| \le |\vec{a}||\vec{b}|$

Solved Example 10

Find the angle between $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$.

1. $\vec{a}\cdot\vec{b} = 2 - 2 + 4 = 4$
2. $|\vec{a}| = \sqrt{4+1+4} = 3$, $|\vec{b}| = \sqrt{1+4+4} = 3$
3. $\cos\theta = \frac{4}{3\times 3} = \frac{4}{9}$

✔ $\theta = \cos^{-1}\left(\frac{4}{9}\right)$

Solved Example 11

Find the projection of $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$.

1. $\vec{a}\cdot\vec{b} = 2 + 6 + 2 = 10$
2. $|\vec{b}| = \sqrt{1+4+1} = \sqrt{6}$
3. Projection $= \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = \frac{10}{\sqrt{6}}$

✔ $\frac{10}{\sqrt{6}} = \frac{5\sqrt{6}}{3}$

Solved Example 13

If $\vec{a}+\vec{b}+\vec{c} = \vec{0}$ with $|\vec{a}| = 3$, $|\vec{b}| = 5$, $|\vec{c}| = 7$, find the angle between $\vec{a}$ and $\vec{b}$.

(contd.)

1. $\vec{a}+\vec{b} = -\vec{c} \Rightarrow |\vec{a}+\vec{b}|^2 = |\vec{c}|^2$
2. $9 + 25 + 2\,\vec{a}\cdot\vec{b} = 49 \Rightarrow \vec{a}\cdot\vec{b} = \frac{15}{2}$
3. $\cos\theta = \frac{15/2}{3\times 5} = \frac{1}{2}$

✔ $\theta = \frac{\pi}{3}$ (i.e. 60°)

Solved Example 12

Find λ if $2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\hat{i} - 2\hat{j} + 3\hat{k}$ are perpendicular.

1. Perpendicular ⇒ dot product $= 0$
2. $2(1) + \lambda(-2) + 1(3) = 0 \Rightarrow 5 - 2\lambda = 0$

✔ $\lambda = \frac{5}{2}$

Solved Example 14

If $\hat{a}$ is a unit vector and $(\vec{x}-\hat{a})\cdot(\vec{x}+\hat{a}) = 12$, find $|\vec{x}|$.

1. $(\vec{x}-\hat{a})\cdot(\vec{x}+\hat{a}) = |\vec{x}|^2 - |\hat{a}|^2$
2. $|\vec{x}|^2 - 1 = 12 \Rightarrow |\vec{x}|^2 = 13$

✔ $|\vec{x}| = \sqrt{13}$

Solved Example 15

Find λ so that $\vec{a} + \lambda\vec{b}$ is perpendicular to $\vec{c}$, where $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}$, $\vec{b} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{c} = 3\hat{i} + 2\hat{j} - \hat{k}$.

1. $\vec{a} + \lambda\vec{b} = (2+\lambda)\hat{i} + (1-\lambda)\hat{j} + (3+2\lambda)\hat{k}$
2. Dot with $\vec{c}$: $3(2+\lambda) + 2(1-\lambda) - (3+2\lambda) = 0$
3. $6 + 3\lambda + 2 - 2\lambda - 3 - 2\lambda = 0 \Rightarrow 5 - \lambda = 0$

✔ $\lambda = 5$

Solved Example 16

If $\hat{a}$ and $\hat{b}$ are unit vectors inclined at an angle θ, prove that $|\hat{a} - \hat{b}| = 2\sin\frac{\theta}{2}$.

1. $|\hat{a} - \hat{b}|^2 = 1 + 1 - 2\,\hat{a}\cdot\hat{b} = 2 - 2\cos\theta$
2. $= 2(1 - \cos\theta) = 2\times 2\sin^2\frac{\theta}{2} = 4\sin^2\frac{\theta}{2}$
3. Take square root ($0 \le \frac{\theta}{2} \le \frac{\pi}{2}$, so $\sin\frac{\theta}{2} \ge 0$)

✔ $|\hat{a} - \hat{b}| = 2\sin\frac{\theta}{2}$ (similarly $|\hat{a} + \hat{b}| = 2\cos\frac{\theta}{2}$)

Solved Example 17

Find the projection VECTOR of $\vec{a} = 3\hat{i} - \hat{j} + 2\hat{k}$ on $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$.

1. $\vec{a}\cdot\vec{b} = 3 - 2 + 4 = 5$, $|\vec{b}|^2 = 1 + 4 + 4 = 9$
2. Projection vector $= \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\vec{b} = \frac{5}{9}(\hat{i} + 2\hat{j} + 2\hat{k})$
3. (Scalar projection $= \frac{5}{3}$)

✔ $\frac{5}{9}\hat{i} + \frac{10}{9}\hat{j} + \frac{10}{9}\hat{k}$

Application: Work done

Work done by a constant force $\vec{F}$ in a displacement $\vec{d}$: $W = \vec{F} \cdot \vec{d} = |\vec{F}||\vec{d}|\cos\theta$. If several forces act, add them first, then dot with $\vec{d}$.

Solved Example 18

Find the projection of $\overrightarrow{AB}$ on $\overrightarrow{CD}$, where $A(1,0,2)$, $B(3,2,3)$, $C(0,1,-1)$, $D(2,0,1)$.

1. $\overrightarrow{AB} = 2\hat{i} + 2\hat{j} + \hat{k}$, $\overrightarrow{CD} = 2\hat{i} - \hat{j} + 2\hat{k}$
2. $\overrightarrow{AB} \cdot \overrightarrow{CD} = 4 - 2 + 2 = 4$, $|\overrightarrow{CD}| = 3$

✔ Projection $= \frac{4}{3}$



Shivam Bhaiya says

Board mein vectors ka **"angle + projection"** wala 2-3 marks ka sawaal almost har saal kisi na kisi form mein aata hai. Dot product ka formula aur $|\vec{a}|^2 = \vec{a} \cdot \vec{a}$ wali trick ratt lo, aadha chapter wahi hai.

6. Cross (Vector) Product

Definition

$\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\sin\theta\,\hat{n}$, where $\hat{n}$ is the unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ (right-hand rule: curl fingers from $\vec{a}$ to $\vec{b}$, thumb gives $\hat{n}$). The answer is a **vector**.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$
$$= (a_2b_3 - a_3b_2)\hat{i} - (a_1b_3 - a_3b_1)\hat{j} + (a_1b_2 - a_2b_1)\hat{k}$$

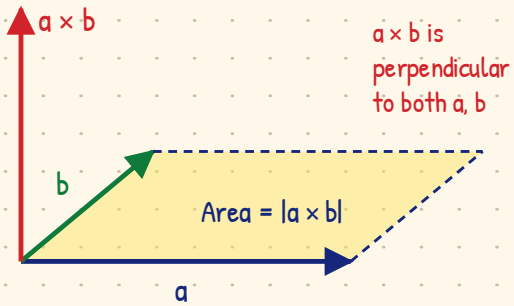


Fig: a x b is perpendicular to the plane of a, b; its length = area of the parallelogram

- $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$; reverse order gives a minus sign
- $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{0}$; $\vec{a} \times \vec{a} = \vec{0}$
- $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$ (NOT commutative)
- $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$
- $\vec{a} \parallel \vec{b} \iff \vec{a} \times \vec{b} = \vec{0}$ (for non-zero vectors)
- $\sin\theta = \dfrac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}$

Yaad karo: i, j, k ki clock (circle)

Circle mein $\hat{i} \to \hat{j} \to \hat{k} \to \hat{i}$ likho. **Clockwise** chalo toh answer + (agla vector): $\hat{j} \times \hat{k} = \hat{i}$. **Ulta** chalo toh −: $\hat{k} \times \hat{j} = -\hat{i}$. Same vector twice $= \vec{0}$.

Formula Box: Cross product uses

- Area of parallelogram with adjacent sides $\vec{a}, \vec{b}$: $|\vec{a} \times \vec{b}|$
- Area of parallelogram with diagonals $\vec{d}_1, \vec{d}_2$: $\frac{1}{2}|\vec{d}_1 \times \vec{d}_2|$
- Area of $\triangle ABC = \frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|$
- Unit vector perpendicular to both $\vec{a}, \vec{b}$: $\pm\dfrac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$
- Lagrange: $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2|\vec{b}|^2$
- A, B, C collinear $\iff \overrightarrow{AB} \times \overrightarrow{AC} = \vec{0}$

Why these formulas work (1-line derivations)

- $\vec{a} \cdot \vec{b} = (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \cdot (b_1\hat{i} + b_2\hat{j} + b_3\hat{k})$: only $\hat{i} \cdot \hat{i}$, $\hat{j} \cdot \hat{j}$, $\hat{k} \cdot \hat{k}$ survive (each $= 1$), so $a_1b_1 + a_2b_2 + a_3b_3$.
- Projection: in the right triangle of the figure, adjacent side $= |\vec{a}|\cos\theta = \dfrac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$.
- Area: parallelogram = base × height = $|\vec{a}|\,(|\vec{b}|\sin\theta) = |\vec{a} \times \vec{b}|$.

Solved Example 19

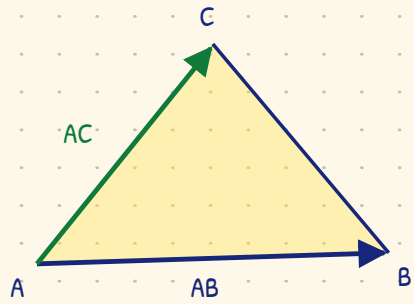
Find a unit vector perpendicular to both $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$.

1. $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & 2 & 3 \end{vmatrix}$
2. $= (3-2)\hat{i} - (3-1)\hat{j} + (2-1)\hat{k} = \hat{i} - 2\hat{j} + \hat{k}$
3. $|\vec{a} \times \vec{b}| = \sqrt{1+4+1} = \sqrt{6}$

✔ $\pm\frac{1}{\sqrt{6}}(\hat{i} - 2\hat{j} + \hat{k})$

Quick check $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{i} - \hat{j}$: $\vec{a} \cdot \vec{b} = 1 - 1 = 0$, so $\vec{a} \perp \vec{b}$; and $\vec{a} \times \vec{b} = -2\hat{k}$.

Quick check Projection of $3\hat{i} + 4\hat{j}$ on $\hat{i}$ is 3; on $\hat{j}$ is 4. Components are projections on the axes.



Area of triangle = half of $|\vec{AB} \times \vec{AC}|$

Fig: triangle ABC is half of the parallelogram on $\vec{AB}, \vec{AC}$

Step marking: "unit vector ⊥ to both" (3M)

1. Write $\vec{a} \times \vec{b}$ as a determinant (1/2 mark)
2. Expand with the $-$ on $\hat{j}$ (1 mark)
3. Find $|\vec{a} \times \vec{b}|$ (1/2 mark)
4. Write $\pm\dfrac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$ with BOTH signs (1 mark)

Solved Example 20

Find the area of the triangle with vertices $A(1,1,2)$, $B(2,3,5)$, $C(1,5,5)$.

1. $\overrightarrow{AB} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\overrightarrow{AC} = 4\hat{j} + 3\hat{k}$
2. $\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 0 & 4 & 3 \end{vmatrix} = -6\hat{i} - 3\hat{j} + 4\hat{k}$
3. $|\overrightarrow{AB} \times \overrightarrow{AC}| = \sqrt{36 + 9 + 16} = \sqrt{61}$

✔ Area $= \frac{\sqrt{61}}{2}$ sq units

Solved Example 21

Find the area of a parallelogram whose diagonals are $\vec{d_1} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{d_2} = \hat{i} - 3\hat{j} + 4\hat{k}$.

1. $\vec{d_1} \times \vec{d_2} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & -2 \\ 1 & -3 & 4 \end{vmatrix}$
2. $= (4-6)\hat{i} - (12+2)\hat{j} + (-9-1)\hat{k} = -2\hat{i} - 14\hat{j} - 10\hat{k}$
3. $|\vec{d_1} \times \vec{d_2}| = \sqrt{4 + 196 + 100} = \sqrt{300} = 10\sqrt{3}$

✔ Area $= \frac{1}{2} \times 10\sqrt{3} = 5\sqrt{3}$ sq units

Solved Example 22

Using cross product, show that $A(2,-1,3)$, $B(3,1,1)$ and $C(5,5,-3)$ are collinear.

1. $\overrightarrow{AB} = \hat{i} + 2\hat{j} - 2\hat{k}$, $\overrightarrow{AC} = 3\hat{i} + 6\hat{j} - 6\hat{k}$
2. $\overrightarrow{AB} \times \overrightarrow{AC} = (-12+12)\hat{i} - (-6+6)\hat{j} + (6-6)\hat{k} = \vec{0}$
3. Area of $\triangle ABC = 0$, so no triangle is formed

✔ A, B, C are collinear

Solved Example 23

If $|\vec{a}| = 2$, $|\vec{b}| = 7$ and $\vec{a} \times \vec{b} = 3\hat{i} + 2\hat{j} + 6\hat{k}$, find the angle between $\vec{a}$ and $\vec{b}$.

1. $|\vec{a} \times \vec{b}| = \sqrt{9 + 4 + 36} = 7$
2. $\sin\theta = \dfrac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|} = \dfrac{7}{2 \times 7} = \dfrac{1}{2}$
3. Cross product gives only $\sin\theta$; since $0 \le \theta \le \pi$, θ can be acute or obtuse

✔ $\theta = \frac{\pi}{6}$ or $\frac{5\pi}{6}$

Solved Example 24

Find λ if the area of the parallelogram with adjacent sides $\vec{a} = \hat{i} + \hat{j} + \lambda\hat{k}$ and $\vec{b} = \hat{i} - \hat{j}$ is $2\sqrt{3}$ sq units.

1. $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & \lambda \\ 1 & -1 & 0 \end{vmatrix} = \lambda\hat{i} + \lambda\hat{j} - 2\hat{k}$
2. $|\vec{a} \times \vec{b}| = \sqrt{2\lambda^2 + 4} = 2\sqrt{3}$
3. $2\lambda^2 + 4 = 12 \Rightarrow \lambda^2 = 4$

✔ $\lambda = \pm 2$

Meme samjho



Exam mein dost ka $\vec{a} \times \vec{b}$ copy kiya, par jaldi mein $\vec{b} \times \vec{a}$ likh diya. Answer ka **sign hi ulta** ho gaya!

Asli baat: $\vec{b} \times \vec{a} = -(\vec{a} \times \vec{b})$. Cross product mein order matters; dot product mein nahi ($\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$).

Solved Example 25

Prove Lagrange identity: $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2|\vec{b}|^2$

.

1. $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2 \sin^2\theta$
2. $(\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2|\vec{b}|^2 \cos^2\theta$
3. Add: $|\vec{a}|^2|\vec{b}|^2(\sin^2\theta + \cos^2\theta)$

✔ $= |\vec{a}|^2|\vec{b}|^2$, proved

Yaad karo: "Dot = Digit, Cross = Cross-wise arrow"

Dot product ka answer ek **digit** (number), **cos** ke saath.
Cross product ka answer ek **arrow** (vector), **sin** ke saath.
$\perp$ check: dot $= 0$; $\parallel$ check: cross $= \vec{0}$.

7. Dot vs Cross and Proofs by Vectors

Point	Dot $\vec{a} \cdot \vec{b}$	Cross $\vec{a} \times \vec{b}$
Result	scalar	vector $\perp$ to both
Formula	$\lvert\vec{a}\rvert\lvert\vec{b}\rvert\cos\theta$	$\lvert\vec{a}\rvert\lvert\vec{b}\rvert\sin\theta\,\hat{n}$
Order	commutative	$\vec{b} \times \vec{a} = -\vec{a} \times \vec{b}$

Point	Dot $\vec{a}\cdot\vec{b}$	Cross $\vec{a}\times\vec{b}$
Zero when	$\vec{a}\perp\vec{b}$	$\vec{a}\parallel\vec{b}$
Same vector	$\vec{a}\cdot\vec{a}=\lvert\vec{a}\rvert^2$	$\vec{a}\times\vec{a}=\vec{0}$
Unit vectors	$\hat{i}\cdot\hat{i}=1,\hat{i}\cdot\hat{j}=0$	$\hat{i}\times\hat{i}=\vec{0},\hat{i}\times\hat{j}=\hat{k}$
Use	angle, projection, work	area, normal direction

Proof questions (3M) use only: $\vec{a}\cdot\vec{a}=\lvert\vec{a}\rvert^2$, dot $=0$ for $\perp$, and head minus tail.

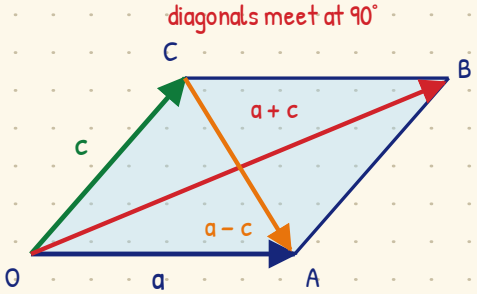


Fig: rhombus OABC; diagonals $\vec{a}+\vec{c}$ and $\vec{a}-\vec{c}$

Solved Example 26

Prove, using vectors, that the diagonals of a rhombus are perpendicular.

1. Let OABC be a rhombus with $\overrightarrow{OA}=\vec{a}$, $\overrightarrow{OC}=\vec{c}$, and $\lvert\vec{a}\rvert=\lvert\vec{c}\rvert$ (all sides equal)
2. Diagonals: $\overrightarrow{OB}=\vec{a}+\vec{c}$, $\overrightarrow{CA}=\vec{a}-\vec{c}$
3. $\overrightarrow{OB}\cdot\overrightarrow{CA}=(\vec{a}+\vec{c})\cdot(\vec{a}-\vec{c})=\lvert\vec{a}\rvert^2-\lvert\vec{c}\rvert^2=0$

✔ Dot product is 0, so the diagonals are perpendicular

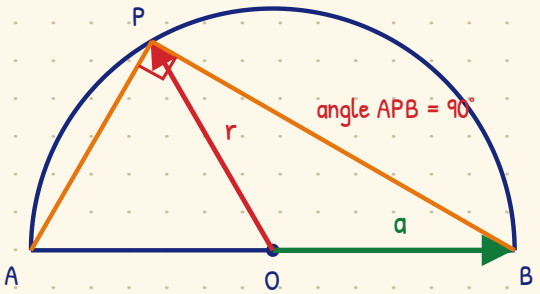


Fig: P on the circle, $\lvert\vec{r}\rvert=\lvert\vec{a}\rvert$ = radius

Solved Example 27

Prove, using vectors, that the angle in a semicircle is a right angle.

1. Centre O, diameter AB, $\overrightarrow{OB}=\vec{a}$, so $\overrightarrow{OA}=-\vec{a}$; P on the circle with $\overrightarrow{OP}=\vec{r}$, $\lvert\vec{r}\rvert=\lvert\vec{a}\rvert$ (radius)
2. $\overrightarrow{AP}=\vec{r}+\vec{a}$, $\overrightarrow{BP}=\vec{r}-\vec{a}$
3. $\overrightarrow{AP}\cdot\overrightarrow{BP}=\lvert\vec{r}\rvert^2-\lvert\vec{a}\rvert^2=0$

✔ $\angle APB=90^\circ$

Quick fact $\hat{k}\times\hat{j}=-\hat{i}$ (anticlockwise gives minus).

Quick fact Projection of $\vec{a}$ on itself $=\lvert\vec{a}\rvert$.

8. Board Special: 3-mark Patterns

Read the data → Need angle / ⊥? DOT → Need area / ⊥ to both? CROSS → Magnitudes only? SQUARE both sides

Solved Example 28

If $\vec{a}=3\hat{i}+\hat{j}-2\hat{k}$ and $\vec{b}=\hat{i}-2\hat{j}+3\hat{k}$, show that $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$ are perpendicular.

1. $\vec{a}+\vec{b}=4\hat{i}-\hat{j}+\hat{k}$, $\vec{a}-\vec{b}=2\hat{i}+3\hat{j}-5\hat{k}$
2. Dot $=8-3-5=0$
3. Shortcut: $(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b})=\lvert\vec{a}\rvert^2-\lvert\vec{b}\rvert^2=14-14=0$

✔ Perpendicular (because $\lvert\vec{a}\rvert=\lvert\vec{b}\rvert$)

Solved Example 29

Find a vector of magnitude $2\sqrt{29}$ perpendicular to both $\vec{a}=\hat{i}+2\hat{j}+\hat{k}$ and $\vec{b}=2\hat{i}-\hat{k}$ which makes an acute angle with the Y-axis.

1. $\vec{a}\times\vec{b}=(-2-0)\hat{i}-(-1-2)\hat{j}+(0-4)\hat{k}=-2\hat{i}+3\hat{j}-4\hat{k}$, magnitude $\sqrt{29}$
2. Required $=\pm2(-2\hat{i}+3\hat{j}-4\hat{k})$
3. Acute with Y-axis $\Rightarrow$ $\hat{j}$ component positive, take $+$

✔ $-4\hat{i}+6\hat{j}-8\hat{k}$

Solved Example 30

Find the angles which $\vec{r}=\hat{i}-\hat{j}+\sqrt{2}\hat{k}$ makes with the coordinate axes.

1. $\lvert\vec{r}\rvert=\sqrt{1+1+2}=2$
2. $\cos\alpha=\frac{1}{2}$, $\cos\beta=-\frac{1}{2}$, $\cos\gamma=\frac{\sqrt{2}}{2}=\frac{1}{\sqrt{2}}$

✔ $\alpha=\frac{\pi}{3},\beta=\frac{2\pi}{3},\gamma=\frac{\pi}{4}$

Solved Example 31

If $\vec{a}+\vec{b}+\vec{c}=\vec{0}$, prove that $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$.

1. Cross with $\vec{a}$: $\vec{a}\times\vec{a}+\vec{a}\times\vec{b}+\vec{a}\times\vec{c}=\vec{0}\Rightarrow\vec{a}\times\vec{b}=-\vec{a}\times\vec{c}=\vec{c}\times\vec{a}$
2. Cross with $\vec{b}$: $\vec{b}\times\vec{a}+\vec{0}+\vec{b}\times\vec{c}=\vec{0}\Rightarrow\vec{b}\times\vec{c}=\vec{a}\times\vec{b}$

✔ $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$

Solved Example 32

Show that $A(0,1,2)$, $B(2,3,2)$, $C(1,1,1)$ form a right-angled triangle.

1. $\overrightarrow{CA}=-\hat{i}+\hat{k}$, $\overrightarrow{CB}=\hat{i}+2\hat{j}+\hat{k}$
2. $\overrightarrow{CA}\cdot\overrightarrow{CB}=-1+0+1=0$
3. Check: $CA^2+CB^2=2+6=8=AB^2$ ($\overrightarrow{AB}=2\hat{i}+2\hat{j}$)

(contd.)

✔ Right angle at C

Solved Example 33

If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and the angle between them is $60°$, find $|2\vec{a} - 3\vec{b}|$.

1. $\vec{a} \cdot \vec{b} = 3 \times 4 \times \frac{1}{2} = 6$
2. $|2\vec{a} - 3\vec{b}|^2 = 4|\vec{a}|^2 + 9|\vec{b}|^2 - 12\,\vec{a} \cdot \vec{b}$
3. $= 36 + 144 - 72 = 108$

✔ $|2\vec{a} - 3\vec{b}| = 6\sqrt{3}$

Solved Example 34

If $|\vec{a} + \vec{b}| = |\vec{a}| = |\vec{b}|$, find the angle between $\vec{a}$ and $\vec{b}$.

1. Let $|\vec{a}| = |\vec{b}| = k$. $|\vec{a} + \vec{b}|^2 = k^2 \Rightarrow k^2 + k^2 + 2\,\vec{a} \cdot \vec{b} = k^2$
2. $\vec{a} \cdot \vec{b} = -\frac{k^2}{2} \Rightarrow \cos\theta = -\frac{1}{2}$

✔ $\theta = \frac{2\pi}{3}$

9. Mixed Board-level Problems

Solved Example 35

$\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$ are adjacent sides of a parallelogram. Find unit vectors along its diagonals and its area.

1. Diagonals: $\vec{a} + \vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$ (magnitude $\sqrt{14}$), $\vec{a} - \vec{b} = \hat{i} + 3\hat{j} - 4\hat{k}$ (magnitude $\sqrt{26}$)
2. Unit vectors: $\frac{1}{\sqrt{14}}(3\hat{i} - \hat{j} + 2\hat{k})$ and $\frac{1}{\sqrt{26}}(\hat{i} + 3\hat{j} - 4\hat{k})$
3. $\vec{a} \times \vec{b} = (3-2)\hat{i} - (6+1)\hat{j} + (-4-1)\hat{k} = \hat{i} - 7\hat{j} - 5\hat{k}$

✔ Area $= \sqrt{1 + 49 + 25} = \sqrt{75} = 5\sqrt{3}$ sq units

Solved Example 36

Write $\vec{b} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ as the sum of a vector parallel to $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and a vector perpendicular to $\vec{a}$.

1. Parallel part $= \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2}\vec{a} = \frac{12}{3}(\hat{i} + \hat{j} + \hat{k}) = 4\hat{i} + 4\hat{j} + 4\hat{k}$
2. Perpendicular part $= \vec{b} - (4\hat{i} + 4\hat{j} + 4\hat{k}) = -\hat{i} + \hat{k}$
3. Check: $(-\hat{i} + \hat{k}) \cdot \vec{a} = -1 + 0 + 1 = 0$ ✓

✔ $\vec{b} = (4\hat{i} + 4\hat{j} + 4\hat{k}) + (-\hat{i} + \hat{k})$

Solved Example 37

Show that $A(1,0,1)$, $B(3,1,3)$, $C(2,3,1)$ form an isosceles triangle and find $\cos B$.

(contd.)

1. $\overrightarrow{AB} = 2\hat{i} + \hat{j} + 2\hat{k}$, $AB = 3$; $\overrightarrow{BC} = -\hat{i} + 2\hat{j} - 2\hat{k}$, $BC = 3$; $\overrightarrow{AC} = \hat{i} + 3\hat{j}$, $AC = \sqrt{10}$
2. $AB = BC$, so isosceles
3. $\overrightarrow{BA} \cdot \overrightarrow{BC} = (-2)(-1) + (-1)(2) + (-2)(-2) = 4$

✔ $\cos B = \frac{4}{3 \times 3} = \frac{4}{9}$

Solved Example 38

Find a vector $\vec{r}$ of magnitude 6 which makes an angle of $\frac{\pi}{3}$ with the X-axis and $\frac{\pi}{4}$ with the Y-axis.

1. $l = \cos\frac{\pi}{3} = \frac{1}{2}$, $m = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$
2. $n^2 = 1 - \frac{1}{4} - \frac{1}{2} = \frac{1}{4} \Rightarrow n = \pm\frac{1}{2}$
3. $\vec{r} = 6(l\hat{i} + m\hat{j} + n\hat{k})$

✔ $\vec{r} = 3\hat{i} + 3\sqrt{2}\hat{j} \pm 3\hat{k}$

Solved Example 39

Prove the Cauchy-Schwarz inequality $|\vec{a} \cdot \vec{b}| \le |\vec{a}||\vec{b}|$ and the triangle inequality $|\vec{a} + \vec{b}| \le |\vec{a}| + |\vec{b}|$.

1. $|\vec{a} \cdot \vec{b}| = |\vec{a}||\vec{b}||\cos\theta| \le |\vec{a}||\vec{b}|$ since $|\cos\theta| \le 1$
2. $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b} \le |\vec{a}|^2 + |\vec{b}|^2 + 2|\vec{a}||\vec{b}|$
3. $= (|\vec{a}| + |\vec{b}|)^2$; take square roots

✔ Both inequalities proved (equality when $\vec{a} \parallel \vec{b}$, same direction)

Solved Example 40

Find the area of the triangle whose two sides are $\vec{a} = 2\hat{i} + 3\hat{j}$ and $\vec{b} = -\hat{i} + 4\hat{j}$.

1. $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 0 \\ -1 & 4 & 0 \end{vmatrix} = (8+3)\hat{k} = 11\hat{k}$
2. For 2D vectors only the $\hat{k}$ part survives

✔ Area $= \frac{11}{2}$ sq units

Solved Example 41

If $|\vec{a}| = |\vec{b}| = |\vec{a} - \vec{b}| = 1$, find $|\vec{a} + \vec{b}|$.

1. $|\vec{a} - \vec{b}|^2 = 1 + 1 - 2\,\vec{a} \cdot \vec{b} = 1 \Rightarrow \vec{a} \cdot \vec{b} = \frac{1}{2}$
2. $|\vec{a} + \vec{b}|^2 = 1 + 1 + 2 \times \frac{1}{2} = 3$

✔ $|\vec{a} + \vec{b}| = \sqrt{3}$

Solved Example 42

Find a unit vector perpendicular to the plane of triangle ABC, where $A(1,0,0)$, $B(0,1,0)$, $C(0,0,1)$, and the area of the triangle.

(contd.)

1. $\overrightarrow{AB} = -\hat{i} + \hat{j}$, $\overrightarrow{AC} = -\hat{i} + \hat{k}$
2. $\overrightarrow{AB} \times \overrightarrow{AC} = (1-0)\hat{i} - (-1-0)\hat{j} + (0+1)\hat{k} = \hat{i} + \hat{j} + \hat{k}$
3. Magnitude $\sqrt{3}$

✔ $\pm\frac{1}{\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})$; area $= \frac{\sqrt{3}}{2}$ sq units

Question says ... so use ...

Words in the question	Use
"unit vector along / in the direction of"	$\vec{a}/\lvert\vec{a}\rvert$
"angle between", "perpendicular", "orthogonal"	dot product
"projection of $\vec{a}$ on $\vec{b}$"	$\vec{a}\cdot\vec{b}/\lvert\vec{b}\rvert$
"parallel", "collinear"	ratios equal or cross $= \vec{0}$
"area", "perpendicular to both", "normal"	cross product
only magnitudes given	square: $\lvert\vec{x}\rvert^2 = \vec{x}\cdot\vec{x}$

Solved Example 43

If $\vec{a} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$, find λ such that $\vec{a} + \lambda\vec{b}$ is parallel to $5\hat{i} - \hat{j} + \hat{k}$.

1. $\vec{a} + \lambda\vec{b} = (1+2\lambda)\hat{i} + (-1+\lambda)\hat{j} + (1-\lambda)\hat{k}$
2. Parallel: $\frac{1+2\lambda}{5} = \frac{-1+\lambda}{-1} = \frac{1-\lambda}{1}$
3. From the 1st and 3rd: $1+2\lambda = 5-5\lambda \Rightarrow \lambda = \frac{4}{7}$; check 2nd: $\frac{-1+4/7}{-1} = \frac{3}{7} = 1 - \frac{4}{7}$ ✓

✔ $\lambda = \frac{4}{7}$

Solved Example 44

Find the angle between the diagonals of the parallelogram with adjacent sides $\vec{a} = 3\hat{i} + \hat{j}$ and $\vec{b} = \hat{i} + 3\hat{j}$.

1. Diagonals: $\vec{a} + \vec{b} = 4\hat{i} + 4\hat{j}$, $\vec{a} - \vec{b} = 2\hat{i} - 2\hat{j}$
2. Dot $= 8 - 8 = 0$ (because $\lvert\vec{a}\rvert = \lvert\vec{b}\rvert = \sqrt{10}$: it is a rhombus)

✔ The diagonals are perpendicular: angle $= \frac{\pi}{2}$

Quick check Area of triangle with sides $2\hat{i}$ and $3\hat{j}$: $\frac{1}{2}\lvert 6\hat{k}\rvert = 3$ sq units.

Quick fact $\vec{a}\cdot\vec{a} = \lvert\vec{a}\rvert^2$: square of a vector means dot with itself.

Formula Box: Full chapter in 7 lines

→ $\lvert\vec{a}\rvert = \sqrt{a_1^2 + a_2^2 + a_3^2}$, $\hat{a} = \frac{\vec{a}}{\lvert\vec{a}\rvert}$

→ DCs: $l = \frac{a_1}{\lvert\vec{a}\rvert}$ etc, $l^2 + m^2 + n^2 = 1$

→ $\overrightarrow{AB} = \vec{b} - \vec{a}$

→ $\vec{a}\cdot\vec{b} = \lvert\vec{a}\rvert\lvert\vec{b}\rvert\cos\theta = a_1b_1 + a_2b_2 + a_3b_3$

→ Projection of $\vec{a}$ on $\vec{b}$: $\frac{\vec{a}\cdot\vec{b}}{\lvert\vec{b}\rvert}$

→ $\lvert\vec{a}\times\vec{b}\rvert = \lvert\vec{a}\rvert\lvert\vec{b}\rvert\sin\theta$

→ $\perp$: $\vec{a}\cdot\vec{b} = 0$; $\parallel$: $\vec{a}\times\vec{b} = \vec{0}$

Common mistakes

→ Writing $\overrightarrow{AB} = \vec{a} - \vec{b}$. It is $\vec{b} - \vec{a}$ (head minus tail).

→ Dividing by $\lvert\vec{a}\rvert$ instead of $\lvert\vec{b}\rvert$ for projection of $\vec{a}$ on $\vec{b}$.

→ Middle term of the cross product: forgetting the minus sign before the $\hat{j}$ cofactor.

→ Taking only $+$ for a unit vector perpendicular to both: there are two, $\pm$.

→ Area of triangle $= \lvert\overrightarrow{AB}\times\overrightarrow{AC}\rvert$: forgot the $\frac{1}{2}$. Diagonals given: also $\frac{1}{2}$.

Most Asked in Boards

1. Unit vector / vector of given magnitude along a vector (1-2M)
2. Angle between two vectors, value of λ for perpendicular / parallel (1-2M)
3. Projection of one vector on another (2M)
4. Unit vector perpendicular to both $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$ (3M)
5. Area of triangle / parallelogram by cross product (2-3M)
6. $\vec{a}+\vec{b}+\vec{c} = \vec{0}$ type: find angle or $\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}$ (3M)
7. Find $\vec{d}$ perpendicular to two vectors with $\vec{c}\cdot\vec{d}$ given (3-5M)
8. Section formula and collinearity (2M)



Shivam Bhaiya says

Cross product ka determinant likhte waqt **beech wale $\hat{j}$ ka minus** sabse zyada log bhoolte hain. Har baar answer ko $\vec{a}$ se dot karke check karo: perpendicular hai toh dot $= 0$ aana chahiye. 10 second ka check, poore marks safe.

Try yourself (10 quick ones, answers below)

1. $|3\hat{i}-4\hat{j}+12\hat{k}|$
2. Unit vector along $\hat{i}+\hat{j}$
3. DCs of $6\hat{i}-2\hat{j}+3\hat{k}$
4. Are $\hat{i}-\hat{j}+2\hat{k}$ and $3\hat{i}+\hat{j}-\hat{k}$ perpendicular?
5. $(2\hat{i}+3\hat{j})\times(\hat{i}-\hat{j})$
6. Projection of $2\hat{i}-\hat{j}+\hat{k}$ on $\hat{i}+\hat{j}+\hat{k}$
7. Mid-point of $A(2,-1,4)$, $B(0,3,2)$
8. λ if $3\hat{i}+\lambda\hat{j}+6\hat{k} \parallel \hat{i}+2\hat{j}+2\hat{k}$
9. Area of triangle with adjacent sides $\hat{i}$ and $\hat{j}$
10. $|\vec{a}|=2$, $|\vec{b}|=3$, $\theta=60°$: find $\vec{a}\cdot\vec{b}$ and $|\vec{a}\times\vec{b}|$

↳ **Answers:** (1) 13 (2) $\frac{1}{\sqrt{2}}(\hat{i}+\hat{j})$ (3) $\frac{6}{7},-\frac{2}{7},\frac{3}{7}$ (4) yes, dot $=0$ (5) $-5\hat{k}$ (6) $\frac{2}{\sqrt{3}}$ (7) $(1,1,3)$ (8) 6 (9) $\frac{1}{2}$ (10) 3 and $3\sqrt{3}$

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. If $|\vec{a}|=4$, $|\vec{b}|=3$ and $\vec{a}\cdot\vec{b}=6\sqrt{3}$, the angle between $\vec{a}$ and $\vec{b}$ is 1M Board fav
(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

Ans. (a) $\cos\theta=\frac{6\sqrt{3}}{12}=\frac{\sqrt{3}}{2}$, so $\theta=\frac{\pi}{6}$.

Q2. $\hat{i}\times(\hat{j}+\hat{k})$ equals 1M
(a) $\hat{j}+\hat{k}$ (b) $\hat{j}-\hat{k}$ (c) $\hat{k}-\hat{j}$ (d) $\vec{0}$

Ans. (c) $\hat{i}\times\hat{j}+\hat{i}\times\hat{k}=\hat{k}-\hat{j}$.

Q3. The direction cosines of $\hat{i}-2\hat{j}+2\hat{k}$ are 1M
(a) $1,-2,2$ (b) $\frac{1}{3},-\frac{2}{3},\frac{2}{3}$ (c) $\frac{1}{3},\frac{2}{3},\frac{2}{3}$ (d) $-\frac{1}{3},\frac{2}{3},-\frac{2}{3}$

Ans. (b) magnitude $=3$, DCs $=\frac{1}{3},-\frac{2}{3},\frac{2}{3}$.

Q4. The projection of $\hat{i}+3\hat{j}+7\hat{k}$ on $2\hat{i}-3\hat{j}+6\hat{k}$ is 1M Most asked
(a) 7 (b) $\frac{35}{\sqrt{59}}$ (c) 35 (d) 5

Ans. (d) $\dfrac{2-9+42}{\sqrt{4+9+36}}=\dfrac{35}{7}=5$.

Q5. The area of the parallelogram with adjacent sides $\hat{i}+\hat{j}$ and $\hat{j}+\hat{k}$ is 1M
(a) $\sqrt{3}$ (b) $\frac{\sqrt{3}}{2}$ (c) 3 (d) $\sqrt{2}$

Ans. (a) $(\hat{i}+\hat{j})\times(\hat{j}+\hat{k})=\hat{i}-\hat{j}+\hat{k}$, magnitude $\sqrt{3}$.

Q6. If $\lambda\vec{a}$ is a unit vector ($\vec{a}$ non-zero), then 1M
(a) $\lambda=1$ (b) $\lambda=|\vec{a}|$ (c) $|\lambda|=\frac{1}{|\vec{a}|}$ (d) $\lambda=-1$

Ans. (c) $|\lambda||\vec{a}|=1\Rightarrow|\lambda|=\frac{1}{|\vec{a}|}$.

Q7. If $\vec{a}\cdot\vec{b}=0$ and $\vec{a}\times\vec{b}=\vec{0}$, then 1M HOTS
(a) $\vec{a}\perp\vec{b}$ (b) $\vec{a}\parallel\vec{b}$ (c) $|\vec{a}|=|\vec{b}|$ (d) $\vec{a}=\vec{0}$ or $\vec{b}=\vec{0}$

Ans. (d) $\perp$ and $\parallel$ together is possible only if one of them is the zero vector.

Q9. The angle between $\hat{i}$ and $\hat{i}+\hat{j}+\hat{k}$ is 1M
(a) $\cos^{-1}\frac{1}{\sqrt{3}}$ (b) $\frac{\pi}{4}$ (c) $\cos^{-1}\frac{1}{3}$ (d) $\frac{\pi}{3}$

(contd.) (a) $\cos\theta=\frac{1}{1\times\sqrt{3}}$, so $\theta=\cos^{-1}\frac{1}{\sqrt{3}}$.

Q8. If $|\vec{a}|=|\vec{b}|=1$ and $|\vec{a}+\vec{b}|=\sqrt{3}$, the angle between $\vec{a}$ and $\vec{b}$ is 1M
(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{2\pi}{3}$

Ans. (b) $1+1+2\cos\theta=3\Rightarrow\cos\theta=\frac{1}{2},\theta=\frac{\pi}{3}$.

Q10. The vector with initial point $(2,-3,5)$ and terminal point $(3,-4,7)$ is 1M
(a) $\hat{i}-\hat{j}+2\hat{k}$ (b) $5\hat{i}-7\hat{j}+12\hat{k}$ (c) $-\hat{i}+\hat{j}-2\hat{k}$ (d) $\hat{i}+\hat{j}+2\hat{k}$

Ans. (a) head minus tail: $(3-2)\hat{i}+(-4+3)\hat{j}+(7-5)\hat{k}$.

Q11. If $|\vec{a}\times\vec{b}|=4$ and $|\vec{a}\cdot\vec{b}|=2$, then $|\vec{a}|^2|\vec{b}|^2$ is 1M
(a) 6 (b) 2 (c) 20 (d) 8

Ans. (c) Lagrange: $16+4=20$.

Q12. The value(s) of p for which $p(\hat{i}+\hat{j}+\hat{k})$ is a unit vector 1M
(a) 0 (b) $\sqrt{3}$ (c) 1 (d) $\pm\frac{1}{\sqrt{3}}$

Ans. (d) $|p|\sqrt{3}=1\Rightarrow p=\pm\frac{1}{\sqrt{3}}$.

Q13. $\hat{a}$, $\hat{b}$ are unit vectors at angle θ. Then $|\hat{a}+\hat{b}|<1$ if 1M HOTS
(a) $\theta=\frac{\pi}{2}$ (b) $\theta<\frac{\pi}{3}$ (c) $\theta=\frac{\pi}{3}$ (d) $\frac{2\pi}{3}<\theta\le\pi$

Ans. (d) $2+2\cos\theta<1\Rightarrow\cos\theta<-\frac{1}{2}\Rightarrow\theta\in\left(\frac{2\pi}{3},\pi\right]$.

Q14. If $(\vec{a}-\vec{b})\cdot(\vec{a}+\vec{b})=0$, then 1M
(a) $\vec{a}=\vec{b}$ (b) $|\vec{a}|=|\vec{b}|$ (c) $\vec{a}\perp\vec{b}$ (d) $\vec{a}\parallel\vec{b}$

Ans. (b) $|\vec{a}|^2-|\vec{b}|^2=0$.

Q15. If $|\vec{a}|=2$, $|\vec{b}|=5$ and $|\vec{a}\times\vec{b}|=8$, then $\vec{a}\cdot\vec{b}$ is 1M
(a) 6 (b) -6 (c) ±6 (d) ±2

Ans. (c) $(\vec{a}\cdot\vec{b})^2=100-64=36$, so ±6.

Q16. A: The projection of $\hat{i}$ on $\hat{j}$ is 0. R: $\hat{i}$ and $\hat{j}$ are perpendicular. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Ans. (a) $\hat{i}\cdot\hat{j}=0$, so projection $=0$. Both true, R explains A.

Q17. The number of unit vectors perpendicular to both $\vec{a}$ and $\vec{b}$ (non-parallel) is 1M
(a) one (b) two (c) three (d) infinitely many

Ans. (b) two: $\pm\dfrac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$.

Q18. Which of these is NOT a unit vector? 1M
(a) $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$ (b) $\frac{\hat{i}-\hat{j}+\hat{k}}{\sqrt{3}}$ (c) $\frac{\hat{i}+\hat{j}+\hat{k}}{3}$ (d) $\frac{2\hat{i}+\hat{j}}{\sqrt{5}}$

Ans. (c) $\left|\frac{\hat{i}+\hat{j}+\hat{k}}{3}\right|=\frac{\sqrt{3}}{3}\neq1$.

Q19. If $|\vec{a}\times\vec{b}|=\vec{a}\cdot\vec{b}$ (both non-zero), the angle between $\vec{a}$ and $\vec{b}$ is 1M Most asked
(a) 0 (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{2}$ (d) π

Ans. (b) $\sin\theta=\cos\theta\Rightarrow\tan\theta=1,\theta=\frac{\pi}{4}$.

Q20. If $\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|$ (non-zero vectors), the angle between them is 1M
(a) 0 (b) $\frac{\pi}{2}$ (c) π (d) $\frac{\pi}{3}$

(contd.) (a) $\cos\theta = 1 \Rightarrow \theta = 0$ (same direction).

Q21. If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $|\vec{a} + \vec{b}| = 5$, then $|\vec{a} - \vec{b}|$ is 1M

(a) 1 (b) 7 (c) 5 (d) $\sqrt{7}$

Ans. (c) $|\vec{a} + \vec{b}|^2 + |\vec{a} - \vec{b}|^2 = 2(9 + 16) = 50 \Rightarrow |\vec{a} - \vec{b}| = 5$.

Q22. **A:** $2\hat{i} - 3\hat{j} + \hat{k}$ and $-4\hat{i} + 6\hat{j} - 2\hat{k}$ are collinear. **R:** Two vectors are collinear if one is a scalar multiple of the other. 1M

(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Ans. (a) $-4\hat{i} + 6\hat{j} - 2\hat{k} = -2(2\hat{i} - 3\hat{j} + \hat{k})$. Both true, R explains A.

Q23. **A:** $\hat{i} \times \hat{j} = \hat{k}$ and $\hat{j} \times \hat{i} = \hat{k}$. **R:** $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Ans. (d) $\hat{j} \times \hat{i} = -\hat{k}$, so A is false; R is true.

Q24. **A:** If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $\vec{a} \cdot \vec{b} = 6$, then $|\vec{a} \times \vec{b}| = 6\sqrt{3}$. **R:** $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2|\vec{b}|^2$. 1M Board fav

(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Ans. (a) $|\vec{a} \times \vec{b}|^2 = 144 - 36 = 108$, so $|\vec{a} \times \vec{b}| = 6\sqrt{3}$. Both true, R explains A.

Very short (2 marks)

Q25. Find a vector of magnitude 6 units in the direction opposite to $2\hat{i} - \hat{j} + 2\hat{k}$. 2M

Ans. $|\vec{a}| = 3$. Required $= -6\,\hat{a} = -2(2\hat{i} - \hat{j} + 2\hat{k}) = -4\hat{i} + 2\hat{j} - 4\hat{k}$.

Q26. Find the angle between $\vec{a}$ and $\vec{b}$ if $|\vec{a}| = \sqrt{3}$, $|\vec{b}| = 2$ and $\vec{a} \cdot \vec{b} = \sqrt{6}$. 2M Most asked

Ans. $\cos\theta = \dfrac{\sqrt{6}}{2\sqrt{3}} = \dfrac{1}{\sqrt{2}}$, so $\theta = \dfrac{\pi}{4}$.

Q27. P and Q have position vectors $\vec{a} + 2\vec{b}$ and $3\vec{a} - \vec{b}$. Find the position vector of R dividing PQ internally in the ratio $2 : 1$. 2M

Ans. $\vec{r} = \dfrac{2(3\vec{a} - \vec{b}) + 1(\vec{a} + 2\vec{b})}{2 + 1} = \dfrac{7\vec{a}}{3}$.

Q28. A unit vector makes angles $\frac{\pi}{4}$ with $\hat{i}$, $\frac{\pi}{3}$ with $\hat{j}$ and an acute angle θ with $\hat{k}$. Find θ and the vector. 2M Board fav

Ans. $\cos^2\frac{\pi}{4} + \cos^2\frac{\pi}{3} + \cos^2\theta = 1 \Rightarrow \frac{1}{2} + \frac{1}{4} + \cos^2\theta = 1 \Rightarrow \cos\theta = \frac{1}{2}, \theta = \frac{\pi}{3}$. Vector $= \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{2}\hat{j} + \frac{1}{2}\hat{k}$.

Q29. Find $|\vec{a}|$ and $|\vec{b}|$ if $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 15$ and $|\vec{a}| = 4|\vec{b}|$. 2M

Ans. $|\vec{a}|^2 - |\vec{b}|^2 = 15 \Rightarrow 16|\vec{b}|^2 - |\vec{b}|^2 = 15 \Rightarrow |\vec{b}| = 1$, $|\vec{a}| = 4$.

Q30. Find the area of the parallelogram whose adjacent sides are $2\hat{i} + \hat{j} - \hat{k}$ and $\hat{i} - \hat{j} + 2\hat{k}$. 2M

(contd.) Cross $= (2 - 1)\hat{i} - (4 + 1)\hat{j} + (-2 - 1)\hat{k} = \hat{i} - 5\hat{j} - 3\hat{k}$. Area $= \sqrt{1 + 25 + 9} = \sqrt{35}$ sq units.

Q31. Find the unit vector in the direction of the sum of $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 2\hat{i} - \hat{j} + 5\hat{k}$. 2M Most asked

Ans. $\vec{a} + \vec{b} = 3\hat{i} + \hat{j} + 2\hat{k}$, magnitude $\sqrt{14}$. Unit vector $= \frac{1}{\sqrt{14}}(3\hat{i} + \hat{j} + 2\hat{k})$.

Q32. Show that $\hat{i} + \hat{j} + \hat{k}$ is equally inclined to the three axes. 2M

Ans. Magnitude $\sqrt{3}$; DCs $= \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$, all equal, so each angle is $\cos^{-1}\frac{1}{\sqrt{3}}$.

Q33. Find the projection of $\vec{b} + \vec{c}$ on $\vec{a}$, where $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + \hat{j} - 3\hat{k}$. 2M

Ans. $\vec{b} + \vec{c} = 3\hat{i} - 2\hat{k}$. Projection $= \dfrac{(\vec{b} + \vec{c}) \cdot \vec{a}}{|\vec{a}|} = \dfrac{3 + 0 - 4}{3} = -\dfrac{1}{3}$.

Q34. If $|\vec{a}| = 2$, $|\vec{b}| = \frac{1}{\sqrt{2}}$ and $\vec{a} \times \vec{b}$ is a unit vector, find the angle between $\vec{a}$ and $\vec{b}$. 2M

Ans. $|\vec{a} \times \vec{b}| = 2 \times \frac{1}{\sqrt{2}} \sin\theta = \sqrt{2}\sin\theta = 1 \Rightarrow \sin\theta = \frac{1}{\sqrt{2}}, \theta = \frac{\pi}{4}$ (or $\frac{3\pi}{4}$).

Q35. If $\vec{a} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{k}$, find $(\vec{a} + \vec{b}) \cdot (2\vec{a} - \vec{b})$. 2M

Ans. $\vec{a} + \vec{b} = 3\hat{i} + 2\hat{j}$, $2\vec{a} - \vec{b} = 4\hat{j} - 3\hat{k}$. Dot $= 0 + 8 + 0 = 8$.

Q36. If $|\vec{a}| = 5$, $|\vec{b}| = 13$ and $|\vec{a} \times \vec{b}| = 25$, find $\vec{a} \cdot \vec{b}$. 2M

Ans. Lagrange: $(\vec{a} \cdot \vec{b})^2 = 25 \times 169 - 625 = 3600$, so $\vec{a} \cdot \vec{b} = \pm 60$.

Q37. Find the projection of $2\hat{i} - \hat{j} + \hat{k}$ on the X-axis and the angle it makes with the X-axis. 2M

Ans. Projection on $\hat{i} = \frac{(2\hat{i} - \hat{j} + \hat{k}) \cdot \hat{i}}{1} = 2$. Magnitude $\sqrt{6}$, so $\cos\alpha = \frac{2}{\sqrt{6}}$, $\alpha = \cos^{-1}\frac{2}{\sqrt{6}}$.

Q38. Find the position vector of the point which divides the join of $A(1, -2, 3)$ and $B(3, 4, -5)$ externally in the ratio $1 : 2$. 2M

Ans. $\vec{r} = \dfrac{1 \times \vec{b} - 2 \times \vec{a}}{1 - 2} = \dfrac{(3 - 2)\hat{i} + (4 + 4)\hat{j} + (-5 - 6)\hat{k}}{-1} = -\hat{i} - 8\hat{j} + 11\hat{k}$.

Q39. Find p if $2\hat{i} - 3\hat{j} + 4\hat{k}$ and $4\hat{i} + p\hat{j} + 8\hat{k}$ are parallel. 2M

Ans. $\frac{4}{2} = \frac{p}{-3} = \frac{8}{4} = 2 \Rightarrow p = -6$.

Q40. If $|\vec{a} \times \vec{b}| = \sqrt{3}\,(\vec{a} \cdot \vec{b})$, find the angle between $\vec{a}$ and $\vec{b}$. 2M

Ans. $|\vec{a}||\vec{b}|\sin\theta = \sqrt{3}|\vec{a}||\vec{b}|\cos\theta \Rightarrow \tan\theta = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3}$.

Q41. Find a vector of magnitude 9 units in the direction of $2\hat{i} - \hat{j} + 2\hat{k}$. 2M

Ans. Magnitude of the given vector $= 3$; required

(contd.) $= \frac{9}{3}(2\hat{i} - \hat{j} + 2\hat{k}) = 6\hat{i} - 3\hat{j} + 6\hat{k}$.

Q42. If $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$, show that $\vec{a} \perp \vec{b}$ ($\vec{a}, \vec{b}$ non-zero). 2M Board fav

Ans. Square: $|\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a}\cdot\vec{b} = |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a}\cdot\vec{b} \Rightarrow 4\,\vec{a}\cdot\vec{b} = 0 \Rightarrow \vec{a}\cdot\vec{b} = 0$, so $\vec{a} \perp \vec{b}$.

Q43. Find λ if $(2\hat{i} + 6\hat{j} + 14\hat{k}) \times (\hat{i} - \lambda\hat{j} + 7\hat{k}) = \vec{0}$. 2M

Ans. Cross product zero $\Rightarrow$ parallel: $\frac{1}{2} = \frac{-\lambda}{6} = \frac{7}{14}$, so $\lambda = -3$.

Q44. Show that $A(1,1,1)$, $B(2,3,0)$, $C(2,1,2)$ are the vertices of a right-angled triangle. 2M

Ans. $\overrightarrow{AB} = \hat{i} + 2\hat{j} - \hat{k}$, $\overrightarrow{AC} = \hat{i} + \hat{k}$. $\overrightarrow{AB}\cdot\overrightarrow{AC} = 1 + 0 - 1 = 0$, so right angle at A. (Check: $AB^2 + AC^2 = 6 + 2 = 8 = BC^2$.)

Short answer (3 marks)

Q45. If $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} + \hat{k}$, find a unit vector perpendicular to both $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$. 3M Most asked

Ans. $\vec{a} + \vec{b} = 3\hat{i} + 4\hat{j}$, $\vec{a} - \vec{b} = \hat{i} - 2\hat{j} - 2\hat{k}$. Cross $= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 0 \\ 1 & -2 & -2 \end{vmatrix} = -8\hat{i} + 6\hat{j} - 10\hat{k}$, magnitude $10\sqrt{2}$.

Unit vector $= \pm\dfrac{-4\hat{i} + 3\hat{j} - 5\hat{k}}{5\sqrt{2}}$.

Q46. Let $\vec{a} = \hat{i} + 2\hat{j} - \hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{c} = \hat{i} + \hat{j} + \hat{k}$. Find $\vec{d}$ perpendicular to both $\vec{a}$ and $\vec{b}$ with $\vec{c}\cdot\vec{d} = 21$. 3M Board fav

Ans. $\vec{a} \times \vec{b} = (2-1)\hat{i} - (1+2)\hat{j} + (-1-4)\hat{k} = \hat{i} - 3\hat{j} - 5\hat{k}$. Let $\vec{d} = \lambda(\hat{i} - 3\hat{j} - 5\hat{k})$. $\vec{c}\cdot\vec{d} = \lambda(1 - 3 - 5) = -7\lambda = 21 \Rightarrow \lambda = -3$. $\vec{d} = -3\hat{i} + 9\hat{j} + 15\hat{k}$.

Q47. Using vectors, find the area of the triangle with vertices $A(1,2,3)$, $B(2,-1,4)$, $C(4,5,-1)$. 3M

Ans. $\overrightarrow{AB} = \hat{i} - 3\hat{j} + \hat{k}$, $\overrightarrow{AC} = 3\hat{i} + 3\hat{j} - 4\hat{k}$. $\overrightarrow{AB} \times \overrightarrow{AC} = (12-3)\hat{i} - (-4-3)\hat{j} + (3+9)\hat{k} = 9\hat{i} + 7\hat{j} + 12\hat{k}$. Area $= \frac{1}{2}\sqrt{81 + 49 + 144} = \frac{\sqrt{274}}{2}$ sq units.

Q48. Find a vector $\vec{r}$ whose dot products with $\hat{i} + \hat{j} + \hat{k}$, $2\hat{i} - \hat{j} + \hat{k}$ and $\hat{i} + \hat{j} - \hat{k}$ are 6, 3 and 0 respectively. 3M

Ans. Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$: $x + y + z = 6$, $2x - y + z = 3$, $x + y - z = 0$. From 1st and 3rd: $z = 3$, $x + y = 3$. Then $2x - y = 0 \Rightarrow x = 1, y = 2$. $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k}$.

Q49. If $\vec{a} \times \vec{b} = \vec{c} \times \vec{d}$ and $\vec{a} \times \vec{c} = \vec{b} \times \vec{d}$, show that $\vec{a} - \vec{d}$ is parallel to $\vec{b} - \vec{c}$ (given $\vec{a} \neq \vec{d}$, $\vec{b} \neq \vec{c}$). 3M HOTS

Ans. $(\vec{a} - \vec{d}) \times (\vec{b} - \vec{c}) = \vec{a} \times \vec{b} - \vec{a} \times \vec{c} - \vec{d} \times \vec{b} + \vec{d} \times \vec{c} = \vec{c} \times \vec{d} - \vec{b} \times \vec{d} + \vec{b} \times \vec{d} - \vec{c} \times \vec{d} = \vec{0}$. Cross product zero, so parallel.

Q50. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - \hat{k}$, find $\vec{c}$ such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a}\cdot\vec{c} = 3$. 3M HOTS

(contd.) Let $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$. $\vec{a} \times \vec{c} = (z - y)\hat{i} + (x - z)\hat{j} + (y - x)\hat{k} = \hat{i} - \hat{k}$ gives $z - y = 1, x = z, y = x - 1$. $\vec{a}\cdot\vec{c} = x + (x-1) + x = 3 \Rightarrow x = \frac{4}{3}$. $\vec{c} = \frac{4}{3}\hat{i} + \frac{1}{3}\hat{j} + \frac{4}{3}\hat{k}$.

Q51. If $|\vec{a}| = 1$, $|\vec{b}| = 2$, $|\vec{c}| = 3$ and each is perpendicular to the sum of the other two, find $|\vec{a} + \vec{b} + \vec{c}|$. 3M Board fav

Ans. $\vec{a}\cdot(\vec{b} + \vec{c}) = 0, \vec{b}\cdot(\vec{c} + \vec{a}) = 0, \vec{c}\cdot(\vec{a} + \vec{b}) = 0$. Adding: $2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}) = 0$. $|\vec{a} + \vec{b} + \vec{c}|^2 = 1 + 4 + 9 + 0 = 14$, so $|\vec{a} + \vec{b} + \vec{c}| = \sqrt{14}$.

Q52. For what values of x is the angle between $\vec{a} = 2x^2\hat{i} + 4x\hat{j} + \hat{k}$ and $\vec{b} = 7\hat{i} - 2\hat{j} + x\hat{k}$ obtuse? 3M HOTS

Ans. Obtuse $\Rightarrow \vec{a}\cdot\vec{b} < 0$: $14x^2 - 8x + x = 7x(2x - 1) < 0 \Rightarrow 0 < x < \frac{1}{2}$. (In this range the vectors are never opposite, so the angle is truly obtuse, not π.)

Q53. Using vectors, find k so that $(k,2,3)$, $(1,0,-1)$ and $(2,1,1)$ are collinear. 3M

Ans. Let $A(1;0,-1)$, $B(2,1,1)$, $C(k,2,3)$. $\overrightarrow{AB} = \hat{i} + \hat{j} + 2\hat{k}$, $\overrightarrow{AC} = (k-1)\hat{i} + 2\hat{j} + 4\hat{k}$. Collinear $\Rightarrow \frac{k-1}{1} = \frac{2}{1} = \frac{4}{2} \Rightarrow k = 3$.

Q54. Find the area of the parallelogram whose diagonals are $2\hat{i} - \hat{j} + \hat{k}$ and $3\hat{i} + 4\hat{j} - \hat{k}$. 3M Most asked

Ans. $\vec{d_1} \times \vec{d_2} = (1-4)\hat{i} - (-2-3)\hat{j} + (8+3)\hat{k} = -3\hat{i} + 5\hat{j} + 11\hat{k}$, magnitude $\sqrt{155}$. Area $= \frac{\sqrt{155}}{2}$ sq units.

Q55. Show that $\vec{a} = \hat{i} + 2\hat{j} - \hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + 3\hat{k}$, $\vec{c} = -3\hat{i} - \hat{j} - 2\hat{k}$ can form the sides of a triangle, and find its area. 3M

Ans. $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, so they form a triangle (head to tail). $\vec{a} \times \vec{b} = (6-1)\hat{i} - (3+2)\hat{j} + (-1-4)\hat{k} = 5\hat{i} - 5\hat{j} - 5\hat{k}$. Area $= \frac{1}{2} \times 5\sqrt{3} = \frac{5\sqrt{3}}{2}$ sq units.

Q56. Find the value of λ for which the vectors $\hat{i} - 2\hat{j} + \lambda\hat{k}$ and $3\hat{i} + \hat{j} - \hat{k}$ make an angle of $\frac{\pi}{2}$, and then find a unit vector perpendicular to both. 3M

Ans. Dot $= 3 - 2 - \lambda = 0 \Rightarrow \lambda = 1$. Cross of $\hat{i} - 2\hat{j} + \hat{k}$ and $3\hat{i} + \hat{j} - \hat{k}$: $(2-1)\hat{i} - (-1-3)\hat{j} + (1+6)\hat{k} = \hat{i} + 4\hat{j} + 7\hat{k}$, magnitude $\sqrt{66}$. Unit vector $= \pm\frac{1}{\sqrt{66}}(\hat{i} + 4\hat{j} + 7\hat{k})$.

Q57. Find a vector of magnitude 9 perpendicular to both $\hat{i} + \hat{j}$ and $\hat{j} + \hat{k}$. 3M

Ans. $(\hat{i} + \hat{j}) \times (\hat{j} + \hat{k}) = \hat{i} - \hat{j} + \hat{k}$, magnitude $\sqrt{3}$. Required $= \pm\frac{9}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k}) = \pm 3\sqrt{3}(\hat{i} - \hat{j} + \hat{k})$.

Q58. If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ with $|\vec{a}| = 2$, $|\vec{b}| = 3$, $|\vec{c}| = 4$, find $\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}$. 3M Most asked

Ans. $|\vec{a} + \vec{b} + \vec{c}|^2 = 0 \Rightarrow 4 + 9 + 16 + 2S = 0 \Rightarrow S = -\frac{29}{2}$.

Q59. Express $\vec{b} = 3\hat{i} + \hat{j} + 2\hat{k}$ as $\vec{b_1} + \vec{b_2}$, where $\vec{b_1} \parallel \vec{a}$ and $\vec{b_2} \perp \vec{a}$, with $\vec{a} = \hat{i} - \hat{j}$. 3M

Ans. $\vec{b_1} = \frac{\vec{b}\cdot\vec{a}}{|\vec{a}|^2}\vec{a} = \frac{2}{2}(\hat{i} - \hat{j}) = \hat{i} - \hat{j}$.

(contd.) $\vec{b_2} = \vec{b} - \vec{b_1} = 2\hat{i} + 2\hat{j} + 2\hat{k}$. Check $\vec{b_2} \cdot \vec{a} = 2 - 2 = 0$ ✓

Q60. Prove the cosine rule $c^2 = a^2 + b^2 - 2ab\cos C$ for triangle ABC using vectors. 3M

Ans. Let $\overrightarrow{CB} = \vec{a}$, $\overrightarrow{CA} = \vec{b}$, so $\overrightarrow{AB} = \vec{a} - \vec{b}$. $c^2 = |\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a}\cdot\vec{b} = a^2 + b^2 - 2ab\cos C$, since the angle between $\vec{a}$ and $\vec{b}$ is C.

Q61. Show that $A(1,0,2)$, $B(2,2,3)$, $C(4,6,5)$ are collinear and find the ratio in which B divides AC. 3M

Ans. $\overrightarrow{AB} = \hat{i} + 2\hat{j} + \hat{k}$, $\overrightarrow{BC} = 2\hat{i} + 4\hat{j} + 2\hat{k} = 2\overrightarrow{AB}$: collinear (B common). $AB : BC = 1 : 2$, so B divides AC internally in the ratio $1 : 2$.

Q62. Show that $(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b}) = 2(\vec{a} \times \vec{b})$. 3M

Ans. Expand: $\vec{a}\times\vec{a} + \vec{a}\times\vec{b} - \vec{b}\times\vec{a} - \vec{b}\times\vec{b} = \vec{0} + \vec{a}\times\vec{b} + \vec{a}\times\vec{b} - \vec{0} = 2(\vec{a}\times\vec{b})$, using $\vec{b}\times\vec{a} = -\vec{a}\times\vec{b}$.

Q63. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors with $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, find $\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}$. 3M Must do

Ans. $|\vec{a} + \vec{b} + \vec{c}|^2 = 0 \Rightarrow 1 + 1 + 1 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}) = 0$. So the value is $-\frac{3}{2}$.

Long answer (5 marks)

Q64. For $\vec{a} = 2\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} - \hat{j} + 2\hat{k}$, find (i) the angle between them (ii) the projection of $\vec{a}$ on $\vec{b}$ (iii) a unit vector perpendicular to both (iv) the area of the parallelogram with these adjacent sides. Verify Lagrange identity. 5M Must do

Ans. (i) $\vec{a}\cdot\vec{b} = 4 - 2 + 2 = 4$, $|\vec{a}| = |\vec{b}| = 3$, $\cos\theta = \frac{4}{9}$. (ii) $\frac{4}{3}$. (iii) $\vec{a}\times\vec{b} = (4+1)\hat{i} - (4-2)\hat{j} + (-2-4)\hat{k} = 5\hat{i} - 2\hat{j} - 6\hat{k}$, magnitude $\sqrt{65}$; unit $= \pm\frac{1}{\sqrt{65}}(5\hat{i} - 2\hat{j} - 6\hat{k})$. (iv) Area $= \sqrt{65}$. Check: $65 + 16 = 81 = 9\times 9$ ✓

Q65. For the triangle with vertices $A(1,2,0)$, $B(3,1,2)$, $C(2,4,3)$, find (i) its area (ii) the length of the altitude from A to BC (iii) $\cos A$. 5M Must do

Ans. $\overrightarrow{AB} = 2\hat{i} - \hat{j} + 2\hat{k}$, $\overrightarrow{AC} = \hat{i} + 2\hat{j} + 3\hat{k}$. (i) $\overrightarrow{AB}\times\overrightarrow{AC} = -7\hat{i} - 4\hat{j} + 5\hat{k}$, magnitude $\sqrt{90} = 3\sqrt{10}$, area $= \frac{3\sqrt{10}}{2}$. (ii) $\overrightarrow{BC} = -\hat{i} + 3\hat{j} + \hat{k}$, $BC = \sqrt{11}$; altitude $= \frac{2\times\text{Area}}{BC} = \frac{3\sqrt{10}}{\sqrt{11}}$. (iii) $\cos A = \frac{\overrightarrow{AB}\cdot\overrightarrow{AC}}{AB\times AC} = \frac{6}{3\sqrt{14}} = \frac{2}{\sqrt{14}}$.

Q66. Points $A(1,2,-1)$, $B(3,3,1)$, $C(4,1,5)$, $D(2,0,3)$ are given. Show that ABCD is a parallelogram and find its area in two ways (using sides and using diagonals). 5M

Ans. $\overrightarrow{AB} = 2\hat{i} + \hat{j} + 2\hat{k} = \overrightarrow{DC}$, so ABCD is a parallelogram. $\overrightarrow{AD} = \hat{i} - 2\hat{j} + 4\hat{k}$. $\overrightarrow{AB}\times\overrightarrow{AD} = (4+4)\hat{i} - (8-2)\hat{j} + (-4-1)\hat{k} = 8\hat{i} - 6\hat{j} - 5\hat{k}$, area $= \sqrt{125} = 5\sqrt{5}$. Diagonals $\overrightarrow{AC} = 3\hat{i} - \hat{j} + 6\hat{k}$, $\overrightarrow{BD} = -\hat{i} - 3\hat{j} + 2\hat{k}$; $\overrightarrow{AC}\times\overrightarrow{BD} = 16\hat{i} - 12\hat{j} - 10\hat{k}$, $\frac{1}{2}\sqrt{500} = 5\sqrt{5}$ ✓

Q67. If $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular vectors of equal magnitude, show that $\vec{a} + \vec{b} + \vec{c}$ is equally inclined to $\vec{a}$, $\vec{b}$ and $\vec{c}$, and find that angle. 5M

Ans. Let $|\vec{a}| = |\vec{b}| = |\vec{c}| = k$; all mutual dot products are 0. $|\vec{a} + \vec{b} + \vec{c}|^2 = 3k^2$, so $|\vec{a} + \vec{b} + \vec{c}| = \sqrt{3}k$. $(\vec{a} + \vec{b} + \vec{c})\cdot\vec{a} = k^2$, so $\cos\alpha = \frac{k^2}{\sqrt{3}k\times k} = \frac{1}{\sqrt{3}}$. Same for $\vec{b}, \vec{c}$. Angle $= \cos^{-1}\frac{1}{\sqrt{3}}$.

Q68. Let $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{c} = 3\hat{i} + \hat{j} + \hat{k}$. (i) Find the angle between $\vec{a}$ and $\vec{b}$. (ii) Find $\vec{d}$ perpendicular to both $\vec{a}$ and $\vec{b}$ with $\vec{c}\cdot\vec{d} = 15$. (iii) Find the projection of $\vec{d}$ on $\vec{c}$. 5M Board fav

Ans. (i) $\vec{a}\cdot\vec{b} = 2 - 1 - 2 = -1$, $|\vec{a}| = |\vec{b}| = \sqrt{6}$, $\cos\theta = -\frac{1}{6}$, $\theta = \cos^{-1}\left(-\frac{1}{6}\right)$. (ii) $\vec{a}\times\vec{b} = (2-1)\hat{i} - (4+1)\hat{j} + (-2-1)\hat{k} = \hat{i} - 5\hat{j} - 3\hat{k}$; $\vec{d} = \lambda(\hat{i} - 5\hat{j} - 3\hat{k})$, $\vec{c}\cdot\vec{d} = \lambda(3 - 5 - 3) = -5\lambda = 15 \Rightarrow \lambda = -3$, $\vec{d} = -3\hat{i} + 15\hat{j} + 9\hat{k}$. (iii) $\frac{\vec{d}\cdot\vec{c}}{|\vec{c}|} = \frac{15}{\sqrt{11}}$.

Q69. Show that the area of a quadrilateral ABCD is $\frac{1}{2}|\overrightarrow{AC}\times\overrightarrow{BD}|$. Hence find the area of ABCD with $A(0,0,0)$, $B(4,0,0)$, $C(5,3,0)$, $D(1,4,0)$. 5M HOTS

Ans. Area $= \triangle ABC + \triangle ACD = \frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}| + \frac{1}{2}|\overrightarrow{AC}\times\overrightarrow{AD}|$; both cross products point the same way (convex, same plane), so area $= \frac{1}{2}|\overrightarrow{AC}\times(\overrightarrow{AD} - \overrightarrow{AB})| = \frac{1}{2}|\overrightarrow{AC}\times\overrightarrow{BD}|$. Here $\overrightarrow{AC} = 5\hat{i} + 3\hat{j}$, $\overrightarrow{BD} = -3\hat{i} + 4\hat{j}$; cross $= (20 + 9)\hat{k} = 29\hat{k}$. Area $= \frac{29}{2}$ sq units.

Q70. Show that $A(0,0,0)$, $B(2,1,2)$, $C(3,3,0)$, $D(1,2,-2)$ are the vertices of a square. Find its area and show that its diagonals are perpendicular. 5M

Ans. $\overrightarrow{AB} = 2\hat{i} + \hat{j} + 2\hat{k} = \overrightarrow{DC}$ (parallelogram). $|\overrightarrow{AB}| = |\overrightarrow{AD}| = 3$ and $\overrightarrow{AB}\cdot\overrightarrow{AD} = 2 + 2 - 4 = 0$, so it is a square. $\overrightarrow{AB}\times\overrightarrow{AD} = -6\hat{i} + 6\hat{j} + 3\hat{k}$, area $= \sqrt{81} = 9$ sq units. $\overrightarrow{AC}\cdot\overrightarrow{BD} = (3,3,0)\cdot(-1,1,-4) = 0$ ✓

Case based (4 marks)

Q71. Drone delivery. A drone takes off from its base O (origin) and flies straight to point $A(2,3,6)$ (units in 100 m), then turns and flies to $B(5,7,6)$. (i) Find the distance OA. (1) (ii) Find the direction cosines of $\overrightarrow{OA}$. (1) (iii) Find the angle between $\overrightarrow{OA}$ and $\overrightarrow{AB}$. (2) 4M

Ans. (i) $\sqrt{4 + 9 + 36} = 7$, i.e. 700 m. (ii) $\frac{2}{7}, \frac{3}{7}, \frac{6}{7}$. (iii) $\overrightarrow{AB} = 3\hat{i} + 4\hat{j}$, $|\overrightarrow{AB}| = 5$; $\overrightarrow{OA}\cdot\overrightarrow{AB} = 6 + 12 = 18$; $\cos\theta = \frac{18}{35}$, $\theta = \cos^{-1}\frac{18}{35}$.

Q72. Pushing a box. A constant force $\vec{F} = 3\hat{i} + 2\hat{j} - 4\hat{k}$ moves a box from $P(1,-1,2)$ to $Q(4,1,3)$. Work done $= \vec{F}\cdot\vec{d}$. (i) Find the displacement $\vec{d} = \overrightarrow{PQ}$. (1) (ii) Find $|\vec{d}|$. (1) (iii) Find the work done and the cosine of the angle between $\vec{F}$ and $\vec{d}$. (2) 4M

Ans. (i) $\vec{d} = 3\hat{i} + 2\hat{j} + \hat{k}$. (ii) $\sqrt{14}$. (iii)

(contd.) $W = 9 + 4 - 4 = 9$ units; $|\vec{F}| = \sqrt{29}$, $\cos\theta = \frac{9}{\sqrt{29}\sqrt{14}} = \frac{9}{\sqrt{406}}$.

Q73. Solar panel. A flat solar panel is a parallelogram with adjacent edges $\vec{a} = 2\hat{i} + \hat{k}$ and $\vec{b} = 2\hat{j} + \hat{k}$ (in m). Sunlight falls along $\vec{d} = -\hat{i} - \hat{j} + \hat{k}$. (i) Find $\vec{a} \times \vec{b}$. (1) (ii) Find the area of the panel. (1) (iii) Find the cosine of the angle between the sunlight and the normal $\vec{a} \times \vec{b}$. (2) 4M

Ans. (i) $\vec{a} \times \vec{b} = (0-2)\hat{i} - (2-0)\hat{j} + (4-0)\hat{k} = -2\hat{i} - 2\hat{j} + 4\hat{k}$. (ii) $\sqrt{4+4+16} = 2\sqrt{6}$ sq m. (iii) $\cos\theta = \dfrac{2+2+4}{2\sqrt{6} \times \sqrt{3}} = \dfrac{8}{6\sqrt{2}} = \dfrac{2\sqrt{2}}{3}$.

Bonus practice (mixed)

Q74. The projection of $\hat{i} + \hat{j}$ on $\hat{k}$ is 1M Must do
(a) 1 (b) 0 (c) $\sqrt{2}$ (d) -1

Ans. (b) 0, since $(\hat{i} + \hat{j}) \cdot \hat{k} = 0$.

Q75. Assertion (A): For $\vec{a} = \hat{i} - \hat{j}$ and $\vec{b} = -2\hat{i} + 2\hat{j}$, $\vec{a} \times \vec{b} = \vec{0}$. **Reason (R):** The cross product of two parallel vectors is the zero vector. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Ans. (a) $\vec{b} = -2\vec{a}$, so they are parallel and R explains A.

Q76. Find the angle between $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + \hat{j} + 2\hat{k}$. 2M Most asked

Ans. $\vec{a} \cdot \vec{b} = 2 - 1 + 2 = 3$, $|\vec{a}| = |\vec{b}| = \sqrt{6}$, $\cos\theta = \frac{3}{6} = \frac{1}{2}$, so $\theta = \frac{\pi}{3}$.

Q77. If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $\vec{a} \cdot \vec{b} = 6$, find $|\vec{a} \times \vec{b}|$. 2M Board fav

Ans. $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 = 144 - 36 = 108$, so $|\vec{a} \times \vec{b}| = 6\sqrt{3}$.

Q78. Find a unit vector perpendicular to both $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$, where $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$. 3M Most asked

Ans. $\vec{a} + \vec{b} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{a} - \vec{b} = -\hat{j} - 2\hat{k}$. Cross product $= -2\hat{i} + 4\hat{j} - 2\hat{k}$, magnitude $\sqrt{24} = 2\sqrt{6}$. Unit vector $= \pm\frac{1}{\sqrt{6}}(-\hat{i} + 2\hat{j} - \hat{k})$.

Q79. Find the area of the triangle with vertices $A(1,2,1)$, $B(2,4,3)$, $C(1,5,4)$. 3M Board fav

Ans. $\overrightarrow{AB} = \hat{i} + 2\hat{j} + 2\hat{k}$, $\overrightarrow{AC} = 3\hat{j} + 3\hat{k}$, $\overrightarrow{AB} \times \overrightarrow{AC} = 0\hat{i} - 3\hat{j} + 3\hat{k}$, magnitude $3\sqrt{2}$. Area $= \frac{3\sqrt{2}}{2}$ sq units.

Q80. Show that $A(1,0,2)$, $B(3,1,2)$, $C(0,2,5)$ are vertices of a right-angled triangle and find its area. 3M Must do

Ans. $\overrightarrow{AB} = 2\hat{i} + \hat{j}$, $\overrightarrow{AC} = -\hat{i} + 2\hat{j} + 3\hat{k}$, $\overrightarrow{AB} \cdot \overrightarrow{AC} = -2 + 2 + 0 = 0$, so right angle at A. Check: $|AB|^2 + |AC|^2 = 5 + 14 = 19 = |BC|^2$ ✓. Area $= \frac{1}{2}\sqrt{5}\sqrt{14} = \frac{\sqrt{70}}{2}$ sq units.

Q81. Find the position vector of the point R dividing the join of $P(2\hat{i} - \hat{j} + 3\hat{k})$ and $Q(5\hat{i} + 2\hat{j} - 3\hat{k})$ internally in the ratio $1:2$. 2M Must do

Ans. $\vec{r} = \frac{1\times(5,2,-3)+2\times(2,-1,3)}{3} = \frac{(9,0,3)}{3}$, so $\vec{r} = 3\hat{i} + \hat{k}$.

Q82. Find p if $\vec{a} = 2\hat{i} + 6\hat{j} + 4\hat{k}$ and $\vec{b} = \hat{i} + p\hat{j} + 2\hat{k}$ are parallel. 2M Board fav

Ans. Parallel $\Rightarrow$ ratios equal: $\frac{2}{1} = \frac{6}{p} = \frac{4}{2}$, so $p = 3$.

Q83. If $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 45$ and $|\vec{a}| = 2|\vec{b}|$, find $|\vec{a}|$ and $|\vec{b}|$. 2M Most asked

Ans. $|\vec{a}|^2 - |\vec{b}|^2 = 45 \Rightarrow 4|\vec{b}|^2 - |\vec{b}|^2 = 45 \Rightarrow |\vec{b}| = \sqrt{15}$, $|\vec{a}| = 2\sqrt{15}$.

Q84. Find the vector of magnitude 10 units in the direction of $3\hat{i} + 4\hat{k}$. 1M

Ans. Unit vector $= \frac{3\hat{i}+4\hat{k}}{5}$, so required vector $= 6\hat{i} + 8\hat{k}$.

Ek nazar mein: Vector Algebra

Type of sawal	Method	Marks
Unit vector, DCs	divide by magnitude	1-2
Angle, $\perp$ condition	dot product	1-2
Projection	$\vec{a} \cdot \vec{b}/\|\vec{b}\|$	2
$\perp$ to both, area	cross product	3
$\vec{a} + \vec{b} + \vec{c} = \vec{0}$	square both sides	3

Last minute: 8 one-liners

1. $\overrightarrow{AB}$ = head minus tail.
2. Unit vector = vector / magnitude; DCs squares add to 1.
3. Dot gives a number, uses cos; cross gives a vector, uses sin.
4. $\perp$: dot $= 0$; $\|$: cross $= \vec{0}$ or ratios equal.
5. Projection of $\vec{a}$ on $\vec{b}$: divide by $|\vec{b}|$.
6. $\hat{i} \to \hat{j} \to \hat{k}$ clockwise is $+$.
7. Triangle area: half of $|\overrightarrow{AB} \times \overrightarrow{AC}|$.
8. Perpendicular unit vector: always write $\pm$.

Apne notes / galtiyan yahan likho

Most asked: shortest distance between skew lines, foot of perpendicular, angle between lines

THREE DIMENSIONAL GEOMETRY

↳Syllabus 2026-27: only **lines in space**. Planes are deleted, so this chapter = DCs/DRs, equation of a line, angle, skew lines and shortest distance.

1. Direction Cosines and Direction Ratios of a Line

If a directed line makes angles α, β, γ with the positive X, Y, Z axes, then $l = \cos\alpha$, $m = \cos\beta$, $n = \cos\gamma$ are its **direction cosines** (DCs). If we reverse the direction, the DCs become $-l, -m, -n$, so a line (not directed) has DCs $\pm(l, m, n)$.

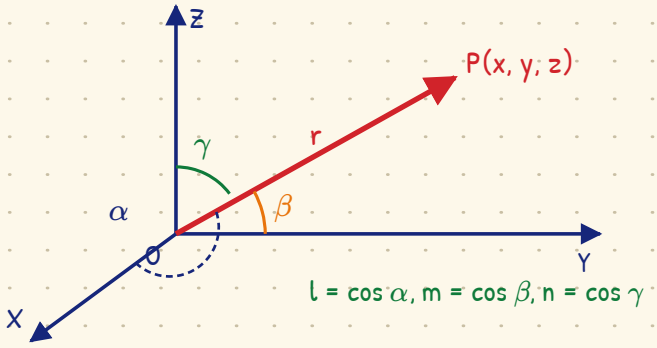


Fig: angles α, β, γ of OP with the axes; $x = lr, y = mr, z = nr$

→ $l^2 + m^2 + n^2 = 1$ (always)

→ **Direction ratios** (DRs) a, b, c: any numbers proportional to l, m, n. A line has infinitely many sets of DRs.

→ From DRs to DCs:

$$l = \frac{\pm a}{\sqrt{a^2+b^2+c^2}}$$

and similarly $m = \frac{\pm b}{\sqrt{a^2+b^2+c^2}}$, $n = \frac{\pm c}{\sqrt{a^2+b^2+c^2}}$ (same sign throughout)

→ Line through $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$: DRs $= x_2 - x_1,\ y_2 - y_1,\ z_2 - z_1$; DCs $= \frac{x_2 - x_1}{PQ}$ etc.

→ DCs of the axes: X-axis $1, 0, 0$; Y-axis $0, 1, 0$; Z-axis $0, 0, 1$.

Yaad karo: "Square karke jodo, 1 aana chahiye"

DCs ke squares ka sum **hamesha 1**. Teen number DCs hain ya nahi, check karna ho toh square karke jodo. 1 nahi aaya? Toh ye sirf DRs hain, DCs nahi.

Quick fact X-axis: $\frac{x}{1} = \frac{y}{0} = \frac{z}{0}$ (DRs $1, 0, 0$).

Quick fact Intersecting lines: SD $= 0$.

Solved Example 1

Find the direction cosines of a line whose direction ratios are $2, -1, -2$.

1. $\sqrt{a^2+b^2+c^2} = \sqrt{4+1+4} = 3$
2. DCs $= \pm\left(\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3}\right)$

✔ $\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3}$ (or all signs reversed)

Solved Example 2

A line makes angles $90°, 135°, 45°$ with the X, Y, Z axes. Find its direction cosines.

1. $l = \cos 90° = 0$, $m = \cos 135° = -\frac{1}{\sqrt{2}}$, $n = \cos 45° = \frac{1}{\sqrt{2}}$
2. Check: $0 + \frac{1}{2} + \frac{1}{2} = 1$ ✓

✔ $0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

Solved Example 3

Find the DRs and DCs of the line through $A(-2, 4, -5)$ and $B(1, 2, 3)$.

1. DRs $= 1-(-2),\ 2-4,\ 3-(-5) = 3, -2, 8$
2. $\sqrt{9+4+64} = \sqrt{77}$

✔ DRs $3, -2, 8$; DCs $\pm\left(\frac{3}{\sqrt{77}}, \frac{-2}{\sqrt{77}}, \frac{8}{\sqrt{77}}\right)$

Solved Example 4

Show that $A(1, 2, 3)$, $B(3, 5, 7)$ and $C(7, 11, 15)$ are collinear.

1. DRs of AB: $2, 3, 4$; DRs of BC: $4, 6, 8$
2. $\frac{4}{2} = \frac{6}{3} = \frac{8}{4} = 2$, so AB $\parallel$ BC
3. B is common to both, so all three lie on one line

✔ A, B, C are collinear

Solved Example 5

If a line makes angles α, β, γ with the axes, prove that $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$.

1. $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 3 - (\cos^2\alpha + \cos^2\beta + \cos^2\gamma)$
2. $= 3 - (l^2 + m^2 + n^2) = 3 - 1$

✔ $= 2$. (Similarly $\cos 2\alpha + \cos 2\beta + \cos 2\gamma = -1$)

Quick check Line $\frac{x-1}{2} = \frac{y}{3} = \frac{z+2}{6}$: point $(1, 0, -2)$, DCs $\frac{2}{7}, \frac{3}{7}, \frac{6}{7}$.

Solved Example 6

Find the DCs of a line which is equally inclined to the three axes.

1. $\alpha = \beta = \gamma \Rightarrow l = m = n$
2. $3l^2 = 1 \Rightarrow l = \pm\frac{1}{\sqrt{3}}$

✔ $\pm\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

2. Equation of a Line in Space

A line is fixed by **one point + one direction**, or by **two points**.

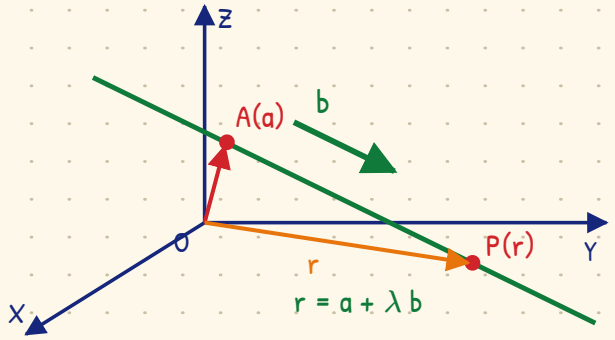


Fig: line through A (position vector $\vec{a}$) parallel to $\vec{b}$; any point $P(\vec{r})$ on it

(a) Through a point, parallel to a vector

Point $A(\vec{a})$, direction $\vec{b} = a\hat{i} + b\hat{j} + c\hat{k}$. Since $\overrightarrow{AP} \parallel \vec{b}$, $\vec{r} - \vec{a} = \lambda\vec{b}$:

$$\vec{r} = \vec{a} + \lambda\vec{b} \quad (\lambda \text{ real})$$

Cartesian form (point (x_1, y_1, z_1), DRs a, b, c):

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

(b) Through two points

$\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$, i.e.

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

(c) General point (most useful!)

Put each fraction $= \lambda$: any point on the line is

$$(x_1 + a\lambda,\ y_1 + b\lambda,\ z_1 + c\lambda)$$

Use it for foot of perpendicular, intersection, and points at a given distance.

Yaad karo: "Standard form = x, y, z ka coefficient +1"

Cartesian equation se DRs padhne se pehle form theek karo: x, y, z **ka coefficient** $+1$ hona chahiye. $\frac{3-x}{2} = \frac{x-3}{-2}$ aur $\frac{2z-6}{3} = \frac{z-3}{3/2}$. Neeche wale number hi DRs hain, upar wale (sign badal ke) point.

Quick fact Skew lines exist only in 3D, never in a plane.

Solved Example 7

Find the vector and Cartesian equations of the line through $(2, -1, 4)$ parallel to $\hat{i} + 3\hat{j} - 2\hat{k}$.

1. $\vec{a} = 2\hat{i} - \hat{j} + 4\hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} - 2\hat{k}$
2. Vector: $\vec{r} = (2\hat{i} - \hat{j} + 4\hat{k}) + \lambda(\hat{i} + 3\hat{j} - 2\hat{k})$

✔ **Cartesian:** $\frac{x-2}{1} = \frac{y+1}{3} = \frac{z-4}{-2}$

Solved Example 8

Find the equations of the line through $(1, 2, -1)$ and $(3, -1, 4)$.

1. Direction $= (3-1)\hat{i} + (-1-2)\hat{j} + (4+1)\hat{k} = 2\hat{i} - 3\hat{j} + 5\hat{k}$
2. Vector: $\vec{r} = (\hat{i} + 2\hat{j} - \hat{k}) + \lambda(2\hat{i} - 3\hat{j} + 5\hat{k})$

✔ **Cartesian:** $\frac{x-1}{2} = \frac{y-2}{-3} = \frac{z+1}{5}$

Solved Example 9

The Cartesian equation of a line is $\frac{3-x}{2} = \frac{y+1}{4} = \frac{2z-6}{3}$. Find its DRs, a point on it, and its vector equation.

1. Standard form: $\frac{x-3}{-2} = \frac{y+1}{4} = \frac{z-3}{3/2}$
2. DRs: $-2, 4, \frac{3}{2}$, i.e. $-4, 8, 3$; point $(3, -1, 3)$

✔ $\vec{r} = (3\hat{i} - \hat{j} + 3\hat{k}) + \lambda(-4\hat{i} + 8\hat{j} + 3\hat{k})$

Solved Example 10

Find the points on the line $\frac{x-1}{2} = \frac{y+2}{-1} = \frac{z-3}{2}$ at a distance of 6 units from the point $(1, -2, 3)$.

1. General point $P(1 + 2\lambda,\ -2 - \lambda,\ 3 + 2\lambda)$
2. Distance from $(1, -2, 3)$: $\sqrt{4\lambda^2 + \lambda^2 + 4\lambda^2} = 3|\lambda| = 6 \Rightarrow \lambda = \pm 2$
3. $\lambda = 2$: $(5, -4, 7)$; $\lambda = -2$: $(-3, 0, -1)$

✔ $(5, -4, 7)$ **and** $(-3, 0, -1)$

Solved Example 11

Does the point $(7, 5, -3)$ lie on the line $\vec{r} = (\hat{i} - \hat{j} + 3\hat{k}) + \lambda(2\hat{i} + 2\hat{j} - 2\hat{k})$?

1. General point: $(1 + 2\lambda,\ -1 + 2\lambda,\ 3 - 2\lambda)$
2. $1 + 2\lambda = 7 \Rightarrow \lambda = 3$; then $y = -1 + 6 = 5$ ✓, $z = 3 - 6 = -3$ ✓

✔ **Yes, it lies on the line (at** $\lambda = 3$**)**

Line	Equation
X-axis	$\frac{x}{1} = \frac{y}{0} = \frac{z}{0}$, i.e. $\vec{r} = \lambda\hat{i}$
Through (x_1, y_1, z_1) parallel to Z-axis	$x = x_1, y = y_1; \vec{r} = \vec{a} + \lambda\hat{k}$
$x = y = z$	through origin, DRs $1, 1, 1$
$ax - p = by - q = cz - r$	divide: DRs $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$

↳A zero in a denominator is allowed in this "symmetric" form: it only means that coordinate is constant ($x = x_1$). Never divide by it in working.

Solved Example 12

Find the equation of the line through $(1, 2, 3)$ which makes equal angles with the coordinate axes.

1. Equal angles $\Rightarrow l = m = n$, so DRs $1, 1, 1$

✔ $\frac{x-1}{1} = \frac{y-2}{1} = \frac{z-3}{1}$, i.e. $x - 1 = y - 2 = z - 3$



Meme samjho

Line $\frac{3-x}{2} = \ldots$ dekh ke DRs $(2, \ldots)$ likh diye? Pehle x ka coefficient $+1$ banao, tab bolna "perfect!"

Asli baat: $\frac{3-x}{2} = \frac{x-3}{-2}$, so the first DR is -2, not 2. The point is $x = 3$.

Formula Box: Equation of a line

→ DCs: $l^2 + m^2 + n^2 = 1$; DRs → DCs: divide by $\sqrt{a^2 + b^2 + c^2}$

→ Line through 2 points: DRs $x_2 - x_1,\ y_2 - y_1,\ z_2 - z_1$

→ Vector: $\vec{r} = \vec{a} + \lambda\vec{b}$; through 2 points: $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$

→ Cartesian: $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$

→ General point: $(x_1 + a\lambda,\ y_1 + b\lambda,\ z_1 + c\lambda)$

3. Angle Between Two Lines

The angle between two lines = angle between their direction vectors. Lines $\vec{r} = \vec{a_1} + \lambda\vec{b_1}$ and $\vec{r} = \vec{a_2} + \mu\vec{b_2}$:

$$\cos\theta = \left|\frac{\vec{b_1} \cdot \vec{b_2}}{|\vec{b_1}||\vec{b_2}|}\right|$$

With DRs a_1, b_1, c_1 and a_2, b_2, c_2:

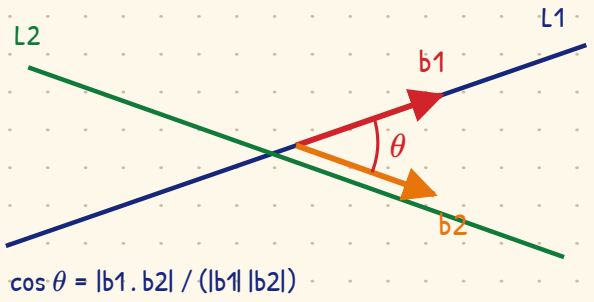
$$\cos\theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

With DCs: $\cos\theta = |l_1l_2 + m_1m_2 + n_1n_2|$.

Lines are	Condition (DRs)
Perpendicular	$a_1a_2 + b_1b_2 + c_1c_2 = 0$
Parallel	$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

↳Modulus lagate hain taaki acute angle mile (two lines make θ and $\pi - \theta$; we report the acute one).

Quick check DRs $1, 2, 3$ and $3, 0, -1$: $3 + 0 - 3 = 0$, so the lines are perpendicular.



$\cos\theta$ = |b1 . b2| / (|b1| |b2|)

Fig: the angle between two lines is the angle between their directions $\vec{b_1}, \vec{b_2}$

Solved Example 13

Find the angle between the lines $\vec{r} = (2\hat{i} + \hat{k}) + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$ and $\vec{r} = (\hat{j} - \hat{k}) + \mu(4\hat{i} + 4\hat{j} + 7\hat{k})$.

1. $\vec{b_1} \cdot \vec{b_2} = 4 + 8 + 14 = 26$
2. $|\vec{b_1}| = 3,\ |\vec{b_2}| = \sqrt{16 + 16 + 49} = 9$
3. $\cos\theta = \frac{26}{27}$

✔ $\theta = \cos^{-1}\left(\frac{26}{27}\right)$

Solved Example 14

Find the angle between the lines $\frac{x-1}{2} = \frac{y}{2} = \frac{z+3}{1}$ and $\frac{x+2}{4} = \frac{y-1}{1} = \frac{z}{8}$.

1. DRs: $2, 2, 1$ and $4, 1, 8$
2. $\cos\theta = \frac{|8 + 2 + 8|}{\sqrt{9}\sqrt{81}} = \frac{18}{27} = \frac{2}{3}$

✔ $\theta = \cos^{-1}\left(\frac{2}{3}\right)$

Solved Example 15

Find k so that the lines $\frac{x-2}{k} = \frac{y+1}{2} = \frac{z-3}{-1}$ and $\frac{x}{3} = \frac{y-4}{k} = \frac{z+1}{4}$ are perpendicular.

1. DRs: $k, 2, -1$ and $3, k, 4$
2. Perpendicular: $3k + 2k - 4 = 0 \Rightarrow 5k = 4$

✔ $k = \frac{4}{5}$

Solved Example 16

Find p so that the lines $\frac{1-x}{2} = \frac{2y-4}{p} = \frac{z+1}{3}$ and $\frac{3-3x}{p} = \frac{y-1}{2} = \frac{2-z}{1}$ are perpendicular.

1. Standard form of line 1: $\frac{x-1}{-2} = \frac{y-2}{p/2} = \frac{z+1}{3}$; DRs $-2, \frac{p}{2}, 3$
2. Line 2: $\frac{x-1}{-p/3} = \frac{y-1}{2} = \frac{z-2}{-1}$; DRs $-\frac{p}{3}, 2, -1$
3. $\frac{2p}{3} + p - 3 = 0 \Rightarrow \frac{5p}{3} = 3$

✔ $p = \frac{9}{5}$

Solved Example 17

Find the angle between the lines whose DCs satisfy $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$.

(contd.)

1. $n = -(l+m)$, so $l^2 + m^2 = (l+m)^2 \Rightarrow 2lm = 0$
2. $l = 0$: then $n = -m$, DRs $0, 1, -1$. $m = 0$: then $n = -l$, DRs $1, 0, -1$
3. $\cos\theta = \dfrac{|0+0+1|}{\sqrt{2}\sqrt{2}} = \dfrac{1}{2}$

✔ $\theta = \frac{\pi}{3}$

Solved Example 18

Find the angle between any two diagonals of a cube.

1. Cube of side a with O at a corner: diagonals OP (to (a,a,a)) and AQ from $A(a,0,0)$ to $(0,a,a)$
2. DRs: $1,1,1$ and $-1,1,1$
3. $\cos\theta = \dfrac{|-1+1+1|}{\sqrt{3}\sqrt{3}} = \dfrac{1}{3}$

✔ $\theta = \cos^{-1}\left(\frac{1}{3}\right)$



Shivam Bhaiya says

Angle wale sawaal mein **sirf direction** ($\vec{b}$ ya DRs) chahiye, point bilkul use nahi hota. Aur perpendicular ka condition (DRs ka dot $= 0$) board mein "find k" ban ke aata hai: pehle dono lines ko standard form mein laao, phir lagao.

4. Skew Lines and Shortest Distance

Pairs of lines in space

- → **Intersecting:** meet at one point (lie in one plane)
- → **Parallel:** same direction, never meet
- → **Skew:** NOT parallel and do NOT meet (they lie in different planes). Example: a road on a flyover and a road below it going across.

The **shortest distance** (SD) between two lines is the length of the segment perpendicular to **both**.

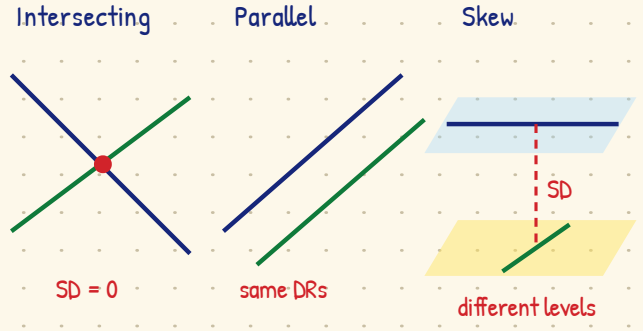


Fig: two lines in space can meet, be parallel, or be skew (like two roads at different levels)

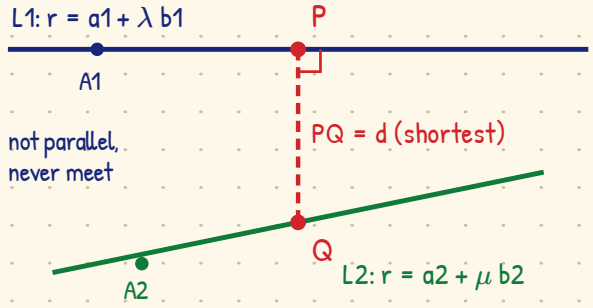


Fig: PQ is perpendicular to both lines; its length is the shortest distance

For $\vec{r} = \vec{a_1} + \lambda\vec{b_1}$ and $\vec{r} = \vec{a_2} + \mu\vec{b_2}$, PQ is along $\vec{b_1} \times \vec{b_2}$, so SD = projection of $\vec{a_2} - \vec{a_1}$ on $\vec{b_1} \times \vec{b_2}$:

$$d = \left| \frac{(\vec{b_1} \times \vec{b_2}) \cdot (\vec{a_2} - \vec{a_1})}{|\vec{b_1} \times \vec{b_2}|} \right|$$

Cartesian form: lines $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$ and $\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$:

$$d = \frac{|D|}{|\vec{b_1} \times \vec{b_2}|}$$

where

$$D = \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}$$

where $|\vec{b_1} \times \vec{b_2}| = \sqrt{(b_1c_2 - b_2c_1)^2 + (c_1a_2 - c_2a_1)^2 + (a_1b_2 - a_2b_1)^2}$.

Parallel lines $\vec{r} = \vec{a_1} + \lambda\vec{b}$, $\vec{r} = \vec{a_2} + \mu\vec{b}$:

$$d = \frac{|\vec{b} \times (\vec{a_2} - \vec{a_1})|}{|\vec{b}|}$$

→ $d = 0 \iff$ the lines intersect (if not parallel).

→ $d \neq 0$ and not parallel $\iff$ skew.

Yaad karo: "Gap dot Cross, upon Cross"

SD $= \dfrac{|\text{Gap} \cdot \text{Cross}|}{|\text{Cross}|}$, jahan **Gap** $= \vec{a_2} - \vec{a_1}$ (dono points ka difference) aur **Cross** $= \vec{b_1} \times \vec{b_2}$ (dono directions ka cross). Parallel lines? Cross banta hi nahi (zero), toh $\frac{|\vec{b} \times \text{Gap}|}{|\vec{b}|}$.

Why the SD formula works

PQ is perpendicular to both lines, so it is along $\vec{n} = \vec{b_1} \times \vec{b_2}$. Take any point A_1 on line 1 and A_2 on line 2. PQ is the projection of $\overrightarrow{A_1A_2} = \vec{a_2} - \vec{a_1}$ on $\vec{n}$: $d = \dfrac{|(\vec{a_2} - \vec{a_1}) \cdot \vec{n}|}{|\vec{n}|}$. That is all.

Step marking: SD of skew lines (5M)

1. Identify $\vec{a_1}, \vec{b_1}, \vec{a_2}, \vec{b_2}$ correctly (1)
2. $\vec{a_2} - \vec{a_1}$ (1/2)
3. $\vec{b_1} \times \vec{b_2}$ by determinant (1 1/2)
4. $|\vec{b_1} \times \vec{b_2}|$ (1/2)
5. Dot product and final d with units (1 1/2)

Solved Example 19

Find the shortest distance between $\vec{r} = (\hat{i} + 2\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = (2\hat{i} + \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 3\hat{k})$.

1. $\vec{a_2} - \vec{a_1} = \hat{i} + \hat{j} - 3\hat{k}$
2. $\vec{b_1} \times \vec{b_2} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 2 & 1 & 3 \end{vmatrix} = -4\hat{i} - \hat{j} + 3\hat{k}$, magnitude $\sqrt{26}$
3. $(\vec{b_1} \times \vec{b_2}) \cdot (\vec{a_2} - \vec{a_1}) = -4 - 1 - 9 = -14$

✔ $d = \frac{14}{\sqrt{26}} = \frac{7\sqrt{26}}{13}$ units

Solved Example 20

Find the shortest distance between the lines $\frac{x-1}{2} = \frac{y+2}{-1} = \frac{z-3}{1}$ and $\frac{x+1}{1} = \frac{y-1}{3} = \frac{z}{-2}$.

1. $(x_1, y_1, z_1) = (1, -2, 3)$, $(x_2, y_2, z_2) = (-1, 1, 0)$; gap $= -2, 3, -3$
2. $\begin{vmatrix} -2 & 3 & -3 \\ 2 & -1 & 1 \\ 1 & 3 & -2 \end{vmatrix} = -2(2-3) - 3(-4-1) - 3(6+1) = 2 + 15 - 21 = -4$
3. Denominator: $\vec{b_1} \times \vec{b_2} = (2-3)\hat{i} - (-4-1)\hat{j} + (6+1)\hat{k} = -\hat{i} + 5\hat{j} + 7\hat{k}$, magnitude $\sqrt{75} = 5\sqrt{3}$

✔ $d = \frac{4}{5\sqrt{3}} = \frac{4\sqrt{3}}{15}$ units

Solved Example 21

Find the distance between the parallel lines $\vec{r} = (\hat{i} - \hat{k}) + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + \mu(\hat{i} + 2\hat{j} + 2\hat{k})$.

1. $\vec{a_2} - \vec{a_1} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$, $|\vec{b}| = 3$
2. $\vec{b} \times (\vec{a_2} - \vec{a_1}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 2 \\ 1 & -1 & 2 \end{vmatrix} = 6\hat{i} - 3\hat{k}$
3. $|6\hat{i} - 3\hat{k}| = \sqrt{45} = 3\sqrt{5}$

✔ $d = \frac{3\sqrt{5}}{3} = \sqrt{5}$ units

Solved Example 22

Show that the lines $\frac{x-1}{1} = \frac{y+1}{1} = \frac{z}{1}$ and $\frac{x-5}{2} = \frac{y}{-1} = \frac{z-4}{2}$ intersect, and find the point of intersection.

(contd.)

1. General points: $(1 + \lambda, -1 + \lambda, \lambda)$ and $(5 + 2\mu, -\mu, 4 + 2\mu)$
2. Equate x and y: $1 + \lambda = 5 + 2\mu$, $-1 + \lambda = -\mu$ $\Rightarrow \mu = -1, \lambda = 2$
3. Check z: $\lambda = 2$ and $4 + 2\mu = 2$ ✓ (equal, so they meet)

✔ They intersect at $(3, 1, 2)$



Meme samjho

Jab poori class skew lines ka SD dekh ke ghabra rahi ho, aur tumhe "**Gap dot Cross, upon Cross**" yaad ho

Asli baat: $d = \frac{|(\vec{b_1} \times \vec{b_2}) \cdot (\vec{a_2} - \vec{a_1})|}{|\vec{b_1} \times \vec{b_2}|}$. If $d = 0$ the lines intersect.

5. Foot of Perpendicular, Image and Distance of a Point

Write the general point Q of the line (λ form)

↓

Find vector PQ = Q minus P

↓

Put (vector PQ) dot (direction of line) $= 0$ and get λ

↓

Put λ back: foot Q; distance $= PQ$; image $= 2Q - P$

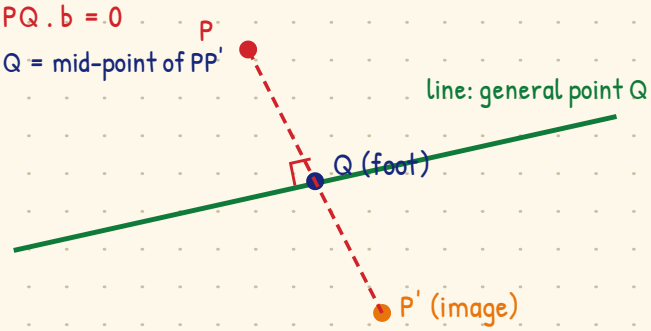


Fig: Q is the foot of the perpendicular from P; image $P' = 2Q - P$

Yaad karo: "G-D-D: General point, Direction se Dot, Done"

Foot of perpendicular ke liye 3 kadam: General point likho, $\overrightarrow{PQ}$ ka Direction $\vec{b}$ se Dot $= 0$, λ nikal ke **Done**. Image chahiye? Foot mid-point hai, toh image $= 2\times$ foot $-$ P.

Quick fact Mid-point of P and its image = foot of perpendicular.

Solved Example 23

Find the foot of the perpendicular from $P(5,4,-3)$ to the line $\frac{x-1}{2}=\frac{y+1}{1}=\frac{z-2}{-1}$. Also find the perpendicular distance and the image of P in the line.

1. General point $Q(1+2\lambda,\ -1+\lambda,\ 2-\lambda)$; $\overrightarrow{PQ}=(2\lambda-4)\hat{i}+(\lambda-5)\hat{j}+(5-\lambda)\hat{k}$
2. $\overrightarrow{PQ}\cdot\vec{b}=2(2\lambda-4)+(\lambda-5)-(5-\lambda)=6\lambda-18=0\Rightarrow\lambda=3$
3. Foot $Q(7,2,-1)$; $PQ=\sqrt{4+4+4}=2\sqrt{3}$
4. Image $=2Q-P=(14-5,\ 4-4,\ -2+3)$

✔ Foot $(7,2,-1)$, distance $2\sqrt{3}$, image $(9,0,1)$

Solved Example 24

Find the equation of the perpendicular from $P(5,4,-3)$ to the line of the previous example.

1. It passes through P and the foot $(7,2,-1)$
2. DRs $=2,-2,2$, i.e. $1,-1,1$

✔ $\frac{x-5}{1}=\frac{y-4}{-1}=\frac{z+3}{1}$

Solved Example 25

Find the equation of the line through $(2,-1,3)$ which is perpendicular to both the lines with DRs $1,2,-2$ and $3,-1,1$.

1. Required direction $\parallel\vec{b_1}\times\vec{b_2}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&2&-2\\3&-1&1\end{vmatrix}$
2. $=(2-2)\hat{i}-(1+6)\hat{j}+(-1-6)\hat{k}=-7\hat{j}-7\hat{k}$, i.e. DRs $0,1,1$

✔ $\vec{r}=(2\hat{i}-\hat{j}+3\hat{k})+\lambda(\hat{j}+\hat{k})$, i.e. $x=2$, $\frac{y+1}{1}=\frac{z-3}{1}$

6. Mixed Board-level Problems

Question says ... so use ...

Words in the question	Use
"direction cosines / ratios"	divide by $\sqrt{a^2+b^2+c^2}$
"angle between the lines", "find k ... perpendicular"	dot of DRs
"shortest distance", "skew"	Gap dot Cross over Cross
"intersect", "point of intersection"	equate general points
"foot", "image", "perpendicular distance from a point"	general point, dot $=0$
"perpendicular to both lines"	direction $=\vec{b_1}\times\vec{b_2}$

Solved Example 26

Show that the line through $(1,2,3)$ and $(3,5,4)$ is parallel to the line through $(0,-1,2)$ and $(4,5,4)$.

1. DRs of first line: $2,3,1$
2. DRs of second line: $4,6,2=2(2,3,1)$

✔ Proportional DRs, so the lines are parallel

Solved Example 27

Show that the line through $(1,0,2)$ and $(3,1,0)$ is perpendicular to the line through $(2,1,1)$ and $(4,3,4)$.

1. DRs: $2,1,-2$ and $2,2,3$
2. $a_1a_2+b_1b_2+c_1c_2=4+2-6=0$

✔ The lines are perpendicular

Solved Example 28

Find the angle between the lines $x=2y=-3z$ and $3x=-y=2z$.

1. $x=2y=-3z\Rightarrow\frac{x}{1}=\frac{y}{1/2}=\frac{z}{-1/3}$; multiply by 6: DRs $6,3,-2$
2. $3x=-y=2z\Rightarrow\frac{x}{1/3}=\frac{y}{-1}=\frac{z}{1/2}$; multiply by 6: DRs $2,-6,3$
3. $\cos\theta=\frac{|12-18-6|}{7\times7}=\frac{12}{49}$

✔ $\theta=\cos^{-1}\left(\frac{12}{49}\right)$

Solved Example 29

Find the Cartesian equation of the line through $(1,0,-2)$ parallel to $\frac{x-2}{3}=\frac{1-y}{2}=\frac{2z+1}{4}$.

1. Standard form: $\frac{x-2}{3}=\frac{y-1}{-2}=\frac{z+\frac{1}{2}}{2}$
2. DRs $3,-2,2$

✔ $\frac{x-1}{3}=\frac{y}{-2}=\frac{z+2}{2}$

Solved Example 30

Find the shortest distance between the lines $\vec{r}=(2+t)\hat{i}+(1-t)\hat{j}+2t\hat{k}$ and $\vec{r}=(1+2s)\hat{i}+(3+s)\hat{j}+(1-s)\hat{k}$.

1. Separate: $\vec{a_1}=2\hat{i}+\hat{j}$, $\vec{b_1}=\hat{i}-\hat{j}+2\hat{k}$; $\vec{a_2}=\hat{i}+3\hat{j}+\hat{k}$, $\vec{b_2}=2\hat{i}+\hat{j}-\hat{k}$
2. $\vec{b_1}\times\vec{b_2}=(1-2)\hat{i}-(-1-4)\hat{j}+(1+2)\hat{k}=-\hat{i}+5\hat{j}+3\hat{k}$, magnitude $\sqrt{35}$
3. Gap $=-\hat{i}+2\hat{j}+\hat{k}$; dot $=1+10+3=14$

✔ $d=\frac{14}{\sqrt{35}}=\frac{2\sqrt{35}}{5}$ units

Solved Example 31

Find the shortest distance between the Z-axis and the line $\vec{r}=(\hat{i}+2\hat{j})+\lambda(\hat{i}-\hat{j}+\hat{k})$.

(contd.)

1. Z-axis: $\vec{r} = \vec{0} + \mu\hat{k}$, so $\vec{a_1} = \vec{0}$, $\vec{b_1} = \hat{k}$; $\vec{a_2} = \hat{i} + 2\hat{j}$, $\vec{b_2} = \hat{i} - \hat{j} + \hat{k}$
2. $\vec{b_1} \times \vec{b_2} = \hat{k} \times \hat{i} - \hat{k} \times \hat{j} = \hat{j} + \hat{i}$, magnitude $\sqrt{2}$
3. $(\hat{i} + \hat{j}) \cdot (\hat{i} + 2\hat{j}) = 3$

✔ $d = \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$ units

Solved Example 32

Prove that the lines $x = ay + b,\ z = cy + d$ and $x = a'y + b',\ z = c'y + d'$ are perpendicular if $aa' + cc' + 1 = 0$.

1. Line 1: $\frac{x-b}{a} = \frac{y}{1} = \frac{z-d}{c}$, DRs $a, 1, c$
2. Line 2: DRs $a', 1, c'$
3. Perpendicular $\iff aa' + 1 + cc' = 0$

✔ Proved

Solved Example 33

Find the foot of the perpendicular from the origin to the line through $(2, 1, 0)$ with direction $\hat{i} - \hat{j} + \hat{k}$.

1. $Q(2 + \lambda,\ 1 - \lambda,\ \lambda)$; $\overrightarrow{OQ} \cdot \vec{b} = (2 + \lambda) - (1 - \lambda) + \lambda = 1 + 3\lambda$
2. $1 + 3\lambda = 0 \Rightarrow \lambda = -\frac{1}{3}$

✔ Foot $\left(\frac{5}{3}, \frac{4}{3}, -\frac{1}{3}\right)$; distance $= \frac{\sqrt{42}}{3}$

Solved Example 34

Find the foot and length of the perpendicular from $P(7, 1, 4)$ to the line $\frac{x-1}{1} = \frac{y+1}{2} = \frac{z}{2}$, and the equation of the perpendicular.

1. $Q(1 + \lambda, -1 + 2\lambda, 2\lambda)$; $\overrightarrow{PQ} = (\lambda - 6,\ 2\lambda - 2,\ 2\lambda - 4)$
2. Dot with $(1, 2, 2)$: $\lambda - 6 + 4\lambda - 4 + 4\lambda - 8 = 9\lambda - 18 = 0 \Rightarrow \lambda = 2$; $Q(3, 3, 4)$
3. $\overrightarrow{PQ} = -4\hat{i} + 2\hat{j}$, length $\sqrt{20} = 2\sqrt{5}$

✔ Foot $(3, 3, 4)$, length $2\sqrt{5}$; perpendicular: $\frac{x-7}{2} = \frac{y-1}{-1}$, $z = 4$

Quick check Distance of (x, y, z) from the X-axis is $\sqrt{y^2 + z^2}$; from the Z-axis $\sqrt{x^2 + y^2}$.

Quick check $\vec{r} = (2\hat{i} - \hat{j}) + \lambda(\hat{i} + \hat{k})$ in Cartesian: $\frac{x-2}{1} = \frac{y+1}{0} = \frac{z}{1}$.

Quick fact $\cos\theta = 1$ for parallel lines, 0 for perpendicular.

Try yourself (8 quick ones, answers below)

1. DCs of a line with DRs $1, -2, 2$
2. DRs of the line through $(1, 0, 2)$ and $(3, -1, 4)$
3. Vector form of $\frac{x-1}{2} = \frac{y+3}{4} = \frac{z}{-1}$
4. Are lines with DRs $2, 1, -1$ and $1, -1, 1$ perpendicular?
5. Angle between lines with DRs $1, 0, 0$ and $0, 0, 1$
6. General point of $\frac{x+1}{3} = \frac{y-2}{-1} = \frac{z}{2}$
7. Are DRs $1, 2, 3$ and $-2, -4, -6$ parallel?
8. DCs of a line at $60°$, $60°$ to X, Y axes, acute to Z

↳ **Answers:** (1) $\pm(\frac{1}{3}, -\frac{2}{3}, \frac{2}{3})$ (2) $2, -1, 2$ (3) $\vec{r} = (\hat{i} - 3\hat{j}) + \lambda(2\hat{i} + 4\hat{j} - \hat{k})$ (4) yes (5) $\frac{\pi}{2}$ (6) $(3\lambda - 1,\ 2 - \lambda,\ 2\lambda)$ (7) yes (8) $\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}}$

7. All Forms Side by Side

Line formulas: vector vs Cartesian

→ **Point + direction:** $\vec{r} = \vec{a} + \lambda\vec{b} \Leftrightarrow$

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

→ **Two points:** $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a}) \Leftrightarrow$

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

→ **Angle:** $\cos\theta = \frac{|\vec{b_1} \cdot \vec{b_2}|}{|\vec{b_1}||\vec{b_2}|}$; Cartesian: same, with $|a_1a_2 + b_1b_2 + c_1c_2|$ on top.

→ **Skew SD:** vector formula of Section 4; Cartesian: $\frac{|D|}{|\vec{b_1} \times \vec{b_2}|}$ with D the 3×3 determinant.

→ **Parallel SD:** $\frac{|\vec{b} \times (\vec{a_2} - \vec{a_1})|}{|\vec{b}|}$, use DRs as $\vec{b}$.

Two lines given → DRs proportional? Parallel: use parallel SD → Else find SD → SD $= 0$: intersect; SD $\neq 0$: skew

Solved Example 35

A, B, C, D are $(1, 0, 2)$, $(3, 3, 1)$, $(0, 1, 1)$, $(4, 7, -1)$. Find the angle between AB and CD.

1. DRs of AB: $2, 3, -1$; DRs of CD: $4, 6, -2$
2. $\frac{4}{2} = \frac{6}{3} = \frac{-2}{-1} = 2$: proportional

✔ AB $\parallel$ CD, so the angle is 0

Quick check Angle with DCs: lines $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0$ and $0, 0, 1$ give $\cos\theta = 0$: $90°$.

Solved Example 36

Find the length of the perpendicular from $P(2,3,4)$ to the X-axis, and its foot.

1. X-axis: general point $Q(\lambda,0,0)$, direction $\hat{i}$
2. $\overrightarrow{PQ}\cdot\hat{i}=\lambda-2=0\Rightarrow\lambda=2$; foot $(2,0,0)$

✔ Length $=\sqrt{0+9+16}=5$ (in general $\sqrt{y^2+z^2}$)

Solved Example 37

The vertices of a triangle are $A(1,2,3)$, $B(3,0,1)$, $C(1,4,-1)$. Find the equation of the median through A.

1. Mid-point of BC: $M\left(\frac{3+1}{2},\frac{0+4}{2},\frac{1-1}{2}\right)=(2,2,0)$
2. DRs of AM: $1,0,-3$

✔ $\frac{x-1}{1}=\frac{y-2}{0}=\frac{z-3}{-3}$, i.e. $y=2,\ \frac{x-1}{1}=\frac{z-3}{-3}$

Formula Box: Angle, SD, foot

- $\cos\theta=\dfrac{|\vec{b_1}\cdot\vec{b_2}|}{|\vec{b_1}||\vec{b_2}|}=\dfrac{|a_1a_2+b_1b_2+c_1c_2|}{\sqrt{\sum a_1^2}\sqrt{\sum a_2^2}}$
- $\perp$: $a_1a_2+b_1b_2+c_1c_2=0$; $\parallel$: $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$
- Skew SD: $\dfrac{|(\vec{b_1}\times\vec{b_2})\cdot(\vec{a_2}-\vec{a_1})|}{|\vec{b_1}\times\vec{b_2}|}$
- Parallel SD: $\dfrac{|\vec{b}\times(\vec{a_2}-\vec{a_1})|}{|\vec{b}|}$
- Foot: $\overrightarrow{PQ}\cdot\vec{b}=0$; image $=2Q-P$

Common mistakes

- Reading DRs from $\frac{3-x}{2}$ as 2: it is -2 (make the coefficient of x equal to $+1$).
- Using the points $(\vec{a_1},\vec{a_2})$ in the angle formula: angle uses only $\vec{b_1},\vec{b_2}$.
- Forgetting the modulus in SD or angle: distance and the acute angle are never negative.
- Using the skew formula for parallel lines: $\vec{b_1}\times\vec{b_2}=\vec{0}$ gives $\frac{0}{0}$; use the parallel formula.
- Intersection: equating only two coordinates and not checking the third.



Shivam Bhaiya says

5 marks ka sawaal is chapter se almost pakka **SD of skew lines** ya **foot of perpendicular + image** hota hai. Dono ka method upar 3-4 steps mein likha hai. Determinant mein calculation slow aur saaf karo, ek sign galat aur poora answer gaya.

Most Asked in Boards

1. DCs/DRs of a line, line through two points (1-2M)
2. Convert Cartesian $\leftrightarrow$ vector form (2M)
3. Find k / p for perpendicular lines (2-3M)
4. Angle between two lines (2-3M)
5. Shortest distance between skew lines, vector or Cartesian (3-5M)
6. Distance between parallel lines (3M)
7. Foot of perpendicular, image of a point, perpendicular distance (5M)
8. Show two lines intersect and find the point (3-5M)

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. The direction cosines of the Y-axis are 1M
(a) $1,0,0$ (b) $0,1,0$ (c) $0,0,1$ (d) $1,1,1$

Ans. (b) It makes $90^\circ,0^\circ,90^\circ$ with the axes: $0,1,0$.

Q2. If a line makes angles α,β,γ with the axes, then $\cos2\alpha+\cos2\beta+\cos2\gamma$ equals 1M Board fav
(a) 2 (b) 1 (c) -2 (d) -1

Ans. (d) $=2(\cos^2\alpha+\cos^2\beta+\cos^2\gamma)-3=2-3=-1$.

Q3. The direction cosines of the line $\frac{x-2}{3}=\frac{y+1}{-2}=\frac{z-5}{6}$ are 1M
(a) $\pm\left(\frac{3}{7},-\frac{2}{7},\frac{6}{7}\right)$ (b) $3,-2,6$ (c) $\frac{2}{7},-\frac{1}{7},\frac{5}{7}$ (d) $\frac{3}{7},\frac{2}{7},\frac{6}{7}$

Ans. (a) DRs $3,-2,6$, $\sqrt{9+4+36}=7$.

Q4. The angle between two lines with direction ratios $1,1,0$ and $0,1,1$ is 1M
(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

Ans. (c) $\cos\theta=\frac{1}{\sqrt{2}\sqrt{2}}=\frac{1}{2}$.

Q5. The lines $\vec{r}=\vec{a_1}+\lambda(2\hat{i}-\hat{j}+3\hat{k})$ and $\vec{r}=\vec{a_2}+\mu(-4\hat{i}+2\hat{j}-6\hat{k})$ (with $\vec{a_2}$ not on the first line) are 1M
(a) parallel (b) perpendicular (c) intersecting (d) skew

Ans. (a) $\vec{b_2}=-2\vec{b_1}$, so parallel (distinct, as $\vec{a_2}$ is not on line 1).

Q6. The vector equation of the line through the origin and $(5,-2,3)$ is 1M
(a) $\vec{r}=5\hat{i}-2\hat{j}+3\hat{k}$ (b) $\vec{r}=\lambda(5\hat{i}-2\hat{j}+3\hat{k})$ (c) $\vec{r}=\hat{i}+\lambda(5\hat{i}-2\hat{j})$ (d) $\vec{r}=\lambda(5\hat{i}+2\hat{j}+3\hat{k})$

Ans. (b) through $\vec{0}$ with direction $5\hat{i}-2\hat{j}+3\hat{k}$.

Q7. The lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-1}{-4}=\frac{y-2}{-6}=\frac{z-3}{-8}$ are 1M HOTS
(a) parallel and distinct (b) coincident (c) skew (d) perpendicular

Ans. (b) Same direction $(-4,-6,-8=-2(2,3,4))$ and the common point $(1,2,3)$: the same line.

Q8. The value of λ for which $\frac{x-1}{2}=\frac{y}{\lambda}=\frac{z+1}{3}$ is perpendicular to $\frac{x}{1}=\frac{y-1}{2}=\frac{z}{-1}$ is 1M

(a) $-\frac{1}{2}$ (b) 1 (c) $\frac{1}{2}$ (d) 2

Ans. (c) $2+2\lambda-3=0\Rightarrow\lambda=\frac{1}{2}$.

Q9. If k,k,k are the direction cosines of a line, then k is 1M

(a) 1 (b) $\frac{1}{3}$ (c) $\pm\frac{1}{\sqrt{3}}$ (d) $\pm\frac{1}{3}$

Ans. (c) $3k^2=1\Rightarrow k=\pm\frac{1}{\sqrt{3}}$.

Q10. The Cartesian equation of the line through $(1,2,3)$ parallel to the Z-axis is 1M

(a) $\frac{x-1}{0}=\frac{y-2}{0}=\frac{z-3}{1}$ (b) $\frac{x-1}{1}=\frac{y-2}{0}=\frac{z-3}{0}$ (c) $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ (d) $x-1=y-2=z-3$

Ans. (a) DRs of Z-axis $0,0,1$.

Q11. The shortest distance between the lines $\vec{r}=\hat{i}+\lambda\hat{j}$ and $\vec{r}=\hat{j}+\mu\hat{k}$ is 1M HOTS

(a) 0 (b) 1 (c) $\sqrt{2}$ (d) $\frac{1}{\sqrt{2}}$

Ans. (b) $\hat{j}\times\hat{k}=\hat{i}$; gap $=-\hat{i}+\hat{j}$; $|(-\hat{i}+\hat{j})\cdot\hat{i}|=1$.

Q12. A line makes equal angles with the three axes. Each angle is 1M

(a) $\frac{\pi}{4}$ (b) $\cos^{-1}\frac{1}{\sqrt{3}}$ (c) $\frac{\pi}{3}$ (d) $\cos^{-1}\frac{1}{3}$

Ans. (b) $l=m=n=\frac{1}{\sqrt{3}}$ (acute), angle $\cos^{-1}\frac{1}{\sqrt{3}}$.

Q13. Which point lies on the line $\frac{x-2}{3}=\frac{y+1}{2}=\frac{z-1}{-1}$? 1M

(a) $(3,2,-1)$ (b) $(2,1,1)$ (c) $(5,1,0)$ (d) $(5,-1,2)$

Ans. (c) $\lambda=1$: $(5,1,0)$.

Q14. The distance of the point $(3,4,5)$ from the Y-axis is 1M

(a) 5 (b) $\sqrt{41}$ (c) $\sqrt{34}$ (d) 4

Ans. (c) Foot $(0,4,0)$; distance $\sqrt{9+25}=\sqrt{34}$.

Q15. The direction ratios of the line $x-1=2y+3=3z$ are 1M Board fav

(a) $1,2,3$ (b) $6,3,2$ (c) $1,-2,3$ (d) $3,2,1$

Ans. (b) $\frac{x-1}{1}=\frac{y+3/2}{1/2}=\frac{z}{1/3}$; multiply by 6: $6,3,2$.

Q16. The direction ratios of the line through $(1,-1,2)$ and $(2,1,-1)$ are 1M

(a) $1,2,-3$ (b) $3,0,1$ (c) $1,-2,3$ (d) $-1,2,3$

Ans. (a) $2-1,\ 1+1,\ -1-2$.

Q17. **A:** The lines $\vec{r}=\vec{a}+\lambda\vec{b}$ and $\vec{r}=\vec{a}+\mu\vec{c}$ $(\vec{b}\nparallel\vec{c})$ intersect. **R:** Both lines pass through the point with position vector $\vec{a}$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. (a) Both true, and the common point $\vec{a}$ explains why they meet.

Q18. The line $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5}$ is parallel to the vector 1M

(a) $2\hat{i}+3\hat{j}+4\hat{k}$ (b) $3\hat{i}+4\hat{j}+5\hat{k}$ (c) $-2\hat{i}-3\hat{j}-4\hat{k}$ (d) $\hat{i}+\hat{j}+\hat{k}$

Ans. (b) its direction is given by the denominators.

Q19. The cosine of the angle between the line $\vec{r}=(\hat{i}+\hat{j})+\lambda(\hat{i}+\hat{j}+\hat{k})$ and the Z-axis is 1M

(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{3}$ (c) 1 (d) 0

Ans. (a) $\frac{(\hat{i}+\hat{j}+\hat{k})\cdot\hat{k}}{\sqrt{3}\times1}=\frac{1}{\sqrt{3}}$.

Q20. Write the Cartesian equation of the line through $(1,2,3)$ parallel to $3\hat{i}+2\hat{j}-2\hat{k}$. 1M

Ans. $\frac{x-1}{3}=\frac{y-2}{2}=\frac{z-3}{-2}$.

Q21. **A:** Lines with DRs $1,2,-1$ and $3,-1,1$ are perpendicular. **R:** Two lines are perpendicular if $a_1a_2+b_1b_2+c_1c_2=0$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. (a) $3-2-1=0$. Both true, R explains A.

Q22. **A:** $1,1,1$ can be the direction cosines of a line. **R:** For direction cosines, $l^2+m^2+n^2=1$. 1M Most asked

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. (d) $1+1+1=3\neq1$, so A is false; R is true.

Q23. **A:** The lines $\vec{r}=(\hat{i}+\hat{j})+\lambda(\hat{i}+2\hat{j}-\hat{k})$ and $\vec{r}=(2\hat{i}+\hat{j})+\mu(-2\hat{i}-4\hat{j}+2\hat{k})$ are parallel. **R:** Their direction vectors satisfy $\vec{b_2}=-2\vec{b_1}$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. (a) Both true and R explains A.

Q24. **A:** Two skew lines can lie in one plane. **R:** Skew lines are neither parallel nor intersecting. 1M

(a) Both true, R explains A (b) Both true, R does not explain A

(c) A true, R false (d) A false, R true

Ans. (d) Lines in one plane either meet or are parallel, so A is false; R is the definition (true).

Very short (2 marks)

Q25. Find the Cartesian equation of the line through $(1,-3,2)$ parallel to $\frac{x-4}{2}=\frac{y+1}{-3}=\frac{z}{5}$. 2M

Ans. Same DRs $2,-3,5$: $\frac{x-1}{2}=\frac{y+3}{-3}=\frac{z-2}{5}$.

Q26. Write the vector equation of the line $\frac{2x-1}{4}=\frac{3-y}{2}=\frac{z+1}{3}$. 2M Most asked

Ans. Standard form: $\frac{x-\frac{1}{2}}{2}=\frac{y-3}{-2}=\frac{z+1}{3}$. $\vec{r}=\left(\frac{1}{2}\hat{i}+3\hat{j}-\hat{k}\right)+\lambda(2\hat{i}-2\hat{j}+3\hat{k})$.

Q27. Find the direction cosines of the line through $(2,-1,3)$ and $(4,1,2)$. 2M

Ans. DRs $2,2,-1$; $\sqrt{4+4+1}=3$. DCs $\pm\left(\frac{2}{3},\frac{2}{3},-\frac{1}{3}\right)$.

Q28. Find the angle which the line $\frac{x-1}{1}=\frac{y-2}{1}=\frac{z-3}{\sqrt{2}}$ makes with the Z-axis. 2M

Ans. DRs $1,1,\sqrt{2}$; $\sqrt{1+1+2}=2$; $n=\cos\gamma=\frac{\sqrt{2}}{2}=\frac{1}{\sqrt{2}}$, so $\gamma=\frac{\pi}{4}$.

Q29. Show that the lines $\frac{x-1}{2}=\frac{y+1}{-3}=\frac{z-2}{4}$ and $\frac{x}{1}=\frac{y-3}{2}=\frac{z+1}{1}$ are perpendicular. 2M

(contd.) $2(1)+(-3)(2)+4(1)=2-6+4=0$, so perpendicular.

Q30. Find the vector equation of the line through $(2,3,-1)$ parallel to the line $x=y=z$. 2M

Ans. $x=y=z$ means $\frac{x}{1}=\frac{y}{1}=\frac{z}{1}$: DRs $1,1,1$. $\vec{r}=(2\hat{i}+3\hat{j}-\hat{k})+\lambda(\hat{i}+\hat{j}+\hat{k})$.

Q31. Check whether the point $(3,1,2)$ lies on the line $\frac{x-1}{1}=\frac{y+1}{1}=\frac{z}{1}$. 2M

Ans. $\frac{3-1}{1}=2$, $\frac{1+1}{1}=2$, $\frac{2}{1}=2$: all equal, so yes (at $\lambda=2$).

Q32. The x-coordinate of a point on the line joining $P(1,3,-2)$ and $Q(4,0,1)$ is 3. Find the point. 2M

Ans. General point $(1+3\lambda,\ 3-3\lambda,\ -2+3\lambda)$; $1+3\lambda=3\Rightarrow\lambda=\frac{2}{3}$. Point $(3,1,0)$.

Q33. Find the direction ratios and direction cosines of the line $6x-2=3y+1=2z-2$. 2M Board fav

Ans. $6\left(x-\frac{1}{3}\right)=3\left(y+\frac{1}{3}\right)=2(z-1)$; divide by 6: $\frac{x-1/3}{1}=\frac{y+1/3}{2}=\frac{z-1}{3}$. DRs $1,2,3$; DCs $\pm\frac{1}{\sqrt{14}}(1,2,3)$.

Q34. Find the vector equation of the line through $(1,2,3)$ parallel to the Z-axis. 2M

Ans. $\vec{r}=(\hat{i}+2\hat{j}+3\hat{k})+\lambda\hat{k}$.

Q35. Find p so that the lines $\frac{x-1}{2}=\frac{y-2}{p}=\frac{z+1}{4}$ and $\frac{x}{1}=\frac{y}{3}=\frac{z}{2}$ are parallel. 2M

Ans. $\frac{2}{1}=\frac{p}{3}=\frac{4}{2}\Rightarrow p=6$.

Q36. Find the vector equation of the line through $A(1,-2,3)$ and $B(4,2,-9)$, and the DCs of AB. 2M

Ans. $\overrightarrow{AB}=3\hat{i}+4\hat{j}-12\hat{k}$, $|\overrightarrow{AB}|=13$. $\vec{r}=(\hat{i}-2\hat{j}+3\hat{k})+\lambda(3\hat{i}+4\hat{j}-12\hat{k})$; DCs $\frac{3}{13},\frac{4}{13},-\frac{12}{13}$.

Q37. Find λ if the line $\frac{x+3}{2}=\frac{y-1}{\lambda}=\frac{z}{5}$ is perpendicular to the line with DRs $3,-4,1$. 2M

Ans. $6-4\lambda+5=0\Rightarrow\lambda=\frac{11}{4}$.

Q38. Find the angle between the lines $2x=3y=-z$ and $4x=-y=-2z$. 2M

Ans. $\frac{x}{1/2}=\frac{y}{1/3}=\frac{z}{-1}$: DRs $3,2,-6$. $\frac{x}{1/4}=\frac{y}{-1}=\frac{z}{-1/2}$: DRs $1,-4,-2$. $\cos\theta=\frac{|3-8+12|}{7\sqrt{21}}=\frac{1}{\sqrt{21}}$.

Q39. If the point $(3,k,m)$ lies on the line $\frac{x-1}{2}=\frac{y+2}{-1}=\frac{z-3}{2}$, find k and m. 2M

Ans. $\frac{3-1}{2}=1=\lambda$. $k=-2-1=-3$, $m=3+2=5$.

Q40. Find the vector equation of the line through $(-1,0,2)$ parallel to the line joining $(1,1,1)$ and $(2,3,0)$. 2M

Ans. Direction $=\hat{i}+2\hat{j}-\hat{k}$. $\vec{r}=(-\hat{i}+2\hat{k})+\lambda(\hat{i}+2\hat{j}-\hat{k})$.

Q41. Show that the line joining the origin to $(1,2,2)$ is perpendicular to the line through $(3,1,0)$ and $(5,0,0)$. 2M

Ans. DRs $1,2,2$ and $2,-1,0$: $2-2+0=0$, so perpendicular.

Q42. Find the direction cosines of the line through the origin and $(2,-6,3)$. 2M

Ans. DRs $2,-6,3$; $\sqrt{4+36+9}=7$; DCs $\pm\left(\frac{2}{7},-\frac{6}{7},\frac{3}{7}\right)$.

Q43. Find the direction cosines of the line $\frac{x+2}{2}=\frac{2y-7}{6}=\frac{5-z}{6}$. 2M Most asked

Ans. $\frac{x+2}{2}=\frac{y-\frac{7}{2}}{3}=\frac{z-5}{-6}$: DRs $2,3,-6$; $\sqrt{4+9+36}=7$; DCs $\pm\left(\frac{2}{7},\frac{3}{7},-\frac{6}{7}\right)$.

Short answer (3 marks)

Q44. Find the shortest distance between $\vec{r}=(\hat{i}-\hat{j})+\lambda(2\hat{i}+\hat{j}+\hat{k})$ and $\vec{r}=(2\hat{j}+\hat{k})+\mu(\hat{i}-\hat{j}+2\hat{k})$. 3M Most asked

Ans. $\vec{a_2}-\vec{a_1}=-\hat{i}+3\hat{j}+\hat{k}$. $\vec{b_1}\times\vec{b_2}=(2+1)\hat{i}-(4-1)\hat{j}+(-2-1)\hat{k}=3\hat{i}-3\hat{j}-3\hat{k}$, magnitude $3\sqrt{3}$. Dot $=-3-9-3=-15$. $d=\frac{15}{3\sqrt{3}}=\frac{5}{\sqrt{3}}$ units.

Q45. Find the distance between the parallel lines $\vec{r}=(\hat{i}+2\hat{j}-\hat{k})+\lambda(2\hat{i}-\hat{j}+2\hat{k})$ and $\vec{r}=(3\hat{i}-\hat{j}+2\hat{k})+\mu(2\hat{i}-\hat{j}+2\hat{k})$. 3M

Ans. $\vec{a_2}-\vec{a_1}=2\hat{i}-3\hat{j}+3\hat{k}$. $\vec{b}\times(\vec{a_2}-\vec{a_1})=(-3+6)\hat{i}-(6-4)\hat{j}+(-6+2)\hat{k}=3\hat{i}-2\hat{j}-4\hat{k}$, magnitude $\sqrt{29}$. $d=\frac{\sqrt{29}}{3}$ units.

Q46. Find the equation of the line through $(1,2,3)$ perpendicular to both lines with DRs $1,-2,3$ and $3,1,-1$. 3M Board fav

Ans. $\vec{b_1}\times\vec{b_2}=(2-3)\hat{i}-(-1-9)\hat{j}+(1+6)\hat{k}=-\hat{i}+10\hat{j}+7\hat{k}$. Line: $\frac{x-1}{-1}=\frac{y-2}{10}=\frac{z-3}{7}$.

Q47. Show that the lines $\vec{r}=(\hat{i}+\hat{k})+\lambda(\hat{i}+2\hat{j}-\hat{k})$ and $\vec{r}=(2\hat{i}+5\hat{j}+\hat{k})+\mu(\hat{i}-\hat{j}-2\hat{k})$ intersect. Find the point. 3M

Ans. Points: $(1+\lambda,2\lambda,1-\lambda)$, $(2+\mu,5-\mu,1-2\mu)$. x: $\lambda=1+\mu$; y: $2\lambda=5-\mu\Rightarrow 2+2\mu=5-\mu\Rightarrow\mu=1,\lambda=2$. z: $1-2=-1$ and $1-2=-1$ ✓. Point $(3,4,-1)$.

Q48. Find the angle between the lines $\vec{r}=(\hat{i}+\hat{j})+\lambda(2\hat{i}+3\hat{j}+6\hat{k})$ and $\vec{r}=(2\hat{i}-\hat{k})+\mu(\hat{i}+2\hat{j}+2\hat{k})$. 3M

Ans. $\vec{b_1}\cdot\vec{b_2}=2+6+12=20$, $|\vec{b_1}|=7$, $|\vec{b_2}|=3$. $\cos\theta=\frac{20}{21}$, $\theta=\cos^{-1}\frac{20}{21}$.

Q49. Find k so that the lines $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{1}$ and $\frac{x-2}{1}=\frac{y-k}{1}=\frac{z-1}{-1}$ intersect. Also find the point. 3M HOTS

Ans. Points: $(1+s,2+2s,3+s)$ and $(2+t,k+t,1-t)$. x: $s=1+t$; z: $3+s=1-t\Rightarrow 4+t=1-t\Rightarrow t=-\frac{3}{2}$, $s=-\frac{1}{2}$. y: $2+2s=1=k+t\Rightarrow k=\frac{5}{2}$. Point $\left(\frac{1}{2},1,\frac{5}{2}\right)$.

Q50. Find the shortest distance between $\vec{r}=(\hat{i}+\hat{j})+\lambda(\hat{i}+2\hat{j}+3\hat{k})$ and $\vec{r}=(2\hat{i}+3\hat{j}+\hat{k})+\mu(2\hat{i}+3\hat{j}+4\hat{k})$. Do the lines meet? 3M

(contd.) $\vec{a_2}-\vec{a_1}=\hat{i}+2\hat{j}+\hat{k}$; $\vec{b_1}\times\vec{b_2}=(8-9)\hat{i}-(4-6)\hat{j}+(3-4)\hat{k}=-\hat{i}+2\hat{j}-\hat{k}$, magnitude $\sqrt{6}$. Dot $=-1+4-1=2$. $d=\frac{2}{\sqrt{6}}=\frac{\sqrt{6}}{3}\neq 0$: skew, they do not meet.

Q51. Using the Cartesian SD formula, show that the lines $\frac{x-1}{1}=\frac{y-2}{-1}=\frac{z-3}{2}$ and $\frac{x-1}{1}=\frac{y}{1}=\frac{z-4}{1}$ intersect, and find the point. 3M

Ans. $D=\begin{vmatrix}0&-2&1\\1&-1&2\\1&1&1\end{vmatrix}=0-(-2)(1-2)+1(1+1)=0$, so SD $=0$ and (not parallel) they intersect. Points $(1+s,2-s,3+2s)=(1+t,t,4+t)$: $s=t$, $2-s=s\Rightarrow s=1$; z: $5=5$ ✓. Point $(2,1,5)$.

Q52. Find the coordinates of the foot of the perpendicular from $P(3,1,-3)$ to the line $\vec{r}=(\hat{i}-\hat{j})+\lambda(\hat{i}+2\hat{j}+2\hat{k})$, and the distance. 3M

Ans. $Q(1+\lambda,-1+2\lambda,2\lambda)$; $\overrightarrow{PQ}=(\lambda-2,2\lambda-2,2\lambda+3)$; dot with $(1,2,2)$: $\lambda-2+4\lambda-4+4\lambda+6=9\lambda=0\Rightarrow\lambda=0$. Foot $(1,-1,0)$; distance $\sqrt{4+4+9}=\sqrt{17}$.

Q53. Show that the lines $\vec{r}=(\hat{i}+\hat{j})+\lambda(2\hat{i}-\hat{j}+\hat{k})$ and $\vec{r}=(2\hat{i}+\hat{j}-\hat{k})+\mu(3\hat{i}-4\hat{j}+2\hat{k})$ are skew, and find the shortest distance. 3M Must do

Ans. Not parallel ($\frac{2}{3}\neq\frac{-1}{-4}$). $\vec{b_1}\times\vec{b_2}=(-2+4)\hat{i}-(4-3)\hat{j}+(-8+3)\hat{k}=2\hat{i}-\hat{j}-5\hat{k}$, magnitude $\sqrt{30}$. Gap $=\hat{i}-\hat{k}$; dot $=2+5=7$. $d=\frac{7}{\sqrt{30}}\neq 0$, so skew.

Q54. Find the equation of the line through $(2,1,3)$ perpendicular to the lines with DRs $1,2,3$ and $-3,2,5$. 3M

Ans. $\vec{b_1}\times\vec{b_2}=(10-6)\hat{i}-(5+9)\hat{j}+(2+6)\hat{k}=4\hat{i}-14\hat{j}+8\hat{k}\parallel 2\hat{i}-7\hat{j}+4\hat{k}$. Line: $\frac{x-2}{2}=\frac{y-1}{-7}=\frac{z-3}{4}$.

Q55. Find the shortest distance between $\frac{x+1}{1}=\frac{y-2}{-1}=\frac{z-1}{2}$ and $\frac{x-2}{2}=\frac{y}{1}=\frac{z+1}{-1}$. 3M

Ans. Gap $=3,-2,-2$. $\vec{b_1}\times\vec{b_2}=(1-2)\hat{i}-(-1-4)\hat{j}+(1+2)\hat{k}=-\hat{i}+5\hat{j}+3\hat{k}$, magnitude $\sqrt{35}$. Dot $=-3-10-6=-19$. $d=\frac{19}{\sqrt{35}}$ units.

Q56. Lines L_1: $x=2$, $\frac{y}{4-\alpha}=\frac{z}{-1}$ and L_2: $x=\alpha$, $\frac{y}{-2}=\frac{z}{3-\alpha}$ are perpendicular. Find α. 3M HOTS

Ans. DRs: L_1: $0,4-\alpha,-1$; L_2: $0,-2,3-\alpha$. $0-2(4-\alpha)-(3-\alpha)=0\Rightarrow 3\alpha-11=0\Rightarrow\alpha=\frac{11}{3}$.

Q57. Find the vector and Cartesian equations of the line through $A(2,-3,1)$ and $B(5,1,-11)$. Also find AB. 3M

Ans. Direction $\overrightarrow{AB}=3\hat{i}+4\hat{j}-12\hat{k}$. $\vec{r}=(2\hat{i}-3\hat{j}+\hat{k})+\lambda(3\hat{i}+4\hat{j}-12\hat{k})$; $\frac{x-2}{3}=\frac{y+3}{4}=\frac{z-1}{-12}$. $AB=\sqrt{9+16+144}=13$.

Q58. Find the equation of the line through $(2,3,-4)$ perpendicular to both the X-axis and the line $\frac{x}{2}=\frac{y}{3}=\frac{z}{-1}$. 3M

(contd.) Direction $\parallel\hat{i}\times(2\hat{i}+3\hat{j}-\hat{k})=3(\hat{i}\times\hat{j})-(\hat{i}\times\hat{k})=3\hat{k}+\hat{j}$. Line: $x=2$, $\frac{y-3}{1}=\frac{z+4}{3}$; vector: $\vec{r}=(2\hat{i}+3\hat{j}-4\hat{k})+\lambda(\hat{j}+3\hat{k})$.

Q59. Find the foot of the perpendicular from $A(5,0,1)$ to the line joining $B(2,1,0)$ and $C(4,3,2)$, and the perpendicular distance. 3M Board fav

Ans. Line BC: $(2+\lambda,1+\lambda,\lambda)$ (direction $1,1,1$). $\overrightarrow{AQ}=(\lambda-3,1+\lambda,\lambda-1)$; dot with $(1,1,1)$: $3\lambda-3=0\Rightarrow\lambda=1$. Foot $(3,2,1)$; distance $\sqrt{4+4+0}=2\sqrt{2}$.

Q60. Find the shortest distance between $\vec{r}=(2\hat{i}+\hat{j}+\hat{k})+\lambda(\hat{i}-2\hat{j}+2\hat{k})$ and $\vec{r}=(-\hat{i}+2\hat{k})+\mu(2\hat{i}+\hat{j}-2\hat{k})$. 3M

Ans. Gap $=-3\hat{i}-\hat{j}+\hat{k}$. $\vec{b_1}\times\vec{b_2}=(4-2)\hat{i}-(-2-4)\hat{j}+(1+4)\hat{k}=2\hat{i}+6\hat{j}+5\hat{k}$, magnitude $\sqrt{65}$. Dot $=-6-6+5=-7$. $d=\frac{7}{\sqrt{65}}$ units.

Long answer (5 marks)

Q61. Find the foot of the perpendicular from $P(5,5,7)$ to the line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$. Hence find the image of P in the line and the perpendicular distance. 5M Must do

Ans. $Q(\lambda,1+2\lambda,2+3\lambda)$; $\overrightarrow{PQ}=(\lambda-5)\hat{i}+(2\lambda-4)\hat{j}+(3\lambda-5)\hat{k}$. $\overrightarrow{PQ}\cdot\vec{b}=\lambda-5+4\lambda-8+9\lambda-15=14\lambda-28=0\Rightarrow\lambda=2$. Foot $(2,5,8)$. Image $=2Q-P=(-1,5,9)$. Distance $PQ=\sqrt{9+0+1}=\sqrt{10}$.

Q62. For the lines $\frac{x-1}{2}=\frac{y+2}{-1}=\frac{z-3}{1}$ and $\frac{x+1}{1}=\frac{y-1}{3}=\frac{z}{-2}$, find (i) the angle between them (ii) the shortest distance. Are they skew? 5M

Ans. (i) $\vec{b_1}\cdot\vec{b_2}=2-3-2=-3$, $|\vec{b_1}|=\sqrt{6}$, $|\vec{b_2}|=\sqrt{14}$; $\cos\theta=\frac{3}{\sqrt{84}}=\frac{3}{2\sqrt{21}}$. (ii) From Solved Example above, $d=\frac{4}{5\sqrt{3}}=\frac{4\sqrt{3}}{15}$. Not parallel and $d\neq 0$, so skew.

Q63. Find the vector equation of the line through $(2,-1,3)$ perpendicular to the lines $\vec{r}=(\hat{i}+\hat{k})+\lambda(\hat{i}+2\hat{j}-2\hat{k})$ and $\vec{r}=(2\hat{j}-\hat{k})+\mu(3\hat{i}-\hat{j}+\hat{k})$. Also find the angle between the two given lines. 5M

Ans. $\vec{b_1}\times\vec{b_2}=-7\hat{j}-7\hat{k}\parallel\hat{j}+\hat{k}$. Line: $\vec{r}=(2\hat{i}-\hat{j}+3\hat{k})+t(\hat{j}+\hat{k})$. Angle: $\vec{b_1}\cdot\vec{b_2}=3-2-2=-1$, $|\vec{b_1}|=3$, $|\vec{b_2}|=\sqrt{11}$; $\cos\theta=\frac{1}{3\sqrt{11}}$.

Q64. Find the foot of the perpendicular and the perpendicular distance from $P(6,2,2)$ to the line through $A(1,1,1)$ with direction ratios $2,-2,1$. Also find the image of P. 5M Board fav

Ans. Line: $(1+2\lambda,1-2\lambda,1+\lambda)$. $\overrightarrow{PQ}=(2\lambda-5)\hat{i}+(-2\lambda-1)\hat{j}+(\lambda-1)\hat{k}$. Dot with $(2,-2,1)$: $4\lambda-10+4\lambda+2+\lambda-1=9\lambda-9=0\Rightarrow\lambda=1$. Foot $(3,-1,2)$, distance $\sqrt{9+9+0}=3\sqrt{2}$, image $(0,-4,2)$.

Q66. Find the points on the line $\frac{x-1}{2}=\frac{y+1}{2}=\frac{z-2}{1}$ at a distance of $\sqrt{3}$ units from $A(4,2,2)$. 5M

(contd.) General point $(1+2\lambda,-1+2\lambda,2+\lambda)$. Distance^2 $=(2\lambda-3)^2+(2\lambda-3)^2+\lambda^2=9\lambda^2-24\lambda+18=3\Rightarrow 3\lambda^2-8\lambda+5=0\Rightarrow\lambda=1$ or $\frac{5}{3}$. Points $(3,1,3)$ and $\left(\frac{13}{3},\frac{7}{3},\frac{11}{3}\right)$.

Q65. Find the image of $P(4,3,4)$ in the line $\frac{x-1}{1}=\frac{y-2}{-1}=\frac{z-3}{1}$. Also write the equation of the line joining P and its image. 5M

Ans. $Q(1+\lambda,2-\lambda,3+\lambda)$; $\overrightarrow{PQ}=(\lambda-3,-1-\lambda,\lambda-1)$; dot with $(1,-1,1)$: $\lambda-3+1+\lambda+\lambda-1=3\lambda-3=0\Rightarrow\lambda=1$. Foot $(2,1,4)$; image $(0,-1,4)$; distance $2\sqrt{2}$. Line through $(4,3,4)$ and $(0,-1,4)$: DRs $1,1,0$: $\frac{x-4}{1}=\frac{y-3}{1}$, $z=4$.

Case based (4 marks)

Q67. Two flight paths. Plane A flies along the line through $(1,2,3)$ with DRs $2,-1,2$; plane B flies along the line through $(3,0,1)$ with DRs $1,2,2$ (units in km). (i) Find the DCs of A's path. (1) (ii) Find the cosine of the angle between the paths. (1) (iii) Find the shortest distance between the paths. (2) 4M

Ans. (i) $\frac{2}{3},-\frac{1}{3},\frac{2}{3}$. (ii) $\frac{2-2+4}{3\times3}=\frac{4}{9}$. (iii) $\vec{b_1}\times\vec{b_2}=(-2-4)\hat{i}-(4-2)\hat{j}+(4+1)\hat{k}=-6\hat{i}-2\hat{j}+5\hat{k}$, magnitude $\sqrt{65}$; gap $=2\hat{i}-2\hat{j}-2\hat{k}$; dot $=-12+4-10=-18$; $d=\frac{18}{\sqrt{65}}$ km.

Q68. Tower and road. A straight road runs along $\frac{x-1}{2}=\frac{y+1}{1}=\frac{z-2}{-1}$ and a mobile tower stands at $P(3,6,1)$. A cable is to be laid from P to the nearest point of the road. (i) Write the general point of the road. (1) (ii) Find λ for the nearest point. (1) (iii) Find the nearest point and the cable length. (2) 4M

Ans. (i) $Q(1+2\lambda,-1+\lambda,2-\lambda)$. (ii) $\overrightarrow{PQ}=(2\lambda-2,\lambda-7,1-\lambda)$; dot with $(2,1,-1)$: $4\lambda-4+\lambda-7-1+\lambda=6\lambda-12=0\Rightarrow\lambda=2$. (iii) $Q(5,1,0)$; length $=\sqrt{4+25+1}=\sqrt{30}$ units.

Q69. Laser show. A laser beam passes through $A(2,-1,3)$ and $B(4,3,-1)$. (i) Find the DRs of the beam. (1) (ii) Write its Cartesian equation. (1) (iii) Does it hit a mirror placed at $C(5,5,-3)$? Find the angle between this beam and a second beam with DRs $2,2,1$. (2) 4M

Ans. (i) $2,4,-4$, i.e. $1,2,-2$. (ii) $\frac{x-2}{1}=\frac{y+1}{2}=\frac{z-3}{-2}$. (iii) $\frac{5-2}{1}=\frac{5+1}{2}=\frac{-3-3}{-2}=3$: yes. $\cos\theta=\frac{|2+4-2|}{3\times3}=\frac{4}{9}$, $\theta=\cos^{-1}\frac{4}{9}$.

Bonus practice (mixed)

Q70. If a line has direction ratios $2,-1,2$, its direction cosines are 1M Must do

(a) $2,-1,2$ (b) $\frac{2}{9},-\frac{1}{9},\frac{2}{9}$ (c) $\pm\left(\frac{2}{3},-\frac{1}{3},\frac{2}{3}\right)$ (d) $\frac{2}{5},-\frac{1}{5},\frac{2}{5}$

Ans. (c) divide by $\sqrt{4+1+4}=3$: $\pm\left(\frac{2}{3},-\frac{1}{3},\frac{2}{3}\right)$.

Q71. Find the angle between two lines with direction ratios $1,1,0$ and $0,1,1$. 2M Board fav

Ans. $\cos\theta=\frac{|0+1+0|}{\sqrt{2}\sqrt{2}}=\frac{1}{2}$, so $\theta=\frac{\pi}{3}$.

Q72. Find the Cartesian equation of the line through $(2,-1,4)$ and $(5,1,3)$. 2M Most asked

Ans. DRs $5-2,\ 1+1,\ 3-4=3,2,-1$. Line: $\frac{x-2}{3}=\frac{y+1}{2}=\frac{z-4}{-1}$.

Q73. Find λ if the lines $\frac{x-1}{2}=\frac{y-2}{\lambda}=\frac{z+1}{3}$ and $\frac{x+2}{1}=\frac{y}{2}=\frac{z-3}{-2}$ are perpendicular. 2M Must do

Ans. $2(1)+\lambda(2)+3(-2)=0\Rightarrow2\lambda=4\Rightarrow\lambda=2$.

Q74. Find the shortest distance between the lines $\vec{r}=(\hat{i}+\hat{j})+\lambda(2\hat{i}-\hat{j}+\hat{k})$ and $\vec{r}=(2\hat{i}+\hat{j}-\hat{k})+\mu(4\hat{i}-2\hat{j}+2\hat{k})$. 3M Board fav

Ans. Parallel, since $4\hat{i}-2\hat{j}+2\hat{k}=2\vec{b}$ with $\vec{b}=2\hat{i}-\hat{j}+\hat{k}$. $\vec{a_2}-\vec{a_1}=\hat{i}-\hat{k}$, $\vec{b}\times(\vec{a_2}-\vec{a_1})=\hat{i}+3\hat{j}+\hat{k}$, magnitude $\sqrt{11}$; $|\vec{b}|=\sqrt{6}$. $d=\frac{\sqrt{11}}{\sqrt{6}}=\frac{\sqrt{66}}{6}$ units.

Ek nazar mein: 3D Geometry (lines)

Type of sawal	Method	Marks
DCs, DRs	divide by root of squares	1-2
Equation of line	point + direction	2
Angle, find k	dot of DRs	2-3
Skew SD	Gap dot Cross / Cross	3-5
Foot, image	general point, dot $=0$	5

Last minute: 8 one-liners

1. $l^2+m^2+n^2=1$; DRs are any multiple.
2. Standard form first: coefficients of x,y,z must be $+1$.
3. $\vec{r}=\vec{a}+\lambda\vec{b}$: $\vec{a}$ point, $\vec{b}$ direction.
4. Angle uses only directions, with modulus.
5. $\perp$: dot of DRs $=0$; $\parallel$: DRs proportional.
6. Skew SD: Gap dot Cross over Cross; $d=0$ means they meet.
7. Parallel SD: $|\vec{b}\times\text{Gap}|/|\vec{b}|$.
8. Foot: general point, $\overrightarrow{PQ}\cdot\vec{b}=0$; image $=2Q-P$.

Apne notes / galtiyan yahan likho

Most asked: graphical method (5M), formulate + solve word problem

Linear Programming

Is chapter mein kya hai (2026-27)

- → LPP terms, mathematical formulation, **graphical method** (corner point method) for problems in two variables with bounded and unbounded feasible regions.
- → Board paper: one 5-mark question (or a 4-mark case study) plus 1-mark MCQs. Unit V carries 5 marks.

1. Basic Terms

A **Linear Programming Problem (LPP)** asks for the maximum or minimum value of a linear function of variables x, y when the variables obey some **linear inequalities**.

Key terms

- → **Objective function:** $Z = ax + by$ (a, b constants), to be maximised or minimised.
- → **Decision variables:** x and y (e.g. number of chairs and tables).
- → **Constraints:** the linear inequalities, like $x + 2y \leq 40$.
- → **Non-negative constraints:** $x \geq 0, y \geq 0$.
- → **Feasible region:** common region of all constraints including $x, y \geq 0$. Every point in it is a **feasible solution**.
- → **Infeasible solution:** a point outside the feasible region.
- → **Optimal solution:** a feasible point giving the max (or min) value of Z.
- → **Bounded region:** can be enclosed in a circle; otherwise **unbounded**.

2. Drawing a Constraint (Half-Plane)

1. Change $\leq$ or $\geq$ to $=$ and draw the line using its two intercepts.
2. **Test point** $(0, 0)$ (if the line does not pass through it): if $(0, 0)$ satisfies the inequality, shade the **origin side**, else the other side.
3. Do this for every constraint; the region common to all, in the first quadrant, is the feasible region.

Quick fact Corner = meeting point of two boundary lines.

Quick fact A feasible region of an LPP is always a convex set.

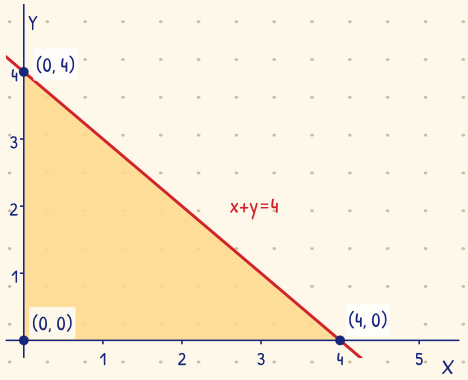


Fig: $x + y \leq 4$: origin satisfies it, so the origin side is shaded

Shading: two quick examples

Line	$x = 0 \Rightarrow$	$y = 0 \Rightarrow$	$(0, 0)$ test
$x + y \leq 4$	$(0, 4)$	$(4, 0)$	$0 \leq 4$ true: origin side
$x + 2y \geq 8$	$(0, 4)$	$(8, 0)$	$0 \geq 8$ false: away side

Yaad karo: "Origin se poocho"

Line kis taraf shade karni hai? **Origin se poocho:** $(0, 0)$ daal ke dekho. Sach nikle to origin wali side, jhooth nikle to doosri side. $\leq$ wali lines mostly origin ki taraf, $\geq$ wali door.

3. Corner Point Method

Theorems (statement only)

- → **Theorem 1:** If the feasible region R is a convex polygon and $Z = ax + by$ has an optimal value, it occurs at a corner point of R.
- → **Theorem 2:** If R is **bounded**, Z has **both** a maximum and a minimum on R, each at a corner point.
- → If R is **unbounded**, a max or min may **not exist**; if it exists, it is at a corner point.

Graph the constraints → shade feasible region → find corner points → table of Z at corners → bounded: answer; unbounded: half-plane check

Quick fact $x \geq 0, y \geq 0$ keeps the region in the first quadrant.

Quick fact A point on a boundary line is still feasible (for $\leq$ or $\geq$).

Unbounded region: half-plane check

→ Let M = largest value of Z at the corners. Draw $ax + by > M$. If this open half-plane has **no point in common** with R, then M is the maximum; otherwise Z has **no maximum.**

→ Let m = smallest value. Draw $ax + by < m$. No common point: m is the minimum; otherwise **no minimum.**

Special cases

→ **Multiple optimal solutions:** if Z has the same optimal value at two adjacent corners, every point on the segment joining them is optimal.

→ **No feasible region** (constraints contradict): the LPP has no solution.

Formula Box: LPP

→ $Z = ax + by$ (objective), constraints $a_1x + b_1y \le c_1$ etc., $x, y \ge 0$

→ Intercepts of $ax + by = c$: $\left(\frac{c}{a}, 0\right)$ and $\left(0, \frac{c}{b}\right)$

→ Corner = intersection of two boundary lines (solve them together)

→ Bounded: max and min both at corners

→ Unbounded max check: $ax + by > M$ must not meet R

→ Unbounded min check: $ax + by < m$ must not meet R

→ $Z = px + qy$ equal at corners (x_1, y_1), (x_2, y_2): $px_1 + qy_1 = px_2 + qy_2$

Yaad karo: "Corner pe hi khazana"

Answer hamesha **corner point** pe milta hai. Poore region mein dhoondhne ki zaroorat nahi: saare corners ki list banao, har corner pe Z nikalo, sabse bada = max, sabse chhota = min.

4. Solved Examples (Graphical Method)

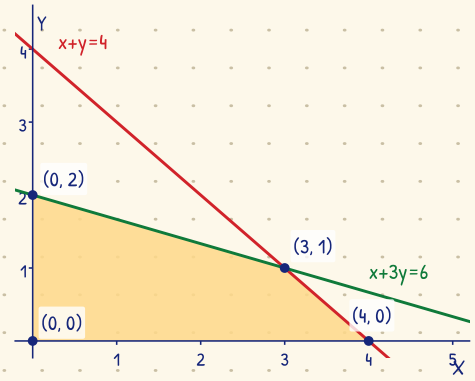


Fig (Example 1): feasible region OABC

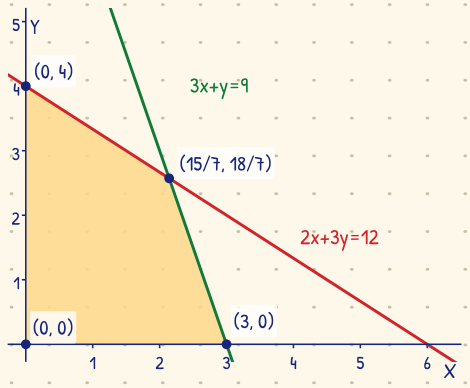


Fig (Example 2): one corner has fractions

Solved Example 1

Maximise $Z = 2x + 5y$ subject to $x + y \le 4$, $x + 3y \le 6$, $x \ge 0$, $y \ge 0$.

1. Lines: $x + y = 4$ through $(4, 0), (0, 4)$; $x + 3y = 6$ through $(6, 0), (0, 2)$. Origin satisfies both: shade towards origin.
2. Intersection: subtract, $2y = 2 \Rightarrow y = 1$, $x = 3$: corner $(3, 1)$.
3. Corners: $O(0, 0)$, $A(4, 0)$, $B(3, 1)$, $C(0, 2)$ (bounded region).

Corner point	$Z = 2x + 5y$
$(0, 0)$	0
$(4, 0)$	8
$(3, 1)$	11 **max**
$(0, 2)$	10

✔ Max $Z = 11$ at $(3, 1)$

Solved Example 2

Maximise $Z = 4x + 5y$ subject to $2x + 3y \le 12$, $3x + y \le 9$, $x, y \ge 0$.

1. Intercepts: $2x + 3y = 12$: $(6, 0), (0, 4)$; $3x + y = 9$: $(3, 0), (0, 9)$.
2. Solve together: $y = 9 - 3x \Rightarrow 2x + 27 - 9x = 12 \Rightarrow x = \frac{15}{7}$, $y = \frac{18}{7}$.
3. Corners: $(0, 0)$, $(3, 0)$, $\left(\frac{15}{7}, \frac{18}{7}\right)$, $(0, 4)$.

Corner point	$Z = 4x + 5y$
$(0, 0)$	0
$(3, 0)$	12
$\left(\frac{15}{7}, \frac{18}{7}\right)$	$\frac{150}{7}$ **max**
$(0, 4)$	20

✔ Max $Z = \frac{150}{7}$ at $\left(\frac{15}{7}, \frac{18}{7}\right)$

Quick fact Max of Z on a bounded region always exists.

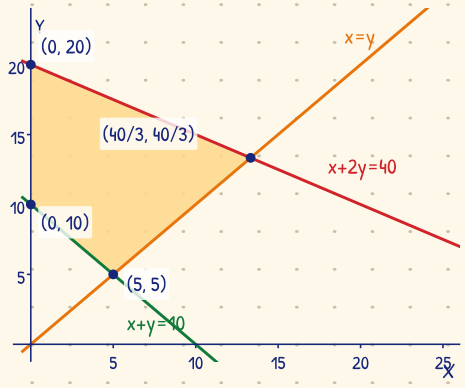


Fig (Example 3): region between $x = y$ and the two lines

Solved Example 3

Find the max and min of $Z = x + 2y$ subject to $x + 2y \leq 40,\ x + y \geq 10,\ x \leq y,\ x, y \geq 0$.

1. $x \leq y$: region above the line $x = y$ (test $(0, 1)$: $0 \leq 1$ true).
2. Corners: $x + y = 10$ and $x = y$ meet at $(5, 5)$; $x + 2y = 40$ and $x = y$ at $\left(\frac{40}{3}, \frac{40}{3}\right)$; on the y-axis $(0, 10)$ and $(0, 20)$.

Corner point	$Z = x + 2y$
$(0, 10)$	20
$(5, 5)$	15 min
$\left(\frac{40}{3}, \frac{40}{3}\right)$	40 max
$(0, 20)$	40 max

✔ Min $Z = 15$ at $(5, 5)$. Max $Z = 40$ at two corners $(0, 20)$ and $\left(\frac{40}{3}, \frac{40}{3}\right)$, so at every point of the segment joining them (multiple optimal solutions)

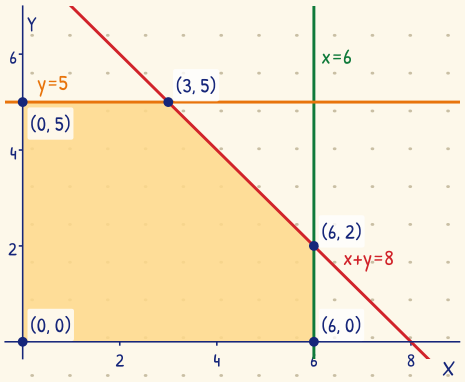


Fig (Example 4): three constraints give 5 corners

Solved Example 4

Maximise $Z = 5x + 3y$ subject to $x + y \leq 8,\ x \leq 6,\ y \leq 5,\ x, y \geq 0$.

1. $x \leq 6$: left of the vertical line $x = 6$; $y \leq 5$: below $y = 5$.
2. $x + y = 8$ meets $x = 6$ at $(6, 2)$ and $y = 5$ at $(3, 5)$.
3. Corners: $(0, 0)$, $(6, 0)$, $(6, 2)$, $(3, 5)$, $(0, 5)$.

Corner point	$Z = 5x + 3y$
$(0, 0)$	0
$(6, 0)$	30
$(6, 2)$	36 max
$(3, 5)$	30
$(0, 5)$	15

✔ Max $Z = 36$ at $(6, 2)$

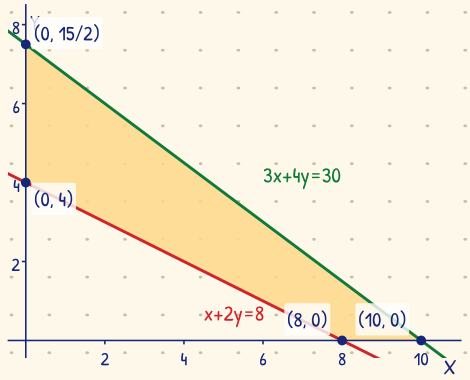


Fig (Example 5): bounded region between two parallel-ish lines

Solved Example 5

Find the min and max of $Z = 200x + 500y$ subject to $x + 2y \geq 8,\ 3x + 4y \leq 30,\ x, y \geq 0$.

1. $x + 2y \geq 8$: away from origin; $3x + 4y \leq 30$: towards origin.
2. The lines do not meet in the first quadrant ($y = -3$ there), so the corners are the axis points.
3. Corners: $(8, 0)$, $(10, 0)$, $\left(0, \frac{15}{2}\right)$, $(0, 4)$ (bounded).

Corner point	$Z = 200x + 500y$
$(8, 0)$	1600 min
$(10, 0)$	2000
$\left(0, \frac{15}{2}\right)$	3750 max
$(0, 4)$	2000

✔ Min $Z = 1600$ at $(8, 0)$; max $Z = 3750$ at $\left(0, \frac{15}{2}\right)$



Meme samjho

Jab LPP ka graph bana aur region **bounded** nikla, sirf 4 corner points: aaj to maza hi aa jayega

Asli baat: Bounded feasible region: Z has both a max and a min, and both occur at corner points (no half-plane check needed).

Quick fact Graph paper pe scale same rakho dono axes pe.

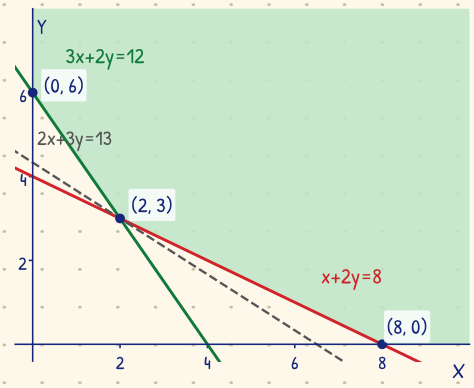


Fig (Example 6): unbounded region (green); dashed line $2x + 3y = 13$

Solved Example 6

Minimise $Z = 2x + 3y$ subject to $x + 2y \geq 8$, $3x + 2y \geq 12$, $x, y \geq 0$.

1. Both are $\geq$: shade **away** from origin. Region is unbounded.
2. Intersection: subtract, $2x = 4 \Rightarrow x = 2, y = 3$.
3. Corners: $(8, 0)$, $(2, 3)$, $(0, 6)$.

Corner point	$Z = 2x + 3y$
$(8, 0)$	16
$(2, 3)$	13 **min**
$(0, 6)$	18

↳Region unbounded hai, isliye half-plane check zaroori. Max nahi milega: Z jitna chaho bada ho sakta hai.

✔ Smallest value 13. Draw $2x + 3y < 13$: this open half-plane has **no point** in common with the region. Hence min $Z = 13$ at $(2, 3)$

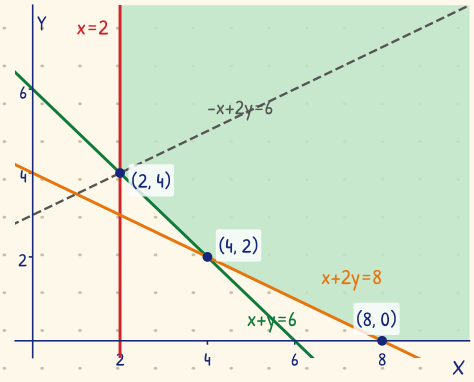


Fig (Example 7): the half-plane $-x + 2y > 6$ meets the region

Solved Example 7

Maximise $Z = -x + 2y$ subject to $x \geq 2$, $x + y \geq 6$, $x + 2y \geq 8$, $y \geq 0$.

Quick fact Strict inequality ($<$) is used only in the half-plane check.

1. Corners: $x = 2$ and $x + y = 6$: $(2, 4)$; $x + y = 6$ and $x + 2y = 8$: $(4, 2)$; $x + 2y = 8$ and $y = 0$: $(8, 0)$.
2. Region is unbounded (it goes on upward and to the right).

Corner point	$Z = -x + 2y$
$(2, 4)$	6 **max**
$(4, 2)$	0
$(8, 0)$	-8

↳Corner pe 6 aaya matlab max 6, aisa nahi! Unbounded region mein pehle check, phir answer.

✔ Largest corner value is 6, but the half-plane $-x + 2y > 6$ does meet the region (e.g. $(2, 5)$ gives $Z = 8$). So Z has **no maximum**

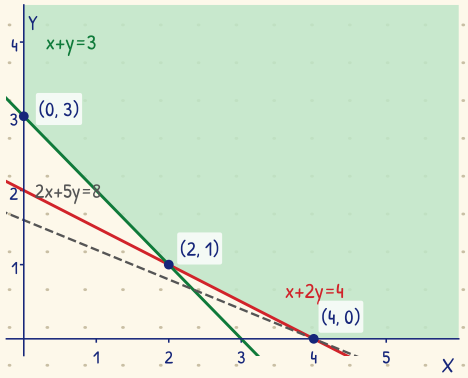


Fig (Example 8): unbounded; dashed $2x + 5y = 8$ touches only at $(4, 0)$

Solved Example 8

Minimise $Z = 2x + 5y$ subject to $x + 2y \geq 4$, $x + y \geq 3$, $x, y \geq 0$.

1. Intersection: subtract, $y = 1$, $x = 2$: corner $(2, 1)$.
2. Corners: $(4, 0)$, $(2, 1)$, $(0, 3)$; region unbounded.

Corner point	$Z = 2x + 5y$
$(4, 0)$	8 **min**
$(2, 1)$	9
$(0, 3)$	15

✔ Smallest value 8. The half-plane $2x + 5y < 8$ has no point in common with the region, so min $Z = 8$ at $(4, 0)$. (Z has no maximum.)

Solved Example 9

Solve: maximise $Z = x + y$ subject to $x + y \leq 2$, $x + y \geq 5$, $x, y \geq 0$.

1. $x + y \leq 2$: below the line $x + y = 2$; $x + y \geq 5$: above the line $x + y = 5$.
2. The two lines are parallel; no point can be below the first and above the second.

✔ No feasible region, so the LPP has **no solution**



Shivam Bhaiya says
Unbounded region dikhe to last line mein **half-plane check** zaroor likho: "$2x + 3y < 13$ has no point in common with the feasible region, hence min = 13". Ye line na likhi to answer adhoora maana jata hai.

5. Formulating Word Problems

1. Name the **decision variables** (x = number of ..., y = number of ...).
2. Write the **objective** Z (profit to max, cost to min).
3. Write each resource limit as an inequality, plus $x, y \geq 0$.
4. Solve graphically (corner point method).

Words	Sign
at most, not more than, available, cannot exceed	$\leq$
at least, minimum requirement, not less than	$\geq$

Yaad karo: "Most = upar ki limit, Least = neeche ki"

at MOST 40 = isse zyada nahi = ≤ 40. **at LEAST** 8 = kam se kam 8 = ≥ 8. Profit/production wale sawal mein mostly $\leq$ (resources limited), diet wale mein $\geq$ (zaroorat poori karni hai).

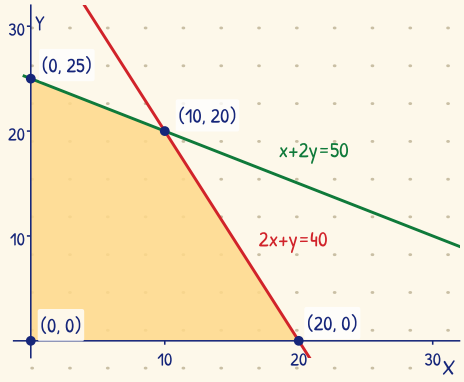


Fig (Example 10): manufacturing problem

Solved Example 10

A factory makes two models A and B. Model A needs 2 h on machine and 1 h of labour; B needs 1 h machine and 2 h labour. Only 40 machine hours and 50 labour hours are available. Profit is Rs 30 on A and Rs 40 on B. How many of each should be made for maximum profit?

1. Let x = number of A, y = number of B. Max $Z = 30x + 40y$.
2. Constraints: $2x + y \leq 40$ (machine), $x + 2y \leq 50$ (labour), $x, y \geq 0$.
3. Corner: solve $2x + y = 40$, $x + 2y = 50$: $x = 10$, $y = 20$. Corners $(0, 0)$, $(20, 0)$, $(10, 20)$, $(0, 25)$.

Corner point	$Z = 30x + 40y$
$(0, 0)$	0
$(20, 0)$	600
$(10, 20)$	1100 **max**
$(0, 25)$	1000

✔ Make 10 of A and 20 of B; maximum profit Rs 1100

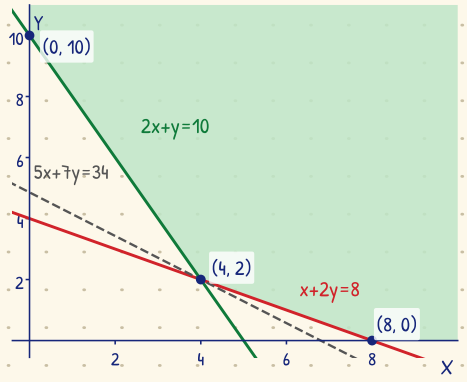


Fig (Example 11): diet problem (unbounded)

Solved Example 11

A diet needs at least 8 units of vitamin and 10 units of mineral. Food F1 gives 1 unit vitamin and 2 units mineral per kg; F2 gives 2 units vitamin and 1 unit mineral per kg. F1 costs Rs 5 and F2 Rs 7 per kg. Find the cheapest mix.

1. Let x kg of F1, y kg of F2. Min $Z = 5x + 7y$.
2. Constraints: $x + 2y \geq 8$ (vitamin), $2x + y \geq 10$ (mineral), $x, y \geq 0$.
3. Corner: $x + 2y = 8$, $2x + y = 10$ give $(4, 2)$. Corners $(8, 0)$, $(4, 2)$, $(0, 10)$.

Corner point	$Z = 5x + 7y$
$(8, 0)$	40
$(4, 2)$	34 **min**
$(0, 10)$	70

✔ Smallest value 34; the half-plane $5x + 7y < 34$ has no common point with the region. Min cost Rs 34 with 4 kg of F1 and 2 kg of F2

Quick fact No common region of constraints: LPP has no solution.

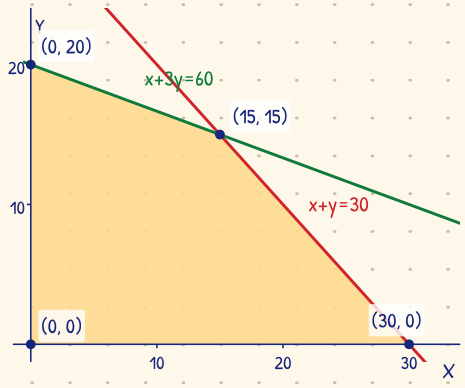


Fig (Example 12): fans and coolers

Solved Example 12

A shop sells fans (cost Rs 2000) and coolers (cost Rs 6000). It can stock at most 30 items and invest at most Rs 1,20,000. Profit is Rs 300 per fan and Rs 700 per cooler. Find the stock for maximum profit.

1. x fans, y coolers: max $Z = 300x + 700y$.
2. $x + y \le 30$; $2000x + 6000y \le 120000 \Rightarrow x + 3y \le 60$; $x, y \ge 0$.
3. Corner: $x + y = 30$, $x + 3y = 60$ give $y = 15$, $x = 15$. Corners $(0,0)$, $(30,0)$, $(15,15)$, $(0,20)$.

Corner point	$Z = 300x + 700y$
$(0,0)$	0
$(30,0)$	9000
$(15,15)$	15000 **max**
$(0,20)$	14000

✔ 15 fans and 15 coolers; maximum profit Rs 15000

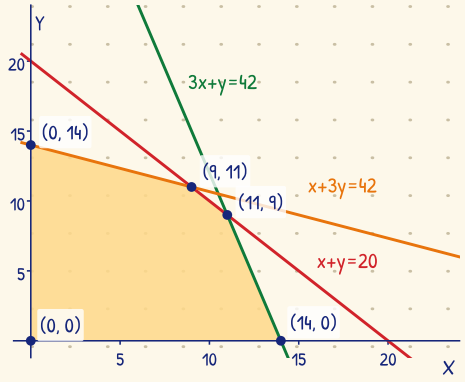


Fig (Example 13): three resource lines, five corners

Solved Example 13

A workshop makes tables (x) and chairs (y). Storage allows at most 20 pieces. A table needs 3 h of carpentry and 1 h of polishing; a chair needs 1 h of carpentry and 3 h of polishing. 42 h of each are available. Profit: Rs 50 per table, Rs 30 per chair. Maximise profit.

Quick fact Corner (x_1, y_1) must satisfy **every** constraint; check it.

1. Max $Z = 50x + 30y$; $x + y \le 20$, $3x + y \le 42$, $x + 3y \le 42$, $x, y \ge 0$.
2. $x + y = 20$ with $3x + y = 42$: $2x = 22$, so $(11, 9)$. With $x + 3y = 42$: $2y = 22$, so $(9, 11)$.
3. Corners: $(0,0)$, $(14,0)$, $(11,9)$, $(9,11)$, $(0,14)$.

Corner point	$Z = 50x + 30y$
$(0,0)$	0
$(14,0)$	700
$(11,9)$	820 **max**
$(9,11)$	780
$(0,14)$	420

✔ 11 tables and 9 chairs; maximum profit Rs 820

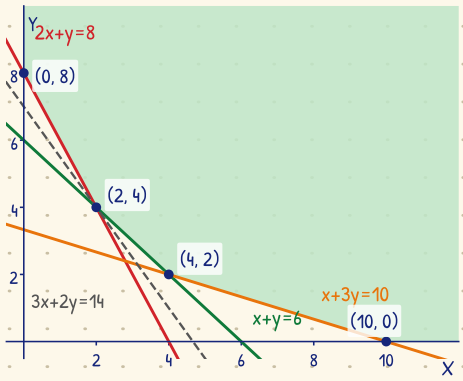


Fig (Example 14): unbounded region with 3 constraints

Solved Example 14

Minimise $Z = 3x + 2y$ subject to $2x + y \ge 8$, $x + y \ge 6$, $x + 3y \ge 10$, $x, y \ge 0$.

1. All $\ge$: region away from origin, unbounded.
2. $2x + y = 8$ and $x + y = 6$: $(2, 4)$; $x + y = 6$ and $x + 3y = 10$: $(4, 2)$.
3. Corners: $(0,8)$, $(2,4)$, $(4,2)$, $(10,0)$.

Corner point	$Z = 3x + 2y$
$(0,8)$	16
$(2,4)$	14 **min**
$(4,2)$	16
$(10,0)$	30

✔ Smallest value 14; $3x + 2y < 14$ has no point in common with the region. Min $Z = 14$ at $(2, 4)$



Shivam Bhaiya says
LPP = 5 marks almost pakke. Graph saaf banao, axes pe scale likho, har line pe uski equation likho, region shade karo, corner table banao. Itna kar diya to poore marks.



Meme samjho

Poore feasible region mein max dhoondhne ki kya zaroorat, jab answer corner points pe hi milta hai

Asli baat: Corner Point Theorem: if an optimal value exists, it occurs at a corner point of the feasible region.

Common mistakes

1. Forgetting $x \geq 0, y \geq 0$: region goes into other quadrants.
2. Shading the wrong side: always test $(0,0)$.
3. Wrong corner: solve the two lines exactly; check the point satisfies **all** constraints.
4. Unbounded region: writing max/min without the half-plane check.
5. Writing only the value of Z; also write the point where it occurs.

Yaad karo: "Unbounded: Max ke liye bada, Min ke liye chhota"

Max check: $Z > M$ (**bada**) wala half-plane. Min check: $Z < m$ (**chhota**) wala half-plane. Region se **mile nahi** to answer pakka, mil gaya to wo max/min exist hi nahi karta.

Most Asked in Boards

1. Solve graphically: max/min with 2-3 constraints (5M)
2. Unbounded region with the half-plane check (5M)
3. Formulate and solve: manufacturing, diet, bakery, transport (5M or 4M case study)
4. Corner points given, find max/min of Z (1M MCQ)
5. Condition on p, q so that max of $px + qy$ occurs at two corners (1M)
6. Which point lies in the feasible region / identify the region (1M)
7. Assertion-Reason on bounded/unbounded regions (1M)

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. The corner points of a feasible region are $(0,3), (2,0), (5,0), (5,6), (0,4)$. For $Z = 3x + 5y$, (max $-$ min) is 1M Most asked

(a) 45 (b) 6 (c) 39 (d) 51

Ans. (c) Z: 15, 6, 15, 45, 20. Max 45, min 6, difference 39.

Q2. Corner points of a bounded region are $(0,0), (5,0), (3,4), (0,5)$. The max of $Z = px + qy$ $(p, q > 0)$ occurs at both $(5,0)$ and $(3,4)$ if 1M Board fav

(a) $p = q$ (b) $p = 2q$ (c) $q = 2p$ (d) $p = 3q$

Ans. (b) $5p = 3p + 4q \Rightarrow p = 2q$

Q3. Which point lies in the feasible region of $2x + 3y \leq 12, x, y \geq 0$? 1M

(a) $(3,3)$ (b) $(5,1)$ (c) $(2,2)$ (d) $(6,1)$

Ans. (c) $2(2) + 3(2) = 10 \leq 12$; the others give 15, 13, 15.

Q4. The objective function of an LPP is 1M

(a) a constant (b) a linear function (c) a quadratic function (d) an inequality

Ans. (b) $Z = ax + by$ is always linear.

Q5. The feasible region of $x + y \leq 4, x \geq 0, y \geq 0$ has how many corner points? 1M

(a) 2 (b) 3 (c) 4 (d) infinitely many

Ans. (b) $(0,0), (4,0), (0,4)$: three.

Q6. In an unbounded region, m is the smallest value of $Z = ax + by$ at the corners. m is the minimum of Z if the open half-plane 1M

(a) $ax + by < m$ has no common point (b) $ax + by > m$ has no common point (c) $ax + by < m$ meets the region (d) none

Ans. (a) $ax + by < m$ has no point in common with the feasible region.

Q7. **A:** If the feasible region is bounded, $Z = ax + by$ has both a maximum and a minimum. **R:** In a bounded region, the optimal values occur at corner points. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) Both true; R (corner point theorem) explains A.

Q8. **A:** If the feasible region is unbounded, a maximum of Z always exists. **R:** An unbounded region extends without limit in some direction. 1M HOTS

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (d) A is false: an unbounded region may have no maximum (check the half-plane); R is true.

Q9. The corner points of the feasible region of $x + 2y \geq 4$, $x + y \geq 3, x, y \geq 0$ are 1M

(a) $(4,0), (2,1), (0,3)$ (b) $(0,0), (4,0), (0,3)$ (c) $(2,1)$ only (d) $(3,0), (0,2)$

Ans. (a) the region is unbounded with corners $(4,0), (2,1), (0,3)$

Q10. The maximum of $Z = 4x + y$ for the region with corners $(0,0), (3,0), (2,2), (0,3)$ is 1M

(a) 10 (b) 3 (c) 12 (d) 14

Ans. (c) Z: 0, 12, 10, 3; max 12.

Q12. For the region with corners $(0,10), (5,5), (15,15), (0,20)$, the minimum of $Z = x + 2y$ is 1M

(a) 20 (b) 15 (c) 0 (d) 45

(contd.) (b) Z: 20, 15, 45, 40; min 15 at $(5,5)$

Q11. **A:** The point $(1,1)$ is a feasible solution of $x+y \le 2$, $x, y \ge 0$. **R:** Points on a boundary line of a $\le$ constraint satisfy it. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Ans. (a) $1+1=2 \le 2$: boundary points are feasible; R explains A.

Q13. Corners of a bounded region are $O(0,0)$, $A(0,4)$, $B(3,3)$, $C(5,0)$. The max of $Z=2x+3y$ is at 1M
(a) A (b) B (c) C (d) O

Ans. (b) Z: 0, 12, 15, 10; max 15 at B

Very short (2 marks)

Q14. Find the corner points of the region $x+y \le 5$, $x \le 3$, $x, y \ge 0$ and the maximum of $Z=2x+y$. 2M

Ans. Corners $(0,0)$, $(3,0)$, $(3,2)$, $(0,5)$; Z: 0, 6, 8, 5. Max $Z=8$ at $(3,2)$

Q15. The corner points of a bounded feasible region are $O(0,0)$, $A(5,0)$, $B(2,3)$, $C(0,4)$. Find the max and min of $Z=3x+4y$. 2M

Ans. Z: 0, 15, 18, 16. Max 18 at $B(2,3)$, min 0 at O

Q16. A dealer has Rs 1500 to buy rice bags (Rs 180 each) and wheat bags (Rs 120 each). He can store at most 10 bags. Profit is Rs 11 per rice bag and Rs 9 per wheat bag. Formulate the LPP. 2M Must do

Ans. x rice, y wheat bags. Max $Z=11x+9y$ subject to $180x+120y \le 1500$ i.e. $3x+2y \le 25$, $x+y \le 10$, $x, y \ge 0$

Q17. Show that the LPP: max $Z=x+2y$ subject to $x+y \le 3$, $x+y \ge 6$, $x, y \ge 0$ has no solution. 2M

Ans. $x+y \le 3$ and $x+y \ge 6$ cannot hold together (parallel lines, opposite sides). No feasible region, so no solution.

Short answer (3 marks)

Q18. Solve graphically: maximise $Z=3x+2y$ subject to $x+2y \le 10$, $3x+y \le 15$, $x, y \ge 0$. 3M

Ans. Corners $(0,0)$, $(5,0)$, $(4,3)$ [from $x+2y=10$, $3x+y=15$], $(0,5)$. Z: 0, 15, 18, 10. Max $Z=18$ at $(4,3)$

Q19. Minimise $Z=x+3y$ subject to $3x+y \ge 6$, $x+3y \ge 6$, $x, y \ge 0$. Show that the minimum occurs at more than two points. 3M Board fav

Ans. Corners $(0,6)$, $\left(\frac{3}{2}, \frac{3}{2}\right)$, $(6,0)$; Z: 18, 6, 6. $x+3y<6$ has no point in common with the (unbounded) region, so min $Z=6$ at every point of the segment joining $\left(\frac{3}{2}, \frac{3}{2}\right)$ and $(6,0)$

Q20. Maximise $Z=5x+2y$ subject to $x+y \le 6$, $x, y \ge 0$. 3M

Ans. Corners $(0,0)$, $(6,0)$, $(0,6)$; Z: 0, 30, 12. Max 30 at

(contd.) $(6,0)$

Q21. Minimise $Z=-2x+3y$ subject to $x+2y \le 10$, $3x+2y \le 18$, $x, y \ge 0$. 3M Board fav

Ans. Corners $(0,0)$, $(6,0)$, $(4,3)$ [subtract: $2x=8$], $(0,5)$; Z: 0, -12, 1, 15. Min $Z=-12$ at $(6,0)$

Q22. Maximise $Z=2x+3y$ subject to $x+y \le 6$, $x \le 4$, $y \le 3$, $x, y \ge 0$. 3M

Ans. Corners $(0,0)$, $(4,0)$, $(4,2)$, $(3,3)$, $(0,3)$; Z: 0, 8, 14, 15, 9. Max 15 at $(3,3)$

Q23. Minimise $Z=5x+4y$ subject to $x+y \ge 4$, $x \ge 1$, $y \ge 1$. 3M HOTS

Ans. Corners $(1,3)$, $(3,1)$ (region unbounded); Z: 17, 19. $5x+4y<17$ has no point in common with the region, so min $Z=17$ at $(1,3)$

Long answer (5 marks)

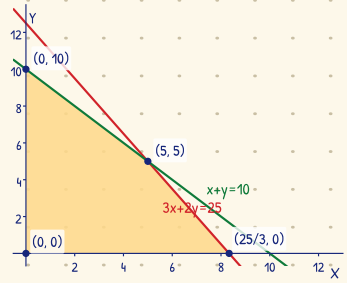


Fig: rice-wheat problem, corners 0, $(25/3, 0)$, $(5,5)$, $(0,10)$

Q24. Solve the rice-wheat problem above graphically: how many bags of each should the dealer buy for maximum profit? 5M Most asked

Ans. Corners of $3x+2y \le 25$, $x+y \le 10$: $(0,0)$, $\left(\frac{25}{3}, 0\right)$, $(5,5)$, $(0,10)$. $Z=11x+9y$: 0, $\frac{275}{3} \approx 91.7$, 100, 90. Max profit Rs 100 with 5 bags of rice and 5 bags of wheat.

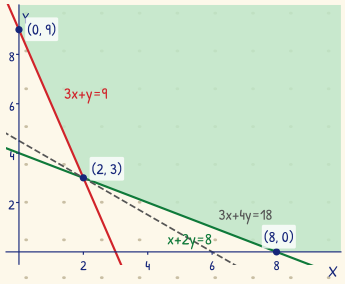


Fig: diet problem, dashed $3x+4y=18$ stays outside

Q25. Food P costs Rs 60/kg and food Q Rs 80/kg. P has 3 units of vitamin A and 1 unit of vitamin B per kg; Q has 1 unit of A and 2 units of B per kg. At least 9 units of A and 8 units of B are needed. Find the least cost. 5M Board fav

Ans. Min $Z=60x+80y$, $3x+y \ge 9$, $x+2y \ge 8$, $x, y \ge 0$. Corners $(0,9)$, $(2,3)$, $(8,0)$; Z: 720, 360, 480. Region unbounded: $60x+80y<360$ (i.e. $3x+4y<18$) has no common point. Least cost Rs 360 with 2 kg of P and 3 kg of Q.

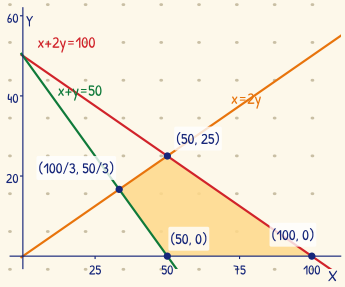


Fig: bounded region with corners on $x = 2y$

Q26. Find the max and min of $Z = 5x + 10y$ subject to $x + 2y \le 100, x + y \ge 50, x - 2y \ge 0, x, y \ge 0$. 5M Must do

Ans. Corners: $\left(\frac{100}{3}, \frac{50}{3}\right), (50, 25), (100, 0), (50, 0)$. Z: $\frac{1000}{3}$, 500, 500, 250. Min 250 at $(50, 0)$; max 500 at $(50, 25)$ and $(100, 0)$, i.e. at all points of the segment joining them.

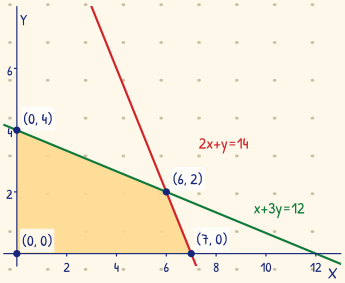


Fig: nuts and bolts, best corner $(6, 2)$

Q27. A factory makes nuts and bolts in packets. A packet of nuts needs 2 h on machine A and 1 h on machine B; a packet of bolts needs 1 h on A and 3 h on B. A runs at most 14 h and B at most 12 h a day. Profit is Rs 10 per packet of nuts and Rs 8 per packet of bolts. Find the daily production for maximum profit. 5M Most asked

Ans. x nuts, y bolts packets. Max $Z = 10x + 8y$, $2x + y \le 14, x + 3y \le 12, x, y \ge 0$. Corners $(0, 0)$, $(7, 0)$, $(6, 2)$, $(0, 4)$; Z: 0, 70, 76, 32. Max Rs 76 with 6 packets of nuts and 2 of bolts.

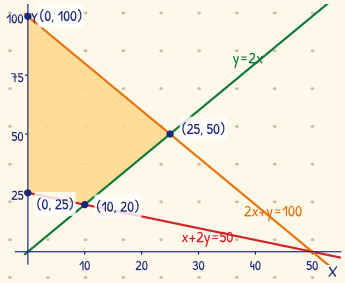


Fig: bounded region between $y = 2x$ and the y-axis

Q28. Find the min and max of $Z = x + 2y$ subject to $x + 2y \ge 50, 2x - y \le 0, 2x + y \le 100, x, y \ge 0$. 5M HOTS

Ans. Corners: $(0, 25), (10, 20)$ [$y = 2x$ with $x + 2y = 50$], $(25, 50)$ [$y = 2x$ with $2x + y = 100$], $(0, 100)$. Z: 50, 50, 125, 200. Min 50 at all points of the segment joining $(0, 25)$ and $(10, 20)$; max 200 at $(0, 100)$

Quick fact Profit/production LPP: usually Max with $\le$ constraints.

Q29. A farmer has 40 hectares for crops A and B. Profit per hectare is Rs 12000 for A and Rs 10000 for B. A needs 30 kg and B needs 20 kg of fertiliser per hectare; at most 1000 kg is available. How should he plant to maximise profit? 5M Board fav

Ans. x ha of A, y ha of B. Max $Z = 12000x + 10000y$, $x + y \le 40, 30x + 20y \le 1000$ i.e. $3x + 2y \le 100$, $x, y \ge 0$. Corners $(0, 0)$, $\left(\frac{100}{3}, 0\right), (20, 20), (0, 40)$; Z: 0, 400000, 440000, 400000. Max profit Rs 4,40,000 with 20 ha of each crop.

Case based (4 marks)

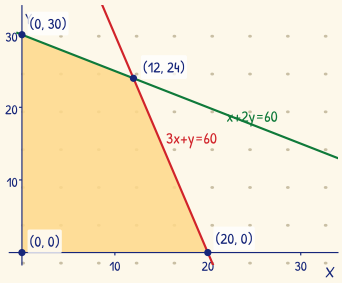


Fig: bakery problem, best corner $(12, 24)$

Q30. Home bakery. Riya bakes two cakes. Cake A needs 300 g flour and 20 g fat; cake B needs 100 g flour and 40 g fat. She has 6 kg flour and 1.2 kg fat. (i) Write the constraints if she bakes x of A and y of B. (1) (ii) Find the corner points of the feasible region. (1) (iii) Find the maximum number of cakes she can bake. (2) 4M Most asked

Ans. (i) $300x + 100y \le 6000 \Rightarrow 3x + y \le 60; 20x + 40y \le 1200 \Rightarrow x + 2y \le 60; x, y \ge 0$. (ii) $(0, 0)$, $(20, 0), (12, 24), (0, 30)$. (iii) $Z = x + y$: 0, 20, 36, 30. Max 36 cakes (12 of A, 24 of B).

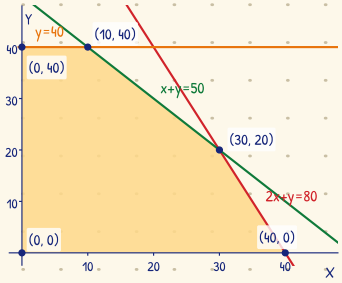


Fig: company region O, A$(40, 0)$, B$(30, 20)$, C$(10, 40)$, D $(0, 40)$

Q31. Company planning. The feasible region of an LPP is the polygon with corners $O(0, 0)$, $A(40, 0)$, $B(30, 20)$, $C(10, 40)$, $D(0, 40)$. (i) Find the max of $Z = 4x + 3y$. (1) (ii) Find the max of $Z = x + 2y$. (1) (iii) If $Z = px + qy$ has its max at both B and C, find the relation between p and q. (2) 4M Board fav

Ans. (i) Z: 0, 160, 180, 160, 120: max 180 at B. (ii) Z: 0, 40, 70, 90, 80: max 90 at C. (iii) $30p + 20q = 10p + 40q \Rightarrow 20p = 20q \Rightarrow p = q$

Quick fact Diet/cost LPP: usually Min with $\ge$ constraints.

Ek nazar mein: Linear Programming

Type of sawal	Method	Marks
Corners given, find max/min	table of Z	1
Bounded, 2-3 constraints	graph + corner table	3-5
Unbounded	corner table + half-plane check	5
Word problem	variables, Z, constraints, graph	4-5

Last minute: 6 one-liners

1. Optimal value (if it exists) is always at a corner point.
2. Bounded region: max and min both exist.
3. Unbounded: check $Z > M$ (max) or $Z < m$ (min) half-plane.
4. Same optimal value at two corners: whole segment is optimal.
5. Test point $(0, 0)$ decides the side to shade.
6. Never forget $x \geq 0, y \geq 0$.

True or False? (concept check)

1. Every LPP has a feasible region. False (constraints may contradict)
2. In a bounded region, Z has both max and min. True
3. The optimal value can occur inside the region but not at a corner. False (if it exists, a corner also gives it)
4. An unbounded region always has a minimum for $Z = ax + by$. False (check the half-plane)
5. Two corners giving the same max make every point between them optimal. True
6. $x \geq 0, y \geq 0$ are also constraints. True (non-negative constraints)

Apne notes / galtiyan yahan likho

Khud karo: 8 quick practice questions

1. Corner points $(0, 0), (6, 0), (4, 4), (0, 5)$; max of $Z = 2x + 3y$.
2. Same corners; min of $Z = x - y$.
3. Does $(3, 2)$ satisfy $2x + y \leq 8$?
4. Intercepts of the line $4x + 5y = 20$.
5. Corner where $x + y = 6$ meets $x - y = 2$.
6. Max of $Z = x + y$ on $x + 2y \leq 8, x, y \geq 0$.
7. Write: "at least 3 units of iron" as an inequality (x kg of food with 2 units/kg).
8. Bounded or unbounded: $x + y \geq 2, x, y \geq 0$?

Answers: 1) 20 at $(4, 4)$ 2) -5 at $(0, 5)$ 3) yes, $8 \leq 8$ 4) $(5, 0), (0, 4)$ 5) $(4, 2)$ 6) 8 at $(8, 0)$ 7) $2x \geq 3$ 8) unbounded

5-mark LPP answer ka format

1. **Let** x = ..., y = ... (only for word problems).
2. **Objective:** Max/Min $Z = ax + by$.
3. **Constraints** with $x \geq 0, y \geq 0$.
4. **Graph:** each line with two points, label it, shade, mark corners A, B, C ...
5. **Corner table** with Z values.
6. **Unbounded?** write the half-plane check line.
7. **Conclusion:** max/min value and the point (in words for word problems).

Khud karo 3: corner point drills

1. Corners $(0, 0), (5, 0), (3, 4), (0, 6)$: max of $Z = 2x + 3y$ [18 at $(0, 6)$]
2. Same corners: max of $Z = 4x + y$ [20 at $(5, 0)$]
3. Corners $(2, 0), (6, 0), (4, 3), (1, 3)$: min of $Z = x + 2y$ [2 at $(2, 0)$]
4. Intersection of $x + y = 8$ and $x - y = 2$ [$(5, 3)$]
5. Is $(3, 3)$ feasible for $x + 2y \leq 8, x \geq 0, y \geq 0$? [No, $9 > 8$]
6. Corners $(0, 2), (3, 0), (6, 0), (0, 5)$: min of $Z = 5x + 4y$ [8 at $(0, 2)$]

Apne notes / galtiyan yahan likho

Most asked: Bayes' theorem (5M), probability distribution + mean

PROBABILITY

1. Quick Recap: Class 11 Basics

Sample space S = set of all outcomes. **Event** = subset of S.

For equally likely outcomes, $P(E) = \frac{n(E)}{n(S)}$ and $0 \le P(E) \le 1$.

- $P(S) = 1$ (sure event), $P(\phi) = 0$ (impossible event)
- **Mutually exclusive:** $E \cap F = \phi$, they cannot happen together.
- **Exhaustive:** $E_1 \cup E_2 \cup \cdots \cup E_n = S$, at least one must happen.

Formula Box 1: Basics

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A') = 1 - P(A)$ (complement, "not A")
- $P(A \cap B') = P(A) - P(A \cap B)$ (only A)
- $P(A' \cap B') = 1 - P(A \cup B)$ (neither)
- Exactly one of A, B: $P(A) + P(B) - 2P(A \cap B)$

Words	Set
A or B (at least one)	$A \cup B$
A and B (both)	$A \cap B$
not A	A'
neither A nor B	$A' \cap B'$

Solved Example 1

If $P(A) = 0.45$, $P(B) = 0.35$ and $P(A \cap B) = 0.15$, find $P(A \cup B)$, $P(A' \cap B')$, $P(\text{only A})$ and $P(\text{exactly one})$.

1. $P(A \cup B) = 0.45 + 0.35 - 0.15 = 0.65$
2. $P(A' \cap B') = 1 - 0.65 = 0.35$
3. Only A: $0.45 - 0.15 = 0.30$
4. Exactly one: $0.45 + 0.35 - 2(0.15) = 0.50$

✔ 0.65, 0.35, 0.30, 0.50

Quick fact $P(E|E) = 1$ and $P(\phi|F) = 0$ for any F with $P(F) > 0$.

Yaad rakho Sure event S is independent of every event; so is the impossible event ϕ.

Recall A pack of 52 cards: 4 suits of 13, 4 aces, 12 face cards (J, Q, K), 26 red and 26 black.

2. Conditional Probability

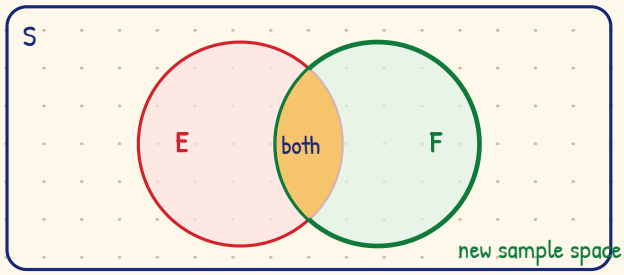


Fig: given F has happened, only F is left; E can now happen only as "E and F"

Definition

If E and F are events of the same sample space and $P(F) \neq 0$, the probability of E given that F has already occurred is

$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$

For equally likely outcomes: $P(E|F) = \frac{n(E \cap F)}{n(F)}$.

Properties of P(E|F)

- $P(S|F) = P(F|F) = 1$
- $P((A \cup B)|F) = P(A|F) + P(B|F) - P((A \cap B)|F)$
- $P(E'|F) = 1 - P(E|F)$
- $0 \le P(E|F) \le 1$

Yaad karo: "Given wala Ghar mein (neeche)"

$P(E|F)$ mein jo **given** hai (F, line ke baad wala) wahi **denominator** mein jaata hai. Upar hamesha "dono saath" $P(E \cap F)$. Ulta likha to poora sawal gaya.

Solved Example 2

If $P(A) = 0.6$, $P(B) = 0.4$ and $P(A \cup B) = 0.7$, find $P(A|B)$ and $P(B|A)$.

1. $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.6 + 0.4 - 0.7 = 0.3$
2. $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.4} = \frac{3}{4}$
3. $P(B|A) = \frac{0.3}{0.6} = \frac{1}{2}$

✔ $P(A|B) = \frac{3}{4}$, $P(B|A) = \frac{1}{2}$

Solved Example 3

Two dice are thrown. Find the probability that the sum is 8, given that the first die shows an even number.

1. F: first die even, $n(F) = 3 \times 6 = 18$
2. E: sum 8 = (2,6), (3,5), (4,4), (5,3), (6,2)
3. $E \cap F$ = (2,6), (4,4), (6,2), so $n(E \cap F) = 3$
4. $P(E|F) = \frac{3}{18}$

✔ $P(E|F) = \frac{1}{6}$

Solved Example 4

A family has two children. Find the probability that both are girls, given that (i) the elder child is a girl (ii) at least one child is a girl.

1. S = {BB, BG, GB, GG} (first letter = elder)
2. (i) F = {GB, GG}; both girls = {GG}: $P = \frac{1}{2}$
3. (ii) F = {BG, GB, GG}; $P = \frac{1}{3}$

✔ (i) $\frac{1}{2}$ (ii) $\frac{1}{3}$

Solved Example 5

A coin is tossed three times. E: at least two heads, F: first toss is a head. Find $P(E|F)$.

1. F = {HHH, HHT, HTH, HTT}, $n(F) = 4$
2. $E \cap F$ = {HHH, HHT, HTH}, $n = 3$

✔ $P(E|F) = \frac{3}{4}$

Solved Example 6

Two dice are thrown. Find the probability that the sum is 6, given that at least one die shows a 4.

1. F: at least one 4. Row 4 (6) + column 4 (6) − (4,4) counted twice: $n(F) = 11$
2. E: sum 6 = (1,5), (2,4), (3,3), (4,2), (5,1)
3. $E \cap F$ = (2,4), (4,2): $n = 2$

✔ $P(E|F) = \frac{2}{11}$

Check trick A Bayes answer can never be more than 1; the numerator is one term of the denominator.

Board tip Two dice = 36 outcomes, three coins = 8 outcomes; write the count before solving.

Quick fact The mean $E(X)$ need not be a value that X actually takes (e.g. $\frac{7}{2}$ for a die).

green = F (a 4 shows): 11 cells

1,1	1,2	1,3	1,4	1,5	1,6
2,1	2,2	2,3	2,4	2,5	2,6
3,1	3,2	3,3	3,4	3,5	3,6
4,1	4,2	4,3	4,4	4,5	4,6
5,1	5,2	5,3	5,4	5,5	5,6
6,1	6,2	6,3	6,4	6,5	6,6

pink = sum 6 only, orange = both: 2 cells

Fig: 36 outcomes of two dice; P(E|F) = 2/11

Solved Example 7

Cards numbered 1 to 10 are put in a box and one card is drawn. Given that the number is more than 3, find the probability that it is even.

1. F = {4, 5, ..., 10}, $n(F) = 7$
2. Even in F: {4, 6, 8, 10}, $n = 4$

✔ $P = \frac{4}{7}$

Solved Example 8

$P(A) = 0.8$, $P(B) = 0.5$ and $P(B|A) = 0.4$. Find $P(A \cap B)$, $P(A|B)$ and $P(A \cup B)$.

1. $P(A \cap B) = P(A)P(B|A) = 0.8 \times 0.4 = 0.32$
2. $P(A|B) = \frac{0.32}{0.5} = 0.64$
3. $P(A \cup B) = 0.8 + 0.5 - 0.32 = 0.98$

✔ 0.32, 0.64, 0.98

Solved Example 9

A class has 25 boys (10 wear glasses) and 35 girls (14 wear glasses). A student is chosen at random. Find (i) $P(\text{girl}|\text{wears glasses})$ (ii) $P(\text{glasses}|\text{boy})$.

1. Glasses total $= 10 + 14 = 24$
2. (i) $\frac{14}{24} = \frac{7}{12}$ (reduced sample space = 24 students with glasses)
3. (ii) $\frac{10}{25} = \frac{2}{5}$

✔ (i) $\frac{7}{12}$ (ii) $\frac{2}{5}$

↳Reduced sample space trick: "given" mila to sirf F ke outcomes gino, baaki sab cross kar do.

3. Multiplication Theorem

From the definition of conditional probability:

Use it for **successive draws without replacement**: the second probability is calculated from what is left after the first draw.

Theorem: Multiplication rule

$$P(E \cap F) = P(E)\,P(F|E) = P(F)\,P(E|F)$$

For three events:

$$P(E \cap F \cap G) = P(E)\,P(F|E)\,P(G|EF)$$

where EF means $E \cap F$.

Solved Example 10

A bag has 5 red and 7 black balls. Two balls are drawn one after the other without replacement. Find the probability that both are red.

1. $P(\text{1st red}) = \dfrac{5}{12}$
2. Now 4 red out of 11 left: $P(\text{2nd red}|\text{1st red}) = \dfrac{4}{11}$
3. $P = \dfrac{5}{12} \times \dfrac{4}{11} = \dfrac{20}{132}$

✔ $P(\text{both red}) = \dfrac{5}{33}$

Solved Example 11

Three cards are drawn one by one without replacement from a well-shuffled pack of 52 cards. Find the probability that the first two are kings and the third is an ace.

1. $P(K_1) = \dfrac{4}{52}$, $P(K_2|K_1) = \dfrac{3}{51}$
2. $P(A_3|K_1K_2) = \dfrac{4}{50}$ (all 4 aces still there)
3. $P = \dfrac{4}{52} \times \dfrac{3}{51} \times \dfrac{4}{50} = \dfrac{48}{132600}$

✔ $P = \dfrac{2}{5525}$

Yaad karo: "Bina wapas = Ghatate jao"

Without replacement mein har draw ke baad **upar (favourable) aur neeche (total) dono 1 se ghatao.** With replacement mein kuch nahi ghatta, har baar same fraction.

Solved Example 13

Two cards are drawn without replacement from a pack of 52. Find the probability that one is a king and the other is a queen.

1. KQ: $\frac{4}{52} \times \frac{4}{51}$; QK: $\frac{4}{52} \times \frac{4}{51}$ (order can be either way)
2. $P = 2 \times \dfrac{16}{2652} = \dfrac{32}{2652}$

✔ $P = \dfrac{8}{663}$

Solved Example 12

A bag has 3 red and 4 white balls. A ball is drawn, its colour noted, and it is put back along with 2 more balls of the same colour. A second ball is then drawn. Find (i) $P(\text{both red})$ (ii) $P(\text{second ball red})$.

1. (i) $P(R_1) = \frac{3}{7}$; then 5 red out of 9: $P(R_2|R_1) = \frac{5}{9}$. $P = \frac{3}{7} \times \frac{5}{9} = \frac{15}{63} = \frac{5}{21}$
2. (ii) Second red after a white: $\frac{4}{7} \times \frac{3}{9} = \frac{12}{63}$
3. $P(R_2) = \frac{15}{63} + \frac{12}{63} = \frac{27}{63}$

✔ (i) $\dfrac{5}{21}$ (ii) $\dfrac{3}{7}$

4. Independent Events

Definition

E and F are **independent** if the occurrence of one does not affect the probability of the other:

$$P(E \cap F) = P(E) \times P(F)$$

Equivalently $P(E|F) = P(E)$ (if $P(F) \neq 0$) and $P(F|E) = P(F)$.

→ If E, F are independent, so are (E, F'), (E', F) and (E', F').

→ E, F, G are **mutually independent** if all three pairs are independent AND $P(E \cap F \cap G) = P(E)P(F)P(G)$.

→ For independent events: $P(\text{at least one}) = 1 - P(E')P(F')$.

Independent	Mutually exclusive
$P(E \cap F) = P(E)P(F)$	$P(E \cap F) = 0$
both can happen together	cannot happen together
about probabilities	about the sets (outcomes)

↳Do events with non-zero probability being mutually exclusive mean independent? NO: then $P(E \cap F) = 0$ but $P(E)P(F) > 0$.

Meme samjho



Jab sab "mutually exclusive = independent" likh rahe hon aur tumhe pata ho ki ye galat hai.

Asli baat: If $P(E), P(F) > 0$ and $E \cap F = \phi$, then $P(E \cap F) = 0 \neq P(E)P(F)$. So they are NOT independent.

Recall ${}^nC_2 = \dfrac{n(n-1)}{2}$: ${}^4C_2 = 6$, ${}^5C_2 = 10$, ${}^6C_2 = 15$.

Solved Example 14

A die is thrown. E: the number is a multiple of 3, F: the number is even. Are E and F independent?

1. E = {3, 6}, $P(E) = \frac{1}{3}$; F = {2, 4, 6}, $P(F) = \frac{1}{2}$
2. $E \cap F$ = {6}, $P(E \cap F) = \frac{1}{6}$
3. $P(E)P(F) = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6} = P(E \cap F)$

✔ Yes, E and F are independent

Solved Example 15

If A and B are independent with $P(A) = 0.3$ and $P(B) = 0.6$, find (i) $P(A \cap B)$ (ii) $P(A \cup B)$ (iii) $P(A' \cap B')$ (iv) $P(A|B)$.

1. (i) $0.3 \times 0.6 = 0.18$
2. (ii) $0.3 + 0.6 - 0.18 = 0.72$
3. (iii) $P(A')P(B') = 0.7 \times 0.4 = 0.28$ (check: $1 - 0.72 = 0.28$)
4. (iv) independent, so $P(A|B) = P(A) = 0.3$

✔ 0.18, 0.72, 0.28, 0.3

Solved Example 16

A problem is given to three students A, B, C whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{5}$, independently. Find the probability that (i) the problem is solved (ii) exactly one of them solves it.

1. Not solving: $\frac{1}{2}$, $\frac{2}{3}$, $\frac{4}{5}$
2. (i) $P(\text{solved}) = 1 - \frac{1}{2} \times \frac{2}{3} \times \frac{4}{5} = 1 - \frac{8}{30} = \frac{11}{15}$
3. (ii) $\frac{1}{2} \times \frac{2}{3} \times \frac{4}{5} + \frac{1}{2} \times \frac{1}{3} \times \frac{4}{5} + \frac{1}{2} \times \frac{2}{3} \times \frac{1}{5}$
4. $= \frac{8}{30} + \frac{4}{30} + \frac{2}{30} = \frac{14}{30}$

✔ (i) $\frac{11}{15}$ (ii) $\frac{7}{15}$

Yaad karo: "At least one = 1 minus None"

"At least one / problem solved / target hit" dikhe to seedha $1 - P(\text{koi nahi})$. Koi nahi = saare **fail** probabilities ka product (independent ho tab).

Yaad rakho With replacement: draws are independent. Without replacement: use conditional probability.

Quick fact $P(A \cap B) \le P(A) \le P(A \cup B)$ always.

Solved Example 17

A and B are independent, $P(A) = 0.4$ and $P(A \cup B) = 0.7$. Find $P(B)$.

1. Let $P(B) = p$. $P(A \cup B) = P(A) + P(B) - P(A)P(B)$
2. $0.7 = 0.4 + p - 0.4p \Rightarrow 0.6p = 0.3$

✔ $P(B) = 0.5$

Solved Example 18

A die is thrown twice. A: first throw is odd, B: the sum is 7. Are A and B independent?

1. $P(A) = \frac{1}{2}$, $P(B) = \frac{6}{36} = \frac{1}{6}$
2. $A \cap B$ = (1,6), (3,4), (5,2): $P(A \cap B) = \frac{3}{36} = \frac{1}{12}$
3. $P(A)P(B) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$, equal

✔ Yes, independent

Solved Example 19

Three coins are tossed. E: first coin shows head, F: exactly two heads. Are E and F independent?

1. $P(E) = \frac{4}{8} = \frac{1}{2}$, F = {HHT, HTH, THH}: $P(F) = \frac{3}{8}$
2. $E \cap F$ = {HHT, HTH}: $P(E \cap F) = \frac{2}{8} = \frac{1}{4}$
3. $P(E)P(F) = \frac{3}{16} \neq \frac{1}{4}$

✔ Not independent

Alternate turns (geometric series)

When two players try one after the other until someone succeeds, the first player wins on turns 1, 3, 5, ... Use the sum of an infinite GP: $a + ar + ar^2 + \cdots = \frac{a}{1-r}$ $(|r| < 1)$.

Solved Example 20

A and B toss a fair coin alternately; the first one to get a head wins. A starts. Find the probability that A wins.

1. A wins on toss 1, 3, 5, ...: $P(A) = \frac{1}{2} + \left(\frac{1}{2}\right)^2 \frac{1}{2} + \left(\frac{1}{2}\right)^4 \frac{1}{2} + \ldots$
2. GP with $a = \frac{1}{2}$, $r = \frac{1}{4}$: $P(A) = \frac{1/2}{1 - 1/4} = \frac{1/2}{3/4}$
3. $P(B) = 1 - \frac{2}{3} = \frac{1}{3}$

✔ $P(A) = \frac{2}{3}$, $P(B) = \frac{1}{3}$

"given" / "if it is known that" → $P(E|F)$ → "both, one after the other" → multiply → "came from which?" ⇒ Bayes

Solved Example 21

A, B and C hit a target independently with probabilities 0.5, 0.6 and 0.8. Find the probability that exactly two of them hit it.

1. Misses: 0.5, 0.4, 0.2
2. Only A, B hit: $0.5 \times 0.6 \times 0.2 = 0.06$
3. Only A, C: $0.5 \times 0.4 \times 0.8 = 0.16$; only B, C: $0.5 \times 0.6 \times 0.8 = 0.24$
4. Sum $= 0.06 + 0.16 + 0.24$

✔ $P(\text{exactly two}) = 0.46$

Result with proof (board 2-3M)

If E and F are independent, then E and F' are independent.

Proof: $E = (E \cap F) \cup (E \cap F')$, disjoint. So $P(E \cap F') = P(E) - P(E \cap F) = P(E) - P(E)P(F) = P(E)\,[1 - P(F)] = P(E)P(F')$. Hence proved.

Solved Example 22

How many times must a fair coin be tossed so that the probability of getting at least one head is more than 0.9?

1. $P(\text{no head in } n \text{ tosses}) = \left(\frac{1}{2}\right)^n$ (independent tosses)
2. Need $1 - \left(\frac{1}{2}\right)^n > 0.9 \Rightarrow \left(\frac{1}{2}\right)^n < 0.1 \Rightarrow 2^n > 10$
3. $2^3 = 8$ (no), $2^4 = 16$ (yes)

✔ At least 4 tosses

Board tip: independence check

Sirf ek line: $P(A \cap B)$ nikalo aur $P(A) \times P(B)$ se compare karo. Dono values likh ke "**hence independent**" (ya "not independent") likhna, wahi 1 mark hai.

5. Total Probability and Bayes' Theorem

Partition of S

Events $E_1, E_2, \ldots, E_n$ form a **partition** of S if they are pairwise disjoint ($E_i \cap E_j = \phi$), exhaustive ($E_1 \cup \cdots \cup E_n = S$) and $P(E_i) > 0$. Example: E and E' always partition S.

Check trick In a distribution table, if any p comes negative or $\sum p \neq 1$, recheck k.

Theorem of Total Probability

If $E_1, \ldots, E_n$ is a partition of S and A is any event, then

$$P(A) = \sum_{i=1}^{n} P(E_i)\,P(A|E_i)$$

Bayes' Theorem

With the same partition, for any i:

$$P(E_i|A) = \frac{P(E_i)\,P(A|E_i)}{\sum_{j=1}^{n} P(E_j)\,P(A|E_j)}$$

$P(E_i)$ = **prior** probabilities (known before), $P(E_i|A)$ = **posterior** (revised after A is seen). E_i are called **hypotheses**.

Bayes in 4 steps (board method)

1. Name the events: $E_1, E_2, \ldots$ (the causes) and A (what is observed).
2. Write $P(E_1), P(E_2), \ldots$ (sum must be 1).
3. Write $P(A|E_1), P(A|E_2), \ldots$ (read from the question).
4. Put in the formula: (wanted branch) / (sum of all branches).

Yaad karo: "Bayes = Mera hissa / Sabka total"

Tree banao, har branch ka product nikalo. Total probability = **sabka total** (saare products ka sum). Bayes = **mera hissa** (jo branch poochi hai) divided by sabka total. Numerator hamesha denominator ka ek term hota hai.

Solved Example 23

Bag I has 4 red and 5 black balls; Bag II has 6 red and 3 black balls. A bag is chosen at random and a ball is drawn. Find (i) the probability that it is red (ii) given that it is red, the probability that it came from Bag II.

1. E_1: Bag I, E_2: Bag II, $P(E_1) = P(E_2) = \frac{1}{2}$; A: red
2. $P(A|E_1) = \frac{4}{9}$, $P(A|E_2) = \frac{6}{9}$
3. (i) $P(A) = \frac{1}{2} \times \frac{4}{9} + \frac{1}{2} \times \frac{6}{9} = \frac{4}{18} + \frac{6}{18} = \frac{10}{18}$
4. (ii) $P(E_2|A) = \dfrac{6/18}{10/18} = \dfrac{6}{10}$

✔ (i) $\frac{5}{9}$ (ii) $\frac{3}{5}$

Board tip In Bayes, write "By Bayes' theorem" before the formula: it carries a step mark.

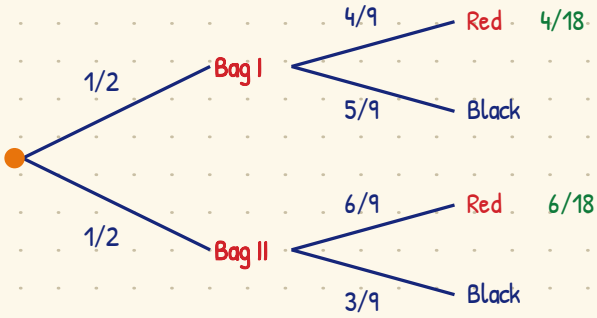


Fig: tree for the Bag I / Bag II example. Multiply along a branch, add the branches you want.

Solved Example 24

Machines A, B, C make 50%, 30% and 20% of the bolts of a factory; 3%, 4% and 5% of their output is defective. A bolt picked at random is defective. Find the probability that it was made by machine B.

1. $P(A) = 0.5$, $P(B) = 0.3$, $P(C) = 0.2$; D: defective
2. $P(D|A) = 0.03$, $P(D|B) = 0.04$, $P(D|C) = 0.05$
3. $P(D) = 0.015 + 0.012 + 0.010 = 0.037$
4. $P(B|D) = \dfrac{0.3 \times 0.04}{0.037} = \dfrac{0.012}{0.037}$

✔ $P(B|D) = \dfrac{12}{37}$

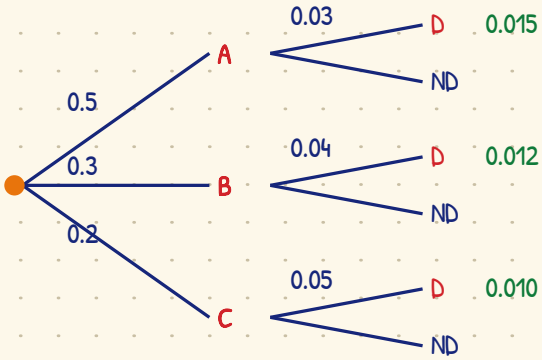


Fig: D = defective, ND = not defective. P(D) = 0.015 + 0.012 + 0.010 = 0.037

Solved Example 25

A speaks the truth 3 out of 5 times. A die is thrown and A reports that it is a six. Find the probability that it was actually a six.

1. E_1: six occurs, $P(E_1) = \frac{1}{6}$; E_2: not six, $P(E_2) = \frac{5}{6}$
2. R: A reports six. $P(R|E_1) = \frac{3}{5}$ (truth), $P(R|E_2) = \frac{2}{5}$ (lie)
3. $P(E_1|R) = \dfrac{\frac{1}{6} \times \frac{3}{5}}{\frac{1}{6} \times \frac{3}{5} + \frac{5}{6} \times \frac{2}{5}} = \dfrac{3}{3+10}$

✔ $P = \dfrac{3}{13}$

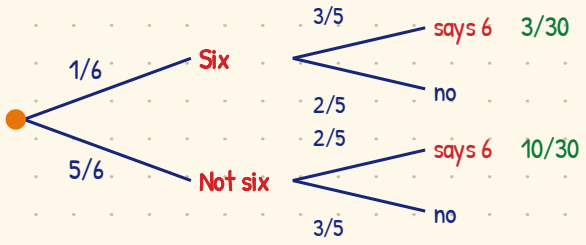


Fig: truth-teller tree. P(six | says six) = 3/(3+10)

Meme samjho



Bayes ka 5 marker dekh ke skip karne ka plan? Study bro, don't waste time!

Asli baat: Bayes is pure formula + tree: 4 fixed steps (events, priors, likelihoods, formula). Practise 5 types: bags, machines, truth-teller, test result, transport/insurance.

Solved Example 26

1% of the people of a town have a certain disease. A test detects the disease in 95% of the people who have it, but also shows positive for 10% of the healthy people. A person tests positive. Find the probability that the person actually has the disease.

1. E_1: has disease, $P(E_1) = 0.01$; E_2: healthy, $P(E_2) = 0.99$
2. A: positive. $P(A|E_1) = 0.95$, $P(A|E_2) = 0.10$
3. $P(E_1|A) = \dfrac{0.01 \times 0.95}{0.01 \times 0.95 + 0.99 \times 0.10} = \dfrac{0.0095}{0.1085}$
4. $= \dfrac{95}{1085}$

✔ $P = \dfrac{19}{217}$ (only about 9%!)



Fig: test tree. Most positives come from the huge healthy group!

Recall $P(\text{at least one head in } n \text{ tosses}) = 1 - \left(\frac{1}{2}\right)^n$.

Quick fact If $P(A|B) = P(A)$, then also $P(B|A) = P(B)$: independence works both ways.

Yaad rakho $P(A|B) + P(A'|B) = 1$, but $P(A|B) + P(A|B')$ need not be 1.

Solved Example 27

An insurance company insured 3000 two-wheeler drivers, 5000 car drivers and 2000 truck drivers. The probabilities of an accident are 0.02, 0.01 and 0.10 respectively. One insured person meets with an accident. Find the probability that he is a two-wheeler driver.

1. Total 10000: $P(E_1) = 0.3$, $P(E_2) = 0.5$, $P(E_3) = 0.2$
2. $P(A|E_1) = 0.02$, $P(A|E_2) = 0.01$, $P(A|E_3) = 0.10$
3. $P(A) = 0.006 + 0.005 + 0.020 = 0.031$
4. $P(E_1|A) = \dfrac{0.006}{0.031}$

✔ $P = \dfrac{6}{31}$

Solved Example 28

In an exam with 4 options per MCQ, a student knows the answer with probability $\frac{3}{5}$ and guesses with probability $\frac{2}{5}$. A guess is correct with probability $\frac{1}{4}$. Given that the answer is correct, find the probability that the student knew it.

1. E_1: knows, $\frac{3}{5}$; E_2: guesses, $\frac{2}{5}$; A: correct
2. $P(A|E_1) = 1$, $P(A|E_2) = \frac{1}{4}$
3. $P(E_1|A) = \dfrac{\frac{3}{5} \times 1}{\frac{3}{5} + \frac{2}{5} \times \frac{1}{4}} = \dfrac{\frac{3}{5}}{\frac{7}{10}}$

✔ $P = \dfrac{6}{7}$

Bayes type	$E_1, E_2, \ldots$ (causes)	A (observed)
Bags / urns	bag chosen	colour of ball
Machines / suppliers	machine	item defective
Truth-teller	event really happened or not	person reports it
Medical test	has disease or not	test positive
Transport / insurance	mode / type of driver	late / accident
Ball transfer	colour transferred	colour drawn later

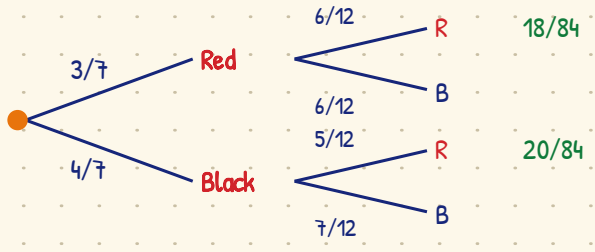


Fig: transfer tree. First branch = colour moved; Bag II then has 12 balls.

Solved Example 29

Bag I has 3 red and 4 black balls; Bag II has 5 red and 6 black balls. One ball is transferred from Bag I to Bag II, and then a ball is drawn from Bag II. It is red. Find the probability that the transferred ball was black.

1. E_1: red transferred, $\frac{3}{7}$; E_2: black transferred, $\frac{4}{7}$
2. Bag II then has 12 balls: $P(R|E_1) = \frac{6}{12}$, $P(R|E_2) = \frac{5}{12}$
3. $P(R) = \frac{3}{7} \times \frac{6}{12} + \frac{4}{7} \times \frac{5}{12} = \frac{18+20}{84} = \frac{38}{84}$
4. $P(E_2|R) = \dfrac{20/84}{38/84} = \dfrac{20}{38}$

✔ $P = \dfrac{10}{19}$

Solved Example 30

There are three coins: one is two-headed, one is biased and shows heads 75% of the time, and one is fair. A coin is chosen at random and tossed; it shows heads. Find the probability that it was the two-headed coin.

1. $P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$
2. $P(H|E_1) = 1$, $P(H|E_2) = \frac{3}{4}$, $P(H|E_3) = \frac{1}{2}$
3. $P(E_1|H) = \dfrac{1}{1 + \frac{3}{4} + \frac{1}{2}} = \dfrac{1}{9/4}$ (the $\frac{1}{3}$ cancels)

✔ $P = \dfrac{4}{9}$

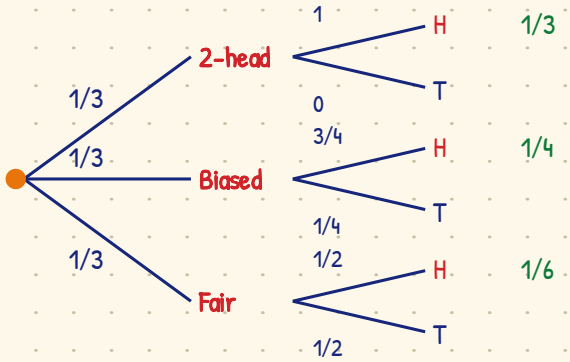


Fig: three coins tree. P(H) = 1/3 + 1/4 + 1/6 = 3/4

Yaad karo: "Cancel the common prior"

Jab saare $P(E_i)$ barabar hon (bag/coin "at random"), to Bayes mein woh cancel ho jaate hain: $P(E_i|A) = \dfrac{P(A|E_i)}{\sum P(A|E_j)}$. Time bachao, galti kam.

Check trick Total probability $P(A)$ must lie between the smallest and largest $P(A|E_i)$.

Recall Leap year: 366 days = 52 weeks + 2 days, so P(53 Sundays) $= \frac{2}{7}$.

Solved Example 31

A man throws a die. If he gets a 6, he tosses a coin three times and notes the number of heads; otherwise he tosses the coin once. He gets exactly one head. Find the probability that he did NOT get a 6 on the die.

1. E_1: six, $\frac{1}{6}$; E_2: not six, $\frac{5}{6}$
2. $P(A|E_1) = P(\text{one head in 3 tosses}) = \frac{3}{8}$; $P(A|E_2) = \frac{1}{2}$
3. $P(E_2|A) = \frac{\frac{5}{6}\times\frac{1}{2}}{\frac{1}{6}\times\frac{3}{8}+\frac{5}{6}\times\frac{1}{2}} = \frac{\frac{5}{12}}{\frac{1}{16}+\frac{5}{12}} = \frac{20}{3+20}$

✔ $P = \frac{20}{23}$

Solved Example 32

A card from a pack of 52 is lost. From the remaining 51 cards, two cards are drawn and both are kings. Find the probability that the lost card was a king.

1. E_1: lost card king, $\frac{4}{52}$; E_2: not king, $\frac{48}{52}$
2. $P(A|E_1) = \frac{3\times2}{51\times50}$ (3 kings left); $P(A|E_2) = \frac{4\times3}{51\times50}$
3. $P(E_1|A) = \frac{4\times6}{4\times6+48\times12} = \frac{24}{600}$

✔ $P = \frac{1}{25}$

Table method (Example on insurance)

E_i	$P(E_i)$	$P(A\|E_i)$	product
2-wheeler	0.3	0.02	0.006
car	0.5	0.01	0.005
truck	0.2	0.10	0.020
Total = P(A)			0.031

Bayes = (row you want) / total. Same as the tree, faster in exams.

Why Bayes works (3-line idea)

$P(E_i \cap A) = P(E_i)P(A|E_i)$ (multiplication rule) and also $= P(A)P(E_i|A)$. So $P(E_i|A) = \frac{P(E_i)P(A|E_i)}{P(A)}$, and $P(A)$ comes from the theorem of total probability.

Board tip Leave answers as simplified fractions unless the question uses decimals.

Quick fact Number of outcomes: n coins $= 2^n$, n dice $= 6^n$.

Solved Example 33

Two groups compete for the board of a company: group I wins with probability 0.55 and group II with 0.45. If group I wins, a new product is launched with probability 0.8; if group II wins, with probability 0.4. (i) Find the probability that the new product is launched. (ii) It was launched; find the probability that group II had won.

1. (i) $P(N) = 0.55\times0.8 + 0.45\times0.4 = 0.44 + 0.18 = 0.62$
2. (ii) $P(\text{II}|N) = \frac{0.18}{0.62} = \frac{18}{62}$

✔ (i) 0.62 (ii) $\frac{9}{31}$

Solved Example 34

In a school, sections A, B and C have 30, 40 and 30 students. 80%, 60% and 90% of them respectively pass a test. A student chosen at random has failed. Find the probability that the student is from section B.

1. $P(A) = 0.3$, $P(B) = 0.4$, $P(C) = 0.3$; F: fails
2. $P(F|A) = 0.2$, $P(F|B) = 0.4$, $P(F|C) = 0.1$
3. $P(F) = 0.06 + 0.16 + 0.03 = 0.25$
4. $P(B|F) = \frac{0.16}{0.25}$

✔ $P = \frac{16}{25}$

Solved Example 35

30% of the emails reaching an inbox are spam. A filter flags 90% of spam emails and also 5% of genuine emails. An email is flagged. Find the probability that it is really spam.

1. E_1: spam, 0.3; E_2: genuine, 0.7; A: flagged
2. $P(A|E_1) = 0.9$, $P(A|E_2) = 0.05$
3. $P(E_1|A) = \frac{0.27}{0.27+0.035} = \frac{0.27}{0.305} = \frac{270}{305}$

✔ $P = \frac{54}{61}$

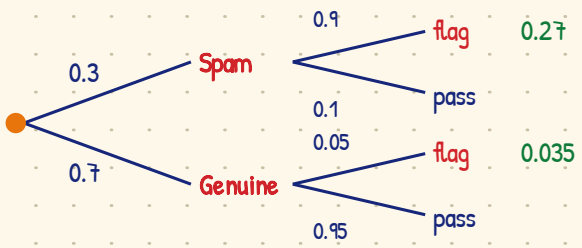


Fig: spam filter tree. P(flagged) = 0.27 + 0.035 = 0.305

Recall $P(A\cup B\cup C) = \sum P(A) - \sum P(A\cap B) + P(A\cap B\cap C)$.

Solved Example 36

Four machines M_1, M_2, M_3, M_4 make 10%, 20%, 30% and 40% of a factory's output, with defect rates 5%, 4%, 3% and 2%. A defective item is found. Find the probability that it was made by M_4.

1. Products: $0.10(0.05) = 0.005$, $0.20(0.04) = 0.008$, $0.30(0.03) = 0.009$, $0.40(0.02) = 0.008$
2. $P(D) = 0.005 + 0.008 + 0.009 + 0.008 = 0.030$
3. $P(M_4|D) = \dfrac{0.008}{0.030} = \dfrac{8}{30}$

✔ $P = \dfrac{4}{15}$

Model board answer (5 marks): how to present Bayes

Q. A factory has two machines. Machine I makes 60% of the items and Machine II makes 40%. 2% of Machine I's items and 5% of Machine II's items are faulty. An item is found faulty. Find the probability that it was made by Machine II.

1. Let E_1: item made by Machine I, E_2: item made by Machine II, A: item is faulty. (1 mark)
2. $P(E_1) = \dfrac{60}{100} = \dfrac{3}{5}$, $P(E_2) = \dfrac{2}{5}$. ($\frac{1}{2}$ mark)
3. $P(A|E_1) = \dfrac{2}{100}$, $P(A|E_2) = \dfrac{5}{100}$. ($\frac{1}{2}$ mark)
4. By Bayes' theorem, $P(E_2|A) = \dfrac{P(E_2)P(A|E_2)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2)}$ (1 mark)
5. $= \dfrac{\frac{2}{5} \times \frac{5}{100}}{\frac{3}{5} \times \frac{2}{100} + \frac{2}{5} \times \frac{5}{100}} = \dfrac{10}{6 + 10}$ (1 mark)
6. $= \dfrac{10}{16} = \dfrac{5}{8}$. (1 mark)



Shivam Bhaiya says
Bayes ka sawal board mein **5 marks ka pakka** aata hai, kabhi LA mein, kabhi case study mein. Events ka naam likhna (E_1, E_2, A) mat bhoolna, aur formula pehle likho, numbers baad mein. Steps ke marks milte hain.

6. Random Variable and its Distribution

How to make a distribution

1. Write the possible values of X (smallest to largest).
2. Find $P(X = x)$ for each value (count outcomes / multiply branches).
3. Check: probabilities add to 1. Then table → mean.

Definitions

→ **Random variable** X: a real-valued function on the sample space (a number attached to each outcome), e.g. X = number of heads.

→ **Probability distribution** of X: the table of its values $x_1, x_2, \ldots, x_n$ with probabilities $p_1, p_2, \ldots, p_n$, where

$$p_i \geq 0 \text{ and } \sum p_i = 1$$

→ **Mean / expectation**: $\mu = E(X) = \sum x_i p_i$

Solved Example 37

A coin is tossed three times. X = number of heads. Find the probability distribution of X and its mean.

1. X can be 0, 1, 2, 3 (8 equally likely outcomes)
2. $P(0) = \frac{1}{8}$ (TTT), $P(1) = \frac{3}{8}$, $P(2) = \frac{3}{8}$, $P(3) = \frac{1}{8}$ (HHH)
3. $E(X) = 0 \times \frac{1}{8} + 1 \times \frac{3}{8} + 2 \times \frac{3}{8} + 3 \times \frac{1}{8} = \frac{12}{8}$

✔ Mean $= \dfrac{3}{2}$

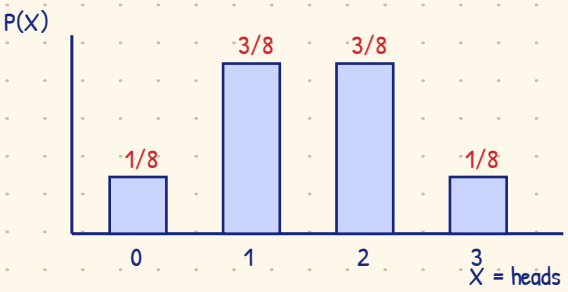


Fig: distribution of X = number of heads in 3 tosses (bars add up to 1)

Solved Example 38

X takes values 0, 1, 2, 3, 4 with probabilities 0.1, k, $2k$, $2k$, k. Find k, $P(X \geq 2)$ and $E(X)$.

1. Sum = 1: $0.1 + 6k = 1 \Rightarrow k = 0.15$
2. $P(X \geq 2) = 2k + 2k + k = 5k = 0.75$
3. $E(X) = 0 + 0.15 + 2(0.30) + 3(0.30) + 4(0.15)$
4. $= 0.15 + 0.60 + 0.90 + 0.60 = 2.25$

✔ $k = 0.15$, $P(X \geq 2) = 0.75$, $E(X) = 2.25$

Yaad rakho "Odds in favour $a : b$" means $P = \dfrac{a}{a+b}$.

Quick fact For any event, $P(E) + P(E') = 1$; for a partition, $\sum P(E_i) = 1$.

Solved Example 39

Two cards are drawn one by one **with replacement** from a pack of 52 cards. Find the probability distribution of the number of aces and its mean.

1. $P(\text{ace}) = \frac{1}{13}$, $P(\text{not ace}) = \frac{12}{13}$ each time
2. $P(0) = \frac{12}{13} \times \frac{12}{13} = \frac{144}{169}$
3. $P(1) = 2 \times \frac{1}{13} \times \frac{12}{13} = \frac{24}{169}$ (AN or NA)
4. $P(2) = \frac{1}{169}$; $E(X) = \frac{24+2}{169} = \frac{26}{169}$

✔ Mean $= \frac{2}{13}$

Solved Example 40

A lot of 10 items has 3 defective ones. Two items are drawn without replacement. X = number of defective items. Find the distribution and mean of X.

1. $P(0) = \frac{7}{10} \times \frac{6}{9} = \frac{42}{90} = \frac{7}{15}$
2. $P(1) = \frac{3}{10} \times \frac{7}{9} + \frac{7}{10} \times \frac{3}{9} = \frac{42}{90} = \frac{7}{15}$
3. $P(2) = \frac{3}{10} \times \frac{2}{9} = \frac{6}{90} = \frac{1}{15}$ (sum = 1 ✓)
4. $E(X) = \frac{7}{15} + \frac{2}{15} = \frac{9}{15}$

✔ Mean $= \frac{3}{5}$

Solved Example 41

Two dice are thrown. X = the larger of the two numbers. Find $P(X = k)$ and $E(X)$.

1. Max $= k$: pairs with both $\le k$ minus both $\le k-1$: $k^2 - (k-1)^2 = 2k - 1$
2. $P(X = k) = \frac{2k-1}{36}$, $k = 1, \ldots, 6$ (1, 3, 5, 7, 9, 11 out of 36)
3. $E(X) = \frac{1+6+15+28+45+66}{36} = \frac{161}{36}$

✔ $E(X) = \frac{161}{36} \approx 4.47$

Solved Example 42

A die is loaded so that each even number is twice as likely as each odd number. X = the number shown. Find $E(X)$.

1. Odd faces: p each, even faces: $2p$ each: $3p + 6p = 1 \Rightarrow p = \frac{1}{9}$
2. $E(X) = \frac{1+3+5}{9} + \frac{2(2+4+6)}{9} = \frac{9+24}{9} = \frac{33}{9}$

✔ $E(X) = \frac{11}{3}$

Board tip For "at least one", always try the complement first.

Solved Example 43

An urn has 5 red and 2 black balls. Two balls are drawn without replacement. X = number of black balls. Find the mean of X.

1. $P(0) = \frac{5}{7} \times \frac{4}{6} = \frac{20}{42}$, $P(1) = 2 \times \frac{5}{7} \times \frac{2}{6} = \frac{20}{42}$
2. $P(2) = \frac{2}{7} \times \frac{1}{6} = \frac{2}{42}$ (total $\frac{42}{42}$ ✓)
3. $E(X) = \frac{20}{42} + 2 \times \frac{2}{42} = \frac{24}{42}$

✔ Mean $= \frac{4}{7}$

Solved Example 44

A coin is tossed twice. X = (number of heads) − (number of tails). Find the distribution and mean of X.

1. HH: $X = 2$; HT, TH: $X = 0$; TT: $X = -2$
2. $P(2) = \frac{1}{4}$, $P(0) = \frac{1}{2}$, $P(-2) = \frac{1}{4}$
3. $E(X) = 2 \times \frac{1}{4} + 0 - 2 \times \frac{1}{4} = 0$

✔ Mean = 0 (symmetric, as expected)

Solved Example 45

A coin is tossed until a head appears or until it has been tossed 3 times. X = number of tosses. Find the distribution and mean of X.

1. $X = 1$: H, $P = \frac{1}{2}$; $X = 2$: TH, $P = \frac{1}{4}$
2. $X = 3$: TTH or TTT, $P = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$
3. $E(X) = 1 \times \frac{1}{2} + 2 \times \frac{1}{4} + 3 \times \frac{1}{4} = \frac{2+2+3}{4}$

✔ Mean $= \frac{7}{4}$

Yaad karo: "Distribution ka DM test"

Distribution valid tabhi jab har $p \ge 0$ aur Dabba total (sum) = 1. Mean = har value ko uski probability se Multiply karke jodo: $\sum xp$. Pehle k nikalo, phir kuch bhi.

Formula Box 2: Chapter at a glance

- $P(E|F) = \frac{P(E \cap F)}{P(F)}$, $P(F) \neq 0$
- $P(E'|F) = 1 - P(E|F)$
- $P(E \cap F) = P(E)P(F|E) = P(F)P(E|F)$
- Independent: $P(E \cap F) = P(E)P(F)$
- At least one (indep.): $1 - P(E')P(F')$
- $P(A) = \sum P(E_i)P(A|E_i)$ (total)
- $P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum P(E_j)P(A|E_j)}$
- $\sum p_i = 1$, $E(X) = \sum x_i p_i$

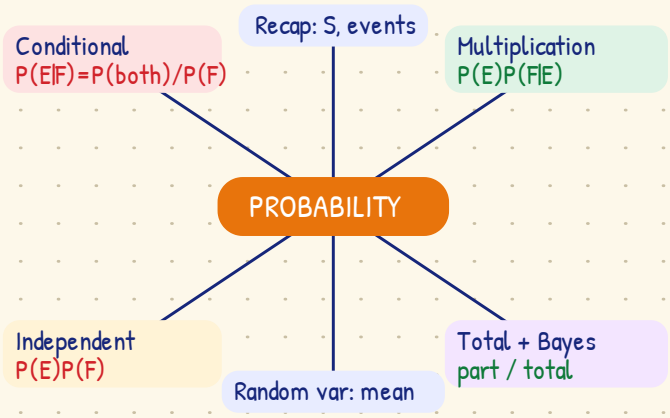


Fig: chapter mind map. Revise in this order.

7. Mixed Board Favourites

Solved Example 46

$P(A) = 0.6$, $P(B) = 0.5$ and $P(A' \cup B') = 0.7$. Are A and B independent?

1. $A' \cup B' = (A \cap B)'$, so $P(A \cap B) = 1 - 0.7 = 0.3$ (De Morgan)
2. $P(A)P(B) = 0.6 \times 0.5 = 0.3$

✔ Equal, so A and B are independent

Solved Example 47

Two dice are thrown. X = sum of the numbers. Find $E(X)$.

1. Number of ways for sum 2, 3, ..., 12: 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1
2. $\sum x \times (\text{ways}) = 2 + 6 + 12 + 20 + 30 + 42 + 40 + 36 + 30 + 22 + 12 = 252$
3. $E(X) = \dfrac{252}{36}$

✔ $E(X) = 7$

Sum	2	3	4	5	6	7	8	9	10	11	12
Ways	1	2	3	4	5	6	5	4	3	2	1

↳ Two dice ka sum table yaad rakho: 7 ke 6 ways, aur dono taraf 1-1 kam hota jaata hai.

Yaad karo: "Unknown bag = sab possibilities"

Bag ke andar kya hai pata nahi? Har possibility (2, 3 ya 4 white) ek hypothesis E_i hai. "Equally likely" diya ho to priors Bayes mein cancel.

Check trick If $P(A|B) > P(A)$, then B makes A more likely (A, B not independent).

Recall Face cards: 12 (J, Q, K of 4 suits); honour cards: 16 (A, J, Q, K).

Solved Example 48

A bag has 4 balls. Two are drawn at random (without replacement) and both are white. Assuming that 2, 3 or 4 white balls in the bag are equally likely, find the probability that all 4 balls are white.

1. E_2, E_3, E_4: bag has 2, 3, 4 white; each $\frac{1}{3}$
2. $P(A|E_2) = \frac{{}^2C_2}{{}^4C_2} = \frac{1}{6}$, $P(A|E_3) = \frac{{}^3C_2}{{}^4C_2} = \frac{3}{6}$, $P(A|E_4) = 1$
3. Equal priors cancel: $P(E_4|A) = \dfrac{1}{\frac{1}{6} + \frac{3}{6} + 1} = \dfrac{6}{10}$

✔ $P = \dfrac{3}{5}$

8. Distribution Practice (board type)

Solved Example 49

X = number of hours a student studies on a day: $P(X = 0) = 0.2$; $P(X = x) = kx$ for $x = 1, 2$; $P(X = x) = k(5 - x)$ for $x = 3, 4$; 0 otherwise. Find k, $P(X \geq 2)$ and $E(X)$.

1. Probabilities: 0.2, k, $2k$, $2k$, k; sum: $0.2 + 6k = 1 \Rightarrow k = \frac{2}{15}$
2. $P(X \geq 2) = 2k + 2k + k = 5k = \frac{10}{15} = \frac{2}{3}$
3. $E(X) = 1(k) + 2(2k) + 3(2k) + 4(k) = 15k = 2$

✔ $k = \dfrac{2}{15}$, $P(X \geq 2) = \dfrac{2}{3}$, $E(X) = 2$ hours

Solved Example 50

A lot of 15 bulbs has 5 defective ones. 3 bulbs are drawn one by one without replacement. Find the mean number of defective bulbs.

1. Total ordered draws: $15 \times 14 \times 13 = 2730$
2. $P(0) = \frac{10 \times 9 \times 8}{2730} = \frac{24}{91}$, $P(1) = \frac{3 \times 5 \times 10 \times 9}{2730} = \frac{45}{91}$
3. $P(2) = \frac{3 \times 5 \times 4 \times 10}{2730} = \frac{20}{91}$, $P(3) = \frac{5 \times 4 \times 3}{2730} = \frac{2}{91}$
4. $E(X) = \frac{45 + 40 + 6}{91} = \frac{91}{91}$

✔ Mean = 1

Solved Example 51

A die has faces marked 1, 1, 2, 2, 2, 3. X = the number on the top face. Find the distribution and mean of X.

1. $P(1) = \frac{2}{6}$, $P(2) = \frac{3}{6}$, $P(3) = \frac{1}{6}$
2. $E(X) = \frac{2 + 6 + 3}{6}$

✔ Mean $= \dfrac{11}{6}$

Solved Example 52

A fair die is rolled. A player wins Rs 20 for a 6, Rs 10 for a 4 or 5, and loses Rs 5 otherwise. Find the expected gain.

1. X = 20, 10, -5 with P = $\frac{1}{6}, \frac{2}{6}, \frac{3}{6}$
2. $E(X) = \frac{20+20-15}{6} = \frac{25}{6}$

✔ Expected gain = Rs $\frac{25}{6} \approx$ Rs 4.17

Last-minute checklist

- → "Given" ⇒ conditional; given event in the denominator.
- → Independent ⇒ multiply; mutually exclusive ⇒ $P(\text{both}) = 0$.
- → "From which bag / machine / person?" ⇒ Bayes (part / total).
- → "Probability that A, overall" with several causes ⇒ total probability.
- → Distribution ⇒ find k first, check $\sum p = 1$, then mean $\sum xp$.

Common mistakes

- → Writing $P(E|F) = \frac{P(E \cap F)}{P(E)}$: the **given** event goes below.
- → Saying mutually exclusive events are independent (they are not, when both probabilities are non-zero).
- → In Bayes, forgetting that $P(E_1) + P(E_2) + \ldots$ must be 1, or using $P(E_i|A)$ where $P(A|E_i)$ is needed.
- → Without replacement but using the same fraction twice.
- → Finding the mean before finding k, or not checking $\sum p = 1$.

Self-practice (answers in brackets)

1. $P(A) = 0.5, P(B) = 0.4, P(A \cap B) = 0.2$: find $P(A|B)$. [0.5]
2. A die is thrown. P(prime | odd)? [$\frac{2}{3}$]
3. Two cards drawn with replacement: P(both hearts)? [$\frac{1}{16}$]
4. A, B independent, $P(A) = 0.2, P(B) = 0.5$: $P(A \cup B)$? [0.6]
5. P(at least one six in two throws of a die)? [$\frac{11}{36}$]
6. X: 1, 2, 3 with P: 0.3, 0.4, k. Find k and $E(X)$. [0.3, 2]
7. Bag I: 1 red 1 black, Bag II: 3 red 1 black; bag at random, red drawn. P(Bag II)? [$\frac{3}{5}$]
8. Three coins tossed. P(all heads | first is head)? [$\frac{1}{4}$]

Quick fact Two events with $P(E) = 0$ or $P(E) = 1$ are independent of every event.

Most Asked in Boards

- → Bayes' theorem: bags / machines / truth-teller / test result / transport (5M or case study)
- → Find $P(A|B)$, $P(A \cup B)$ from given values (1M/2M)
- → Check whether two events are independent (2M)
- → Problem solved / target hit by at least one person (2M/3M)
- → Find k, then $P(X \le a)$ and mean of a distribution (3M/5M)
- → Distribution of number of aces / defectives / heads, with mean (3M/5M)
- → Case study on total probability + Bayes (4M)

Yaad karo: "Mutually exclusive = Milte hi nahi, Independent = Ignore karte hain"

Mutually exclusive events milte hi nahi (intersection khaali, $P = 0$). Independent events ek doosre ko ignore karte hain (ek hone se doosre ki probability nahi badalti), isliye multiply.



Shivam Bhaiya says

Distribution wale sawal mein table banana compulsory samjho: X ki values upar, P(X) neeche, aur last mein $\sum p = 1$ check. Mean ka answer fraction mein chhodo, decimal tabhi jab question mein decimal ho.

MOST IMPORTANT QUESTIONS

Board pattern | answers + solutions green mein

MCQ + Assertion-Reason (1 mark)

Q1. If $P(A) = 0.4$ and $P(B|A) = 0.6$, then $P(A \cap B)$ is 1M

(a) 0.2 (b) 0.24 (c) 0.3 (d) 0.6

Ans. (b) $P(A)P(B|A) = 0.4 \times 0.6 = 0.24$

Q2. A and B are independent with $P(A) = 0.3$ and $P(B) = 0.4$. Then $P(A \cup B)$ is 1M Board fav

(a) 0.7 (b) 0.12 (c) 0.58 (d) 0.82

Ans. (c) $0.3 + 0.4 - 0.12 = 0.58$

Q3. If A and B are mutually exclusive with $P(B) > 0$, then $P(A|B)$ is 1M

(a) 0 (b) 1 (c) $P(A)$ (d) $P(B)$

Ans. (a) $A \cap B = \phi$, so $P(A|B) = \frac{0}{P(B)} = 0$

Q4. A die is thrown. The probability that the number is 4, given that it is even, is 1M

(a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) $\frac{1}{3}$ (d) $\frac{2}{3}$

Ans. (c) $\frac{1/6}{3/6} = \frac{1}{3}$

Q5. X takes values 1, 2, 3 with $P(X=x)=kx$. The value of k is 1M Most asked

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) 1 (d) $\frac{1}{6}$

Ans. (d) $k+2k+3k=1 \Rightarrow k=\frac{1}{6}$

Q6. If A, B are independent, $P(A)=\frac{1}{2}$ and $P(B)=\frac{1}{3}$, then $P(A' \cap B')$ is 1M

(a) $\frac{1}{6}$ (b) $\frac{1}{3}$ (c) $\frac{2}{3}$ (d) $\frac{5}{6}$

Ans. (b) $\frac{1}{2} \times \frac{2}{3}=\frac{1}{3}$

Q7. X = number shown on a fair die. $E(X)$ is 1M

(a) $\frac{7}{2}$ (b) 3 (c) 4 (d) $\frac{21}{2}$

Ans. (a) $\frac{1+2+\cdots+6}{6}=\frac{21}{6}=\frac{7}{2}$

Q8. If $P(A|B)=0.3$ and $P(B)=0.5$, then $P(A \cap B)$ is 1M

(a) 0.6 (b) 0.8 (c) 0.15 (d) 0.2

Ans. (c) $0.3 \times 0.5=0.15$

Q9. If $P(E|F)=0.35$, then $P(E'|F)$ is 1M

(a) 0.35 (b) 0.5 (c) 0.65 (d) 1

Ans. (c) $1-0.35=0.65$

Q10. E and F are independent, $P(E)=0.3$ and $P(E \cup F)=0.5$. Then $P(E|F)-P(F|E)$ is 1M HOTS

(a) $\frac{1}{70}$ (b) $\frac{2}{7}$ (c) $\frac{3}{35}$ (d) $\frac{1}{7}$

Ans. (a) $0.3+p-0.3p=0.5 \Rightarrow p=\frac{2}{7}$. $P(E|F)=0.3, P(F|E)=\frac{2}{7}$; difference $=\frac{3}{10}-\frac{2}{7}=\frac{1}{70}$

Q11. A and B are independent with $P(A)=0.6$ and $P(B)=0.3$. Then $P(A \cap B')$ is 1M

(a) 0.18 (b) 0.42 (c) 0.28 (d) 0.12

Ans. (b) $0.6 \times 0.7=0.42$

Q12. Bag I has 2 red and 3 black balls, bag II has 4 red and 1 black ball. A bag is chosen at random and a ball is drawn. P(red) is 1M

(a) $\frac{2}{5}$ (b) $\frac{1}{2}$ (c) $\frac{3}{5}$ (d) $\frac{4}{5}$

Ans. (c) $\frac{1}{2}\left(\frac{2}{5}+\frac{4}{5}\right)=\frac{3}{5}$

Q13. X takes values 0, 1, 2 with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. $E(X)$ is 1M

(a) $\frac{1}{2}$ (b) 1 (c) $\frac{3}{2}$ (d) 2

Ans. (b) $0+\frac{1}{2}+\frac{2}{4}=1$

Q14. $P(A)=0.2$, $P(B)=0.3$ and $P(A \cup B)=0.5$. Then A and B are 1M

(a) independent (b) mutually exclusive (c) both (d) neither

Ans. (b) $P(A \cap B)=0.2+0.3-0.5=0$, so mutually exclusive (not independent, as $P(A)P(B)=0.06$).

Q15. $P(A)=0.5$, $P(B)=0.4$ and $P(A|B)=0.5$. Then $P(B|A)$ is 1M

(a) 0.4 (b) 0.5 (c) 0.2 (d) 0.8

Ans. (a) $P(A \cap B)=0.5 \times 0.4=0.2; P(B|A)=\frac{0.2}{0.5}=0.4$

Recall Two dice: P(doublet) $=\frac{6}{36}=\frac{1}{6}$; P(sum 7) $=\frac{1}{6}$ too.

Q16. Two cards are drawn one by one with replacement from a pack of 52. The probability that both are aces is 1M

(a) $\frac{1}{221}$ (b) $\frac{2}{13}$ (c) $\frac{1}{26}$ (d) $\frac{1}{169}$

Ans. (d) $\frac{1}{13} \times \frac{1}{13}=\frac{1}{169}$

Q17. For a random variable X, $P(X=0)=0.4$, $P(X=1)=0.6$. Then $E(X)$ is 1M

(a) 0.4 (b) 0.5 (c) 0.6 (d) 1

Ans. (c) $0 \times 0.4+1 \times 0.6=0.6$

Q18. If $P(B)=\frac{3}{5}$, $P(A|B)=\frac{1}{2}$ and $P(A \cup B)=\frac{4}{5}$, then $P(A)$ is 1M Board fav

(a) $\frac{3}{10}$ (b) $\frac{1}{2}$ (c) $\frac{1}{5}$ (d) $\frac{3}{5}$

Ans. (b) $P(A \cap B)=\frac{1}{2} \times \frac{3}{5}=\frac{3}{10}; P(A)=\frac{4}{5}-\frac{3}{5}+\frac{3}{10}=\frac{1}{2}$

Q19. A die is thrown. E = {2, 4, 6}, F = {1, 2}. Then E and F are 1M

(a) independent (b) mutually exclusive (c) exhaustive (d) equally likely

Ans. (a) $P(E \cap F)=\frac{1}{6}=\frac{1}{2} \times \frac{1}{3}=P(E)P(F)$

Q20. $P(X=x)=\frac{x}{6}$ for $x=1,2,3$. Then $E(X)$ is 1M

(a) 2 (b) $\frac{3}{2}$ (c) $\frac{13}{6}$ (d) $\frac{7}{3}$

Ans. (d) $\frac{1+4+9}{6}=\frac{14}{6}=\frac{7}{3}$

Q21. **A:** If E and F are independent, then E' and F' are also independent. **R:** $P(E' \cap F')=1-P(E \cup F)=P(E')P(F')$ when $P(E \cap F)=P(E)P(F)$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (a) Both true; R is exactly the proof of A.

Q22. **A:** Two mutually exclusive events with non-zero probabilities are independent. **R:** For independent events, $P(E \cap F)=P(E)P(F)$. 1M Board fav

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Ans. (d) A false ($0 \neq P(E)P(F)$), R true.

Very short (2 marks)

Q23. Given $P(A)=\frac{5}{11}$, $P(B)=\frac{6}{11}$ and $P(A \cup B)=\frac{8}{11}$, find $P(A|B)$ and $P(B|A)$. 2M Most asked

Ans. $P(A \cap B)=\frac{5+6-8}{11}=\frac{3}{11}$. $P(A|B)=\frac{3/11}{6/11}=\frac{1}{2}$, $P(B|A)=\frac{3}{5}$.

Q24. A box has 10 bulbs, of which 3 are defective. Two bulbs are drawn one by one without replacement. Find the probability that both are good. 2M

Ans. $\frac{7}{10} \times \frac{6}{9}=\frac{42}{90}=\frac{7}{15}$.

Q25. $P(A)=0.4$, $P(B)=0.5$ and $P(A \cup B)=0.7$. Are A and B independent? 2M Board fav

Ans. $P(A \cap B)=0.4+0.5-0.7=0.2$ and $P(A)P(B)=0.2$. Equal, so A and B are independent.

Q26. X takes values 0, 1, 2, 3 with probabilities $k, 3k, 3k, k$. Find k and $P(X<2)$. 2M

(contd.) $8k = 1 \Rightarrow k = \frac{1}{8}$. $P(X < 2) = k + 3k = \frac{4}{8} = \frac{1}{2}$.

Q27. Two coins are tossed. Find the probability of two heads, given that at least one head appears. 2M

Ans. F = {HH, HT, TH}, E ∩ F = {HH}. $P = \frac{1}{3}$.

Q28. A die is rolled. E = {1, 3, 5} and F = {2, 3}. Find $P(E|F)$ and $P(F|E)$. 2M

Ans. $E \cap F$ = {3}. $P(E|F) = \frac{1/6}{2/6} = \frac{1}{2}$, $P(F|E) = \frac{1/6}{3/6} = \frac{1}{3}$.

Q29. X takes values 1, 2, 3 with probabilities 0.2, 0.5, 0.3. Find the mean of X. 2M

Ans. $E(X) = 0.2 + 1.0 + 0.9 = 2.1$.

Q30. Two balls are drawn with replacement from a bag of 3 white and 5 black balls. Find the probability that the first is white and the second is black. 2M

Ans. Independent draws: $\frac{3}{8} \times \frac{5}{8} = \frac{15}{64}$.

Q31. If A and B are independent, prove that $P(A \cup B) = 1 - P(A')P(B')$. 2M

Ans. $P(A \cup B) = 1 - P((A \cup B)') = 1 - P(A' \cap B')$. A, B independent $\Rightarrow A', B'$ independent, so $P(A' \cap B') = P(A')P(B')$. Hence proved.

Q32. A die is thrown. Find the probability that the number is greater than 4, given that it is odd. 2M

Ans. F = {1, 3, 5}; greater than 4 in F: {5}. $P = \frac{1}{3}$.

Q33. A bag has 3 red and 5 black balls. Two balls are drawn one by one without replacement. Find the probability that the second ball is red, given that the first is black. 2M

Ans. After a black ball: 3 red out of 7 left. $P = \frac{3}{7}$.

Q34. X has distribution: X = -1, 0, 1 with P = 0.3, 0.4, a. Find a and $E(X)$. 2M

Ans. $a = 1 - 0.7 = 0.3$. $E(X) = -0.3 + 0 + 0.3 = 0$.

Q35. If $P(A) = \frac{3}{8}$, $P(B) = \frac{1}{2}$ and $P(A \cap B) = \frac{1}{4}$, find $P(A'|B')$. 2M Board fav

Ans. $P(A \cup B) = \frac{3}{8} + \frac{4}{8} - \frac{2}{8} = \frac{5}{8}$; $P(A' \cap B') = \frac{3}{8}$; $P(B') = \frac{1}{2}$. $P(A'|B') = \frac{3/8}{1/2} = \frac{3}{4}$.

Q36. Find the probability of getting a sum of 9 in a throw of two dice, given that the first die shows 5 or more. 2M

Ans. F: first die 5 or 6, $n(F) = 12$. Sum 9 in F: (5,4), (6,3). $P = \frac{2}{12} = \frac{1}{6}$.

Short answer (3 marks)

Q37. A and B throw a die alternately till one of them gets a 6 and wins. A starts. Find the probability of each winning. 3M HOTS

Ans. A wins on throws 1, 3, 5, ...: $P(A) = \frac{1}{6} + \left(\frac{5}{6}\right)^2 \frac{1}{6} + \left(\frac{5}{6}\right)^4 \frac{1}{6} + \ldots = \frac{1/6}{1 - 25/36} = \frac{6}{11}$. $P(B) = 1 - \frac{6}{11} = \frac{5}{11}$.

Q38. A bag has 4 white and 6 black balls. Three balls are drawn one by one without replacement. Find the probability that all three are black. 3M

Ans. $\frac{6}{10} \times \frac{5}{9} \times \frac{4}{8} = \frac{120}{720} = \frac{1}{6}$.

Q39. A die is thrown twice. Find the mean of the number of sixes. 3M Most asked

Ans. $P(0) = \frac{25}{36}$, $P(1) = 2 \times \frac{1}{6} \times \frac{5}{6} = \frac{10}{36}$, $P(2) = \frac{1}{36}$. $E(X) = \frac{10+2}{36} = \frac{1}{3}$.

Q40. A and B shoot at a target independently with probabilities $\frac{1}{3}$ and $\frac{1}{4}$ of hitting. Find the probability that (i) the target is hit (ii) exactly one of them hits it. 3M

Ans. (i) $1 - \frac{2}{3} \times \frac{3}{4} = 1 - \frac{1}{2} = \frac{1}{2}$. (ii) $\frac{1}{3} \times \frac{3}{4} + \frac{2}{3} \times \frac{1}{4} = \frac{3}{12} + \frac{2}{12} = \frac{5}{12}$.

Q41. In a class, 60% of the students are girls. 30% of the girls and 20% of the boys chose Maths as an elective. A student chosen at random has chosen Maths. Find the probability that the student is a girl. 3M

Ans. $P(M) = 0.6 \times 0.3 + 0.4 \times 0.2 = 0.18 + 0.08 = 0.26$. $P(G|M) = \frac{0.18}{0.26} = \frac{9}{13}$.

Q42. Two numbers are chosen at random, without replacement, from 1, 2, 3, 4, 5. X = the smaller of the two. Find the distribution and mean of X. 3M

Ans. 10 equally likely pairs. $P(X = 1) = \frac{4}{10}$, $P(2) = \frac{3}{10}$, $P(3) = \frac{2}{10}$, $P(4) = \frac{1}{10}$. $E(X) = \frac{4+6+6+4}{10} = 2$.

Q43. Bag A has 2 white and 3 red balls; bag B has 4 white and 1 red ball. One ball is moved from A to B, and then a ball is drawn from B. Find the probability that it is white. 3M

Ans. Moved white ($\frac{2}{5}$): B has 5W, 1R, $P = \frac{5}{6}$. Moved red ($\frac{3}{5}$): B has 4W, 2R, $P = \frac{4}{6}$. $P(W) = \frac{2}{5} \times \frac{5}{6} + \frac{3}{5} \times \frac{4}{6} = \frac{10+12}{30} = \frac{11}{15}$.

Q44. Two cards are drawn without replacement from a pack of 52. Find the mean of the number of kings. 3M Most asked

Ans. $P(0) = \frac{48}{52} \times \frac{47}{51} = \frac{188}{221}$, $P(1) = 2 \times \frac{4}{52} \times \frac{48}{51} = \frac{32}{221}$, $P(2) = \frac{4}{52} \times \frac{3}{51} = \frac{1}{221}$. $E(X) = \frac{32+2}{221} = \frac{2}{13}$.

Q45. $P(A) = 0.4$, $P(B) = 0.3$ and $P(A \cap B) = 0.2$. Find $P(A'|B')$. 3M

Ans. $P(A' \cap B') = 1 - P(A \cup B) = 1 - 0.5 = 0.5$; $P(B') = 0.7$. $P(A'|B') = \frac{0.5}{0.7} = \frac{5}{7}$.

Q46. $P(A) = 0.6$, $P(B) = 0.5$ and $P(B|A) = 0.5$. Find $P(A \cup B)$ and $P(A|B)$. 3M

Ans. $P(A \cap B) = 0.6 \times 0.5 = 0.3$. $P(A \cup B) = 0.6 + 0.5 - 0.3 = 0.8$. $P(A|B) = \frac{0.3}{0.5} = 0.6$.

Q47. A family has three children. Find the probability that all are boys, given that at least one is a boy. 3M

Ans. 8 equally likely outcomes; F (at least one boy) has 7. Only BBB is all boys. $P = \frac{1}{7}$.

Q48. Two numbers are drawn without replacement from 1, 2, 3, 4, 5, 6. X = the larger number. Find $E(X)$. 3M HOTS

(contd.) 15 equally likely pairs; $P(X=k)=\frac{k-1}{15}$ for $k=2,\ldots,6$. $E(X)=\frac{2(1)+3(2)+4(3)+5(4)+6(5)}{15}=\frac{70}{15}=\frac{14}{3}$.

Q49. Box I has 2 gold and 1 silver coin; box II has 1 gold and 2 silver coins. A box is chosen at random and a coin is drawn from it; the coin is gold. Find the probability that box I was chosen. 3M

Ans. $P(G|\mathrm{I})=\frac{2}{3}$, $P(G|\mathrm{II})=\frac{1}{3}$, priors equal. $P(\mathrm{I}|G)=\frac{2/3}{2/3+1/3}=\frac{2}{3}$.

Q50. A and B are independent, $P(A)=0.3$ and $P(B)=0.2$. Find the probability that exactly one of them occurs. 3M

Ans. $P(A)P(B')+P(A')P(B)=0.3\times0.8+0.7\times0.2=0.24+0.14=0.38$.

Q51. $P(A)=0.4$, $P(B)=0.5$, $P(A\cap B)=0.2$. Find $P\big(A|(A\cup B)\big)$. 3M HOTS

Ans. $A\cap(A\cup B)=A$. $P(A\cup B)=0.4+0.5-0.2=0.7$. $P=\frac{0.4}{0.7}=\frac{4}{7}$.

Q52. A bag has 5 white, 7 red and 8 black balls. Four balls are drawn one by one with replacement. Find the probability that none is white. 3M

Ans. $P(\text{not white})=\frac{15}{20}=\frac{3}{4}$ each draw, independent. $P=\left(\frac{3}{4}\right)^4=\frac{81}{256}$.

Q53. A and B are independent, $P(A)=0.3$ and $P(B)=0.4$. Find $P(A\cup B')$. 3M

Ans. A and B' are independent. $P(A\cup B')=0.3+0.6-0.3\times0.6=0.9-0.18=0.72$.

Q54. X takes values 1, 2, 3, 4, 5 with probabilities k, $2k$, $3k$, $2k$, $2k$. Find k, $P(X\le3)$ and $E(X)$. 3M

Ans. $10k=1\Rightarrow k=\frac{1}{10}$. $P(X\le3)=\frac{6}{10}=\frac{3}{5}$. $E(X)=\frac{1+4+9+8+10}{10}=3.2$.

Q55. A card is drawn from a pack of 52. E: the card is a spade, F: the card is an ace. Show that E and F are independent. 3M

Ans. $P(E)=\frac{13}{52}=\frac{1}{4}$, $P(F)=\frac{4}{52}=\frac{1}{13}$, $P(E\cap F)=\frac{1}{52}$ (ace of spades) $=\frac{1}{4}\times\frac{1}{13}$. Hence independent.

Q56. A couple has two children. Find the probability that both are boys, given that at least one of them is a boy. Also find the probability that both are boys if the younger child is a boy. 3M

Ans. F = {BB, BG, GB}: $P=\frac{1}{3}$. Younger is a boy: {BB, GB}: $P=\frac{1}{2}$.

Long answer (5 marks)

Q57. Urn I has 3 white and 2 red balls, Urn II has 2 white and 4 red balls, Urn III has 4 white and 1 red ball. An urn is chosen at random and two balls are drawn from it without replacement. Both are white. Find the probability that they came from Urn III. 5M Board fav

Ans. $P(E_i)=\frac{1}{3}$. $P(A|E_1)=\frac{{}^3C_2}{{}^5C_2}=\frac{3}{10}$, $P(A|E_2)$ (contd.) $=\frac{{}^2C_2}{{}^6C_2}=\frac{1}{15}$, $P(A|E_3)=\frac{{}^4C_2}{{}^5C_2}=\frac{6}{10}$. $P(E_3|A)=\frac{\frac{6}{10}}{\frac{3}{10}+\frac{1}{15}+\frac{6}{10}}=\frac{18/30}{29/30}=\frac{18}{29}$.

Q58. A man goes to office by car, metro or bus with probabilities $\frac{1}{5}$, $\frac{1}{2}$ and $\frac{3}{10}$. The probabilities that he is late are $\frac{1}{4}$, $\frac{1}{10}$ and $\frac{1}{3}$ respectively. (i) Find the probability that he is late on a given day. (ii) He was late one day; find the probability that he came by metro. 5M Most asked

Ans. (i) $P(L)=\frac{1}{5}\times\frac{1}{4}+\frac{1}{2}\times\frac{1}{10}+\frac{3}{10}\times\frac{1}{3}=\frac{1}{20}+\frac{1}{20}+\frac{1}{10}=\frac{1}{5}$. (ii) $P(\text{metro}|L)=\frac{1/20}{1/5}=\frac{1}{4}$.

Q59. $P(X=x)=kx$ for $x=1,2,3$; $P(X=4)=2k$; $P(X=x)=0$ otherwise. Find k, $P(X\le2)$, $P(X>2)$ and $E(X)$. 5M Must do

Ans. $k+2k+3k+2k=8k=1\Rightarrow k=\frac{1}{8}$. $P(X\le2)=\frac{3}{8}$, $P(X>2)=\frac{5}{8}$. $E(X)=\frac{1(1)+2(2)+3(3)+4(2)}{8}=\frac{22}{8}=\frac{11}{4}$.

Q60. A box has 2 gold coins and 3 silver coins; another box has 4 gold and 1 silver coin. A coin is tossed: heads, a coin is taken from box 1; tails, from box 2. The coin taken is gold. Find the probability that it came from box 2. Also find the probability that the coin is silver. 5M

Ans. $P(G)=\frac{1}{2}\times\frac{2}{5}+\frac{1}{2}\times\frac{4}{5}=\frac{6}{10}=\frac{3}{5}$. $P(B_2|G)=\frac{4/10}{6/10}=\frac{2}{3}$. $P(\text{silver})=1-\frac{3}{5}=\frac{2}{5}$.

Q61. Three candidates A, B, C apply for the post of manager; their chances of selection are in the ratio 2 : 3 : 5. If selected, the probabilities that they introduce a new policy are 0.6, 0.4 and 0.2. A new policy was introduced. Find the probability that it was due to C. 5M Most asked

Ans. $P(A)=0.2$, $P(B)=0.3$, $P(C)=0.5$. $P(N)=0.2(0.6)+0.3(0.4)+0.5(0.2)=0.12+0.12+0.10=0.34$. $P(C|N)=\frac{0.10}{0.34}=\frac{5}{17}$.

Q62. A die is thrown twice. "Success" means a number greater than 4. Find the probability distribution of the number of successes and its mean. 5M

Ans. $P(\text{success})=\frac{2}{6}=\frac{1}{3}$ each throw. $P(0)=\frac{2}{3}\times\frac{2}{3}=\frac{4}{9}$, $P(1)=2\times\frac{1}{3}\times\frac{2}{3}=\frac{4}{9}$, $P(2)=\frac{1}{9}$. Mean $=\frac{4}{9}+\frac{2}{9}=\frac{2}{3}$.

Q63. A shop gets bulbs from factories A (50%), B (30%) and C (20%). The probability of a bulb lasting over 2 years is 0.9, 0.8 and 0.6 respectively. (i) Find the probability that a bulb chosen at random lasts over 2 years. (ii) A bulb failed before 2 years; find the probability that it came from C. 5M Most asked

Ans. (i) $0.5(0.9)+0.3(0.8)+0.2(0.6)=0.45+0.24+0.12=0.81$. (ii) Fail: $0.5(0.1)+0.3(0.2)+0.2(0.4)=0.05+0.06+0.08=0.19$; $P(C|\text{fail})=\frac{0.08}{0.19}=\frac{8}{19}$.

Q64. A bag has one fair coin and one biased coin with $P(H) = \frac{2}{3}$. A coin is taken at random and tossed twice; both tosses show heads. Find the probability that it is the biased coin. 5M

> **Ans.** $P(HH|\text{fair}) = \frac{1}{4}$, $P(HH|\text{biased}) = \frac{4}{9}$, priors $\frac{1}{2}$ each (cancel). $P(\text{biased}|HH) = \frac{4/9}{1/4+4/9} = \frac{16}{25}$.

Q65. 2% of a population is infected with a virus. A test is positive for 98% of infected people and for 3% of healthy people. A person tests positive. Find the probability that the person is infected. Comment on the result. 5M

> **Ans.** $P(+) = 0.02 \times 0.98 + 0.98 \times 0.03 = 0.0196 + 0.0294 = 0.049$. $P(I|+) = \frac{0.0196}{0.049} = \frac{2}{5}$. Only 40%: since healthy people are so many, a confirmatory test is needed.

Case based (4 marks)

Q66. Case: LED bulbs. A shop gets LED bulbs from suppliers X, Y, Z in the ratio 5 : 3 : 2. Their bulbs are defective with probabilities 2%, 3% and 5%. (i) Find the probability that a bulb is from Y and defective. (1) (ii) Find the probability that a bulb is defective. (1) (iii) A bulb is defective; find the probability that it is from Z. (2) 4M Board fav

> **Ans.** (i) $0.3 \times 0.03 = 0.009$. (ii) $0.010 + 0.009 + 0.010 = 0.029$. (iii) $\frac{0.010}{0.029} = \frac{10}{29}$.

Q67. Case: Fair spinner. A spinner has 8 equal sectors marked 1, 1, 1, 2, 2, 3, 3, 5. X = the number it stops on. (i) Find $P(X = 1)$. (1) (ii) Find $P(X \geq 2)$. (1) (iii) Find the probability distribution and the mean of X. (2) 4M

> **Ans.** (i) $\frac{3}{8}$. (ii) $1 - \frac{3}{8} = \frac{5}{8}$. (iii) X: 1, 2, 3, 5 with P: $\frac{3}{8}, \frac{2}{8}, \frac{2}{8}, \frac{1}{8}$; $E(X) = \frac{3+4+6+5}{8} = \frac{18}{8} = \frac{9}{4}$.

Q68. Case: Survey. In a survey of 200 students, 120 like cricket, 90 like football and 50 like both. One student is picked at random. (i) Find $P(\text{cricket}|\text{football})$. (1) (ii) Find $P(\text{football}|\text{cricket})$. (1) (iii) Are "likes cricket" and "likes football" independent? Justify. (2) 4M

> **Ans.** (i) $\frac{50}{90} = \frac{5}{9}$. (ii) $\frac{50}{120} = \frac{5}{12}$. (iii) $P(C)P(F) = \frac{120}{200} \times \frac{90}{200} = \frac{27}{100}$, $P(C \cap F) = \frac{50}{200} = \frac{1}{4} = \frac{25}{100}$; not equal, so NOT independent.

Q69. Case: Parcel delivery. An online store uses courier partners P, Q and R for 40%, 35% and 25% of its parcels. They deliver late 5%, 4% and 8% of the time. (i) Find the probability that a parcel is sent by Q and is late. (1) (ii) Find the probability that a parcel is late. (1) (iii) A parcel is late; find the probability that it was sent by R. (2) 4M Board fav

> **Ans.** (i) $0.35 \times 0.04 = 0.014$. (ii) $0.020 + 0.014 + 0.020 = 0.054$. (iii) $\frac{0.020}{0.054} = \frac{10}{27}$.

Apne notes / galtiyan yahan likho

Q71. Case: Quiz show. A contestant answers 2 questions; each is answered correctly with probability 0.7, independently. Each correct answer wins Rs 100. X = winnings. (i) Find P(both correct). (1) (ii) Find P(at least one correct). (1) (iii) Find the distribution of X and the expected winnings. (2) 4M Must do

> (contd.) (i) $0.7 \times 0.7 = 0.49$. (ii) $1 - 0.3 \times 0.3 = 0.91$. (iii) X = 0, 100, 200 with P = 0.09, 0.42, 0.49; $E(X) = 42 + 98 =$ Rs 140.

Q70. Case: Rain and the match. The probability of rain tomorrow is 0.3. If it rains, the school match is held with probability 0.2; if it does not rain, with probability 0.9. (i) Find the probability that it rains and the match is not held. (1) (ii) Find the probability that the match is held. (1) (iii) The match was held; find the probability that it had rained. (2) 4M

> **Ans.** (i) $0.3 \times 0.8 = 0.24$. (ii) $0.3 \times 0.2 + 0.7 \times 0.9 = 0.06 + 0.63 = 0.69$. (iii) $\frac{0.06}{0.69} = \frac{6}{69} = \frac{2}{23}$.

Q72. Case: Coaching batches. A coaching centre has a morning batch (60% of its students) and an evening batch (40%). 70% of the morning students and 50% of the evening students score above 80 in a test. A student is picked at random. (i) Find P(student is from the evening batch and scores above 80). (1) (ii) Find P(student scores above 80). (1) (iii) The student scored above 80; find the probability that the student is from the morning batch. (2) 4M

> **Ans.** (i) $0.4 \times 0.5 = 0.20$. (ii) $0.6 \times 0.7 + 0.20 = 0.42 + 0.20 = 0.62$. (iii) $\frac{0.42}{0.62} = \frac{21}{31}$.

Khud karo: quick probability drills

1. $P(A) = 0.6$, $P(B) = 0.5$, independent. $P(A \cup B)$? [0.8]
2. $P(A \cap B) = 0.12$, $P(B) = 0.3$. $P(A|B)$? [0.4]
3. Card from a pack of 52. P(king | face card)? [$\frac{1}{3}$]
4. X: 0, 1, 2 with $p = \frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. $E(X)$? [1]
5. Coin tossed 3 times. P(exactly 2 heads | first is head)? [$\frac{1}{2}$]
6. Bag 3 red, 2 blue; two drawn with replacement. P(both blue)? [$\frac{4}{25}$]

Yaad rakho Mean is also called expectation: $E(X) = \mu = \sum x_i p_i$.

Board tip Name events with capital letters and define them in words before using them.

Quick fact $P(A|B)$ and $P(B|A)$ are equal only when $P(A) = P(B)$ (or $P(A \cap B) = 0$).

Apne notes / galtiyan yahan likho

FORMULA SHEET All 13 chapters

Exam se pehle: har box ek baar padho, ek baar likh ke dekho

Saare Formulas, Ek Jagah

1. Relations and Functions

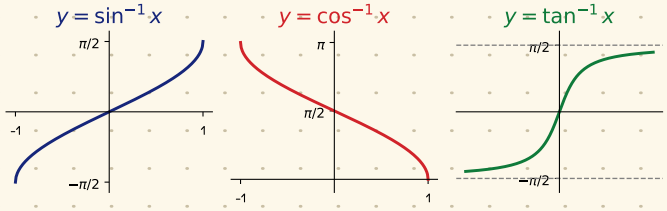
- Relation from A to B: any subset of $A \times B$; on set A: subset of $A \times A$
- Reflexive: $(a, a) \in R$ for every $a \in A$
- Symmetric: $(a, b) \in R \Rightarrow (b, a) \in R$
- Transitive: $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$
- Equivalence = reflexive + symmetric + transitive; equivalence class $[a] = \{x : (x, a) \in R\}$; classes are disjoint and cover A
- Empty relation $R = \phi$; universal relation $R = A \times A$
- One-one (injective): $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$
- Onto (surjective): range = codomain (every y has a pre-image)
- Bijective = one-one + onto
- If $n(A) = m, n(B) = n$: number of relations $A \to B = 2^{mn}$; relations on $A = 2^{m^2}$; functions $A \to B = n^m$
- One-one functions $A \to B$ $(m \le n) = {}^nP_m$; bijections $A \to A = m!$
- Finite set A, $f : A \to A$: one-one $\Leftrightarrow$ onto
- Strictly increasing/decreasing function on $\mathbb{R}$ is one-one; polynomial of even degree $\mathbb{R} \to \mathbb{R}$ is never onto

2. Inverse Trigonometric Functions

Function	Domain	Principal range
$\sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$
$\sec^{-1} x$	$\lvert x\rvert \ge 1$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$
$\csc^{-1} x$	$\lvert x\rvert \ge 1$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

2. Inverse Trig: key results

- Negative inputs: $\sin^{-1}(-x) = -\sin^{-1} x$, $\tan^{-1}(-x) = -\tan^{-1} x$, $\csc^{-1}(-x) = -\csc^{-1} x$
- $\cos^{-1}(-x) = \pi - \cos^{-1} x$, $\cot^{-1}(-x) = \pi - \cot^{-1} x$, $\sec^{-1}(-x) = \pi - \sec^{-1} x$
- $\sin^{-1}(\sin x) = x$ only for $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$; else shift into the principal range first
- Values to know: $\sin^{-1}\frac{1}{2} = \frac{\pi}{6}$, $\cos^{-1}\frac{1}{2} = \frac{\pi}{3}$, $\tan^{-1} 1 = \frac{\pi}{4}$, $\tan^{-1}\sqrt{3} = \frac{\pi}{3}$, $\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$



3. Matrices

- Order $m \times n$: m rows, n columns; $A = [a_{ij}]$
- Square $(m = n)$, diagonal, scalar, identity I, zero O, row and column matrices
- $A + B$ needs same order; AB needs (columns of A) = (rows of B); order of AB = $m \times p$
- $AB \neq BA$ in general; $AB = O$ does not mean $A = O$ or $B = O$
- $(A')' = A$, $(kA)' = kA'$, $(A + B)' = A' + B'$, $(AB)' = B'A'$
- Symmetric: $A' = A$; skew-symmetric: $A' = -A$ (diagonal elements all 0)
- $A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$ = symmetric + skew-symmetric
- A invertible if $AB = BA = I$; inverse is unique; $(AB)^{-1} = B^{-1}A^{-1}$, $(A^{-1})' = (A')^{-1}$
- Number of matrices of order $m \times n$ with entries 0 or 1 $= 2^{mn}$

4. Determinants

- $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$; 3×3: expand along any row/column with signs $\begin{matrix} + & - & + \\ - & + & - \\ + & - & + \end{matrix}$
- Minor M_{ij} = det after deleting row i, column j; cofactor $A_{ij} = (-1)^{i+j} M_{ij}$
- $|A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$; elements of one row $\times$ cofactors of another row $= 0$
- adj A = transpose of cofactor matrix; $A(\text{adj}A) = (\text{adj}A)A = |A|I$

→ $A^{-1} = \frac{1}{|A|}\operatorname{adj} A$, exists iff $|A| \neq 0$ (non-singular)

→ 2×2: $\operatorname{adj}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

→ $|kA| = k^n|A|$, $|AB| = |A||B|$, $|A'| = |A|$, $|A^{-1}| = \frac{1}{|A|}$, $|\operatorname{adj} A| = |A|^{n-1}$

→ Area of triangle $= \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|$; area 0 $\Leftrightarrow$ collinear

→ $AX = B$: $|A| \neq 0 \Rightarrow$ unique solution $X = A^{-1}B$ (consistent)

→ $|A| = 0$: find $(\operatorname{adj} A)B$; $\neq O \Rightarrow$ no solution (inconsistent); $= O \Rightarrow$ no solution or infinitely many

5. Continuity and Differentiability

→ Continuous at a: $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = f(a)$ (LHL = RHL = value)

→ Differentiable at a: $Lf'(a) = Rf'(a)$, where $Rf'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$

→ Differentiable $\Rightarrow$ continuous (converse false: $|x|$ at 0)

→ Sum, difference, product, quotient (denominator $\neq 0$) and composition of continuous functions are continuous

→ Polynomials, $\sin x$, $\cos x$, e^x continuous everywhere; $\log x$ for $x > 0$

→ Chain rule: $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$

→ Parametric: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$; $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}$

→ Log differentiation for $y = [u(x)]^{v(x)}$: $\log y = v \log u$, then differentiate

→ Implicit: differentiate both sides w.r.t. x, treat y as a function of x, collect $\frac{dy}{dx}$

Derivatives you must know

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
x^n	nx^{n-1}	$\sin x$	$\cos x$
e^x	e^x	$\cos x$	$-\sin x$
a^x	$a^x \log a$	$\tan x$	$\sec^2 x$
$\log x$	$\frac{1}{x}$	$\cot x$	$-\csc^2 x$
$\log_a x$	$\frac{1}{x \log a}$	$\sec x$	$\sec x \tan x$
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	$\csc x$	$-\csc x \cot x$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}$
$\tan^{-1} x$	$\frac{1}{1+x^2}$	$\cot^{-1} x$	$\frac{-1}{1+x^2}$
$\sec^{-1} x$	$\frac{1}{\lvert x\rvert\sqrt{x^2-1}}$	$\csc^{-1} x$	$\frac{-1}{\lvert x\rvert\sqrt{x^2-1}}$

→ Product: $(uv)' = u'v + uv'$; quotient: $\left(\frac{u}{v}\right)' = \frac{u'v-uv'}{v^2}$

6. Application of Derivatives

→ Rate of change of y w.r.t. $x = \frac{dy}{dx}$; related rates: $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$

→ $f'(x) > 0$ on (a, b): strictly increasing; $f'(x) < 0$: strictly decreasing; $f'(x) = 0$ on interval: constant

→ Critical points: $f'(c) = 0$ or $f'(c)$ does not exist

→ First derivative test: f' changes $+ \to -$: local max; $- \to +$: local min; no change: point of inflection

→ Second derivative test ($f'(c) = 0$): $f''(c) < 0$ max, $f''(c) > 0$ min, $f''(c) = 0$ test fails

→ Absolute max/min on $[a, b]$: compare f at critical points and at end points a, b

→ Useful: circle $A = \pi r^2$, sphere $V = \frac{4}{3}\pi r^3$, $S = 4\pi r^2$; cylinder $V = \pi r^2 h$, CSA $= 2\pi rh$; cone $V = \frac{1}{3}\pi r^2 h$, $l^2 = r^2 + h^2$; cube $V = x^3$, $S = 6x^2$

7. Integrals: standard results

$\int f(x)\,dx$	$=$
$\int x^n\,dx\ (n \neq -1)$	$\frac{x^{n+1}}{n+1} + C$
$\int \frac{1}{x}\,dx$	$\log\lvert x\rvert + C$
$\int e^x\,dx$; $\int a^x\,dx$	$e^x + C$; $\frac{a^x}{\log a} + C$
$\int \sin x\,dx$; $\int \cos x\,dx$	$-\cos x + C$; $\sin x + C$
$\int \sec^2 x\,dx$; $\int \csc^2 x\,dx$	$\tan x + C$; $-\cot x + C$
$\int \sec x \tan x\,dx$; $\int \csc x \cot x\,dx$	$\sec x + C$; $-\csc x + C$
$\int \tan x\,dx$	$\log\lvert\sec x\rvert + C$
$\int \cot x\,dx$	$\log\lvert\sin x\rvert + C$
$\int \sec x\,dx$	$\log\lvert\sec x + \tan x\rvert + C$

$\int f(x)\,dx$	$=$
$\int \csc x\,dx$	$\log\lvert\csc x-\cot x\rvert+C$

7. Integrals: special forms

- $\int \frac{dx}{x^2-a^2}=\frac{1}{2a}\log\left\lvert\frac{x-a}{x+a}\right\rvert+C$
- $\int \frac{dx}{a^2-x^2}=\frac{1}{2a}\log\left\lvert\frac{a+x}{a-x}\right\rvert+C$
- $\int \frac{dx}{x^2+a^2}=\frac{1}{a}\tan^{-1}\frac{x}{a}+C$
- $\int \frac{dx}{\sqrt{a^2-x^2}}=\sin^{-1}\frac{x}{a}+C$
- $\int \frac{dx}{\sqrt{x^2-a^2}}=\log\left\lvert x+\sqrt{x^2-a^2}\right\rvert+C$
- $\int \frac{dx}{\sqrt{x^2+a^2}}=\log\left\lvert x+\sqrt{x^2+a^2}\right\rvert+C$
- $\int \sqrt{a^2-x^2}\,dx=\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\frac{x}{a}+C$
- $\int \sqrt{x^2-a^2}\,dx=\frac{x}{2}\sqrt{x^2-a^2}-\frac{a^2}{2}\log\left\lvert x+\sqrt{x^2-a^2}\right\rvert+C$
- $\int \sqrt{x^2+a^2}\,dx=\frac{x}{2}\sqrt{x^2+a^2}+\frac{a^2}{2}\log\left\lvert x+\sqrt{x^2+a^2}\right\rvert+C$
- ax^2+bx+c in denominator: complete the square; $(px+q)$ on top: write $px+q=A\frac{d}{dx}(ax^2+bx+c)+B$

7. Integrals: methods

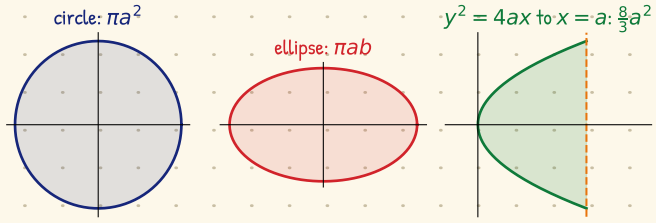
- Substitution: $\int f(g(x))g'(x)\,dx=\int f(t)\,dt$, $t=g(x)$
- $\int \frac{f'(x)}{f(x)}dx=\log\lvert f(x)\rvert+C$; $\int[f(x)]^n f'(x)\,dx=\frac{[f(x)]^{n+1}}{n+1}+C$
- By parts: $\int u\,v\,dx=u\int v\,dx-\int\left(u'\int v\,dx\right)dx$; choose u by ILATE (Inverse trig, Log, Algebraic, Trig, Exponential)
- $\int e^x[f(x)+f'(x)]\,dx=e^x f(x)+C$
- Partial fractions: $\frac{px+q}{(x-a)(x-b)}=\frac{A}{x-a}+\frac{B}{x-b}$; $\frac{px+q}{(x-a)^2}=\frac{A}{x-a}+\frac{B}{(x-a)^2}$; $\frac{px^2+qx+r}{(x-a)(x^2+bx+c)}=\frac{A}{x-a}+\frac{Bx+C}{x^2+bx+c}$
- Improper fraction (degree top $\geq$ bottom): divide first
- Trig: $\sin^2x=\frac{1-\cos 2x}{2}$, $\cos^2x=\frac{1+\cos 2x}{2}$, $\sin^3x=\frac{3\sin x-\sin 3x}{4}$, $2\sin A\cos B=\sin(A+B)+\sin(A-B)$

7. Definite integrals: properties

- FTC: $\int_a^b f(x)\,dx=F(b)-F(a)$ where $F'=f$
- $\int_a^b f(x)\,dx=-\int_b^a f(x)\,dx$; $\int_a^a f=0$; $\int_a^b f(x)\,dx=\int_a^b f(t)\,dt$
- $\int_a^b f=\int_a^c f+\int_c^b f$ (use for $\lvert x\rvert$, $[x]$, piecewise)
- King: $\int_a^b f(x)\,dx=\int_a^b f(a+b-x)\,dx$; $\int_0^a f(x)\,dx=\int_0^a f(a-x)\,dx$
- $\int_0^{2a} f(x)\,dx=2\int_0^a f(x)\,dx$ if $f(2a-x)=f(x)$; $=0$ if $f(2a-x)=-f(x)$
- $\int_{-a}^a f(x)\,dx=2\int_0^a f(x)\,dx$ (f even); $=0$ (f odd)
- $\int_0^{\pi/2}\log\sin x\,dx=\int_0^{\pi/2}\log\cos x\,dx=-\frac{\pi}{2}\log 2$
- $\int_0^{\pi/2}\frac{\sin^n x}{\sin^n x+\cos^n x}dx=\frac{\pi}{4}$ (king property)

8. Application of Integrals

- Area between curve, x-axis, $x=a$, $x=b$: $A=\int_a^b\lvert y\rvert\,dx$
- Area between curve, y-axis, $y=c$, $y=d$: $A=\int_c^d\lvert x\rvert\,dy$
- Part below the axis gives a negative integral: take modulus of each part separately
- Circle $x^2+y^2=a^2$: area $=\pi a^2$ (by integration: $4\int_0^a\sqrt{a^2-x^2}\,dx$)
- Ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$: area $=\pi ab=4\int_0^a\frac{b}{a}\sqrt{a^2-x^2}\,dx$
- Parabola $y^2=4ax$ and latus rectum $x=a$: area $=\frac{8}{3}a^2$
- Steps: draw the figure, find intersection points, use symmetry, then integrate





Meme samjho

Jab exam se pehle raat ko Formula Sheet poori revise ho gayi

Asli baat: Formula yaad ho to aadha paper wahi solve kar deta hai. Raat ko 2 baar padho aur 7 ghante ki neend lo.

9. Differential Equations

- Order = order of the highest derivative; degree = power of the highest derivative (DE must be a polynomial in derivatives, else degree not defined)

- General solution: arbitrary constants = order; particular solution: no arbitrary constant
- Variables separable: $\frac{dy}{dx} = f(x)g(y) \Rightarrow \int \frac{dy}{g(y)} = \int f(x)\,dx + C$
- Homogeneous: $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$ (check $F(\lambda x, \lambda y) = \lambda^0 F(x,y)$): put $y = vx$, $\frac{dy}{dx} = v + x\frac{dv}{dx}$
- If $\frac{dx}{dy} = F\left(\frac{x}{y}\right)$: put $x = vy$
- Linear: $\frac{dy}{dx} + Py = Q$; I.F. $= e^{\int P\,dx}$; solution $y \times \text{I.F.} = \int (Q \times \text{I.F.})\,dx + C$
- Linear in x: $\frac{dx}{dy} + P_1 x = Q_1$; I.F. $= e^{\int P_1\,dy}$; $x \times \text{I.F.} = \int (Q_1 \times \text{I.F.})\,dy + C$
- $e^{\log f(x)} = f(x)$, $e^{-\log x} = \frac{1}{x}$ (I.F. shortcut)

10. Vector Algebra

- $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, $|\vec{r}| = \sqrt{x^2+y^2+z^2}$; unit vector $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$
- $\overrightarrow{AB} = \vec{b} - \vec{a}$ (position vector of B minus that of A)
- Direction cosines $l = \frac{x}{r}, m = \frac{y}{r}, n = \frac{z}{r}$; $l^2 + m^2 + n^2 = 1$
- Section formula: internal $\frac{m\vec{b}+n\vec{a}}{m+n}$; external $\frac{m\vec{b}-n\vec{a}}{m-n}$; midpoint $\frac{\vec{a}+\vec{b}}{2}$
- Dot product: $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta = a_1b_1 + a_2b_2 + a_3b_3$
- $\vec{a} \perp \vec{b} \Leftrightarrow \vec{a}\cdot\vec{b} = 0$; $\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$; $\vec{a}\cdot\vec{a} = |\vec{a}|^2$
- Projection of $\vec{a}$ on $\vec{b} = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$
- Cross product: $\vec{a}\times\vec{b} = |\vec{a}||\vec{b}|\sin\theta\,\hat{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$
- $\vec{a}\times\vec{b} = -\vec{b}\times\vec{a}$; $\vec{a} \parallel \vec{b} \Leftrightarrow \vec{a}\times\vec{b} = \vec{0}$; $\hat{i}\times\hat{j} = \hat{k}, \hat{j}\times\hat{k} = \hat{i}, \hat{k}\times\hat{i} = \hat{j}$
- Unit vector perpendicular to both: $\pm\frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$
- Area of parallelogram (adjacent sides) $= |\vec{a}\times\vec{b}|$; (diagonals) $= \frac{1}{2}|\vec{d}_1 \times \vec{d}_2|$; triangle $= \frac{1}{2}|\vec{a}\times\vec{b}|$
- $|\vec{a}+\vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a}\cdot\vec{b}$; $|\vec{a}\times\vec{b}|^2 + (\vec{a}\cdot\vec{b})^2 = |\vec{a}|^2|\vec{b}|^2$

11. Three Dimensional Geometry (lines)

- DRs of line through $(x_1,y_1,z_1), (x_2,y_2,z_2)$: $x_2-x_1, y_2-y_1, z_2-z_1$
- DCs from DRs a, b, c: $\pm\frac{a}{\sqrt{a^2+b^2+c^2}}$, etc.
- Line through $\vec{a}$ parallel to $\vec{b}$: $\vec{r} = \vec{a} + \lambda\vec{b}$; Cartesian $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$
- Through two points: $\vec{r} = \vec{a} + \lambda(\vec{b}-\vec{a})$; $\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$
- Angle between lines: $\cos\theta = \left|\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}\right|$
- Perpendicular: $a_1a_2 + b_1b_2 + c_1c_2 = 0$; parallel: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
- Skew lines: neither parallel nor intersecting
- Shortest distance: $d = \left|\frac{(\vec{b}_1\times\vec{b}_2)\cdot(\vec{a}_2-\vec{a}_1)}{|\vec{b}_1\times\vec{b}_2|}\right|$; $d = 0 \Rightarrow$ lines intersect
- Parallel lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}$, $\vec{r} = \vec{a}_2 + \mu\vec{b}$: $d = \left|\frac{\vec{b}\times(\vec{a}_2-\vec{a}_1)}{|\vec{b}|}\right|$
- Foot of perpendicular from P to a line: general point Q on line, then $\overrightarrow{PQ}\cdot\vec{b} = 0$

12. Linear Programming

- Objective function $Z = ax + by$ (to maximise/minimise); constraints = linear inequalities; $x \geq 0, y \geq 0$ non-negativity
- Feasible region = common region of all constraints; each point in it = feasible solution
- Corner point theorem: optimal value (if it exists) occurs at a corner point of the feasible region
- Bounded region: max and min both exist at corner points
- Unbounded region: if max is M at a corner, check whether $ax + by > M$ has any common point with the region; if none, M is the max (similarly $ax + by < m$ for min)
- Two corners give the same optimal value: every point on the segment joining them is optimal
- Steps: draw lines (intercept form), shade, list corners, make Z table, conclude

13. Probability

- $P(A|B) = \frac{P(A\cap B)}{P(B)}$, $P(B) \neq 0$; $0 \leq P(A|B) \leq 1$; $P(A'|B) = 1 - P(A|B)$
- Multiplication: $P(A\cap B) = P(A)P(B|A) = P(B)P(A|B)$
- Independent: $P(A\cap B) = P(A)P(B)$ (then A', B' etc. also independent)

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; $P(\text{at least one}) = 1 - P(A')P(B')$ for independent
- Mutually exclusive: $P(A \cap B) = 0$ (not the same as independent!)
- Total probability: $P(A) = \sum P(E_i)P(A|E_i)$ (E_i partition the sample space)
- Bayes: $P(E_i|A) = \dfrac{P(E_i)P(A|E_i)}{\sum_j P(E_j)P(A|E_j)}$
- Random variable X: $\sum p_i = 1$, $0 \le p_i \le 1$
- Mean (expectation) $E(X) = \mu = \sum x_i p_i$

Trigonometry you will need

- $\sin^2 x + \cos^2 x = 1$, $1 + \tan^2 x = \sec^2 x$, $1 + \cot^2 x = \csc^2 x$
- $\sin 2x = 2\sin x \cos x = \frac{2\tan x}{1+\tan^2 x}$; $\cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1 = \frac{1-\tan^2 x}{1+\tan^2 x}$
- $\tan 2x = \frac{2\tan x}{1-\tan^2 x}$; $\sin 3x = 3\sin x - 4\sin^3 x$; $\cos 3x = 4\cos^3 x - 3\cos x$
- $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$; $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
- $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$
- $1 + \cos x = 2\cos^2 \frac{x}{2}$, $1 - \cos x = 2\sin^2 \frac{x}{2}$, $\sin x = 2\sin \frac{x}{2} \cos \frac{x}{2}$

Algebra + limits you will need

- $\log(mn) = \log m + \log n$, $\log \frac{m}{n} = \log m - \log n$, $\log m^n = n \log m$, $e^{\log x} = x$
- $\lim_{x\to 0} \frac{\sin x}{x} = 1$, $\lim_{x\to 0} \frac{\tan x}{x} = 1$, $\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$
- $\lim_{x\to 0} \frac{e^x - 1}{x} = 1$, $\lim_{x\to 0} \frac{\log(1+x)}{x} = 1$, $\lim_{x\to a} \frac{x^n - a^n}{x - a} = na^{n-1}$
- $(a+b)^2$, $(a-b)^2$, $a^2 - b^2 = (a-b)(a+b)$, $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$
- ${}^nC_r = \frac{n!}{r!(n-r)!}$ (for counting in probability)

Trig values table

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	not defined

- $\sin(\pi - x) = \sin x$, $\cos(\pi - x) = -\cos x$, $\tan(\pi - x) = -\tan x$
- $\sin\left(\frac{\pi}{2} - x\right) = \cos x$, $\cos\left(\frac{\pi}{2} - x\right) = \sin x$, $\tan\left(\frac{\pi}{2} - x\right) = \cot x$

Yaad karo: kaunsa integration method?

- **Function + uska derivative** dikh raha hai: substitution (t = andar wala)
- **Rational function**, denominator factor ho raha: partial fractions
- **Do alag type ke functions ka product** ($x \sin x$, xe^x, $\log x$ akela): by parts, ILATE
- $e^x(\dots)$ aur bracket mein $f + f'$: seedha $e^x f(x)$
- $\sqrt{}$ **ya quadratic** neeche: complete the square, special form table
- **Limits** 0 **to** a **ya** $-a$ **to** a: pehle king property / odd-even check karo

Definite integrals: ready answers

- $\int_0^{\pi/2} \sin x\,dx = \int_0^{\pi/2} \cos x\,dx = 1$
- $\int_0^{\pi/2} \sin^2 x\,dx = \int_0^{\pi/2} \cos^2 x\,dx = \frac{\pi}{4}$
- $\int_0^{\pi} \sin x\,dx = 2$; $\int_0^{2\pi} \sin x\,dx = 0$
- $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}}\,dx = \frac{\pi}{4}$
- $\int_0^{\pi/4} \log(1 + \tan x)\,dx = \frac{\pi}{8}\log 2$
- $\int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4}$; $\int_0^1 \frac{dx}{\sqrt{1-x^2}} = \frac{\pi}{2}$
- $\int_{-1}^{1} x^3 \cos x\,dx = 0$ (odd function)
- $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x}\,dx = \frac{\pi^2}{4}$

DE: type pehchano (identify the type)

Look	Type	First step
x terms and y terms alag ho sakte	Variables separable	separate, integrate
Har term same degree ($x^2 + y^2$, xy)	Homogeneous	$y = vx$
$\frac{dy}{dx} + Py = Q$, y ki power 1	Linear in y	I.F. $= e^{\int P\,dx}$
$\frac{dx}{dy} + P_1 x = Q_1$	Linear in x	I.F. $= e^{\int P_1\,dy}$

Matrices + determinants: quick results

- A is $n \times n$: $|\text{adj}A| = |A|^{n-1}$, $\text{adj}(\text{adj}A) = |A|^{n-2}A$, $|kA| = k^n|A|$
- $A^2 - 5A + 7I = O$ type: multiply by A^{-1} to get $A^{-1} = \frac{1}{7}(5I - A)$

→ $(A^{-1})^{-1} = A$; $(kA)^{-1} = \frac{1}{k}A^{-1}$; $I^{-1} = I$
→ Skew-symmetric of odd order: $|A| = 0$; diagonal matrix: $|A|$ = product of diagonal entries
→ $(AB)' = B'A'$ (order reverses!); $A'A$ and AA' are always symmetric

Vectors + 3D: quick results

→ Collinear points A, B, C: $\overrightarrow{AB} = \lambda\overrightarrow{AC}$
→ $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ with $|\vec{a}|, |\vec{b}|, |\vec{c}|$ given: square it to get $\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a} = -\frac{1}{2}(|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2)$
→ Angle with axes α, β, γ: $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$, $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$
→ Line $\frac{x-1}{2} = \frac{y+3}{-1} = \frac{z}{4}$: point $(1, -3, 0)$, DRs $2, -1, 4$ (sign of x, y, z coefficient must be $+1$)
→ Image of a point in a line: foot F first, then image $= 2F - P$

Probability: question patterns

→ "Given that", "if it is known": conditional probability $P(A|B)$
→ "Found to be defective, find probability it came from machine A": Bayes
→ "Probability that the item is defective" (from several sources): total probability
→ "Both hit / at least one hits" with independent shooters: multiply, or $1-$ (none)
→ Two cards / balls **without replacement**: second probability changes; **with replacement**: same
→ Probability distribution table: check $\sum p_i = 1$ first, then find k, then mean $\sum x_i p_i$

LPP: draw karne ke steps

1. Har inequality ko equation banao, intercepts nikalo ($x = 0$, $y = 0$ rakh ke)
2. Test point $(0, 0)$ daal ke side decide karo (line $(0, 0)$ se guzre to $(1, 0)$ lo)
3. Common shaded region = feasible region; $x \geq 0, y \geq 0$ means first quadrant
4. Corner points: axes intercepts + do lines ka intersection (solve simultaneously)
5. Table banao: corner point, value of Z; max/min likho, unbounded ho to half-plane check

Quick check

Area kabhi negative nahi, probability kabhi 1 se zyada nahi, $|\text{DCs}| \leq 1$: answer aisa aaye to dobara check karo.

Continuity + differentiability: question patterns

→ "Find k so that f is continuous at $x = a$": LHL = RHL = $f(a)$, solve for k
→ Two unknowns a, b at two points: make two equations, solve together
→ $|x - a|$ or $[x]$ (greatest integer): check LHL and RHL separately at the break point
→ $y = x^x$, $(\sin x)^{\cos x}$, $x^y = e^{x-y}$: logarithmic differentiation
→ $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ type: put $x = \tan\theta$ first, simplify, then differentiate
→ Prove $(1 - x^2)y_2 - xy_1 = 0$ type: square and differentiate twice, do not substitute early

AOD word problems: set up fast

→ Write the quantity to optimise (area, volume, cost) in ONE variable using the given condition
→ Open box from square sheet side a, cut x: $V = x(a - 2x)^2$, max at $x = \frac{a}{6}$
→ Rectangle of given perimeter: max area when it is a square
→ Closed cylinder of given volume, min surface area: $h = 2r$ (height = diameter)
→ Always confirm with f'' (sign) or sign change of f', and write the final answer with units

By parts: classic results

→ $\int \log x\,dx = x\log x - x + C$; $\int xe^x\,dx = e^x(x-1) + C$
→ $\int x\sin x\,dx = \sin x - x\cos x + C$; $\int x\cos x\,dx = \cos x + x\sin x + C$
→ $\int \tan^{-1} x\,dx = x\tan^{-1} x - \frac{1}{2}\log(1 + x^2) + C$
→ $\int \sin^{-1} x\,dx = x\sin^{-1} x + \sqrt{1 - x^2} + C$
→ $\int e^x(\sin x + \cos x)\,dx = e^x\sin x + C$; $\int e^x\left(\frac{1}{x} - \frac{1}{x^2}\right)dx = \frac{e^x}{x} + C$

Board copy mein formula ka use

→ Pehle formula likho, phir values daalo: step marks milte hain
→ Integration mein $+C$ aur definite mein limits har line pe
→ Answer ke saath units: sq units, cubic units, cm/s

Sign trick

$\cos^{-1}, \cot^{-1}, \sec^{-1}$ of $(-x) = \pi - (\ldots)$; baaki teen mein sirf minus bahar.

MOST ASKED QUESTIONS

↳ Har sawal ek **type** hai jo board paper mein baar baar aata hai. Pehle khud solve karo, phir green answer se match karo. Full method chapter mein hai.

Board paper pattern (80 marks)

Section	Questions	Marks
A	18 MCQ + 2 Assertion-Reason	1 each
B	5 very short	2 each
C	6 short	3 each
D	4 long	5 each
E	3 case based (1+1+2)	4 each

Unit-wise weightage

Unit	Chapters	Marks
Relations and Functions	1, 2	8
Algebra	3, 4	10
Calculus	5 to 9	35
Vectors and 3D	10, 11	14
Linear Programming	12	5
Probability	13	8

1. Relations and Functions

Q1. Check relation type: On $A=\{1,2,3\}$, is $R=\{(1,1),(2,2),(3,3),(1,2)\}$ reflexive, symmetric, transitive? 2M

Ans. Reflexive and transitive, not symmetric ($(1,2)\in R$ but $(2,1)\in R$).

Q2. Equivalence relation: Show that $R=\{(a,b):5 \text{ divides } (a-b)\}$ on $\mathbb{Z}$ is an equivalence relation; find $[2]$. 5M

Ans. Reflexive ($a-a=0$), symmetric ($5\mid a-b \Rightarrow 5\mid b-a$), transitive (add). $[2]=\{\dots,-8,-3,2,7,12,\dots\}$.

Q3. Equivalence class: $A=\{1,2,\dots,9\}$, $R=\{(a,b):|a-b| \text{ is even}\}$. Find $[4]$. 2M

Ans. $[4]=\{2,4,6,8\}$.

Q4. Counting relations: Number of relations and of reflexive relations on a set with 3 elements. 1M

Ans. $2^9=512$; reflexive: $2^6=64$.

Q5. Bijective: Show $f:\mathbb{R}\to\mathbb{R}$, $f(x)=4x-3$ is one-one and onto. 3M

Ans. $f(a)=f(b)\Rightarrow a=b$; for any y, $x=\frac{y+3}{4}$ gives $f(x)=y$. Bijective.

Q6. Counting bijections: Number of bijective functions from a set of 4 elements to itself. 1M

Ans. $4!=24$.

Q7. Onto count: Functions and onto functions from $\{1,2,3\}$ to $\{a,b\}$. 1M

Ans. $2^3=8$ functions; onto $=8-2=6$.

Q8. One-one not onto: Show $f:\mathbb{N}\to\mathbb{N}$, $f(x)=2x$ is one-one but not onto. 2M

Ans. $2a=2b\Rightarrow a=b$; odd numbers (e.g. 3) have no pre-image.

Q9. Check R, S, T: $R=\{(a,b):a\le b^2\}$ on $\mathbb{R}$. 3M

Ans. Not reflexive ($\frac{1}{2}>\frac{1}{4}$); not symmetric ($(1,2)$ in, $(2,1)$ not); not transitive ($(3,2),(2,1.5)$ in, $(3,1.5)$ not).

Q10. Equivalence count: Number of equivalence relations on $\{1,2,3\}$ containing $(1,2)$ and $(2,1)$. 1M

Ans. 2 (one with classes $\{1,2\},\{3\}$; one with $\{1,2,3\}$).

2. Inverse Trigonometric Functions

Q1. Principal value: $\sin^{-1}\left(-\frac{1}{2}\right)$ 1M

Ans. $-\frac{\pi}{6}$

Q2. Principal value: $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$ 1M

Ans. $\frac{5\pi}{6}$

Q3. Expression: $\tan^{-1}\sqrt{3}-\sec^{-1}(-2)$ 2M

Ans. $\frac{\pi}{3}-\frac{2\pi}{3}=-\frac{\pi}{3}$

Q4. Domain: Domain of $\sin^{-1}(2x-1)$ 2M

Ans. $-1\le 2x-1\le 1\Rightarrow 0\le x\le 1$, i.e. $[0,1]$.

Q5. Principal branch: $\sin^{-1}\left(\sin\frac{3\pi}{5}\right)$ 1M

Ans. $\frac{3\pi}{5}\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$; $=\sin^{-1}\left(\sin\frac{2\pi}{5}\right)=\frac{2\pi}{5}$.

Q6. Principal branch: $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$ 1M

Ans. $\cos\frac{7\pi}{6}=\cos\frac{5\pi}{6}$, so $\frac{5\pi}{6}$.

Q7. Range: Range of the principal branch of $\cot^{-1}x$; value of $\cot^{-1}1$. 1M

Ans. $(0,\pi)$; $\frac{\pi}{4}$.

Q8. Expression: $\tan^{-1}1+\cos^{-1}\left(-\frac{1}{2}\right)+\sin^{-1}\left(-\frac{1}{2}\right)$ 2M

Ans. $\frac{\pi}{4}+\frac{2\pi}{3}-\frac{\pi}{6}=\frac{3\pi}{4}$

Q9. Domain: Domain of $\cos^{-1}(x^2-4)$ 2M

Ans. $3\le x^2\le 5$: $[-\sqrt{5},-\sqrt{3}]\cup[\sqrt{3},\sqrt{5}]$

3. Matrices

Q1. Possible orders: How many orders are possible for a matrix with 18 elements? 1M

Ans. 6 ($1\times18, 2\times9, 3\times6, 6\times3, 9\times2, 18\times1$).

Q2. Construct: Construct $A_{2\times3}$ with $a_{ij}=\frac{(i+2j)^2}{2}$. 2M

(contd.) $\begin{bmatrix} \frac{9}{2} & \frac{25}{2} & \frac{49}{2} \\ 8 & 18 & 32 \end{bmatrix}$

Q3. Equality: Find x, y if $\begin{bmatrix} x+y & 2 \\ 5 & x-y \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 2 \end{bmatrix}$. 1M

Ans. $x = 4,\ y = 2$.

Q4. Sym + skew: Express $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$ as $P + Q$, P symmetric, Q skew. 3M

Ans. $P = \begin{bmatrix} 2 & 2 \\ 2 & 4 \end{bmatrix}, Q = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$.

Q5. Matrix equation: For $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$ show $A^2 - 5A + 7I = O$; find A^{-1}. 3M

Ans. $A^2 = \begin{bmatrix} 3 & 5 \\ -5 & 8 \end{bmatrix}$, so it holds. $A^{-1} = \frac{1}{7}(5I - A) = \frac{1}{7}\begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}$.

Q6. Skew-symmetric: Complete the 3×3 skew-symmetric matrix with $a_{12} = 4, a_{13} = -2, a_{23} = 5$. 2M

Ans. $\begin{bmatrix} 0 & 4 & -2 \\ -4 & 0 & 5 \\ 2 & -5 & 0 \end{bmatrix}$

Q7. Order of product: A is 3×4, B is 4×2. Order of AB? Is BA defined? 1M

Ans. AB is 3×2; BA not defined.

Q8. Polynomial in A: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$; find $A^2 - 5A$. 2M

Ans. $A^2 = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$, so $A^2 - 5A = 2I$.

Q9. Sym/skew property: A symmetric, B skew-symmetric (same order). Is $AB - BA$ symmetric? 2M

Ans. $(AB - BA)' = B'A' - A'B' = -BA + AB$: yes, symmetric.

Q10. Matrix equation: Find X if $2A + X = O, A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$. 1M

Ans. $X = -2A = \begin{bmatrix} -2 & 4 \\ -6 & 0 \end{bmatrix}$

4. Determinants

Q1. Evaluate: $\begin{vmatrix} 2 & -1 & 3 \\ 1 & 4 & -2 \\ 3 & 0 & 1 \end{vmatrix}$ 1M

Ans. -21

Q2. Area of triangle: Area with vertices $(1,2), (4,6), (7,3)$. 2M

Ans. $\frac{21}{2}$ sq units.

Q3. Find k from area: Area of triangle $(k,0), (4,0), (0,2)$ is 6. Find k. 2M

Ans. $\frac{1}{2}|8 - 2k| = 6 \Rightarrow k = -2$ or 10.

Formula flash $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

Q4. Minor, cofactor: For $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}$, find M_{12} and A_{12}. 1M

Ans. $M_{12} = 40 - 42 = -2, A_{12} = 2$.

Q5. Adjoint: adj of $\begin{bmatrix} 3 & -4 \\ 2 & 5 \end{bmatrix}$ 1M

Ans. $\begin{bmatrix} 5 & 4 \\ -2 & 3 \end{bmatrix}$

Q6. Property: A is 3×3 with $|A| = 5$. Find $|\text{adj } A|$ and $|3A|$ if $|A| = 2$. 1M

Ans. $|\text{adj } A| = 5^2 = 25; |3A| = 3^3 \times 2 = 54$.

Q7. Solve by inverse: $x + y + z = 6, x - y + z = 2, 2x + y - z = 1$ using $X = A^{-1}B$. 5M

Ans. $|A| = 6 \neq 0; x = 1,\ y = 2,\ z = 3$.

Q8. Consistency: Is $2x + 3y = 5, 4x + 6y = 7$ consistent? 2M

Ans. $|A| = 0$ and $(\text{adj } A)B \neq O$: inconsistent.

Q9. Equal determinants: Find x: $\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$ 1M

Ans. $x^2 - 36 = 0 \Rightarrow x = \pm 6$

Q10. Inverse 3x3: Find A^{-1} for $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 2 \end{bmatrix}$. 5M

Ans. $|A| = 4; A^{-1} = \frac{1}{4}\begin{bmatrix} 2 & -4 & 2 \\ 1 & 2 & -1 \\ -1 & 2 & 1 \end{bmatrix}$

Q11. Inverse property: A is 2×2 with $|A| = 4$. Find $|A^{-1}|$. 1M

Ans. $\frac{1}{|A|} = \frac{1}{4}$

5. Continuity and Differentiability

Q1. Find k: $f(x) = kx + 1\ (x \leq 5), 3x - 5\ (x > 5)$ is continuous at $x = 5$. 2M

Ans. $5k + 1 = 10 \Rightarrow k = \frac{9}{5}$.

Q2. Find a, b: $f(x) = 5\ (x \leq 2), ax + b\ (2 < x < 10), 21\ (x \geq 10)$ is continuous. 3M

Ans. $2a + b = 5, 10a + b = 21 \Rightarrow a = 2,\ b = 1$.

Q3. Continuity vs differentiability: $f(x) = |x| + |x - 1|$ at $x = 1$. 3M

Ans. Continuous at $x = 1$; LHD $= 0$, RHD $= 2$, so not differentiable.

Q4. Chain rule: $\frac{d}{dx}\sin^3(2x + 1)$ 2M

Ans. $6\sin^2(2x + 1)\cos(2x + 1)$

Q5. Implicit: $x^2 + xy + y^2 = 7$; find $\frac{dy}{dx}$. 2M

Ans. $-\frac{2x + y}{x + 2y}$

Q6. Inverse trig: $y = \tan^{-1}\frac{3x - x^3}{1 - 3x^2}, |x| < \frac{1}{\sqrt{3}}$. 3M

(contd.) Put $x = \tan\theta$: $y = 3\tan^{-1}x$, so $\frac{dy}{dx} = \frac{3}{1+x^2}$.

Q7. **Log differentiation:** $y = x^{\sin x}$ $(x > 0)$ 3M

Ans. $x^{\sin x}\left(\cos x \log x + \frac{\sin x}{x}\right)$

Q8. **Parametric:** $x = 2\cos^3 t, y = 2\sin^3 t$ 2M

Ans. $\frac{dy}{dx} = -\tan t$

Q9. **Second order:** $y = 5\cos(\log x) - 2\sin(\log x)$; show $x^2y'' + xy' + y = 0$. 3M

Ans. $xy' = -5\sin(\log x) - 2\cos(\log x)$; differentiate again: $x(xy')' = -y$. Holds.

Q10. **Second order:** $y = e^{2x}(a + bx)$; show $y'' - 4y' + 4y = 0$. 3M

Ans. $y' = 2y + be^{2x}, y'' = 2y' + 2be^{2x} = 4y' - 4y$. Holds.

Q11. **Log differentiation:** $y = (\sin x)^x, 0 < x < \pi$ 3M

Ans. $(\sin x)^x(\log\sin x + x\cot x)$

Q12. **Second order:** $y = \sin^{-1}x$; show $(1 - x^2)y'' - xy' = 0$. 3M

Ans. $\sqrt{1-x^2}\,y' = 1$; differentiate: $\sqrt{1-x^2}\,y'' - \frac{x}{\sqrt{1-x^2}}y' = 0$. Multiply by $\sqrt{1-x^2}$.

Q13. **Differentiability:** $f(x) = |x - 3|$ at $x = 3$ 2M

Ans. Continuous; LHD $= -1$, RHD $= 1$: not differentiable.

Q14. **Chain rule:** $y = \log(\log x), x > 1$ 1M

Ans. $\frac{1}{x\log x}$

Q15. **Parametric second order:** $x = t^2, y = t^3$; find $\frac{d^2y}{dx^2}$. 3M

Ans. $\frac{dy}{dx} = \frac{3t}{2}$; $\frac{d^2y}{dx^2} = \frac{3/2}{2t} = \frac{3}{4t}$

6. Application of Derivatives

Q1. **Rate of change:** Radius of a circle grows at 0.5 cm/s. Rate of area at $r = 4$ cm? 2M

Ans. $2\pi r\frac{dr}{dt} = 4\pi$ cm^2/s

Q2. **Related rates:** Cube volume grows at 9 cm^3/s. Rate of surface area when edge is 10 cm? 3M

Ans. $\frac{dx}{dt} = \frac{9}{300}$; $\frac{dS}{dt} = 12x\frac{dx}{dt} = 3.6$ cm^2/s

Q3. **Increasing, decreasing:** $f(x) = 2x^3 - 9x^2 + 12x + 5$ 3M

Ans. $f' = 6(x-1)(x-2)$: increasing on $(-\infty, 1) \cup (2, \infty)$, decreasing on $(1, 2)$.

Q4. **Local max, min:** $f(x) = x^3 - 6x^2 + 9x + 1$ 3M

Ans. Local max 5 at $x = 1$; local min 1 at $x = 3$.

Q5. **Absolute max, min:** $f(x) = x^3 - 3x + 2$ on $[0, 3]$ 3M

Ans. Max 20 at $x = 3$; min 0 at $x = 1$.

Q6. **Word problem:** $x + y = 24, x, y > 0$; maximise xy^2. 5M

Ans. $y = 16, x = 8$; maximum 2048.

Q7. **Open box:** Squares of side x cut from corners of an 18 cm square sheet; maximise the box volume. 5M

Ans. $V = x(18 - 2x)^2$, max at $x = 3$ cm, $V = 432$ cm^3.

Q8. **Inscribed rectangle:** Largest rectangle inscribed in a circle of radius 5. 5M

(contd.) A square of side $5\sqrt{2}$; area 50 sq units.

Q9. **Max of function:** Maximum value of $\frac{\log x}{x}$ $(x > 0)$. 3M

Ans. At $x = e$; maximum $\frac{1}{e}$.

Q10. **Increasing, decreasing:** $f(x) = \sin x + \cos x, 0 \le x \le 2\pi$ 3M

Ans. Increasing on $\left[0, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right]$; decreasing on $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$.

Q11. **Always increasing:** Show $f(x) = x^3 - 3x^2 + 3x - 100$ is increasing on $\mathbb{R}$. 2M

Ans. $f'(x) = 3(x-1)^2 \ge 0$ for all x.

Q12. **Absolute max, min:** $f(x) = 2x^3 - 15x^2 + 36x + 1$ on $[1, 5]$ 3M

Ans. $f(1) = 24, f(2) = 29, f(3) = 28, f(5) = 56$: max 56 at $x = 5$, min 24 at $x = 1$.

Q13. **Rate of change:** Radius of a balloon grows at 2 cm/s. Rate of volume at $r = 5$ cm? 2M

Ans. $4\pi r^2\frac{dr}{dt} = 200\pi$ cm^3/s

7. Integrals

Q1. **Direct:** $\int \frac{2x+3}{x^2+3x+1}\,dx$ 1M

Ans. $\log|x^2 + 3x + 1| + C$

Q2. **Trig power:** $\int \sin^3 x\,dx$ 2M

Ans. $-\cos x + \frac{\cos^3 x}{3} + C$

Q3. **Special form:** $\int \frac{dx}{x^2+6x+13}$ 2M

Ans. $\frac{1}{2}\tan^{-1}\frac{x+3}{2} + C$

Q4. **Special form:** $\int \frac{dx}{\sqrt{5-4x-x^2}}$ 2M

Ans. $\sin^{-1}\frac{x+2}{3} + C$

Q5. **Partial fractions:** $\int \frac{dx}{(x-1)(x+2)}$ 3M

Ans. $\frac{1}{3}\log\left|\frac{x-1}{x+2}\right| + C$

Q6. **Partial fractions:** $\int \frac{x\,dx}{(x+1)(x^2+1)}$ 5M

Ans. $-\frac{1}{2}\log|x+1| + \frac{1}{4}\log(x^2+1) + \frac{1}{2}\tan^{-1}x + C$

Q7. **By parts:** $\int x^2e^x dx$ 3M

Ans. $e^x(x^2 - 2x + 2) + C$

Q8. $e^x[f + f']$: $\int e^x(\tan x + \sec^2 x)\,dx$ 2M

Ans. $e^x\tan x + C$

Q9. **Root form:** $\int \sqrt{9-x^2}\,dx$ 3M

Ans. $\frac{x}{2}\sqrt{9-x^2} + \frac{9}{2}\sin^{-1}\frac{x}{3} + C$

Q10. **Property P4:** $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x}\,dx$ 3M

Ans. $\frac{\pi}{4}$

Q11. **Modulus:** $\int_{-1}^{2}|x^3 - x|\,dx$ 5M

Ans. $\frac{11}{4}$

Q13. **Odd function:** $\int_{-\pi/2}^{\pi/2}\sin^7 x\,dx$ 1M

(contd.) 0 (odd function)

Q12. Property + sub: $\int_0^{\pi} \frac{x \sin x}{1+\cos^2 x}\,dx$ 5M

Ans. $\frac{\pi^2}{4}$

Q14. Special form: $\int \frac{dx}{x^2-16}$ 2M

Ans. $\frac{1}{8}\log\left|\frac{x-4}{x+4}\right| + C$

Q15. By parts: $\int x \log x\,dx$ 2M

Ans. $\frac{x^2}{2}\log x - \frac{x^2}{4} + C$

Q16. Definite: $\int_0^1 xe^x dx$ 2M

Ans. $[(x-1)e^x]_0^1 = 1$

Q17. Definite: $\int_0^{\pi/4} \tan^2 x\,dx$ 2M

Ans. $[\tan x - x]_0^{\pi/4} = 1 - \frac{\pi}{4}$

Q18. Trig identity: $\int \frac{dx}{1+\cos x}$ 2M

Ans. $\int \frac{1}{2}\sec^2\frac{x}{2}\,dx = \tan\frac{x}{2} + C$

Q19. Substitution: $\int \frac{\sec^2 x}{1+\tan x}\,dx$ 1M

Ans. $\log|1+\tan x| + C$

Q20. Modulus: $\int_0^2 |x-1|\,dx$ 2M

Ans. $\frac{1}{2} + \frac{1}{2} = 1$

Q21. Partial fractions: $\int \frac{dx}{x(x^2+1)}$ 3M

Ans. $\log|x| - \frac{1}{2}\log(x^2+1) + C$

8. Application of Integrals

Q1. Parabola and line: Area bounded by $y^2 = 4x$ and the line $x = 4$. 3M

Ans. $2\int_0^4 2\sqrt{x}\,dx = \frac{64}{3}$ sq units

Q2. Circle: Area of $x^2 + y^2 = 16$ in the first quadrant. 3M

Ans. 4π sq units

Q3. Ellipse: Area of $\frac{x^2}{9} + \frac{y^2}{4} = 1$. 5M

Ans. $\pi ab = 6\pi$ sq units

Q4. Under curve: Area under $y = x^2$ from $x = 1$ to $x = 3$. 2M

Ans. $\frac{26}{3}$ sq units

Q5. About y-axis: Area bounded by $x^2 = 4y$, $y = 1$, $y = 4$ and the y-axis (first quadrant). 3M

Ans. $\int_1^4 2\sqrt{y}\,dy = \frac{28}{3}$ sq units

Q6. Part of ellipse: Area bounded by $\frac{x^2}{16} + \frac{y^2}{9} = 1$, $x = 0$, $x = 2$ in the first quadrant. 5M

Ans. $\frac{3}{4}\int_0^2 \sqrt{16-x^2}\,dx = \frac{3\sqrt{3}}{2} + \pi$ sq units

Q7. Under curve: Area under $y = \sqrt{x}$ from $x = 0$ to $x = 4$. 2M

Ans. $\frac{16}{3}$ sq units

Q8. Sign change: Area bounded by $y = x^3$, the x-axis, $x = -1$ and $x = 2$. 3M

(contd.) $\frac{1}{4} + 4 = \frac{17}{4}$ sq units

9. Differential Equations

Q1. Order, degree: $(y'')^3 + (y')^2 + \sin(y') + 1 = 0$ 1M

Ans. Order 2, degree not defined.

Q2. I.F.: $x\frac{dy}{dx} + 3y = x^2$ 1M

Ans. x^3

Q3. Separable: $\frac{dy}{dx} = (1+y^2)e^x$ 2M

Ans. $\tan^{-1} y = e^x + C$

Q4. Homogeneous: $x\frac{dy}{dx} = 2x + y$ 3M

Ans. $y = 2x\log|x| + Cx$

Q5. Homogeneous: $\frac{dy}{dx} = \frac{x-y}{x+y}$ 3M

Ans. $x^2 - 2xy - y^2 = C$

Q6. Linear, particular: $\frac{dy}{dx} + y = x, y(0) = 1$ 3M

Ans. $y = x - 1 + 2e^{-x}$

Q7. Linear in x: $y\,dx = (x+y^2)\,dy$ 3M

Ans. $x = y^2 + Cy$

Q8. Separable, particular: $\frac{dy}{dx} = 2x(y+1), y(0) = 1$ 3M

Ans. $y = 2e^{x^2} - 1$

Q9. Constants: Number of arbitrary constants in the general solution of a 4th order DE. 1M

Ans. 4

Q10. Verify: Verify $y = Ae^{3x} + Be^{-3x}$ solves $y'' = 9y$. 2M

Ans. $y'' = 9Ae^{3x} + 9Be^{-3x} = 9y$.

Q11. Separable: $\frac{dy}{dx} = x + xy^2$ 2M

Ans. $\tan^{-1} y = \frac{x^2}{2} + C$

10. Vector Algebra

Q1. Unit vector: $\vec{a} = 2\hat{i} - 3\hat{j} + 6\hat{k}$ 1M

Ans. $\frac{1}{7}(2\hat{i} - 3\hat{j} + 6\hat{k})$

Q2. Direction cosines: $\hat{i} + 2\hat{j} + 2\hat{k}$ 1M

Ans. $\frac{1}{3}, \frac{2}{3}, \frac{2}{3}$

Q3. Section formula: Point dividing $P(2,3,-1), Q(5,-3,8)$ internally in $1:2$. 2M

Ans. $(3,1,2)$

Q4. Projection: Projection of $2\hat{i} + 3\hat{j} + 2\hat{k}$ on $\hat{i} + 2\hat{j} + \hat{k}$. 2M

Ans. $\frac{10}{\sqrt{6}}$

Q5. Angle: Angle between $\hat{i} + \hat{j} - \hat{k}$ and $\hat{i} - \hat{j} + \hat{k}$. 2M

Ans. $\cos\theta = -\frac{1}{3}, \theta = \cos^{-1}\left(-\frac{1}{3}\right)$

Q6. Perpendicular: $2\hat{i} + \lambda\hat{j} + \hat{k} \perp \hat{i} - 2\hat{j} + 3\hat{k}$; find λ. 1M

Ans. $\lambda = \frac{5}{2}$

Q7. Parallelogram area: Adjacent sides $3\hat{i}+\hat{j}+4\hat{k}$ and $\hat{i}-\hat{j}+\hat{k}$. 2M

Ans. $|\vec{a}\times\vec{b}|=\sqrt{42}$

Q8. Triangle area: $A(1,1,2), B(2,3,5), C(1,5,5)$ 3M

Ans. $\frac{1}{2}|\vec{AB}\times\vec{AC}|=\frac{\sqrt{61}}{2}$

Q9. Angle from dot: $|\vec{a}|=3, |\vec{b}|=4, \vec{a}\cdot\vec{b}=6$. Angle? 1M

Ans. $\cos\theta=\frac{1}{2}, \theta=\frac{\pi}{3}$

Q10. Magnitude: $|\vec{a}|=3, |\vec{b}|=4, \vec{a}\perp\vec{b}$. Find $|\vec{a}+\vec{b}|$. 1M

Ans. $\sqrt{9+16}=5$

Q11. Unit normal: Unit vector perpendicular to $\hat{i}+\hat{j}+\hat{k}$ and $\hat{i}+2\hat{j}+3\hat{k}$. 2M

Ans. $\pm\frac{1}{\sqrt{6}}(\hat{i}-2\hat{j}+\hat{k})$

Q12. Triangle area: Adjacent sides $2\hat{i}+\hat{j}$ and $\hat{i}+3\hat{j}$. 2M

Ans. $\frac{1}{2}|5\hat{k}|=\frac{5}{2}$ sq units

11. Three Dimensional Geometry

Q1. Cartesian form: Line through $(1,-2,3)$ parallel to $2\hat{i}+\hat{j}-2\hat{k}$. 1M

Ans. $\frac{x-1}{2}=\frac{y+2}{1}=\frac{z-3}{-2}$

Q2. Vector form: Line through $(2,1,0)$ and $(3,4,5)$. 2M

Ans. $\vec{r}=2\hat{i}+\hat{j}+\lambda(\hat{i}+3\hat{j}+5\hat{k})$

Q3. Angle between lines: Direction ratios $2,2,1$ and $3,-6,2$. 2M

Ans. $\cos\theta=\frac{4}{21}$

Q4. Perpendicular lines: Lines with d.r. $k,2,-1$ and $1,k,3$ are perpendicular; find k. 2M

Ans. $k+2k-3=0\Rightarrow k=1$

Q5. Shortest distance: $\vec{r}=(\hat{i}+\hat{j})+\lambda(2\hat{i}-\hat{j}+\hat{k})$ and $\vec{r}=(3\hat{i}+2\hat{j}-\hat{k})+\mu(3\hat{i}-5\hat{j}+2\hat{k})$ 5M

Ans. $\vec{b}_1\times\vec{b}_2=3\hat{i}-\hat{j}-7\hat{k}$; SD $=\frac{12}{\sqrt{59}}$

Q6. Parallel lines: $\vec{r}=\hat{i}+2\hat{j}+\lambda(2\hat{i}+3\hat{j}+6\hat{k})$ and $\vec{r}=3\hat{i}+3\hat{j}-5\hat{k}+\mu(2\hat{i}+3\hat{j}+6\hat{k})$ 3M

Ans. $d=\frac{|\vec{b}\times(\vec{a}_2-\vec{a}_1)|}{|\vec{b}|}=\frac{\sqrt{941}}{7}$

Q7. Equally inclined: Direction cosines of a line equally inclined to the axes. 1M

Ans. $\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}$

Q8. Angle between lines: $\vec{r}=2\hat{i}-5\hat{j}+\hat{k}+\lambda(3\hat{i}+2\hat{j}+6\hat{k})$ and $\vec{r}=7\hat{i}-6\hat{k}+\mu(\hat{i}+2\hat{j}+2\hat{k})$ 2M

Ans. $\cos\theta=\frac{19}{21}$

Q9. Intersect?: Do $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-4}{5}=\frac{y-1}{2}=\frac{z}{1}$ intersect? 3M

Ans. $(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)=0$: SD $=0$, they intersect.

Q10. Direction cosines: Line through the origin and $(2,-3,6)$. 1M

Ans. $\frac{2}{7}, -\frac{3}{7}, \frac{6}{7}$

Formula flash SD of skew lines $=\frac{|(\vec{a}_2-\vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)|}{|\vec{b}_1\times\vec{b}_2|}$

12. Linear Programming

Q1. Bounded, max: Max $Z=3x+2y$: $x+y\le 4, x+3y\le 6, x,y\ge 0$. 3M

Ans. Corners $(0,0),(4,0),(3,1),(0,2)$; max 12 at $(4,0)$.

Q2. Bounded, min: Min $Z=300x+400y$: $x+2y\ge 10, 3x+4y\le 24, x,y\ge 0$. 3M

Ans. Corners $(0,5),(4,3),(0,6)$; min 2000 at $(0,5)$.

Q3. Unbounded: Min $Z=3x+5y$: $x+3y\ge 3, x+y\ge 2, x,y\ge 0$. 5M

Ans. Min 7 at $(\frac{3}{2},\frac{1}{2})$; $3x+5y<7$ has no point in the region.

Q4. Corner points given: Corners $(0,2),(3,0),(6,0),(6,8),(0,5)$; $Z=4x+6y$. 2M

Ans. Max 72 at $(6,8)$; min 12 at every point of the segment joining $(0,2)$ and $(3,0)$.

Q5. No feasible region: Max $Z=x+y$: $x-y\le -1, -x+y\le 0, x,y\ge 0$. 2M

Ans. No common region ($y\ge x+1$ and $y\le x$): no solution.

Q6. Word problem: Profit Rs 5 on A, Rs 4 on B; $2x+y\le 10, x+2y\le 8, x,y\ge 0$. 5M

Ans. Max profit Rs 28 at $(4,2)$.

Q7. Bounded, max: Max $Z=4x+3y$: $3x+5y\le 15, 5x+2y\le 10, x,y\ge 0$. 3M

Ans. Corners $(0,0),(2,0),(\frac{20}{19},\frac{45}{19}),(0,3)$; max $\frac{215}{19}$ at $(\frac{20}{19},\frac{45}{19})$.

Q8. Condition on Z: Corners $(0,0),(4,0),(2,3),(0,4)$; $Z=px+qy$ is max at both $(4,0)$ and $(2,3)$. 2M

Ans. $4p=2p+3q\Rightarrow 2p=3q$

Q9. Three constraints: Max $Z=2x+3y$: $x+y\le 6, x\le 4, y\le 5, x,y\ge 0$. 3M

Ans. Corners $(0,0),(4,0),(4,2),(1,5),(0,5)$; max 17 at $(1,5)$.

13. Probability

Q1. Conditional: $P(A)=0.6, P(B)=0.3, P(A\cap B)=0.2$. Find $P(A|B), P(B|A)$. 2M

Ans. $\frac{2}{3}, \frac{1}{3}$

Q2. Independent: A, B independent, $P(A)=0.4, P(B)=0.5$. Find $P(A\cup B)$. 1M

Ans. $0.4+0.5-0.2=0.7$

Q3. Dice: Two dice; first shows 4. P(sum ≥ 9)? 2M

Ans. Second must be 5 or 6: $\frac{1}{3}$

Q4. Coins: Coin tossed 3 times. P(at least 2 heads | first is head)? 2M

Ans. $\frac{3}{4}$

Q5. Total probability: Bag I: 3 red, 4 black; Bag II: 5 red, 6 black. Random bag, one ball. P(red)? 3M

Ans. $\frac{1}{2}\times\frac{3}{7}+\frac{1}{2}\times\frac{5}{11}=\frac{34}{77}$

Q6. Bayes': In the last question the ball is red. P(it came from Bag II)? 5M

(contd.) $\frac{5/22}{34/77} = \frac{35}{68}$

Q7. Bayes' (test): 1% have a disease; test is positive for 95% of patients and 10% of healthy people. P(disease | positive)? 5M

Ans. $\frac{0.0095}{0.0095+0.099} = \frac{19}{217}$

Q8. Distribution: X = number of heads in 2 tosses. Distribution and mean. 3M

Ans. $0, 1, 2$ with $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$; mean 1

Q9. Find k: $P(X = x) = kx, x = 1, 2, 3, 4$. Find k and $E(X)$. 3M

Ans. $k = \frac{1}{10}$; $E(X) = 3$

Q10. Expectation: Mean number of sixes in two throws of a die. 3M

Ans. $\frac{10}{36} + 2 \times \frac{1}{36} = \frac{1}{3}$

Q11. Independence test: $P(A) = \frac{1}{2}, P(B) = \frac{1}{3}, P(A \cap B) = \frac{1}{6}$. Independent? 1M

Ans. Yes, $P(A)P(B) = \frac{1}{6} = P(A \cap B)$.

Q12. Die: A die is thrown. E: even, F: multiple of 3. Find $P(E|F)$. 1M

Ans. $\frac{P(\{6\})}{P(\{3,6\})} = \frac{1}{2}$

Q13. Problem solved: A and B solve a problem independently with probabilities $\frac{1}{2}, \frac{1}{3}$. P(solved)? 2M

Ans. $1 - \frac{1}{2} \times \frac{2}{3} = \frac{2}{3}$

Q14. Mean: $P(X = 0) = 0.2, P(X = 1) = 0.5, P(X = 2) = 0.3$. Find $E(X)$. 1M

Ans. $0.5 + 0.6 = 1.1$

Q15. Without replacement: Bag: 4 red, 6 black. Two balls drawn one by one without replacement. P(both red)? 2M

Ans. $\frac{4}{10} \times \frac{3}{9} = \frac{2}{15}$

14. Extra practice: board favourites

↳ Mixed revision round: **Calculus first** (sabse zyada marks), phir Vectors/3D, Matrices, LPP, Probability. Timer lagao, 2 min per mark.

Q1. Continuity (find k): $f(x) = \dfrac{1 - \cos 4x}{8x^2}$ for $x \neq 0$, $f(0) = k$. Find k if f is continuous at 0. 3M

Ans. $\lim_{x \to 0} \dfrac{2 \sin^2 2x}{8x^2} = 1$, so $k = 1$

Q2. Continuity (a, b): $f(x) = 3$ for $x \leq 1$, $ax + b$ for $1 < x < 4$, 15 for $x \geq 4$. Continuous everywhere: find a, b. 3M

Ans. $a + b = 3, 4a + b = 15$: $a = 4, b = -1$

Q3. Log differentiation: $y = (\log x)^x, x > 1$. Find $\dfrac{dy}{dx}$. 3M

Ans. $(\log x)^x \left(\log(\log x) + \dfrac{1}{\log x} \right)$

Q4. Implicit: $x^2 + 3xy + 2y^2 = 12$. Find $\dfrac{dy}{dx}$. 2M

Ans. $-\dfrac{2x + 3y}{3x + 4y}$

Formula flash Area of ellipse $= \pi ab$; circle $= \pi r^2$.

Q5. Parametric: $x = 2(t - \sin t), y = 2(1 - \cos t)$. Find $\dfrac{dy}{dx}$ at $t = \dfrac{\pi}{2}$. 3M

Ans. $\dfrac{dy}{dx} = \dfrac{\sin t}{1 - \cos t} = \cot \dfrac{t}{2}$; at $t = \dfrac{\pi}{2}$: 1

Q6. Parametric: $x = 3t^2, y = 6t$. Find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$. 3M

Ans. $\dfrac{1}{t}$ and $-\dfrac{1}{6t^3}$

Q7. Second order: $y = 3e^{2x} + 2e^{3x}$. Show $y'' - 5y' + 6y = 0$. 3M

Ans. $y' = 6e^{2x} + 6e^{3x}$, $y'' = 12e^{2x} + 18e^{3x}$; substitute: 0 ✓

Q8. Second order: $y = e^{2 \sin^{-1} x}$. Show $(1 - x^2)y'' - xy' - 4y = 0$. 3M

Ans. $\sqrt{1 - x^2}\, y' = 2y$; differentiate again and simplify ✓

Q9. Second order: $y = e^x \cos x$. Find $\dfrac{d^2y}{dx^2}$. 2M

Ans. $-2e^x \sin x$

Q10. Inverse trig derivative: $y = \tan^{-1}\left(\dfrac{2x}{1 - x^2} \right), -1 < x < 1$. Find $\dfrac{dy}{dx}$. 2M

Ans. $y = 2 \tan^{-1} x$, so $\dfrac{dy}{dx} = \dfrac{2}{1 + x^2}$

Q11. Rate of change: Radius of a circle grows at 0.5 cm/s. How fast is the area growing when $r = 8$ cm? 2M

Ans. $\dfrac{dA}{dt} = 2\pi r \times 0.5 = 8\pi$ cm^2/s

Q12. Increasing/decreasing: Find intervals where $f(x) = x^3 - 12x + 7$ is increasing or decreasing. 3M

Ans. $f' = 3(x - 2)(x + 2)$: inc on $(-\infty, -2] \cup [2, \infty)$, dec on $[-2, 2]$

Q13. Increasing/decreasing: Same for $f(x) = x^4 - 8x^2$. 3M

Ans. $f' = 4x(x - 2)(x + 2)$: inc on $(-2, 0) \cup (2, \infty)$, dec on $(-\infty, -2) \cup (0, 2)$

Q14. Local max/min: Find local maximum and minimum values of $f(x) = 2x^3 - 9x^2 + 12x + 4$. 3M

Ans. Local max 9 at $x = 1$; local min 8 at $x = 2$

Q15. Absolute max/min: $f(x) = 2x^3 - 24x + 5$ on $[-3, 3]$. 3M

Ans. $f(-3) = 23, f(-2) = 37, f(2) = -27, f(3) = -13$: max 37, min -27

Q16. Optimisation: Square sheet of side 24 cm; equal squares cut from corners to make an open box. Side of cut square for max volume? 5M

Ans. $V = x(24 - 2x)^2, V' = 0 \Rightarrow x = 4$ cm; $V_{\max} = 1024$ cm^3

Q17. Optimisation: Two positive numbers have sum 16. Find them when the sum of their cubes is least. 3M

Ans. 8 and 8 (least sum 1024)

Q18. Substitution: $\int \frac{2x}{1+x^4}\,dx$ 2M

Ans. $\tan^{-1}(x^2)+C$

Q19. Substitution: $\int \frac{e^{\tan^{-1}x}}{1+x^2}\,dx$ 2M

Ans. $e^{\tan^{-1}x}+C$

Q20. Substitution: $\int \frac{(\log x)^2}{x}\,dx$ 2M

Ans. $\frac{(\log x)^3}{3}+C$

Q21. Special form: $\int \frac{dx}{x^2+4x+20}$ 2M

Ans. $\frac{1}{4}\tan^{-1}\frac{x+2}{4}+C$

Q22. Substitution: $\int \frac{\sec^2 x}{1+\tan x}\,dx$ 2M

Ans. $\log|1+\tan x|+C$

Q23. Special form: $\int \frac{dx}{\sqrt{x^2-4x+8}}$ 2M

Ans. $\log\left|x-2+\sqrt{x^2-4x+8}\right|+C$

Q24. Partial fractions: $\int \frac{3x+5}{(x+1)(x+3)}\,dx$ 3M

Ans. $\log|x+1|+2\log|x+3|+C$

Q25. Partial fractions (repeated): $\int \frac{x}{(x-1)^2(x+2)}\,dx$ 5M

Ans. $\frac{2}{9}\log\left|\frac{x-1}{x+2}\right|-\frac{1}{3(x-1)}+C$

Q26. By parts: $\int x^2\cos x\,dx$ 3M

Ans. $x^2\sin x+2x\cos x-2\sin x+C$

Q27. By parts: $\int x\tan^{-1}x\,dx$ 3M

Ans. $\frac{x^2+1}{2}\tan^{-1}x-\frac{x}{2}+C$

Q28. e^x form: $\int e^x\left(\frac{1}{x}-\frac{1}{x^2}\right)dx$ 2M

Ans. $\frac{e^x}{x}+C$

Q29. Definite: $\int_1^e \log x\,dx$ 2M

Ans. $[x\log x-x]_1^e=1$

Q30. King property: $\int_0^{\pi/2}\frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx$ 3M

Ans. $2I=\frac{\pi}{2}$, so $I=\frac{\pi}{4}$

Q31. King property: $\int_0^{\pi}\frac{x}{1+\sin x}\,dx$ 5M

Ans. $I=\frac{\pi}{2}\int_0^{\pi}\frac{dx}{1+\sin x}=\frac{\pi}{2}\times 2=\pi$

Q32. King property: $\int_0^{\pi/4}\log(1+\tan x)\,dx$ 5M

(contd.) $\frac{\pi}{8}\log 2$

Q33. Odd/even: $\int_{-2}^{2}(x^5+3x^2)\,dx$ 2M

Ans. Odd part gives 0; $2\int_0^2 3x^2dx=16$

Q34. Definite: $\int_0^{\pi/2}\cos^2 x\,dx$ 2M

Ans. $\int_0^{\pi/2}\frac{1+\cos 2x}{2}dx=\frac{\pi}{4}$

Q35. Modulus: $\int_0^4|x-1|\,dx$ 3M

Ans. $\frac{1}{2}+\frac{9}{2}=5$

Q36. Area: Area bounded by $y=x^2$, the x-axis and the lines $x=1, x=3$. 3M

Ans. $\int_1^3 x^2dx=\frac{26}{3}$ sq units

Q37. Area (ellipse): Area enclosed by $\frac{x^2}{16}+\frac{y^2}{9}=1$. 5M

Ans. $4\int_0^4\frac{3}{4}\sqrt{16-x^2}\,dx=12\pi$ sq units

Q38. Area (parabola): Area bounded by $y^2=9x, x=4$ and the x-axis, first quadrant. 3M

Ans. $\int_0^4 3\sqrt{x}\,dx=16$ sq units

Q39. Order, degree: $\left(\frac{d^2y}{dx^2}\right)^3+\left(\frac{dy}{dx}\right)^4+y=0$ 1M

Ans. Order 2, degree 3

Q40. Separable: $\frac{dy}{dx}=2xy, y(0)=3$. 3M

Ans. $y=3e^{x^2}$

Q41. Separable: $(1+e^{2x})\,dy+(1+y^2)e^x\,dx=0$, $y(0)=1$. 3M

Ans. $\tan^{-1}y+\tan^{-1}e^x=\frac{\pi}{2}$

Q42. Homogeneous: $2xy\,dy=(x^2+y^2)\,dx$. 5M

Ans. $y=vx$: $x^2-y^2=Cx$

Q43. Homogeneous: $x\,dy-y\,dx=\sqrt{x^2+y^2}\,dx, x>0$. 5M

Ans. $y+\sqrt{x^2+y^2}=Cx^2$

Q44. Linear: $\frac{dy}{dx}+2y=e^{-x}$. 3M

Ans. IF e^{2x}: $y=e^{-x}+Ce^{-2x}$

Q45. Linear: $x\frac{dy}{dx}+2y=x^2, y(1)=1$. 5M

Ans. IF x^2: $y=\frac{x^2}{4}+\frac{3}{4x^2}$

Q46. Linear: $\frac{dy}{dx}+y\tan x=\sec x$. 3M

Ans. IF $\sec x$: $y=\sin x+C\cos x$

Q47. Projection: Projection of $4\hat{i}+2\hat{j}+\hat{k}$ on $\hat{i}+2\hat{j}-2\hat{k}$. 2M

Ans. $\dfrac{4+4-2}{3}=2$

Q48. Area (cross product): Area of triangle $A(1,2,3)$, $B(2,-1,4)$, $C(4,5,-1)$. 3M

Ans. $\overrightarrow{AB}\times\overrightarrow{AC}=9\hat{i}+7\hat{j}+12\hat{k}$; area $\dfrac{\sqrt{274}}{2}$

Q49. Parallelogram: Diagonals $3\hat{i}+\hat{j}-2\hat{k}$ and $\hat{i}-3\hat{j}+4\hat{k}$. Area? 2M

Ans. $\dfrac{1}{2}|\vec{d}_1\times\vec{d}_2|=\dfrac{10\sqrt{3}}{2}=5\sqrt{3}$

Q50. Shortest distance: $\vec{r}=(\hat{i}+2\hat{j}+\hat{k})+\lambda(\hat{i}-\hat{j}+\hat{k})$ and $\vec{r}=(2\hat{i}-\hat{j}-\hat{k})+\mu(2\hat{i}+\hat{j}+2\hat{k})$. 5M

Ans. $\vec{b}_1\times\vec{b}_2=-3\hat{i}+3\hat{k}$; $d=\dfrac{9}{3\sqrt{2}}=\dfrac{3}{\sqrt{2}}$

Q51. Angle between lines: $\dfrac{x-2}{2}=\dfrac{y-1}{5}=\dfrac{z+3}{-3}$ and $\dfrac{x+2}{-1}=\dfrac{y-4}{8}=\dfrac{z-5}{4}$. 2M

Ans. $\cos\theta=\dfrac{26}{9\sqrt{38}}$

Q52. Parallel lines: Distance between $\vec{r}=(\hat{i}+\hat{j})+\lambda(2\hat{i}-\hat{j}+\hat{k})$ and $\vec{r}=(2\hat{i}+\hat{j}-\hat{k})+\mu(4\hat{i}-2\hat{j}+2\hat{k})$. 3M

Ans. $d=\dfrac{|\vec{b}\times(\vec{a}_2-\vec{a}_1)|}{|\vec{b}|}=\sqrt{\dfrac{11}{6}}$

Q53. Inverse: $A=\begin{bmatrix}3&-2\\4&-2\end{bmatrix}$. Find A^{-1}. 2M

Ans. $|A|=2$; $A^{-1}=\dfrac{1}{2}\begin{bmatrix}-2&2\\-4&3\end{bmatrix}$

Q54. Matrix equation: $A=\begin{bmatrix}4&1\\-2&1\end{bmatrix}$. Show $A^2-5A+6I=O$. 3M

Ans. $A^2=\begin{bmatrix}14&5\\-10&-1\end{bmatrix}$; $A^2-5A=-6I$ ✓

Q55. $X=A^{-1}B$: Solve $x+2y+z=8$, $2x-y+z=3$, $x+y-z=0$. 5M

Ans. $|A|=9$; $x=1, y=2, z=3$

Q56. Determinant: $\begin{vmatrix}2x&5\\8&x\end{vmatrix}=\begin{vmatrix}6&-2\\7&3\end{vmatrix}$. Find x. 1M

Ans. $2x^2-40=32$: $x=\pm6$

Q57. Area by determinant: Area of triangle with vertices $(1,0)$, $(6,0)$, $(4,3)$. 2M

Ans. $\dfrac{15}{2}$ sq units

Q58. LPP (bounded): Max $Z=3x+5y$: $x+y\le8$, $x+3y\le12$, $x,y\ge0$. 3M

Ans. Corners $(0,0)$, $(8,0)$, $(6,2)$, $(0,4)$; max 28 at $(6,2)$

Q59. LPP (unbounded): Min $Z=4x+6y$: $x+2y\ge8$, $3x+2y\ge12$, $x,y\ge0$. 3M

Ans. Corners $(0,6)$, $(2,3)$, $(8,0)$: $36, 26, 32$. $4x+6y<26$ misses the region: min 26 at $(2,3)$

Q60. Conditional: $P(A)=0.7$, $P(B)=0.5$, $P(A|B)=0.6$. Find $P(A\cup B)$ and $P(B|A)$. 2M

Ans. $P(A\cap B)=0.3$; 0.9 and $\dfrac{3}{7}$

Q61. Independent: A, B independent, $P(A)=\dfrac{1}{4}$, $P(B)=\dfrac{1}{3}$. P(neither)? 1M

Ans. $\dfrac{3}{4}\times\dfrac{2}{3}=\dfrac{1}{2}$

Q62. Bayes' (coin + bags): Head: draw from Bag I (2 white, 3 red); tail: Bag II (4 white, 1 red). Ball is white. P(Bag I)? 3M

Ans. $\dfrac{\frac{1}{2}\times\frac{2}{5}}{\frac{1}{2}\times\frac{2}{5}+\frac{1}{2}\times\frac{4}{5}}=\dfrac{1}{3}$

Q63. Bayes' (machines): Machines A, B, C make 50%, 30%, 20% of items; defective rates 2%, 3%, 4%. A defective item is picked. P(from C)? 5M

Ans. $\dfrac{0.008}{0.027}=\dfrac{8}{27}$

Q64. Distribution: Two cards drawn with replacement from 52. X = number of aces. Distribution and mean. 3M

Ans. $\dfrac{144}{169}, \dfrac{24}{169}, \dfrac{1}{169}$; mean $\dfrac{2}{13}$

Exam day plan (3 hours)

1. First 15 min: read the full paper, mark the sure-shot questions.
2. Section A (20 MCQ/AR): about 25 min; do not get stuck on one MCQ.
3. Sections B and C: about 55 min; write the formula first, then steps.
4. Sections D and E: about 70 min; diagrams for LPP, areas and 3D lines.
5. Last 15 min: check signs, $+C$, units and that every part is attempted.

Last-week must-do checklist

- → Equivalence relation proof + one equivalence class
- → Principal values of all six inverse trig functions
- → Sym + skew split; $A^2-kA+lI=O$ type
- → System of 3 equations by $X=A^{-1}B$
- → Continuity at a point with unknown k or a, b
- → Log, implicit, parametric, second derivative
- → Increasing/decreasing intervals; one optimisation word problem
- → Partial fractions, by parts, $\int_0^a f(a-x)$ property
- → Area of circle/ellipse/parabola region
- → Homogeneous DE and linear DE with condition
- → Projection, cross product area, shortest distance
- → LPP corner table, unbounded region check
- → Bayes' theorem and probability distribution mean

Formula flash $|\text{adj } A|=|A|^{n-1}$, $|kA|=k^n|A|$ for order n.

LAST-NIGHT REVISION

How to use these 4 pages

Ek chapter padho, aankh band karke uske 3 points bolo, phir aage. Last mein rapid fire ke answers dhak ke try karo. Naya kuch mat padho, jo aata hai use pakka karo.

Exam hall strategy (80 marks, 3 hours)

Section	Marks	Time plan
A: 18 MCQ + 2 A-R (1M)	20	25 to 30 min
B: 5 VSA (2M)	10	15 to 20 min
C: 6 SA (3M)	18	35 to 40 min
D: 4 LA (5M)	20	45 to 50 min
E: 3 case-based (4M)	12	25 min
Final check		10 to 15 min

- → Use the 15 min reading time to pick internal choices and spot the easy 5-markers.
- → Do the questions you are sure of first; mark doubtful MCQs and come back.
- → Write the formula first, then substitute: a correct formula often earns a step mark.
- → Draw figures for LPP, area and 3D questions; box the final answer with units.
- → Never leave an MCQ blank: there is no negative marking.

Unit-wise marks (80)

Unit	Chapters	Marks
I Relations and Functions	1, 2	8
II Algebra	3, 4	10
III Calculus	5 to 9	35
IV Vectors and 3D	10, 11	14
V Linear Programming	12	5
VI Probability	13	8

↳ Calculus = 35 marks: integrals aur derivatives ke questions sabse pehle pakke karo.

1. Relations and Functions

- → Equivalence = Reflexive + Symmetric + Transitive; classes are disjoint and cover the set.
- → Relations on n elements: 2^{n^2}; reflexive: 2^{n^2-n}; symmetric: $2^{n(n+1)/2}$.
- → One-one: $f(x_1)=f(x_2)\Rightarrow x_1=x_2$. Onto: solve $y=f(x)$, check x is in the domain.
- → Trap: x^2 is one-one on $\mathbb{N}$, not on $\mathbb{R}$; transitivity fails needs a specific counter-example.

2. Inverse Trigonometric Functions

- → Ranges: $\sin^{-1}$, $\tan^{-1}$, $\operatorname{cosec}^{-1}$ in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ type; $\cos^{-1}$, $\cot^{-1}$, $\sec^{-1}$ in $[0,\pi]$ type.
- → $\cos^{-1}(-x)=\pi-\cos^{-1}x$; $\sin^{-1}(-x)=-\sin^{-1}x$.
- → $\sin^{-1}(\sin x)=x$ only for $x\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$.
- → Trap: $\cos^{-1}\left(-\frac{1}{2}\right)=\frac{2\pi}{3}$, never $-\frac{\pi}{3}$.

3. Matrices

- → AB exists if columns of A = rows of B; $AB\neq BA$ in general.
- → $(AB)'=B'A'$; $A+A'$ symmetric, $A-A'$ skew-symmetric; $A=\frac{1}{2}(A+A')+\frac{1}{2}(A-A')$.
- → Skew-symmetric: all diagonal entries 0. Inverse is unique; $(AB)^{-1}=B^{-1}A^{-1}$.
- → Trap: $(A+B)^2=A^2+AB+BA+B^2$, not $A^2+2AB+B^2$.

4. Determinants

- → $|kA|=k^n|A|$ for order n; $|AB|=|A||B|$; $|A'|=|A|$.
- → $A^{-1}=\frac{1}{|A|}\operatorname{adj}A$ (needs $|A|\neq 0$); $|\operatorname{adj}A|=|A|^{n-1}$; $A(\operatorname{adj}A)=|A|I$.
- → Area of triangle $=\frac{1}{2}|\Delta|$; area 0 means collinear.
- → $AX=B$: $|A|\neq 0\Rightarrow$ unique solution $X=A^{-1}B$; $|A|=0$: check $(\operatorname{adj}A)B$ (zero: infinitely many or none, non-zero: no solution).

5. Continuity and Differentiability

- → Continuous at a: LHL = RHL = $f(a)$. Differentiable $\Rightarrow$ continuous (not conversely: $|x|$ at 0).
- → Chain rule; $\frac{d}{dx}\sin^{-1}x=\frac{1}{\sqrt{1-x^2}}$, $\frac{d}{dx}\tan^{-1}x=\frac{1}{1+x^2}$.
- → x^x type or long products: take log first. Parametric: $\frac{dy}{dx}=\frac{dy/dt}{dx/dt}$.
- → Trap: $\frac{d^2y}{dx^2}$ in parametric form is $\frac{d}{dt}\left(\frac{dy}{dx}\right)\div\frac{dx}{dt}$, not $\frac{y''(t)}{x''(t)}$.

6. Application of Derivatives

- → Rate of change: differentiate w.r.t. t with the chain rule ($\frac{dA}{dt}=2\pi r\frac{dr}{dt}$).

→ $f'(x) > 0$ increasing, $f'(x) < 0$ decreasing; split at points where $f'(x) = 0$.
→ Local max: $f' = 0$, $f'' < 0$; local min: $f'' > 0$. Absolute on $[a, b]$: compare critical points and end points.
→ Trap: tangents, normals and approximations are out of syllabus this year.

7. Integrals

→ $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}$, $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a}$, $\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right|$.
→ By parts: ILATE, $\int uv\,dx = u\int v\,dx - \int\left(u'\int v\,dx\right)dx$; $\int e^x(f+f')\,dx = e^x f + C$.
→ $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$; odd function on $[-a, a]$ gives 0.
→ Trap: $+C$ in every indefinite integral; change limits when you substitute in a definite one.

8. Application of Integrals

→ Area under $y = f(x)$ from a to $b = \int_a^b |y|\,dx$; about the y-axis use $\int x\,dy$.
→ Circle $x^2 + y^2 = a^2$: area πa^2; ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$: area πab.
→ Always draw a rough sketch and shade the region first.
→ Trap: area is never negative; area between two curves is out this year.

9. Differential Equations

→ Order = highest derivative; degree = its power when the DE is a polynomial in derivatives (else not defined).
→ Variables separable: $f(y)dy = g(x)dx$, integrate both sides.
→ Homogeneous: put $y = vx$, $\frac{dy}{dx} = v + x\frac{dv}{dx}$.
→ Linear: $\frac{dy}{dx} + Py = Q$, I.F. $= e^{\int P\,dx}$, $y \times \text{I.F.} = \int Q \times \text{I.F.}\,dx + C$.

10. Vector Algebra

→ $|\vec{a}| = \sqrt{x^2+y^2+z^2}$; unit vector $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$; d.c.s satisfy $l^2 + m^2 + n^2 = 1$.
→ $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta$; projection of $\vec{a}$ on $\vec{b} = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$; $\perp \iff \vec{a}\cdot\vec{b} = 0$.
→ $|\vec{a}\times\vec{b}|$ = area of parallelogram; triangle $= \frac{1}{2}|\vec{a}\times\vec{b}|$; $\parallel \iff \vec{a}\times\vec{b} = \vec{0}$.
→ Section formula (internal): $\frac{m\vec{b}+n\vec{a}}{m+n}$.

11. Three Dimensional Geometry

→ Line: $\vec{r} = \vec{a} + \lambda\vec{b}$; Cartesian $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$.
→ Angle: $\cos\theta = \frac{|a_1a_2+b_1b_2+c_1c_2|}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}$.
→ Shortest distance (skew): $\frac{|(\vec{b}_1\times\vec{b}_2)\cdot(\vec{a}_2-\vec{a}_1)|}{|\vec{b}_1\times\vec{b}_2|}$; parallel: $\frac{|\vec{b}\times(\vec{a}_2-\vec{a}_1)|}{|\vec{b}|}$.
→ Trap: planes are out of syllabus; SD $= 0$ means the lines intersect.

12. Linear Programming

→ Draw constraints, shade the feasible region, list corner points, evaluate Z at each.
→ Bounded region: max and min both exist at corners.
→ Unbounded region: check the open half-plane $Z > M$ (for max) or $Z < m$ (for min) has no common point with the region.
→ Trap: do not forget $x \geq 0$, $y \geq 0$ and the corner on the axes.

13. Probability

→ $P(A|B) = \frac{P(A\cap B)}{P(B)}$; independent: $P(A\cap B) = P(A)P(B)$.
→ Total probability: $P(A) = \sum P(E_i)P(A|E_i)$; Bayes: $P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum P(E_j)P(A|E_j)}$.
→ Random variable: $\sum p_i = 1$; mean $E(X) = \sum x_i p_i$.
→ Trap: independent is not the same as mutually exclusive; binomial and variance are out this year.



Shivam Bhaiya says
Raat ko naya chapter **bilkul mat** kholna. Ye 4 page + formula sheet padho, rapid fire karo, aur so jao. Paper mein pehle woh questions karo jo pakka aate hain.



Meme samjho
Raat ko 11 baje: "sab yaad hai". Raat ko 2 baje naya chapter khola: **sab bhool gaya.**
Asli baat: Exam ki raat sirf revision karo aur 6-7 ghante soyo; neend mein hi memory pakki hoti hai.

Common mistakes in the exam hall

→ Copying a sign wrong from the question: re-read the question before the final answer.
→ Forgetting $+C$, units (sq units, cm/s), or the domain restriction in the final answer.
→ Leaving $\sqrt{8}$ or $\frac{6}{8}$ unsimplified; write $2\sqrt{2}$, $\frac{3}{4}$.
→ Not writing "Hence proved" or the final statement in proofs.
→ In 3D, using the formula for skew lines when the lines are parallel.
→ In probability, adding when events need multiplying (and vice versa).
→ Spending 15 minutes on one MCQ: skip and return.

5-markers to practise once more

- Ch 1: show a relation is an equivalence relation + find a class; prove a function bijective.
- Ch 3-4: solve 3 equations by $X = A^{-1}B$; A^{-1} by adjoint; product AB then use B^{-1}.
- Ch 5: logarithmic differentiation; second derivative proofs like $y'' + y = 0$ type.
- Ch 6: maxima-minima word problems (box from a sheet, cylinder in sphere, fencing).
- Ch 7: partial fractions; $\int_0^{\pi}$ with the $f(a+b-x)$ property; $\int \sqrt{a^2 - x^2}\,dx$ type.
- Ch 8: area of circle/ellipse part or region under a parabola and a line from the x-axis.
- Ch 9: linear DE with initial condition; homogeneous DE with $y = vx$.
- Ch 10-11: shortest distance between skew lines; foot of perpendicular from a point to a line.
- Ch 12: LPP graph with 3 constraints and corner table.
- Ch 13: Bayes' theorem with 2 or 3 bags/machines; probability distribution and mean.

Night before and morning of

- Pack: admit card, 2 blue pens, pencil, eraser, scale, water bottle.
- Read the formula sheet and these pages once; no new topic.
- Sleep 6 to 7 hours; light breakfast; reach the centre early.
- First 15 minutes: read the whole paper, choose internal options, plan the order.

Derivative and integral pairs (read both ways)

$f(x)$	$f'(x)$	$\int f(x)\,dx$
x^n	nx^{n-1}	$\frac{x^{n+1}}{n+1} + C\ (n \neq -1)$
e^x	e^x	$e^x + C$
a^x	$a^x \log a$	$\frac{a^x}{\log a} + C$
$\sin x$	$\cos x$	$-\cos x + C$
$\cos x$	$-\sin x$	$\sin x + C$
$\tan x$	$\sec^2 x$	$\log\lvert\sec x\rvert + C$
$\log x$	$\frac{1}{x}$	$x\log x - x + C$
$\sec x$	$\sec x \tan x$	$\log\lvert\sec x + \tan x\rvert + C$

Quick fact $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

Quick fact $\sin^{-1} x$ domain $[-1, 1]$, $\tan^{-1} x$ domain $\mathbb{R}$.

Formulas used in almost every paper

- $\sin^2 x + \cos^2 x = 1,\ 1 + \tan^2 x = \sec^2 x$
- $\sin 2x = 2\sin x \cos x,\ \cos 2x = 1 - 2\sin^2 x$
- $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2|\vec{b}|^2$
- $A^{-1} = \frac{1}{|A|}\text{adj}A$
- $\frac{d}{dx}(uv) = u'v + uv',\ \frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$
- $P(A \cap B) = P(A)P(B|A)$

14. Rapid Fire: 1-line Questions

↳ Answers dhak ke bolo; galti wale par tick lagao aur subah ek baar phir dekho.

Ch 1 to 2

#	Rapid fire	Answer
1	Relations on a 2-element set?	$2^4 = 16$
2	Bijections from a 5-element set to itself?	$5! = 120$
3	Is $R = \{(1,1),(2,2)\}$ on $\{1,2,3\}$ reflexive?	No, $(3,3)$ missing
4	Onto functions from $\{1,2,3\}$ to $\{a,b\}$?	$2^3 - 2 = 6$
5	Principal value of $\tan^{-1}(-\sqrt{3})$?	$-\frac{\pi}{3}$
6	Principal value of $\sec^{-1}(-2)$?	$\frac{2\pi}{3}$
7	Range of $\cot^{-1} x$?	$(0, \pi)$
8	$\sin^{-1}\left(\sin\frac{5\pi}{6}\right)$?	$\frac{\pi}{6}$

Ch 3 to 4

#	Rapid fire	Answer
9	Order of AB if A is 2×3, B is 3×4?	2×4
10	Diagonal entries of a skew-symmetric matrix?	All 0
11	Number of 3×3 matrices with entries 0 or 1?	$2^9 = 512$
12	If A is symmetric, $A - A' =$?	O
13	$\lvert 3A\rvert$ if A is 3×3 with $\lvert A\rvert = 2$?	$27 \times 2 = 54$
14	$\lvert\text{adj}A\rvert$ if A is 3×3, $\lvert A\rvert = 5$?	$5^2 = 25$
15	$\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix}$?	$10 - 12 = -2$
16	A^{-1} exists iff?	$\lvert A\rvert \neq 0$

Ch 5 to 6

#	Rapid fire	Answer
17	$\frac{d}{dx}(\sin^{-1} x)$?	$\frac{1}{\sqrt{1-x^2}}$
18	$\frac{d}{dx}(e^{3x})$?	$3e^{3x}$

#	Rapid fire	Answer
19	$\frac{d}{dx}(\log \sin x)$?	$\cot x$
20	Is $\lvert x\rvert$ differentiable at 0?	No (continuous only)
21	$\frac{d}{dx}(x^x)$?	$x^x(1+\log x)$
22	If $x=t^2, y=t^3$, $\frac{dy}{dx}$?	$\frac{3t}{2}$
23	$f(x)=x^3$: increasing on?	$\mathbb{R}$
24	Local min value of x^2-4x+7?	3 (at $x=2$)
25	Circle radius increases at 2 cm/s; $\frac{dA}{dt}$ at $r=5$?	20π cm^2/s
26	Max value of $\sin x+\cos x$?	$\sqrt{2}$

Ch 7 to 9

#	Rapid fire	Answer
27	$\int \frac{dx}{1+x^2}$?	$\tan^{-1}x+C$
28	$\int e^x(\sin x+\cos x)dx$?	$e^x\sin x+C$
29	$\int_{-1}^{1} x^3\,dx$?	0 (odd)
30	$\int_0^{\pi/2}\sin x\,dx$?	1
31	$\int \frac{1}{x}dx$?	$\log\lvert x\rvert+C$
32	$\int_0^1 x\,dx$?	$\frac{1}{2}$
33	Area of ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$?	6π
34	Area of circle $x^2+y^2=16$?	16π
35	Area under $y=x^2$ from 0 to 3?	9
36	Order and degree of $\left(\frac{d^2y}{dx^2}\right)^3+\frac{dy}{dx}=0$?	2 and 3
37	I.F. of $\frac{dy}{dx}+\frac{y}{x}=x^2$?	x
38	Solution of $\frac{dy}{dx}=\frac{y}{x}$?	$y=Cx$
39	Degree of $\frac{dy}{dx}+\sin\left(\frac{dy}{dx}\right)=0$?	Not defined

Ch 10 to 11

#	Rapid fire	Answer
40	$\lvert 2\hat{i}-\hat{j}+2\hat{k}\rvert$?	3
41	$\hat{i}\cdot\hat{i}$ and $\hat{i}\times\hat{j}$?	1 and $\hat{k}$
42	Angle if $\vec{a}\cdot\vec{b}=0$ (non-zero vectors)?	90°
43	Projection of $\hat{i}+\hat{j}$ on $\hat{i}$?	1
44	Can $\frac{1}{2},\frac{1}{2},\frac{1}{2}$ be d.c.s?	No: sum of squares $\frac{3}{4}\neq 1$
45	D.r.s of $\frac{x-1}{2}=\frac{y+3}{-1}=\frac{z}{4}$?	$2,-1,4$
46	Lines with d.r.s $1,2,3$ and $3,0,-1$: angle?	90° (dot $=0$)
47	SD between intersecting lines?	0

#	Rapid fire	Answer
48	In LPP, optimum of a bounded region lies at?	A corner point

Ch 12 to 13

#	Rapid fire	Answer
49	$Z=3x+2y$ at $(2,5)$?	16
50	$P(A\lvert B)$ if $P(A\cap B)=0.2, P(B)=0.5$?	0.4
51	A, B independent, $P(A)=0.3, P(B)=0.5$: $P(A\cap B)$?	0.15
52	Mean of X: values 0, 1, 2 with P = 0.25, 0.5, 0.25?	1
53	$P(\text{sum}=7)$ with two dice?	$\frac{1}{6}$
54	If $\sum p_i=1$ with $p=k,2k,3k,4k,k$?	$\frac{1}{10}$

Mixed bag

#	Rapid fire	Answer
55	Number of reflexive relations on $\{1,2,3\}$?	$2^6=64$
56	$\cos^{-1}\left(\cos\frac{4\pi}{3}\right)$?	$\frac{2\pi}{3}$
57	If $A^2=A$ (idempotent), $(I-A)^2=$?	$I-A$
58	$\frac{d}{dx}(\tan^{-1}x)$ at $x=1$?	$\frac{1}{2}$
59	$\int_0^2(2x+1)\,dx$?	6
60	$P(A\cup B)$ if $P(A)=0.4, P(B)=0.3, P(A\cap B)=0.1$?	0.6

Quick fact $(A')'=A$, $(kA)'=kA'$, $(A+B)'=A'+B'$.

Quick fact $P(\bar{A})=1-P(A)$; $P(A\cup B)=P(A)+P(B)-P(A\cap B)$.

Quick fact Bayes: $P(E_i|A)=\frac{P(E_i)P(A|E_i)}{\sum_j P(E_j)P(A|E_j)}$

Apne notes / galtiyan yahan likho

Mock Paper 1

Max marks 80 | Time 3 hours | General instructions

1. This paper has 38 questions in 5 sections; all are compulsory.
2. **Section A:** Q1 to Q18 MCQs and Q19, Q20 Assertion-Reason, 1 mark each.
3. **Section B:** Q21 to Q25, very short answer, 2 marks each.
4. **Section C:** Q26 to Q31, short answer, 3 marks each.
5. **Section D:** Q32 to Q35, long answer, 5 marks each.
6. **Section E:** Q36 to Q38, case-based (1 + 1 + 2), 4 marks each.
7. Internal choice is given in some questions (marked OR). Calculators are not allowed.

↳ Pehle poora paper 10 min mein padh lo, phir apne strong section se shuru karo. Time plan: A 25 min, B 20, C 40, D 50, E 30, revision 15.

Marks blueprint (2026-27)

Unit	Chapters	Marks
I	Relations and Functions, Inverse Trigonometric Functions	8
II	Matrices, Determinants	10
III	Continuity and Differentiability, Application of Derivatives, Integrals, Application of Integrals, Differential Equations	35
IV	Vector Algebra, Three Dimensional Geometry	14
V	Linear Programming	5
VI	Probability	8
Total		**80**

Section-wise strategy

- → **A:** 1 mark each, no working needed; eliminate options smartly.
- → **B, C:** write the formula first, then substitute; box the final answer.
- → **D:** 5 marks = 5 clear steps; draw the figure in area and LPP questions.
- → **E:** read the case once, then answer the parts; part (iii) carries 2 marks.

Section A (1 mark each)

Q1. The relation $R = \{(1,1),(2,2),(3,3),(1,2),(2,1)\}$ on the set $\{1,2,3\}$ is 1M
(a) reflexive only (b) symmetric only (c) an equivalence relation (d) transitive only

Q2. The value of $\sin^{-1}\left(-\frac{1}{2}\right) + \cos^{-1}\left(\frac{1}{2}\right)$ is 1M
(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{3}$ (d) 0

Q3. If A is a 3×4 matrix and B is a 4×2 matrix, the order of AB is 1M
(a) 4×4 (b) 3×2 (c) 2×3 (d) not defined

Q4. If $A = \begin{bmatrix} 1 & x \\ 3 & 2 \end{bmatrix}$ is symmetric, then x is 1M
(a) 1 (b) 2 (c) 3 (d) 6

Q5. If A is a square matrix such that $A^2 = A$, then $(I + A)^2 - 3A$ is 1M
(a) I (b) A (c) 2A (d) O

Q6. If $\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$, then x is 1M
(a) 6 (b) -6 (c) ± 6 (d) 0

Q7. If A is a 3×3 matrix with $|A| = 4$, then $|\text{adj } A|$ is 1M
(a) 4 (b) 16 (c) 64 (d) 12

Q8. $f(x) = \dfrac{\sin 3x}{x}$ for $x \neq 0$ and $f(0) = k$ is continuous at $x = 0$. Then k is 1M
(a) 1 (b) 0 (c) 3 (d) $\frac{1}{3}$

Q9. $\dfrac{d}{dx}\left(e^{x^2}\right)$ is 1M
(a) e^{x^2} (b) $2xe^{x^2}$ (c) $x^2e^{x^2}$ (d) $2e^x$

Q10. $f(x) = x^3 - 3x$ is strictly decreasing in 1M
(a) $(-\infty, -1)$ (b) $(-1, 1)$ (c) $(1, \infty)$ (d) $\mathbb{R}$

Q11. $\int e^x(\sin x + \cos x)\,dx$ is 1M
(a) $e^x \cos x + C$ (b) $e^x \sin x + C$ (c) $-e^x \sin x + C$ (d) $e^x(\sin x - \cos x) + C$

Q12. The sum of the order and degree of $\left(\dfrac{d^2y}{dx^2}\right)^3 + \left(\dfrac{dy}{dx}\right)^4 + y = 0$ is 1M
(a) 4 (b) 5 (c) 6 (d) 7

Q13. $\int_0^{\pi/2} \sin^2 x\,dx$ is 1M
(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) 1 (d) π

Q14. If $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $\vec{a} \cdot \vec{b} = 3$, the angle between $\vec{a}$ and $\vec{b}$ is 1M
(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

Q15. The direction ratios of the line $\dfrac{x-1}{2} = \dfrac{y+3}{-1} = \dfrac{1-z}{3}$ are 1M
(a) 2, −1, 3 (b) 2, −1, −3 (c) 1, −3, 1 (d) 2, 1, 3

Q16. The corner points of a feasible region are (0, 0), (5, 0), (3, 4), (0, 5). The maximum of $Z = 3x + 2y$ is 1M
(a) 15 (b) 10 (c) 17 (d) 12

Q17. In a linear programming problem, the objective function is always 1M
(a) linear (b) quadratic (c) cubic (d) constant

Q18. A and B are independent with $P(A) = 0.4$ and $P(B) = 0.5$. Then $P(A \cup B)$ is 1M
(a) 0.9 (b) 0.2 (c) 0.7 (d) 0.6

Q19. **A:** $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ are perpendicular. **R:** Two non-zero vectors are perpendicular if and only if $\vec{a} \cdot \vec{b} = 0$. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Q20. **A:** The lines $\vec{r} = (\hat{i} + \hat{j}) + \lambda(2\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = (2\hat{i} + \hat{k}) + \mu(4\hat{i} - 2\hat{j} + 2\hat{k})$ are parallel. **R:** Two lines are parallel if their direction vectors are proportional. 1M
(a) Both true, R explains A (b) Both true, R does not explain A
(c) A true, R false (d) A false, R true

Section B (2 marks each)

Q21. Find the value of $\tan^{-1}\sqrt{3} - \sec^{-1}(-2)$.
OR
Find the domain of $\sin^{-1}(2x - 1)$. 2M

Q22. If $y = \log(\cos e^x)$, find $\dfrac{dy}{dx}$.
OR
If $x = 2\cos t$ and $y = 3\sin t$, find $\dfrac{dy}{dx}$. 2M

Q23. The radius of a circle is increasing at 0.5 cm/s. Find the rate at which its area is increasing when the radius is 4 cm.
OR
Find the intervals in which $f(x) = 2x^3 - 9x^2 + 12x + 15$ is strictly increasing. 2M

Q24. Find $\displaystyle\int \frac{x}{(x+1)(x+2)}\,dx$.
OR
Find $\displaystyle\int xe^x\,dx$. 2M

Q25. Find a unit vector perpendicular to both $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$.
OR
Find the projection of $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. 2M

Section C (3 marks each)

Q26. If $y = x^{\sin x}$, $x > 0$, find $\dfrac{dy}{dx}$.
OR
If $y = e^{a\cos^{-1}x}$, $-1 < x < 1$, show that $(1 - x^2)\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} - a^2y = 0$. 3M

Q27. Find $\displaystyle\int \frac{3x + 5}{x^2 - x - 6}\,dx$.
OR
Find $\displaystyle\int x^2e^x\,dx$. 3M

Q28. Solve the differential equation $\dfrac{dy}{dx} + 2y = e^{-x}$.
OR
Solve $x\dfrac{dy}{dx} = y + x\tan\dfrac{y}{x}$. 3M

Q29. Find the area of the triangle with vertices A(1, 1, 1), B(1, 2, 3) and C(2, 3, 1), using vectors.
OR
Using vectors, show that A(1, 2, 7), B(2, 6, 3) and C(3, 10, −1) are collinear. 3M

Q30. Solve graphically: maximise $Z = 4x + y$ subject to $x + y \le 50$, $3x + y \le 90$, $x \ge 0$, $y \ge 0$.
OR
Solve graphically: minimise $Z = 5x + 7y$ subject to $2x + y \ge 8$, $x + 2y \ge 10$, $x \ge 0$, $y \ge 0$. 3M

Q31. A bag has 4 red and 6 black balls. Two balls are drawn at random without replacement. Find the probability distribution of the number of red balls and its mean.
OR
Bag I has 2 white and 3 red balls; bag II has 4 white and 5 red balls. A bag is chosen at random and a ball drawn from it is red. Find the probability that it came from bag II. 3M

Section D (5 marks each)

Q32. Using the matrix method, solve: $x + y + z = 6$, $x - y + z = 2$, $2x + y - z = 1$.
OR
Find A^{-1} for $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & -1 & 2 \\ 3 & 2 & -1 \end{bmatrix}$ and hence solve the system $2x + y + z = 3$, $x - y + 2z = 6$ and $3x + 2y - z = -1$. 5M

Q33. Evaluate $\displaystyle\int_0^{\pi} \frac{x\sin x}{1 + \cos^2 x}\,dx$.
OR
Evaluate $\displaystyle\int_1^4 \left(|x - 1| + |x - 2| + |x - 3|\right)dx$. 5M

Q34. Using integration, find the area enclosed by the ellipse $\dfrac{x^2}{16} + \dfrac{y^2}{9} = 1$.
OR
Using integration, find the area of the region bounded by $y^2 = 4x$, the lines $x = 1$, $x = 4$ and the x-axis, in the first quadrant. 5M

Exam tip Write question numbers clearly in the margin, exactly as in the paper.

Q35. Find the shortest distance between the lines $\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j} - \hat{k}) + \mu(2\hat{i} + \hat{j} + 2\hat{k})$.

OR

Find the vector and Cartesian equations of the line through $(2, -1, 3)$ parallel to $\frac{x-1}{3} = \frac{y+2}{2} = \frac{z-4}{-1}$. Also find the angle between this line and a line with direction ratios 1, 1, 1. 5M

Section E (case based, 4 marks each)

Q36. Case: Toll pricing. A highway toll system uses $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 3x - 5$ to convert a code x into a charge. (i) Show that f is one-one. (1) (ii) Find x such that $f(x) = 10$. (1) (iii) Show that f is onto, and hence bijective. (2) **OR** (iii) Is $g : \mathbb{R} \to \mathbb{R}$, $g(x) = x^2$ one-one? Justify. (2) 4M

Q37. Case: Water tank. An open-top tank with a square base of side x m must hold 32 m^3 of water. The sheet used is the base plus the four walls. (i) Express the sheet area S in terms of x. (1) (ii) Find $\frac{dS}{dx}$. (1) (iii) Find the dimensions that make S least, and show it is a minimum. (2) **OR** (iii) If the tank must hold 108 m^3, find the side of the base for the least sheet area. (2) 4M

Q38. Case: Pen factory. Machines A and B make 60% and 40% of the pens of a factory; 2% of A's pens and 4% of B's pens are defective. (i) Find the probability that a pen is made by A and is defective. (1) (ii) Find the probability that a pen is defective. (1) (iii) A pen is found defective; find the probability that it was made by B. (2) **OR** (iii) Find the probability that a pen is made by A and is NOT defective. (2) 4M



Meme samjho

Paper khatam karke jab saare answers key se match ho jaayein: "The presentation is done!"

Asli baat: Mock paper ko 3 hours ke timer ke saath likho, phir key se check karo. Jo galat hua, usi chapter ka MIQ dobara karo.

Answer Key (with key steps)

Section A

Q1. Ans. (c) reflexive (all (a, a)), symmetric ((1,2) and (2,1)), transitive: equivalence

Q2. Ans. (a) $-\frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi}{6}$

Q3. Ans. (b) $(3 \times 4)(4 \times 2)$ gives 3×2

Q4. Ans. (c) symmetric: $a_{12} = a_{21} \Rightarrow x = 3$

Q5. Ans. (a) $I + 2A + A^2 - 3A = I + 2A + A - 3A = I$

Q6. Ans. (c) $x^2 - 36 = 36 - 36 = 0 \Rightarrow x = \pm 6$

Q7. Ans. (b) $|\text{adj}\, A| = |A|^{n-1} = 4^2 = 16$

Q8. Ans. (c) $\lim_{x\to 0} \frac{\sin 3x}{x} = 3 = k$

Q9. Ans. (b) chain rule: $e^{x^2} \times 2x$

Q10. Ans. (b) $f'(x) = 3(x^2 - 1) < 0$ on $(-1, 1)$

Q11. Ans. (b) form $\int e^x(f + f')\,dx = e^x f(x) + C$ with $f = \sin x$

Q12. Ans. (b) order 2, degree 3, sum 5

Q13. Ans. (b) $\int_0^{\pi/2} \frac{1-\cos 2x}{2}\,dx = \frac{\pi}{4}$

Q14. Ans. (c) $\cos\theta = \frac{3}{2\times 3} = \frac{1}{2}$

Q15. Ans. (b) $\frac{1-z}{3} = \frac{z-1}{-3}$, so $2, -1, -3$

Q16. Ans. (c) Z at corners: 0, 15, 17, 10; max 17

Q17. Ans. (a) objective function is linear by definition

Q18. Ans. (c) $0.4 + 0.5 - 0.4 \times 0.5 = 0.7$

Q19. Ans. (a) $\vec{a} \cdot \vec{b} = 2 - 1 - 1 = 0$; R explains A

Q20. Ans. (a) $4\hat{i} - 2\hat{j} + 2\hat{k} = 2(2\hat{i} - \hat{j} + \hat{k})$; R explains A

Section B

Q21. Ans. $\frac{\pi}{3} - \frac{2\pi}{3} = -\frac{\pi}{3}$ (principal value of $\sec^{-1}(-2)$ is $\frac{2\pi}{3}$). **OR:** $-1 \le 2x - 1 \le 1 \Rightarrow 0 \le x \le 1$, domain $[0, 1]$.

Q22. Ans. $\frac{dy}{dx} = \frac{1}{\cos e^x} \times (-\sin e^x) \times e^x = -e^x \tan(e^x)$. **OR:** $\frac{dy}{dx} = \frac{3\cos t}{-2\sin t} = -\frac{3}{2}\cot t$.

Q23. Ans. $A = \pi r^2$, $\frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2\pi(4)(0.5) = 4\pi$ cm^2/s. **OR:** $f'(x) = 6x^2 - 18x + 12 = 6(x-1)(x-2) > 0$ for $x < 1$ or $x > 2$: strictly increasing on

(contd.) $(-\infty, 1) \cup (2, \infty)$.

Q24.

Ans. $\frac{x}{(x+1)(x+2)} = \frac{-1}{x+1} + \frac{2}{x+2}$, so $-\log|x+1| + 2\log|x+2| + C$. **OR:** by parts: $xe^x - e^x + C = (x-1)e^x + C$.

Q25.

Ans. $\vec{a} \times \vec{b} = \hat{i} - 2\hat{j} + \hat{k}$, magnitude $\sqrt{6}$. Unit vector $= \pm\frac{1}{\sqrt{6}}(\hat{i} - 2\hat{j} + \hat{k})$. **OR:** $\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = \frac{2+6+2}{\sqrt{6}} = \frac{10}{\sqrt{6}} = \frac{5\sqrt{6}}{3}$.

Section C

Q26.

Ans. $\log y = \sin x \log x \Rightarrow \frac{1}{y}\frac{dy}{dx} = \cos x \log x + \frac{\sin x}{x}$. So $\frac{dy}{dx} = x^{\sin x}\left(\cos x \log x + \frac{\sin x}{x}\right)$. **OR:** $y_1 = \frac{-ay}{\sqrt{1-x^2}}$, so $\sqrt{1-x^2}\,y_1 = -ay$. Differentiate: $\sqrt{1-x^2}\,y_2 - \frac{x\,y_1}{\sqrt{1-x^2}} = -ay_1 = \frac{a^2 y}{\sqrt{1-x^2}}$. Multiply by $\sqrt{1-x^2}$: $(1-x^2)y_2 - xy_1 - a^2 y = 0$.

Q27.

Ans. $x^2 - x - 6 = (x-3)(x+2)$; $\frac{3x+5}{(x-3)(x+2)} = \frac{14/5}{x-3} + \frac{1/5}{x+2}$. Answer: $\frac{14}{5}\log|x-3| + \frac{1}{5}\log|x+2| + C$. **OR:** By parts twice: $x^2e^x - 2\int xe^x\,dx = x^2e^x - 2(x-1)e^x + C = e^x(x^2 - 2x + 2) + C$.

Q28.

Ans. IF $= e^{\int 2dx} = e^{2x}$. $ye^{2x} = \int e^{-x}e^{2x}dx = e^x + C$, so $y = e^{-x} + Ce^{-2x}$. **OR:** put $y = vx$: $v + x\frac{dv}{dx} = v + \tan v \Rightarrow \cot v\,dv = \frac{dx}{x}$. $\log|\sin v| = \log|x| + \log C$, so $\sin\frac{y}{x} = Cx$.

Q29.

Ans. $\vec{AB} = \hat{j} + 2\hat{k}$, $\vec{AC} = \hat{i} + 2\hat{j}$. $\vec{AB} \times \vec{AC} = -4\hat{i} + 2\hat{j} - \hat{k}$, magnitude $\sqrt{21}$. Area $= \frac{\sqrt{21}}{2}$ sq units. **OR:** $\vec{AB} = \hat{i} + 4\hat{j} - 4\hat{k} = \vec{BC}$: parallel with common point B, so collinear.

Q30.

Ans. Corner points (0, 0), (30, 0), (20, 30), (0, 50) [from $x + y = 50$ and $3x + y = 90$]. Z = 0, 120, 110, 50. Max Z = 120 at (30, 0). **OR:** Corners (0, 8), (2, 4), (10, 0); Z = 56, 38, 50. Region is unbounded; the half-plane $5x + 7y < 38$ has no point in common with it, so min Z = 38 at (2, 4).

Q31.

Ans. X = 0, 1, 2. $P(0) = \frac{6}{10}\times\frac{5}{9} = \frac{1}{3}$, $P(1) = 2\times\frac{4}{10}\times\frac{6}{9} = \frac{8}{15}$, $P(2) = \frac{4}{10}\times\frac{3}{9} = \frac{2}{15}$. Mean $= \frac{8}{15} + \frac{4}{15} = \frac{4}{5}$. **OR:** $P(R|\mathrm{I}) = \frac{3}{5}$, $P(R|\mathrm{II}) = \frac{5}{9}$, priors equal: $P(\mathrm{II}|R) = \frac{5/9}{3/5+5/9} = \frac{25}{52}$.

Section D

Q32.

Ans. $A = \begin{bmatrix}1 & 1 & 1\\ 1 & -1 & 1\\ 2 & 1 & -1\end{bmatrix}$., $|A| = 6 \neq 0$. $\mathrm{adj}\,A$

(contd.) $= \begin{bmatrix}0 & 2 & 2\\ 3 & -3 & 0\\ 3 & 1 & -2\end{bmatrix}$. $X = A^{-1}B =$

$\frac{1}{6}\begin{bmatrix}0 & 2 & 2\\ 3 & -3 & 0\\ 3 & 1 & -2\end{bmatrix}\begin{bmatrix}6\\2\\1\end{bmatrix} = \begin{bmatrix}1\\2\\3\end{bmatrix}$. So $x = 1, y = 2, z = 3$.

OR: $|A| = 6$. $\mathrm{adj}\,A = \begin{bmatrix}-3 & 3 & 3\\ 7 & -5 & -3\\ 5 & -1 & -3\end{bmatrix}$, $A^{-1} = \frac{1}{6}\mathrm{adj}\,A$.

$X = A^{-1}\begin{bmatrix}3\\6\\-1\end{bmatrix} = \begin{bmatrix}1\\-1\\2\end{bmatrix}$: $x = 1, y = -1, z = 2$.

Q33.

Ans. $I = \int_0^\pi \frac{(\pi-x)\sin x}{1+\cos^2 x}dx$ (property $f(a-x)$). Add: $2I = \pi\int_0^\pi \frac{\sin x}{1+\cos^2 x}dx$. Put $\cos x = t$: $2I = \pi\int_{-1}^{1}\frac{dt}{1+t^2} = \pi\times\frac{\pi}{2}$. $I = \frac{\pi^2}{4}$. **OR:** split at 2 and 3: $\int_1^2(4-x)dx + \int_2^3 x\,dx + \int_3^4(3x-6)dx = \frac{5}{2} + \frac{5}{2} + \frac{9}{2} = \frac{19}{2}$.

Q34.

Ans. Area $= 4\int_0^4 \frac{3}{4}\sqrt{16-x^2}\,dx =$ $3\left[\frac{x}{2}\sqrt{16-x^2} + 8\sin^{-1}\frac{x}{4}\right]_0^4 = 3\times 8\times\frac{\pi}{2} = 12\pi$ sq units. **OR:** $\int_1^4 2\sqrt{x}\,dx = \frac{4}{3}\left[x^{3/2}\right]_1^4 = \frac{4}{3}(8-1) = \frac{28}{3}$ sq units.

Q35.

Ans. $\vec{a}_2 - \vec{a}_1 = \hat{i} - 3\hat{j} - 2\hat{k}$, $\vec{b}_1 \times \vec{b}_2 = -3\hat{i} + 0\hat{j} + 3\hat{k}$, $|\vec{b}_1 \times \vec{b}_2| = 3\sqrt{2}$. SD $= \frac{|(-3)(1)+0+(3)(-2)|}{3\sqrt{2}} = \frac{9}{3\sqrt{2}} = \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$ units. **OR:** $\vec{r} = (2\hat{i} - \hat{j} + 3\hat{k}) + \lambda(3\hat{i} + 2\hat{j} - \hat{k})$; $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-3}{-1}$. $\cos\theta = \frac{|3+2-1|}{\sqrt{14}\sqrt{3}} = \frac{4}{\sqrt{42}}$, so $\theta = \cos^{-1}\frac{4}{\sqrt{42}}$.

Section E

Q36.

Ans. (i) $3x_1 - 5 = 3x_2 - 5 \Rightarrow x_1 = x_2$, one-one. (ii) $3x - 5 = 10 \Rightarrow x = 5$. (iii) For any real y, $x = \frac{y+5}{3}$ is real and $f(x) = y$: onto. One-one + onto = bijective. **OR:** (iii) $g(1) = g(-1) = 1$ but $1 \neq -1$, so not one-one.

Q37.

Ans. (i) Height $h = \frac{32}{x^2}$; $S = x^2 + 4xh = x^2 + \frac{128}{x}$. (ii) $\frac{dS}{dx} = 2x - \frac{128}{x^2}$. (iii) $\frac{dS}{dx} = 0 \Rightarrow x^3 = 64$, $x = 4$; $\frac{d^2S}{dx^2} = 2 + \frac{256}{x^3} = 6 > 0$, minimum. Base 4 m by 4 m, height 2 m, S = 48 m^2. **OR:** (iii) $S = x^2 + \frac{432}{x}$, $S' = 2x - \frac{432}{x^2} = 0 \Rightarrow x^3 = 216$, $x = 6$ m ($S'' = 2 + \frac{864}{x^3} > 0$).

Q38.

Ans. (i) $0.6\times 0.02 = 0.012$. (ii) $0.012 + 0.4\times 0.04 = 0.012 + 0.016 = 0.028$. (iii) $\frac{0.016}{0.028} = \frac{4}{7}$. **OR:** (iii) $0.6\times 0.98 = 0.588$.

Exam tip In an OR question, attempt only one option; the first one is checked.

Recall IF of $\frac{dy}{dx} + Py = Q$ is $e^{\int P\,dx}$.

Recall $|\mathrm{adj}A| = |A|^{n-1}$ and $|kA| = k^n|A|$ for an $n \times n$ matrix.

Traps in this paper (where students lose marks)

- Q15: $\frac{1-z}{3}$ is NOT in standard form; the DR of z is -3.
- Q21: principal value of $\sec^{-1}(-2)$ is $\frac{2\pi}{3}$, not $-\frac{\pi}{3}$.
- Q12: degree is defined only when the DE is a polynomial in derivatives; here it is 3.
- Q30 OR: unbounded region, so write the half-plane check before giving the minimum.
- Q33: use $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ first; direct by parts gets stuck.
- Q37: always show $\frac{d^2S}{dx^2} > 0$; "minimum" without it loses a mark.

Formulas used in this paper

- $\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$
- $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$
- SD between skew lines $= \left|\frac{(\vec{b}_1\times\vec{b}_2)\cdot(\vec{a}_2-\vec{a}_1)}{|\vec{b}_1\times\vec{b}_2|}\right|$
- $X = A^{-1}B$, $A^{-1} = \frac{1}{|A|}\text{adj}A$
- Projection of $\vec{a}$ on $\vec{b} = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$
- Bayes: $P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum P(E_j)P(A|E_j)}$
- Area of ellipse $= \pi ab$; $\frac{dA}{dt} = 2\pi r\frac{dr}{dt}$

Question map (chapter-wise)

Chapter	Q no.	Chapter	Q no.
Relations, Functions	1, 36	Integrals	11, 13, 24, 27, 33
Inverse Trig	2, 21	Appl. of Integrals	34
Matrices	3, 4, 5	Diff. Equations	12, 28
Determinants	6, 7, 32	Vectors	14, 19, 25, 29
Cont. and Diff.	8, 9, 22, 26	3D Geometry	15, 20, 35
Appl. of Derivatives	10, 23, 37	LPP	16, 17, 30
		Probability	18, 31, 38

Bonus practice (after Mock Paper 1): 8 quick checks

1. If $y = x^x$, find $\frac{dy}{dx}$. **Ans:** $x^x(1+\log x)$
2. Evaluate $\int_0^1 xe^x\,dx$. **Ans:** 1
3. A is 3×3 with $|A| = 4$. Find $|\text{adj}A|$. **Ans:** 16
4. Order and degree of $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^4 + y = 0$. **Ans:** order 2, degree 3
5. Projection of $2\hat{i}+3\hat{j}+\hat{k}$ on $\hat{i}+2\hat{j}+2\hat{k}$. **Ans:** $\frac{10}{3}$
6. $P(A) = 0.4$, $P(B|A) = 0.5$. Find $P(A\cap B)$. **Ans:** 0.2
7. Area under $y = x^2$ from $x = 0$ to $x = 3$ (above x-axis). **Ans:** 9 sq units
8. Corners $(0,0), (4,0), (2,2), (0,3)$. Max of $Z = 3x+4y$? **Ans:** 14 at $(2,2)$

After the paper: check yourself

- Mark your paper honestly with the key; give step marks the way the key shows.
- Note every wrong question's chapter; redo that chapter's Formula Box and MIQ.
- Section A wrong? Revise formulas. Section D wrong? Practise full written solutions.
- Time took more than 3 hours? Next time skip and come back to the hardest question.

Mock Paper 1: score tracker

Section	Max	My score	Time taken
A (MCQ + A-R)	20		
B (2 marks)	10		
C (3 marks)	18		
D (5 marks)	20		
E (case based)	12		
Total	**80**		

Recall Principal ranges: $\sin^{-1}$: $[-\frac{\pi}{2}, \frac{\pi}{2}]$, $\cos^{-1}$: $[0,\pi]$, $\tan^{-1}$: $(-\frac{\pi}{2}, \frac{\pi}{2})$.

Exam tip Leave 15 minutes at the end to recheck signs and units.

Apne notes / galtiyan yahan likho

Mock Paper 2

Max marks 80 | Time 3 hours | General instructions

1. This paper has 38 questions in 5 sections; all are compulsory.
2. **Section A:** Q1 to Q18 MCQs and Q19, Q20 Assertion-Reason, 1 mark each.
3. **Section B:** Q21 to Q25, very short answer, 2 marks each.
4. **Section C:** Q26 to Q31, short answer, 3 marks each.
5. **Section D:** Q32 to Q35, long answer, 5 marks each.
6. **Section E:** Q36 to Q38, case-based (1 + 1 + 2), 4 marks each.
7. Internal choice is given in some questions (marked OR). Calculators are not allowed.

↳ Pehle poora paper 10 min mein padh lo, phir apne strong section se shuru karo. Time plan: A 25 min, B 20, C 40, D 50, E 30, revision 15.

Marks blueprint (2026-27)

Unit	Chapters	Marks
I	Relations and Functions, Inverse Trigonometric Functions	8
II	Matrices, Determinants	10
III	Continuity and Differentiability, Application of Derivatives, Integrals, Application of Integrals, Differential Equations	35
IV	Vector Algebra, Three Dimensional Geometry	14
V	Linear Programming	5
VI	Probability	8
Total		**80**

Section-wise strategy (same as Mock 1)

- → **A:** 1 mark each, no working needed; eliminate options smartly.
- → **B, C:** write the formula first, then substitute; box the final answer.
- → **D:** 5 marks = 5 clear steps; draw the figure in area and LPP questions.
- → **E:** read the case once, then answer the parts; part (iii) carries 2 marks.

Section A (1 mark each)

Q1. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ is 1M
(a) one-one but not onto (b) onto but not one-one (c) neither one-one nor onto (d) bijective

Q2. The range of the principal value branch of $\cos^{-1} x$ is 1M
(a) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (b) $[0, \pi]$ (c) $(0, \pi)$ (d) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Q3. If $\begin{bmatrix} x+y & 2 \\ 5 & x-y \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 2 \end{bmatrix}$, then x is 1M
(a) 2 (b) 3 (c) 4 (d) 6

Q4. The diagonal elements of a skew-symmetric matrix are 1M
(a) all 1 (b) all 0 (c) equal (d) any real numbers

Q5. For matrices A and B for which AB is defined, $(AB)'$ equals 1M
(a) $A'B'$ (b) $B'A'$ (c) AB (d) BA

Q6. The area of the triangle with vertices (1, 0), (6, 0) and (4, 3) is 1M
(a) 15 (b) 7.5 (c) 10 (d) 5

Q7. If $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, then $|A^{-1}|$ is 1M
(a) 6 (b) $\frac{1}{6}$ (c) $\frac{1}{5}$ (d) 5

Q8. $f(x) = |x|$ at $x = 0$ is 1M
(a) continuous and differentiable (b) continuous but not differentiable (c) not continuous (d) differentiable but not continuous

Q9. If $y = \tan^{-1} x$, then $\frac{dy}{dx}$ at $x = 1$ is 1M
(a) 1 (b) $\frac{1}{2}$ (c) $\frac{\pi}{4}$ (d) 2

Q10. The maximum value of $\sin x + \cos x$ is 1M
(a) 1 (b) 2 (c) $\sqrt{2}$ (d) $\frac{1}{\sqrt{2}}$

Q11. $\int \frac{\sec^2 x}{\tan x}\, dx$ is 1M
(a) $\log|\sec x| + C$ (b) $\log|\tan x| + C$ (c) $\tan^2 x + C$ (d) $\log|\sin x| + C$

Q12. The integrating factor of $\frac{dy}{dx} - \frac{y}{x} = x^2$ $(x > 0)$ is 1M
(a) x (b) $\frac{1}{x}$ (c) e^x (d) $\log x$

Q13. $\int_0^1 xe^x\, dx$ is 1M
(a) 1 (b) e (c) $e - 1$ (d) 0

Q14. The projection of $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$ on $\vec{b} = 2\hat{i} + \hat{j} + 2\hat{k}$ is 1M
(a) 6 (b) 2 (c) $\frac{6}{\sqrt{6}}$ (d) 3

Q15. The angle between lines with direction ratios 1, 1, 0 and 0, 1, 1 is 1M
(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

Q16. Which point does NOT lie in the half-plane $2x + 3y \le 12$? 1M
(a) (1, 1) (b) (2, 2) (c) (3, 3) (d) (0, 4)

Recall IF of $\frac{dy}{dx} + Py = Q$ is $e^{\int P\, dx}$.

Q17. Corner points of a feasible region are (0, 2), (3, 0), (6, 0), (6, 8), (0, 5). The minimum of $Z = 4x + 6y$ is 1M

(a) 12, only at (0, 2) (b) 12, at every point of the segment joining (0, 2) and (3, 0) (c) 0 (d) 24

Q18. X takes values 0, 1, 2 with probabilities 0.3, 0.5, 0.2. Then $E(X)$ is 1M

(a) 1 (b) 0.9 (c) 0.7 (d) 1.2

Q19. **A:** $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|$ when $\vec{a} \perp \vec{b}$. **R:** $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Q20. **A:** Lines with direction ratios 1, 2, 3 and $-2, -4, -6$ are perpendicular. **R:** Two lines are parallel if their direction ratios are proportional. 1M

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false (d) A false, R true

Section B (2 marks each)

Q21. Find the principal value of $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) + \tan^{-1}(1)$.

OR

Find the principal value of $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) + \cot^{-1}\left(\frac{1}{\sqrt{3}}\right)$. 2M

Q22. If $x = at^2$ and $y = 2at$, find $\frac{dy}{dx}$.

OR

Find $\frac{dy}{dx}$ if $x^2 + xy + y^2 = 7$. 2M

Q23. Show that $f(x) = x^3 - 6x^2 + 12x - 18$ is increasing on $\mathbb{R}$.

OR

Find the interval in which $f(x) = x^2 - 4x + 6$ is strictly decreasing. 2M

Q24. Find $\int \frac{\cos 2x}{(\sin x + \cos x)^2}\,dx$.

OR

Find $\int \frac{dx}{x^2 + 4x + 8}$. 2M

Q25. Find λ if $\vec{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$ are perpendicular.

OR

Find the area of the parallelogram with adjacent sides $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = \hat{j} + \hat{k}$. 2M

Exam tip Write question numbers clearly in the margin, exactly as in the paper.

Section C (3 marks each)

Q26. If $y = (\tan^{-1} x)^2$, show that $(x^2+1)^2 \frac{d^2y}{dx^2} + 2x(x^2+1)\frac{dy}{dx} = 2$.

OR

Find $\frac{dy}{dx}$ if $x^y = e^{x-y}$. 3M

Q27. Find $\int x^2 \log x\,dx$.

OR

Find $\int \frac{x^2+1}{(x-1)^2(x+3)}\,dx$. 3M

Q28. Solve the differential equation $x\frac{dy}{dx} = x + y$.

OR

Find the particular solution of $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$, given $y = 1$ when $x = 0$. 3M

Q29. If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, $|\vec{a}| = 3$, $|\vec{b}| = 5$ and $|\vec{c}| = 7$, find the angle between $\vec{a}$ and $\vec{b}$.

OR

Using vectors, find the area of the triangle with vertices (1, 0, 0), (0, 1, 0) and (0, 0, 1). 3M

Q30. Solve graphically: minimise $Z = 2x + 3y$ subject to $x + 2y \geq 4$, $x + y \geq 3$, $x \geq 0$, $y \geq 0$.

OR

Solve graphically: maximise $Z = 3x + 2y$ subject to $x + 2y \leq 10$, $3x + y \leq 15$, $x \geq 0$, $y \geq 0$. 3M

Q31. $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{2}$ and $P(A \cap B) = \frac{1}{8}$. Find $P(A' \cap B')$ and check whether A and B are independent.

OR

Two cards are drawn one by one with replacement from a pack of 52. Find the probability distribution of the number of spades, and its mean. 3M

Section D (5 marks each)

Q32. Using the matrix method, solve: $x + 2y + z = 7$, $2x - y + z = 6$, $x + y - z = 0$.

OR

For $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$, find AB and use it to solve $x - y + z = 2$, $2x + y - 3z = -5$, $x + y + z = 6$. 5M

Q33. Evaluate $\int_0^{\pi/2} \frac{\sin^4 x}{\sin^4 x + \cos^4 x}\,dx$.

OR

Find $\int \frac{2x}{(x^2+1)(x^2+3)}\,dx$. 5M

Q34. Using integration, find the area of the region in the first quadrant bounded by the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$, the x-axis and the lines $x=0$ and $x=3$. 5M

Q35. Find the shortest distance between the lines $\vec{r}=(\hat{i}+\hat{j})+\lambda(2\hat{i}-\hat{j}+\hat{k})$ and $\vec{r}=(2\hat{i}+\hat{j}-\hat{k})+\mu(3\hat{i}-5\hat{j}+2\hat{k})$.

OR

Find the distance between the parallel lines $\vec{r}=(\hat{i}+\hat{j})+\lambda(2\hat{i}-\hat{j}+2\hat{k})$ and $\vec{r}=(2\hat{i}+\hat{j}-\hat{k})+\mu(2\hat{i}-\hat{j}+2\hat{k})$. 5M

Section E (case based, 4 marks each)

Q36. Case: Number game. In a class game, the relation $R=\{(a,b):|a-b| \text{ is even}\}$ is defined on $A=\{1,2,3,4,5,6\}$. (i) Is $(2,4)\in R$? (1) (ii) Is R reflexive? (1) (iii) Show that R is an equivalence relation and write the equivalence class [1]. (2) **OR** (iii) Write all the equivalence classes of R. (2) 4M

Q37. Case: Riverside field. A farmer has 200 m of fencing for a rectangular field along a straight river; no fence is needed on the river side. Let x m be each side perpendicular to the river. (i) Express the area A in terms of x. (1) (ii) Find $\frac{dA}{dx}$. (1) (iii) Find the maximum area and the dimensions. (2) **OR** (iii) Show that A is maximum when the side along the river is twice x. (2) 4M

Q38. Case: Test preparation. In a class, 70% of the students prepared for a test. A prepared student passes with probability 0.9 and an unprepared one with probability 0.4. (i) Find P(student prepared and passed). (1) (ii) Find P(student passed). (1) (iii) A student passed; find the probability that the student had prepared. (2) **OR** (iii) A student failed; find the probability that the student had prepared. (2) 4M

Meme samjho



Jab mock mein $\frac{1-z}{3}$ wala DR ka sign galat ho gaya aur poora 1 mark chala gaya.

Asli baat: Line ko standard form $\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$ mein laake hi DRs padho. Har mock ke baad apni galtiyon ki list banao.

Answer Key (with key steps)

Section A

Q1.
Ans. (c) $f(1)=f(-1)$, so not one-one; -1 has no pre-image, so not onto

Q2.
(contd.) (b) $[0,\pi]$

Q3.
Ans. (c) $x+y=6, x-y=2\Rightarrow x=4$

Q4.
Ans. (b) $a_{ii}=-a_{ii}\Rightarrow a_{ii}=0$

Q5.
Ans. (b) reversal law $(AB)'=B'A'$

Q6.
Ans. (b) $\frac{1}{2}|1(0-3)-0+1(18-0)|=\frac{15}{2}$

Q7.
Ans. (b) $|A^{-1}|=\frac{1}{|A|}=\frac{1}{6}$

Q8.
Ans. (b) sharp corner at 0: LHD $=-1$, RHD $=1$

Q9.
Ans. (b) $\frac{1}{1+x^2}=\frac{1}{2}$

Q10.
Ans. (c) $\sqrt{2}\sin\left(x+\frac{\pi}{4}\right)\le\sqrt{2}$

Q11.
Ans. (b) put $t=\tan x$: $\int\frac{dt}{t}$

Q12.
Ans. (b) $e^{\int-\frac{1}{x}dx}=\frac{1}{x}$

Q13.
Ans. (a) $[(x-1)e^x]_0^1=0-(-1)=1$

Q14.
Ans. (b) $\frac{2+2+2}{3}=2$

Q15.
Ans. (c) $\cos\theta=\frac{1}{\sqrt{2}\sqrt{2}}=\frac{1}{2}$

Q16.
Ans. (c) $6+9=15>12$

Q17.
Ans. (b) Z = 12, 12, 24, 72, 30: same min at two corners

Q18.
Ans. (b) $0+0.5+0.4=0.9$

Q19.
Ans. (a) $\sin 90^\circ=1$; R explains A

Q20.
Ans. (d) A false (they are parallel), R true

Section B

Q21.
Ans. $\frac{5\pi}{6}+\frac{\pi}{4}=\frac{13\pi}{12}$. **OR:** $-\frac{\pi}{6}+\frac{\pi}{3}=\frac{\pi}{6}$.

Q22.
Ans. $\frac{dy}{dt}=2a, \frac{dx}{dt}=2at$, so $\frac{dy}{dx}=\frac{1}{t}$. **OR:** $2x+y+xy'+2yy'=0\Rightarrow y'=-\frac{2x+y}{x+2y}$.

Q23.

(contd.) $f'(x) = 3x^2 - 12x + 12 = 3(x-2)^2 \geq 0$ for all real x (zero only at $x = 2$), so f is increasing on $\mathbb{R}$. **OR:** $f'(x) = 2x - 4 < 0 \Rightarrow x < 2$: strictly decreasing on $(-\infty, 2)$.

Q24.

Ans. $\frac{\cos 2x}{(\sin x + \cos x)^2} = \frac{\cos x - \sin x}{\cos x + \sin x}$; answer $\log|\sin x + \cos x| + C$. **OR:** $\int \frac{dx}{(x+2)^2+4} = \frac{1}{2}\tan^{-1}\frac{x+2}{2} + C$.

Q25.

Ans. $\vec{a}\cdot\vec{b} = 2 - 2\lambda + 3 = 0 \Rightarrow \lambda = \frac{5}{2}$. **OR:** $\vec{a}\times\vec{b} = \hat{i} - \hat{j} + \hat{k}$; area $= \sqrt{3}$ sq units.

Section C

Q26.

Ans. $y_1 = \frac{2\tan^{-1}x}{1+x^2}$, so $(1+x^2)y_1 = 2\tan^{-1}x$. Differentiate: $(1+x^2)y_2 + 2xy_1 = \frac{2}{1+x^2}$. Multiply by $(1+x^2)$: $(x^2+1)^2 y_2 + 2x(x^2+1)y_1 = 2$. **OR:** $y\log x = x - y \Rightarrow y = \frac{x}{1+\log x}$, $\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$.

Q27.

Ans. By parts ($u = \log x$): $\frac{x^3}{3}\log x - \int \frac{x^3}{3}\times\frac{1}{x}dx = \frac{x^3}{3}\log x - \frac{x^3}{9} + C$. **OR:** $\frac{x^2+1}{(x-1)^2(x+3)} = \frac{3/8}{x-1} + \frac{1/2}{(x-1)^2} + \frac{5/8}{x+3}$; answer $\frac{3}{8}\log|x-1| - \frac{1}{2(x-1)} + \frac{5}{8}\log|x+3| + C$.

Q28.

Ans. Homogeneous: $y = vx$, $v + x\frac{dv}{dx} = 1 + v \Rightarrow dv = \frac{dx}{x}$, $v = \log|x| + C$, so $y = x\log|x| + Cx$. **OR:** $\tan^{-1}y = \tan^{-1}x + C$; $x = 0, y = 1$ gives $C = \frac{\pi}{4}$; $y = \frac{1+x}{1-x}$.

Q29.

Ans. $\vec{a} + \vec{b} = -\vec{c} \Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a}\cdot\vec{b} = |\vec{c}|^2$: $9 + 25 + 2\vec{a}\cdot\vec{b} = 49$, $\vec{a}\cdot\vec{b} = \frac{15}{2}$. $\cos\theta = \frac{15/2}{3\times5} = \frac{1}{2}$, $\theta = \frac{\pi}{3}$. **OR:** $\vec{AB} = -\hat{i} + \hat{j}$, $\vec{AC} = -\hat{i} + \hat{k}$, $\vec{AB}\times\vec{AC} = \hat{i} + \hat{j} + \hat{k}$; area $= \frac{\sqrt{3}}{2}$ sq units.

Q30.

Ans. Corners (4, 0), (2, 1), (0, 3); Z = 8, 7, 9. Region unbounded; $2x + 3y < 7$ has no common point with it. Min Z = 7 at (2, 1). **OR:** Corners (0, 0), (5, 0), (4, 3), (0, 5); Z = 0, 15, 18, 10. Max Z = 18 at (4, 3).

Q31.

Ans. $P(A\cup B) = \frac{1}{4} + \frac{1}{2} - \frac{1}{8} = \frac{5}{8}$, so $P(A'\cap B') = \frac{3}{8}$. $P(A)P(B) = \frac{1}{8} = P(A\cap B)$: independent. **OR:** $p = \frac{1}{4}$: $P(0) = \frac{9}{16}$, $P(1) = \frac{6}{16}$, $P(2) = \frac{1}{16}$; mean $= \frac{6+2}{16} = \frac{1}{2}$.

Section D

Q32.

Ans. $|A| = 9 \neq 0$. $\operatorname{adj}A = \begin{bmatrix} 0 & 3 & 3 \\ 3 & -2 & 1 \\ 3 & 1 & -5 \end{bmatrix}$, $A^{-1} = \frac{1}{9}\operatorname{adj}A$. $X = A^{-1}B = \frac{1}{9}(18,\ 9,\ 27)$, so $x = 2$, $y = 1$, $z = 3$. **OR:** $AB = 10I$, so $A^{-1} = \frac{1}{10}B$; $X = \frac{1}{10}B(2,\ -5,\ 6)^T = \frac{1}{10}(10,\ 20,\ 30)$: $x = 1, y = 2$, (contd.) $z = 3$.

Q33.

Ans. Using $f\left(\frac{\pi}{2} - x\right)$: $I = \int_0^{\pi/2}\frac{\cos^4 x}{\cos^4 x + \sin^4 x}dx$. Add: $2I = \int_0^{\pi/2} 1\,dx = \frac{\pi}{2}$, so $I = \frac{\pi}{4}$. **OR:** put $t = x^2$: $\int\frac{dt}{(t+1)(t+3)} = \frac{1}{2}\log\left|\frac{t+1}{t+3}\right| + C = \frac{1}{2}\log\frac{x^2+1}{x^2+3} + C$.

Q34.

Ans. $y = \frac{4}{5}\sqrt{25 - x^2}$. Area $= \frac{4}{5}\int_0^3\sqrt{25-x^2}dx = \frac{4}{5}\left[\frac{x}{2}\sqrt{25-x^2} + \frac{25}{2}\sin^{-1}\frac{x}{5}\right]_0^3 = \frac{4}{5}\left(6 + \frac{25}{2}\sin^{-1}\frac{3}{5}\right) = \frac{24}{5} + 10\sin^{-1}\frac{3}{5}$ sq units.

Q35.

Ans. $\vec{b}_1\times\vec{b}_2 = 3\hat{i} - \hat{j} - 7\hat{k}$, $|\vec{b}_1\times\vec{b}_2| = \sqrt{59}$, $\vec{a}_2 - \vec{a}_1 = \hat{i} - \hat{k}$. SD $= \frac{|3+0+7|}{\sqrt{59}} = \frac{10}{\sqrt{59}}$ units. **OR:** $d = \frac{|\vec{b}\times(\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|} = \frac{|\hat{i}+4\hat{j}+\hat{k}|}{3} = \frac{\sqrt{18}}{3} = \sqrt{2}$ units.

Section E

Q36.

Ans. (i) $|2-4| = 2$, even: yes. (ii) $|a-a| = 0$ is even for all a: reflexive. (iii) Symmetric: $|a-b| = |b-a|$. Transitive: $a - c = (a-b) + (b-c)$ is even. So R is an equivalence relation; $[1] = \{1, 3, 5\}$. **OR:** (iii) $\{1, 3, 5\}$ and $\{2, 4, 6\}$.

Q37.

Ans. (i) $A = x(200 - 2x) = 200x - 2x^2$. (ii) $\frac{dA}{dx} = 200 - 4x$. (iii) $x = 50$; $\frac{d^2A}{dx^2} = -4 < 0$, maximum. Field 50 m by 100 m, area 5000 m^2. **OR:** at the maximum, river side $= 200 - 2(50) = 100 = 2x$.

Q38.

Ans. (i) $0.7\times0.9 = 0.63$. (ii) $0.63 + 0.3\times0.4 = 0.75$. (iii) $\frac{0.63}{0.75} = \frac{21}{25}$. **OR:** (iii) $P(\text{fail}) = 0.07 + 0.18 = 0.25$; $\frac{0.07}{0.25} = \frac{7}{25}$.

Traps in this paper (where students lose marks)

- → Q8: $|x|$ is continuous at 0 but NOT differentiable (LHD ≠ RHD).
- → Q17: equal minimum at two corners means every point of the joining segment is optimal.
- → Q20: DRs 1, 2, 3 and $-2, -4, -6$ are proportional, so the lines are parallel, not perpendicular.
- → Q21: $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \frac{5\pi}{6}$, not $-\frac{\pi}{6}$ (range is $[0, \pi]$).
- → Q30: unbounded region: write the half-plane check before stating the minimum.
- → Q34: area of only part of the ellipse; do not use πab here.

Exam tip In an OR question, attempt only one option; the first one is checked.

Formulas used in this paper

- $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$
- $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$
- Distance between parallel lines $= \dfrac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$
- Area of parallelogram $= |\vec{a} \times \vec{b}|$
- Area of triangle $= \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$
- Linear DE: $y \times \text{IF} = \int Q \times \text{IF}\,dx + C$
- $P(A' \cap B') = 1 - P(A \cup B)$; mean $= \sum x_i p_i$

Question map (chapter-wise)

Chapter	Q no.	Chapter	Q no.
Relations, Functions	1, 36	Integrals	11, 13, 24, 27, 33
Inverse Trig	2, 21	Appl. of Integrals	34
Matrices	3, 4, 5	Diff. Equations	12, 28
Determinants	6, 7, 32	Vectors	14, 19, 25, 29
Cont. and Diff.	8, 9, 22, 26	3D Geometry	15, 20, 35
Appl. of Derivatives	10, 23, 37	LPP	16, 17, 30
		Probability	18, 31, 38

After the paper: check yourself

- Mark your paper honestly with the key; give step marks the way the key shows.
- Note every wrong question's chapter; redo that chapter's Formula Box and MIQ.
- Section A wrong? Revise formulas. Section D wrong? Practise full written solutions.
- Time took more than 3 hours? Next time skip and come back to the hardest question.

Bonus practice (after Mock Paper 2): 8 quick checks

1. If $y = \log(\sin x)$, find $\frac{dy}{dx}$. **Ans:** $\cot x$
2. Evaluate $\int_{-\pi/2}^{\pi/2} \sin^3 x\,dx$. **Ans:** 0 (odd function)
3. Find A^{-1} for $A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$. **Ans:** $\begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$
4. I.F. of $\frac{dy}{dx} + \frac{y}{x} = x^2$ $(x > 0)$. **Ans:** x
5. $|\vec{a}| = 2$, $|\vec{b}| = 3$, angle $\frac{\pi}{6}$. Find $|\vec{a} \times \vec{b}|$. **Ans:** 3
6. Two dice are thrown. P(sum is 7)? **Ans:** $\frac{1}{6}$
7. Principal value of $\sin^{-1}\left(-\frac{1}{2}\right)$. **Ans:** $-\frac{\pi}{6}$
8. Interval where $f(x) = x^2 - 4x$ is strictly increasing. **Ans:** $(2, \infty)$

Mock Paper 2: score tracker

Section	Max	My score	Time taken
A (MCQ + A-R)	20		
B (2 marks)	10		
C (3 marks)	18		
D (5 marks)	20		
E (case based)	12		
Total	**80**		

Yaad karo: last 3 days ki priority

Priority	Chapters	Why
1	Integrals, Continuity and Differentiability, Application of Derivatives	Calculus = 35 marks, har section mein
2	Vectors, 3D lines, Differential Equations	fixed-pattern 5 and 3 markers
3	Matrices, Determinants, Probability (Bayes)	sure-shot long answers
4	Relations, Inverse Trig, LPP, Area	short, scoring, quick revision



Shivam Bhaiya says

Dono mock 3 ghante ke timer pe, bina answer dekhe. Galat sawal chapter mein wapas jaake solve karo. Board se pehle ye book 3 baar: samjho, solve karo, revise karo. All the best!

Apne notes / galtiyan yahan likho

Class 12 Maths One Shot

Khatam in 199 Pages | Handwritten Notes | CBSE Session 2026-27

Is notebook mein kya hai

- ✓ All 13 chapters of CBSE Class 12 Maths, 2026-27 rationalised syllabus
- ✓ Handwritten-style notes, definitions, results and clean graphs, 3D and LPP figures
- ✓ Memes + "Shivam Bhaiya says" tips to keep it light
- ✓ Formula Box in every chapter + a full Formula Sheet
- ✓ Solved examples with steps, "Yaad karo" tricks, common mistakes
- ✓ Most Important Questions (MCQ, A-R, short, long, case based) with solutions
- ✓ Formula Sheet, Most Asked Questions Bank, last-night revision + 2 full mock papers with answers



Shivam Bhaiya says

Class 12 Maths mein 95+ practice se aata hai. Is notebook ko 3 baar karo: pehli baar samjho, doosri baar saare examples khud solve karo, teesri baar sirf Formula Sheet, "Yaad karo" boxes aur questions. All the best!

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